THE BEATO BOOK 4.0

A CREATIVE APPROACH TO MUSIC THEORY AND IMPROVISATION FOR CONTEMPORARY INSTRUMENTS

BY NICK BEATO

THE BEATO BOOK

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This book is dedicated to my Mom and Dad.

Special thanks to my wife Nina and our three beautiful children Dylan, Lennon and Layla; my brothers and sisters Pat, Nancy, Mike, Lou, Ray and Jon; and my teachers and friends Steve Brown, Mick Goodrick, Glen Cummings, Paul Smith, Pat Ryan, Tom Wadsworth, Keith Williams and Ken Lanyon.

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I would also like to thank all of you out there that have supported me through this journey!

Rick

Author's Note

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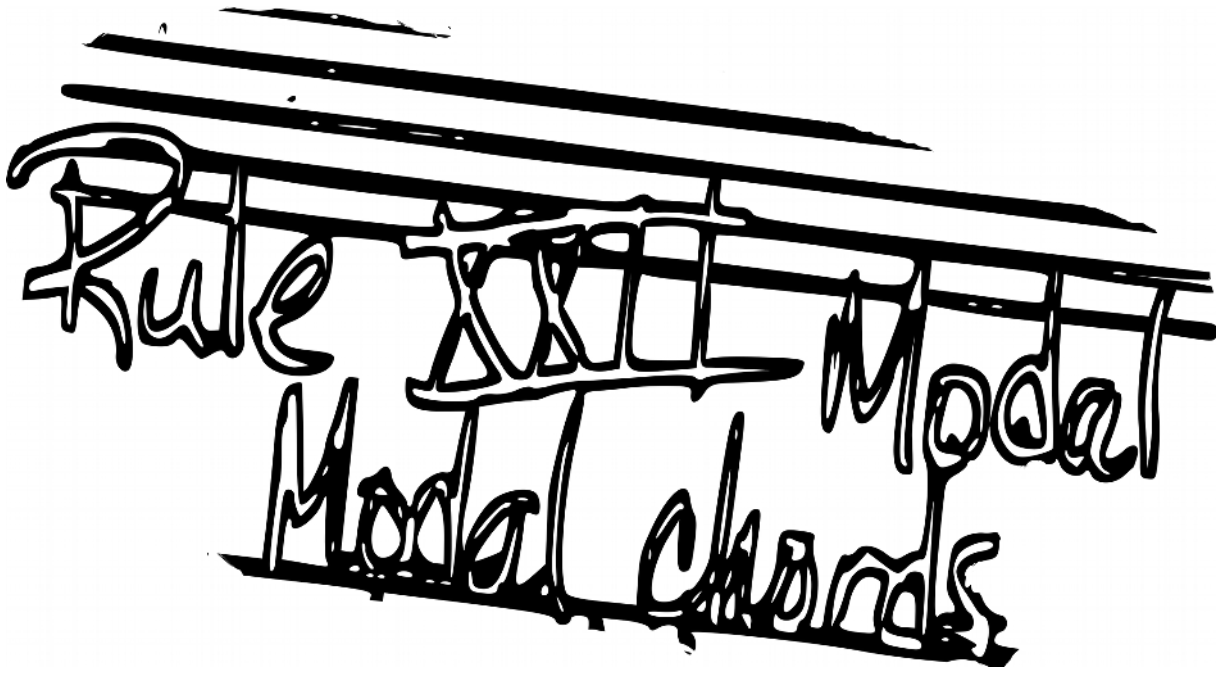
Chapter 1

Theory and Harmony



Chapter 1: Theory and Harmony

A thorough understanding of intervals is of the utmost importance in studying all types of music, as intervals are the building blocks of polyphonic music.

Each interval has vastly different sound characteristics and must be committed to memory aurally and visually.

Intervals can be sounded together (harmonically) or sequentially (melodically).

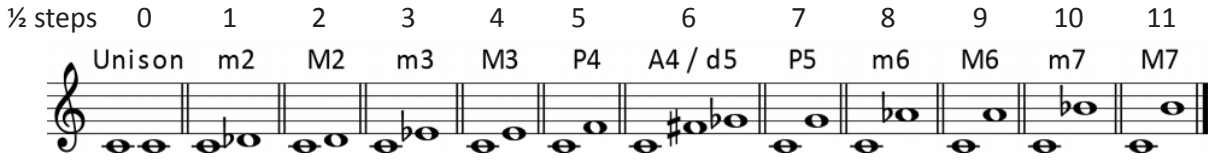
There are twelve intervals in the space of an octave.

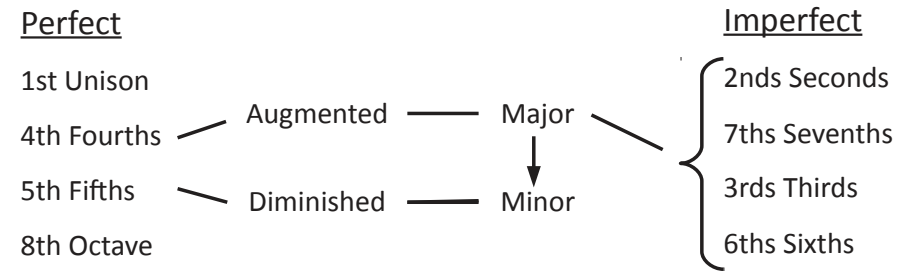
Interval Names and Symbols

	Unison	m2	M2	m3	M3	P4	A4/d5	P5	m6	M6	m7	M7	P8
½ steps	0	1	2	3	4	5	6	7	8	9	10	11	12

Key: m = minor
M = Major
A = Augmented
d = diminished
P = Perfect

Ex. 1





Perfect Intervals: Octave, Unison, Fifth, Fourth

become augmented when enlarged by a half step
become diminished when reduced by a half step

Imperfect Intervals: Second, Seventh, Third, Sixth

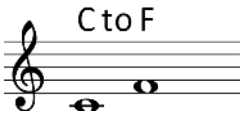
1) Major Intervals
   become augmented when enlarged by a half step
   become minor when reduced by a half step

2) Minor Intervals
   become major when enlarged by a half step
   become diminished when reduced by a half step

Naming Intervals

To precisely identify an interval, generic interval classification must be made based on the number of letter names spanned. For Example:

Ex. 2



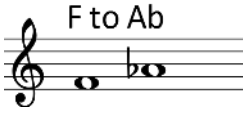
letter names

C	D	E	F
1	2	3	4

a fourth = generic classification

4 letters spanned

Ex. 3



letter names

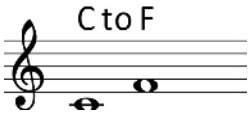
F	G	A
1	2	3

a third = generic classification

3 letters spanned

Once the generic classification has been determined, a more precise description (e.g., perfect, major, minor, diminished or augmented) can be made, based on the number of steps between the two pitches.

Ex. 4



C C# D D# E F

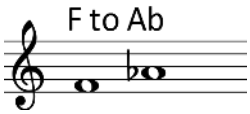
v v v v v

1 2 3 4 5

½ steps spanned

5 half steps = P4 perfect fourth

Ex. 5



F F# G Ab

v v v

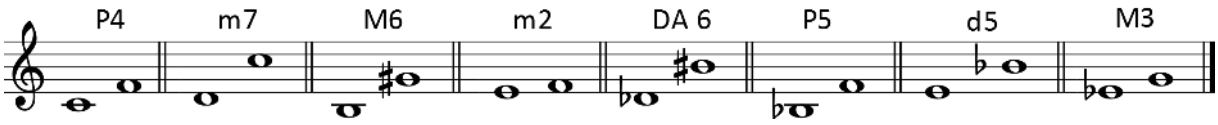
1 2 3

½ steps spanned

3 half steps = m3 minor third

Ex. 6

Here are some more examples of intervals:

(D=Doubly)

Enharmonic Intervals

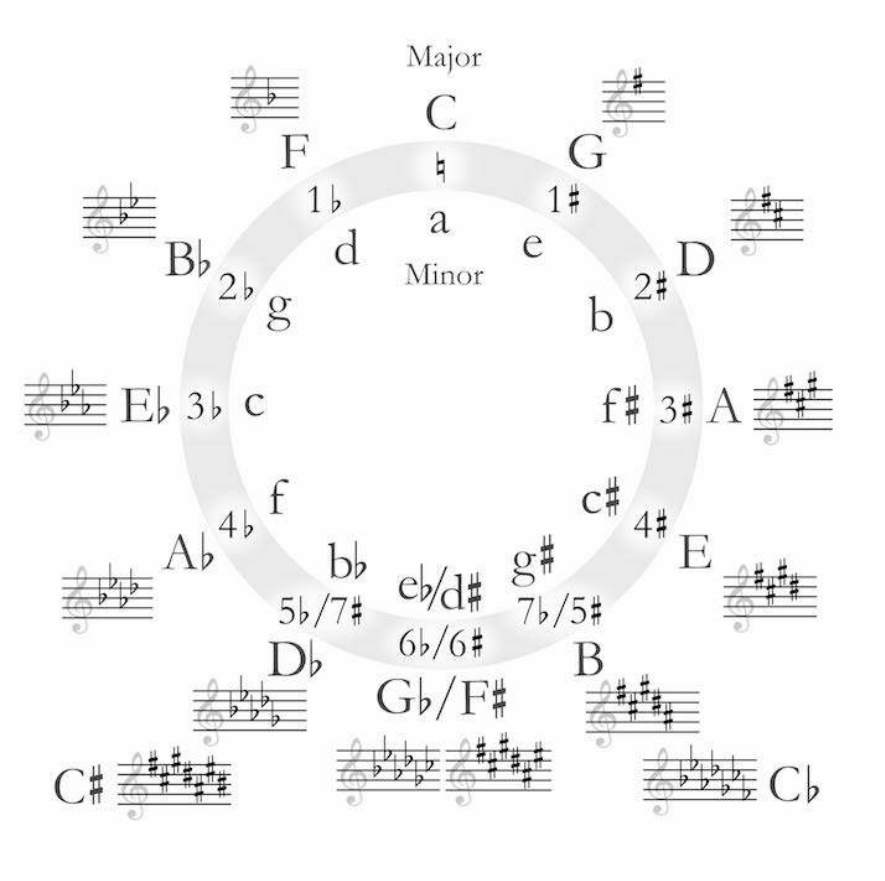
Interval types that contain an equivalent number of half steps but have different names (e.g., Augmented Second and Minor Third) are **enharmonically equivalent**. In other words, they are aurally identical but theoretically different.

Interval Characteristics

In Western Music, the tonal relationships within the range of one octave have been reduced to twelve equal intervals, the twelve notes of which comprise the chromatic scale, through the system of equal temperament. A chart of intervals comprised in an octave is below:

Interval	Characteristic
m2	Sharp Dissonance
M2	Mild Dissonance
m3	Soft Consonance
M3	Soft Consonance
P4	Consonance or Dissonance
TT	Neutral or Restless
P5	Open Consonance
m6	Soft Consonance
M6	Soft Consonance
m7	Mild Dissonance
M7	Sharp Dissonance
P8	Open Consonance

The Circle of Fifths



Order of Flats

Number of Flats - 0 1 2 3 4 5 6 7
Key - C F Bb Eb Ab Db Gb Cb

Order of Sharps

Number of Sharps - 0 1 2 3 4 5 6 7
Key - C G D A E B F# C#

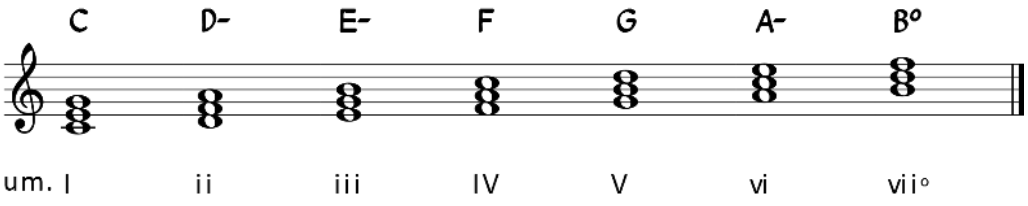
Chords and Their Formulas

Though there are many systems from which to build chords, tertian (third intervals) harmony is the most common in western music.

The stacking of two third intervals produces triads, the fundamental structures in tertian harmony. If these triads originate from one particular key center (signature) they are called diatonic triads. Triads may be built on every note of a particular scale. A seven note scale yields the same number of diatonic triads, not including suspensions which we will discuss later.

Diatonic Triads in C major

Ex. 8



Roman Numeral Symbols

upper case = major triad

lower case = minor triad

° = diminished

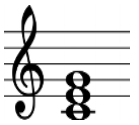
\+ = augmented

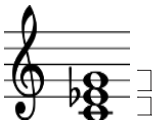
The Roman Numeral system employed has evolved for analysis of chord progressions and their relationship to a relative tonal center. Roman Numerals will be thoroughly discussed in an ensuing segment.

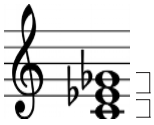
Building Diatonic Triads

The four basic diatonic triad categories are: major, minor, diminished and augmented

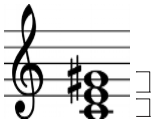
Interval Composition

Ex. 9

= C major = 1 3 5

Ex. 10

= C minor = 1 b3 5

Ex. 11

= C dim = 1 b3 b5

Ex. 12

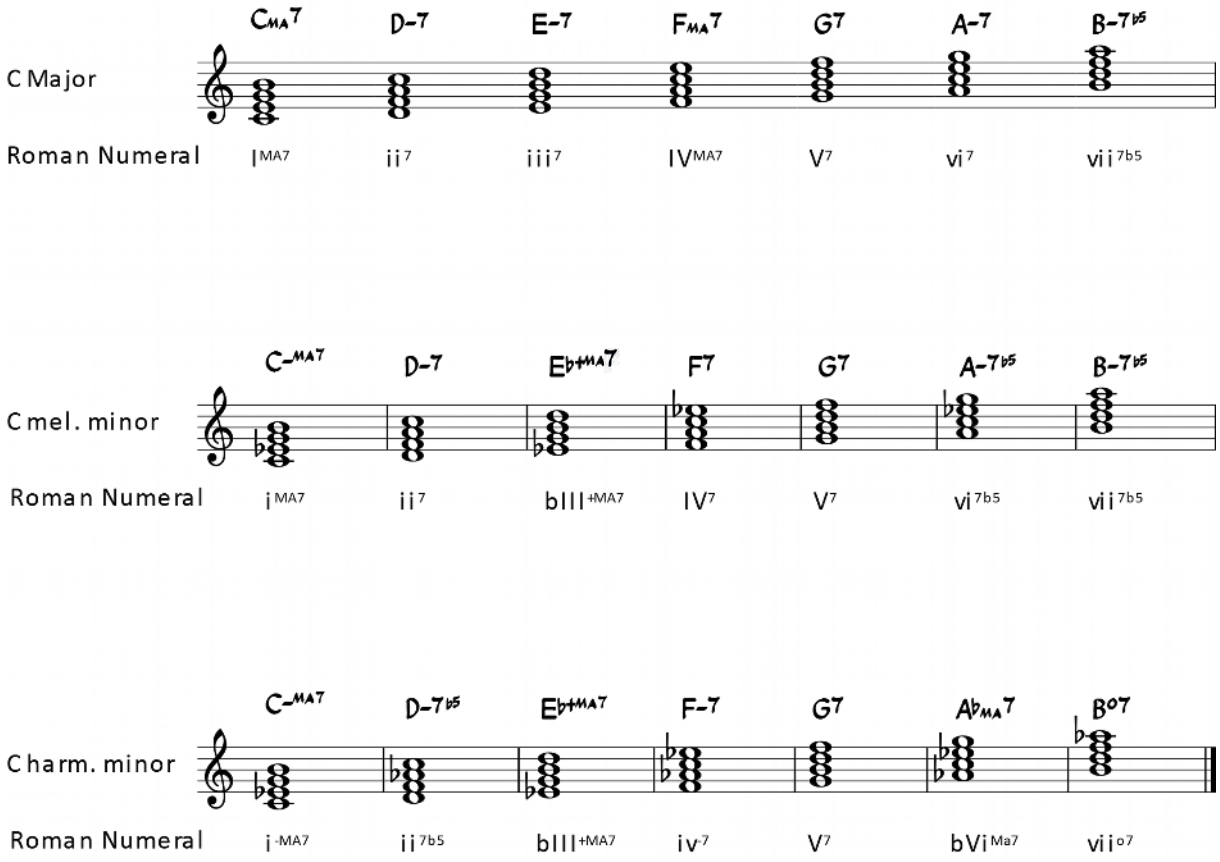


= C aug = 1 3 #5

Building Diatonic Seventh Chords

Seventh Chords are composed of three thirds stacked vertically.

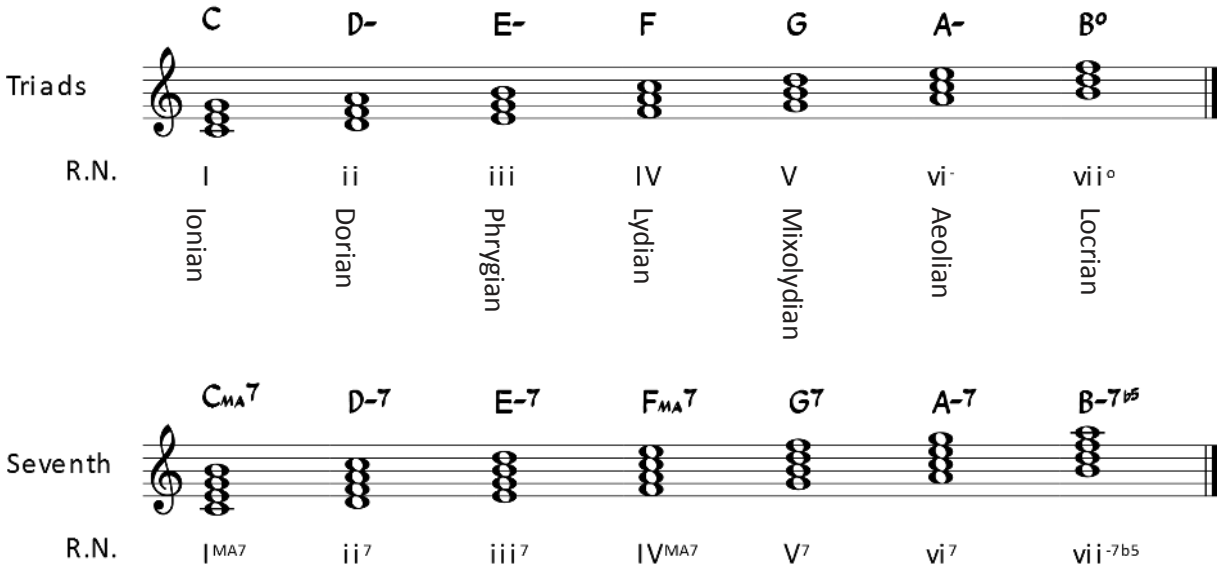
Ex. 13



The following reference chart includes both triad and seventh chords from major, melodic minor and harmonic minor. Roman numeral description and modal scale relationships have been included for study. "C" tonalities have been used in all instances.

Major Scale (Triads & Sevenths)

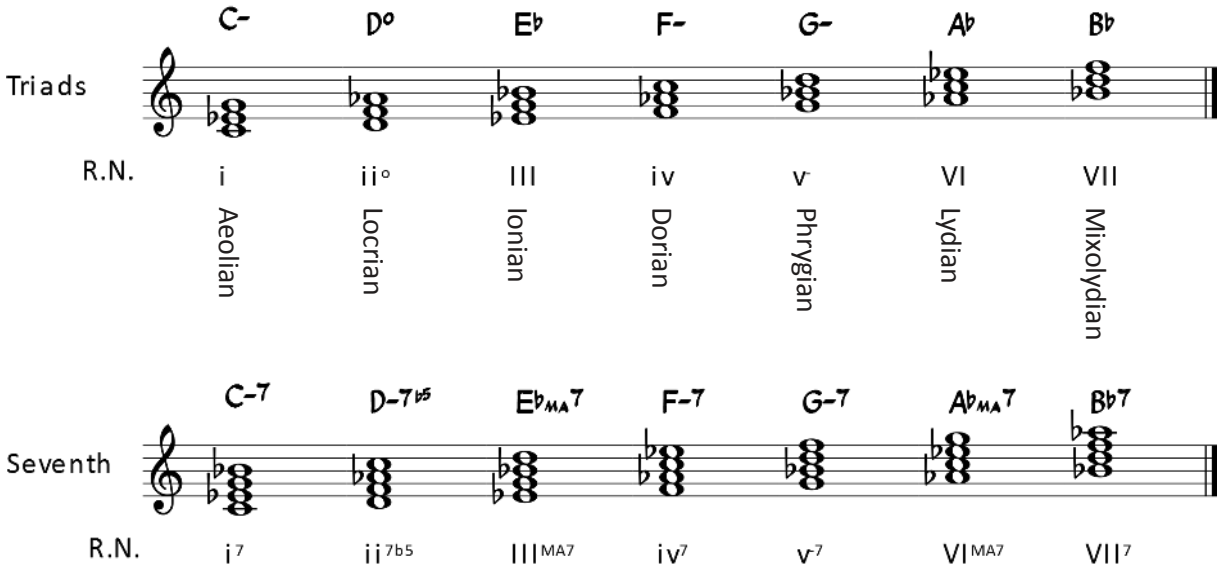
Ex. 14a



Triads	maj	=	I, IV, V
	min	=	ii, iii, vi
	dim	=	vii^{o}
Seventh	maj^{7}	=	I^{maj7}, IV^{maj7}
	min^{7}	=	ii^{7}, iii^{7}, vi^{7}
	dom^{7}	=	V^{7}
	min^{7b5}	=	vii^{7b5}

Natural Minor Scale

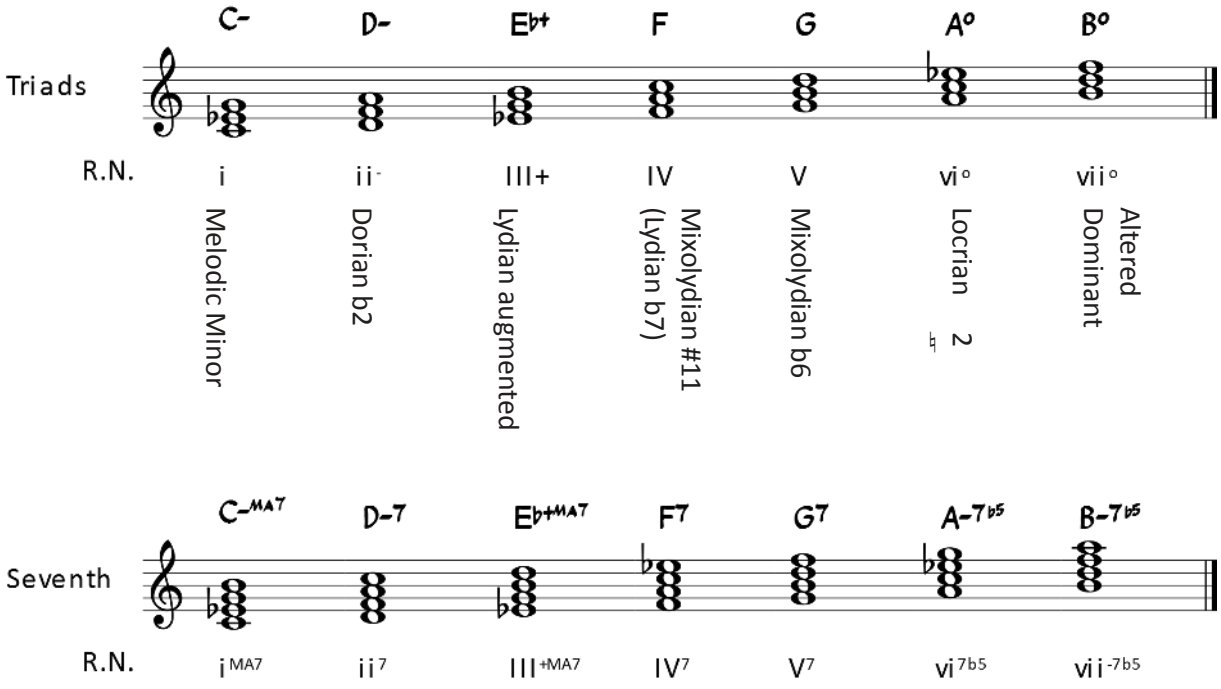
Ex. 14b



Triads	maj	=	III, VI, VII
	min	=	i, iv, v
	dim	=	ii^{o}
Seventh	maj^{7}	=	III^{maj7}, VI^{maj7}
	min^{7}	=	i^{7}, iv^{7}, v^{7}
	dom^{7}	=	VII^{7}
	min^{7b5}	=	ii^{7b5}

Melodic Minor Scale

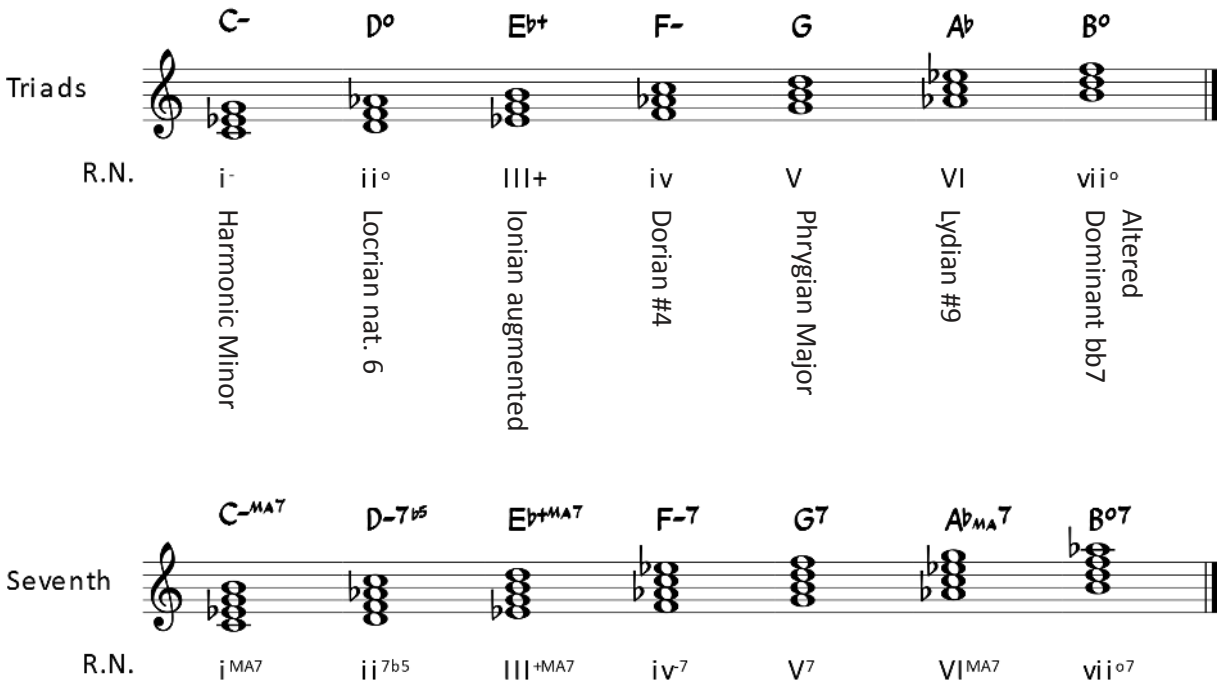
Ex. 15



Triads	maj	=	IV, V
	min	=	i, ii
	dim	=	vi^{o}, vii^{o}
	aug	=	III^{+}
Seventh	min^{maj7}	=	i^{maj7}
	min^{7}	=	ii
	dom^{7}	=	$IV^{7,}$, V^{7}
	min^{7b5}	=	vi^{7b5}, vii^{7b5}
	aug^{maj7}	=	III^{+maj7}

Harmonic Minor Scale

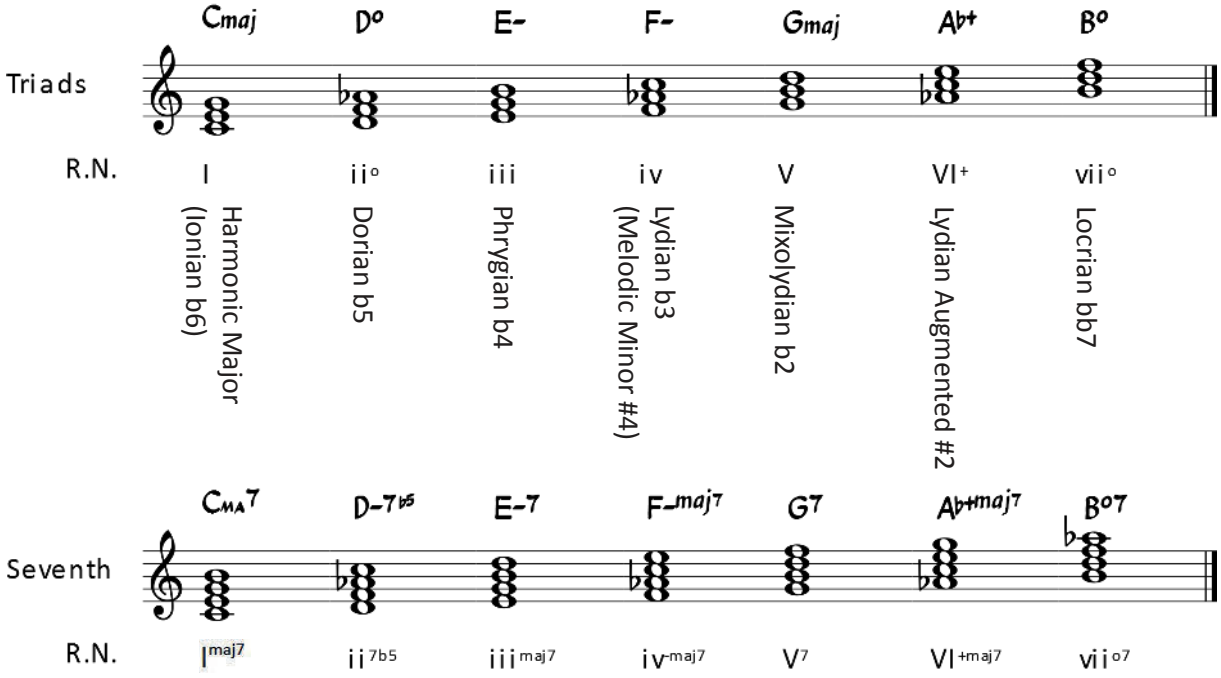
Ex. 16



Triads	maj	=	V, VI
	min	=	i, iv
	dim	=	ii°, vii°
	aug	=	III+
Seventh	maj7	=	VImaj7
	minmaj7	=	imaj7
	min7	=	iv7
	dom7	=	V7
	min7b5	=	ii7b5
	augmaj7	=	III+maj7
	dim7	=	vii°7

Harmonic Major Scale

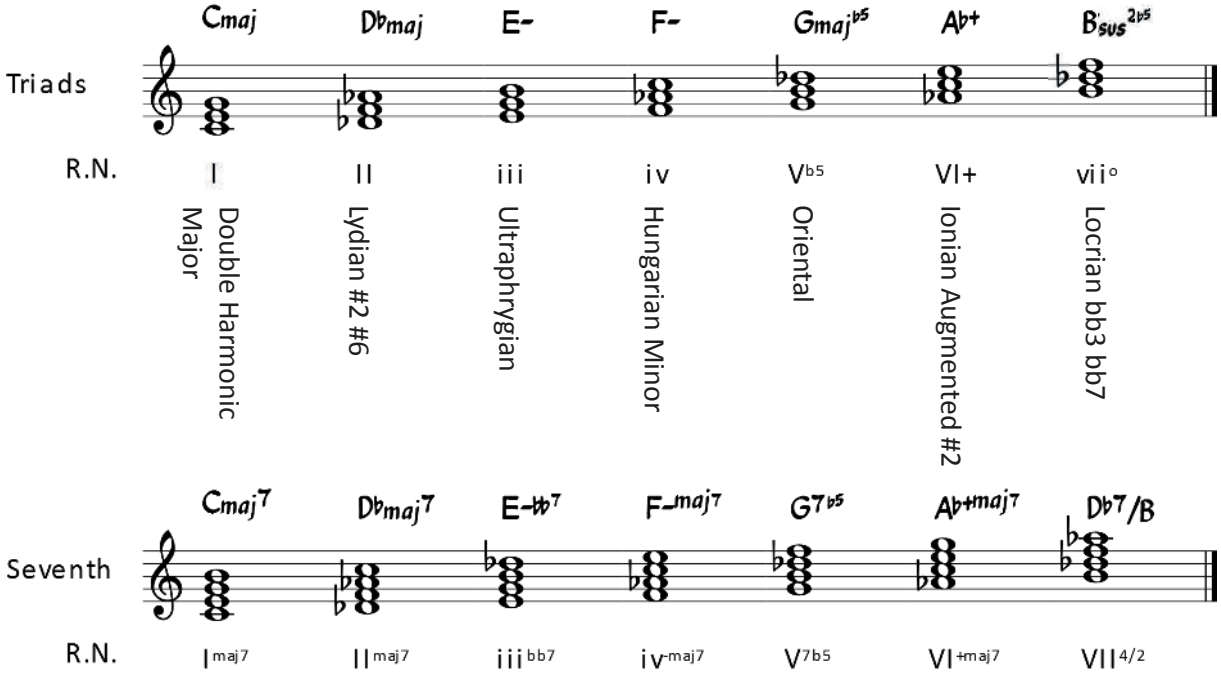
Ex. 17



Triads	maj	=	I, V
	min	=	iii, iv
	dim	=	ii°, vii°
	aug	=	VI+
Seventh	maj7	=	Imaj7
	minmaj7	=	ivmaj7
	min7	=	iii7
	dom7	=	V7
	min7b5	=	ii7b5
	augmaj7	=	VI+maj7
	dim7	=	vii°7

Double Harmonic Major Scale

Ex. 18

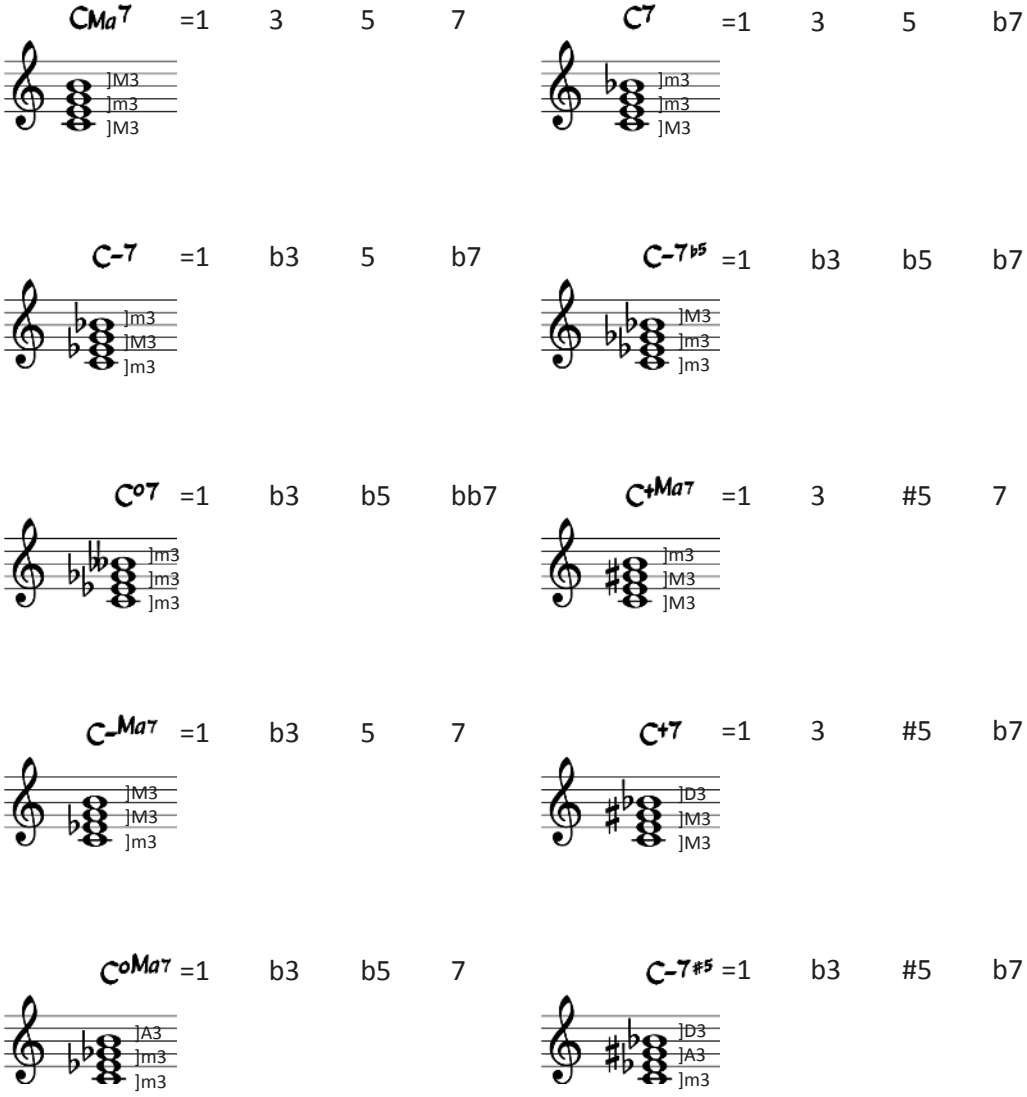


Triads	maj	=	I, II
	maj^{b5}	=	V^{b5}
	min	=	iii, iv
	dim	=	vii^{o}
	aug	=	VI^{+}
Seventh	maj^{7}	=	I^{maj7}, II^{maj7}
	min^{maj7}	=	iv^{maj7}
	min^{bb7}	=	iii^{bb7}
	dom4/2	=	$VII^{4/2}$
	dom^{7b5}	=	V^{7b5}
	aug^{maj7}	=	VI^{+maj7}

More About Seventh Chords

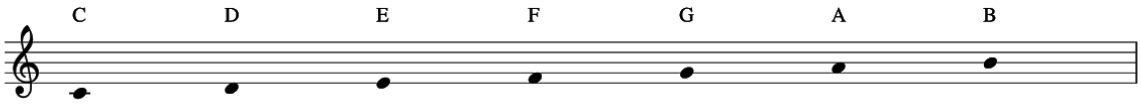
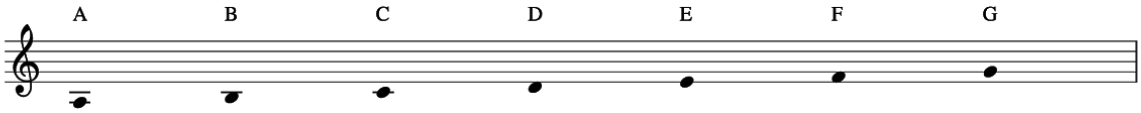
In case you did not understand some of the seventh chords already presented, let´s review some basic seventh chord formulas.

Ex. 19



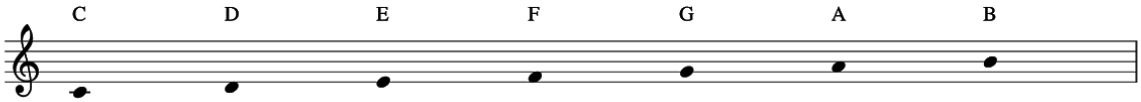
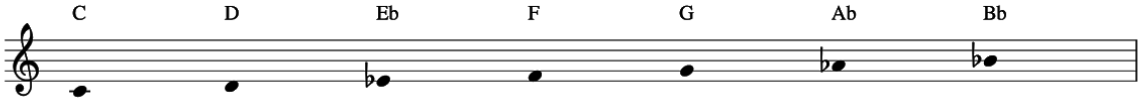
Relative Major/Minor

Major and Minor scales are **Relative** when they share all seven pitches.

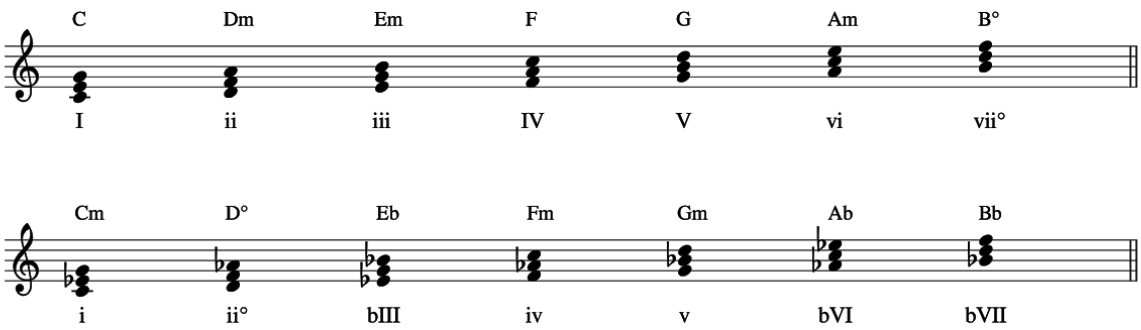
C Major...

...and it's relative minor, A Minor

Likewise, C Major is the relative major of A Minor.

Parallel Major/Minor

Major and Minor scales are **Parallel** when they share the same Root.

C Major…

...and it's parallel minor, C Minor

Likewise, C Major is the parallel major of C Minor.

Diatonic Triads from C Major and C Minor



Chords for Songwriters

An expanded harmonic vocabulary includes secondary dominants, and chords derived from parallel minor scales.

Triads

Primary Chords	C I	Dm ii	Em iii	F IV	G V	Am vi	B° vii°
Sec. V		A V/ii	B V/iii	C V/IV	D V/V	E V/vi	
Sec. viiº		C#° vii°/ii	D#° vii°/iii	E° vii°/IV	F#° vii°/V	G#° vii°/vi	
Parallel Natural Minor	Cm i	D° ii°	Eb bIII	Fm iv	Gm v	Ab bVI	Bb bVII
Parallel Melodic Minor	Cm i	Dm ii	Eb+ bIII+	F IV	G V	Aº vi°	B° vii°
Parallel Harmonic Minor	Cm i	D° ii°	Eb+ bIII+	Fm iv	G V	Ab bVI	B° vii°

Sevenths

Primary Chords	C^{Maj7} I^{Maj7}	Dm^{7} ii^{7}	Em^{7} iii^{7}	F^{Maj7} IV^{Maj7}	G^{7} V^{7}	Am^{7} vi^{7}	$B^{Ø7}$ vii^{7b5}
Sec. V^{7}		A^{7} V^{7}/ii	B^{7} V^{7}/iii	C^{7} V^{7}/IV	D^{7} V^{7}/V	E^{7} V^{7}/vi	
Sec. $vii^{ø7}$		$C\#^{ø7}$ $vii^{ø7}/ii$	$D\#^{ø7}$ $vii^{ø7}/iii$	$E^{ø7}$ $vii^{ø7}/IV$	$F\#^{ø7}$ $vii^{ø7}/V$	$G\#^{ø7}$ $vii^{ø7}/vi$	
Parallel Natural Minor	Cm^{7} i^{7}	$D^{Ø7}$ ii^{7b5}	Eb^{Maj7} $bIII^{Maj7}$	Fm^{7} iv^{7}	Gm^{7} v^{7}	Ab^{Maj7} bVI^{Maj7}	Bb^{7} $bVII^{7}$
Parallel Melodic Minor	Cm^{Maj7} i^{Maj7}	Dm^{7} ii^{7}	$Eb^{Maj7\#5}$ $bIII^{Maj7\#5}$	F^{7} IV^{7}	G^{7} V^{7}	$A^{Ø7}$ vi^{7b5}	$B^{Ø7}$ vii^{7b5}
Parallel Harmonic Minor	Cm^{Maj7} i^{Maj7}	$D^{Ø7}$ ii^{7b5}	$Eb^{Maj7\#5}$ $bIII^{Maj7\#5}$	Fm^{7} iv^{7}	G^{7} V^{7}	Ab^{Maj7} bVI^{Maj7}	$B^{ø7}$ $vii^{ø7}$

Summary of Chords from Major, Secondary Dominants, and Parallel Minor Scales

Triads

Primary Chords	C I	Dm ii	Em iii	F IV	G V	Am vi	B° vii°
Sec. V		A V/ii	B V/iii		D V/V	E V/vi	
Sec. viiº		C#° vii°/ii	D#° vii°/iii	E° vii°/IV	F#° vii°/V	G#° vii°/vi	
Parallel Natural Minor	Cm i	D° ii°	Eb bIII	Fm iv	Gm v	Ab bVI	Bb bVII
Parallel Melodic Minor			Eb+ bIII+			Aº vi°	

Sevenths

Primary Chords	C^{Maj7} I^{Maj7}	Dm^{7} ii^{7}	Em^{7} iii^{7}	F^{Maj7} IV^{Maj7}	G^{7} V^{7}	Am^{7} vi^{7}	$B^{Ø7}$ vii^{7b5}
Sec. V^{7}		A^{7} V^{7}/ii	B^{7} V^{7}/iii	C^{7} V^{7}/IV	D^{7} V^{7}/V	E^{7} V^{7}/vi	
Sec. $vii^{ø7}$		$C\#^{ø7}$ $vii^{ø7}/ii$	$D\#^{ø7}$ $vii^{ø7}/iii$	$E^{ø7}$ $vii^{ø7}/IV$	$F\#^{ø7}$ $vii^{ø7}/V$	$G\#^{ø7}$ $vii^{ø7}/vi$	
Parallel Natural Minor	Cm^{7} i^{7}	$D^{Ø7}$ ii^{7b5}	Eb^{Maj7} $bIII^{Maj7}$	Fm^{7} iv^{7}	Gm^{7} v^{7}	Ab^{Maj7} bVI^{Maj7}	Bb^{7} $bVII^{7}$
Parallel Melodic Minor	Cm^{Maj7} i^{Maj7}	Dm^{7} ii^{7}	$Eb^{Maj7\#5}$ $bIII^{Maj7\#5}$	F^{7} IV^{7}		$A^{Ø7}$ vi^{7b5}	
Parallel Harmonic Minor							$B^{ø7}$ $vii^{ø7}$

Modal Mixture for Songwriters

There are 24 Major and Minor triads. These tables show how they each relate to C.

Borrowed from C Major

Major Triad	Relative Minor	Roman Numerals
C	Am	I / vi
F	Dm	IV / ii
G	Em	V / iii

Borrowed from C Minor

Major Triad	Relative Minor	Roman Numerals
Eb	Cm	bIII / i
Ab	Fm	bVI / iv
Bb	Gm	bVII / v

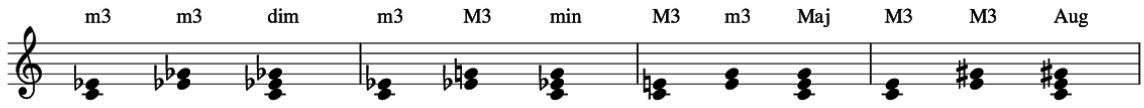
Borrowed from Parallel Modes

Major Triad	Relative Minor	Roman Numerals	Borrowed from Parallel _____
Db	Bbm	bII / bvii	Phrygian
D	Bm	II / vii	Lydian
Gb	Ebm	bV / biii	Locrian

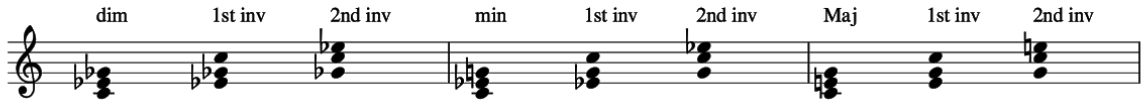
Secondary V chords and their Relative Minors

Major Triad	Relative Minor	Roman Numerals	V/x
E	C#m / Dbm	III / bii	V/vi
A	F#m / Gbm	VI / bii	V/ii
B	G#m / Abm	VII / bvi	V/iii

Basic Triads

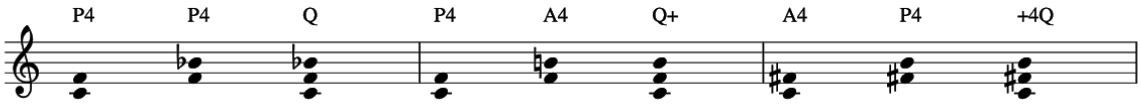
Triads contain three different pitches. Basic Triads are built by stacking pairs of third intervals. The lowest pitch in one of these stacks is called the Root. The pitch a third above the Root is called the Third, and the pitch a fifth above the Root is called the Fifth.

Triad Type	Interval: Root and Third	Interval: Third and Fifth	Interval: Root and Fifth	Formula
Diminished, d, °	m3	m3	d5	1 b3 b5
Minor, m	m3	M3	P5	1 b3 5
Major, M	M3	m3	P5	1 3 5
Augmented, A, +	M3	M3	A5	1 3 #5

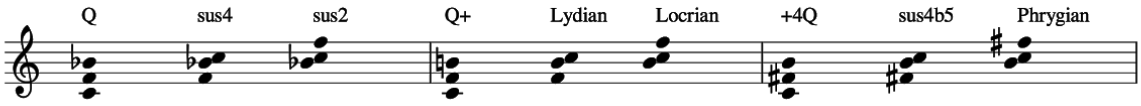
Inversions are generated by changing which pitch appears in the low voice. A triad is in first inversion when the Third is in the low voice. A triad is in second inversion when the Fifth is in the low voice. A consequence of inversion is that the various intervals between the three voices (Low, Middle, and High) change.



Triad Type	Inversion	Interval: Low and Middle	Interval: Middle and High	Interval: Low and High	Formula
Diminished	1st	m3	A4	M6	b3 b5 1
Diminished	2nd	A4	m3	M6	b5 1 b3
Minor	1st	M3	P4	M6	b3 5 1
Minor	2nd	P4	m3	m6	5 1 b3
Major	1st	m3	P4	m6	3 5 1
Major	2nd	P4	M3	M6	5 1 3

Advanced Triads

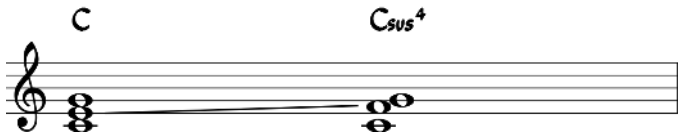
Advanced Triads are built by stacking pairs of fourth intervals. These are called Quartal triads.

Triad Type	Interval: Low and Middle	Interval: Middle and High	Interval: Low and High
Q	P4	P4	m7
Q+	P4	A4	M7
+4Q	A4	P4	M7

Each inversion is treated as its own unique triad, and its lowest note is reinterpreted as a new Root.

Triad Type	Inversion	Name	Formula	Interval: Low and Middle	Interval: Middle and High	Interval: Low and High
Q	1st	sus4	1 4 5	P4	M2	P5
Q	2nd	sus2	1 2 5	M2	P4	P5
Q+	1st	Lydian	1 #4 5	A4	m2	P5
Q+	2nd	Locrian	1 b2 b5	m2	P4	d5
+4Q	1st	sus4b5	1 4 b5	P4	m2	d5
+4Q	2nd	Phrygian	1 b2 5	m2	A4	P5

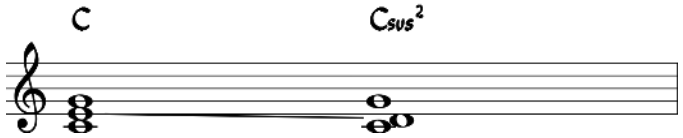
Suspended Triads and Other Three Note Structures

Suspended triads are triads in which one member, usually the third, has been raised or lowered one scale step.

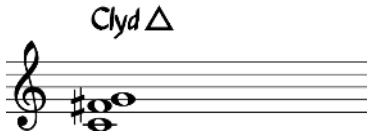
Ex. 20a



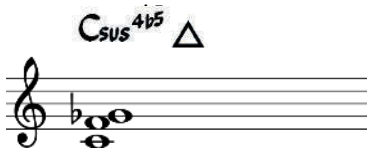
Ex. 20b



In example Ex. 20a the (E) moved one scale step (major scale) up to (F) creating a C suspended fourth triad. Example Ex. 20b the (E) moved down to a (D) creating a C suspended second triad. There are two other types of triads commonly used.

Ex. 21

The (E) of a C major triad is raised a whole step to (F#) to accommodate the #4 of a Lydian scale.

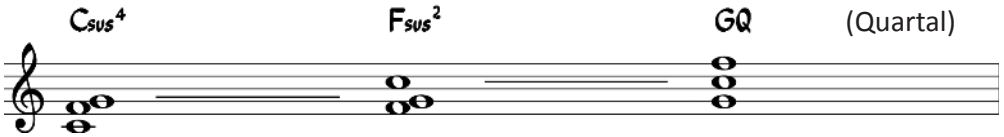
Ex. 22



The Sus4b5 triad is slightly different in that the fourth (F) is raised and the fifth (Gb) is flatted as in a Locrian scale.

In reality, certain suspensions such as sus2 or quartal voicing are simply inversions of sus4, Lydian or Locrian triads. It is because of their distinct sounds that I have chosen to name them in relation to their actual voicing.

Ex. 23

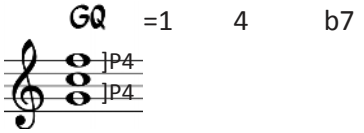


F^{sus2} is a 1st inversion of C^{sus4}

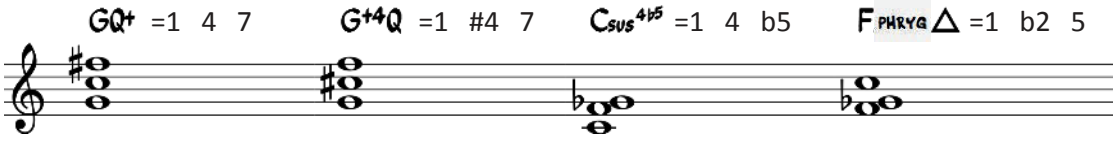
GQ is a 2nd inversion of C^{sus4}

The Q (Quartal) represents a triad containing two perfect fourths.

Ex. 24



Here are some other symbols to know.

Ex. 25

Chord Scales

Major Scale Modes

Scale Degree	Mode	Relation to Major Scale							Most Common Use
I, I^{maj7}	Ionian	1	2	3	4	5	6	7	Maj^{7}
ii, ii^{7}	Dorian	1	2	b3	4	5	6	b7	$Min^{7\ (nat.6)}$
iii, iii^{7}	Phrygian	1	b2	b3	4	5	b6	b7	Min^{7}, MajΔ/7
IV, IV^{maj7}	Lydian	1	2	3	#4	5	6	7	$Maj^{7(\#11)}$
V, V^{7}	Mixolydian	1	2	3	4	5	6	b7	Dom^{7}
vi, vi^{7}	Aeolian	1	2	b3	4	5	b6	b7	$Min^{7(b6)}$
vii^{o}, vii^{7b5}	Locrian	1	b2	b3	4	b5	b6	b7	Min^{7b5}

Melodic Minor Modes

Scale Degree	Mode	Relation to Major Scale							Most Common Use
i, i^{maj7}	Melodic Minor	1	2	b3	4	5	6	7	Min^{maj7}
ii, ii^{7}	Dorian b2	1	b2	b3	4	5	6	b7	$Min^{7\ sus4b9}$
$bIII^{+}$, $bIII^{+maj7}$	Lydian Aug.	1	2	3	#4	#5	6	7	$Maj^{7\#4\#5}$, MajΔ/b6
IV, IV^{7}	Mixolydian #11	1	2	3	#4	5	6	b7	Dom^{7b5}
V, V^{7}	Mixolydian b6	1	2	3	4	5	b6	b7	Dom^{7b6}
vi^{o}, vi^{7b5}	Locrian Nat.9	1	2	b3	4	b5	b6	b7	Min^{9b6}
vii^{o}, vii^{7b5}	Altered Dominant	1	b2	b3	b4	b5	b6	b7	$Dom^{7b9,\#9,b5,\#5}$

Harmonic Minor Modes

Scale Degree	Mode	Relation to Major Scale							Most Common Use
i, i^{maj7}	Harmonic Minor	1	2	b3	4	5	b6	7	Min^{maj7}, oΔ/b7
ii^{o}, ii^{7b5}	Locrian Nat.6	1	b2	b3	4	b5	6	b7	Min^{7b5}
$bIII^{+}$, $bIII^{+maj7}$	Ionian Aug.	1	2	3	4	#5	6	7	$Maj^{7sus4,\#5}$
iv, iv^{7}	Dorian #4	1	2	b3	#4	5	6	b7	$Min^{7(\#11)}$
V, V^{7}	Phrygian Major	1	b2	3	4	5	b6	b7	$Dom^{7sus4,b9,\ \#5}$
Vi, Vi^{maj7}	Lydian #9	1	#2	3	#4	5	6	7	$Maj^{7\#9,\#11}$, MajΔ/b9
vii^{o}, vii^{o7}	Altered Dominant bb7	1	b2	b3	b4	b5	b6	bb7	Dim^{o7}

Miscellaneous Scales

Scale	Relation to Major Scale								Most Common Use
Tonic Diminished	1	2	b3	4	b5	b6	bb7	7	Dim^{7}, MajΔ/b9
Dominant Diminished	1	b2	b3	b4	b5	5	6	b7	$Dom^{13,b9,\#9,b5}$
Whole Tone	1	2	3	#4	#5	b7			$Dom^{7,\#5,b5}$
Augmented	1	#2	3	5	#5	7			AugΔ/7AugΔ
Major Pentatonic	1	2	3	5	6				$Maj^{(6,7)}$
Minor Pentatonic	1	b3	4	5	b7				$Min^{(7,11)}$
Major Blues	1	2	b3	3	5	6			Dom^{7}, $Maj^{(6,7)}$
Minor Blues	1	b3	4	b5	5	b7			$Min^{7,}$, $Dom^{7\#9}$

Chord Families and Their Scales

Major^7 Type

Maj$^{(7)}$ Type	Relation to Major Scale							Characteristics
Ionian	1	2	3	4	5	6	7	sus4
Lydian	1	2	3	#4	5	6	7	#4 (#11)
Lydian aug	1	2	3	#4	#5	6	7	#4, #5
Ionian aug	1	2	3	4	#5	6	7	sus4, #5
Major Petatonic	1	2	3	5	6			no 4 or 7
Major Blues	1	2	b3	3	5	6		b3, no4 or7
Augmented	1	b3	3	5	#5	7		b3, #5

Minor^7 Type

Min$^{(7)}$ Type	Relation to Major Scale							Characteristics
Dorian	1	2	b3	4	5	6	b7	6
Phrygian	1	b2	b3	4	5	b6	b7	b2, 5, b6
Aeolian	1	2	b3	4	5	b6	b7	b6
Melodic Minor	1	2	b3	4	5	6	7	6, 7
Dorian b2	1	b2	b3	4	5	6	b7	b2, 6
Harmonic Minor	1	2	b3	4	5	b6	7	b6, 7
Dorian #4	1	2	b3	#4	5	6	b7	#4, 6
Minor Pentatonic	1	b3	4	5	b7			4(11)
Minor Blues	1	b3	4	#4	5	b7		4, #4 (11, #11)

Minor^{7b5} Type

Min$^{(7b5)}$ Type	Relation to Major Scale							Characteristics
Locrian	1	b2	b3	4	b5	b6	b7	b2, b5
Locrian nat2	1	2	b3	4	b5	b6	b7	2, b5
Locrian nat6	1	b2	b3	4	b5	6	b7	b2, b5, 6

Dom^{7} Type

Dom^{7} Type	Relation to Major Scale							Characteristics
Mixolydian	1	2	3	4	5	6	b7	sus4, b7
Mixolydian #11	1	2	3	#4	5	6	b7	#4 (b5), b7
Mixolydian b6	1	2	3	4	5	b6	b7	sus4, b6 (#5), b7
Altered Dominant	1	b2	b3	b4	b5	b6	b7	b9, #9, b5, #5
Phrygian Major	1	b2	3	4	5	b6	b7	sus4, #5, 5
Dominant Diminished	1	b2	b3	b4	b5	5	6	b7	b9, #9, b5, 5, 13
Whole Tone	1	2	3	#4	#5	b7		#4, #5
Major Pentatonic	1	2	3	5	6			no 4 or b7
Minor Pentatonic	1	b3	4	5	b7			#9, no b7
Major Blues	1	2	b3	3	5	6		b3, no b7
Minor Blues	1	b3	4	#4	5	b7		#9, sus4, b5

Diminished Type

Dim Type	Relation to Major Scale								Characteristics
Tonic Diminished	1	2	b3	4	b5	b6	bb7	7	9, 11, b13, 7
Altered Dominant bb7	1	b2	b3	b4	b5	b6	bb7		b9, 3, b13

Augmented Type

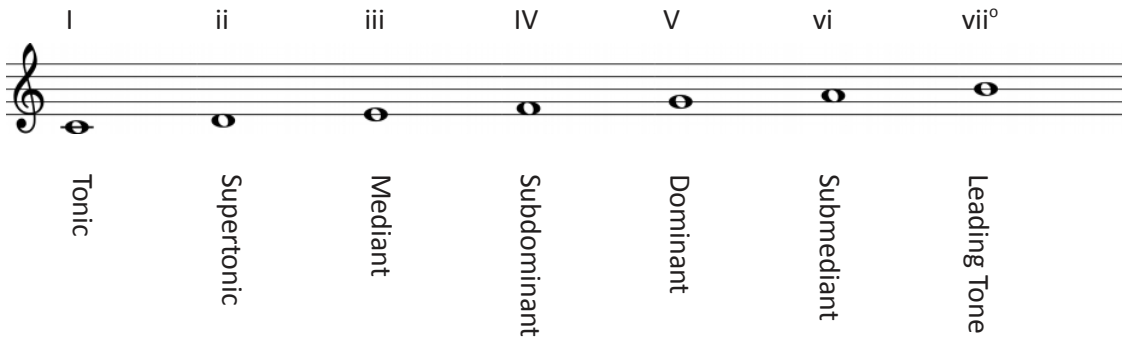
Aug Type	Relation to Major Scale						Characteristics
Whole Tone	1	2	3	#4	#5	b7	#4, #5
Augmented	1	#2	3	5	#5	7	#2, 5, #5, 7

Scale Degree Names and Basic Progressions

Scale degree names are commonly used in traditional harmonic analysis. It would be a good idea to become familiar with these terms for future reference in this text.

Ex. 26

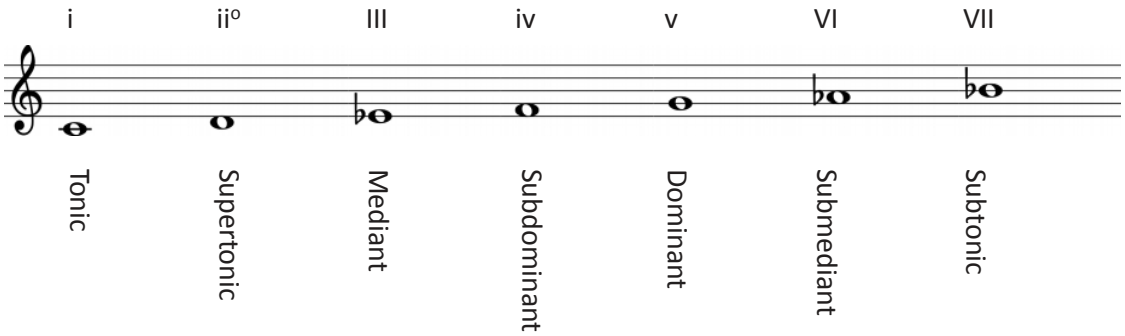
In C major



These scale degree terms apply to all seven note scales with leading tones. In scales containing lowered seventh degrees, the Leading Tone is replaced by the Subtonic.

Ex. 27

In C Aeolian

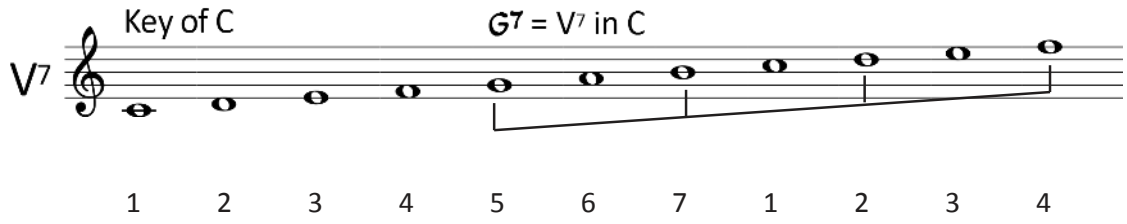


Basic Progressions

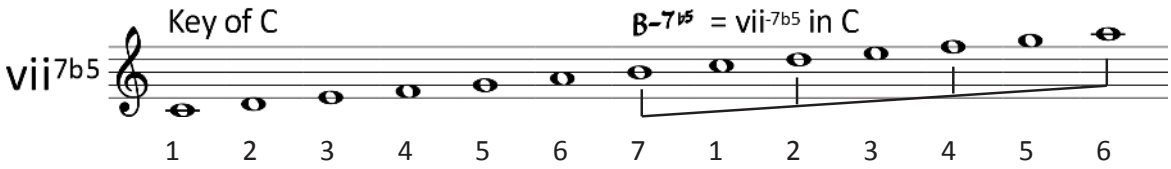
To better understand functional harmony, it should be divided into three basic subheadings from the terms above.

1. **Dominant Chords**: chords which contain the 5th and 7th degrees of the scale. V^7, vii^{7b5}.

Ex. 28



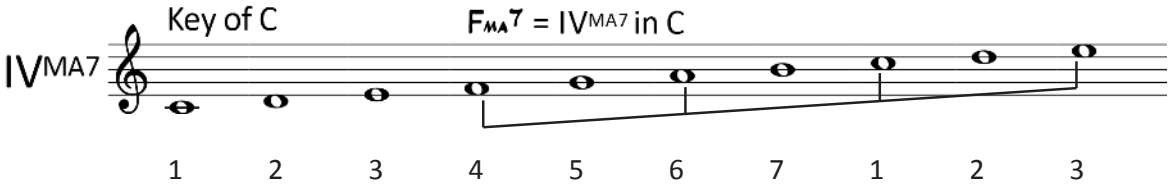
Ex. 29



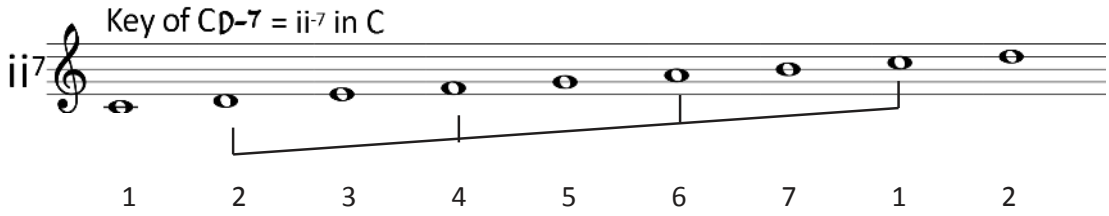
Note: the 4th and 7th degrees are the least stable and therefore must be resolved.

2. **Pre-Dominant Chords:** chords which contain only 4th degree of the scale

Ex. 30

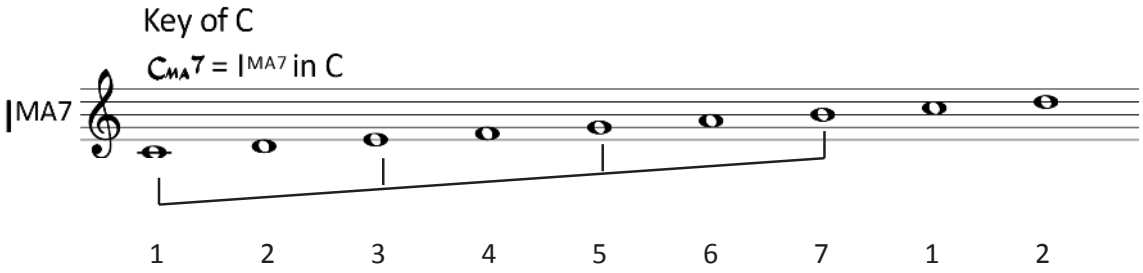


Ex. 31

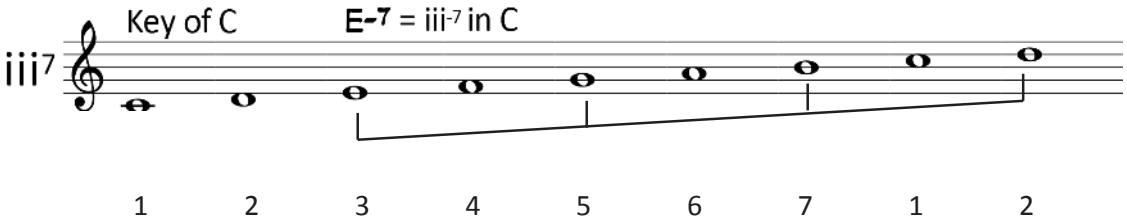


3. **Tonic Chords:** chords which do not contain the 4th degree of the scale

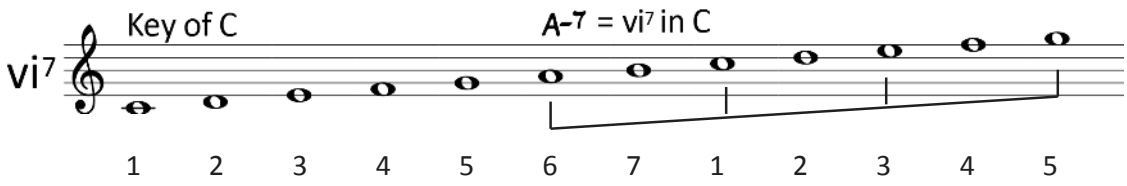
Ex. 32



Ex. 33

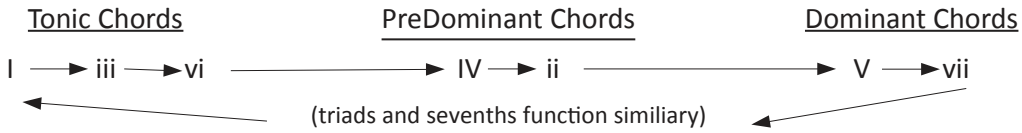


Ex. 34



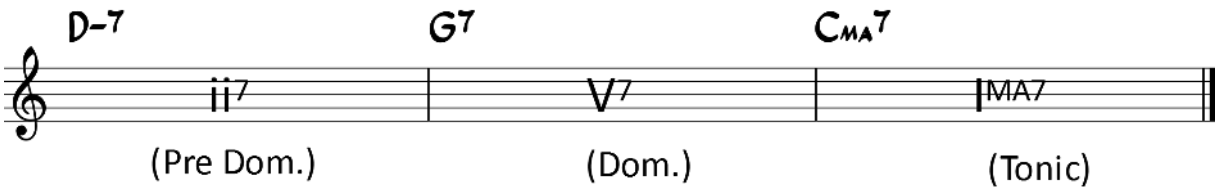
Here is a clearer chart (major key).

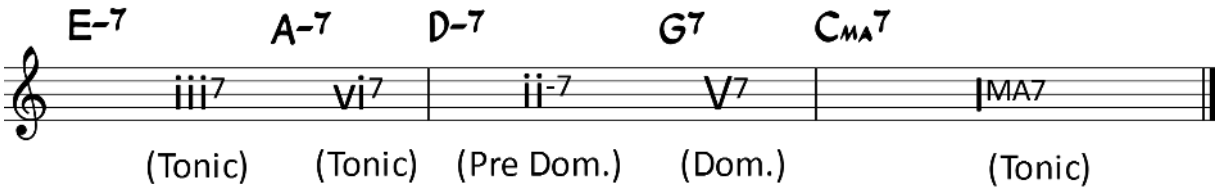
Ex. 35



In traditional harmony, the strongest and most common cadence patterns move clockwise through each group.

Ex. 36



Ex. 37

In Ex. 35 there are two (or three if the progression is repeated) consecutive tonic chords. The particular sequence of chords within group should also be noted in Ex. 37 (I - iii - vi)

More About Roman Numeral Analysis

Roman Numerals are commonly used to describe chords within the diatonic chord systems. Here are definitions of the two elements used in Roman Numeral Analysis.

Exact Distance: Distinctions between basic triads are indicated by the following symbols:

Uppercase Roman Numerals for Major

Lowercase Roman Numerals for minor

(^{o}) and lower case Roman Numerals for diminished

(+) and upper case Roman Numerals for Augmented

Seventh chords use these symbols:

(Maj7) or (MA7) for chords containing a major seventh interval above the root and

(min7) indicating a minor seventh or (b7) interval above the root.

Secondary Dominant and Diminished Seventh:

The terms Secondary Dominant and Secondary Diminished Seventh chords refer to chords outside the diatonic key which momentarly modulate to another key.

Secondary Dominant Chords:

Example 38 shows the Secondary Dominant chords from the key of C major.

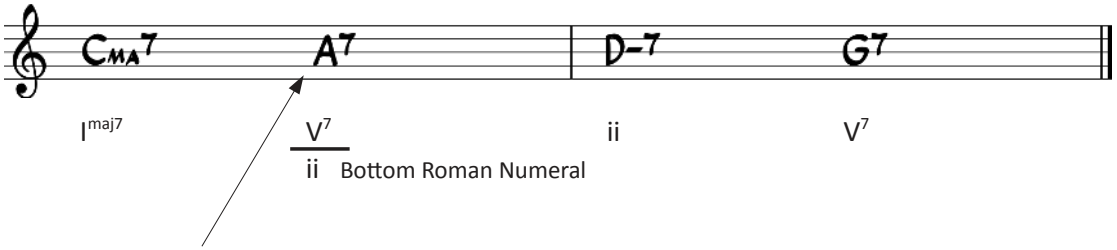
Ex. 38

	I	ii	iii	IV	V	vi	vii^{o}
Cmajor =	C	D-	E-	F	G	A-	B^{o}
Sec. Dom.´s		A^7	B^7	C^7	D^7	E^7	
		$\frac{V^7}{ii}$	$\frac{V^7}{iii}$	$\frac{V^7}{IV}$	$\frac{V^7}{V}$	$\frac{V^7}{vi}$	

not a stable enough resolution chord to have a Sec. Dom.

Basically, the Secondary Dom.^{7} chord is the Dom.^{7} chord in the key of the bottom Roman Numeral.

Ex. 39

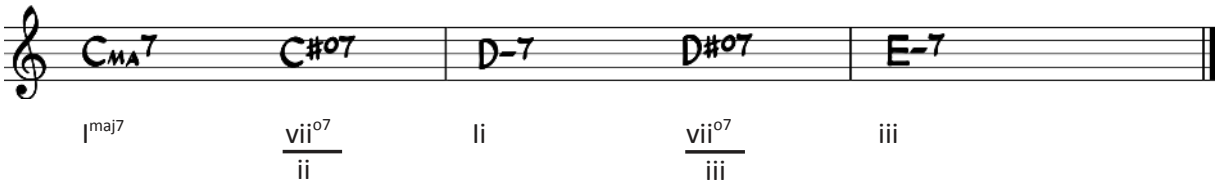


A^7 is the V^7 chord in the key of D minor. So in reality, D^{-7} functions dually as the i^7 chord in D minor and as the ii^7 in major

Secondary Diminished Seventh Chords

Secondary Diminished Seventh Chords are leading tone seventh chords from the key of the denominator.

Ex. 40



The Secondary Diminished Seventh chords in the key of C are:

Ex. 41

	I	ii	iii	IV	V	vi	vii^o
Cmajor =	C	D-	E-	F	G	A-	B^o
Sec. Dim^{o7}		C#^{o7}	D#^{o7}	E^{o7}	F#^{o7}	G#^{o7}	
		vii^{o7} / ii	vii^{o7} / iii	vii^{o7} / VI	vii^{o7} / V	vii^{o7} / vi	

not a stable enough resolution chord to have a Sec. Dom.

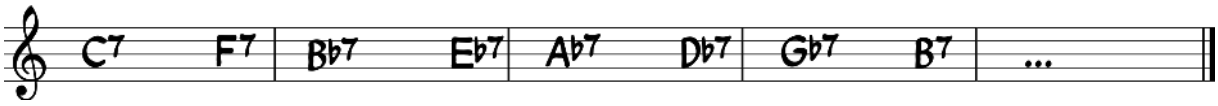
Cycle of Fifth and Backcycling

The terms Cycle of Fifths and Backcycling are frequently mentioned in jazz theory. Here are the definitions and examples of each.

Cycle of Fifths

A progression that is most commonly associated with Dom7 chords (V7) whose roots move in descending fifths pattern.

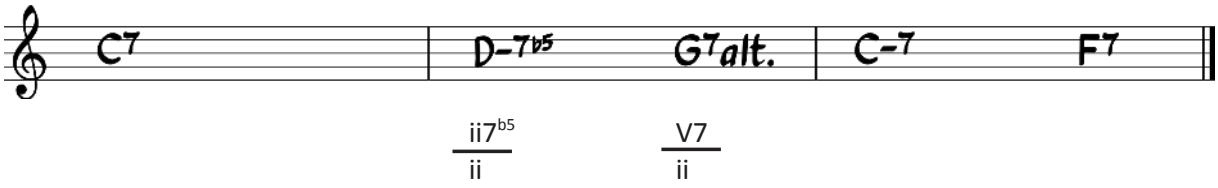
Ex. 42



Back Cycling

Harmonic movement, usually from a minor key to the key center a fifth above.

Ex. 43

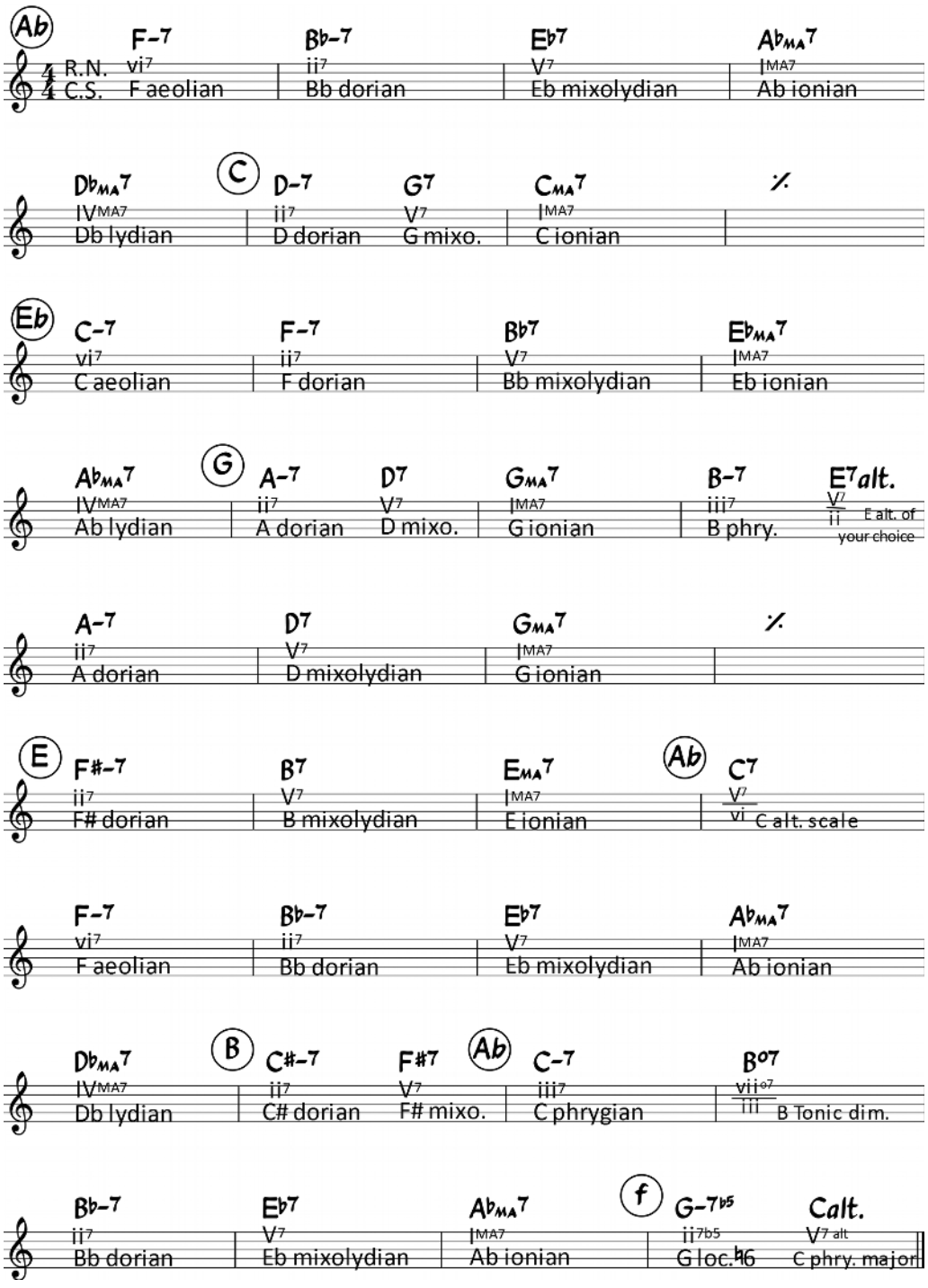


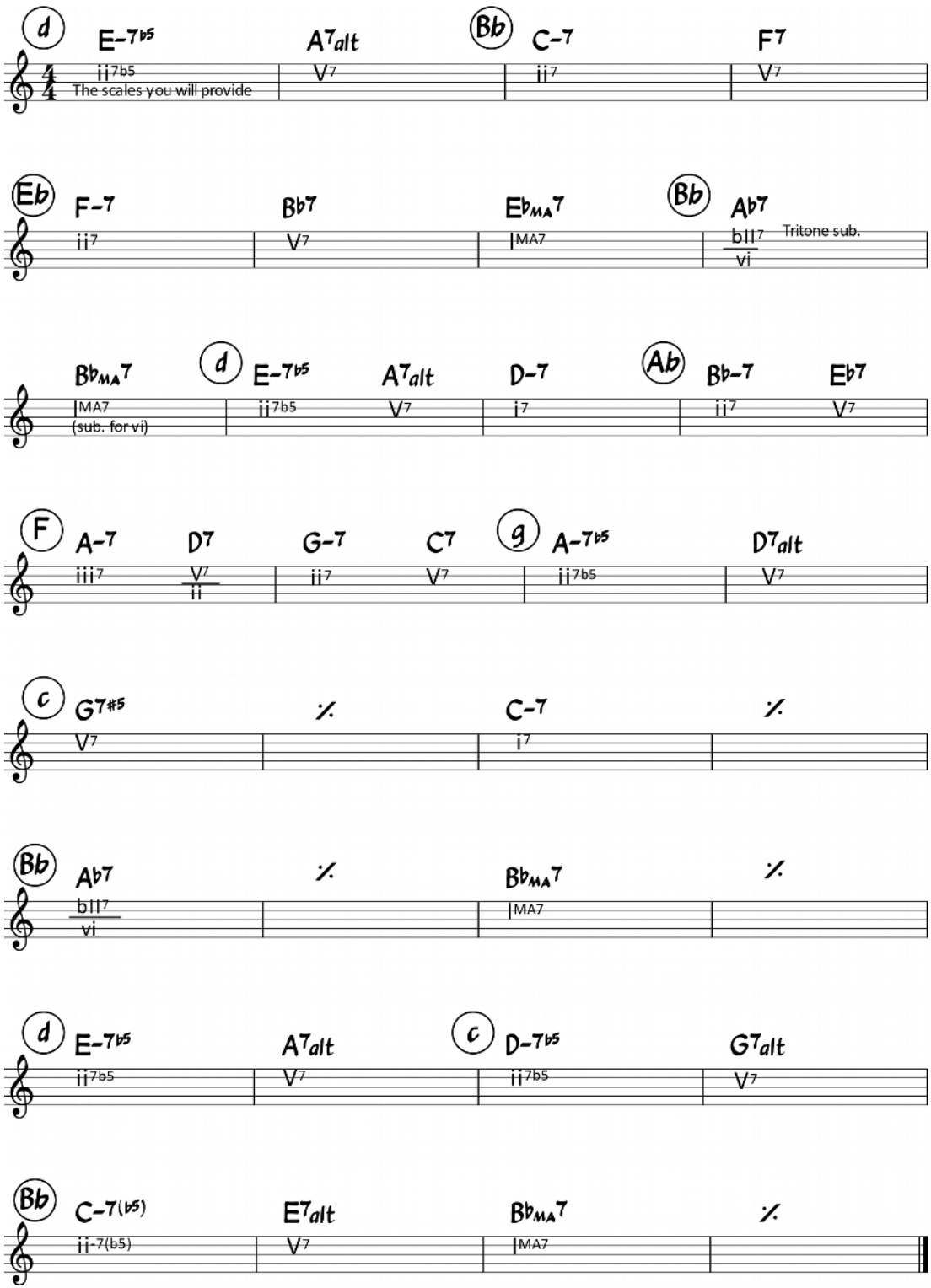
Back Cycles and the Cycle of Fifths are terms that are commonly used. Keep in mind, however, that both devices can be analyzed in other ways (secondary dominants).

Roman Numeral Analysis and Chord Scale Choice

I have included two common harmonic progressions for purposes of Roman Numeral Analysis and Chord Scale Choice. The key centers are circled. Capital letters are Major keys and subscript letters are minor keys.

Ex. 44a



Ex. 44b

You may have noticed the Ab^7 as being bII. The reason for this is because it (Ab^7) is a tritone substitute (to be explained later, so don't panic) for D^7 which is $\frac{V^7}{vi}$ in Bb Major. bII7 chords are interchangeable for V7 chords when resolving to their respective Tonic Major or Minor Chords (we will discuss this later in the book).

Basic Rules for Chord Substitutions

Following is a list of some basic rules for chord substitutions. The subsequent chapters contain additional rules to accommodate the flow of new material.

All chord substitutions may be broken down into three catagories.

1. The bass note stays the same and the chord quality changes.

Ex. 45

$F^{maj7} \longrightarrow F^{7}$

2. The bass note changes but the chord quality stays the same.

Ex. 46

$F^{7/\#5/b9/b5} \longrightarrow B^{9}$

3. Complete chord replacement.

Ex. 47

$B^{maj7} \longrightarrow G\text{-}^{maj7}$

Rule I:

For basic chords, any modal extension may be added.

Ex. 48

1. $G\text{-}^{7\,(dor.)} = G\text{-}^{9} \longrightarrow G\text{-}^{11} \longrightarrow G\text{-}^{13}$
2. $D^{7\,(mix.)} = D^{9} \longrightarrow D^{11} \longrightarrow D^{13}$
3. $Bb^{maj7\,(lyd.)} = Bb^{maj9} \longrightarrow Bb^{maj7\#11} \longrightarrow Bb^{maj13}$

Any of these groups may also combine extensions such as $\longrightarrow G\text{-}^{9/11/13}$.

Rule II: Major Chords

Substitute Major chords own mediant or submediant chord.

Ex. 49

A^{maj7}	sub.	$C\#\text{-}^{7}$ (mediant)	$C\#\text{-}^{7}$	$= A^{maj9}$
		$F\#\text{-}^{7}$ (submediant)	$F\#\text{-}^{7}$	$= A^{6}$

Rule III: Minor Chords

Substitute Relative Major or Dominant Minor

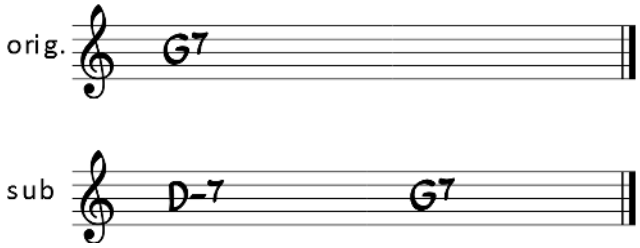
Ex. 50

$D\text{-}^{7}$	sub.	F^{maj7} (relative major)	F^{maj7}	$= D\text{-}^{9}$
		$A\text{-}^{7}$ (dom. Minor)	$A\text{-}^{7}$	$= D\text{-}^{9/11}$

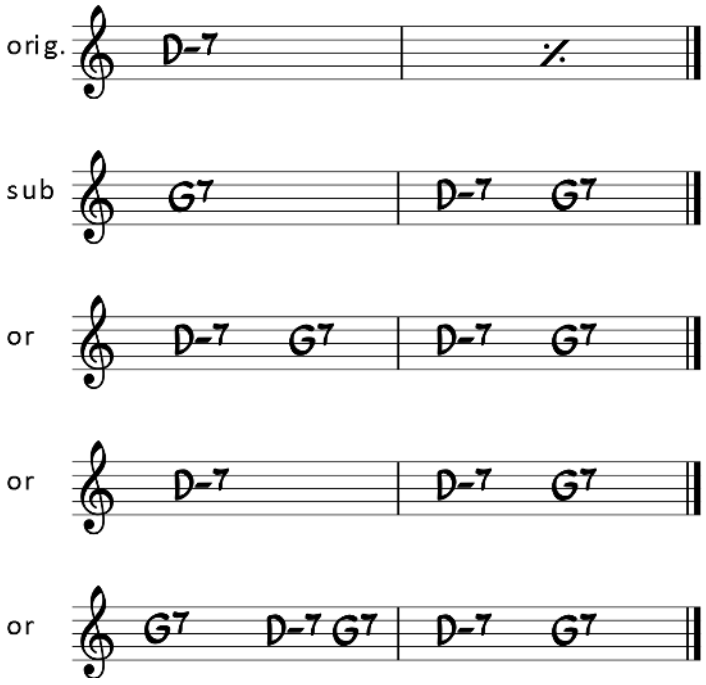
Rule IV: Dom.^{7} Chords (V^{7})

Substitute Dom. minor for Dom. major (must return to Dom. major). This is also called "twoing the five“ because you are putting the ii^{7} chord before the V^{7}. This is one of the most common substitutions found in Jazz (Bebop).

Ex. 51

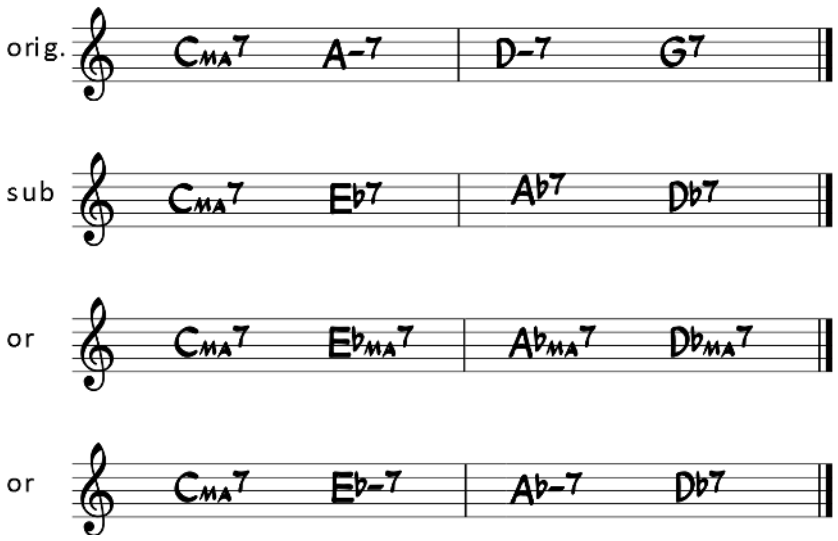


Also reverse works:

Ex. 52

Rule V: All Chords

Substitute any chord which has a root a tritone away from the original chord.

Ex. 53

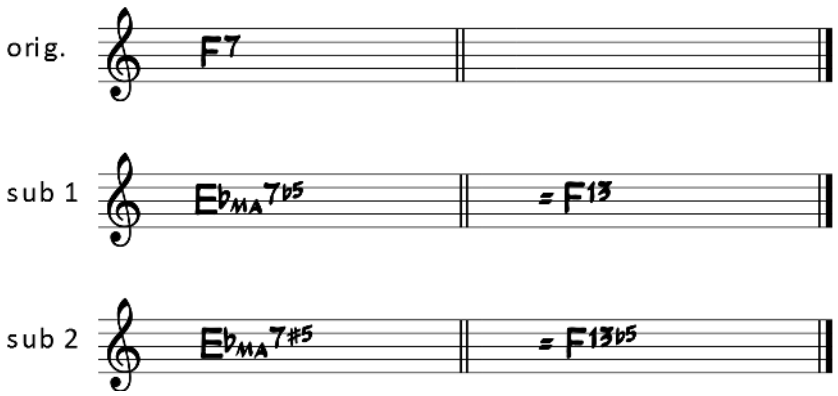


Or combine any lines but do not substitute for tonic chord (not yet!).

Rule VI: Dom^{7} Chords (V^{7})

Substitute maj^{7b5} or maj$^{7\#5}$ built on the subtonic (bvii) of the Dom. chord.

Ex. 54

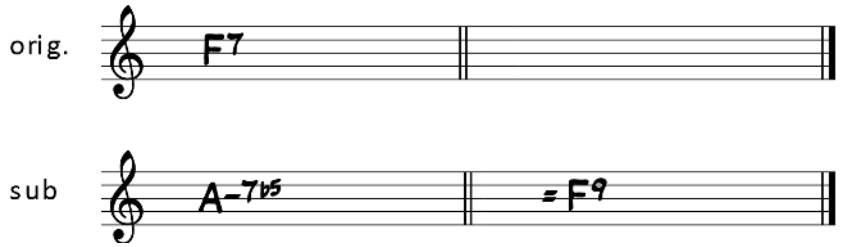


Both should be voiced in higher registers.

Rule VII: Dom^7 Chords (V^7)

Substitute min^{7b5} chord built on the mediant of the Dom^7 chord.

Ex. 55

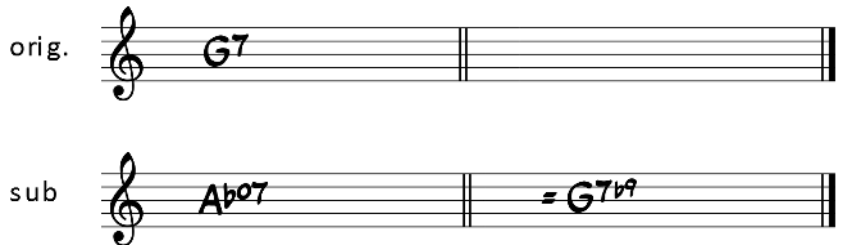


(Chord should also be voiced in a higher register)

Rule VIII: Dom^7 Chords (V^7)

Substitute Dim^7 chord built on bII of the Dom^7 chord.

Ex. 56



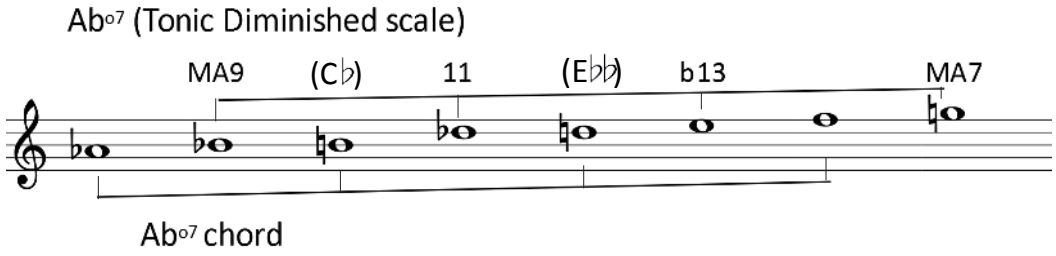
Also the other 3 symmetric dim^7 chords and their extensions

Ex. 57

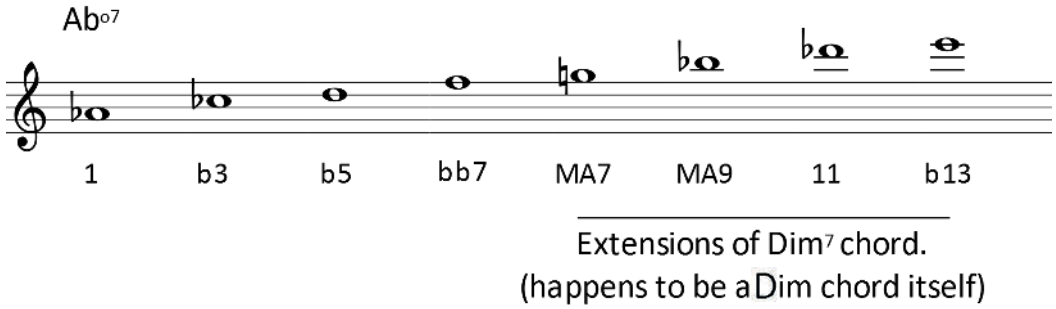
G^{7b9} = Ab^{o7}, B^{o7}, D^{o7}, F^{o7}

Extensions of Dim7 Chords

Ex. 58



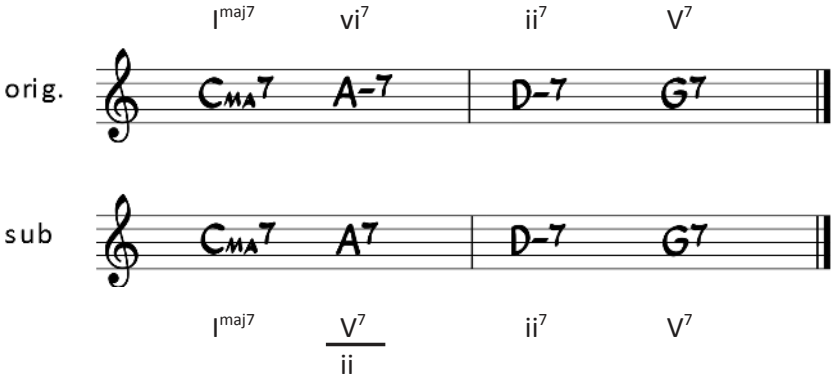
Ex. 59



Rule IX:

V^7 chords may replace minors (secondary dominants) to set up a stronger harmonic cadence (tonicization).

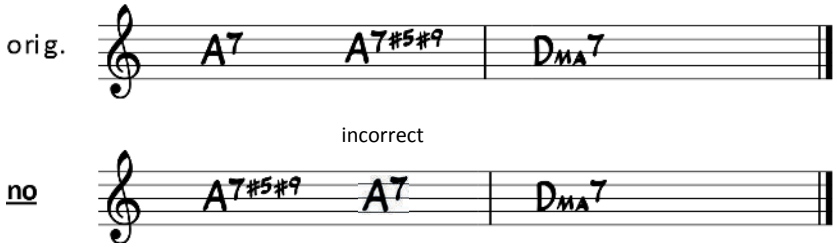
Ex. 60



Rule X: Dom^7 (V^7) chords

Altered (#9, b9, #5, b5) Dom^7 chords may follow unaltered chords when resolving to their tonic. The reverse is not true. When moving towards the tonic, you need to go from less tension to more tension.

Ex. 61



Rule XI: Dom7 (V7) Chords

Altered Dom7 chords can be used (even if not indicated) when the root of the chord is:

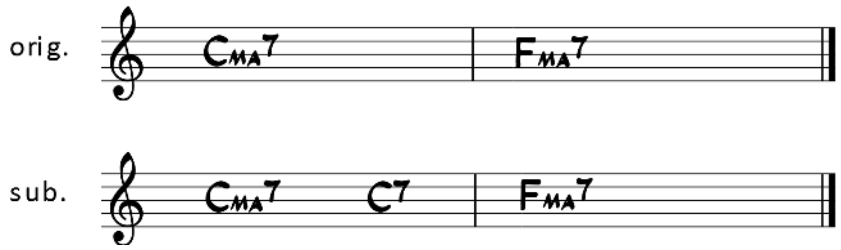
1. a 4th higher (5th lower): $A^{7\ alt}$ D^{maj7}
2. ½ step lower $A^{7\ alt}$ Ab^{maj7}
3. a minor type with the same root $A^{7\ alt}$ $A\text{-}^{7}$
4. ½ step higher $A^{7\ alt}$ Bb^{maj7} In

all other instances use Mixolydian or Mixolydian #11 scales.

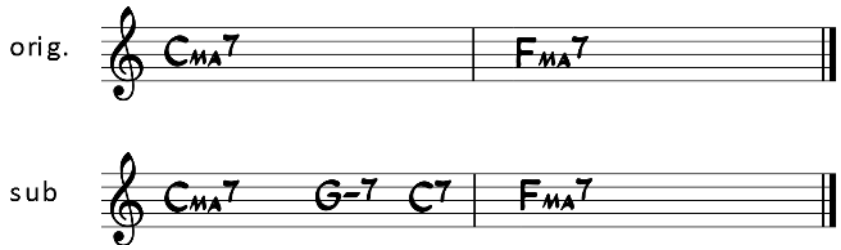
Rule XII:

A) When a Maj or Min chord is followed by a Maj, Min or Dom^7 chord whose root is a 4^{th} higher, you may insert a Dom^7 of the same root for half the duration.

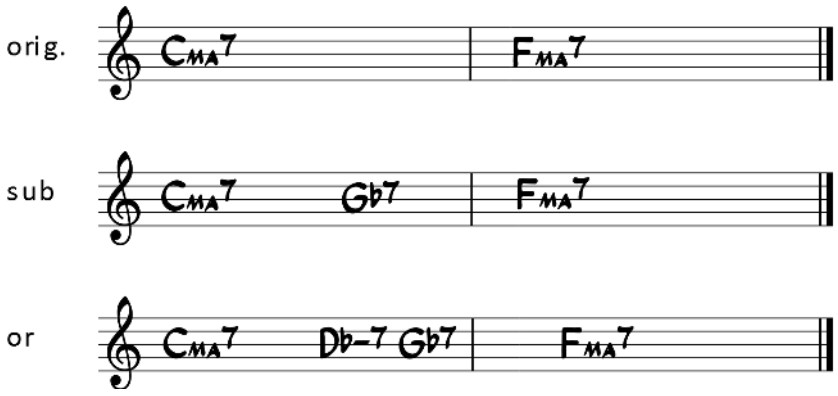
Ex. 62



You may also insert the ii^7 chord of the V^7 sub

Ex. 63

B) You may also insert a Dom^{7} chord a tritone away for the same duration.

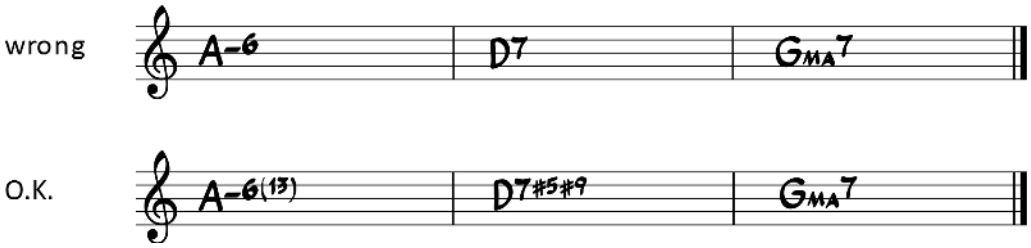
Ex. 64



Rule XIII

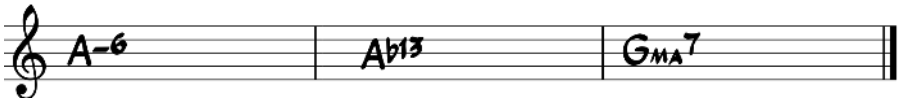
A min^{6} chord cannot be substituted for a min^{7} (ii-function) unless an altered Dom^{7} follows, thus increasing the harmonic tension of the progression.

Ex. 65



Or in combination with tritone substitution

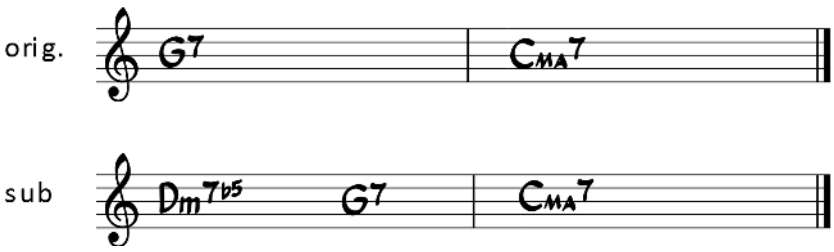
Ex. 66



Rule XIV:

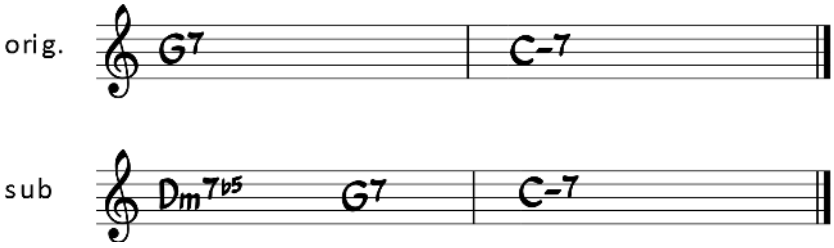
Min^{7b5} chords may be inserted before Dom^{7} (V^{7}) chords in major or minor keys.

Ex. 67a



or

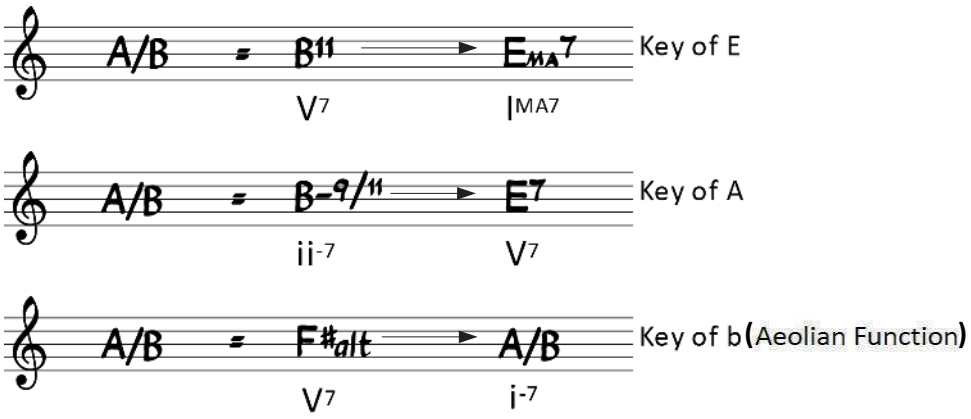
Ex. 67b



Rule XV:

Dom^{11} (V^{7}) chords can function as dom^{7}'s or min^{7}'s (dorian/aeolian)

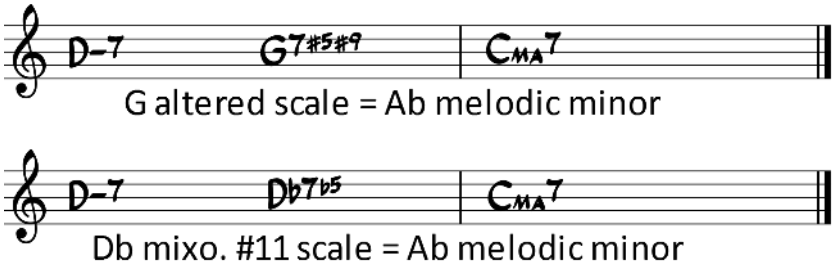
Ex. 68



Rule XVI:

A bII^{7b5} substitution for V$^{7\ alt}$ works because they both share the same scale.

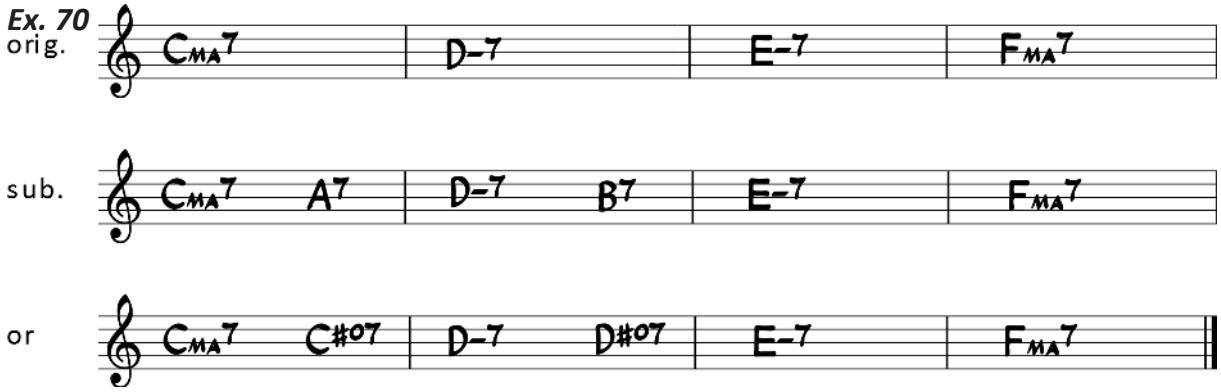
Ex. 69



Rule XVII:

Secondary Dom^7 and Dim^7 chords may be inserted before their respective resolution chords.

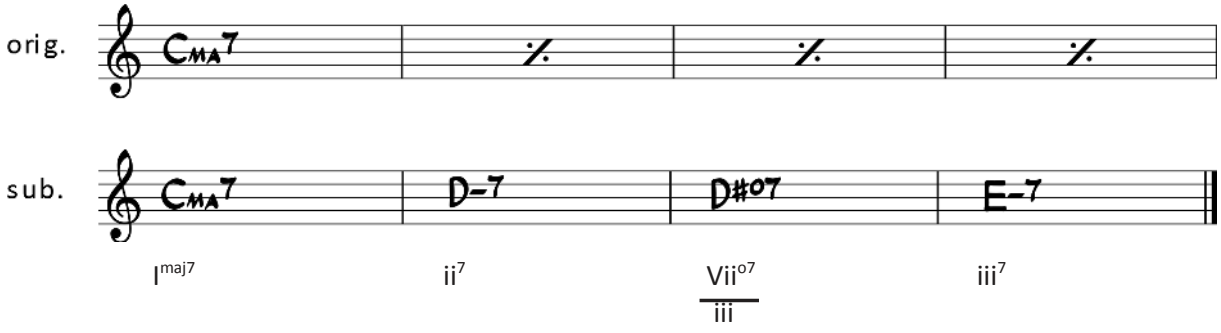
Ex. 70



Rule XVIII:

It is common to insert two diatonic chords seperated by a secondary dim^7 chord in places of little harmonic motion.

Ex. 71



Rule XIX:

Maj$^{7\#5}$ and Maj^{7b5} chords can be subbed for any other chords from the same melodic minor scale.

Ex. 72	Eb$^{maj7\#5}$	for	C-^{maj7}	(3, 5, 7, 9) of C-^{maj7}
	Eb^{maj7b5}	for	F^{13}	(b7, 9, 3, 13) of F^{13}

Rule XX: Symetrical Scales

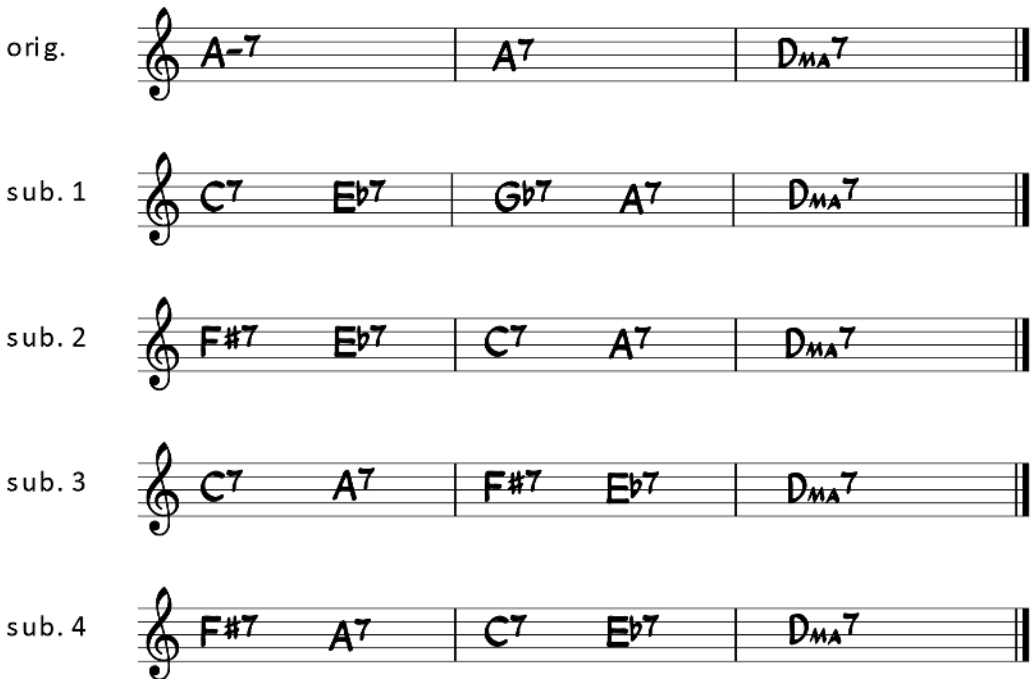
A) Diminished chords repeat every min3rd (4 frets) interval. Therefore, any of the four in the cycle may be substituted for another.

B) Augmented Chords repeat every maj3rd (5 frets) interval, so the same rule applies.

Rule XXI: Dom^7 Chords (V^7)

Dom^7 chords can be moved in min3rd cycles to intersect at cadence points.

Ex. 73



Rule XXII: Modal Chords

Modal Chords can be interchanged (Modal Mixture) as long as the basic integrity of the chord remains intact.

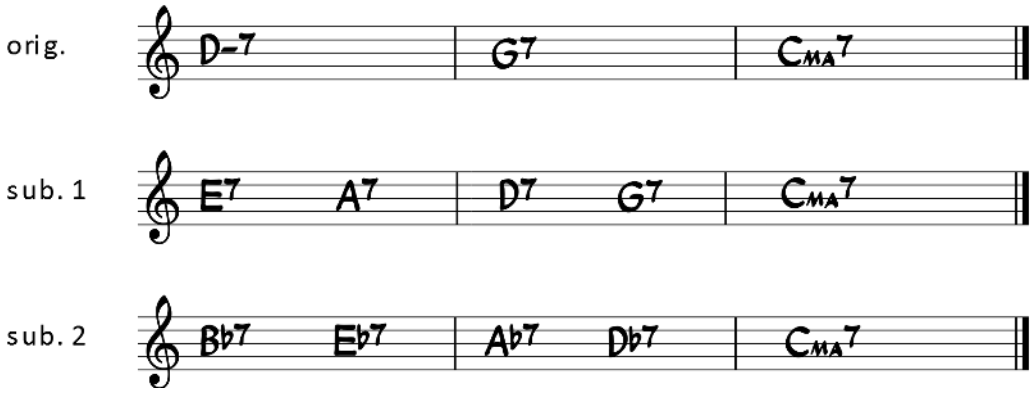
Ex. 74

G^{maj7} (Ionian) orig.	$G^{maj7/6/9}$
G^{maj7} (Lydian) sub.	$G^{maj7/\#11}$
$Bb\text{-}^{7}$ (Dorian) orig.	$Bb\text{-}^{6/9}$
$Bb\text{-}^{7}$ (Aeolian) sub.	$Bb\text{-}^{7/b6}$

Rule XXIII:

The cycle of 5^{th}'s (descending) may be inserted for any length in a progression as long as the intersecting chord has either a V^7 or a bII^7 relationship to the resolution chord.

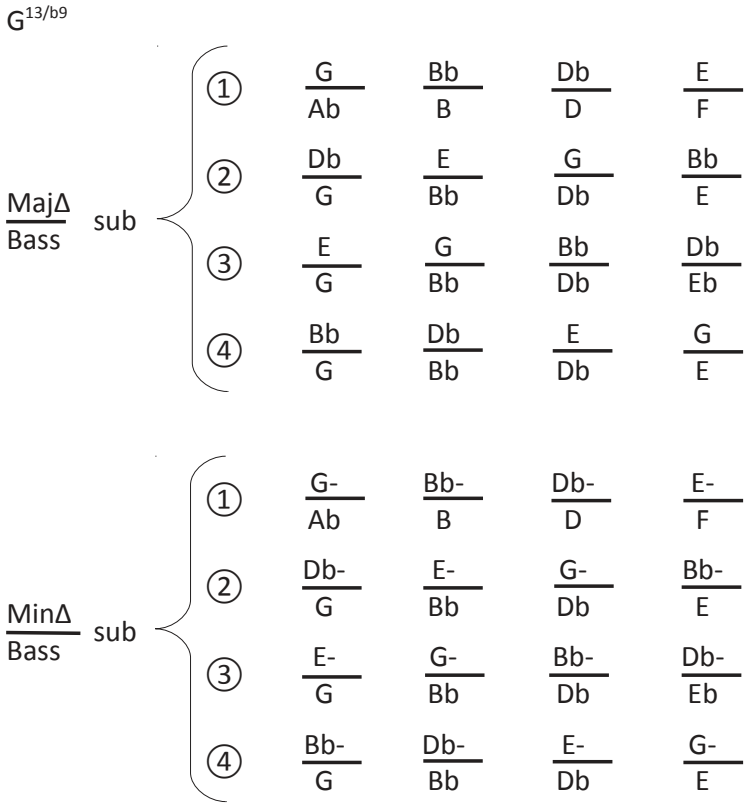
Ex. 75



Rule XXIV

Bitonal chords from the Dominant Diminished scale can be subbed for one another.

Ex. 76

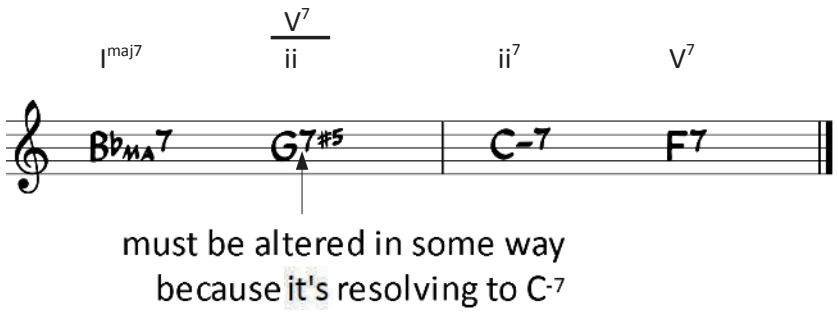


Rule XXV:

A) When in a min ii^{7b5} V^7 i^7 progression, an altered Dom7 chord must be used.

B) All secondary Dom^7 (V^7) chords resolving to minor must have an altered V^7 chord

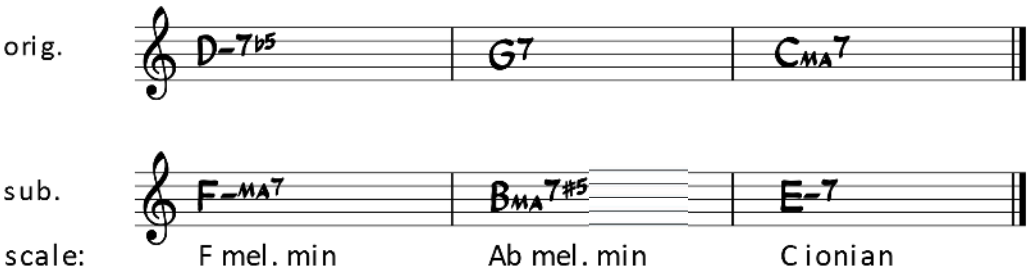
Ex. 77



Rule XXVI:

Any chord can be replaced by another chord from its family.

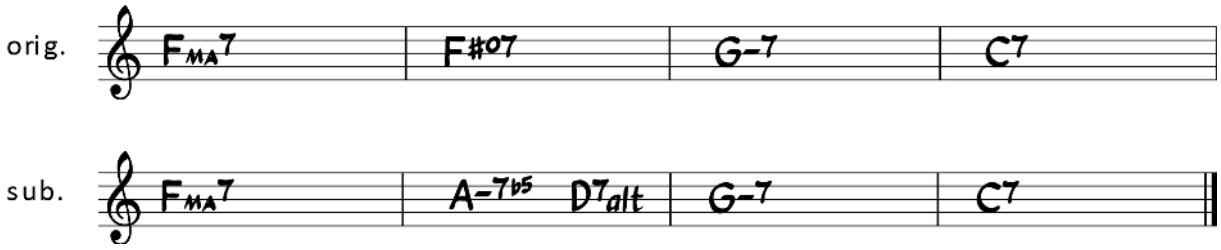
Ex. 78



Rule XXVII

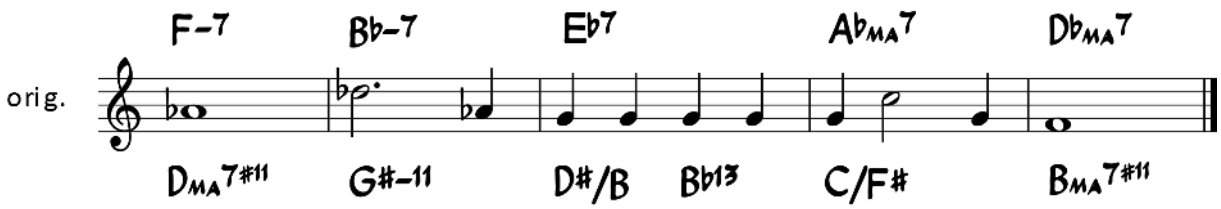
Dim^{7} chords that resolve to min^{7} chords one ½ step above can be replaced with a ii^{7b5} V^{7alt} progression from the key of the min^{7} chord.

Ex. 79



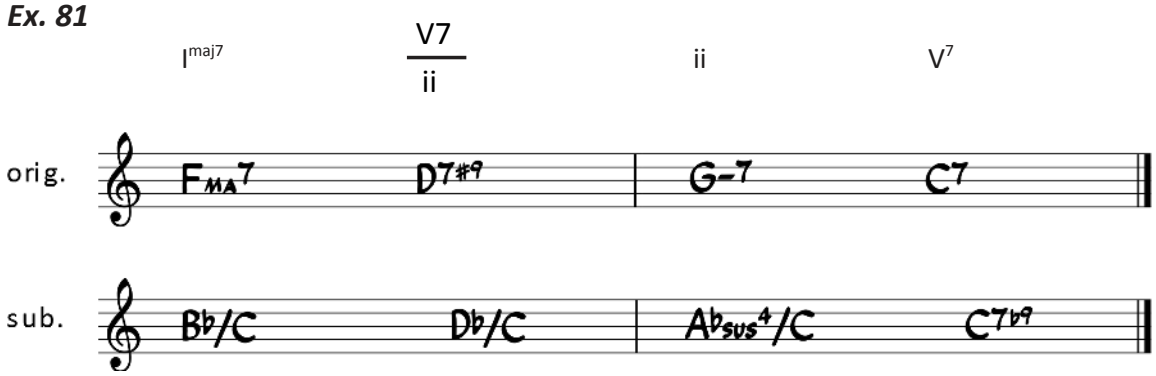
Rule XXVIII:

Any chord whose chord scale contains the melody notes of a particular measure can be substituted.

Ex. 80

Rule XXIX

Triads and Sevenths (V^7) over Dom or Tonic bass notes are very common.

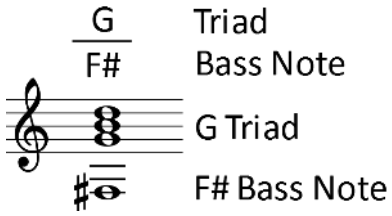
Ex. 81

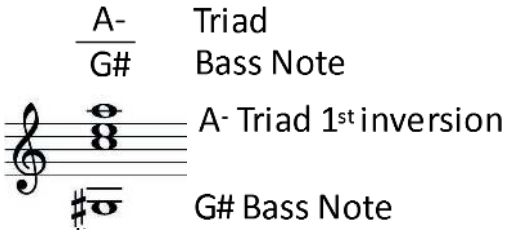


Triads Over Bass Notes

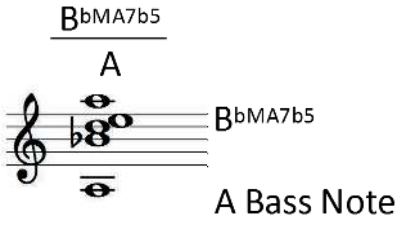
Triads over bass notes are used extensivly in contemporary jazz and classical music. The triad can be of any quality (maj, min, dim, aug, sus, lyd or loc) and inversion.

Ex. 82a



Ex. 82b

The seventh chord over a bass note looks like

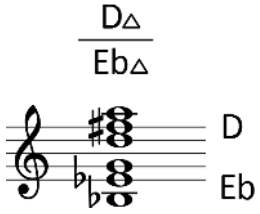
this: ***Ex. 83***



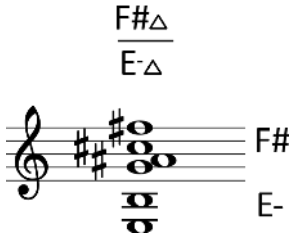
The seventh chord may be in any inversion as well.

The polychord differs from the triad over bass note by the presence of a triad or seventh chord on both the top and bottom.

Ex. 84a



Ex. 84b

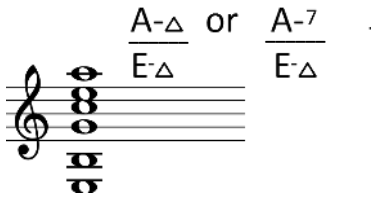


To indicate the presence of a triad, a Δ (triad) symbol should be used.

Seventh chords over triads or other seventh chords are impossible to sound simultaneously on the guitar because of the six string limit (on most guitars).

In some instances their effect can be achieved with a triad over triad or seventh chord over bass note.

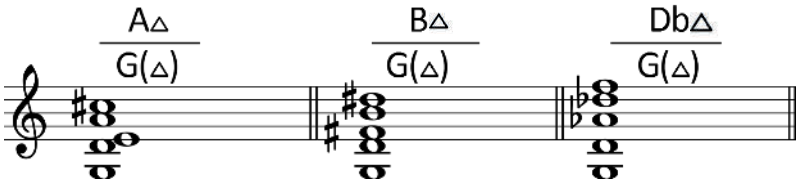
Ex. 85



The G of the E min chord dually functions as the third of Emin and the b7 of A-7.

Many times the root and fifth are sufficient in giving the sound of the bottom triad.

Ex. 86



Triads and Seventh Chords Over Bass Notes and Polychords

Key of C: C triads over C bass notes not included, because they are simply triads. The last row of generic chord names will help you when encountering these same relationships in other keys.

Chord Name	Chord Type (Seventh or Mode)	Triad / Bass Interval Relationship	Scale(s)	Generic Chord Name or Interval above Top Triad
Db / C	$\frac{Db^{maj7}}{7}$ phrygian	b9, 11, b6	Phrygian Phrygian Major	MajΔ / Maj7
Db- / c	$\frac{Db\text{-}^{maj7}}{7}$ altered dominant	b9, 3, #5	Phrygian Major Altered Dominant	MinΔ / Maj7
Db^{o} / C	$\frac{Db^{omaj7}}{7}$ dominant diminished	b9, 3, 5	Phrygian Major Dominant Diminished	DimΔ / Maj7
Db^{+} / C	$\frac{Db^{maj7\#5}}{7}$ dorian b2	b9, 11, 6	Dorian b2 Locrian nat 6	AugΔ / Maj7
Db^{sus4} / C	$\frac{Db^{maj7sus4}}{7}$ locrian or altered dom.	b9, b5, b6	Locrian Altered Dominant	sus4Δ / Maj7
Db^{lyd} / C	$\frac{Db^{lydmaj7}}{7}$ phrygian	b9, 5, b6	Phrygian Phrygian Major	Lyd.Δ / Maj7
$Db^{loc.}$ / C	C Dom.Dim	b9, #11, 5		Loc.Δ / Maj7
D / C	$\frac{D^{7}}{b7}$	9, #11, 13	Lydian (Augmented) Mixolydian #11 Dorian #9 Tonic Diminished	MajΔ / b7
D- / C	$\frac{D\text{-}^{7}}{b7}$	9, 11, 13	Ionian Dorian Mixolydian Meloldic Minor Ionian Augmented Tonic Diminished	MinΔ / b7
D^{o} / C	$\frac{D\text{-}^{7b5}}{b7}$	9, 11, b6	Aeolian Mixolydian b6 Locrian nat 2 Harmonic Minor Tonic Diminished	DimΔ / b7
D^{+} / C	$C^{9/b5\ no3rd}$	9, #11, b7	Mixolydian #11 Dorian #4 Whole Tone	AugΔ / b7

Chord Name	Chord Type (Seventh or Mode)		Triad / Bass Interval Relationship	Scale(s)	Generic Chord Name or Interval above Top Triad
$\frac{D^{sus4}}{C}$	$C^{6/9\ no\ 3rd}$		9, 5, 13	Ionian Dorian Lydian Mixolydian Meloldic Minor Mixolydian #11 Dorian #4 Pentatonic (Major)	$\frac{sus4\Delta}{b7}$
$\frac{D^{lyd}}{C}$	$C^{6/9/\#11\ no\ 3rd}$		9, #5, 13	Lydian Augmented Ionian Augmented	$\frac{Lyd.\Delta}{b7}$
$\frac{D^{loc.}}{C}$	$\frac{Ab^{maj7/b5}}{3^{rd}}$		9, 5, b6		$\frac{Majb5}{3^{rd}}$
$\frac{Eb}{C}$	$C\text{-}^{7}$		b3, 5, b7	All minor scales without ^{maj7}	$\frac{Maj\Delta}{6}$
$\frac{Eb\text{-}}{C}$	$C\text{-}^{7b5}$		b3, b5, b7	Locrian Locrian nat 2 Locrian nat 6	$\frac{Min\Delta}{6}$
$\frac{Eb^{o}}{C}$	C^{o7}	Dim^{7}	b3, b5, bb7	Tonic Diminished (Altered Dom. bb7)	Dim^{7}
$\frac{Eb^{+}}{C}$	$C\text{-}^{maj7}$	Min^{maj7}	b3, 5, 7	Meloldic Minor Harmonic Minor	Min^{maj7}
$\frac{Eb^{sus4}}{C}$	$C\text{-}^{7b6}$	aeolian	b3, b6, b7	Aeolian Altered Dominant	$Min^{7/b6/(\#5)}$
$\frac{Eb^{lyd}}{C}$	$C\text{-}^{13}$	dorian	b3, 6, b7	Dorian Type Meloldic Minor	Min^{13}
$\frac{Eb^{loc.}}{C}$	$C^{o/b13}$		b3, #5, 13	Tonic Diminished	Dim^{b13}
$\frac{E}{C}$	$C^{maj7/\#5}$	Aug^{maj7}	3, #5, 7	Lydian Augmented	$Maj^{7/\#5}$
$\frac{E\text{-}}{C}$	C^{maj7}	Maj^{7}	3, 5, 7	Ionian Lydian Major Pentatonic Major Blues	Maj^{7}
$\frac{E^{o}}{C}$	C^{7}	Dom^{7}	3, 5, b7	Mixolydian Mixolydian #11 Major Pentatonic Major Blues	Dom^{7}
$\frac{E^{+}}{C}$	C^{+}		1, 3, #5	Whole Tone	$Aug\Delta$

Chord Name	Chord Type (Seventh or Mode)	Triad / Bass Interval Relationship	Scale(s)	Generic Chord Name or Interval above Top Triad
$\frac{E^{sus4}}{C}$	$C^{maj7/6}$	3, 6, 7	Ionian Lydian Lydian Augmented	Maj^{13}
$\frac{E^{lyd}}{C}$	Twelve tone	3, b7, 7	Chromatic Scale	
$\frac{E^{loc.}}{C}$	$C^{13\ (no\ 5)}$	3, 6, b7	Mixolydian Mixolydian #11 Dominant Diminished	Dom^{13}
$\frac{F}{C}$	$\frac{F}{5^{th}}$	3, 5, 7	Major Type	$\frac{MajΔ}{P5}$
$\frac{F\text{-}}{C}$	$\frac{F\text{-}}{5^{th}}$ Dom Tonic	b3, 5	Minor Type	$\frac{MinΔ}{P5}$
$\frac{F^{o}}{C}$	$C^{maj7/\#5/sus4}$	4, #5, 7	Aeolian Harmonic Minor	$\frac{DimΔ}{P5}$
$\frac{F^{+}}{C}$	$C^{dor/b2}$	4, 6, b9	Dorian b2 Phrygian Phrygian Major	$\frac{AugΔ}{P5}$
$\frac{F^{sus4}}{C}$	CQ	4, b7	Dorian Mixolydian	Quartal
$\frac{F^{lyd}}{C}$	CQ+	4, 7	Ionian	Q. Aug
$\frac{F^{loc.}}{C}$	C^{loc}	4, b5	Locrian	
$\frac{F\#}{C}$	$C^{7/b9/b5}$	b5, b7, b9	Dominant Diminished Altered Dominant	$Dom^{b9/b5}$
$\frac{F\#\text{-}}{C}$	$C^{13/b9/b5}$	b9, 13, b5	Dominant Diminished	$Dom^{13/b9/b5}$
$\frac{F\#^{o}}{C}$	C^{o}	1, b3, b5	Tonic Diminished	DimΔ
$\frac{F\#^{+}}{C}$	$C^{9/b5}$	b5, b7, 9	Whole Tone	Dom^{9b5}
$\frac{F\#^{sus4}}{C}$	Twelve tone	b5, 7, b9	Chromatic Scale	
$\frac{F\#^{lyd}}{C}$	$C^{7/b9/b5}$	b5, 1, b9	Dominant Diminished Altered Dominant	$Dom^{7/b9/b5}$
$\frac{F\#^{loc.}}{C}$	$C^{maj7/\#11}$	b5, 7, 1	Lydian Lydian Augmented	$Maj^{7/\#11}$

Chord Name	Chord Type (Seventh or Mode)	Triad / Bass Interval Relationship	Scale(s)	Generic Chord Name or Interval above Top Triad
G / C	$C^{MA9\ no5th}$	5, 7, 9	Ionian Lydian	$Maj^{9\ no\ 5th}$
G- / C	C^{9}	5, b7, 9	Mixolydian Mixolydian #11	Dom^{9}
G^{o} / C	$C^{7/b9}$	5, b7, b9	Dominant Diminished	$Dom^{7/b9}$
G^{+} / C	$C\text{-}^{maj7}$	5, 7, b3	Meloldic Minor Harmonic Minor	Min^{maj7}
G^{sus4} / C	C^{sus2}	5, 1, 2	Ionian Lydian	Sus2
G^{lyd} / C	Twelve tone	5, b9, 9	Chromatic Scale	
$G^{loc.}$ / C	$C^{7/b9}$	5, 1, b9	Dominant Diminished	$Dom^{7/b9}$
Ab / C	$C\text{-}^{b6}$	b6, 1, b3	Aeolian	$Min^{7/b6}$
Ab- / C	$C^{omaj7/b13}$	b13, 7, b3	Tonic Diminished	$Dim^{maj7/b13}$
Ab^{o} / C	$C^{omaj7/9/b13}$	b13, 7, 9	Tonic Diminished	$Dim^{maj7/9/b13}$
Ab^{+} / C	$C^{MA\#5}$	#5, 1, 3	Whole Tone Lydian Augmented Ionian Augmented	$Maj^{\#5}$
Ab^{sus4} / C	$C^{(7)/b9/\#9/\#5}$	#5, b9, #9	Altered Dominant	$Dom^{7/\#5/b9/\#9}$
Ab^{lyd} / C	$C\text{-}^{9/b6}$	b6, 9, b3	Aeolian	$Min^{9/b6}$
$Ab^{loc.}$ / C	Twelve tone	b6, b9, 9	Chromatic Scale	
A / C	$C^{13/b9}$	13, b9, 3	Dominant Diminished	$Dom^{13/b9}$
A- / C	C^{13}	13, 1, 3	Mixolydian	Dom^{13}
A^{o} / C	C^{o}	1, b3, b5	Tonic Diminished	DimΔ
A^{+} / C	$C^{13/sus4/b9}$	13, b9, 11	Dorian b2	(Min) $Dom^{13/sus4/b9}$

Chord Name	Chord Type (Seventh or Mode)	Triad / Bass Interval Relationship	Scale(s)	Generic Chord Name or Interval above Top Triad
$\frac{A^{sus4}}{C}$	$C^{6/9}$	6, 9, 3	Ionian Lydian	$Maj^{6/9}$
$\frac{A^{lyd}}{C}$	$C^{13/\#9}$	13, #9, 3	Dominant Diminished	$Dom^{13/\#9}$
$\frac{A^{loc.}}{C}$	$C\text{-}^{6/9}$	6, 9, b3	Dorian Meloldic Minor	$Min^{6/9}$
$\frac{Bb}{C}$	$C^{9/11}$	b7, 9, 11	Mixolydian	$Dom^{9/11}$
$\frac{Bb\text{-}}{C}$	$C^{7/sus4/b9}$	b7, b9, 11	Dorian b2 Phrygian Phrygian Major	$Dom^{7/sus4/b9}$
$\frac{Bb^{o}}{C}$	C^{7b9}	b7, b9, 3	Dominant Diminished	$Dom^{7/b9}$
$\frac{Bb^{+}}{C}$	$C^{9/b5}$	b7, 9, #11	Mixolydian #11 Whole Tone	$Dom^{9/b5}$
$\frac{Bb^{sus4}}{C}$	$C\text{-}^{11}$	b7, b3, 11	Dorian Phrygian Aeolian	Min^{11}
$\frac{Bb^{lyd}}{C}$	$C^{7/sus4/3}$	b7, 3, 4	Mixolydian	$Dom^{7/sus4/3}$
$\frac{Bb^{loc.}}{C}$	$C^{7/\#9}$	b7, #9, 3	Dominant Diminished Altered Dominant	$Dom^{7/\#9}$
$\frac{B}{C}$	C^{omaj7}	7, b3, b5	Tonic Diminished	Dim^{maj7} $\frac{MajΔ}{b9}$
$\frac{B\text{-}}{C}$	$C^{MA9/\#11}$	7, b3, b5	Lydian	$Maj^{9/\#11}$
$\frac{B^{o}}{C}$	$C^{maj7/9/sus4}$	7, 9, 11	Harmonic Minor	$\frac{DimΔ}{b9}$
$\frac{B^{+}}{C}$	$C\text{-}^{maj7}$	7, b3, 5	Meloldic Minor Harmonic Minor	Min^{maj7}
$\frac{B^{sus4}}{C}$	$C^{maj7/\#11}$	7, 3, #11	Lydian Lydian Augmented	$Maj^{7/\#11}$
$\frac{B^{lyd}}{C}$	Chromatic scale	7, 11, #11	Chromatic Scale	
$\frac{B^{loc.}}{C}$	$C^{maj7/sus4}$	7, 3, 4	Ionian	$Maj^{7/sus4}$

Ex. 87

$$\frac{E}{C} = \frac{MajΔ}{b6}$$ (min6 above E)

What we have is a major triad with its b6 in the bass. The b6 is in relation to the note E (above it). The interval on the bottom will always be that interval above the tonic of the triad (E in this case).

Now that you are thoroughly confused, let´s look at it in a few more keys.

Ex. 88

$$\frac{E}{C} = \frac{MajΔ}{b6}$$ Maj triad over its b6 in bass

$$\frac{D}{Bb} = \frac{A}{F} = \frac{F\#}{D} = \frac{MajΔ}{b6}$$

These also all happen to be $Maj^{7\#5}$ chords as we can see.

Ex. 89

$$\frac{E}{C} = \frac{E, G\#, B}{C} = \begin{matrix} C & E & G\# & E \\ 1 & 3 & \#5 & 7 \end{matrix} = C^{maj7\#5}$$

In many cases I have indicated a mode name in the Chord Type column. The $\frac{Db}{C}$ for chord example is named in two ways:

1. as a Db^{maj7} chord with C in the bass
2. as a C phrygian chord

Although I personally believe that all chords should be named in relation to their bass note (in the case of $\frac{Db}{C}$ it is a type of a C chord, not a Db^{maj7} chord in inversion) the other approach should also be recognized. The scales in the chord scale section are related to the bass note.

Ex. 90

$$\frac{Db}{C} =$$ C phrygian

C phrygian major

Reharmonization and Chord Substitutions

Before going into some actual reharmonizations, let´s briefly discuss a few traditional substitute patterns.

Throughout the course of Jazz history, certain compositions have produced a variety of harmonic substitute patterns. Works such as "Giant Steps", "Countdown", "Ladybird" and "Blues for Alice" are all tunes which contain such harmonic substitutes.

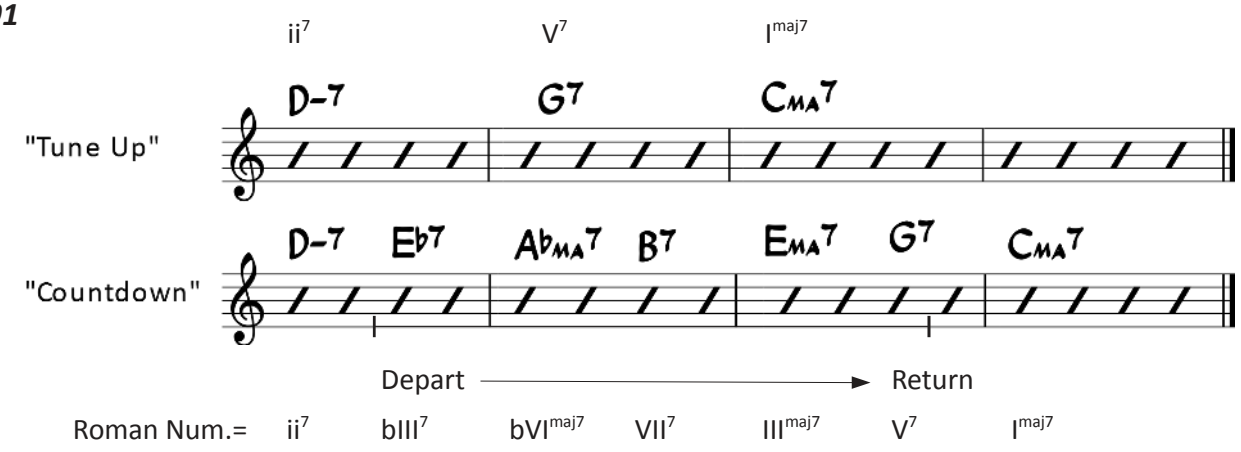
The terms "Countdown Changes" or "Bird Blues" are examples of Jazz lingo referring to specific substitute patterns in todays jazz scene. Let´s examine some traditional substitute patterns.

"Countdown Changes"

The tune "Countdown" by John Coltrane was first released on the album "Giant Steps" in 1959 on Atlantic SD-1311.

"Countdown" is based on the harmonic progression of a Miles Davis composition entitled "Tune Up". The basic principle of chord substitution is as follows:

Ex. 91

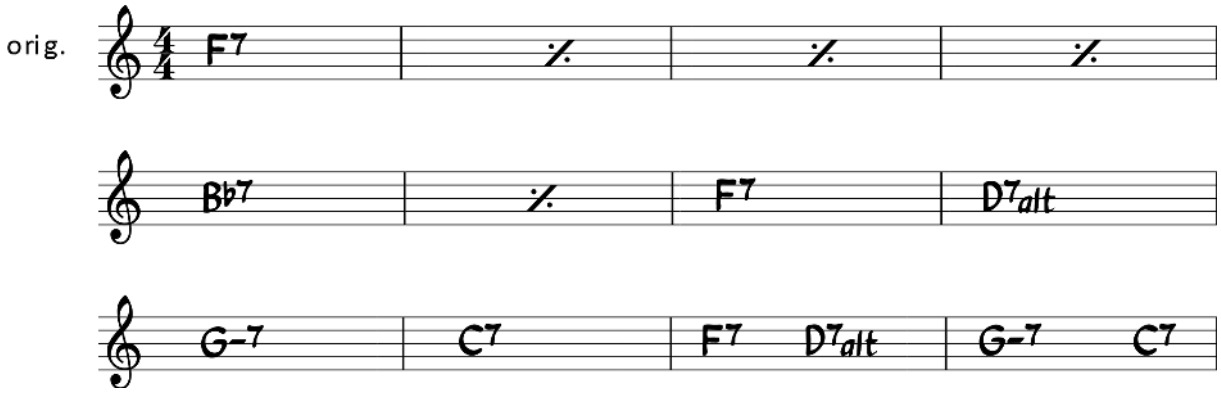


This Progression may be inserted in any tune with a four bar ii-V^7-I^{maj7} progression regardless of what the rhythm section is playing under it.

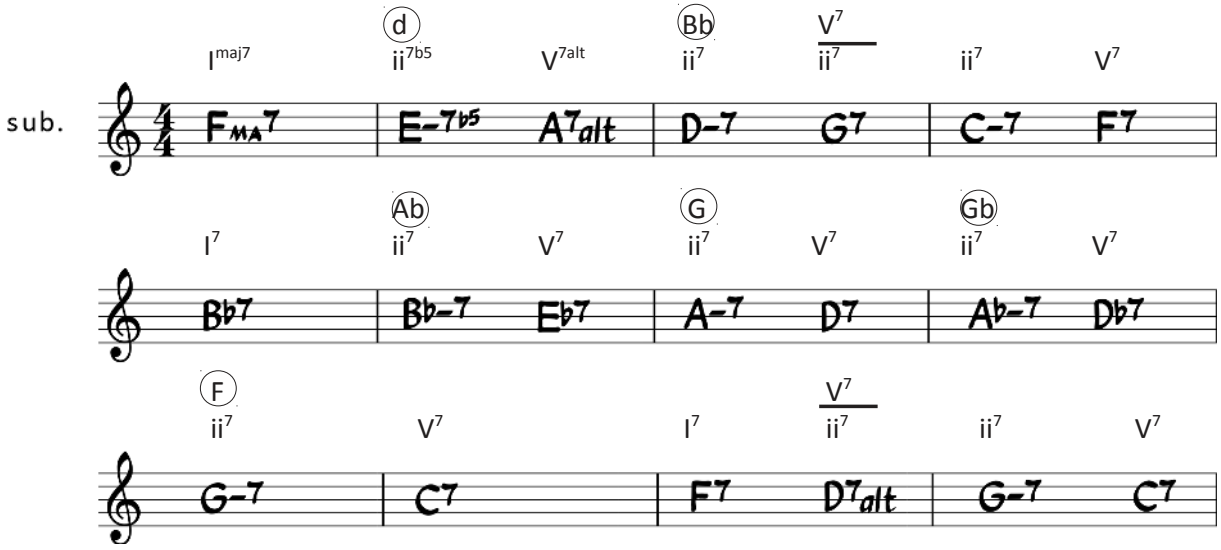
"Bird Blues"

A "Bird Blues" is a chord progression derived from the tune "Blues for Alice" by Charlie Parker. It is based on a twelve bar blues and uses a series of ii-V substitution.

Ex. 92a



Ex. 92b

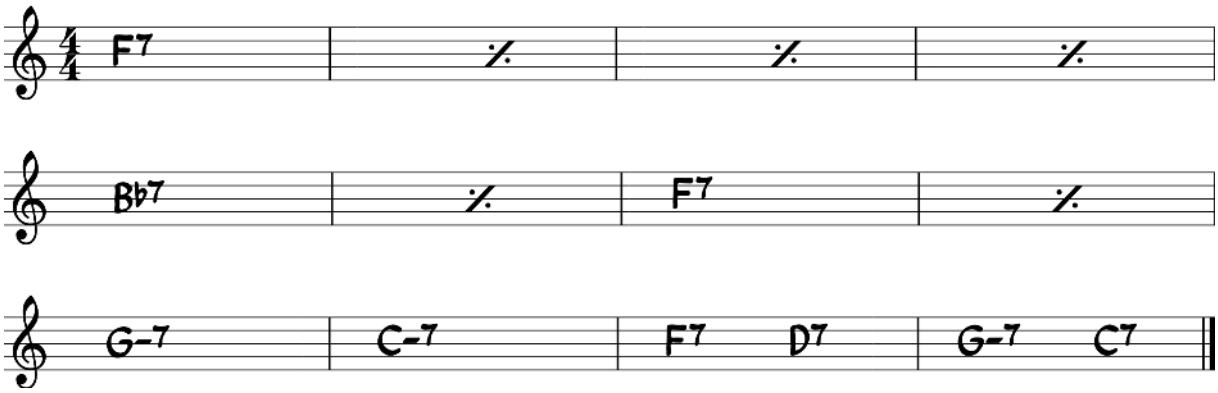


Advanced Reharmonizations

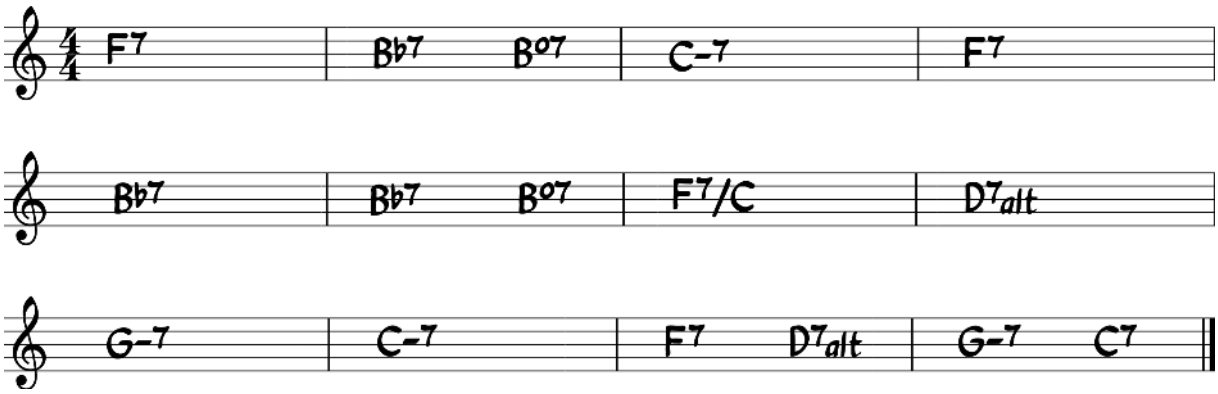
To demonstrate advanced reharmonizations, I have tried to incorporate as many substitution rules as possible.

Additional rules will be presented throughout this section in order to address the new harmonic situations encountered. The type of tunes used for reharmonization fall into the categories of Blues, Rhythm Changes and two "Standard" Jazz forms. Because of the high concentration of Blues-form tunes, I will begin with more of its variations.

Blues

F Blues Basic

Substitution Characteristics

F Blues ①

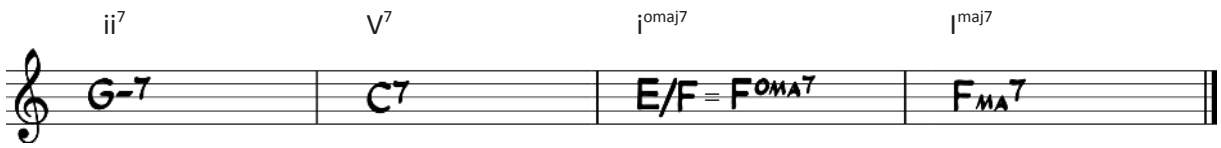


1. Basic Bebop Blues

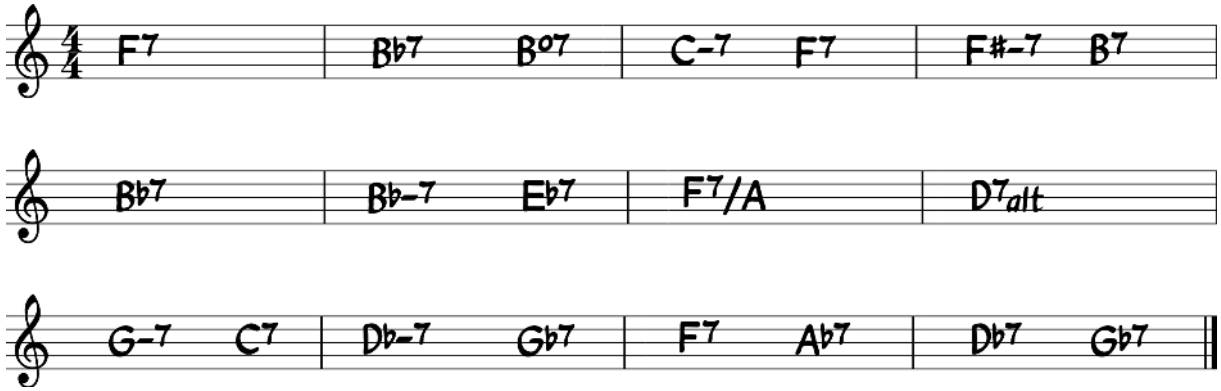
   - B^{o7} in bar 2 is $\frac{vii^{o7}}{ii}$ going to the ii^7 chord ($c\text{-}^7$) in next measure. Rule XVII

   - B^{o7} in bar 5 is actually an F^{o7} resolving to $\frac{F^7}{C}$. This is a common delayed resolution technique and is used many times in ii – V^7 – I^{maj7} for the first 2 beats (or bars) of the I^{maj7} chord as demonstrated in this example.

Ex. 93



F Blues ②



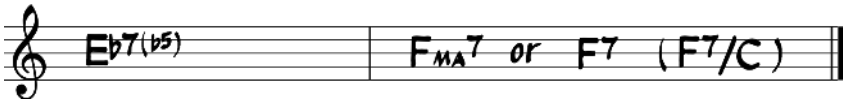
2. The second Blues is still a Bebop type blues with a few modifications.

   - F#-^{7} to B^{7} in bar 3 is a tritone substitution in the key of Bb with its ii^{7} chord in front of it. Rule XII part2

   - Bb^{-7} to Eb^{7} in bar 6 is actually an Eb^{7} to F^{7} resolution with the supertonic chord (Bb-^{7}) in front of the dominant. This brings up a new rule.

Rule XXX: Dom^{7} chords

Dom^{7} chords that resolve to Dom^{7} or Maj^{7} chord a whole step above.

Ex. 94a

F mixolydian = Bb maj

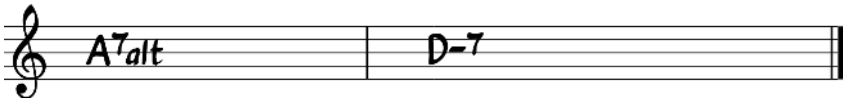
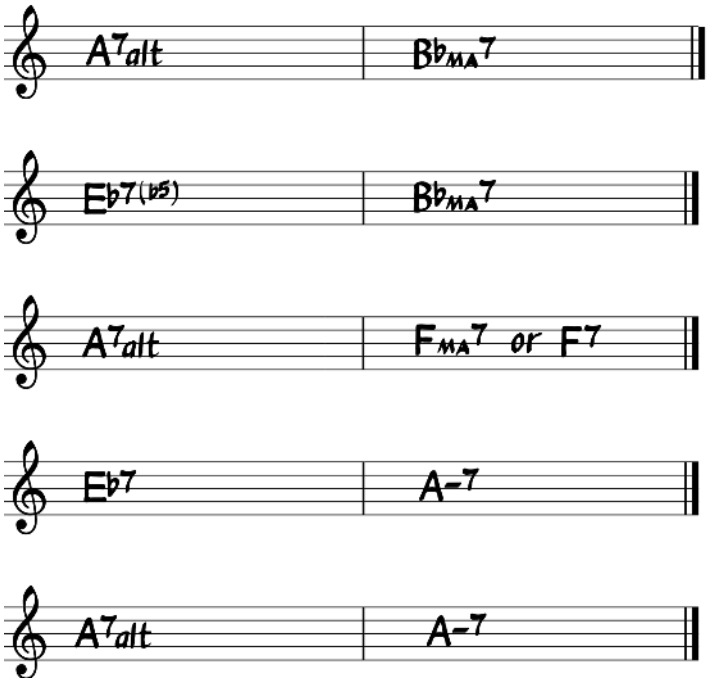
Eb mix#11 = Bb melodic min

A alt. dom = Bb melodic min

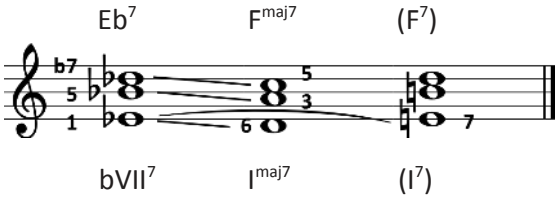
F ionian =

D aeolian =

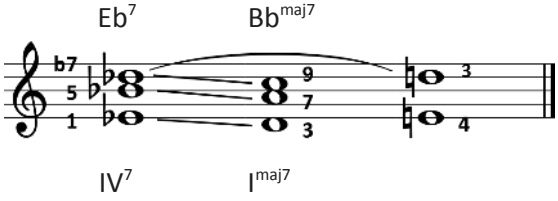
} F maj

D phrygian = Bb maj

Ex. 94b

The main reason that these progressions work is because of the strong resolution tendencies of the Db, Bb and Eb (b7th, 5th and root) of the Eb^7

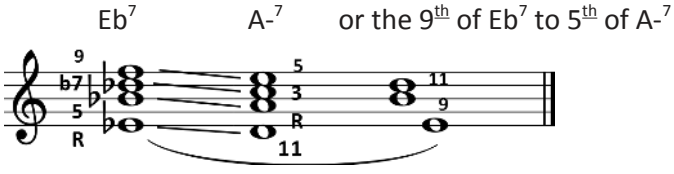
Ex. 95a



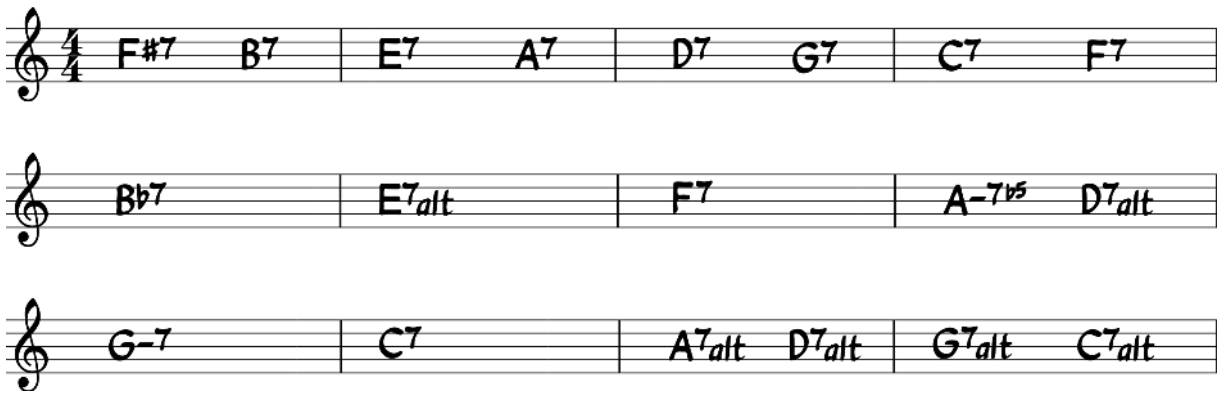
Ex. 95b



Ex. 95c



F Blues ③



3. The third Blues begins on the bII^{7} chord employing a cycle of 5$^{\underline{ths}}$ substitution (Rule XXIII) and is intersecting with the Bb^{7} chord in bar 5.

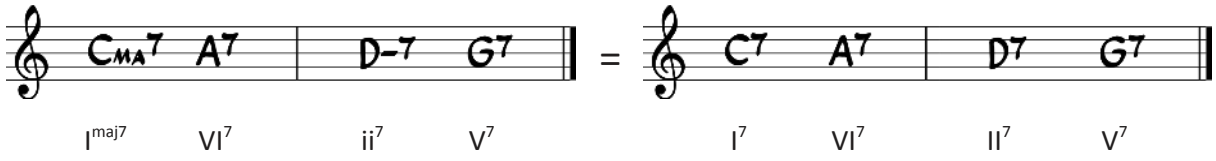
   - E^{7alt} is used in bar 6 as a tritone substitution for Bb$^{7(b5)}$

   - the turnaround A^{7alt} D^{7alt} G^{zalt} C^{7alt} is a III^{7} VI^{7} II^{7} V^{7} Dom^{7} cycle substitution for the usual I^{7} VI^{7} ii^{7} V^{7} progression.

Rule XXXI

Dom^{7} chords can replace either major or minors.

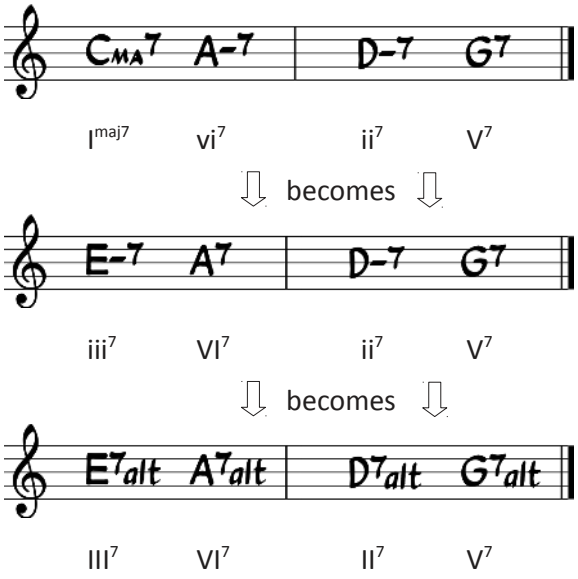
Ex. 96

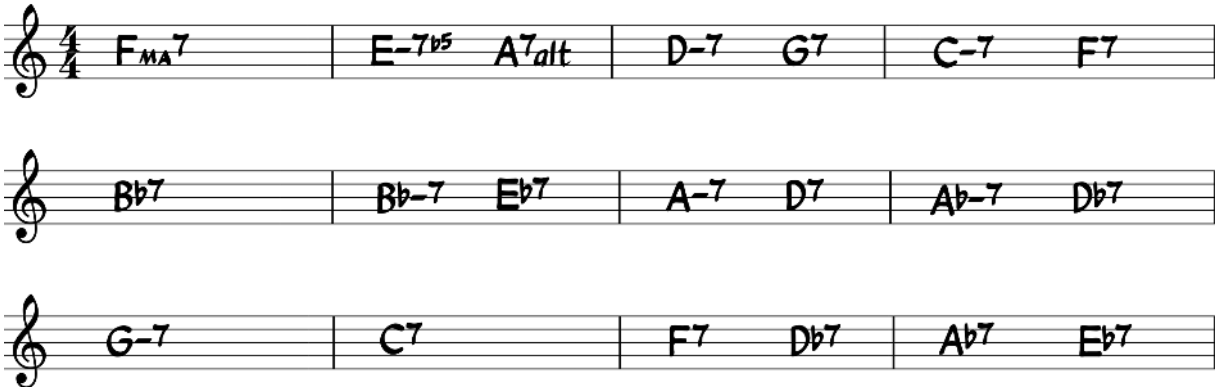


I^{maj7} VI^{7} ii^{7} V^{7} = I^{7} VI^{7} II^{7} V^{7}

Rule XXXI cont.

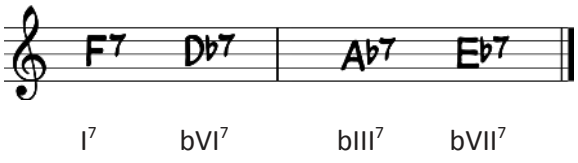
Ex. 97



F Blues ④

4. Blues #4 is commonly called a "Parker (Bird) Blues“ or "Altered Blues“ as we discussed earlier.

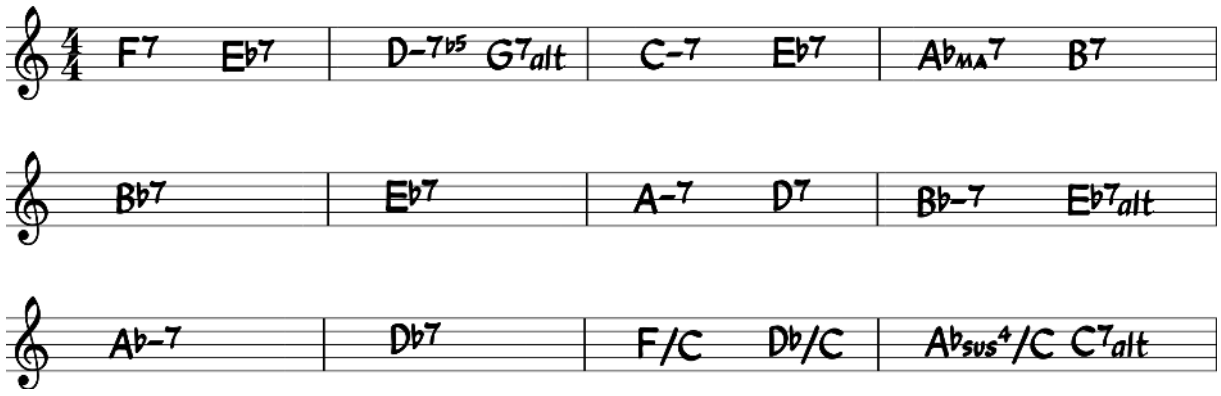
   The following turnaround is common in modern jazz.

Ex. 98



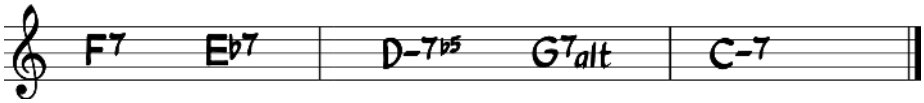
Any or all of the chords may be made maj^7 if desired.

F Blues ⑤



5. Blues #5 uses some interesting devices beginning with the cadence:

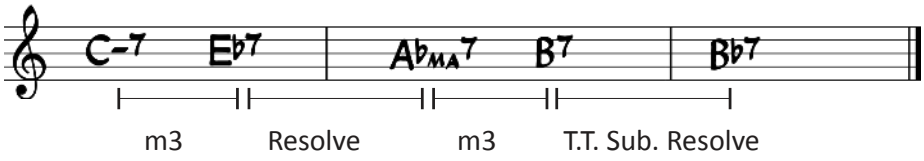
Ex. 99



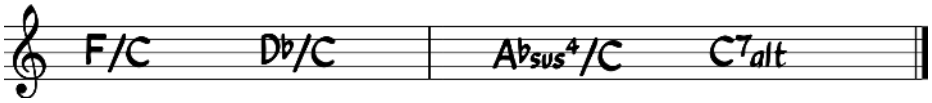
At first it looks deceiving but at a closer glance it makes more sense. The Eb^{7} chord in bar one acts as a bII^{7} tritone substitute to D^{-7b5} (which is actually a substitute for $Bb^{7(9)}$ in bar [Rule VII]).

The D^{-7b5} moves to G^{7alt} and then to C^{-7}, a substitute for F^{7} (Rule IV). The next substitute is based on John Coltrane's "Countdown Substitutions“ discussed earlier.

Ex. 100

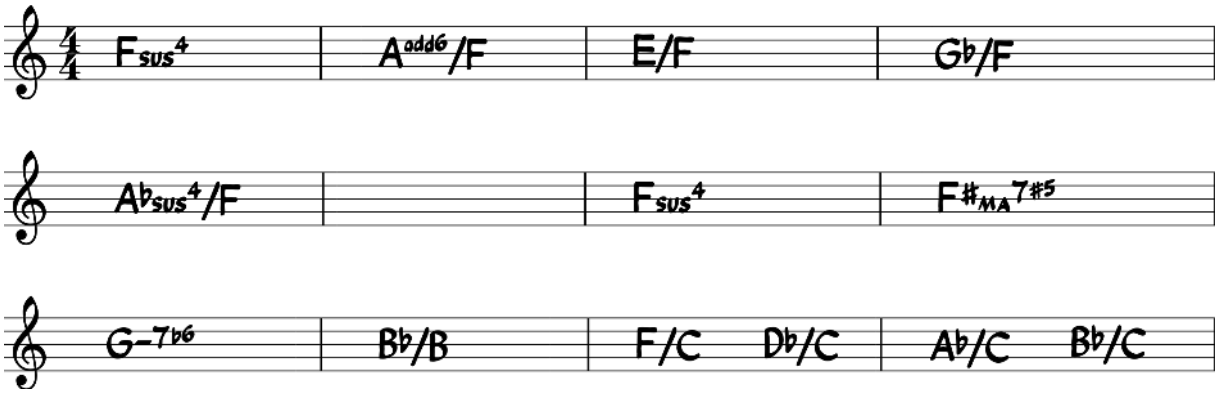


Ex. 101

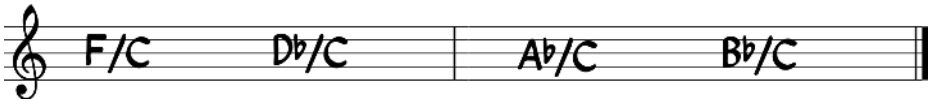


This turnaround is a dominant pedal point that works on a modal modulation principle.

F Blues ⑥

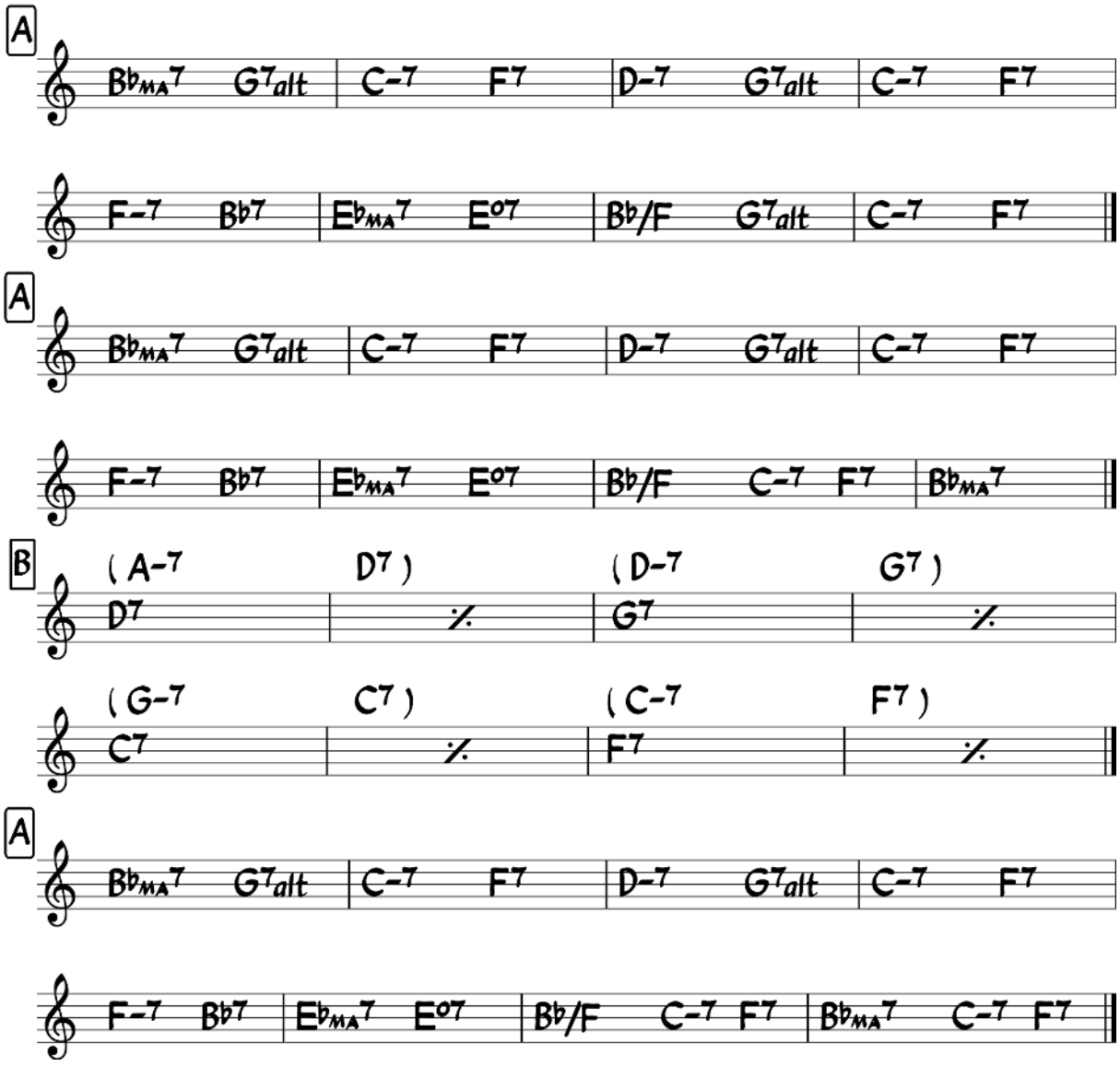


6. Blues #6 uses a tonic pedal (bars 1-7) as well as a dominant pedal for the turnaround

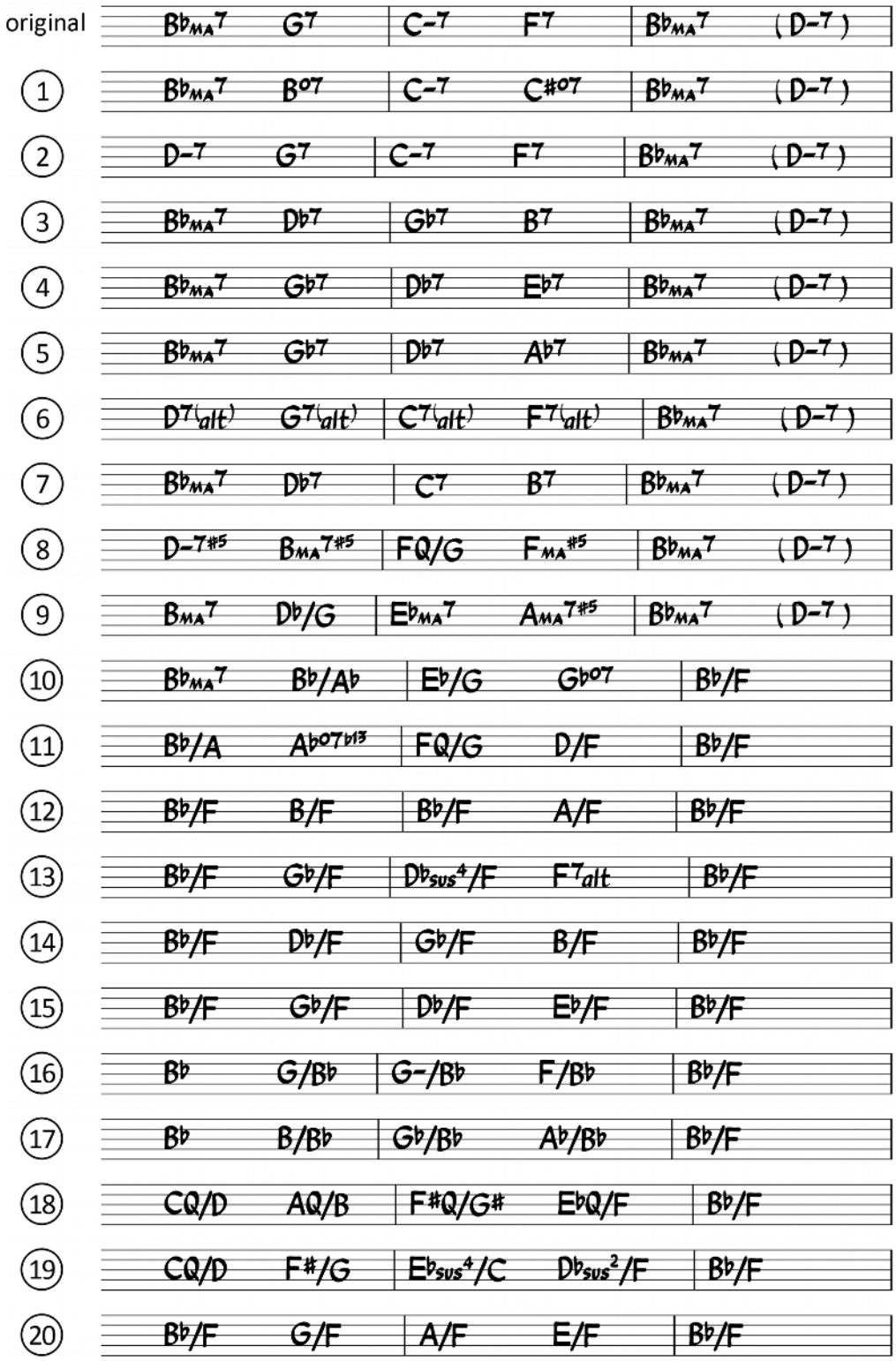
Ex. 102

You should play through every progression to gain an aural familiarity of each. The smart musician will learn how to combine the progressions in different ways. In addition he / she will create new reharmonizations based on the rules previously stated. Remember, all of the measures between different progressions are interchangeable, provided the bassline and voice leading are logical.

Rhythm Changes

Rhythm Changes is a term coined for the chord progression of George Gershwin´s "I Got Rhythm“. The progression has provided a basis for many standard Bebop tunes. The most common harmonization of Rhythm Changes is:

Ex. 103

As you probably have noticed, the I^{maj7} VI^7 ii^7 V^7 and iii VI^7 ii^7 V^7 progressions comprise a large portion of this harmonic progression. That is why this progression is commonly refered to as a "Turnaround" progression. Here are some examples of substitute turnarounds which may be employed in the first four bars of any A section in Rhythm Changes.

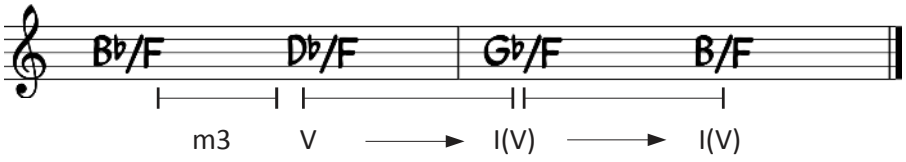
Ex. 104



Rule XXXII

When creating triad over bass note substitutes, pay particular attention to the movement and voiceleading of the upper triads. They shoud move in the strongest direction of resolution as if there were no bass note below.

Ex. 105



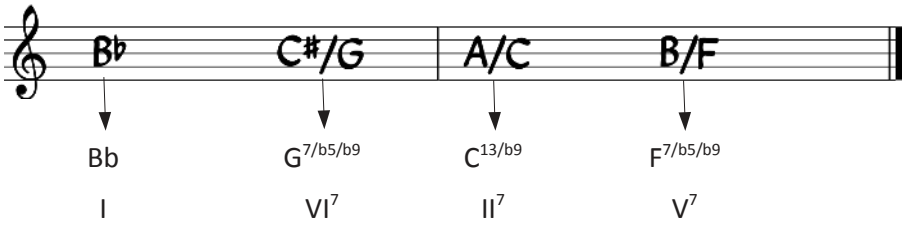
Db → Gb = V I

Gb → B = V I

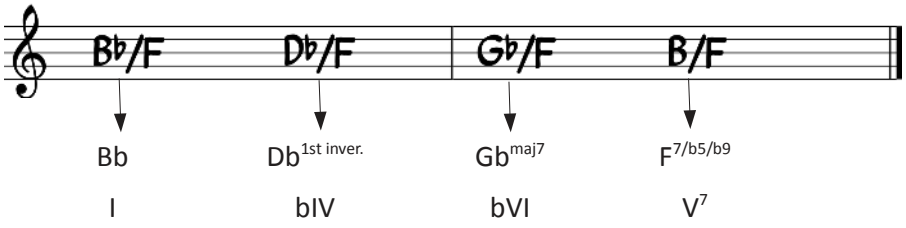
strong cadence of upper triads

You should also keep in mind the relationship of the bass note to the chords and how they move in terms of cadential direction.

Ex. 106a



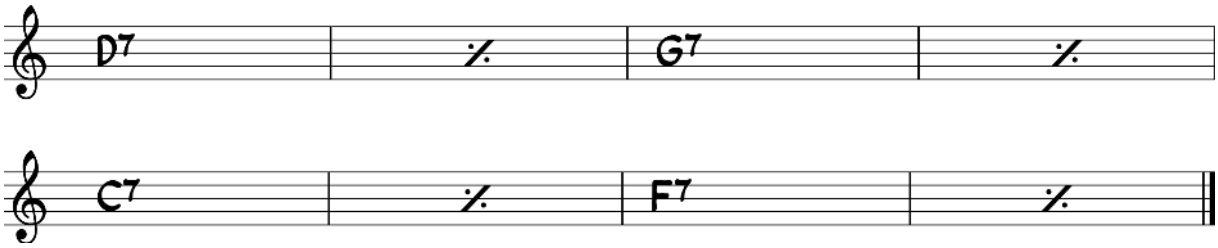
Ex. 106b



The Bridge of Rhythm Changes

The Bridge of Rhythm Changes uses a simple four chord cycle pattern over eight measures.

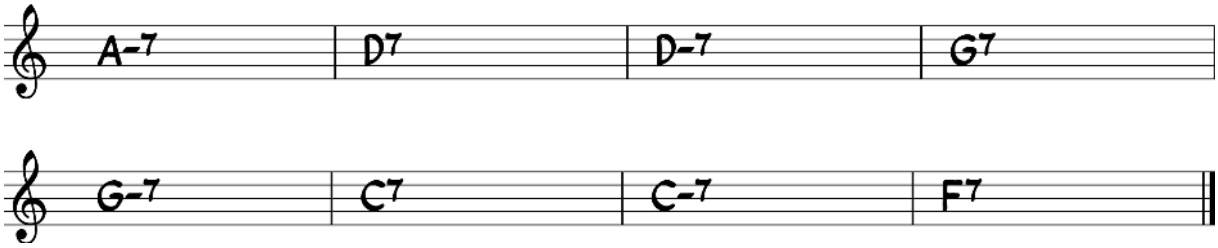
Ex. 107



Since there are many approaches to this progression, I have isolated a few of the most common.

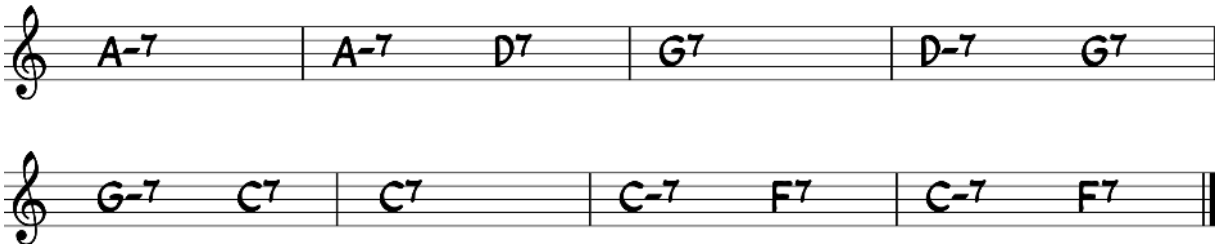
The ii^7 V^7 approach

Ex. 108a



also

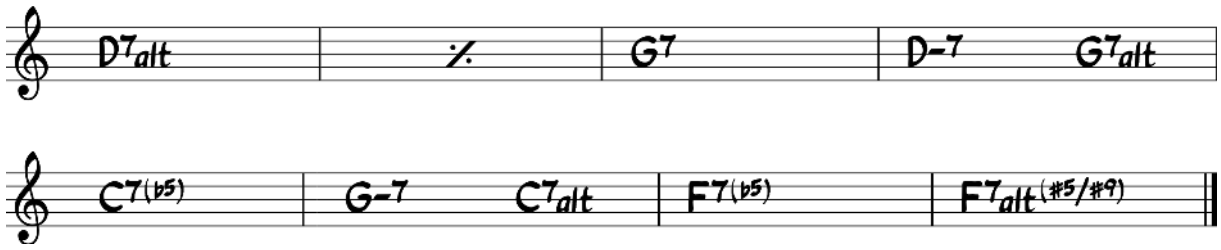
Ex. 108b



You may use any combination of ii^7 V^7 as long as the V^7 chord is returned to (Rule IV)

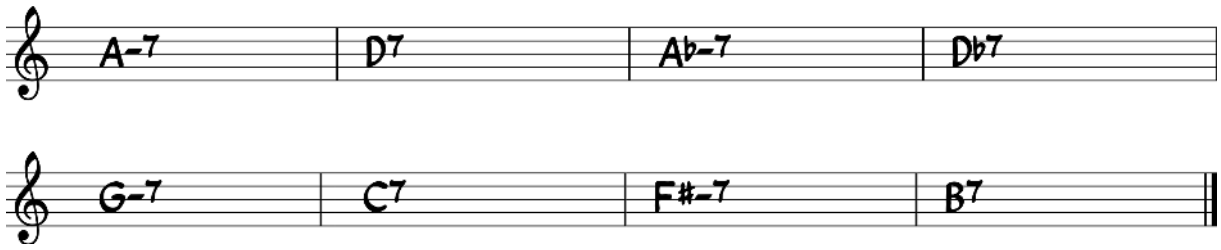
Next is an example which uses altered Dom^7 chords.

Ex. 109



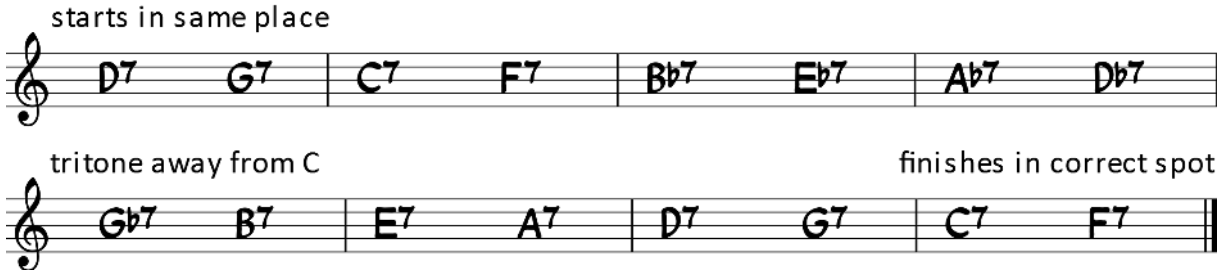
Using Tritone Substitutions

Ex. 110



Using the Cycle

Ex. 111



Any of these chords will also work.

Ex. 112

Chords	Mixolydian	Mixolydian #11	Altered Dominant
	$\underline{D^{7}}$	$\underline{D^{7b5}}$	$\underline{D^{7/\#5/b5/\#9/b9}}$
	$F\#\text{-}^{7b5}$	$A\text{-}^{maj7}$	$Eb\text{-}^{maj7}$
	$A\text{-}^{7}$	$B\text{-}^{13b9}$	$F\text{-}^{13b9}$
	C^{maj7}	$C^{maj7\#5}$	$Gb^{maj7\#5}$
		E^{7b6}	Ab^{7b5}
		$F\#\text{-}^{9b5}$	Bb^{7b6}
		$G\#^{7alt}$	$C\text{-}^{9b5}$

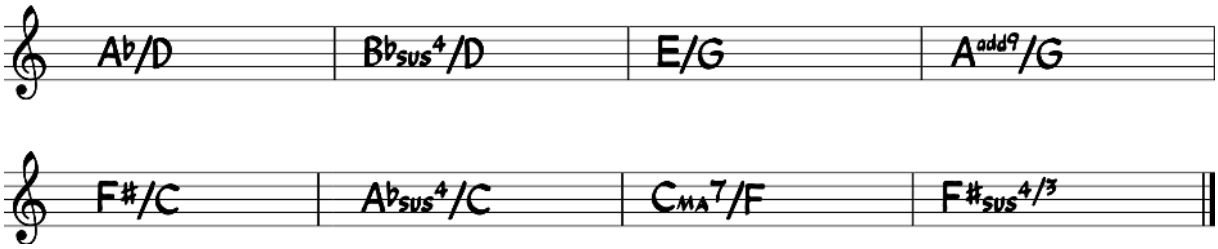
Ex. 113

G^{7}	G^{7b5}	$G^{7/\#5/b5/\#9/b9}$
$B\text{-}^{7b5}$	$D\text{-}^{maj7}$	$Ab\text{-}^{maj7}$
$D\text{-}^{7}$	$E\text{-}^{13b9}$	$Bb\text{-}^{13b9}$
F^{maj7}	$F^{maj7\#5}$	$Cb^{maj7\#5}$
	A^{7b6}	Db^{7b5}
	$B\text{-}^{9b5}$	Eb^{7b6}
	$C\#^{7alt}$	$F\text{-}^{9b5}$

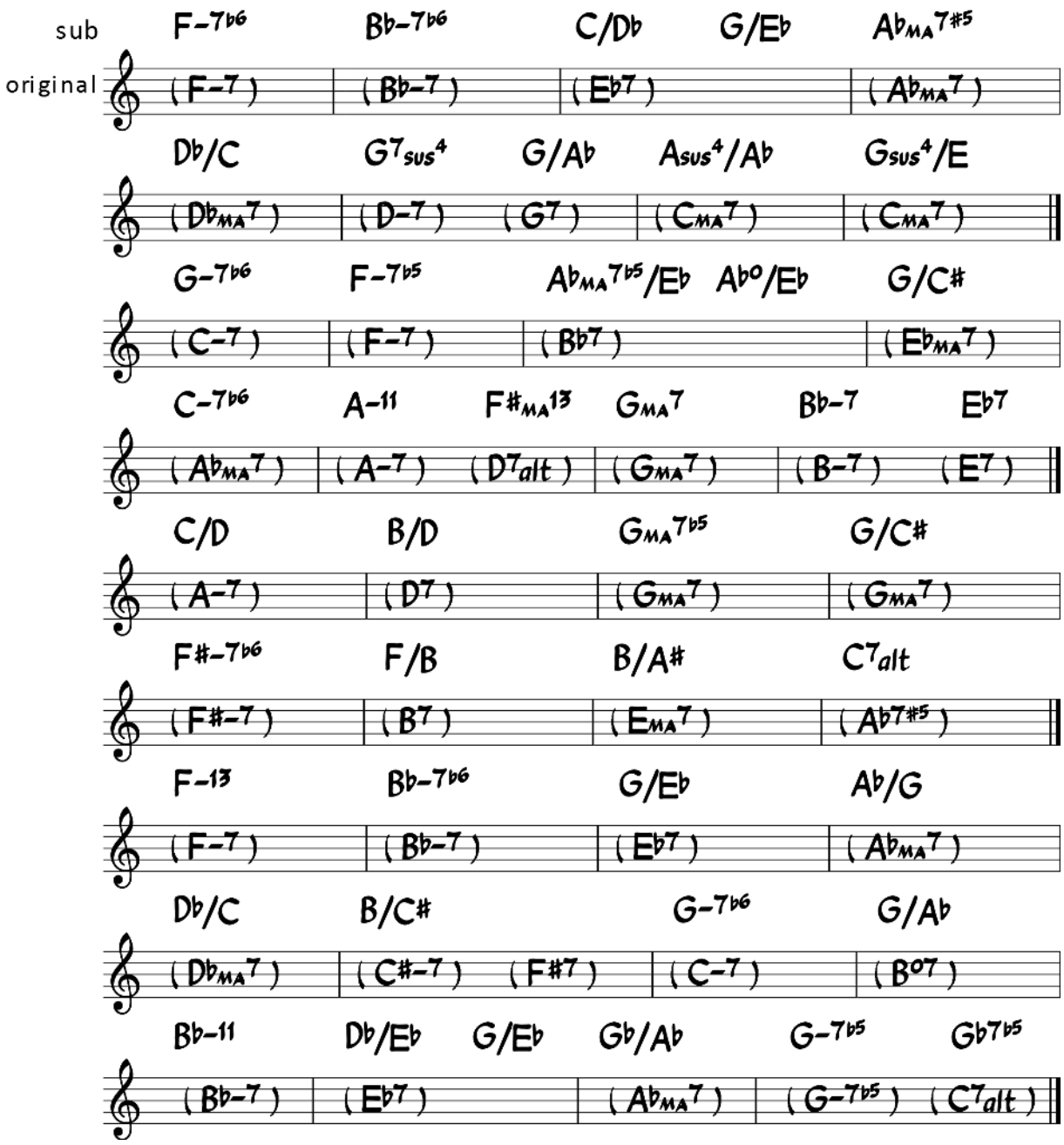
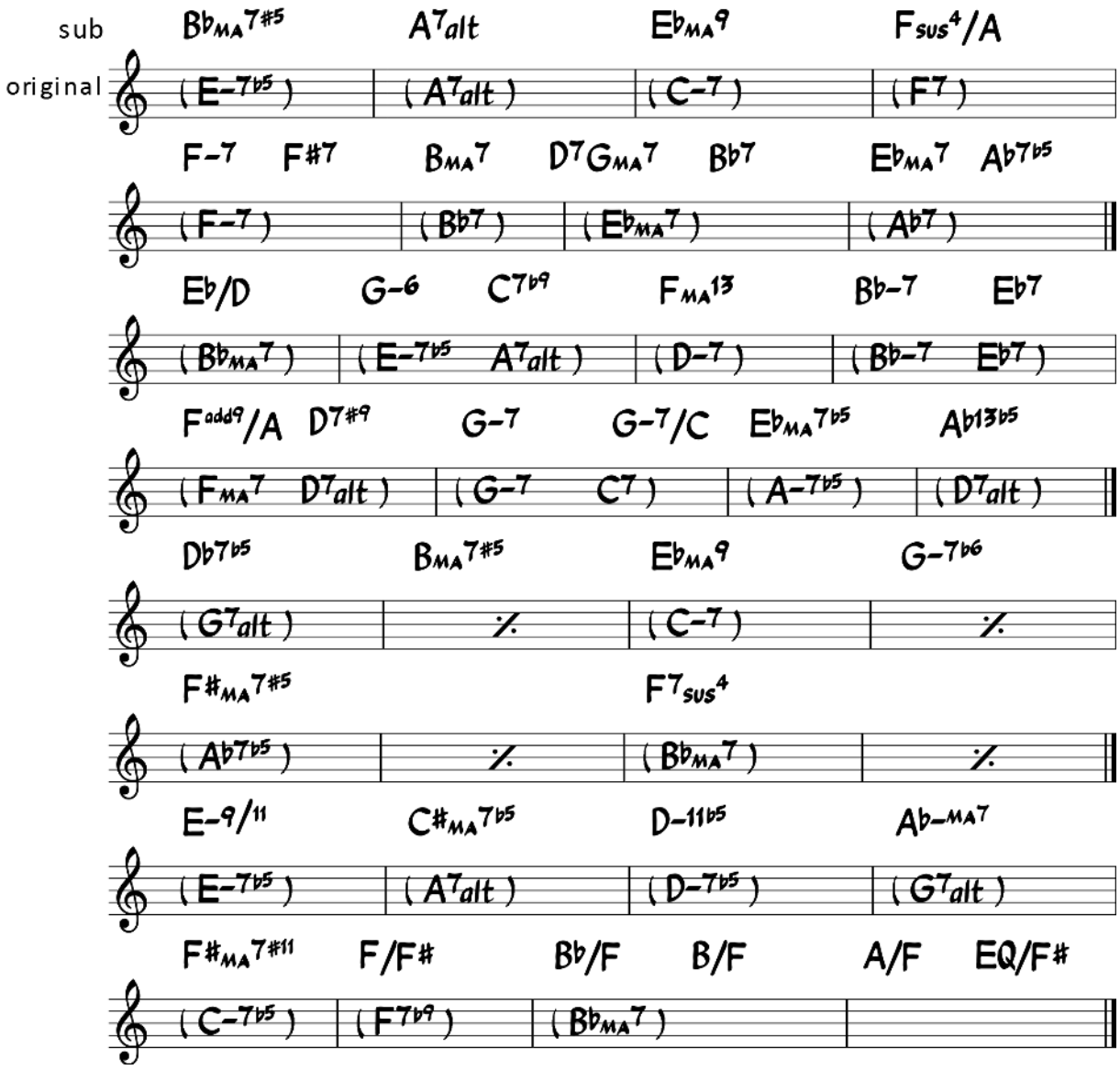
It´s your job to figure out the substitute changes from the modal scales of both C^{7} and F^{7}.

Using Triads Over Bass Notes

Ex. 114



Reharmonization of Standard Forms

Reharmonization No.1

Reharmonization No.2

Common Tone Reharmonization

Rule XXXII

Any melody note may be retained as a common tone (in any voice) for the purpose of reharmonization.

To demonstrate this, I have compiled a short list of possible harmonizations of the note C. Remember the note C may be in any voice.

Ex. 115

Root											
C	C	C-	C^{o}	C^{+}	C^{sus4}	C^{lyd}	C^{loc}	C^{maj7}	C^{-7}	C^{7}	C^{-maj7}
	$C^{7(sus4)}$	C^{-7b5}	C^{o7}	C^{+7}	C^{omaj7}	C^{+maj7}	$C^{-7\#5}$				
	C/Db	C/D	C/Eb	C/E	C/F	C/F#	C/G	C/Ab	C/A	C/Bb	C/B
	C-/Db	C-/D	C-/Eb	C-/E	C-/F	C-/F#	C-/G	C-/Ab	C-/A	C-/Bb	C-/B
	C^{o}/Db	C^{o}/D	C^{o}/Eb	C^{o}/E	C^{o}/F	C^{o}/F#	C^{o}/G	C^{o}/Ab	C^{o}/A	C^{o}/Bb	C^{o}/B
	C^{+}/Db	C^{+}/D	C^{+}/Eb	C^{+}/E	C^{+}/F	C^{+}/F#	C^{+}/G	C^{+}/Ab	C^{+}/A	C^{+}/Bb	C^{+}/B
	C^{sus4}/Db	C^{sus4}/D	C^{sus4}/Eb	C^{sus4}/E	C^{sus4}/F	C^{sus4}/F#	C^{sus4}/G	C^{sus4}/Ab	C^{sus4}/A	C^{sus4}/Bb	C^{sus4}/B
	C^{lyd}/Db	C^{lyd}/D	C^{lyd}/Eb	C^{lyd}/E	C^{lyd}/F	C^{lyd}/F#	C^{lyd}/G	C^{lyd}/Ab	C^{lyd}/A	C^{lyd}/Bb	C^{lyd}/B
	C^{loc}/Db	C^{loc}/D	C^{loc}/Eb	C^{loc}/E	C^{loc}/F	C^{loc}/F#	C^{loc}/G	C^{loc}/Ab	C^{loc}/A	C^{loc}/Bb	C^{loc}/B

Db	Db^{maj7}	C/Db	F/Db	Ab/Db	C-/Db	F-/Db	C^{o}/Db	C^{+}/Db	C^{sus4}/Db	C^{lyd}/Db	C^{loc}/Db
	F^{sus4}/Db	G^{sus4}/Db	F^{lyd}/Db	$F^{\#lyd}$/Db	$F^{\#loc}$/Db	G^{loc}/Db	A^{o}/Db	$F\#^{o}$/Db			

D

D^{7}	$D\text{-}^{7}$	$D\text{-}^{7b5}$	D^{+7}	C/D	F/D	Ab/D	C-/D	F-/D	A-/D	C^{o}/D
C^{+}/D	A^{o}/D	$F\#^{o}$/D	C^{sus4}/D	F^{sus4}/D	G^{sus4}/D	C^{lyd}/D	F^{lyd}/D	$F\#^{lyd}$/D	$F\#^{loc}$/D	G^{loc}/D

Eb

Eb^{MA13}	$Eb\text{-}^{6}$	Eb^{13}	C/Eb	F/Eb	Ab/Eb	Eb^{o7}	C-/Eb	F-/Eb	A-/Eb	C^{o}/Eb
C^{+}/Eb	A^{o}/Eb	$F\#^{o}$/Eb	C^{sus4}/Eb	F^{sus4}/Eb	G^{sus4}/Eb	C^{lyd}/Eb	F^{lyd}/Eb	$F\#^{lyd}$/Eb	C^{loc}/Eb	$F\#^{loc}$/Eb
G^{loc}/Eb										

E

$E\text{-}^{7b6}$	$E^{7\#5}$	C/E	F/E	Ab/E	C-/E	F-/E	A-/E	C^{o}/E	C^{+}/E	A^{o}/E
$F\#^{o}$/E	C^{sus4}/E	F^{sus4}/E	G^{sus4}/E	C^{lyd}/E	F^{lyd}/E	$F\#^{lyd}$/E	C^{loc}/E	$F\#^{loc}$/E	G^{loc}/E	

F

F^{maj7}	$F\text{-}^{7}$	F^{7}	C/F	Ab/F	C-/F	A-/F	C^{o}/F	C^{+}/F	A^{o}/F	$F\#^{o}$/F
C^{sus4}/F	F^{Sus4}	G^{sus4}/F	C^{lyd}/F	F^{lyd}	$F\#^{lyd}$/F	C^{loc}/F	$F\#^{loc}$/F	G^{loc}/F		

F#

$F\#^{maj7\#11}$	$F\#\text{-}^{7b5}$	$F\#^{o7}$	$F\#^{7b5}$	C/F#	Ab/F#	F/F#	C-/F#	A-/F#	F-/F#	A^{o}/F#
C^{sus4}/F#	F^{sus4}/F#	G^{sus4}/F#	C^{lyd}/F#	F^{lyd}/F#	$F\#^{lyd}$	C^{loc}/F#	G^{loc}/F#			

G

$G^{maj7sus4}$	$G\text{-}^{11}$	$G\text{-}^{11b5}$	G^{7sus4}	C/G	Ab/G	F/G	C-/G	A-/G	F-/G	C^{o}/G
A^{o}/G	$F\#^{o}$/G	C^{+}/G	C^{sus4}/G	F^{sus4}/G	G^{sus4}	C^{lyd}/G	F^{lyd}/G	$F\#^{lyd}$/G	C^{loc}/G	$F\#^{loc}$/G

Ab

Ab^{maj7}	Ab^{7}	Ab^{+7}	C/Ab	F/Ab	C-/Ab	A-/Ab	C^{o}/Ab	A^{o}/Ab	$F\#^{o}$/Ab	C^{+}/Ab
C^{sus4}/Ab	F^{sus4}/Ab	G^{sus4}/Ab	C^{lyd}/Ab	F^{lyd}/Ab	$F\#^{lyd}$/Ab	C^{loc}/Ab	$F\#^{loc}$/Ab	G^{loc}/Ab		

A

$A\text{-}^{7}$	$A^{7\#9}$	A^{o7}	$A\text{-}^{7b5}$	C/A	Ab/A	F/A	C-/A	C^{+}/A	C^{sus4}/A	F^{sus4}/A
G^{sus4}/A	C^{lyd}/A	F^{lyd}/A	$F\#^{lyd}$/A	C^{loc}/A	$F\#^{loc}$/A	G^{loc}/A				

Bb

Bb^{MA9}	$Bb-^{9}$	Bb^{9}	$Bb-^{9b5}$	Bb^{o7}	C/Bb	F/Bb	Ab/Bb	C-/Bb	A-/Bb	F-/Bb
C^{o}/Bb	A^{o}/Bb	$F\#^{o}$/Bb	C^{+}/Bb	C^{sus4}/Bb	F^{sus4}/Bb	G^{sus4}/Bb	C^{lyd}/Bb	F^{lyd}/Bb	$F\#^{lyd}$/Bb	C^{loc}/Bb
$F\#^{loc}$/Bb	G^{loc}/Bb									

B

B^{7b9}	C/B	$B-^{7/b5/b9}$	F/B	Ab/B	A-/B	C-/B	F-/B	C^{o}/B	A^{o}/B	$F\#^{o}$/B
C^{+}/B	C^{sus4}/B	F^{sus4}/B	G^{sus4}/B	C^{lyd}/B	F^{lyd}/B	$F\#^{lyd}$/B	C^{loc}/B	$F\#^{loc}$/B	G^{loc}/B	

Remember, this is just a list of easily nameable chord voicings.

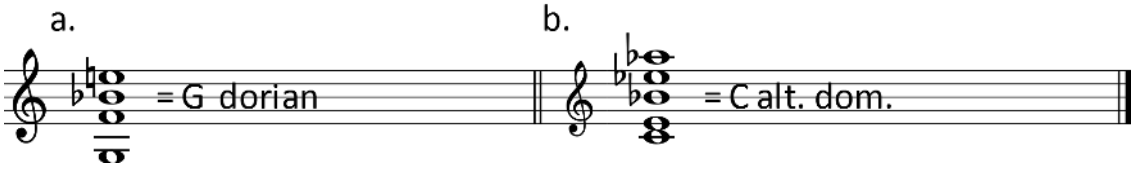
As you can see, it is mind-boggling how many there are. Try to experiment voicing each one of these chords with the note C on top. This will give you a greater understanding of voice leading and increase your chord vocabulary tenfold.

Modal Harmony and Theory

I would like to begin this section on modal harmony and theory by recapping some points already covered.

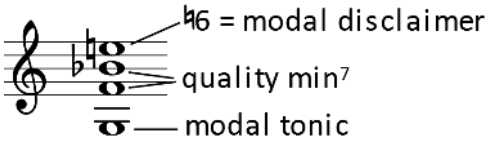
1. In theory, any combination of modal tones over its tonic produces a modal voicing.

Ex. 116



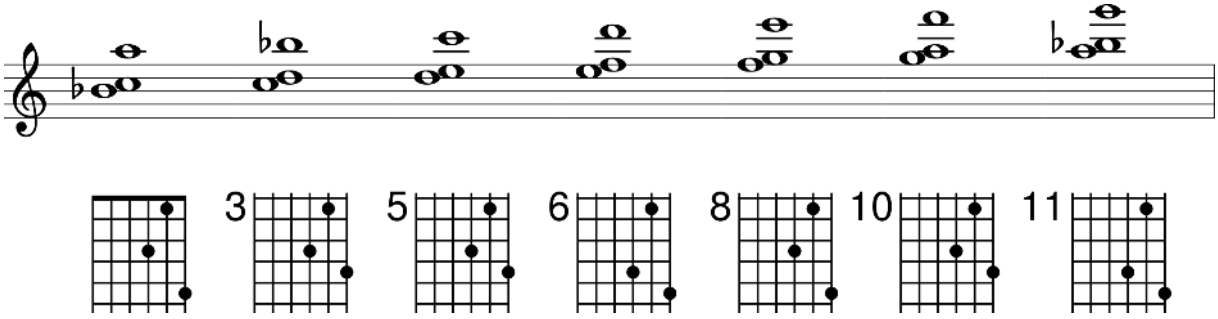
2. The construction of a modal voicing is dependent on the presence of two factors:
    a) A clear definition of the chord's quality (maj, min, etc.) including the seventh if desired.
    b) Emphasis of one or more distinguishing chord tones.

Ex. 117



3. For comping purposes voicing families should be learned horizontally on the fingerboard.

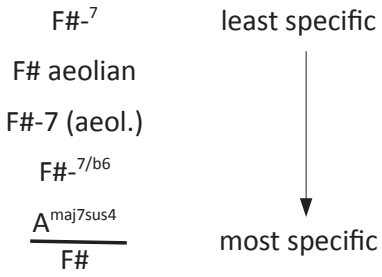
Ex. 118



Each note of the voicing moves up the scale diatonically.

4. Labeling chords by mode is helpful to the improviser in that it provides the appropriate scale choice and a more descriptive harmonic interpretation.

Ex. 119

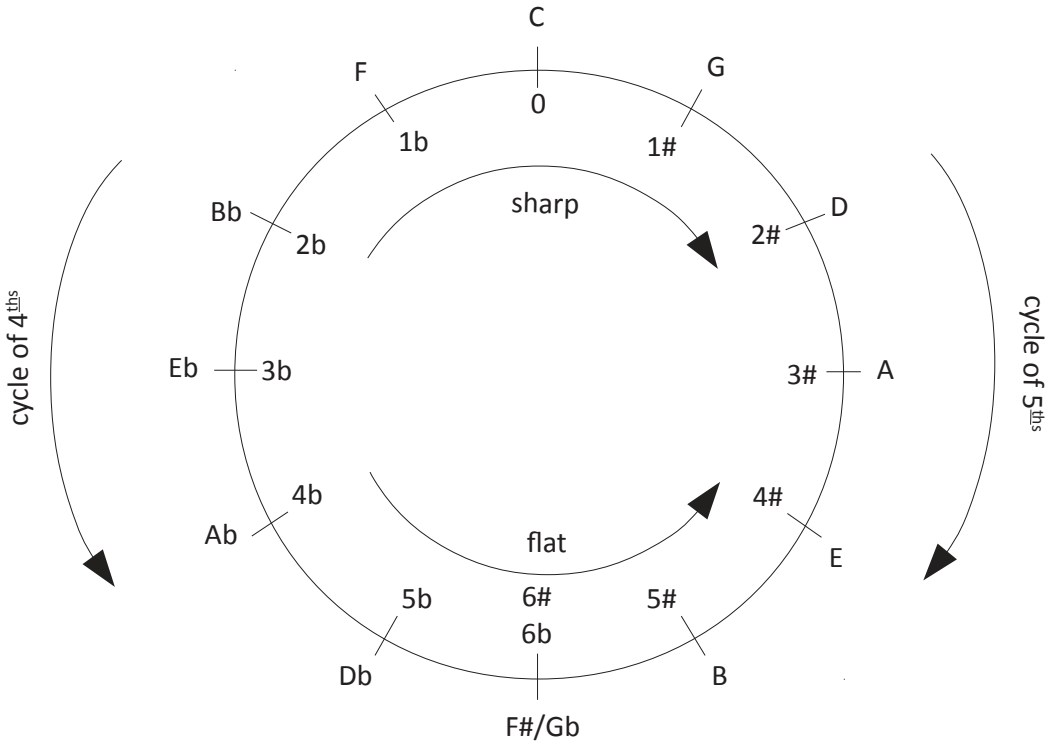


Sharp and Flat Direction Modal Modulation

Sharp and flat direction modal modulations, also known as upwards (sharp) and downwards (flat), occur when moving directly from one modal center to another. It can be better understood by using this chart:

Modal modulations can be used in place of more common modulation devices such as secondary Dom^7, secondary Dim^7 or chromatic mediant modulations.

Ex. 120



Sharp direction / upward modulation:

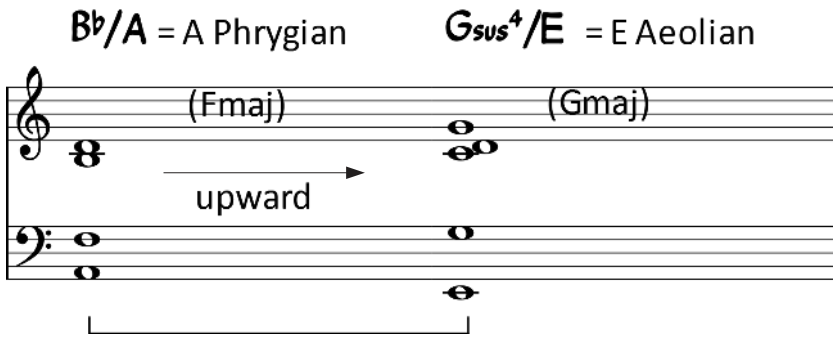
Moving modal center around the circle or adding sharps to naturals.

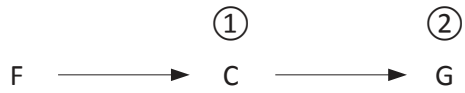
Flat direction / downward modulation:

Moving modal center counterclockwise or adding flats to naturals.

Upward Modulation

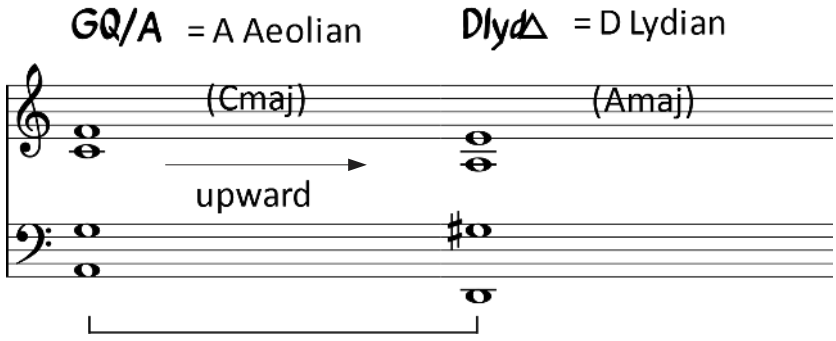
Ex. 121a





2 keys in a sharp direction

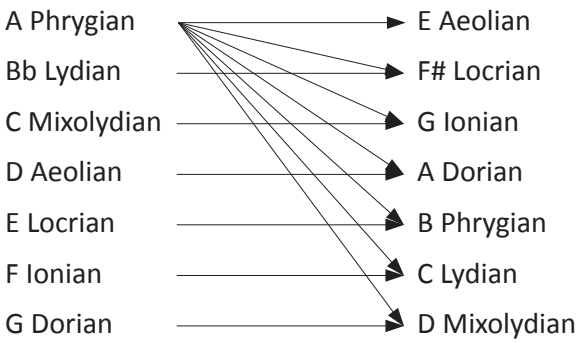
Ex. 121b



3 keys in a sharp direction

Modulating from A Phrygian to E Aeolian (2 keys in a sharp direction) we can generate many more harmonic resolutions between the parent groups.

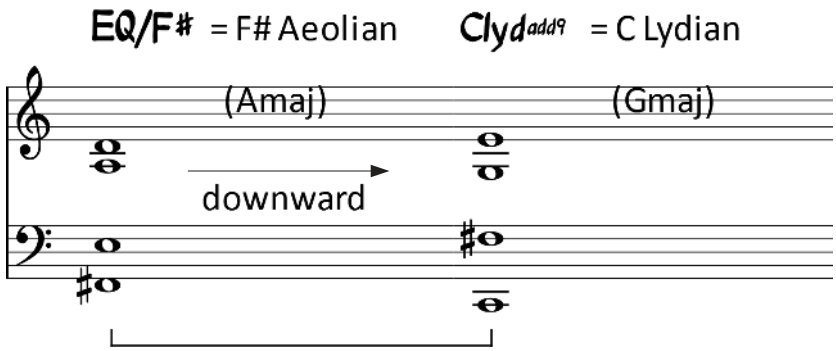
Ex. 122

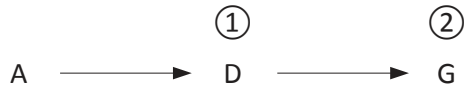


Also any combination of left to right movement as indicated from A Phrygian.

Downward Modulations

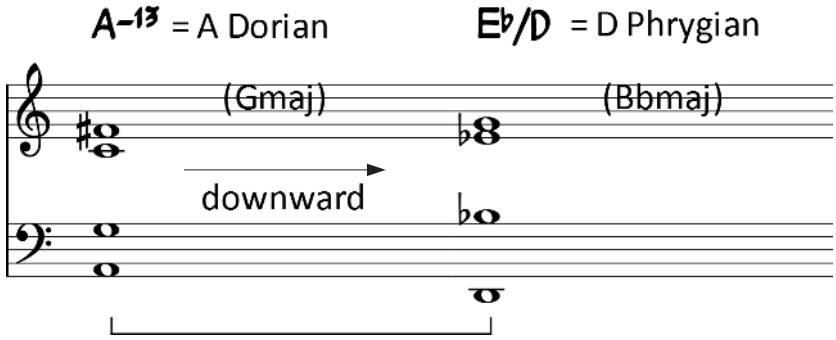
Ex. 123a





2 keys in a flat direction

Ex. 123b

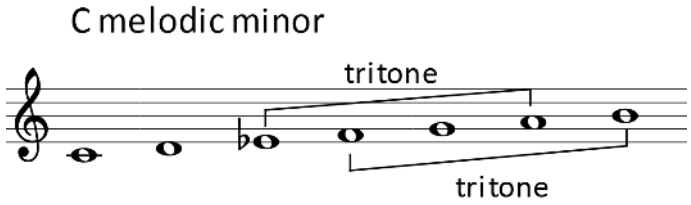


3 keys in a flat direction

Before we go on, I want to remind you that these resolution tendencies are dependent on clear modal voicings.

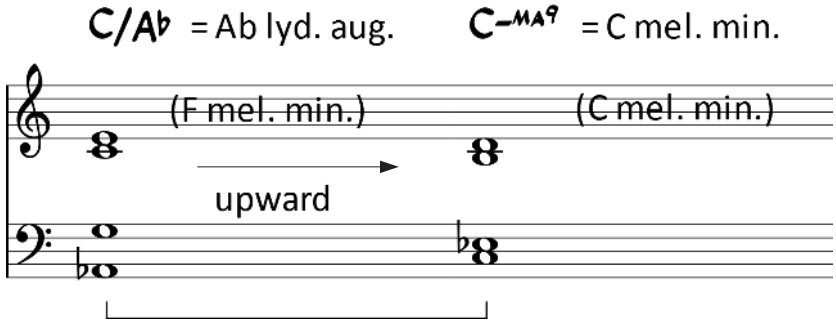
Modal modulation between melodic minor scales tends to be more ambiguious than the major. This is caused by its inherent interval properties, chiefly the pair of tritones.

Ex. 124



Sharp Direction (Melodic Minor)

Ex. 125



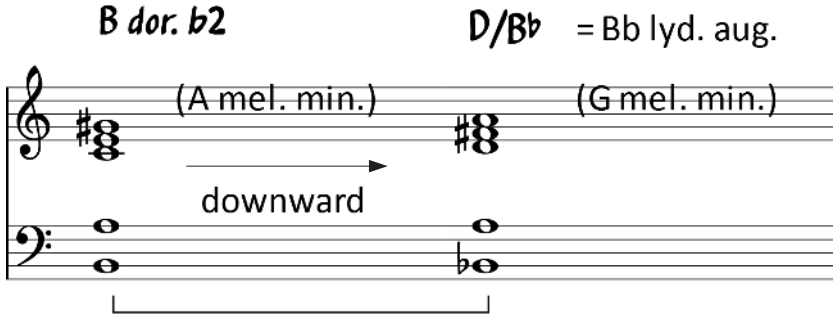
①

F mel. min. → C mel. min.

1 key in a sharp direction

Flat Direction (Melodic Minor)

Ex. 126



① ②

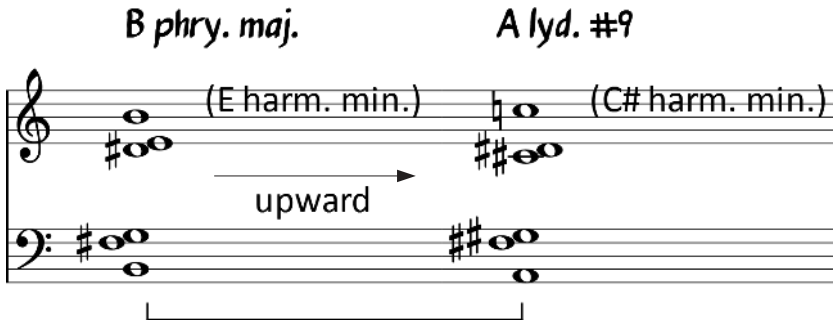
A mel.min. → D mel. min. → G mel. min.

2 keys in a flat direction

Harmonic minor modal resolution is similarly ambiguous.

Sharp Direction (Harmonic Minor)

Ex. 127



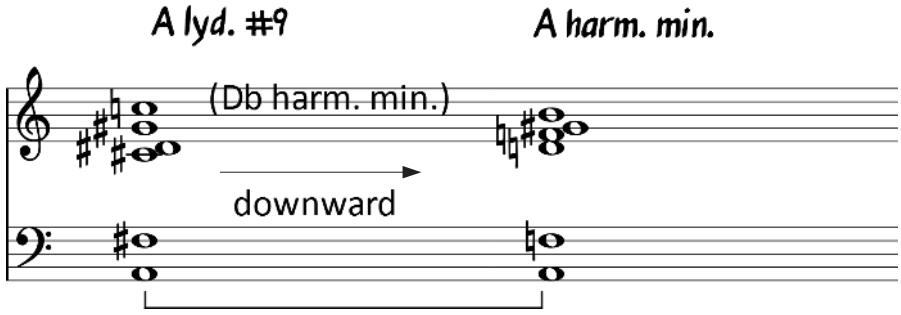
① ② ③

E harm. min. → B harm. min. → F# harm. min. → C# harm. min.

3 keys in a sharp direction

Flat Direction (Harmonic Minor)

Ex. 128



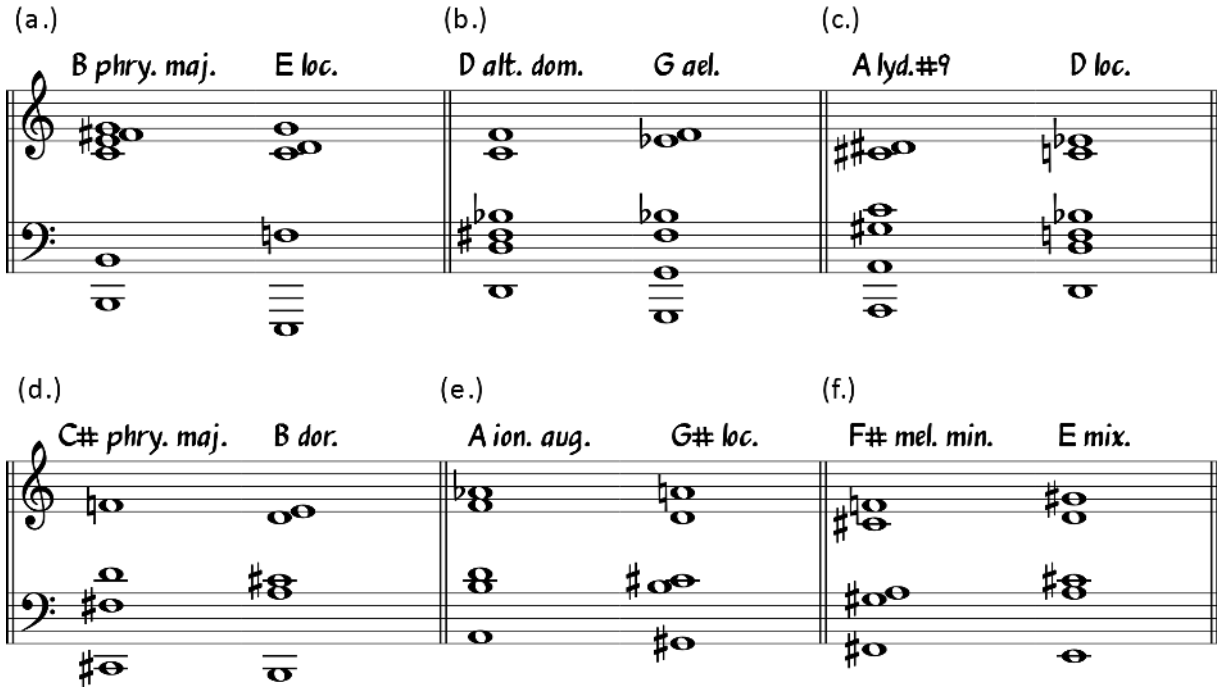
① ② ③ ④

Db harm. min. → Gb harm. min. → B harm. min. → E harm. min. → A harm. min.

4 keys in a flat direction

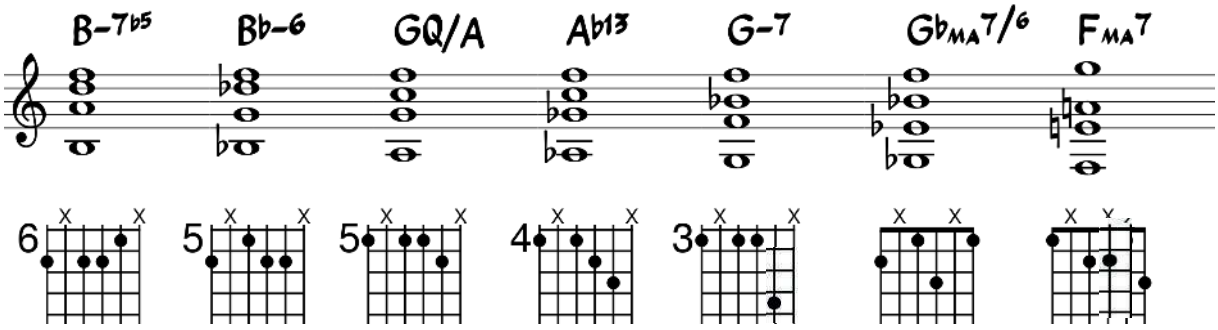
Resolution Tendencies Between Modal Groups

Resolution tendencies between modal groups (major to melodic minor for example) do not fit into a standard formula. They are related to voice leading rather than the internal structure of the two parent scales. Try these on the piano.

Ex. 129

Voice Leading

Voice leading is the interval movement of like voices between two chords. Good voice leading involves either common tone or stepwise motion of the voices while following natural resolution tendencies.

Ex. 130



Smooth voice leading is difficult to achieve on the guitar because of the large number of left hand chord forms involved.

Resolution

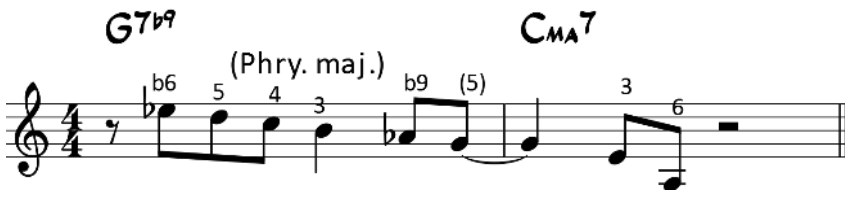
The term resolution implies a change from one chord to another. Half step resolution should be used whenever possible. Here is a chart of V^7 to I^{maj7} resolutions.

Ex. 131

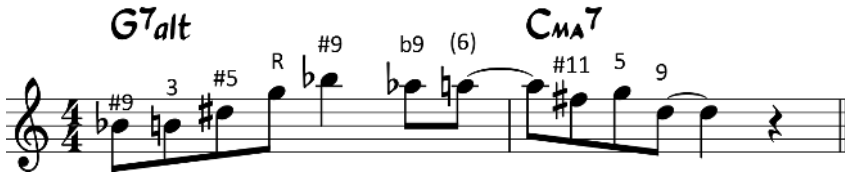
V^7	I^{maj7}
R →	#5, #4, (6)
3 →	R, 6
5 →	R, 3
b7 →	3, #4
9 →	#5, 7, (5)
11 →	9, 7 both weak
13 →	9, #4
b9 →	5, 6
#9 →	7, 6
b5 →	R, 9
#5 →	9, 3

Here is a linear resolution example:

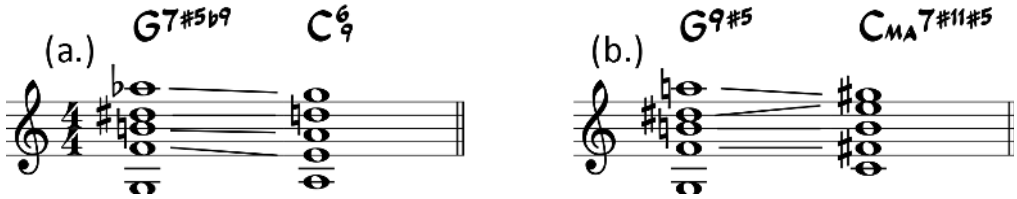
Ex. 132a



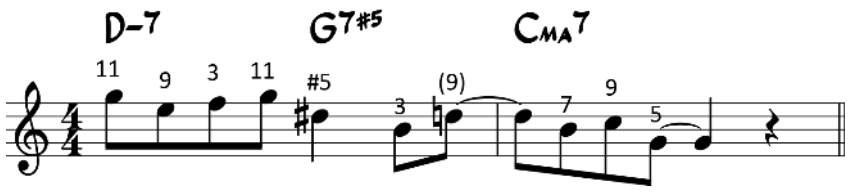
Ex. 132b



Harmonic resolution

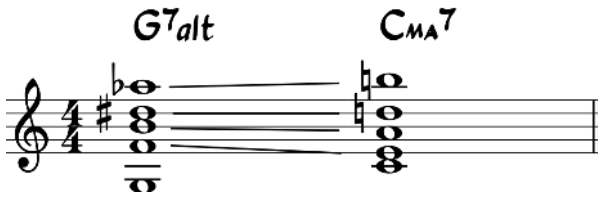
Ex. 133

Resolution by skip is also effective when approached correctly.

Ex. 134

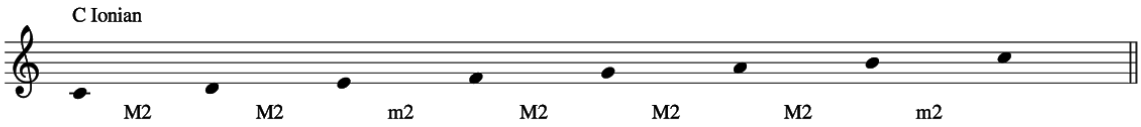
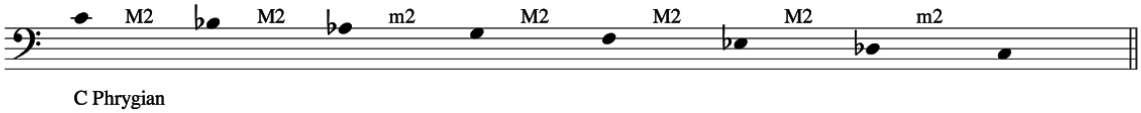


Harmonic resolution by skip

Ex. 135

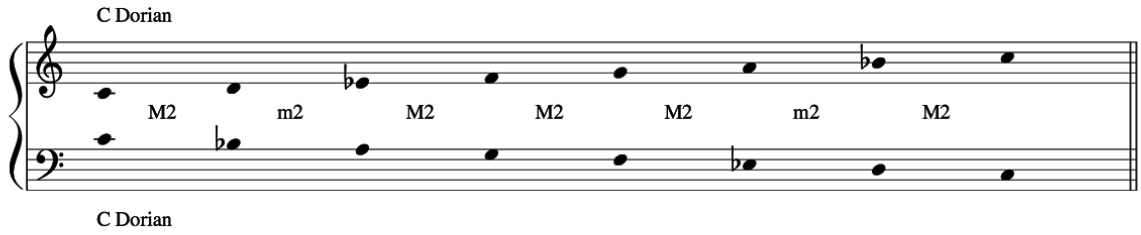


Mirror Modal Equivalents

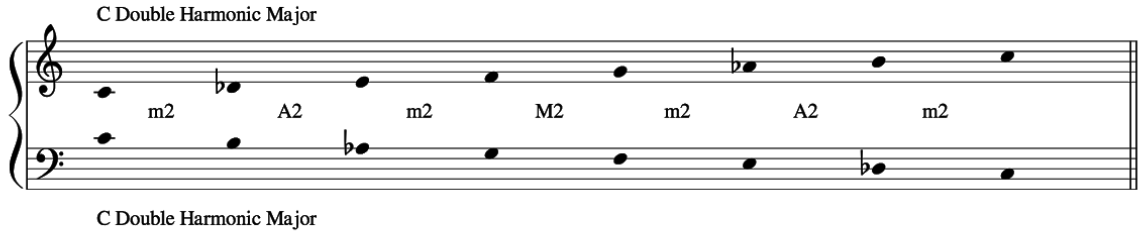
Here is an ascending C Ionian scale. It is constructed out of a specific pattern of intervals.

A mirror image of the ascending C Ionian scale is created by constructing a descending scale from C using the same series of intervals. This mirror image is a descending C Phrygian scale.

This unique relationship makes Ionian and Phrygian **Mirror Modal Equivalents**.

This process can be extended to every mode of the Major scale, resulting in three pairs of modes. They are:
Ionian - Phrygian
Lydian - Locrian
Mixolydian - Aeolian

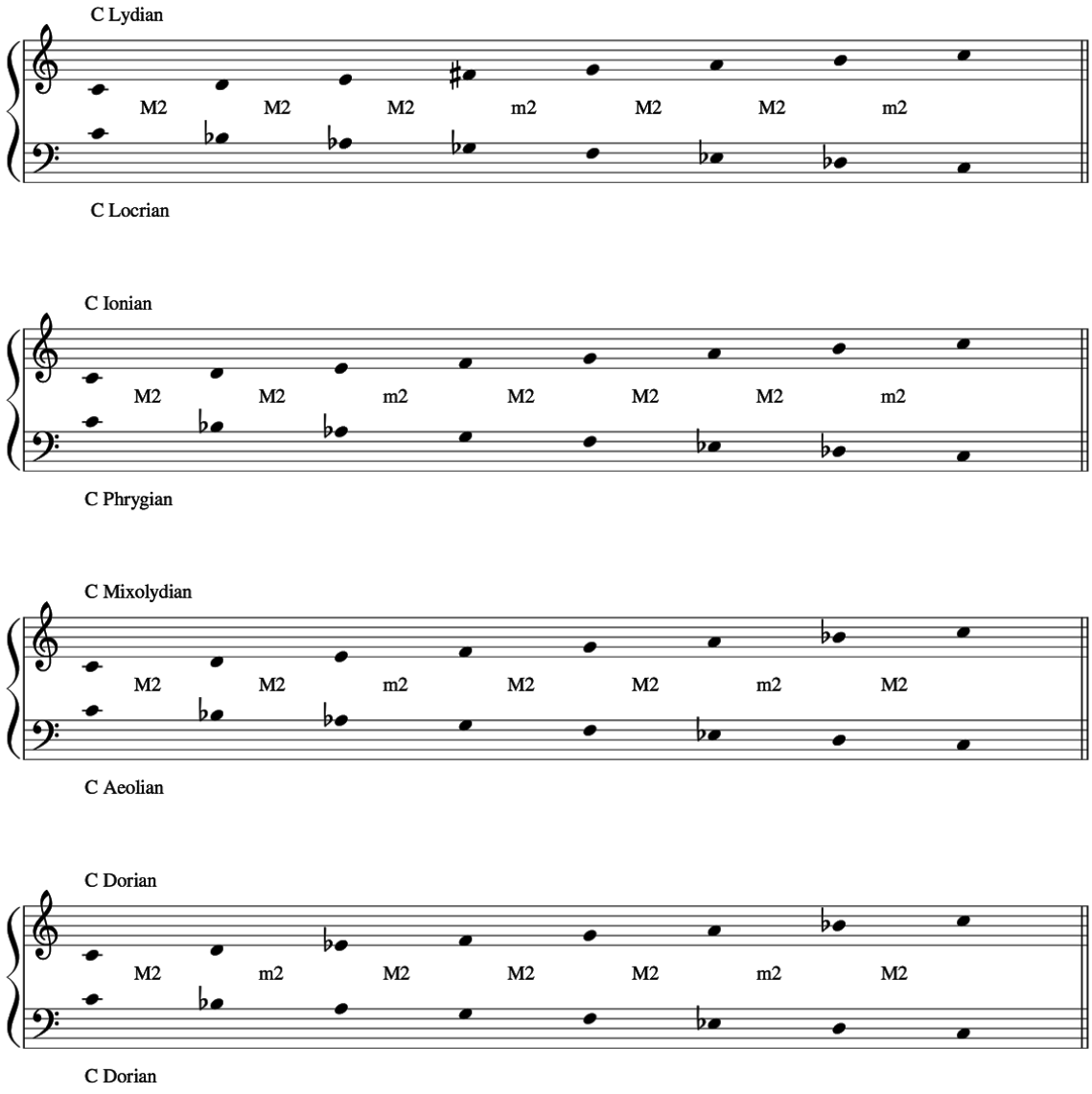
The remaining mode, Dorian, is **reflective** because of its palindromic interval pattern. This causes Dorian to be its own Mirror Image when it is reflected.



Another example of a reflective scale is the Double Harmonic Major scale.



Here the seven modes of the Major scale are paired with their Mirror Modal Equivalent, and arranged in order of brightness. The ascending modes move from bright (Lydian) to dark (Dorian). The descending modes move from dark (Locrian) to bright (Dorian).



Bright → Dark

Lydian	Ionian	Mixolydian	Dorian
Locrian	Phrygian	Aeolian	Dorian

Dark → Bright

Melodic Minor: Mirror Modal Equivalents

The modes of Melodic Minor also include three pairs of mirror modal equivalents, and one reflective mode.

Melodic Minor	Dorian b2
Lydian Augmented	Altered Dominant
Mixolydian #11	Locrian Natural 2
Mixolydian b6	Mixolydian b6

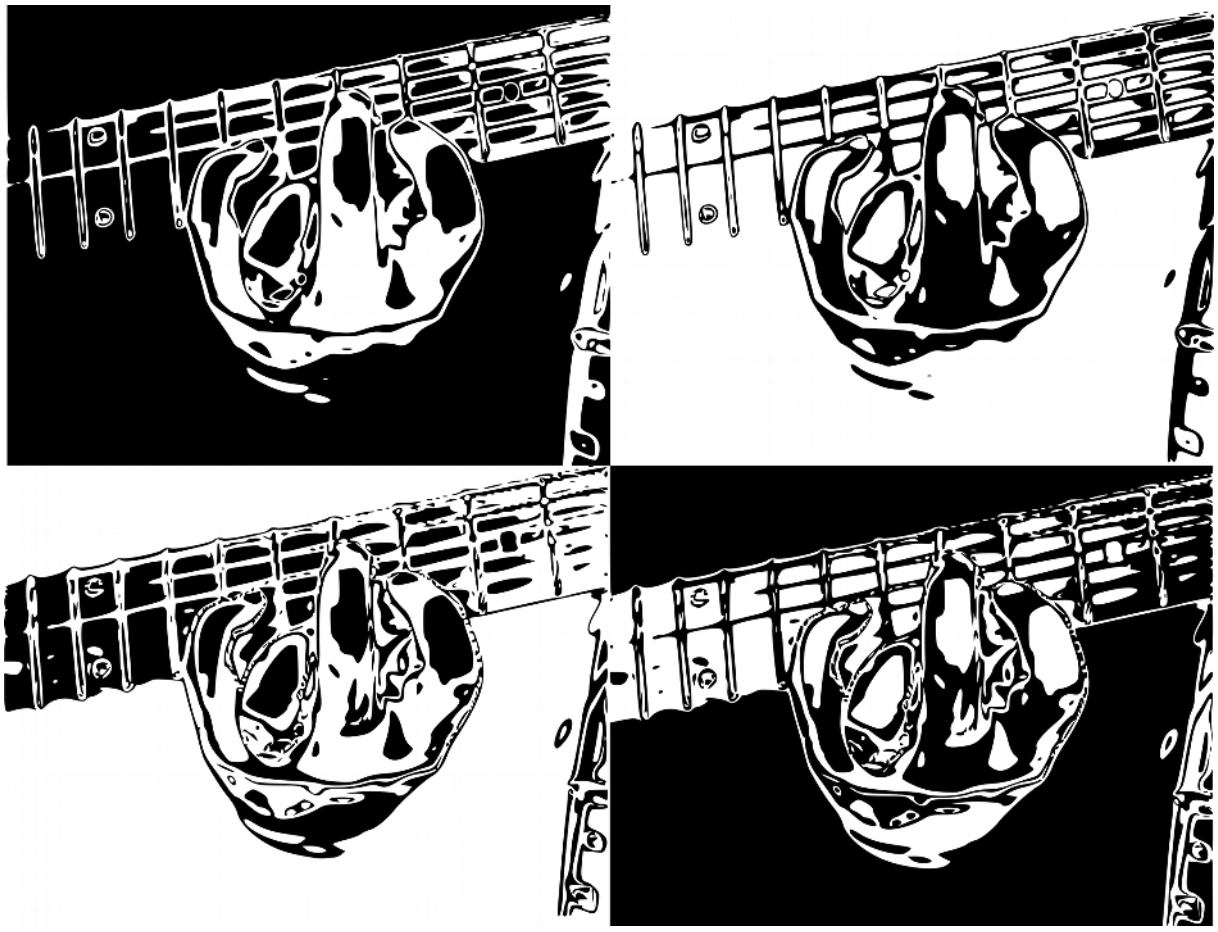
Harmonic Minor and Harmonic Major: Mirror Modal Equivalents

The modes of Harmonic Minor transform into the modes of Harmonic Major when reflected, and vice-versa.

Harmonic Minor Modes	Harmonic Major Modes
Harmonic Minor	Mixolydian b2
Locrian Natural 6	Lydian b3 (Melodic Minor #4)
Ionian Augmented	Phrygian b4
Dorian #4	Dorian b5
Phrygian Major	Harmonic Major
Lydian #9	Locrian bb7
Altered Dominant bb7	Lydian Augmented #2

Chapter 2

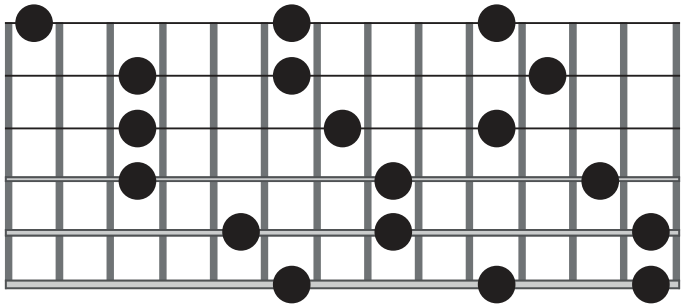
Chord Forms

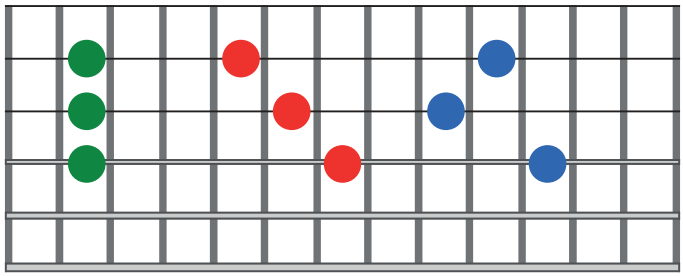


Chapter 2: Chord Forms

Triad (and Suspension) Chord Forms

The diagrams of triadic shapes are intended for harmonic and melodic use. Practice moving horizontally and vertically between shapes. This will acclimate your left hand to the fret spacing of the individual shapes.





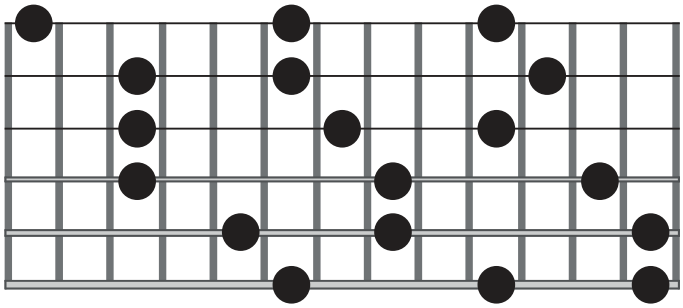
● : Root position

● : 1. Inversion

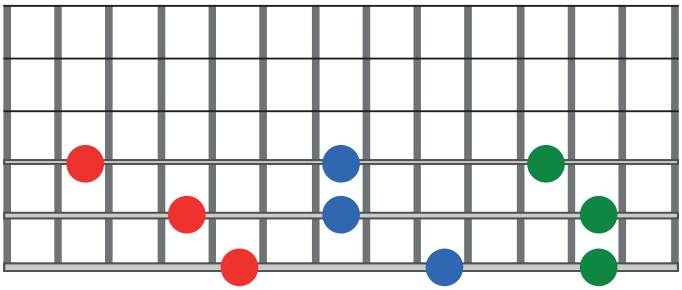
● : 2. Inversion

Major Triads

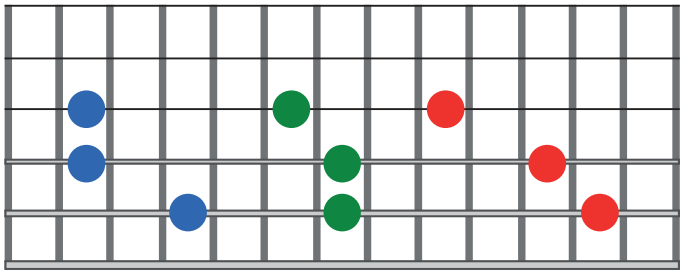
Major Triads



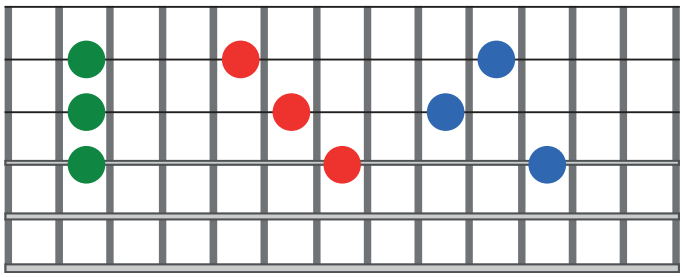
stringset E-A-D



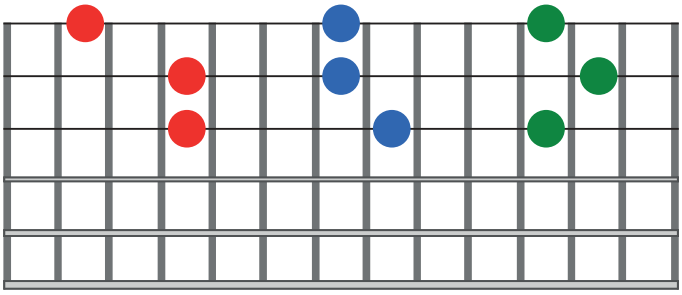
stringset A-D-G



stringset D-G-B

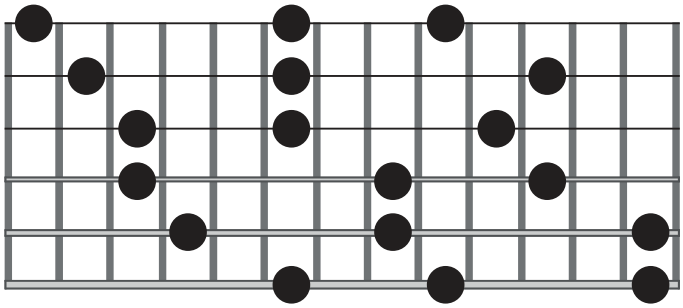


stringset G-B-E

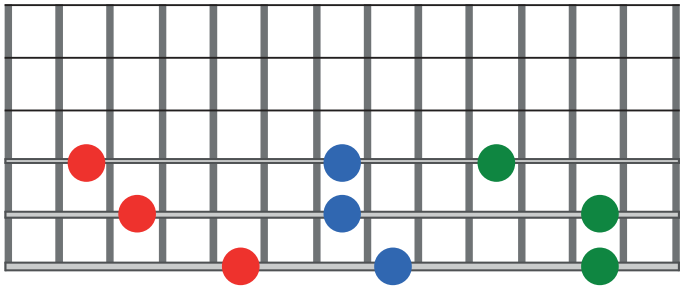


Minor Triads

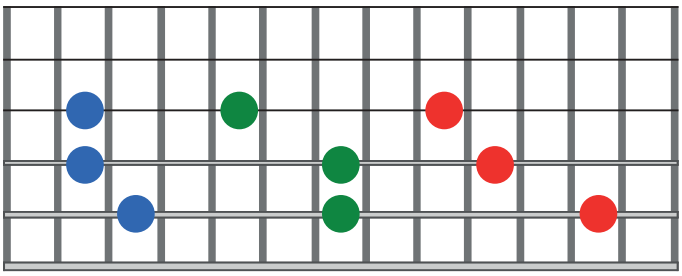
Minor Triads



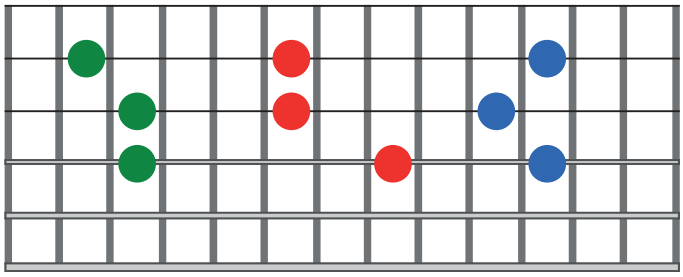
stringset E-A-D



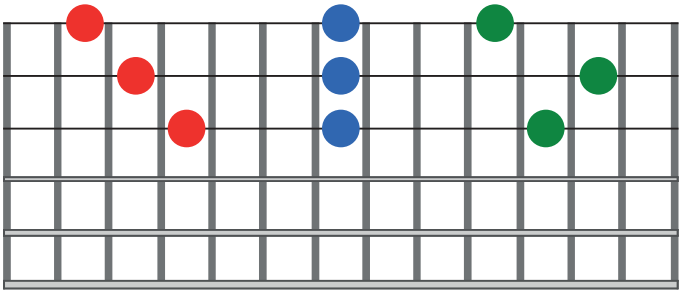
stringset A-D-G



stringset D-G-B

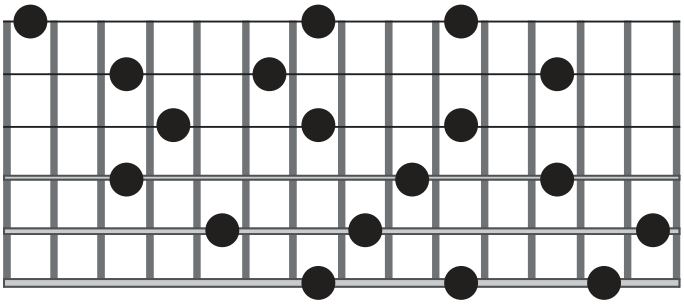


stringset G-B-E

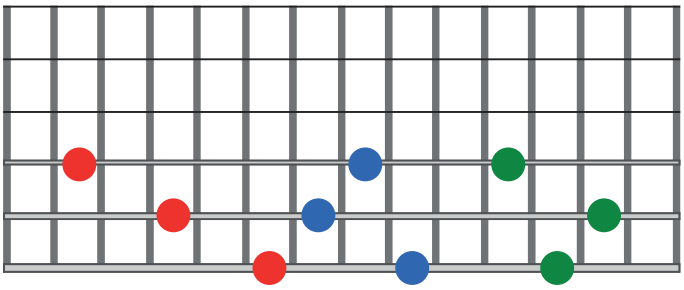


Diminished Triads

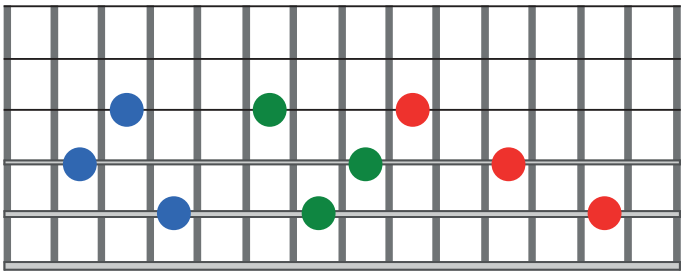
Diminished Triads



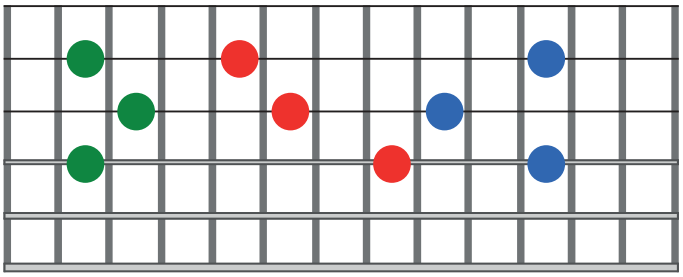
stringset E-A-D



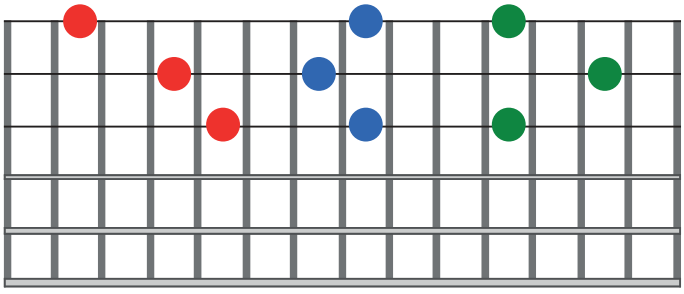
stringset A-D-G



stringset D-G-B

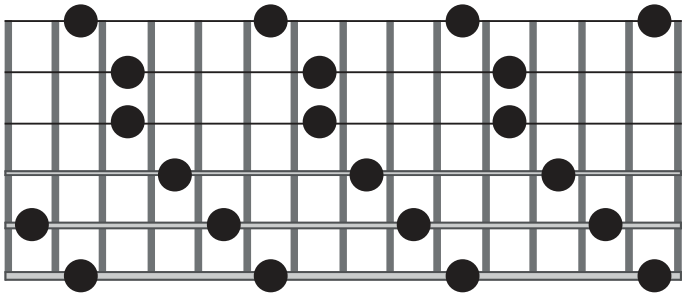


stringset G-B-E

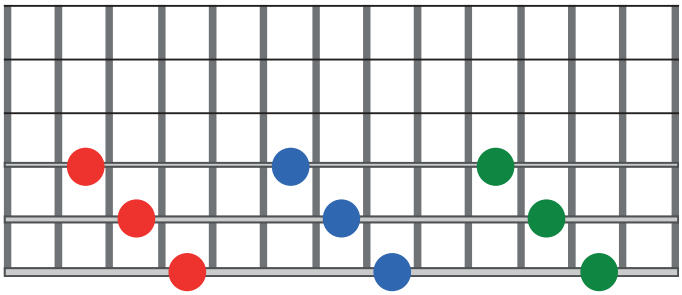


Augmented Triads

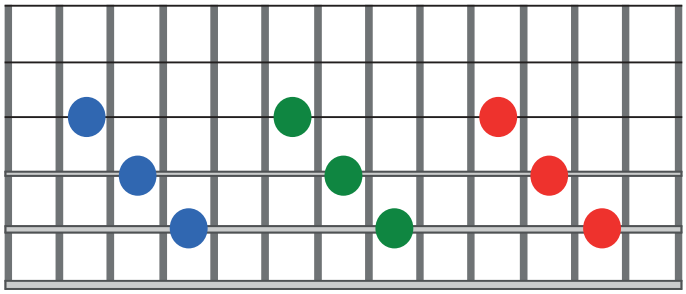
Augmented Triads



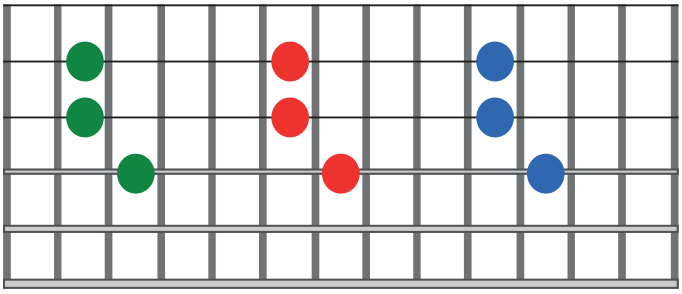
stringset E-A-D



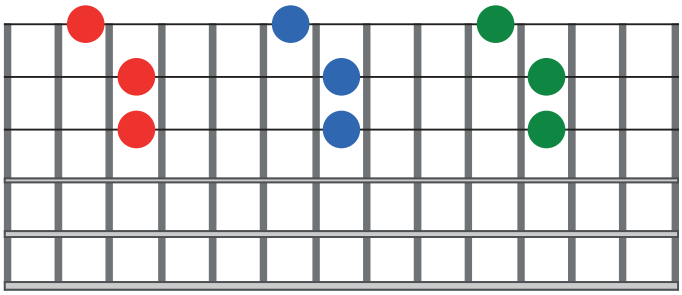
stringset A-D-G



stringset D-G-B

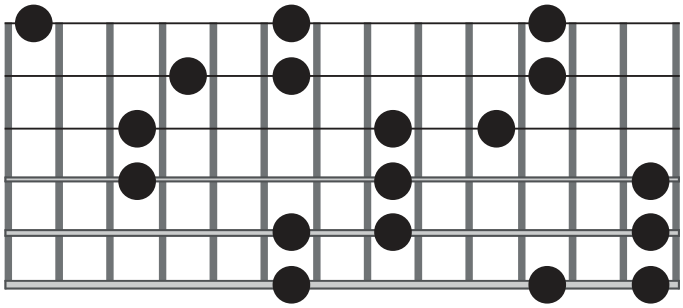


stringset G-B-E

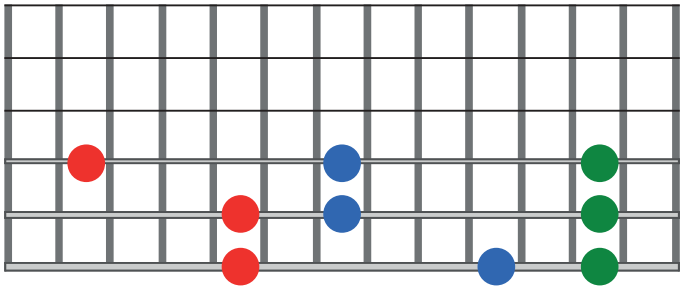


Suspended Triads

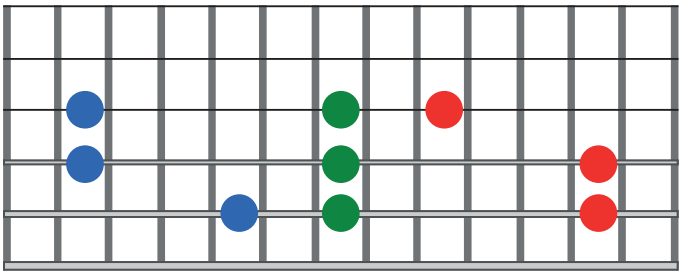
Suspended Triads



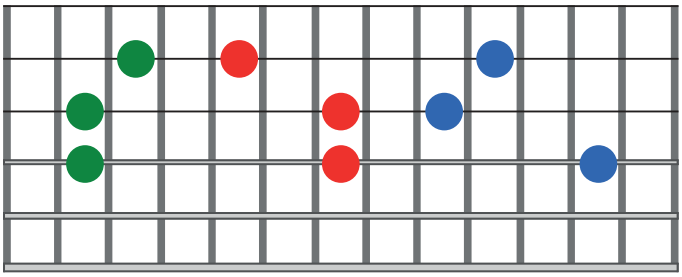
stringset E-A-D



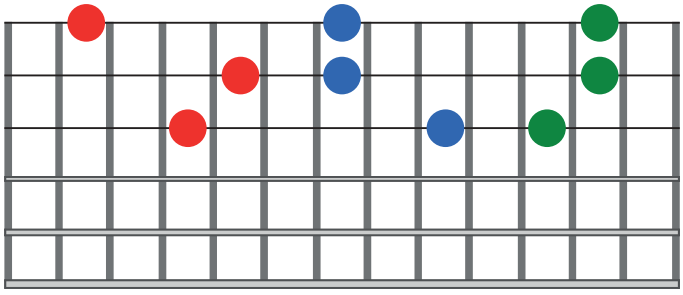
stringset A-D-G



stringset D-G-B

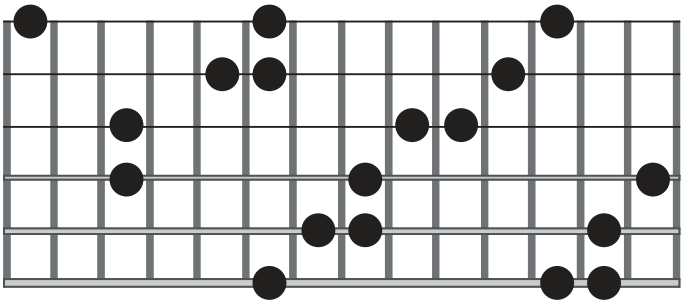


stringset G-B-E

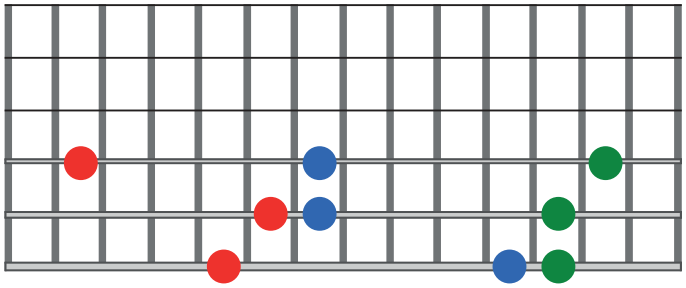


Lydian Triads

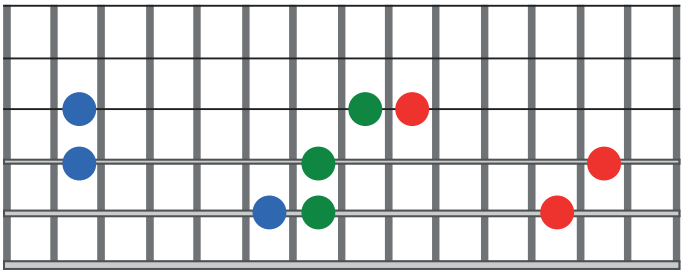
Lydian Triads



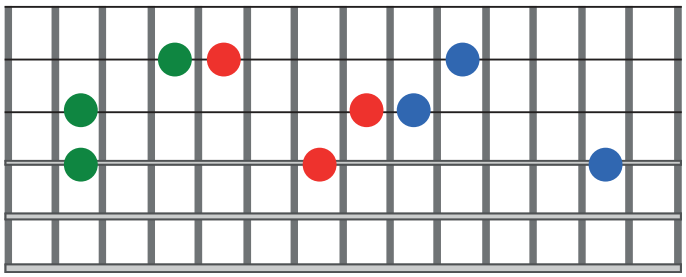
stringset E-A-D



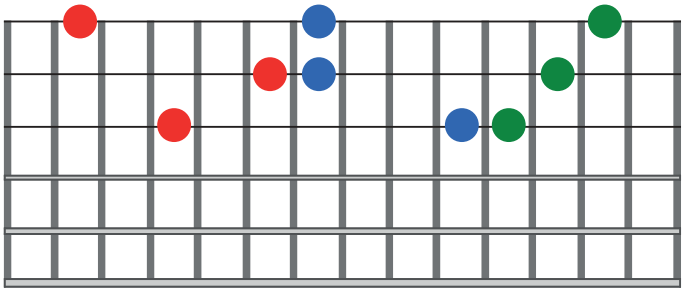
stringset A-D-G



stringset D-G-B

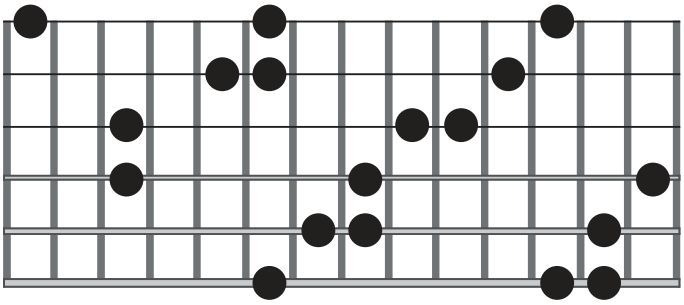


stringset G-B-E

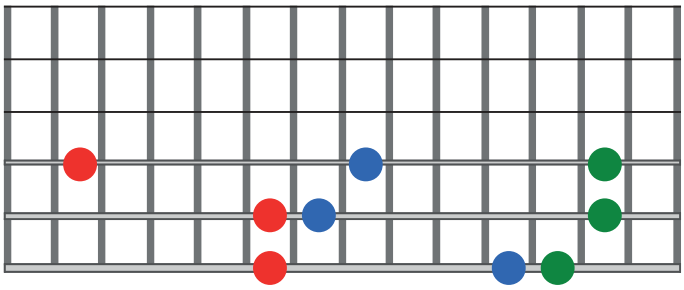


Locrian Triads

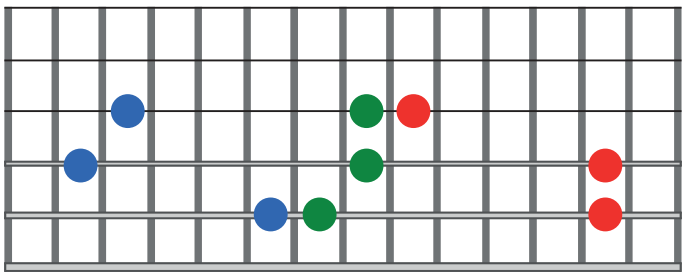
Locrian Triads



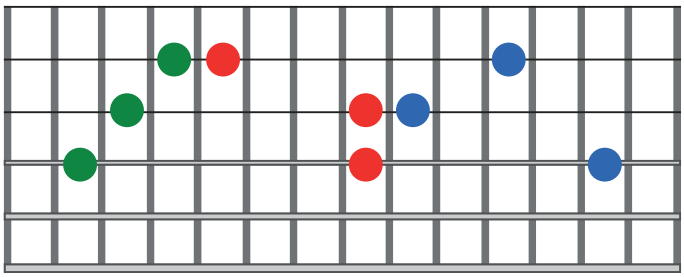
stringset E-A-D



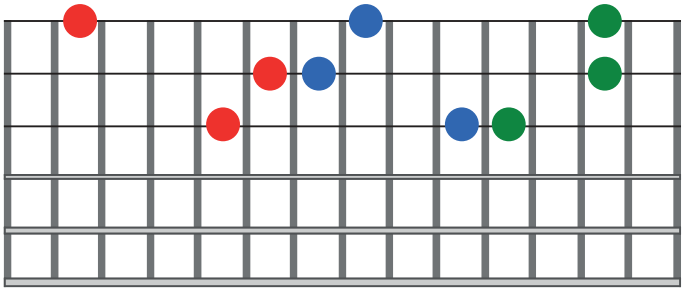
stringset A-D-G



stringset D-G-B



stringset G-B-E

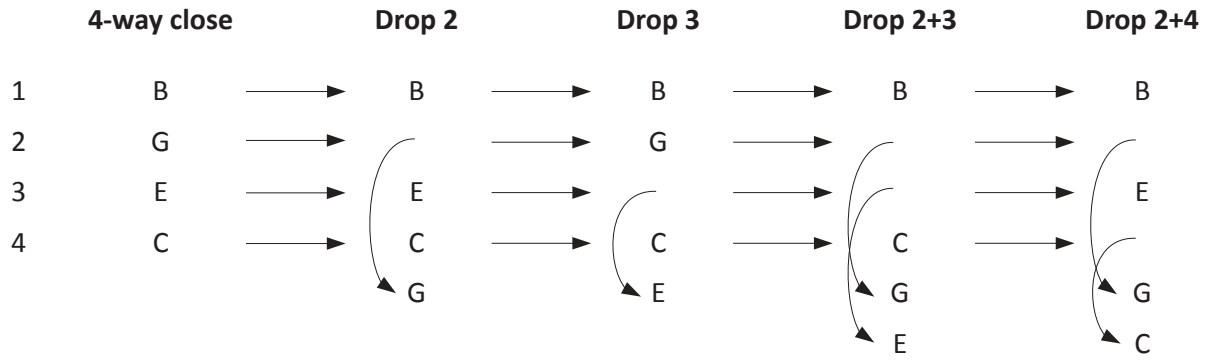


Drop Voicings

Drop Voicings are conmonly used on the guitar because its tuning does not permit easily reached 4-way close stuructures. A Drop voicing reorders the chord providing a more usable inversion.

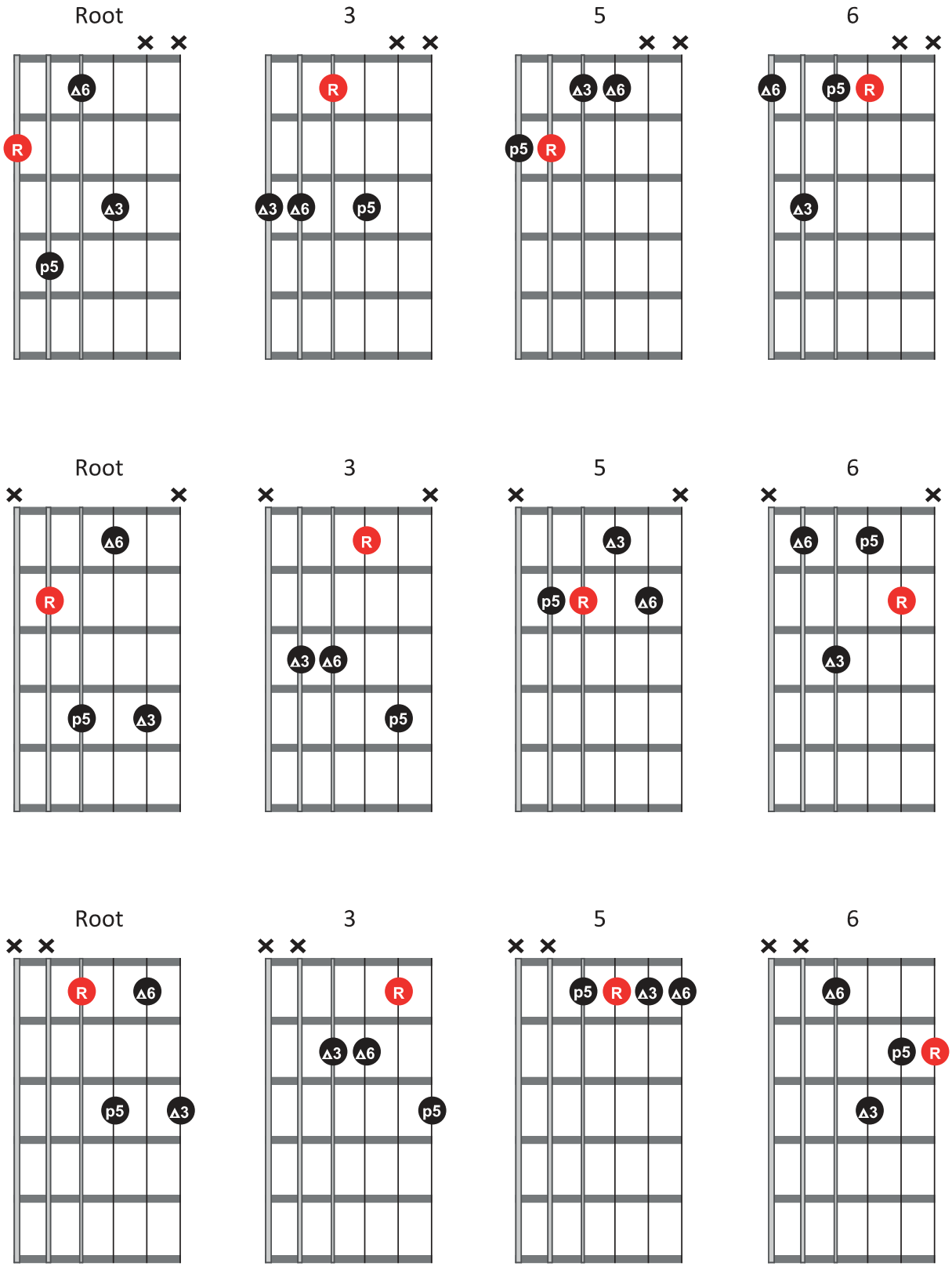
Voicings of $Cmaj^7$

Ex. 136

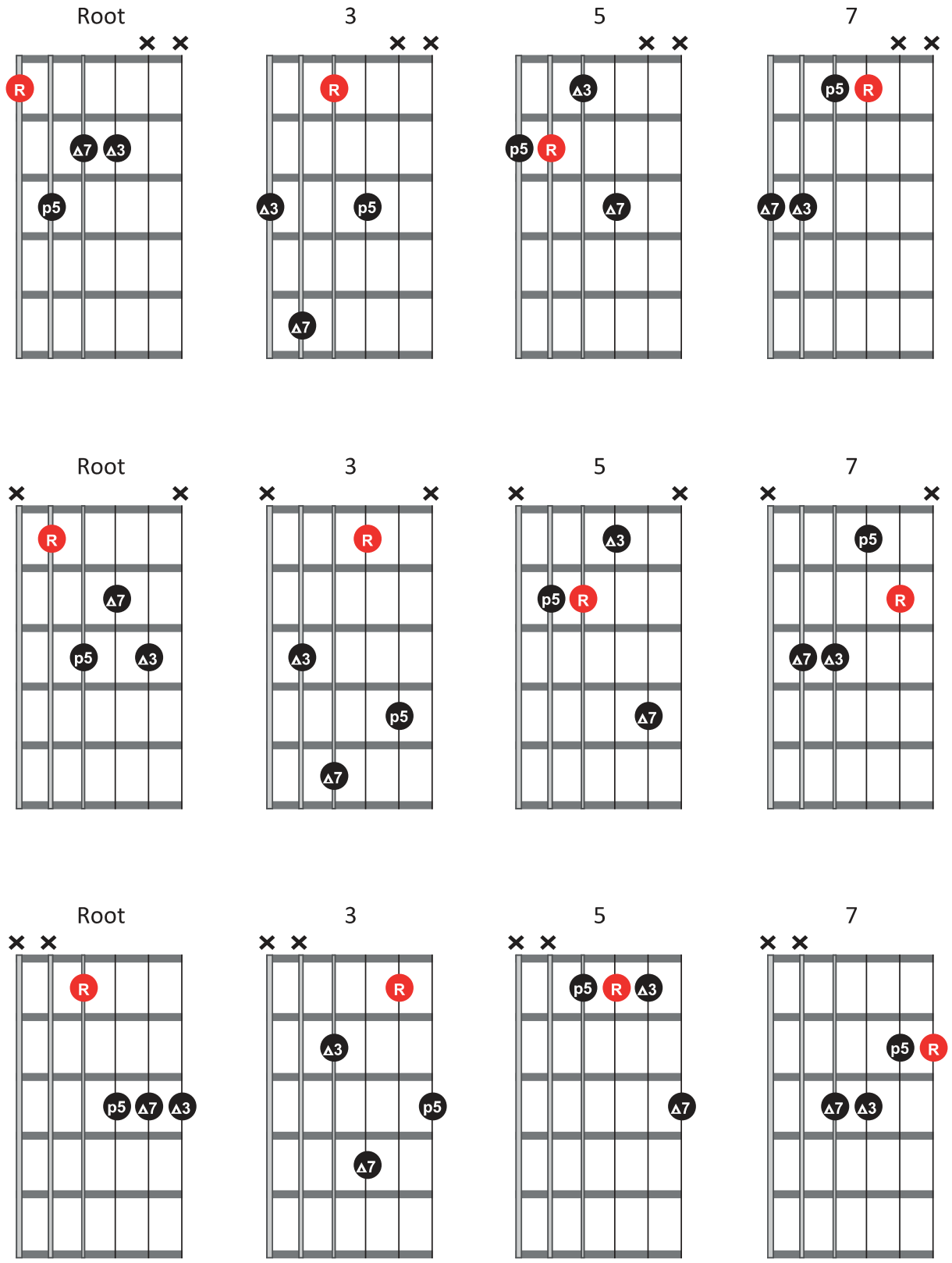


Drop 2

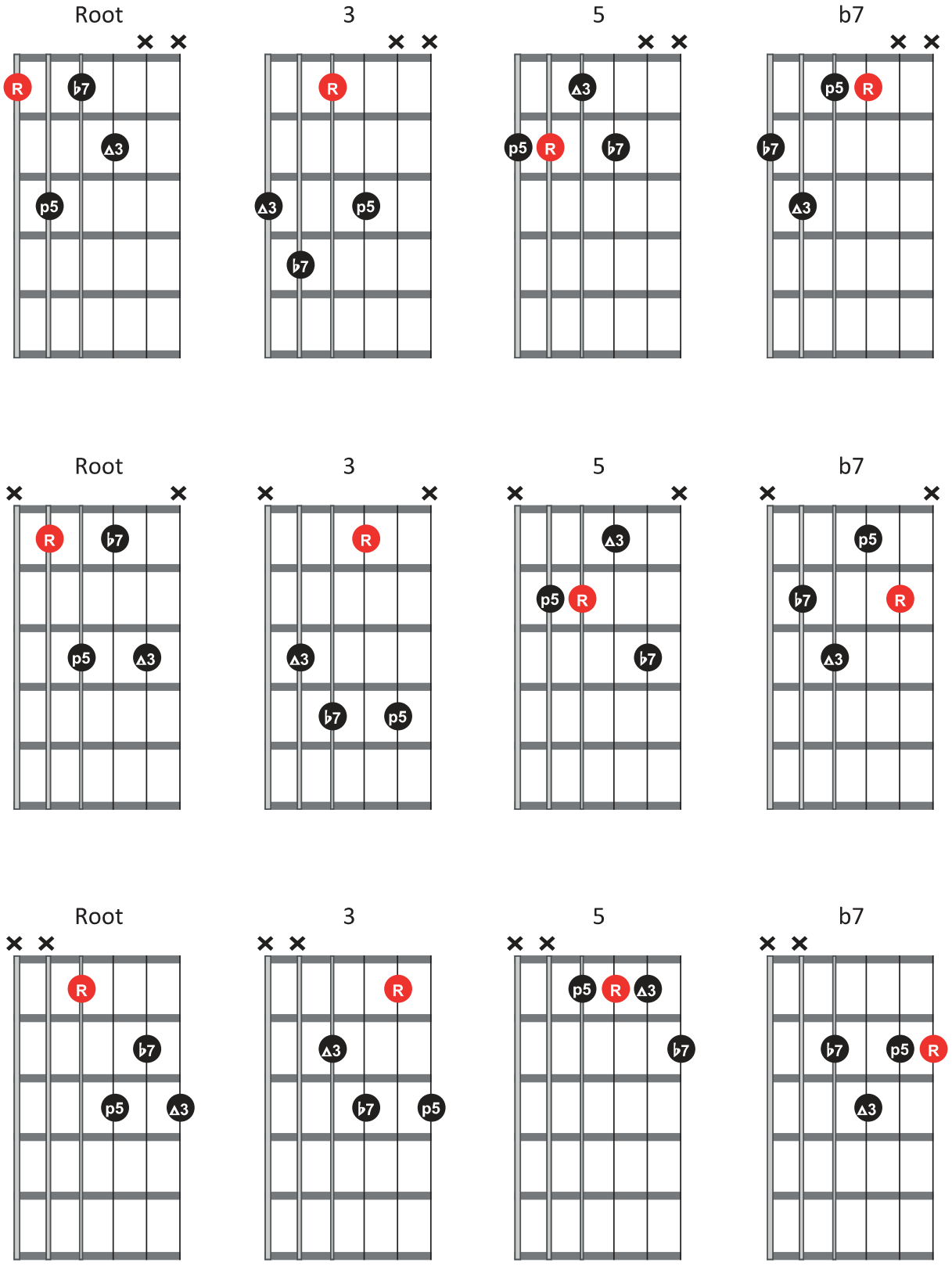
***Maj*^{6}**



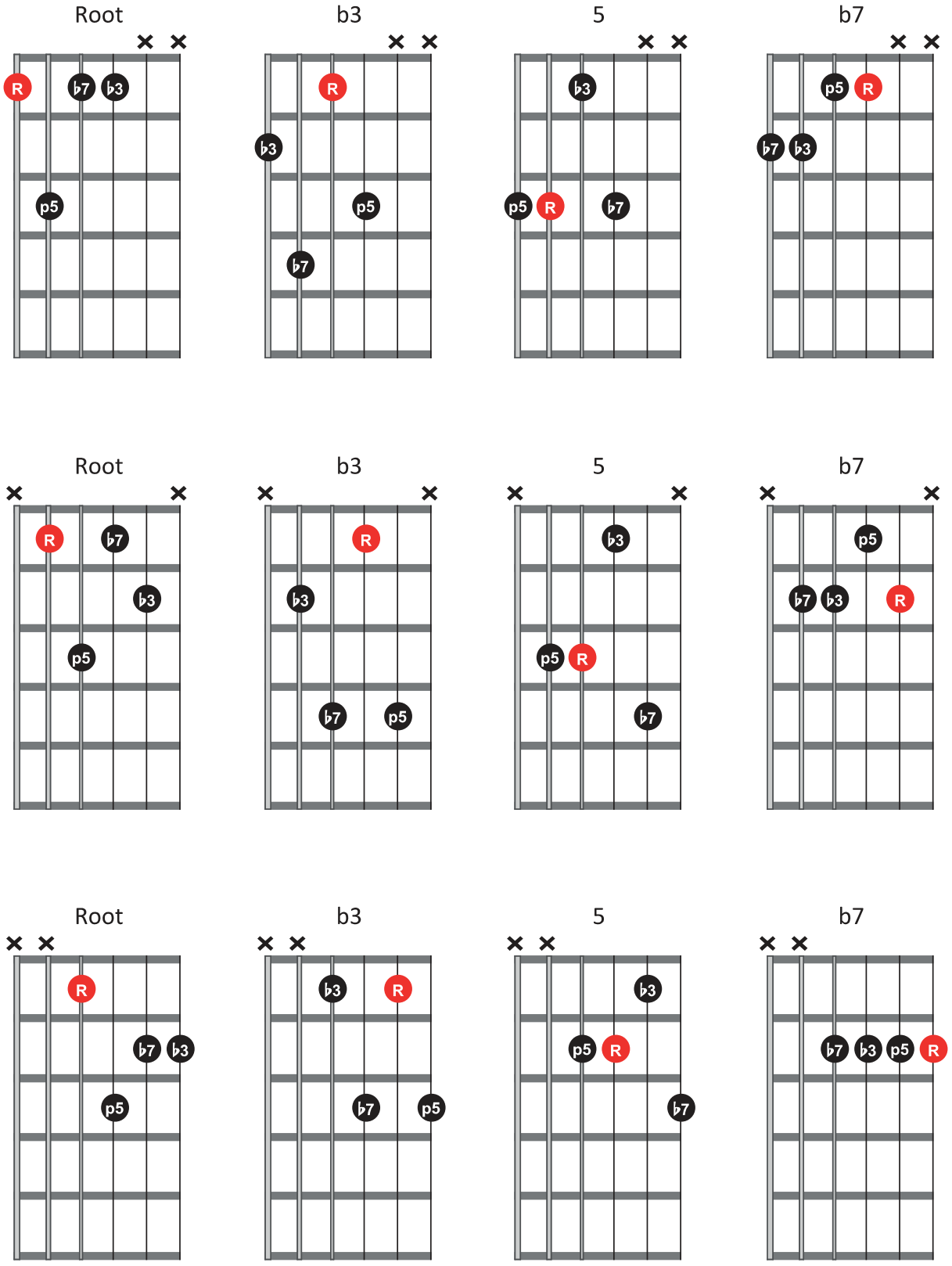
Maj 7



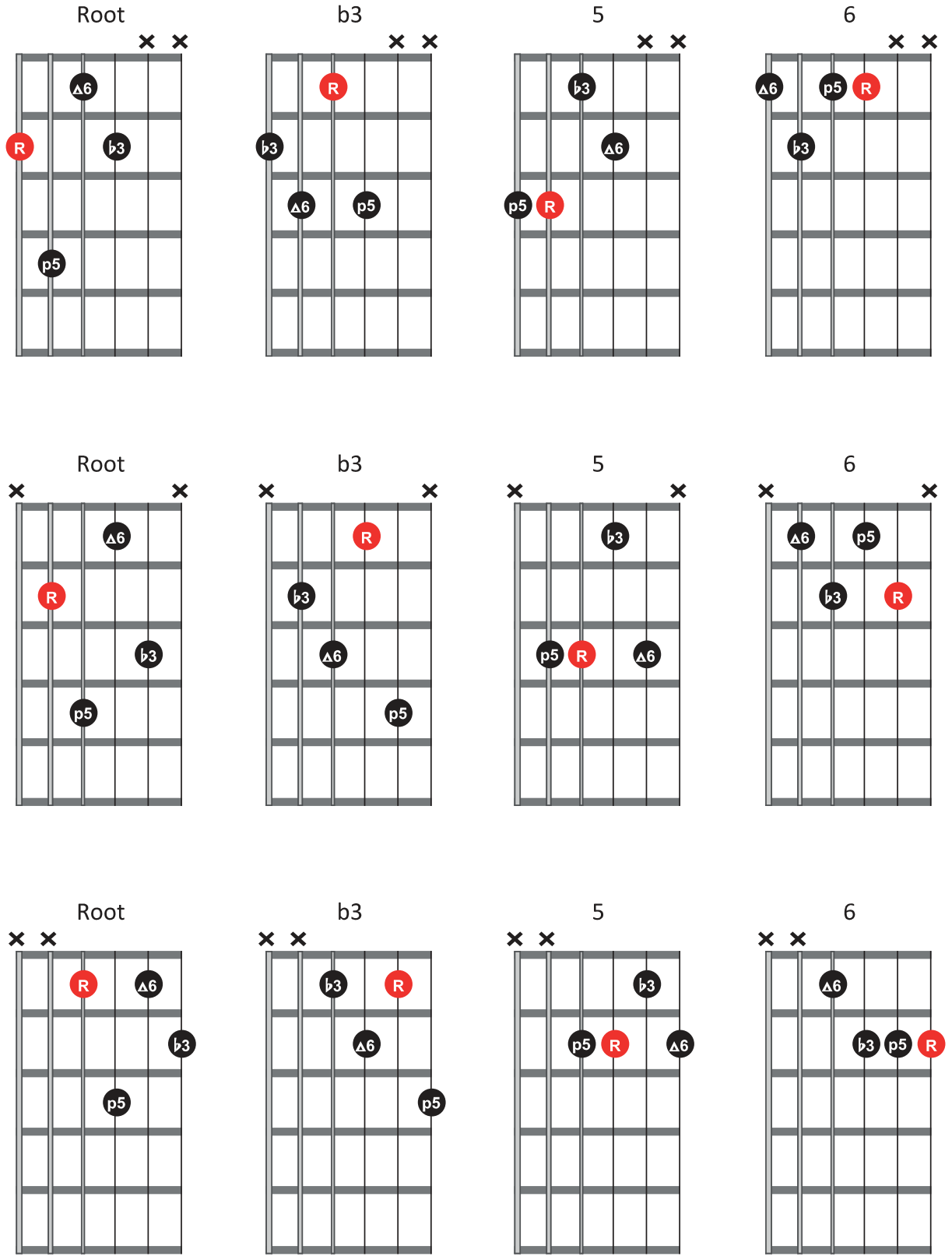
Dom^{7}



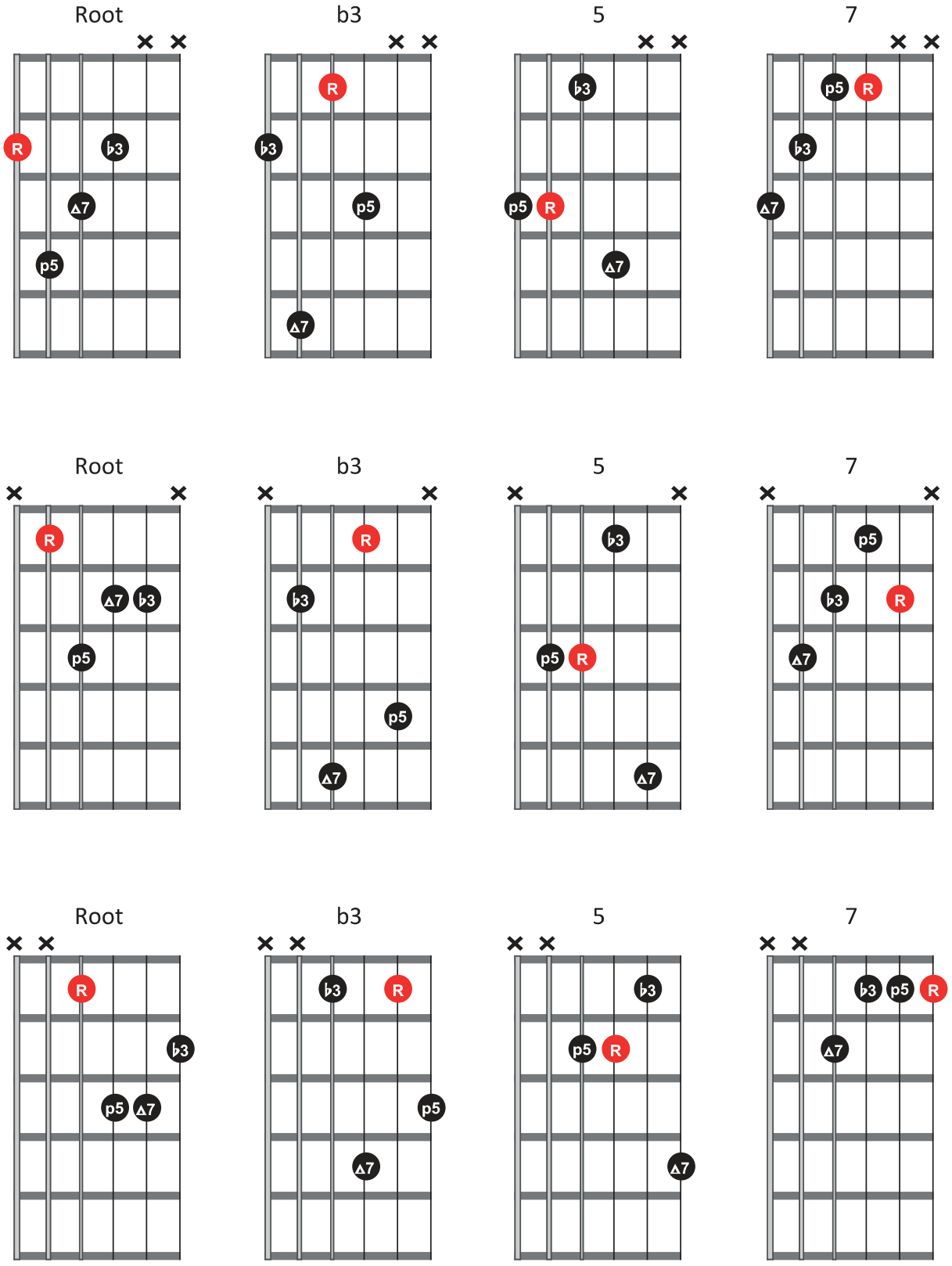
Min^{7}



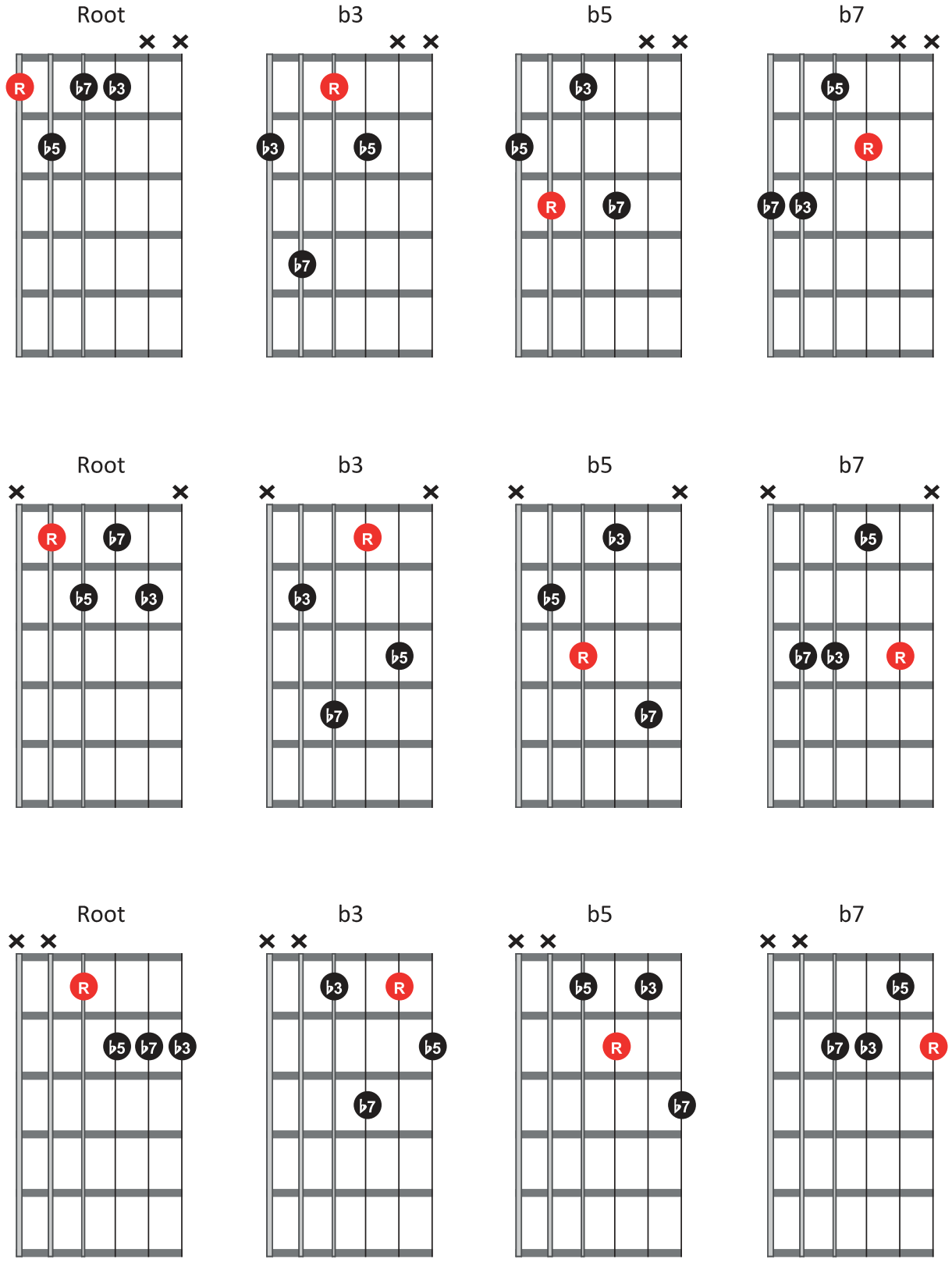
Min^{6}



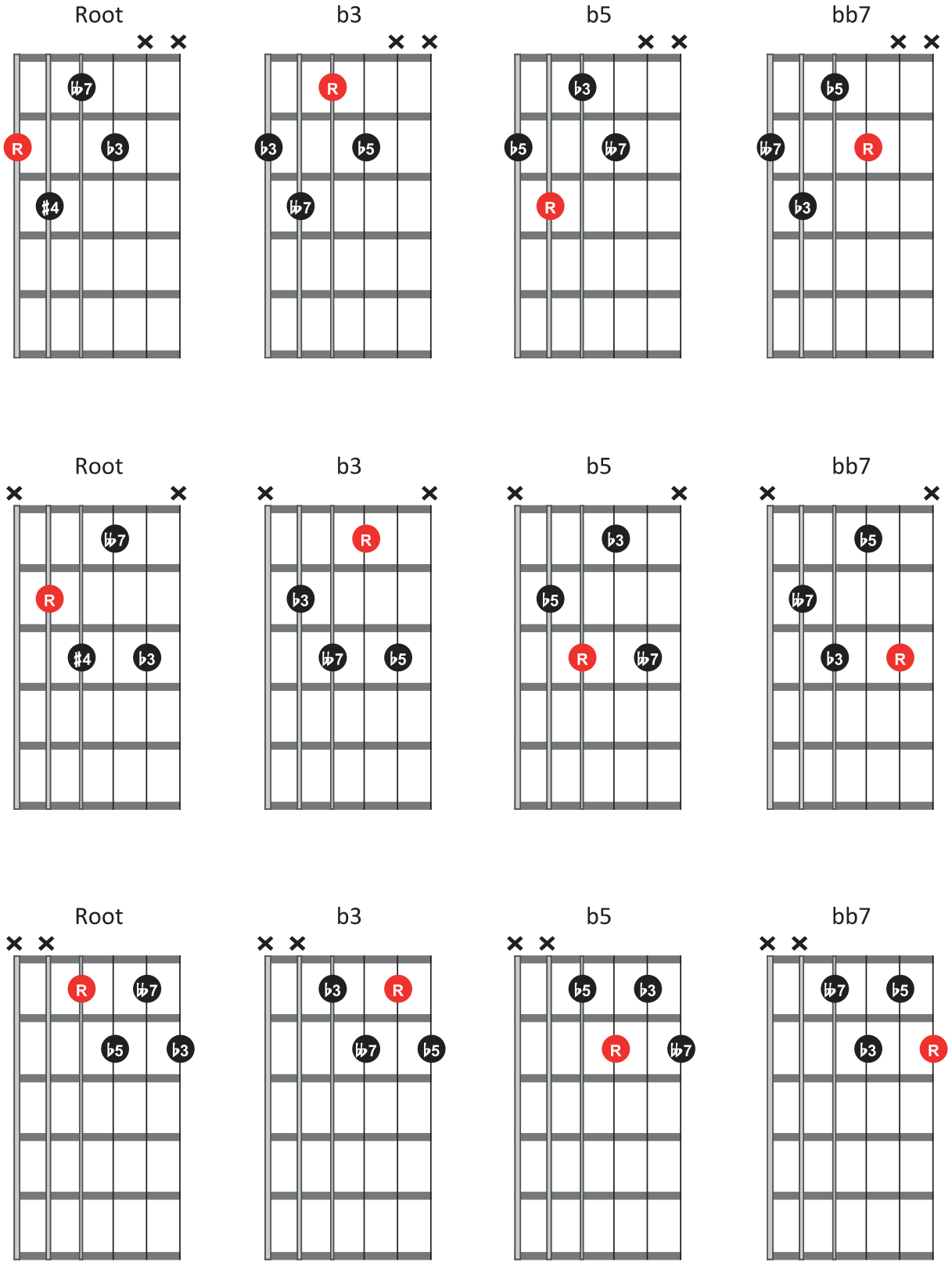
Min^{maj7}



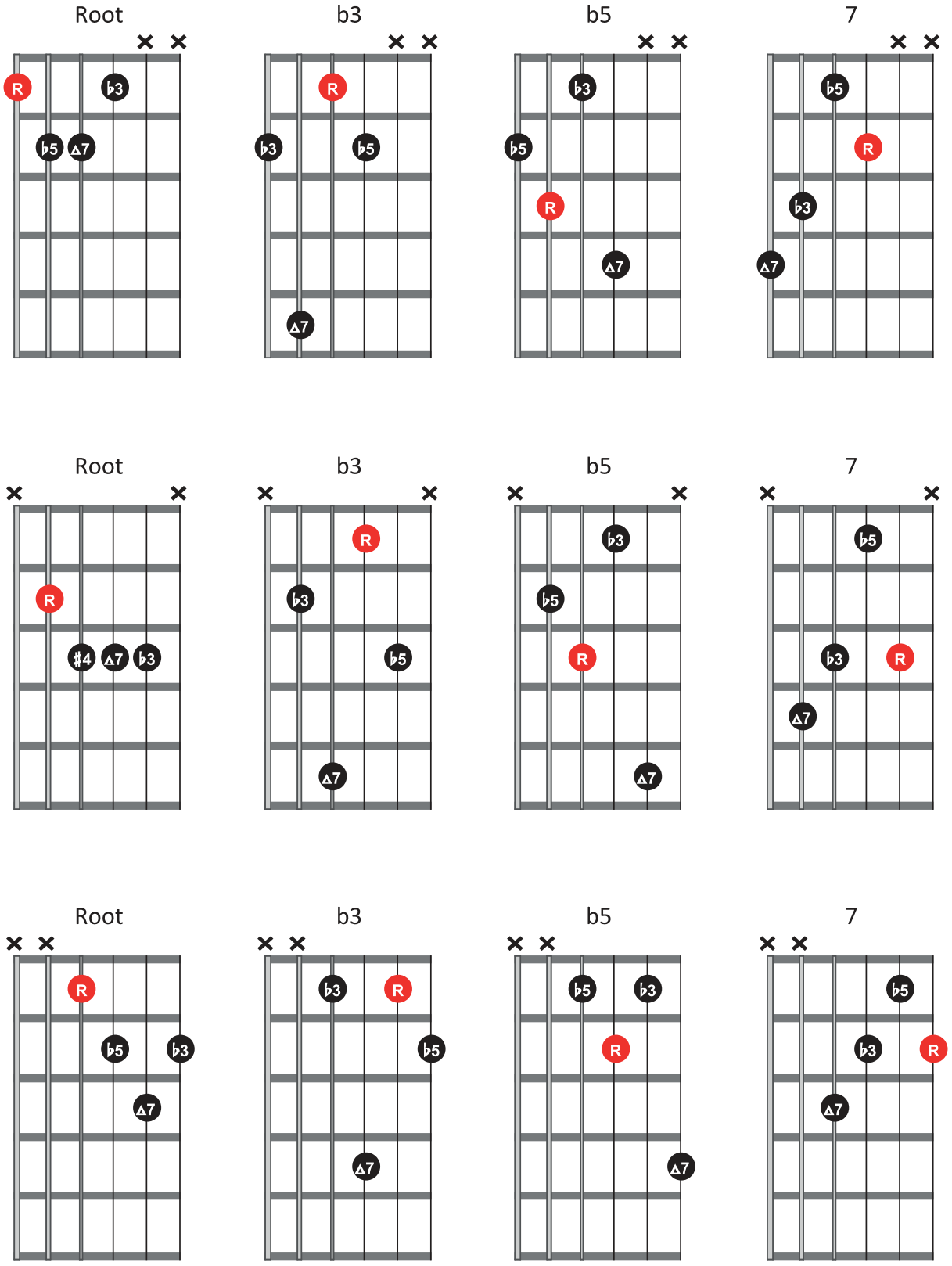
Min^{7b5}



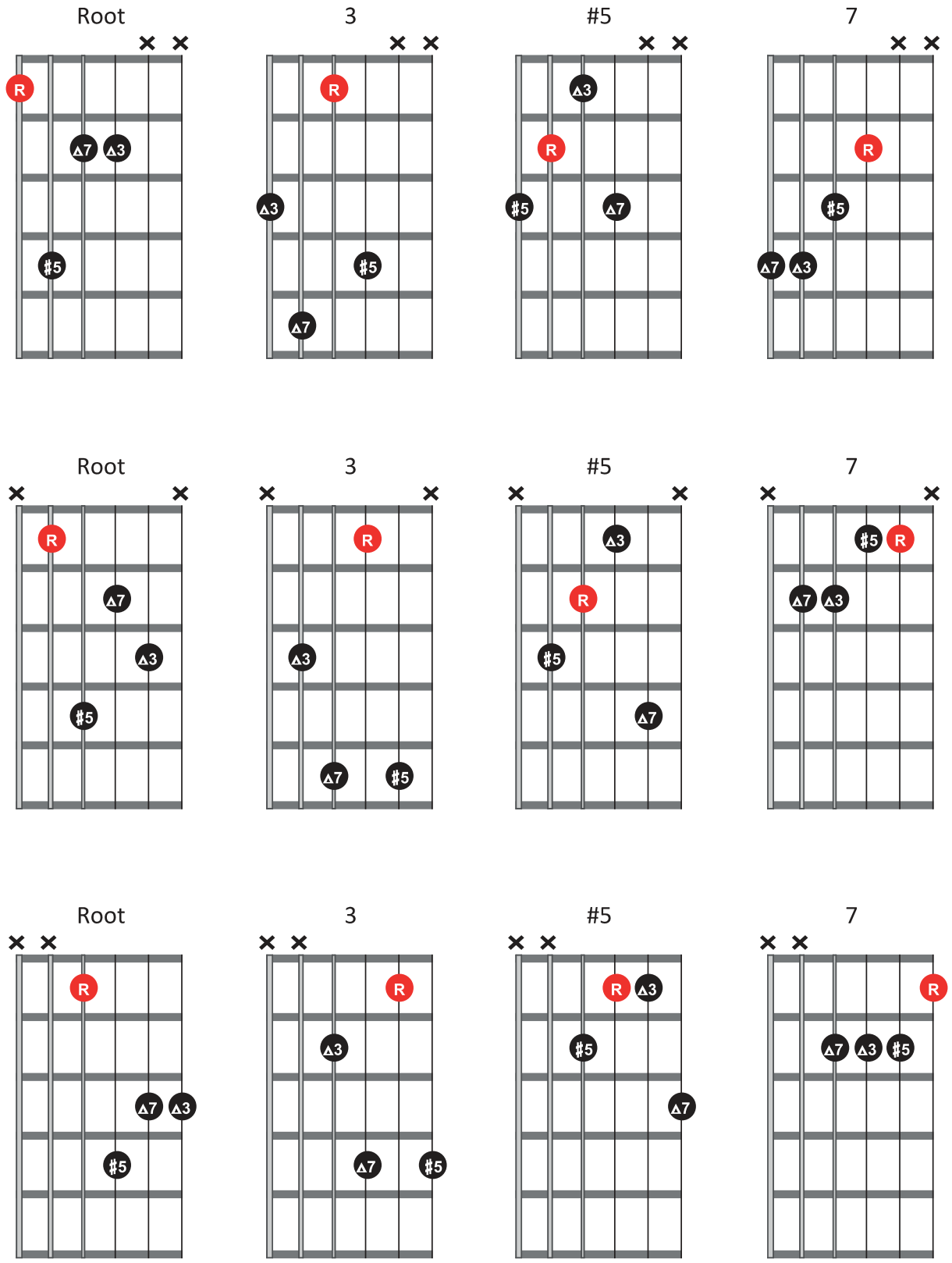
Dim^{7}



Dim^{maj7}

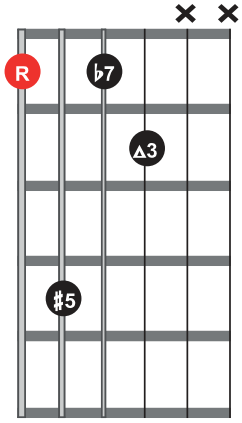


Aug^{maj7}



Aug^{7}

Root



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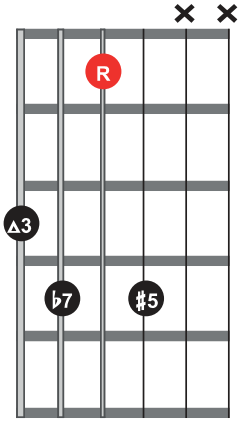

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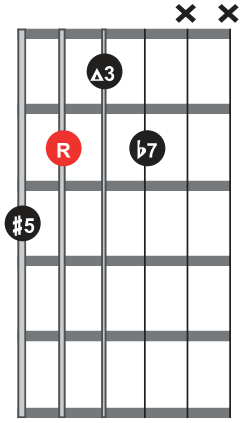
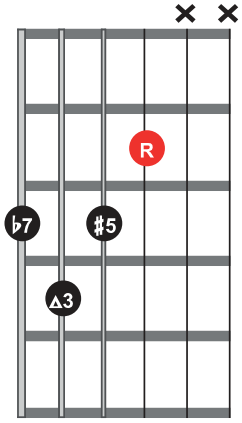
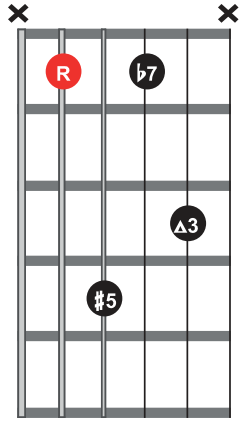


b7



Root



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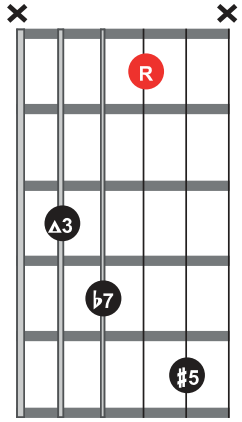

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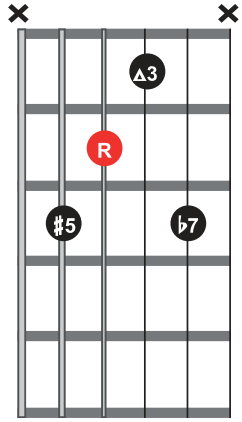

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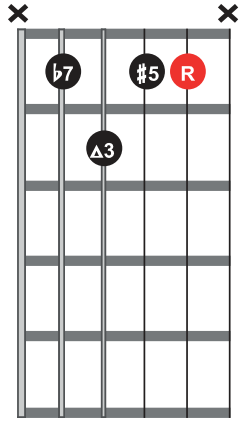

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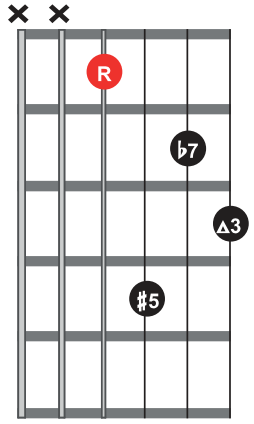

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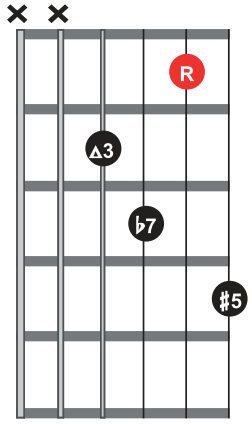

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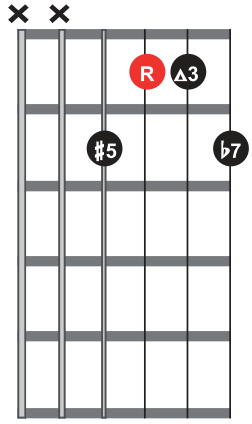
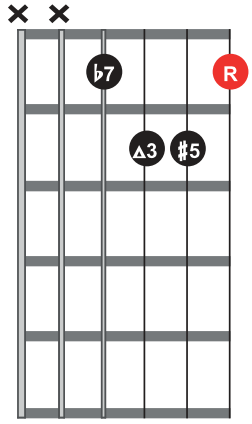
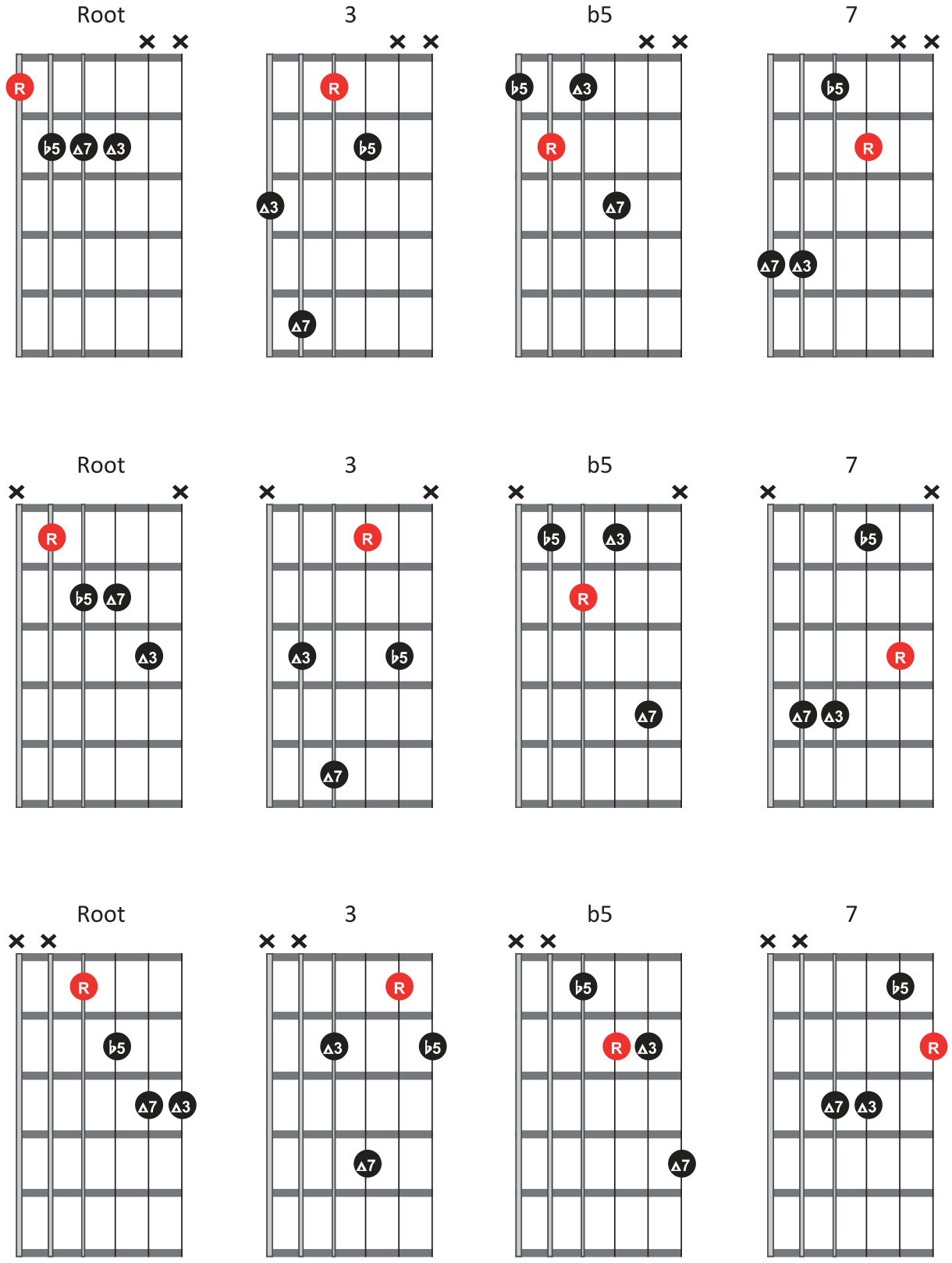


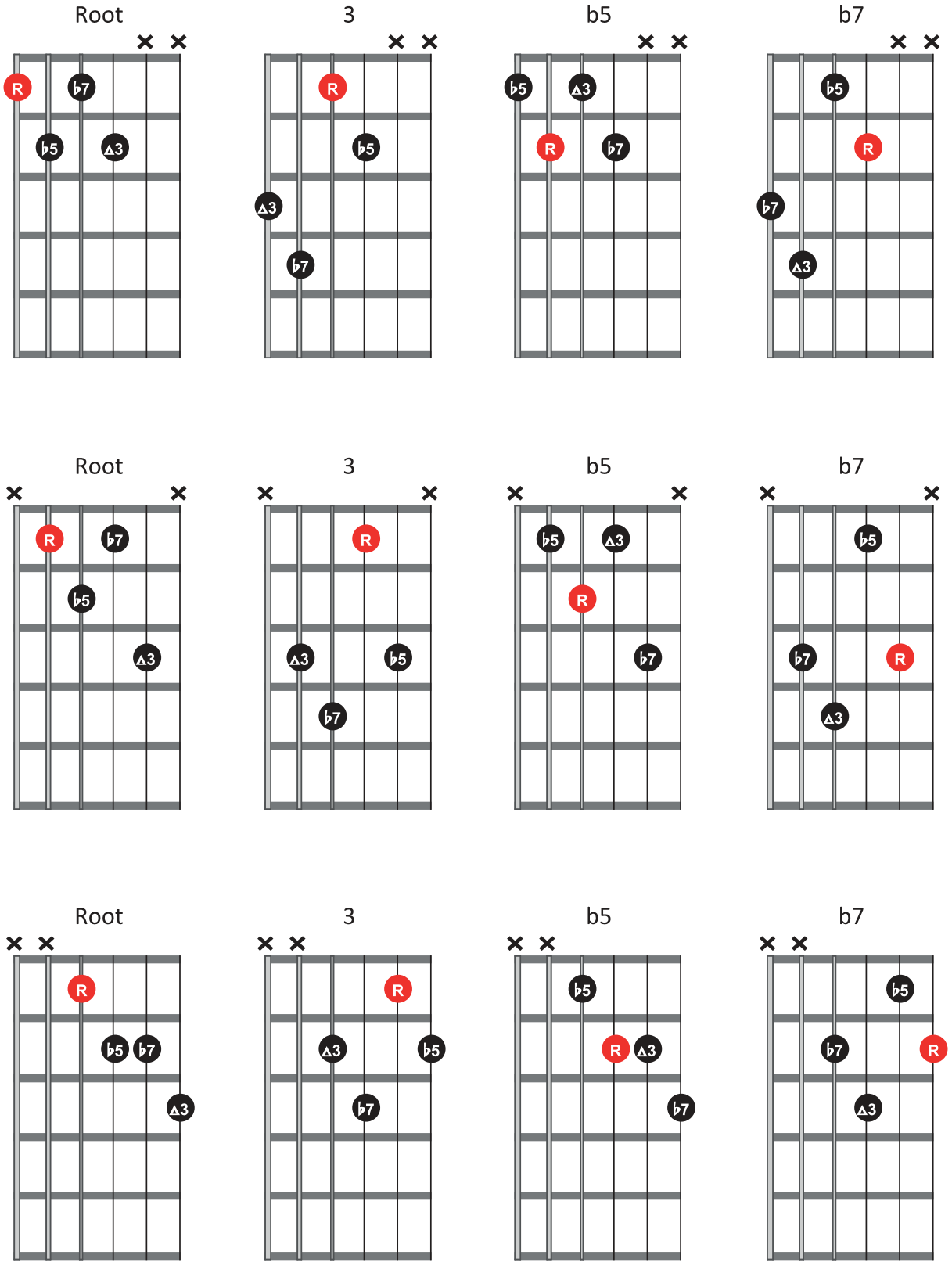
b7



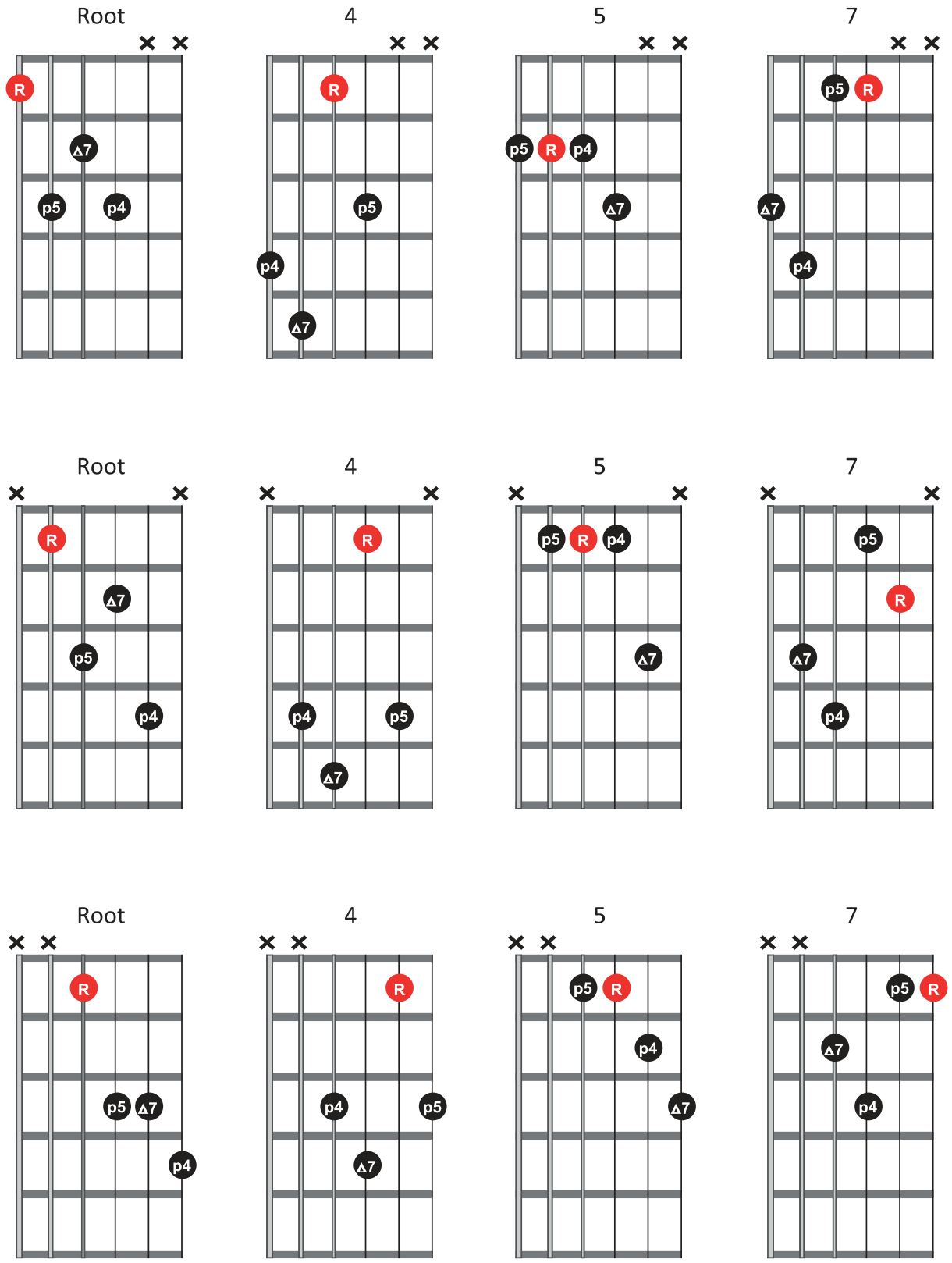
Maj^{7b5}



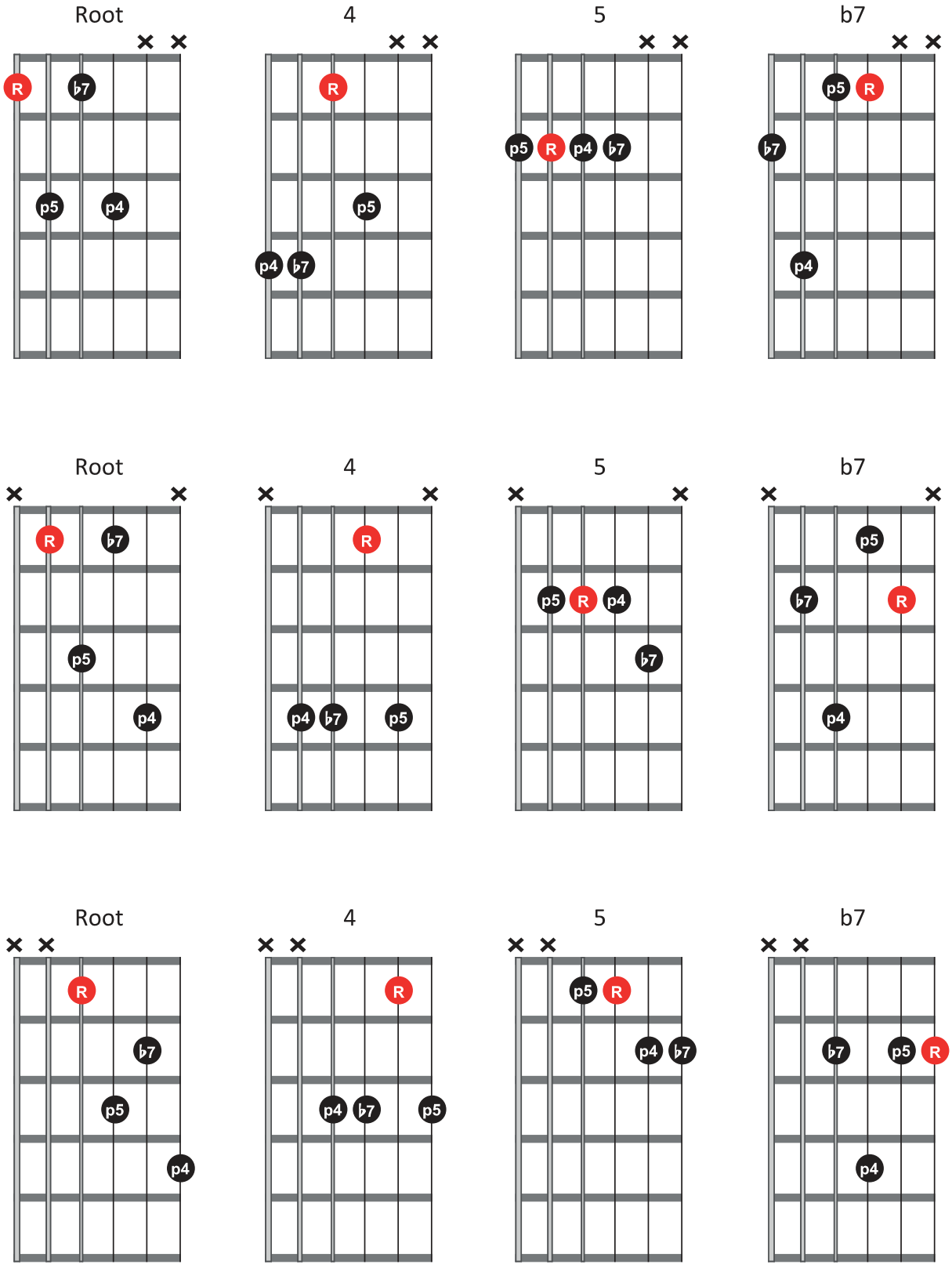
Dom^{7b5}



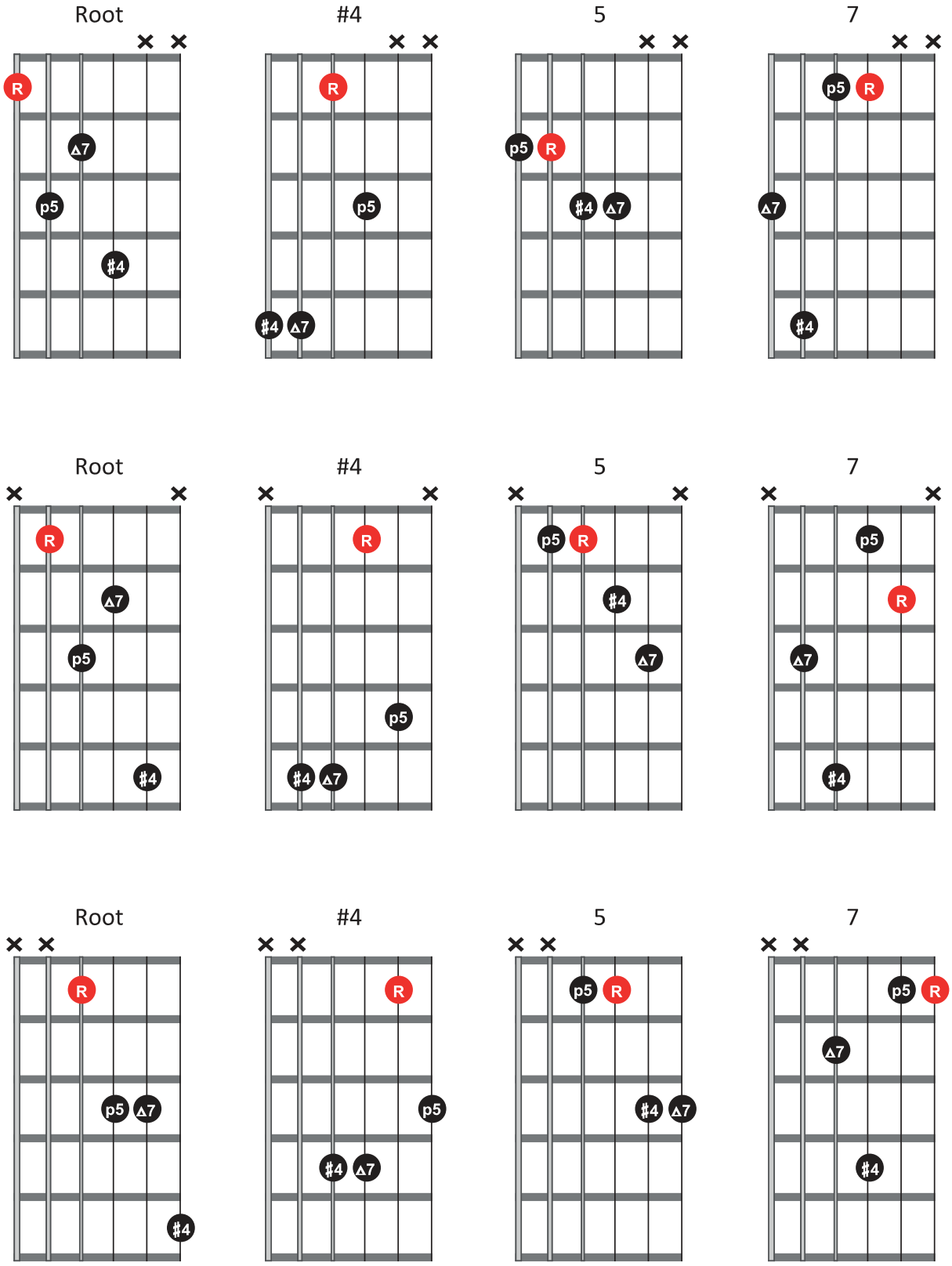
Maj^{7sus4}



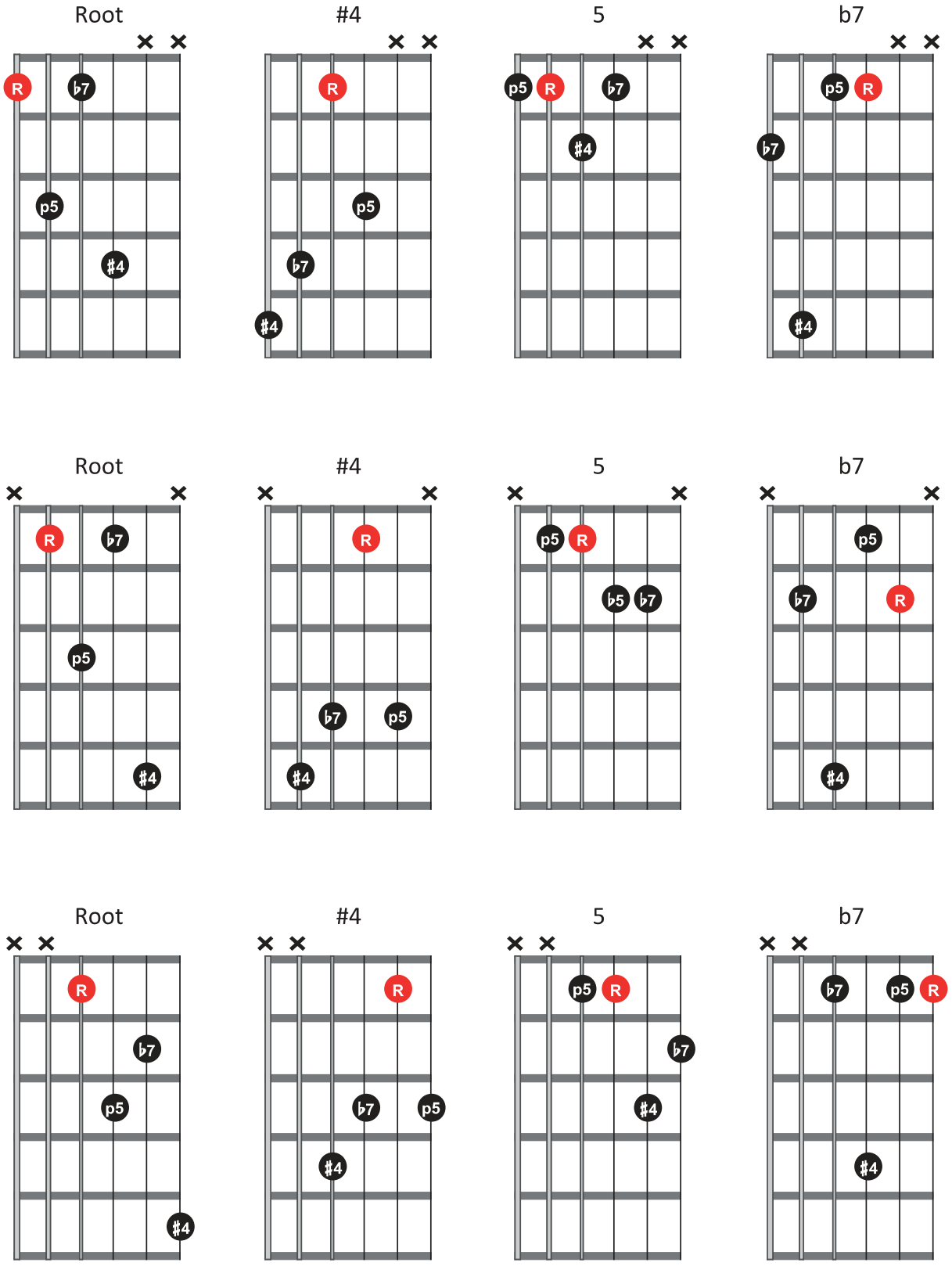
Dom^{7sus4}



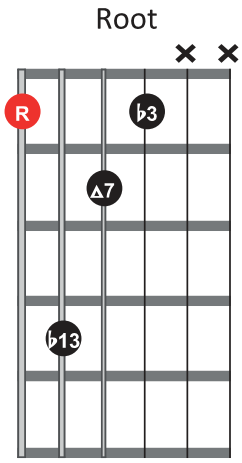
Lydian^{maj7}

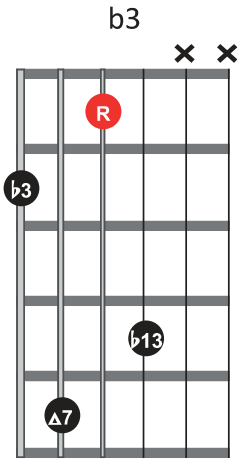


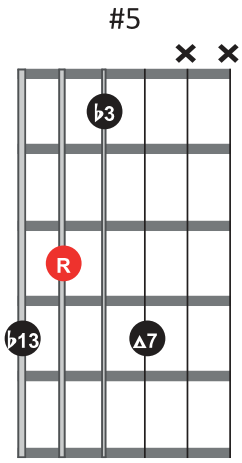
Lydian^{Dom7}



Dim^{maj7b13}


Root
R
♭3
♭13


b3
R
♭3
♭13
♭13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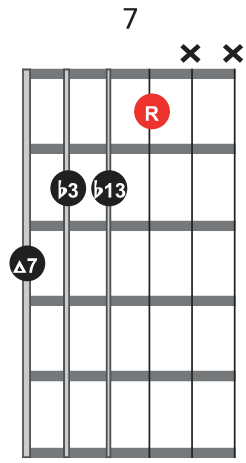
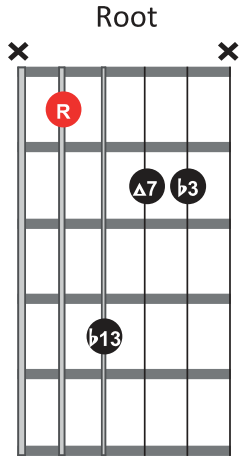
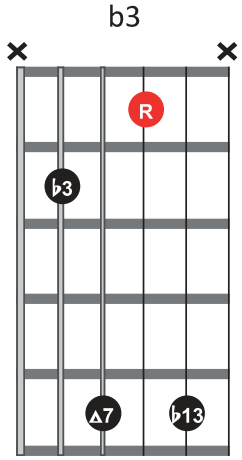
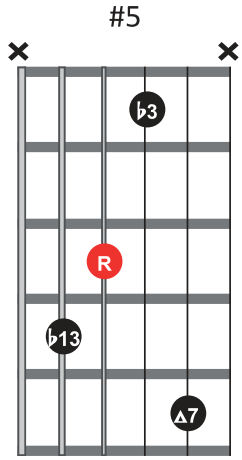
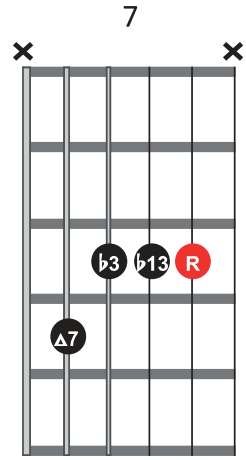
♭3
R
♭13

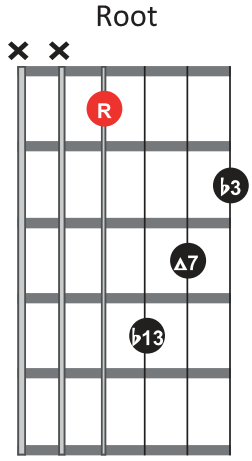

7
R
♭3
♭13

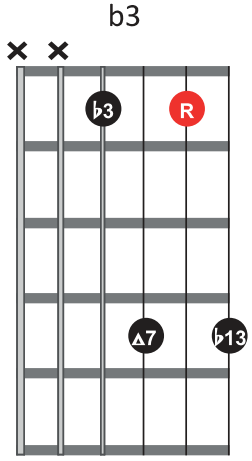

Root
R
♭13

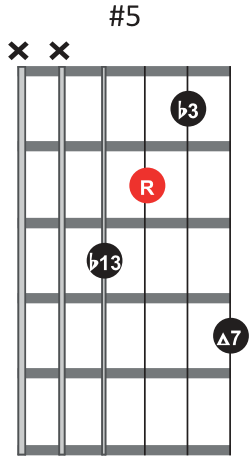

b3
R
♭3
♭13

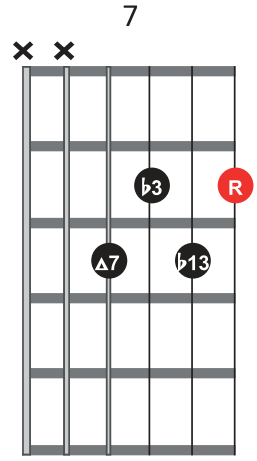

#5
♭3
R
♭13


7
♭3
♭13
R


Root
R
♭3
♭13

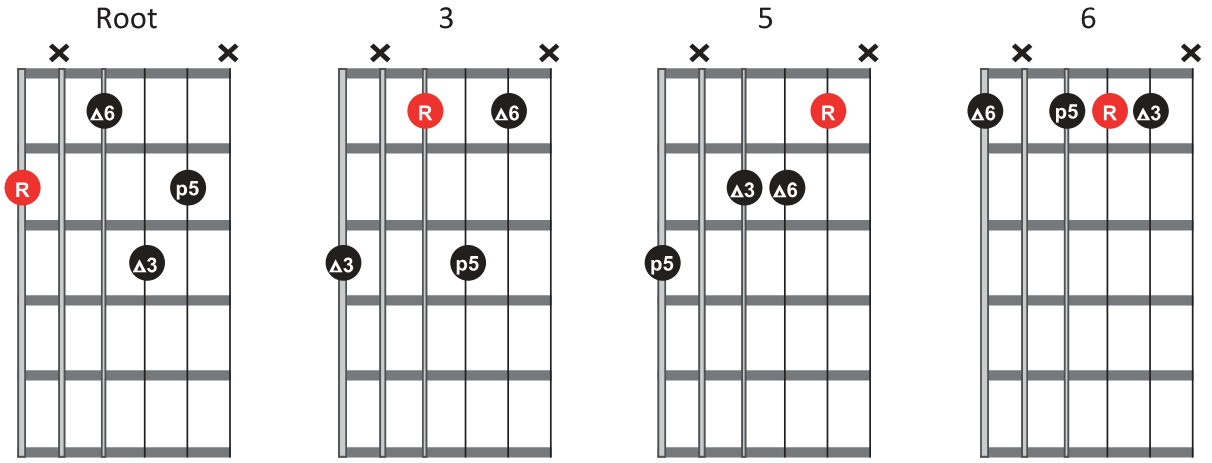

b3
♭3
R
♭13

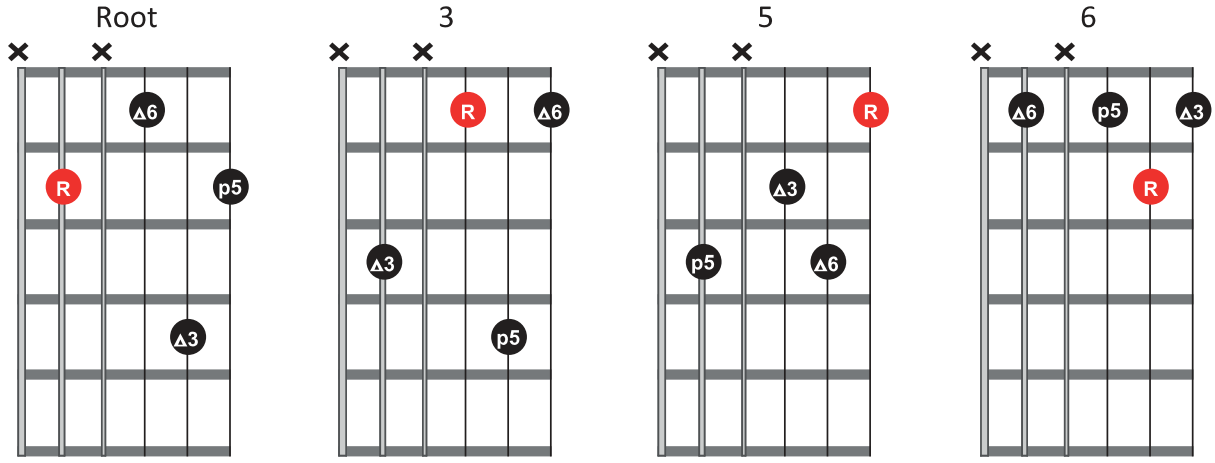

#5
♭3
R
♭13


7
♭3
R
♭13

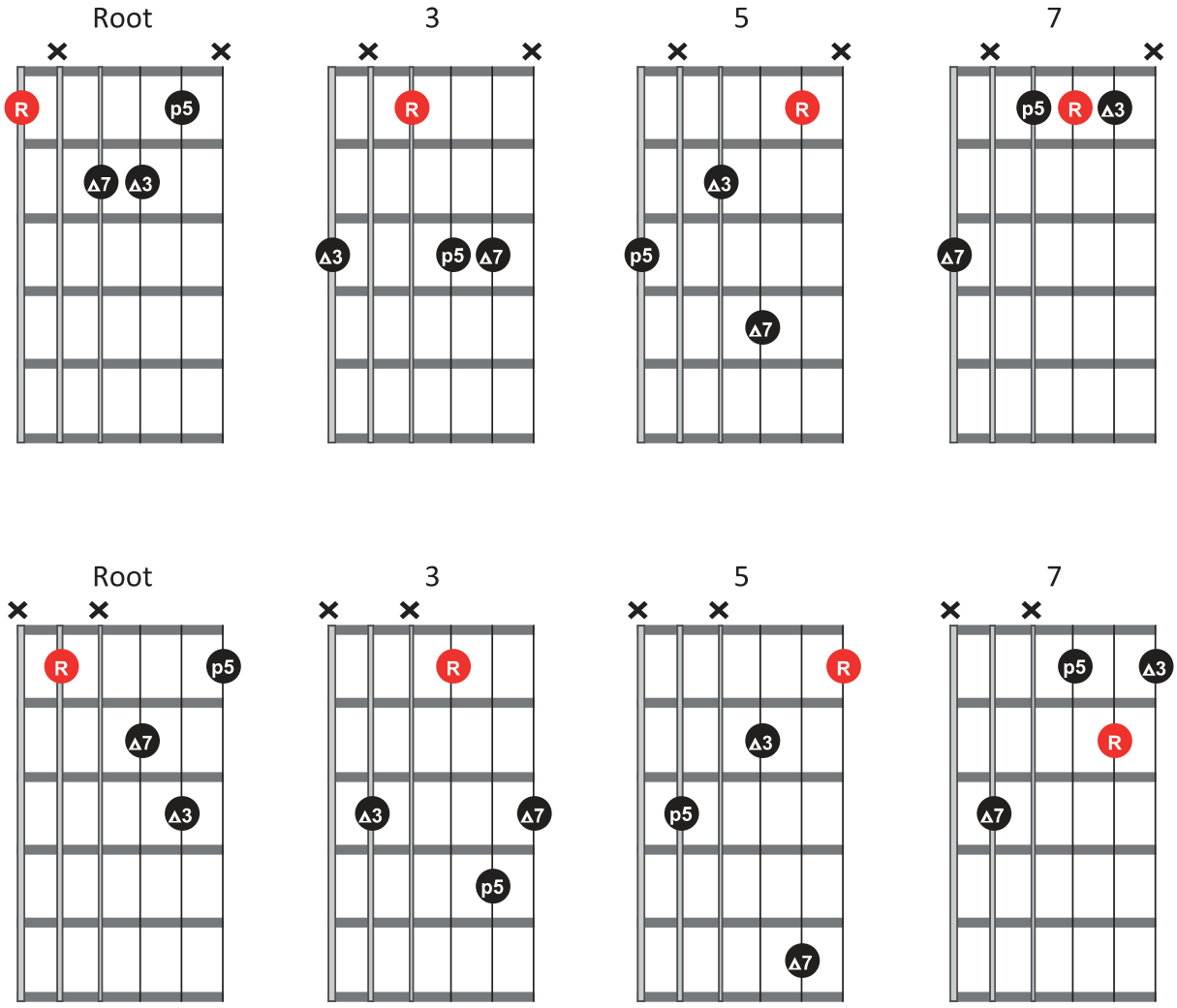
Drop 3

***Maj*^{6}**

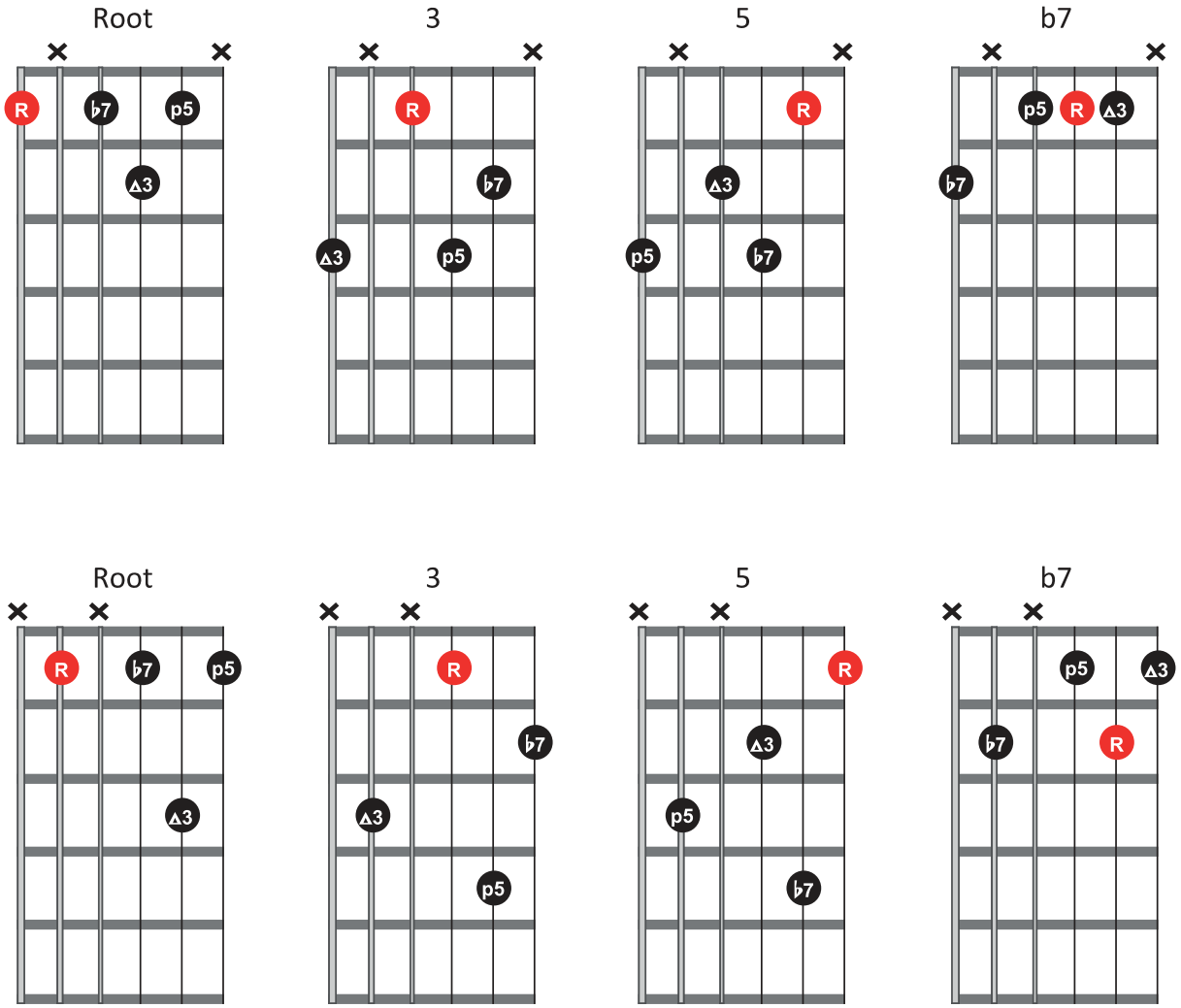




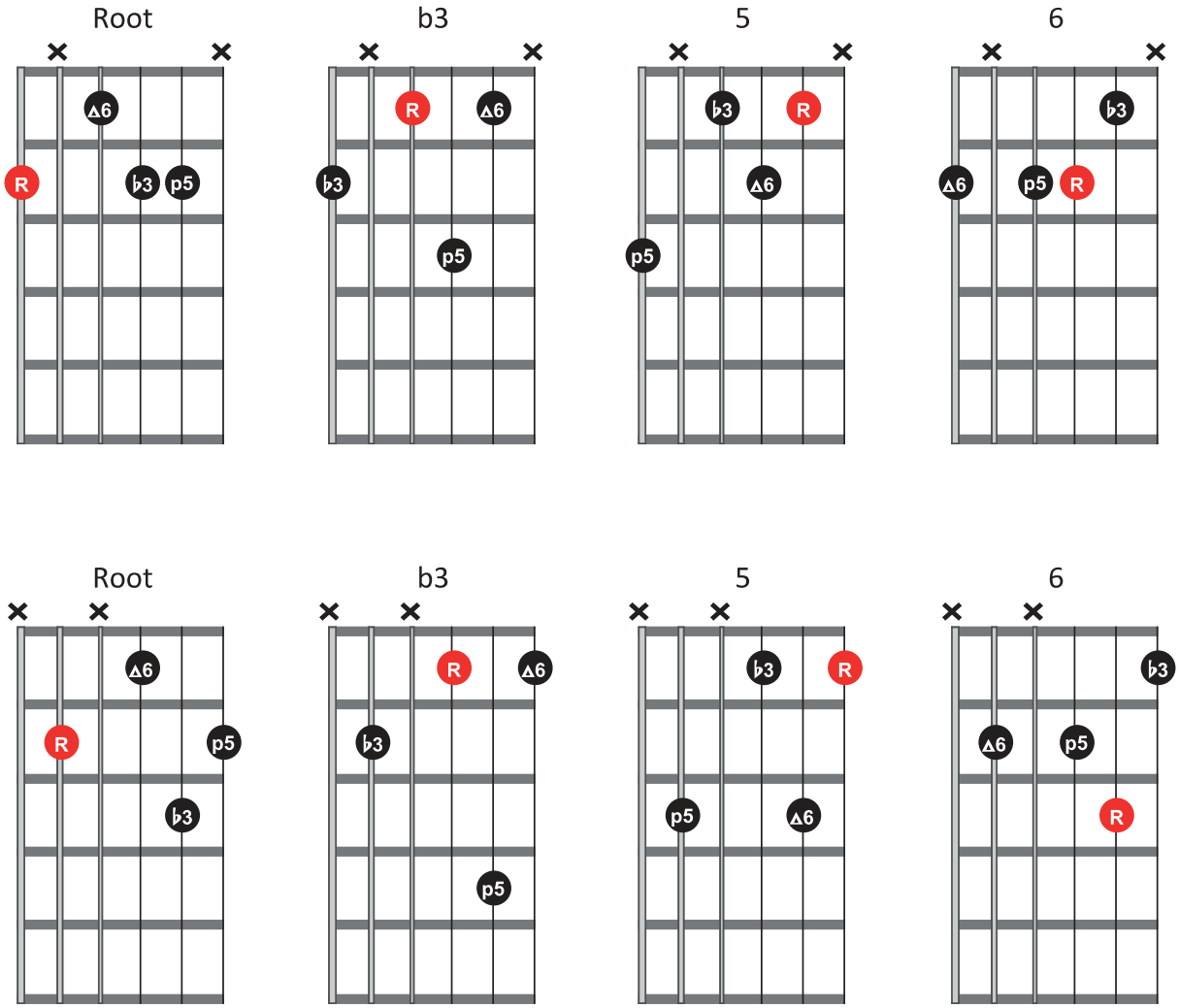
Maj^{7}



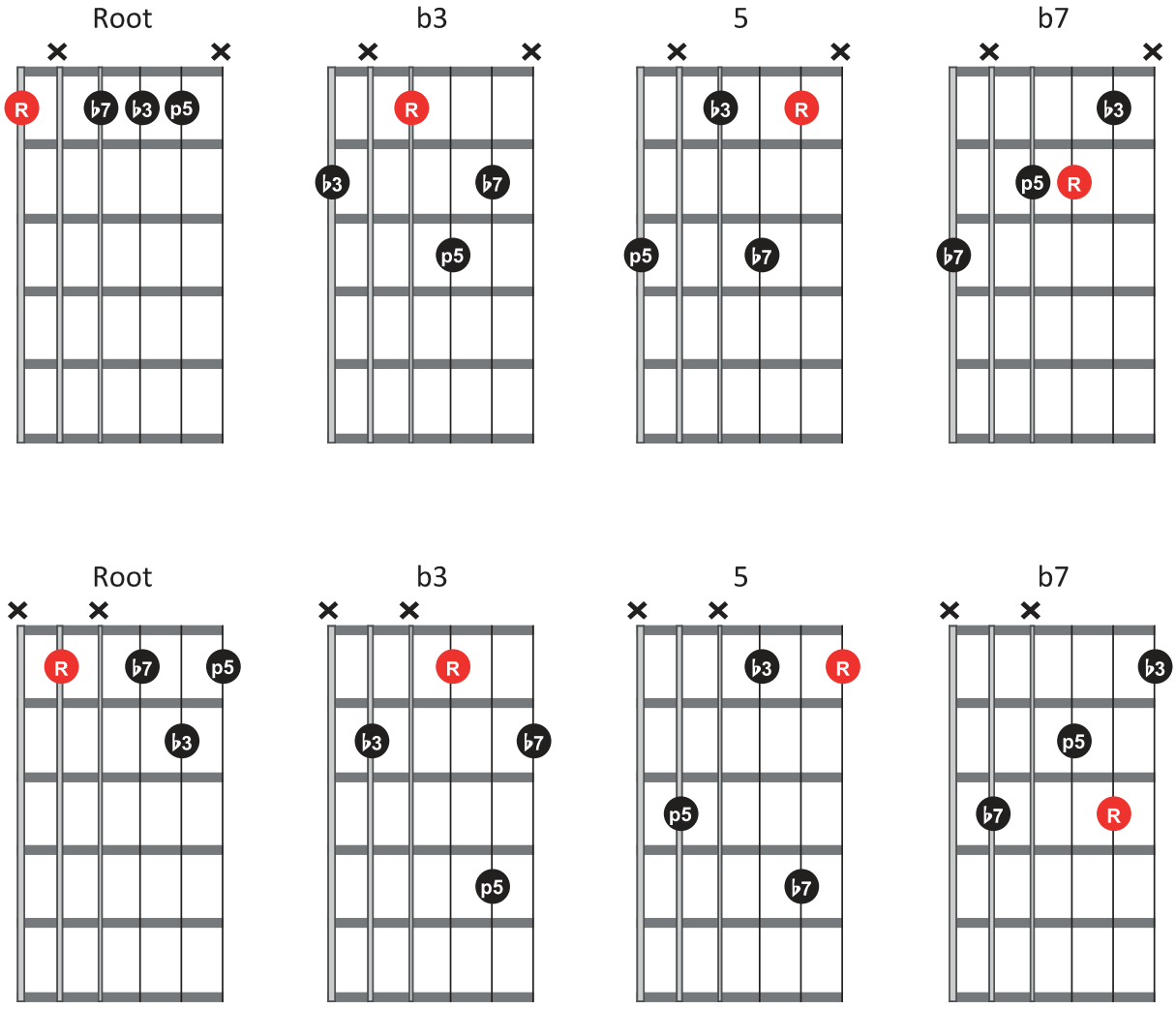
Dom^{7}



Min^{6}



Min^{7}


Root
R
♭7
♭3
p5
b3
R
♭3
♭7
p5
5
♭3
R
p5
♭7
b7
♭3
p5
R
♭7
Root
R
♭7
p5
♭3
b3
R
♭3
♭7
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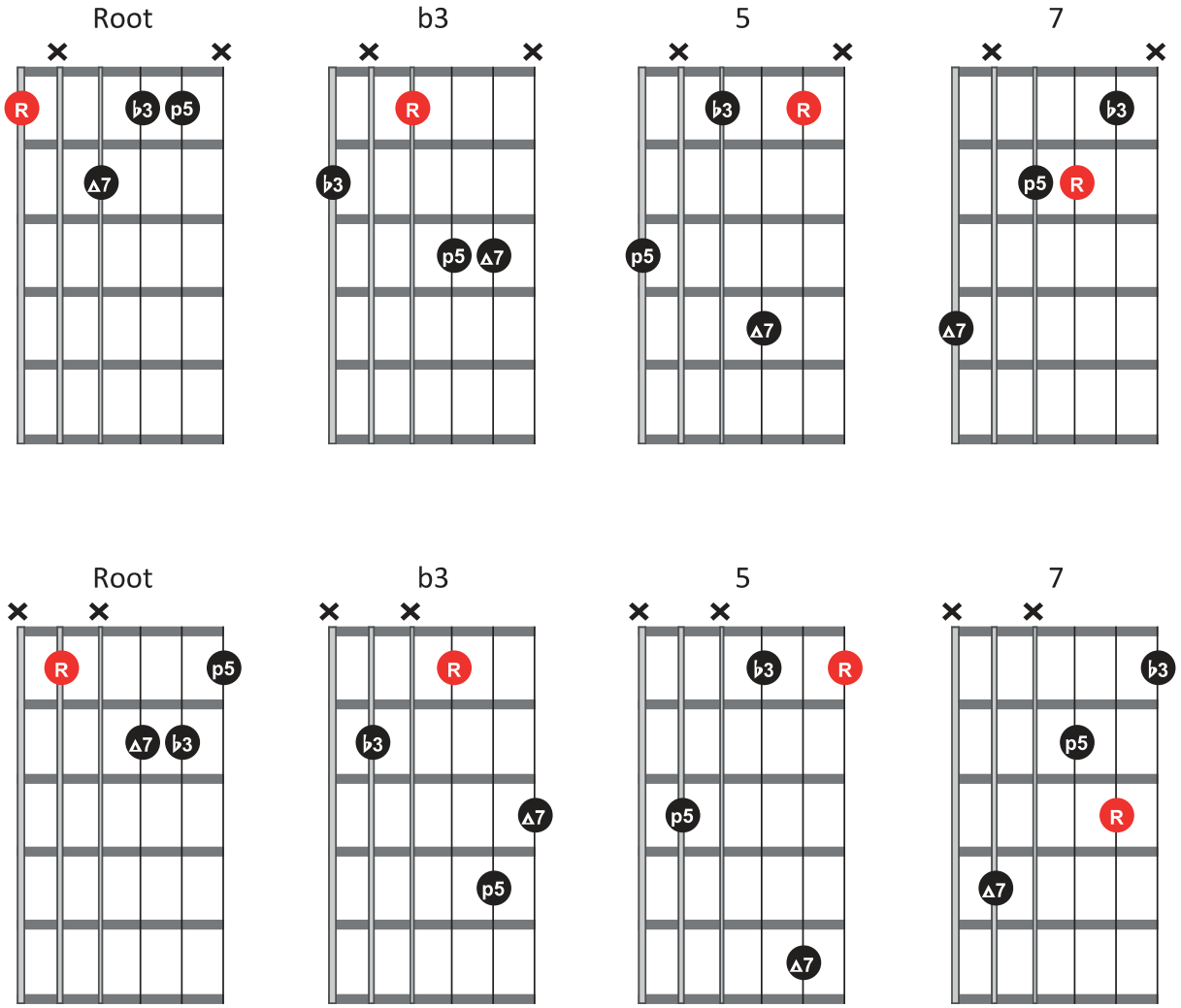
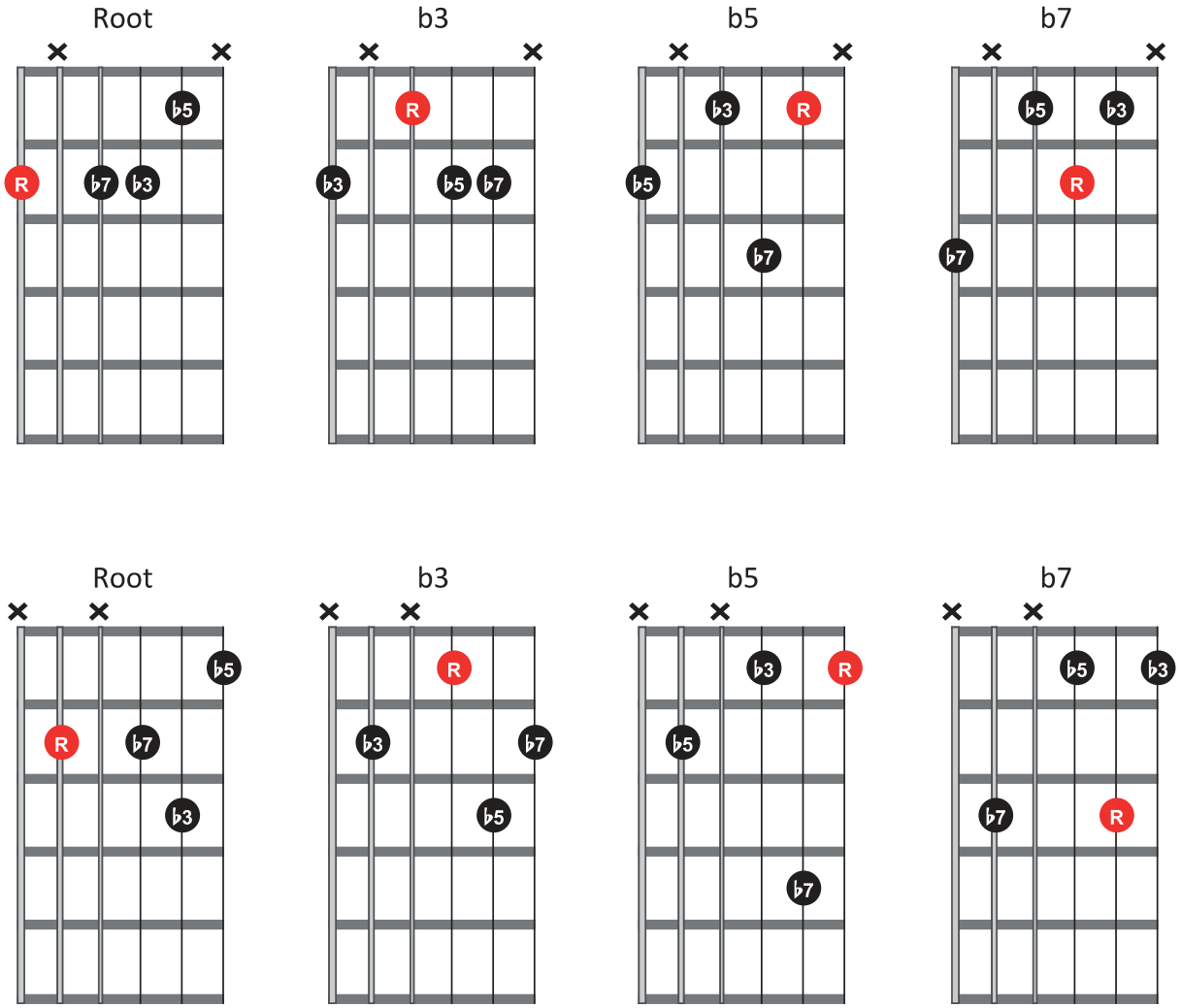
♭3
R
p5
♭7
b7
♭3
p5
♭7
R

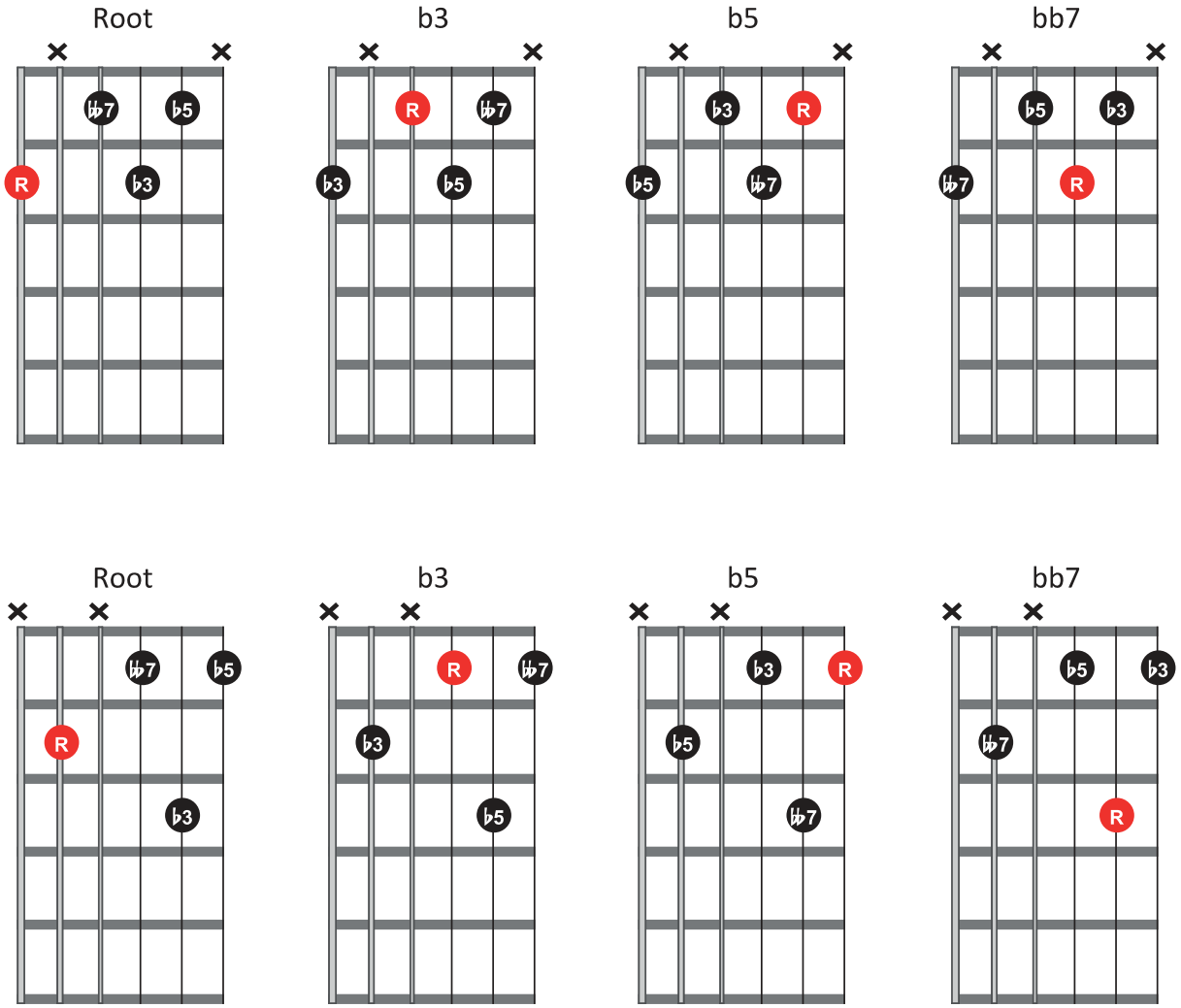
Min^{maj7}



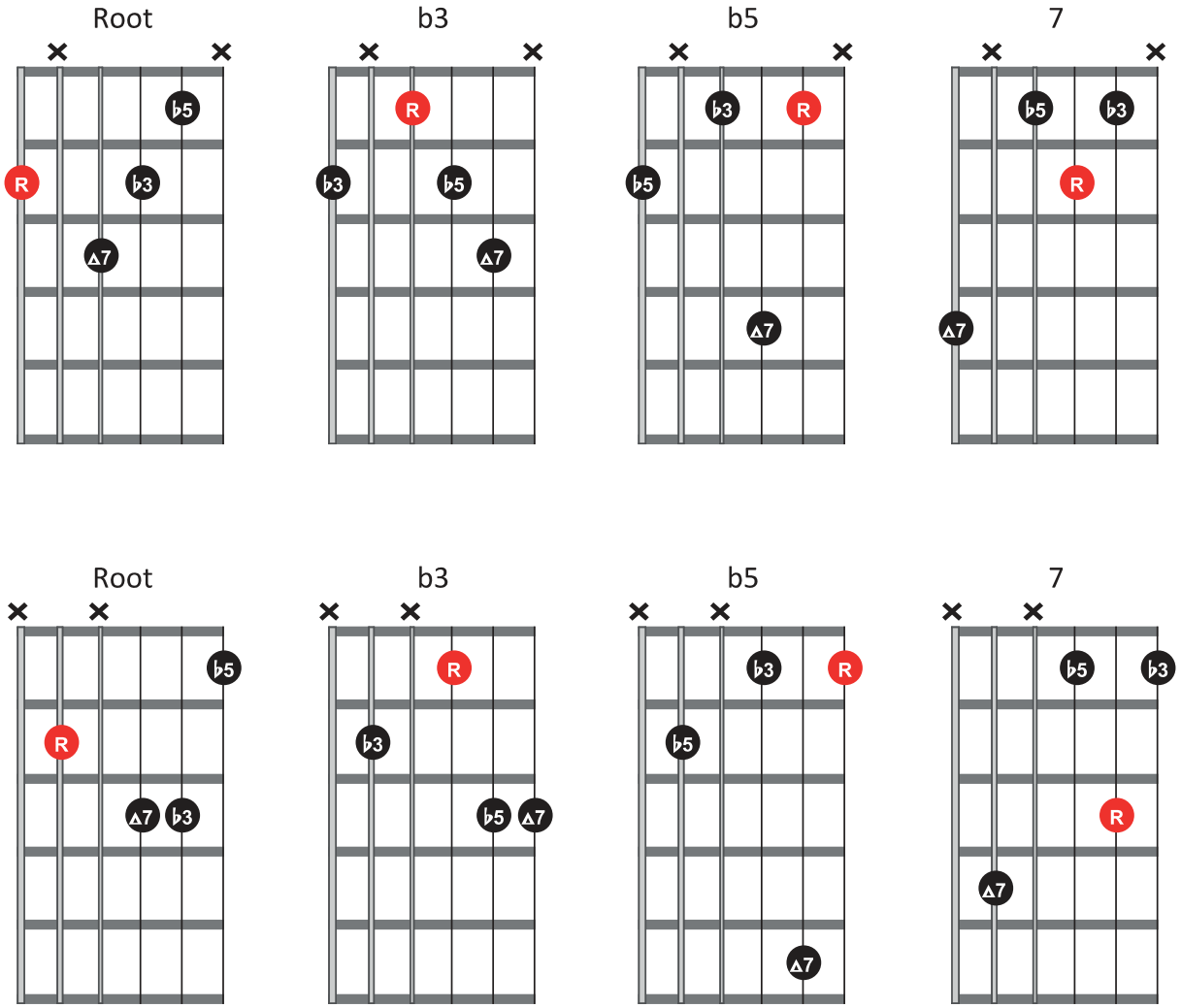
Min^{7b5}



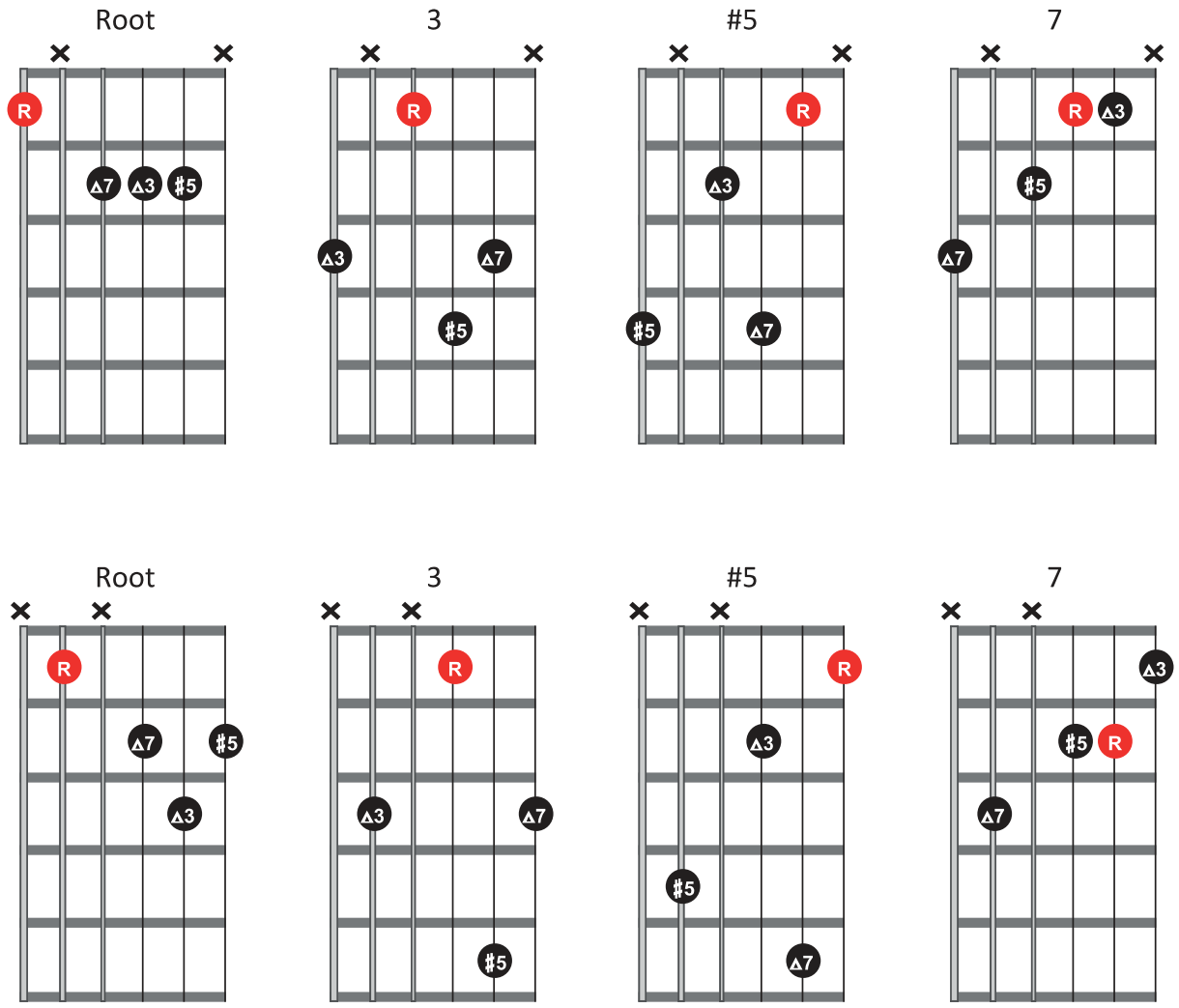
Dim^{7}



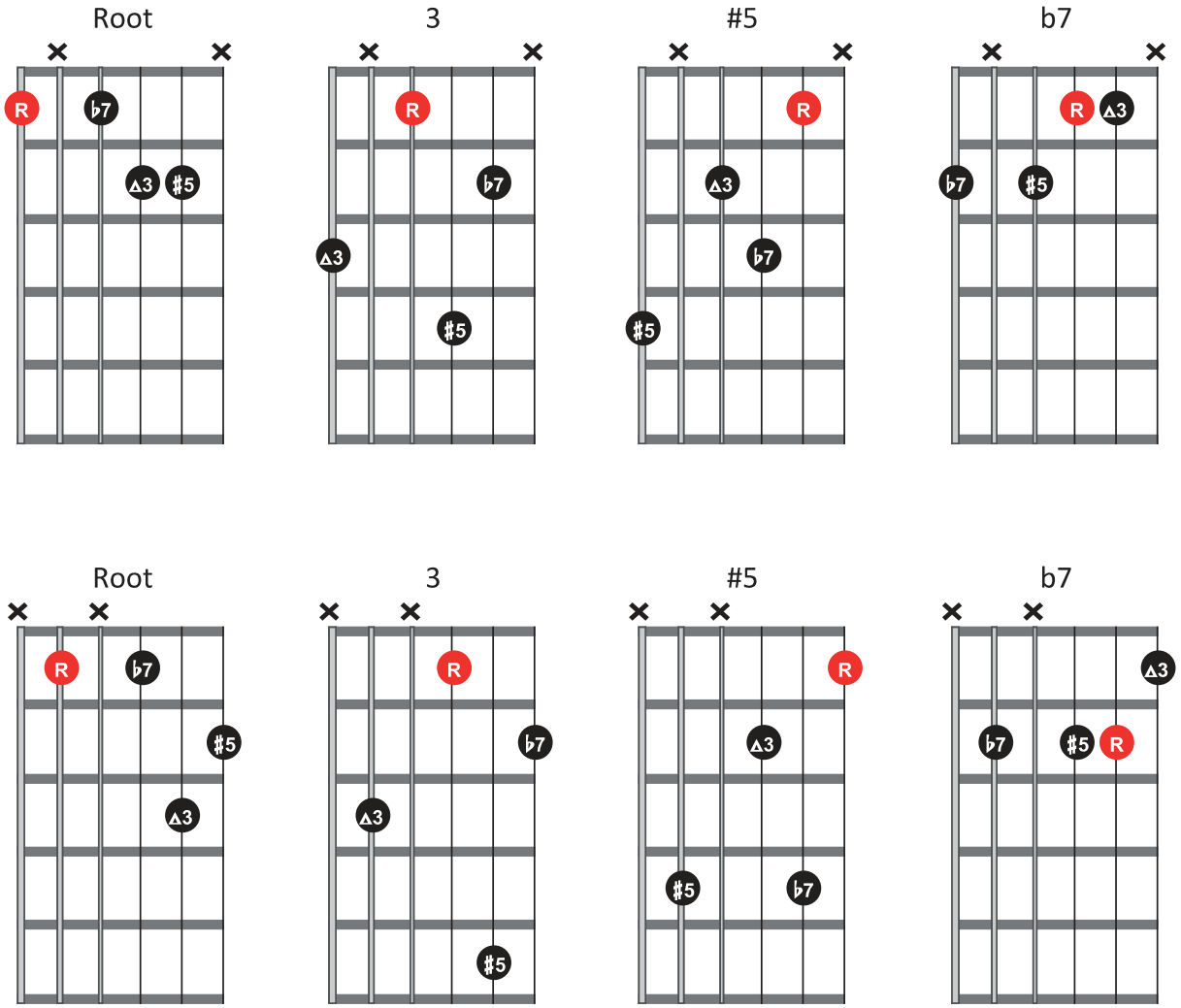
Dim^{maj7}



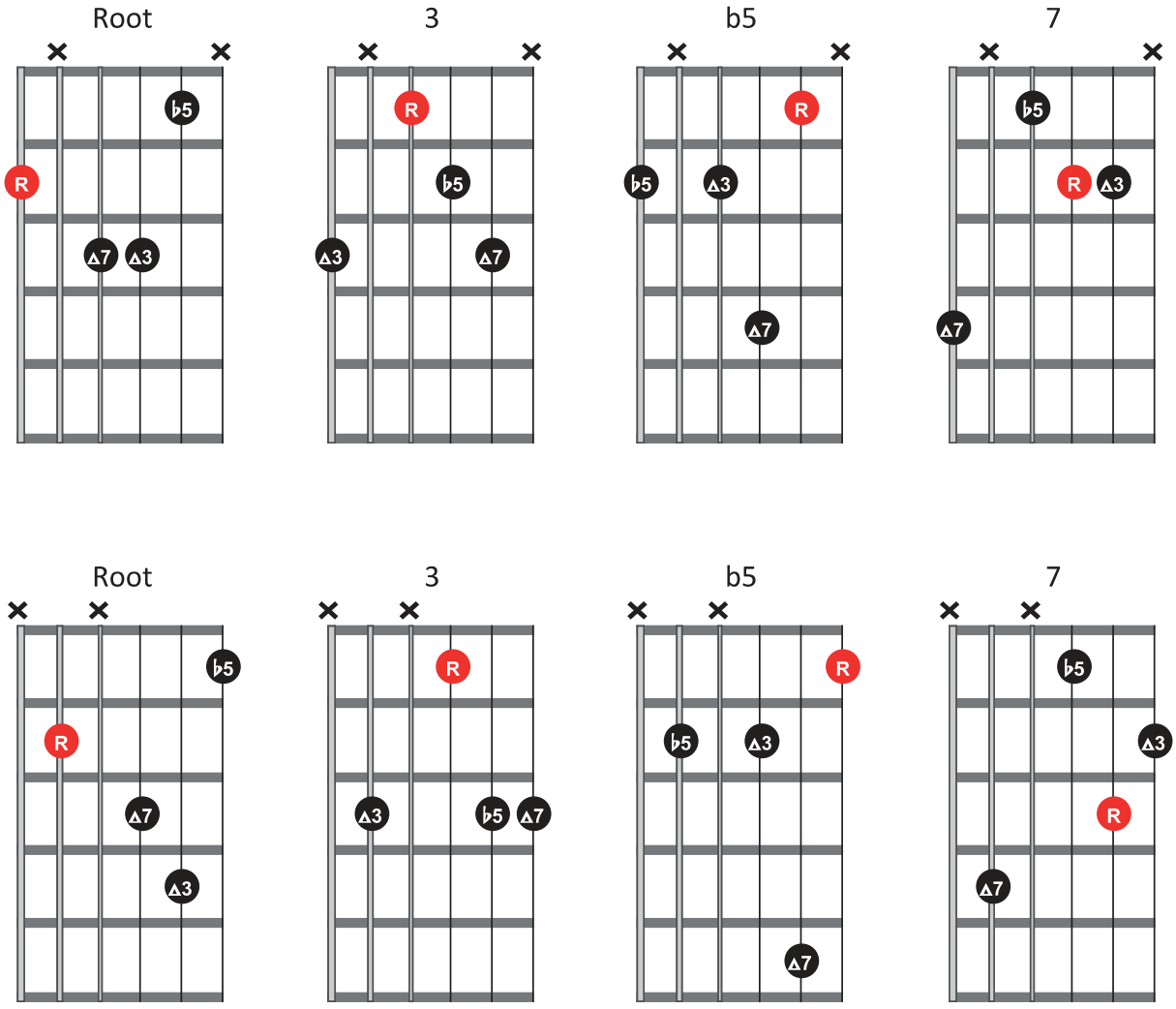
Aug^{maj7}



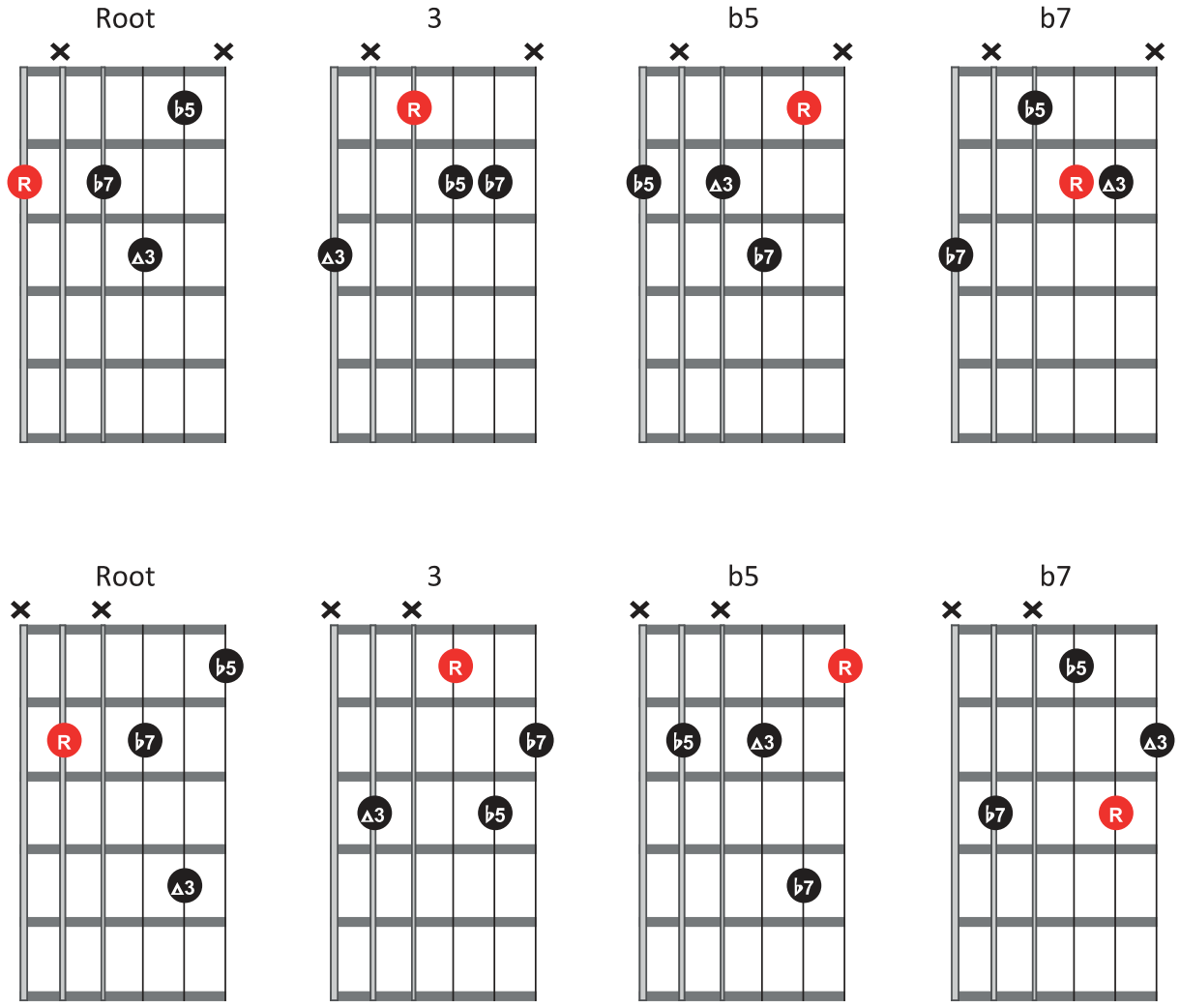
Aug^{7}



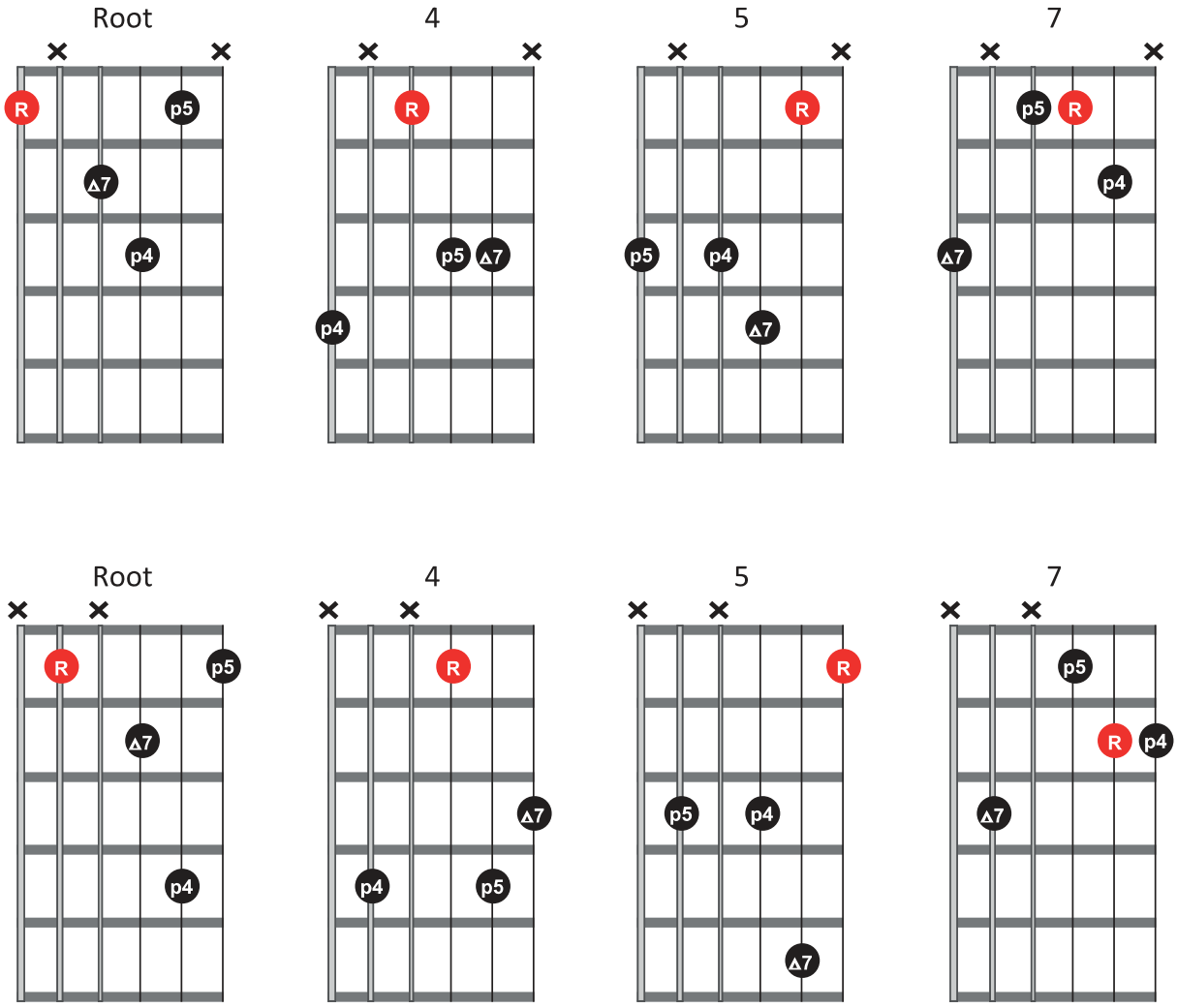
Maj^{7b5}



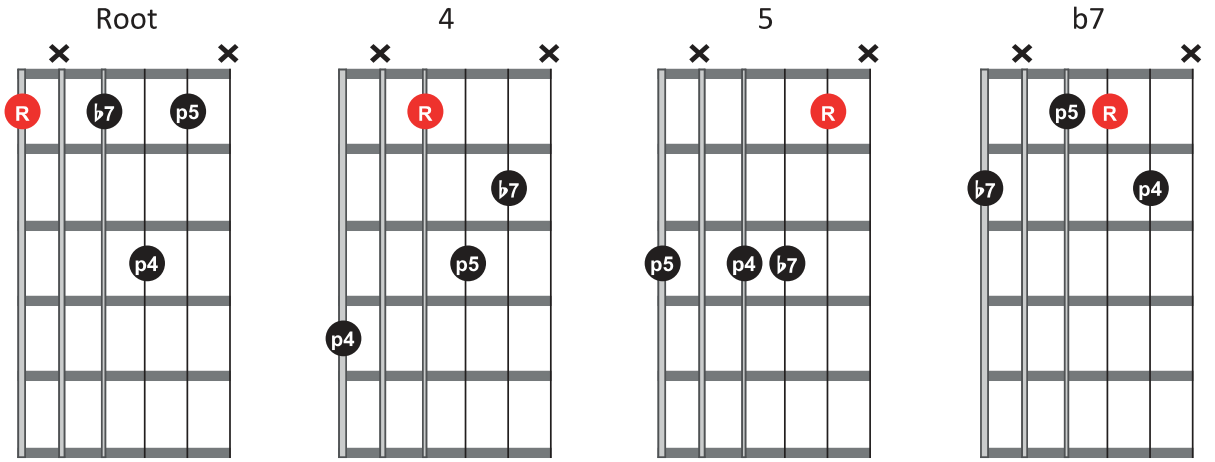
Dom^{7b5}

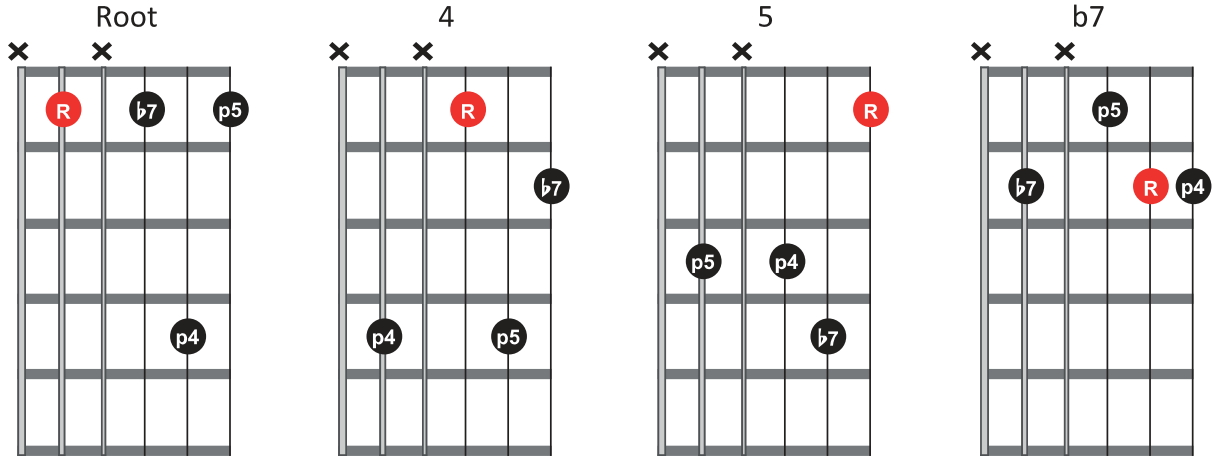


Maj^{7sus4}

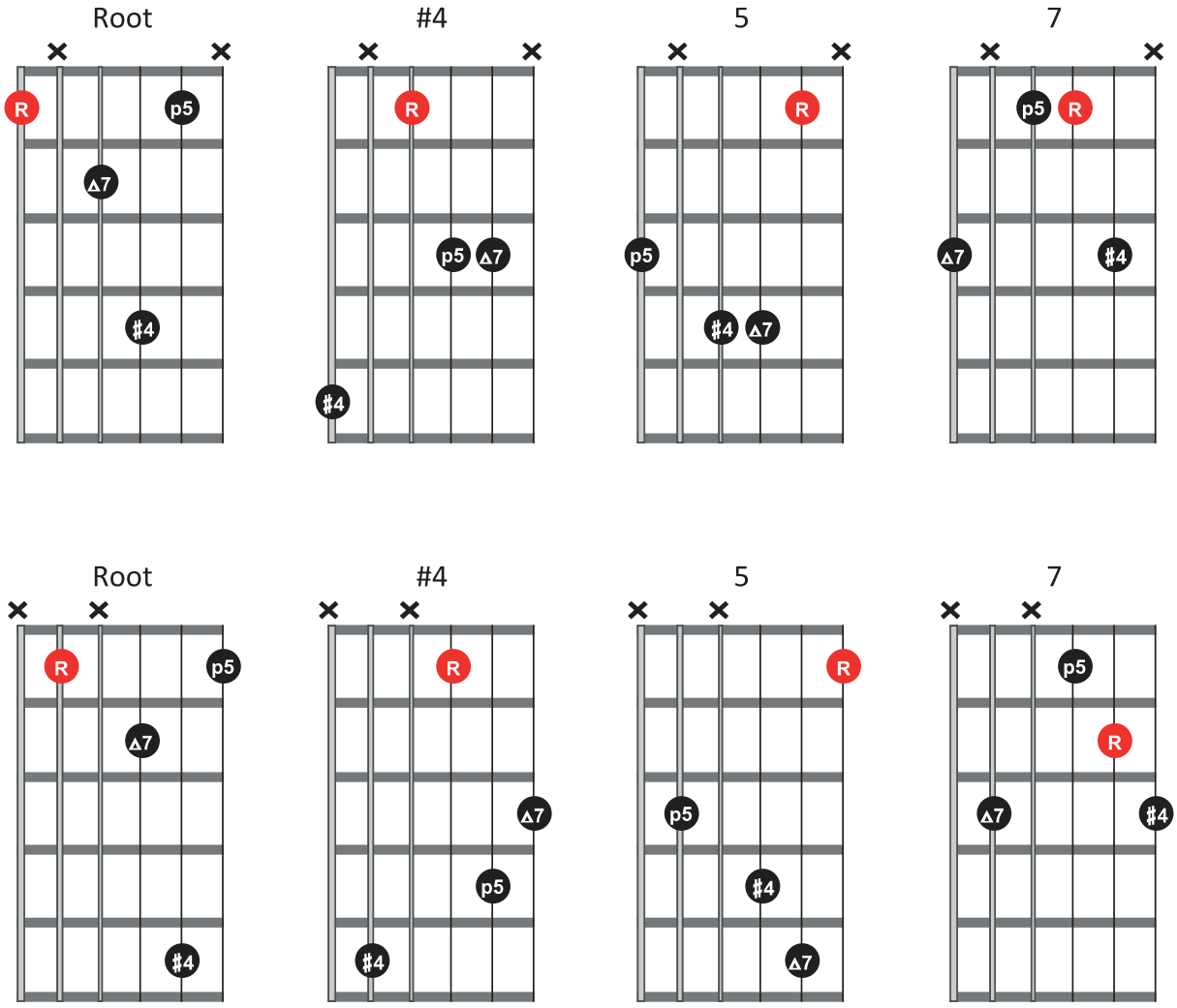


Dom^{7sus4}

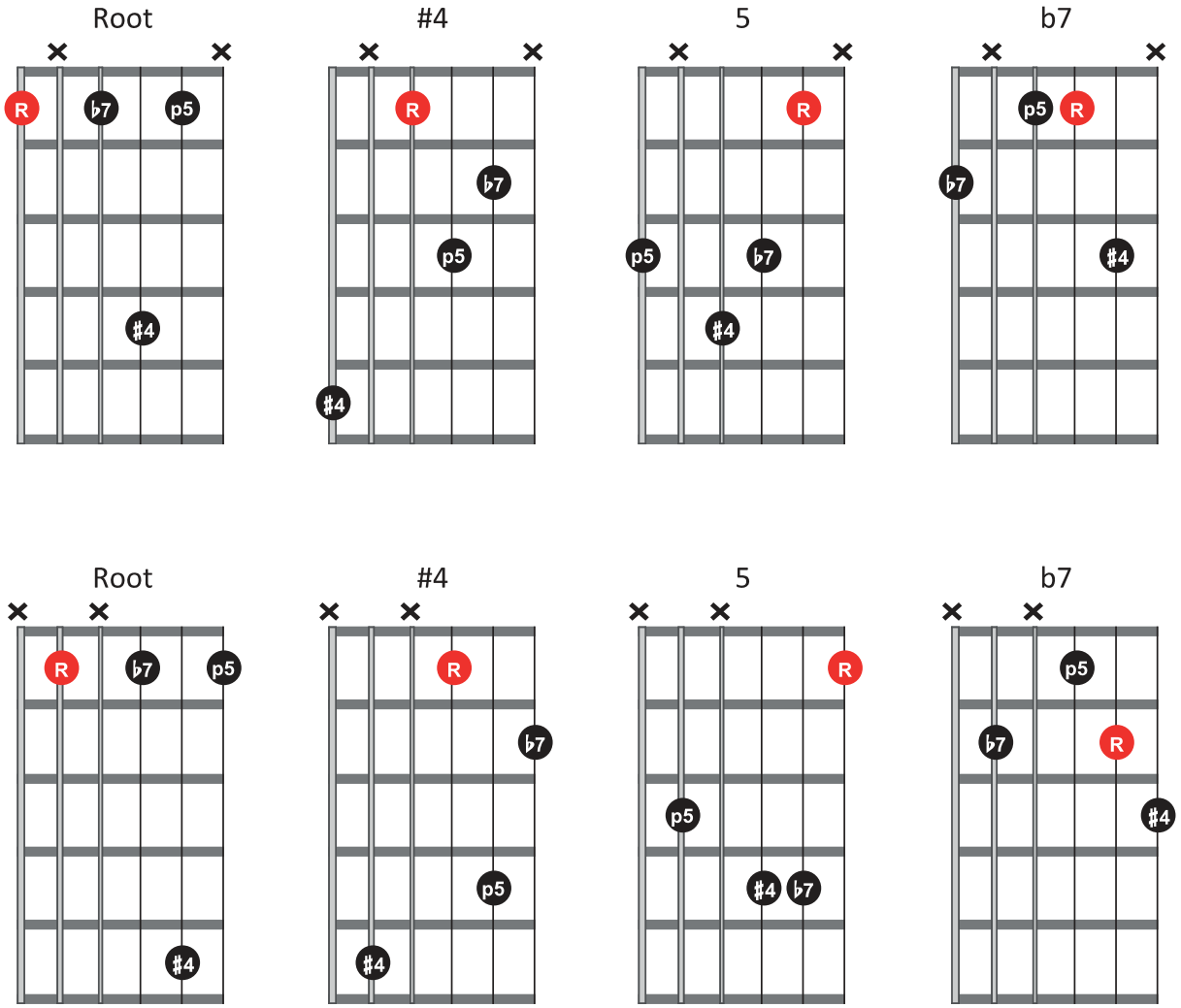




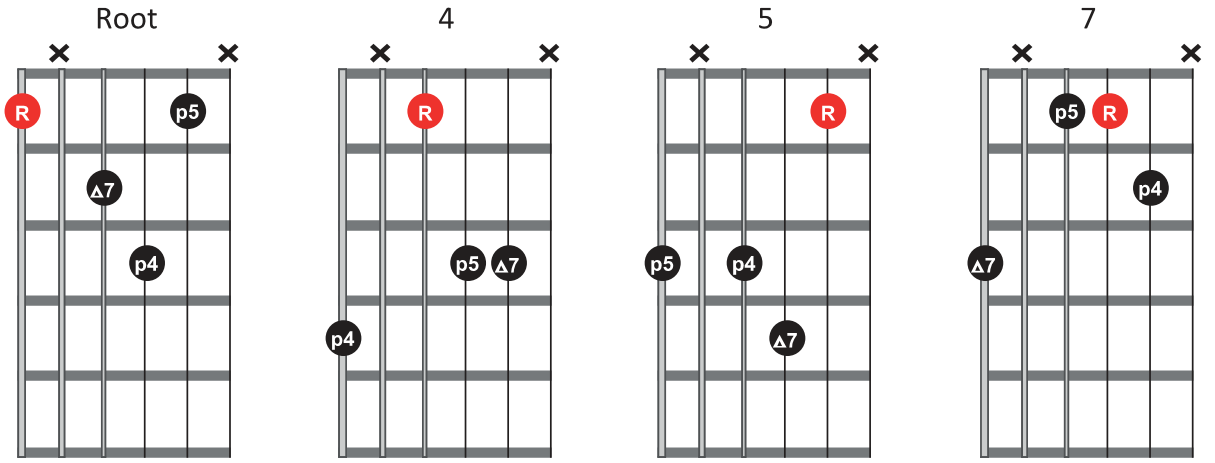
Lydian^{maj7}

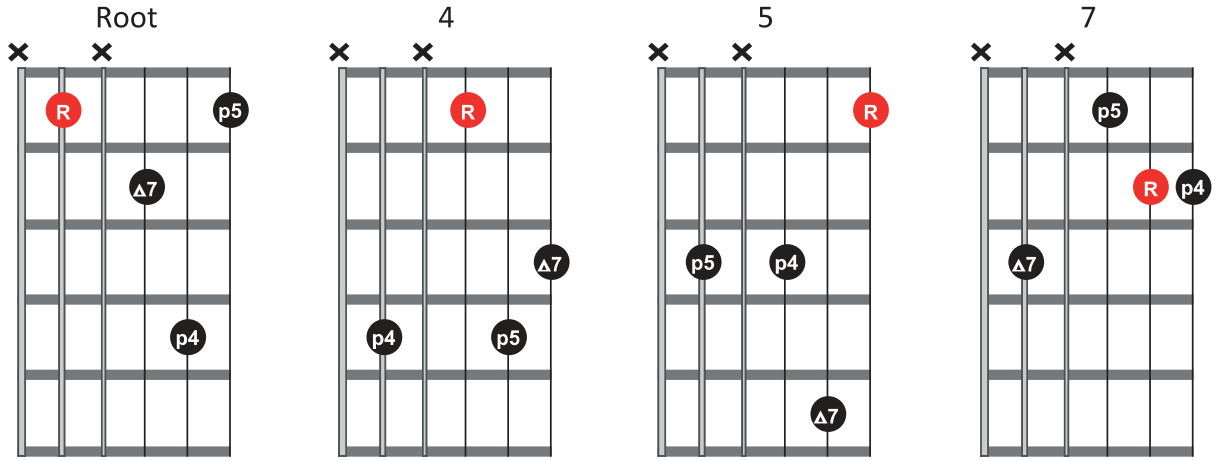


Lydian^{Dom7}



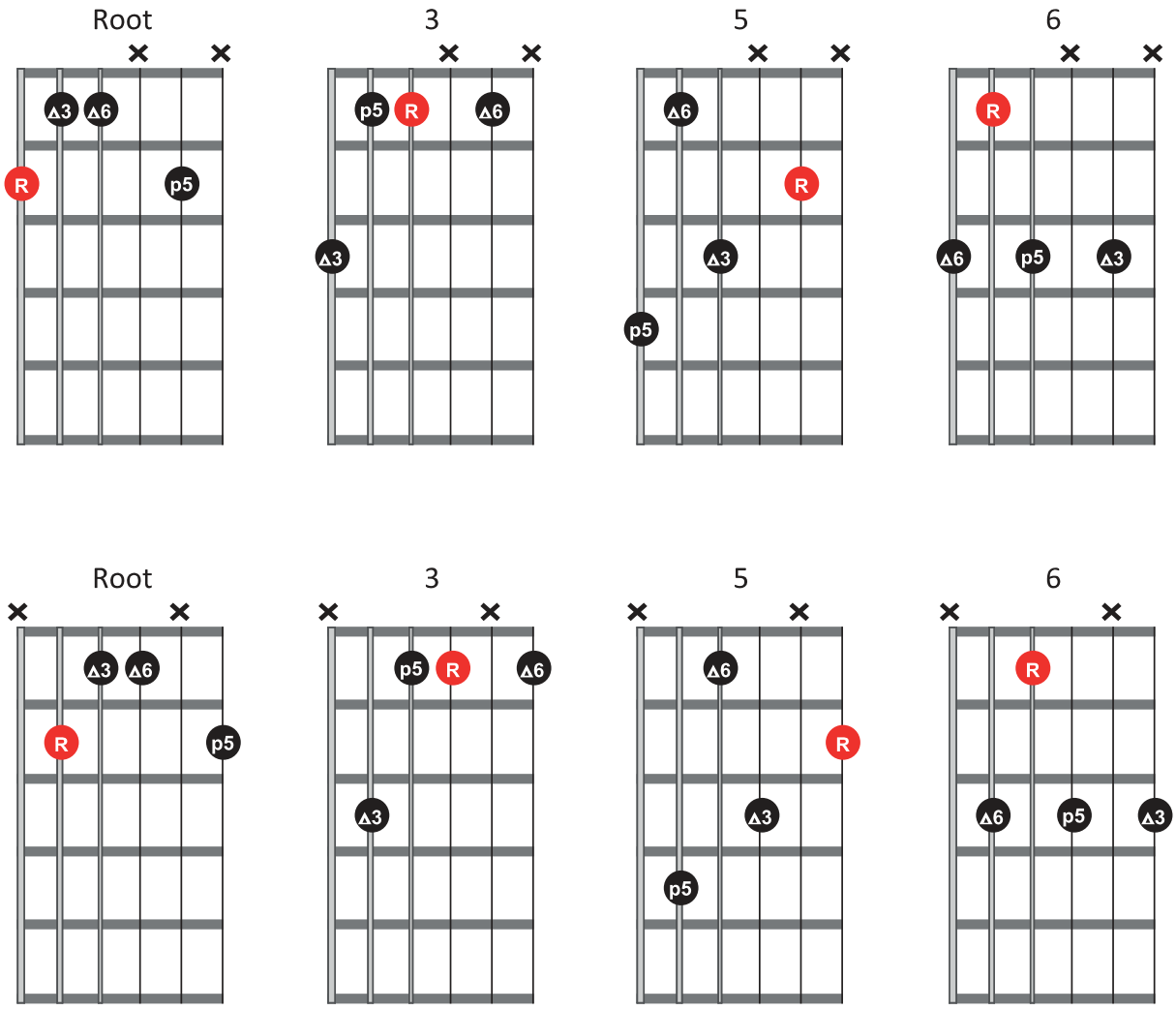
Dim^{maj7b13}



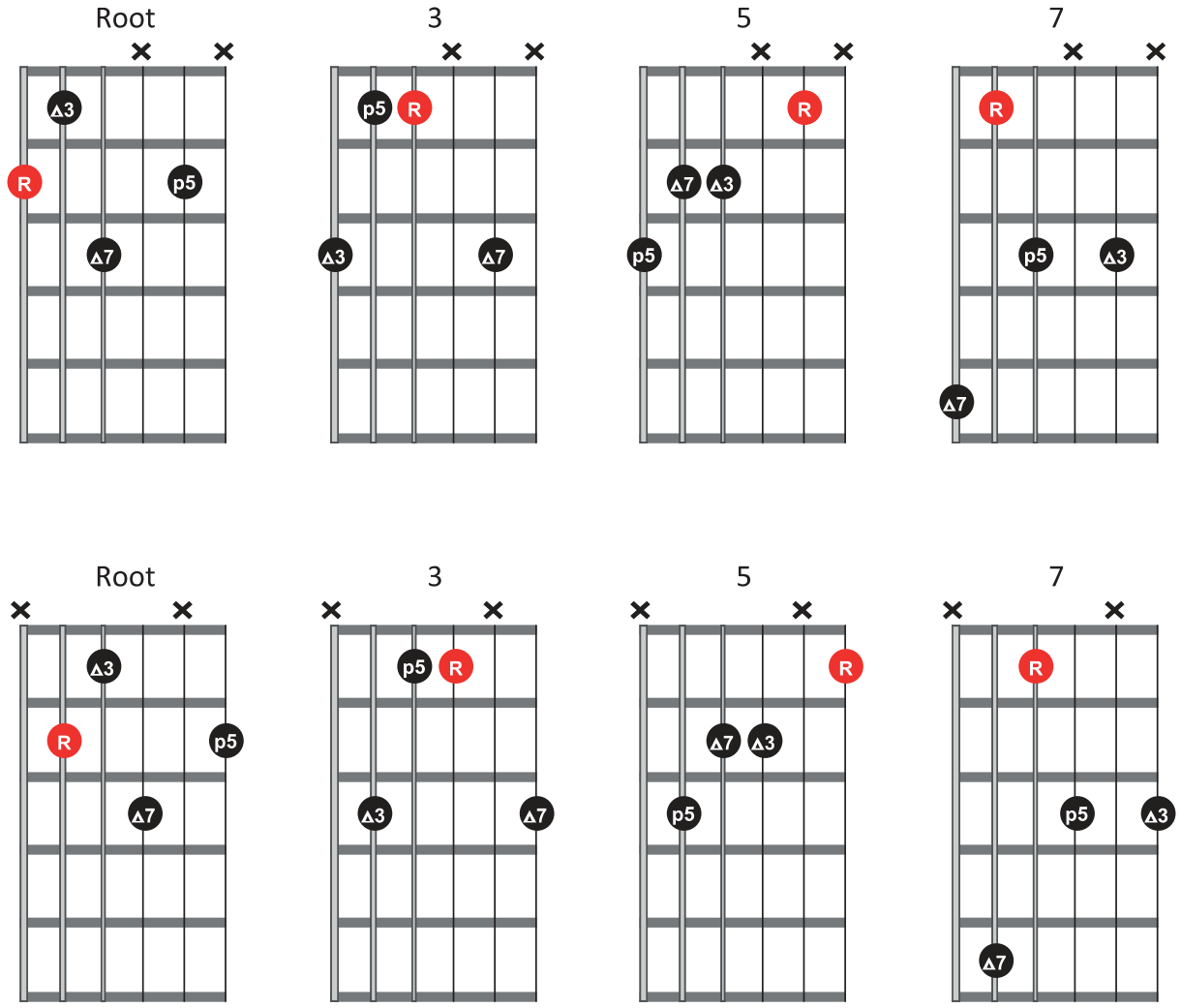


Drop 2 + 3

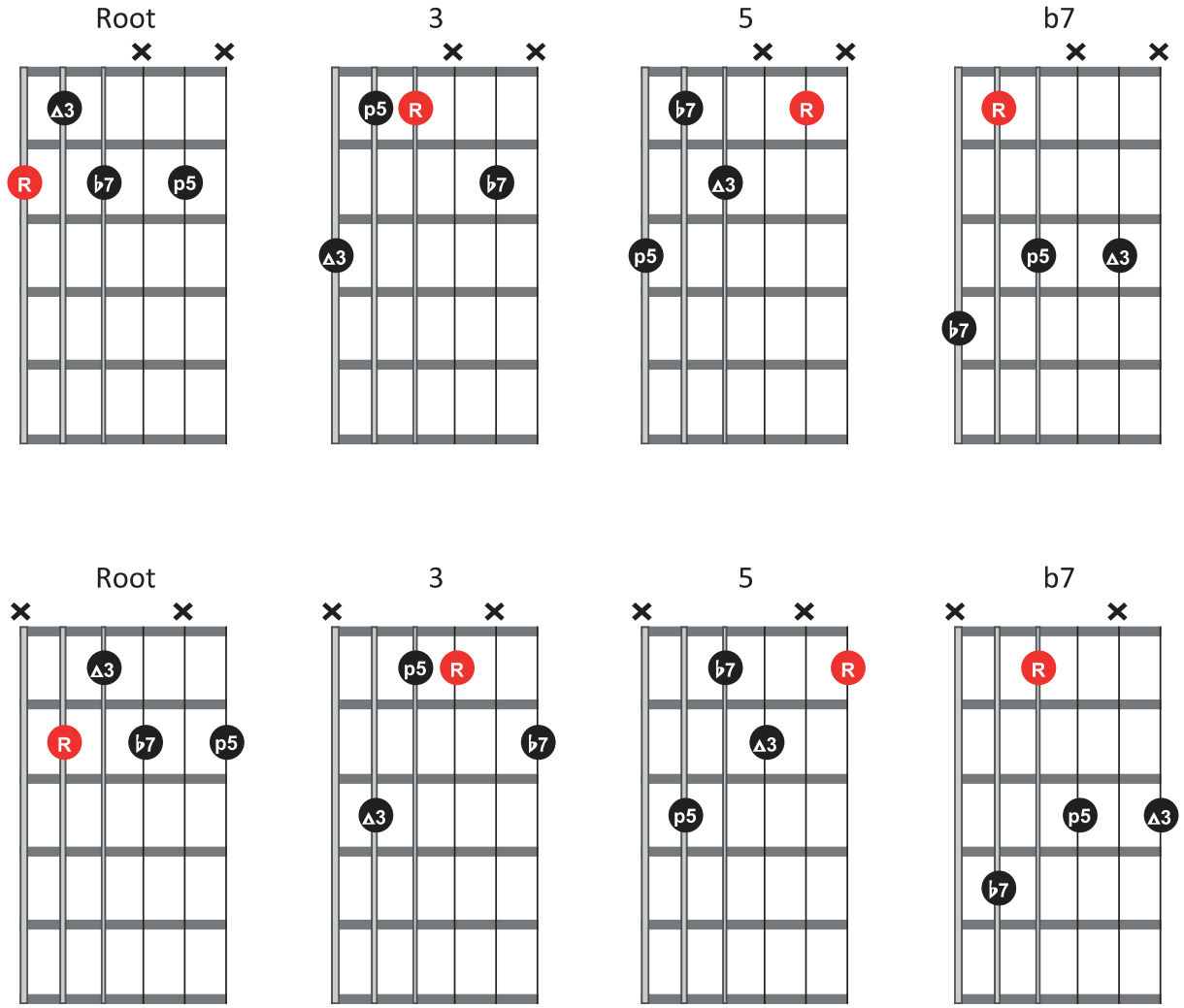
***Maj*^{6}**



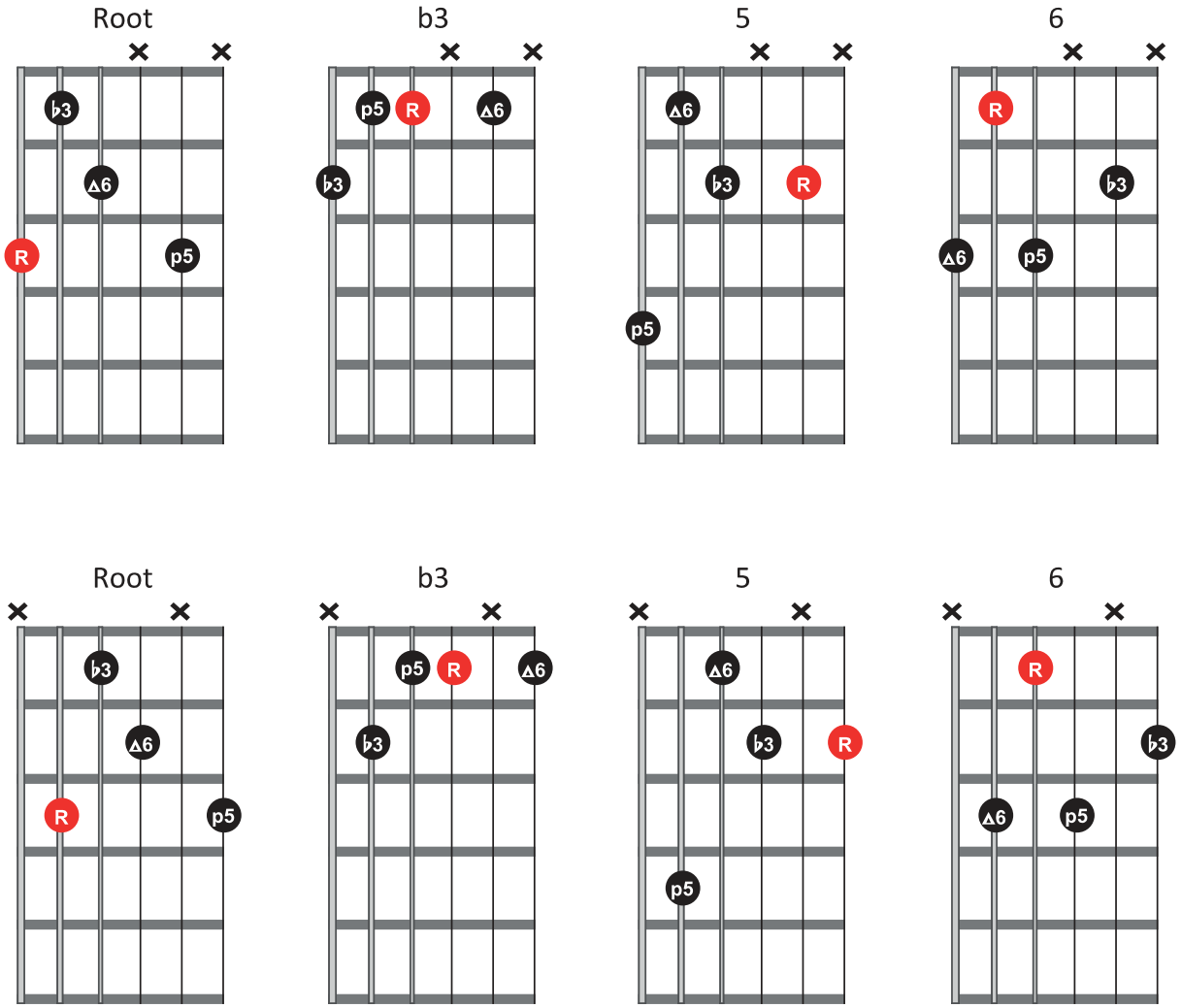
Maj^{7}



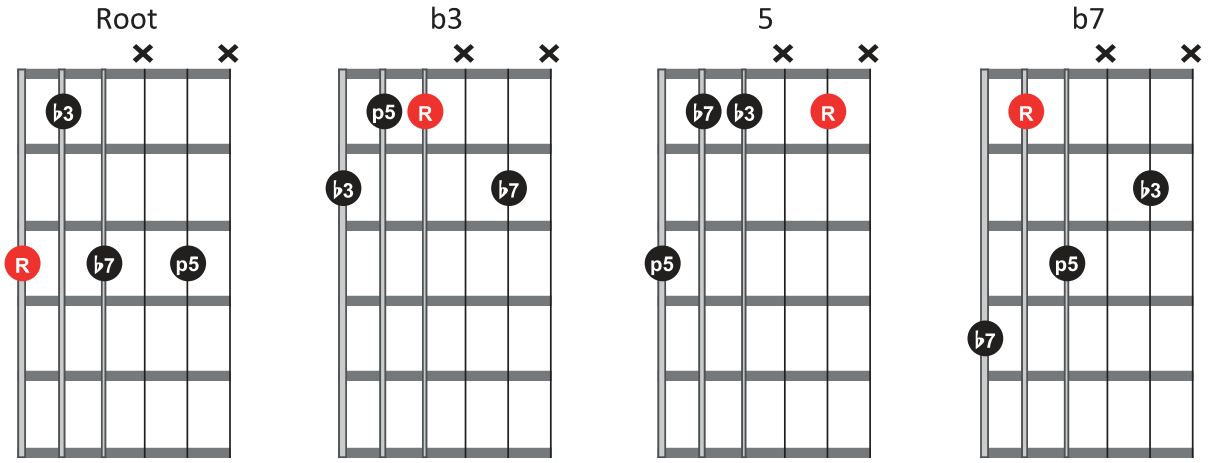
Dom^{7}

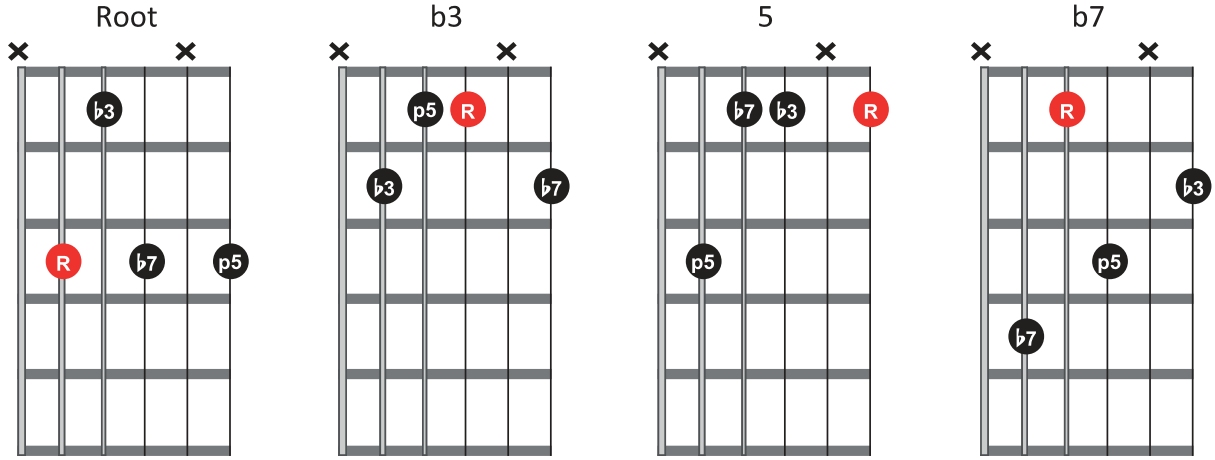


Min^{6}

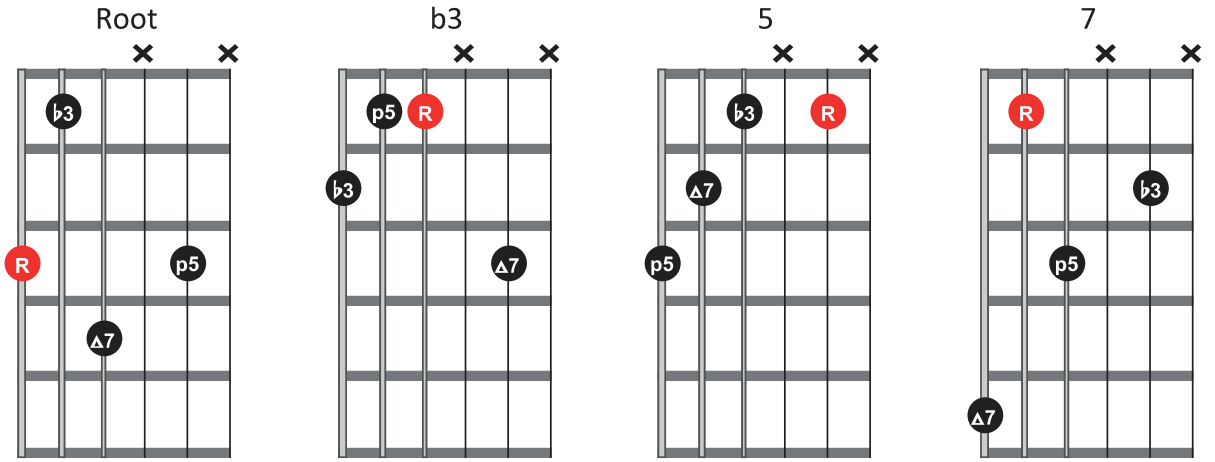


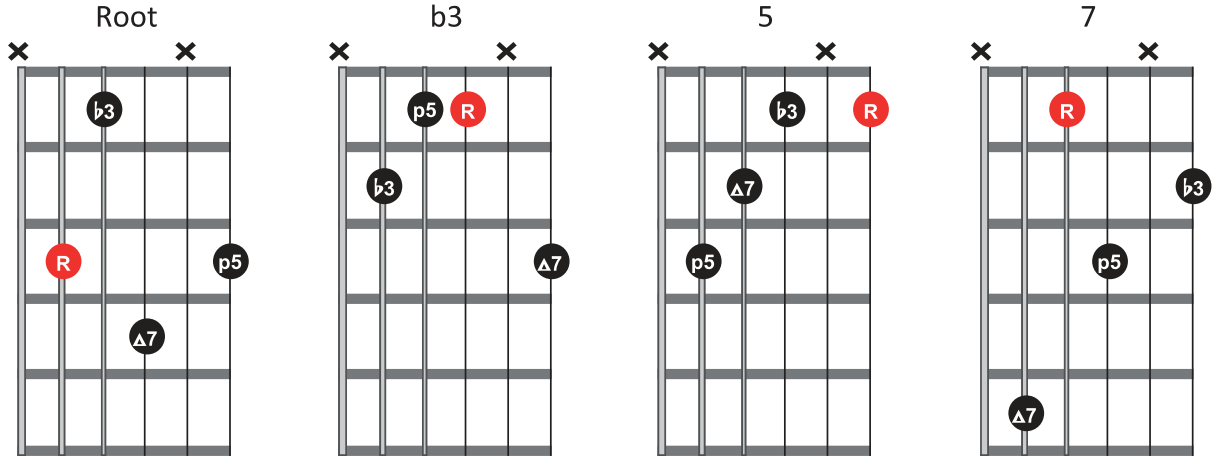
Min^{7}



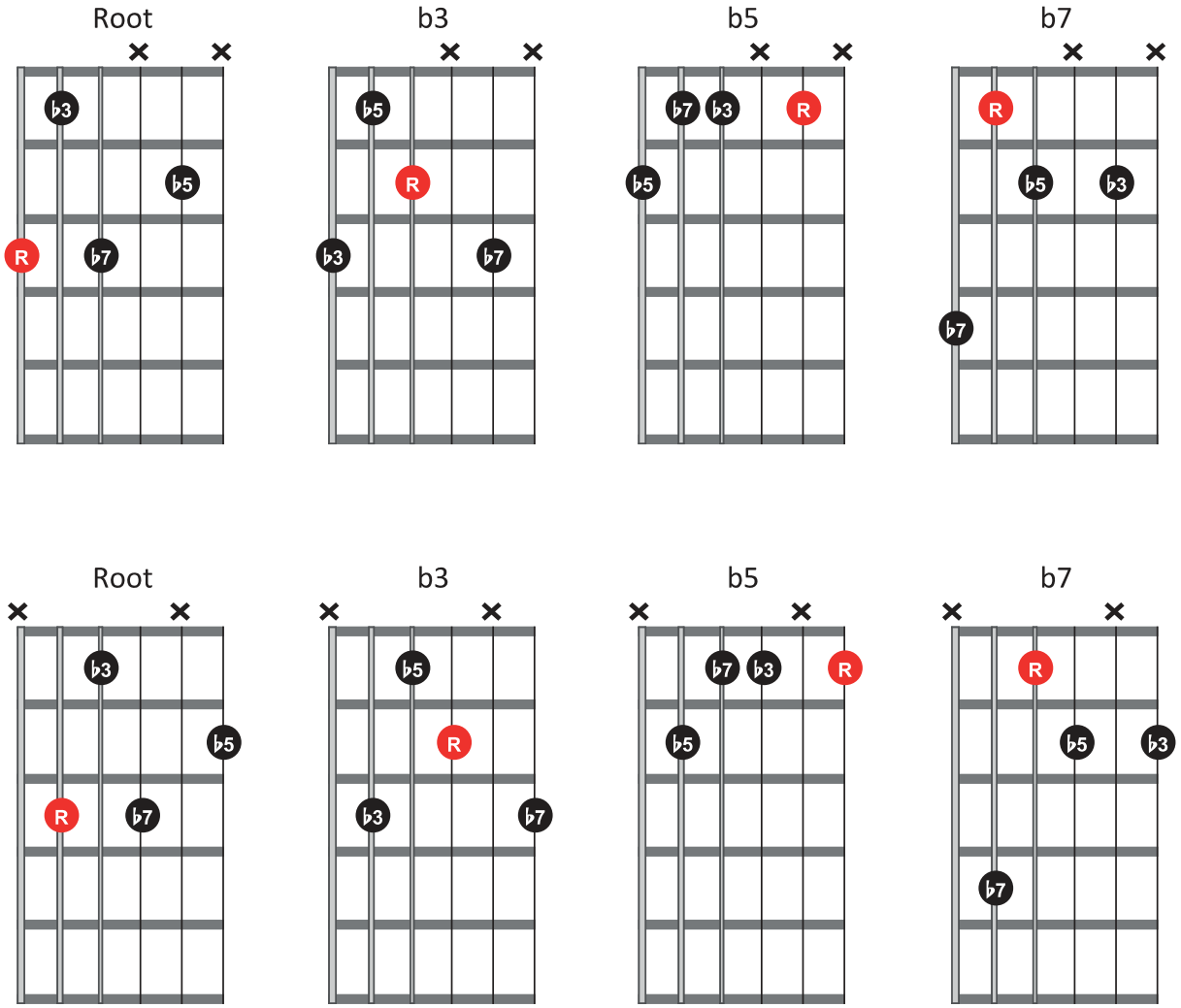


Min^{maj7}

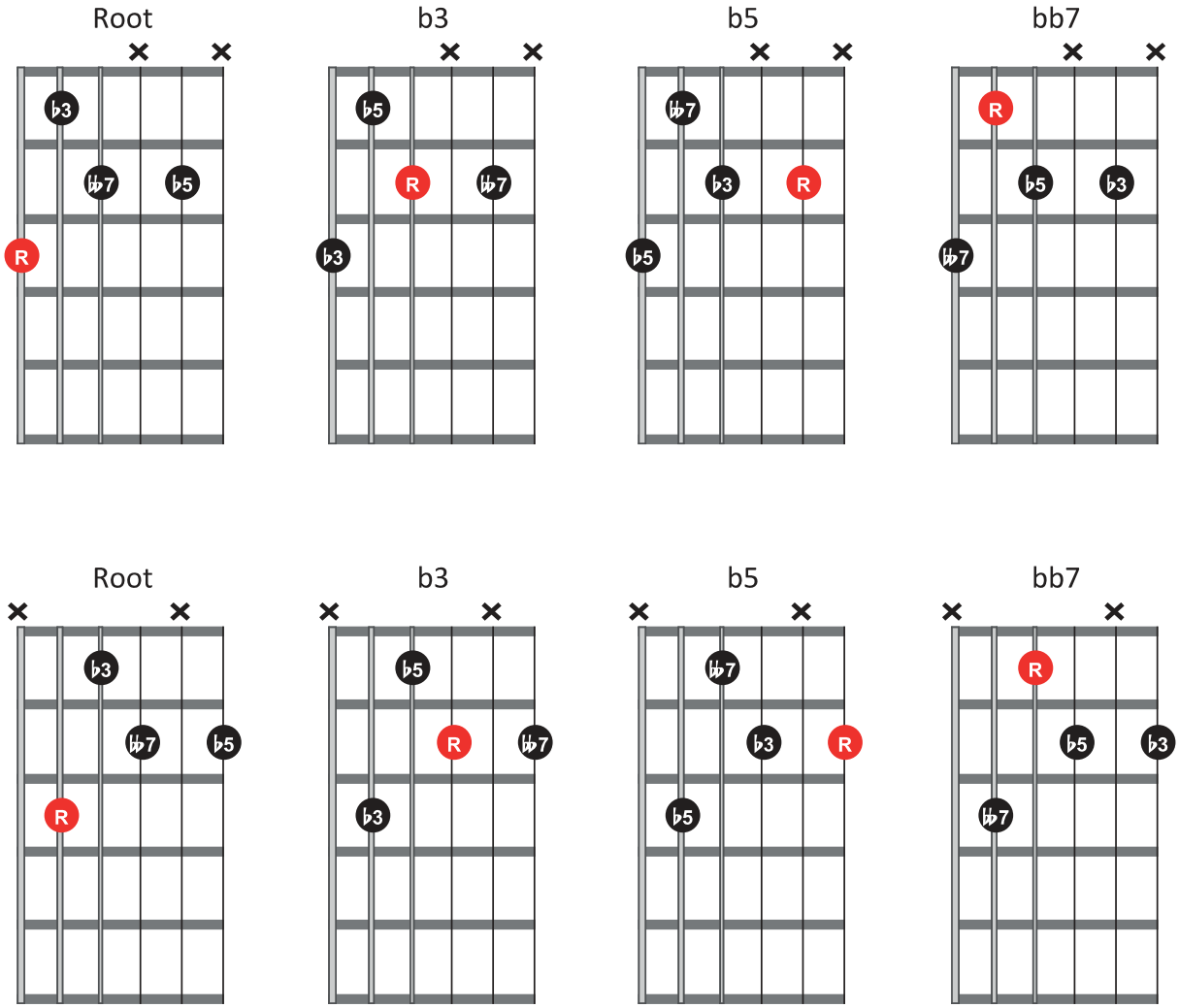




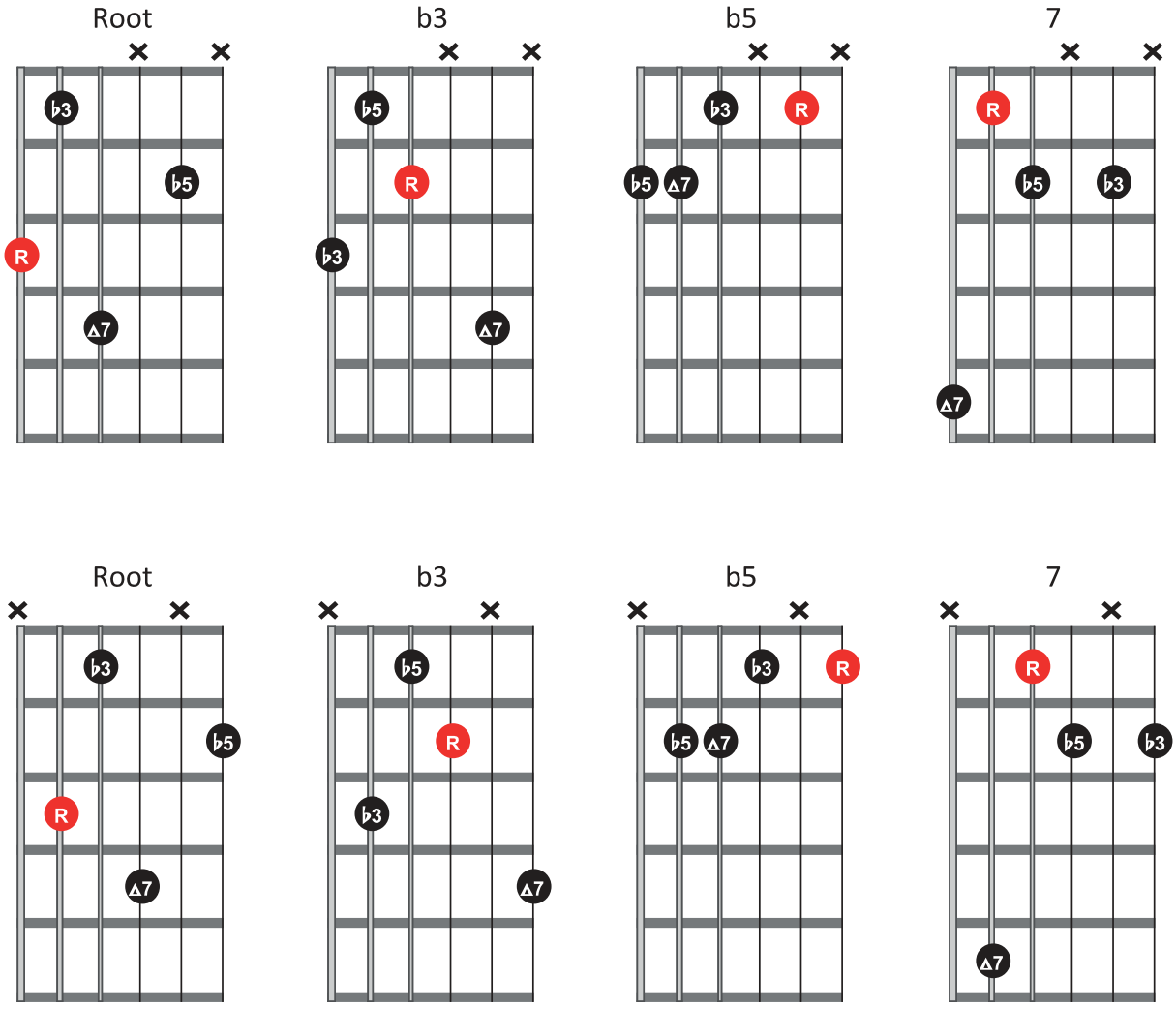
Min^{7b5}



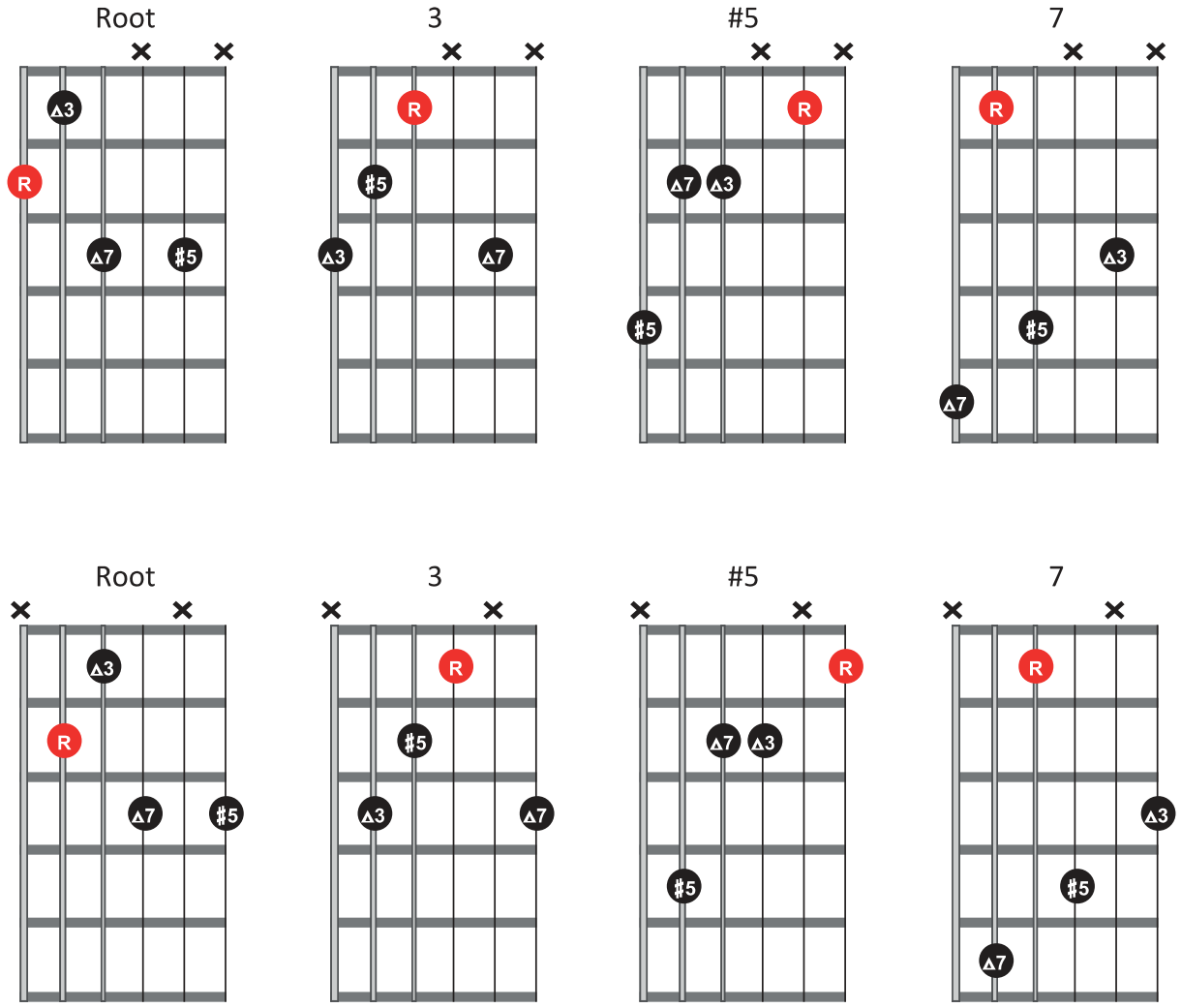
Dim^{7}



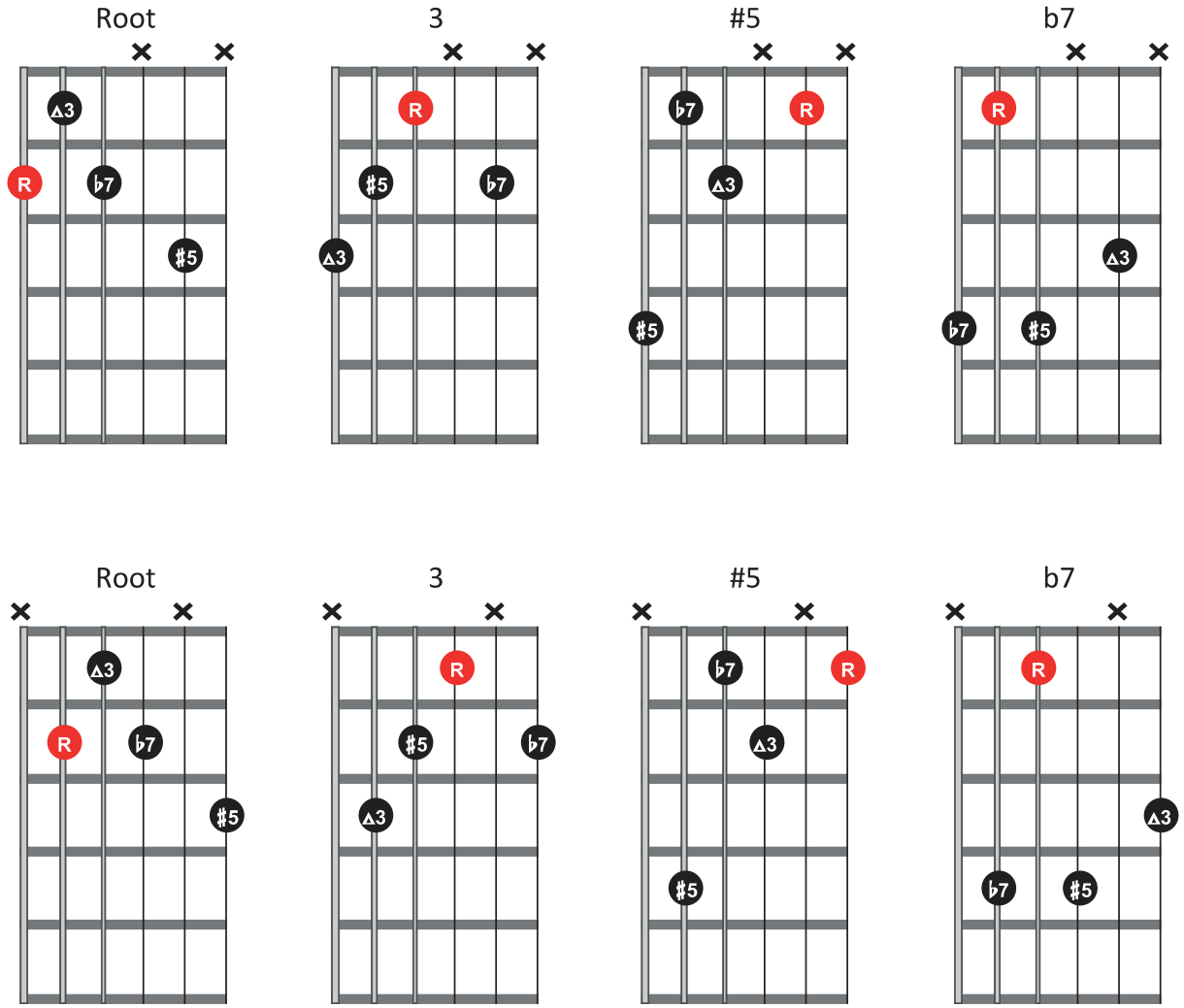
Dim^{maj7}



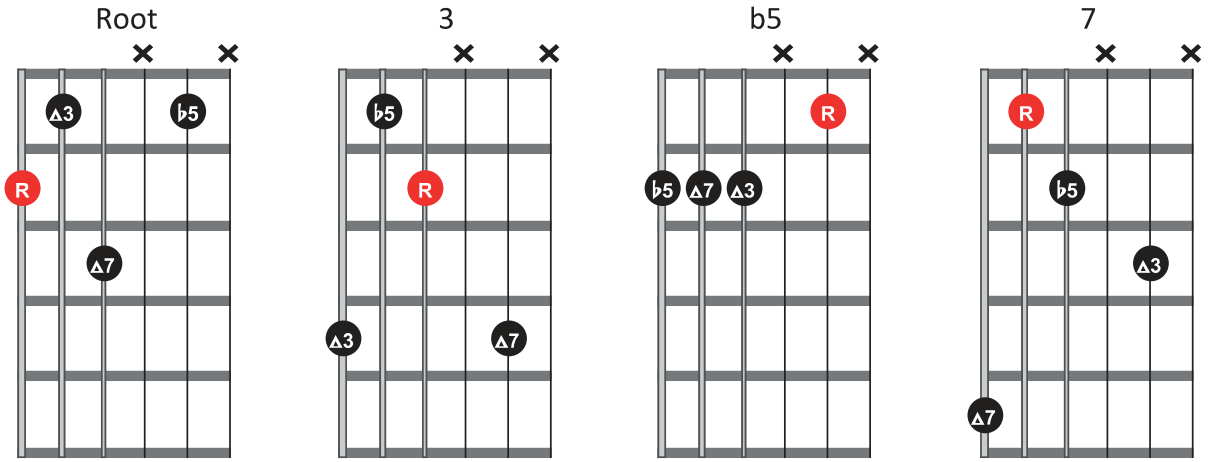
Aug^{maj7}

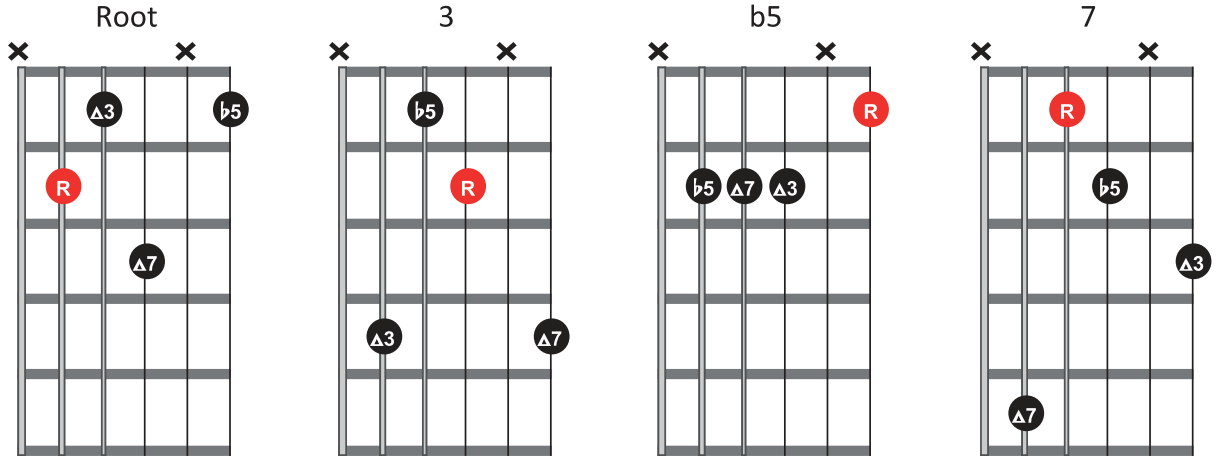


Aug^{7}

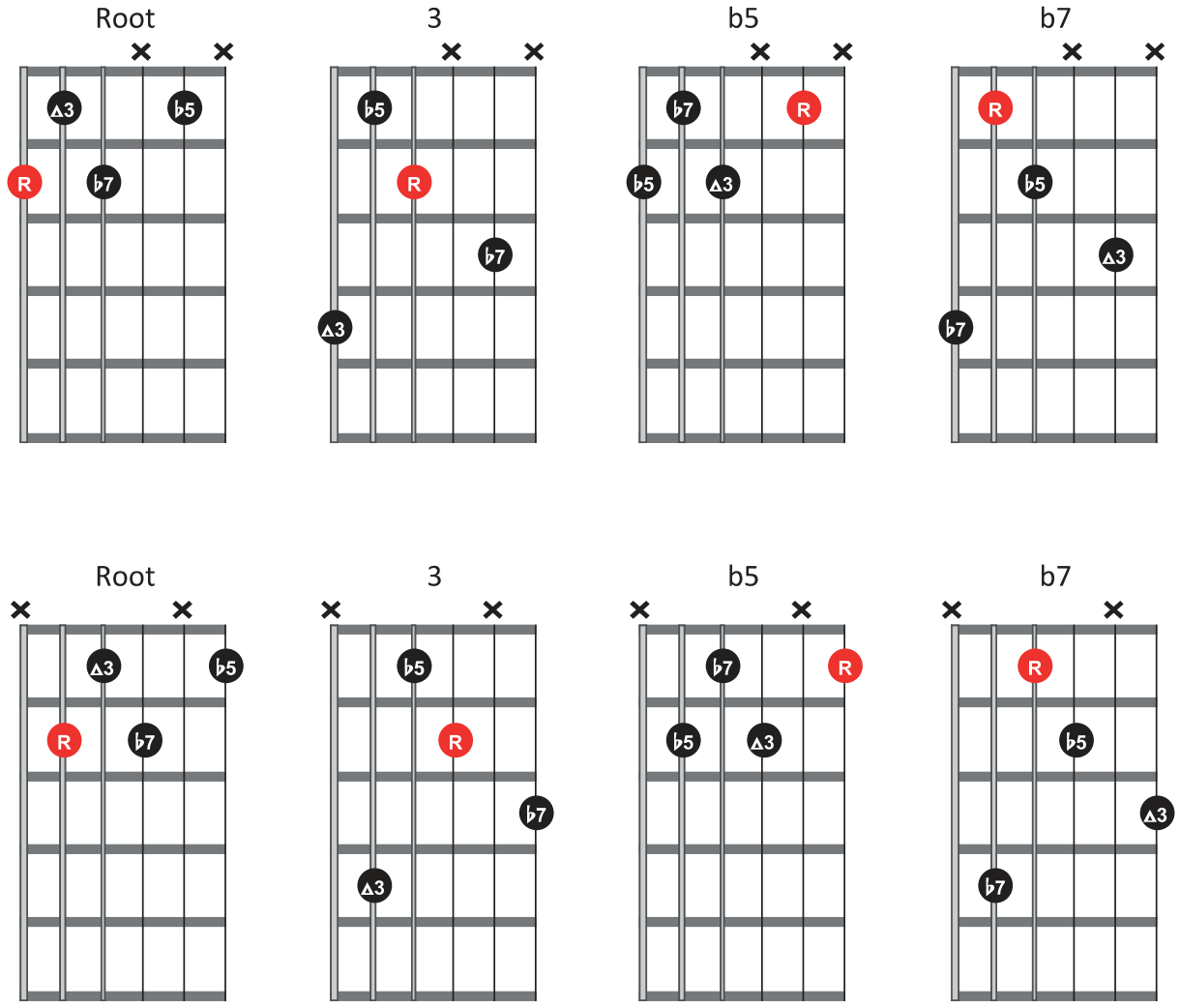


Maj^{7b5}

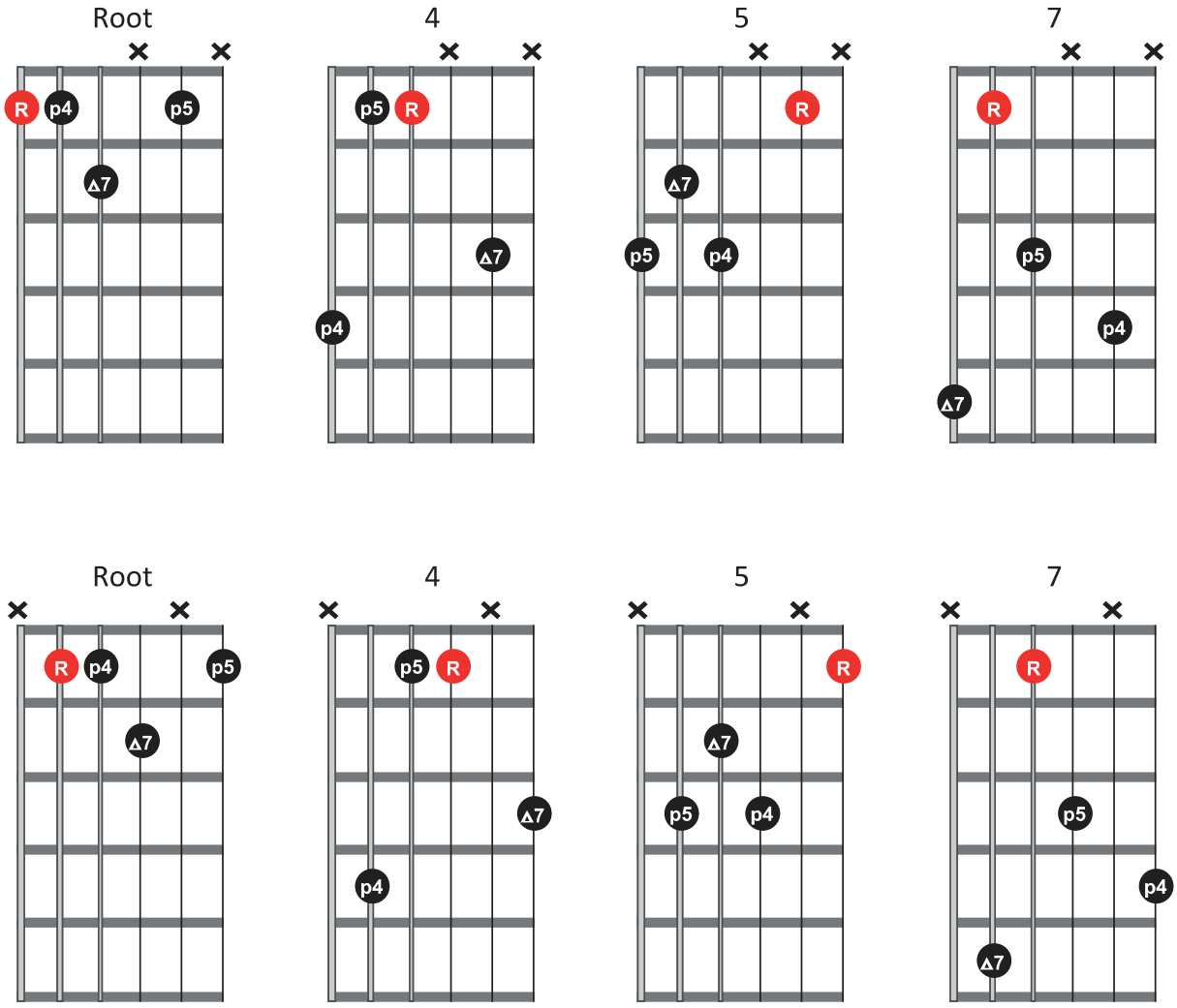




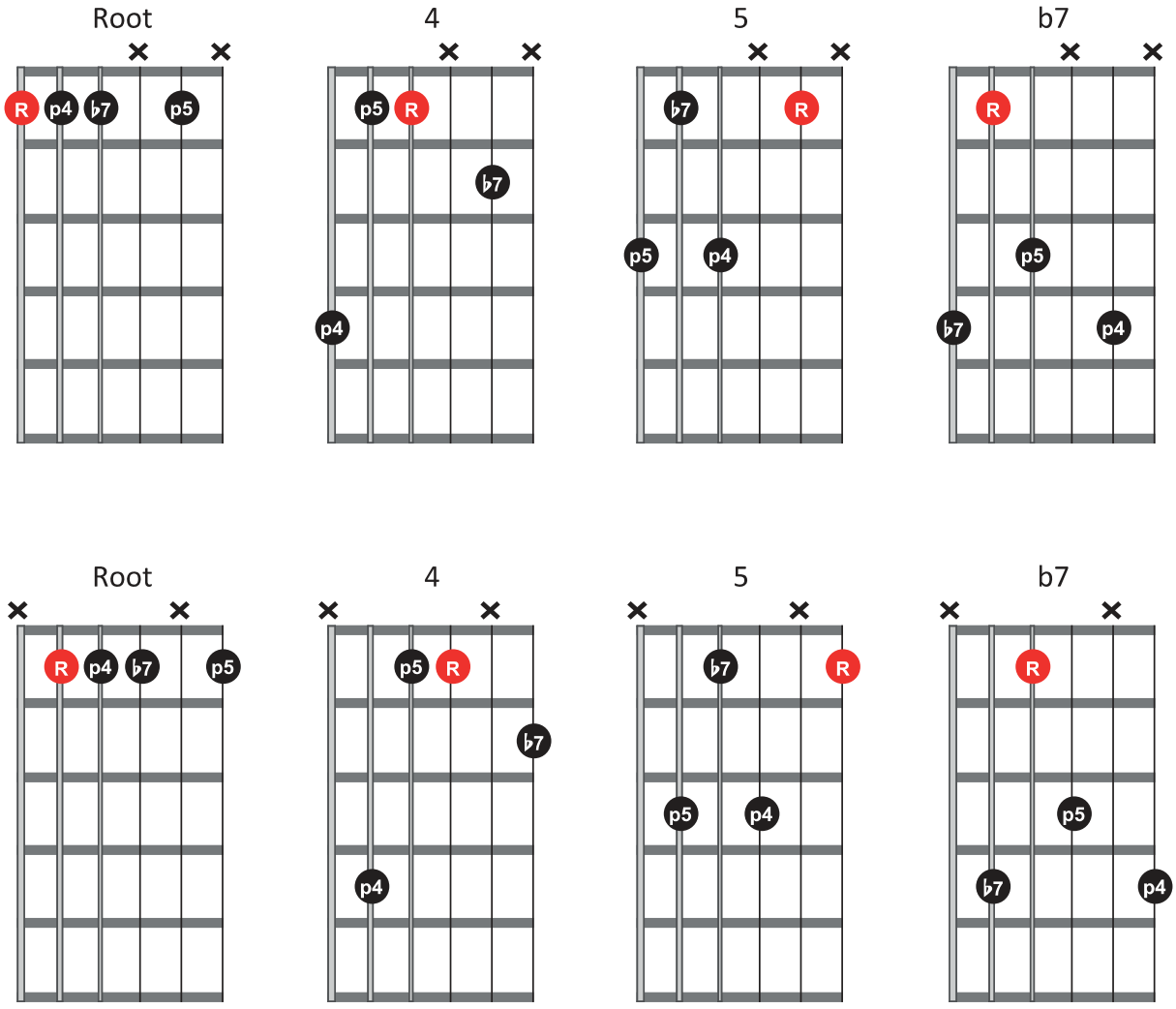
Dom^{7b5}



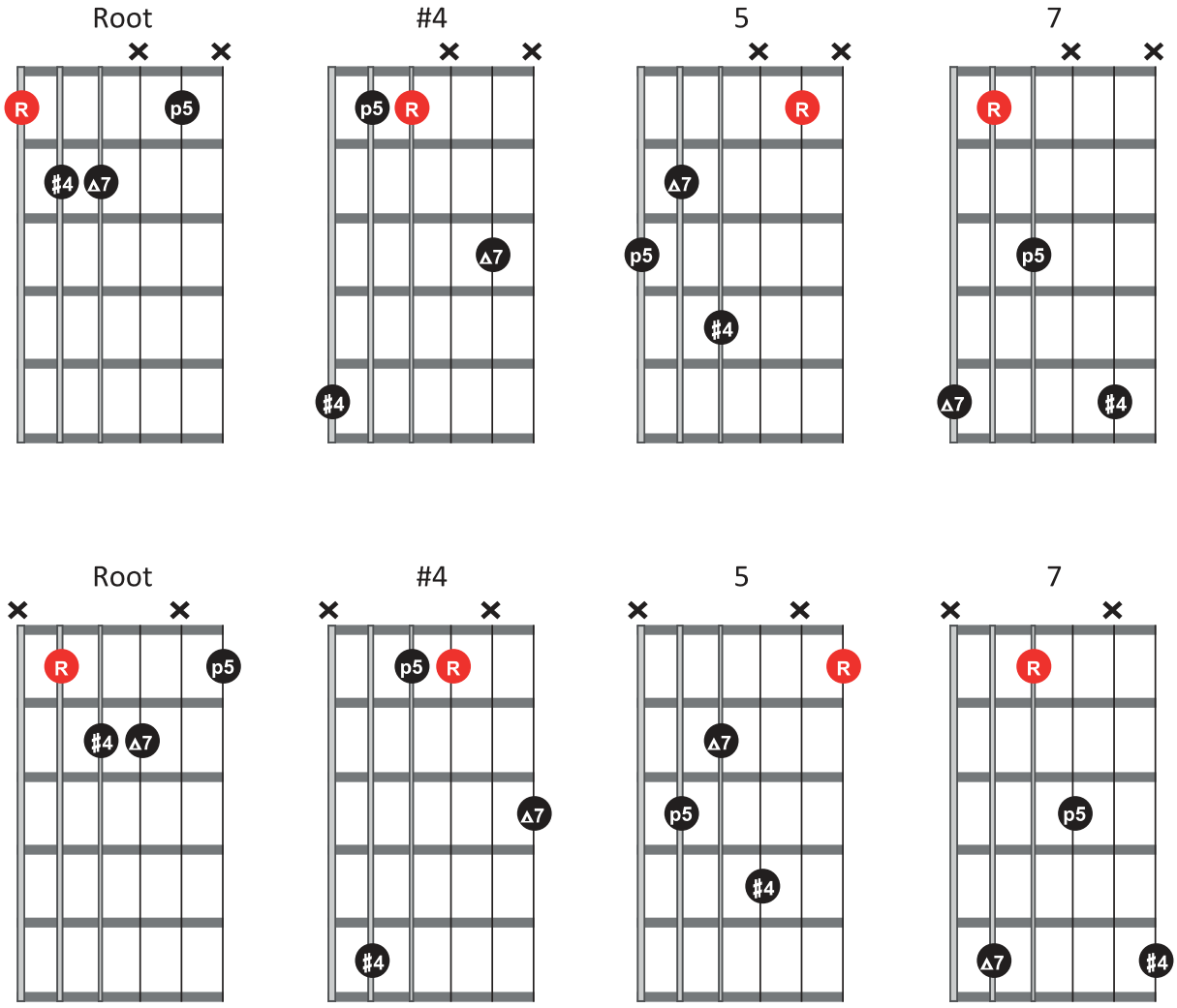
Maj^{7sus4}



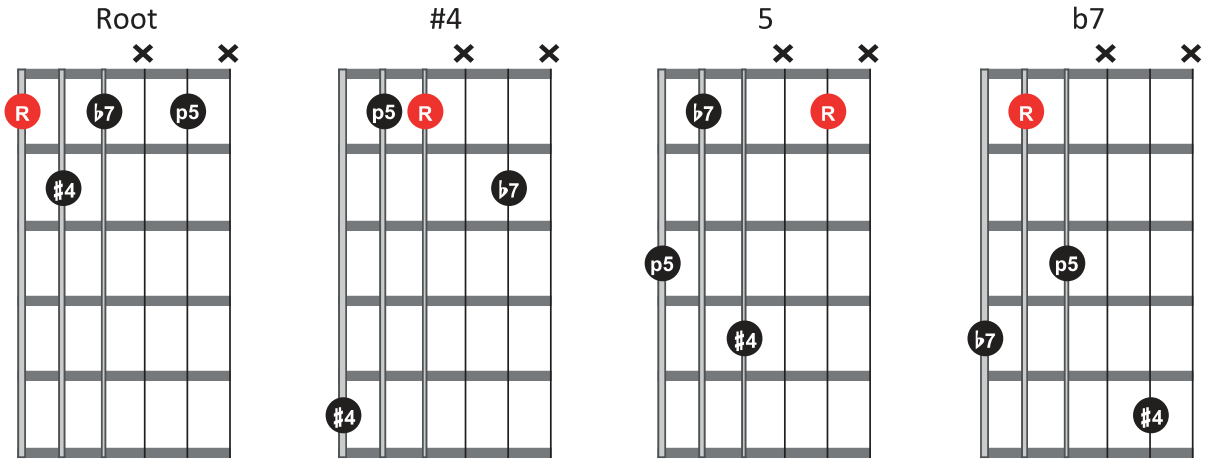
Dom^{7sus4}

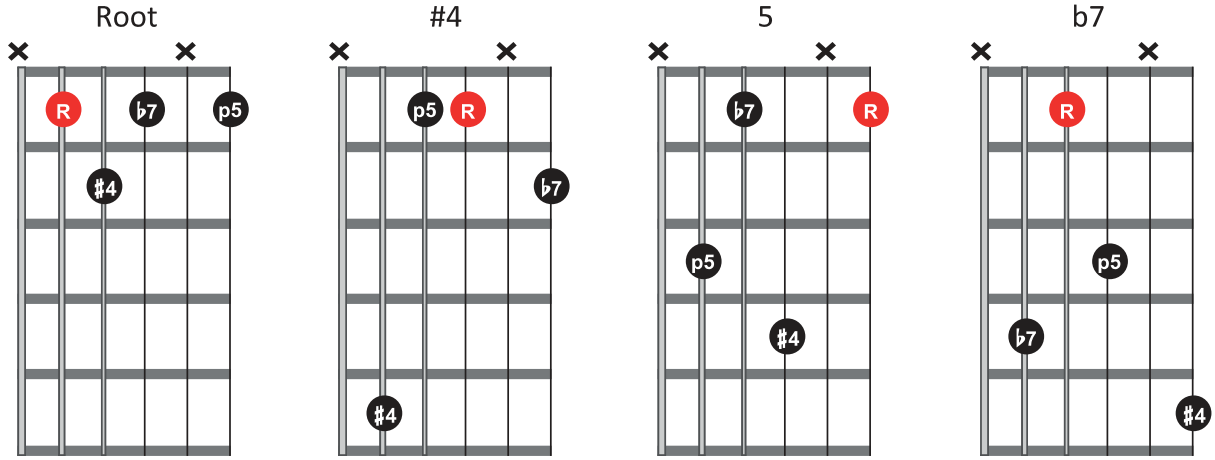


Lydian^{maj7}

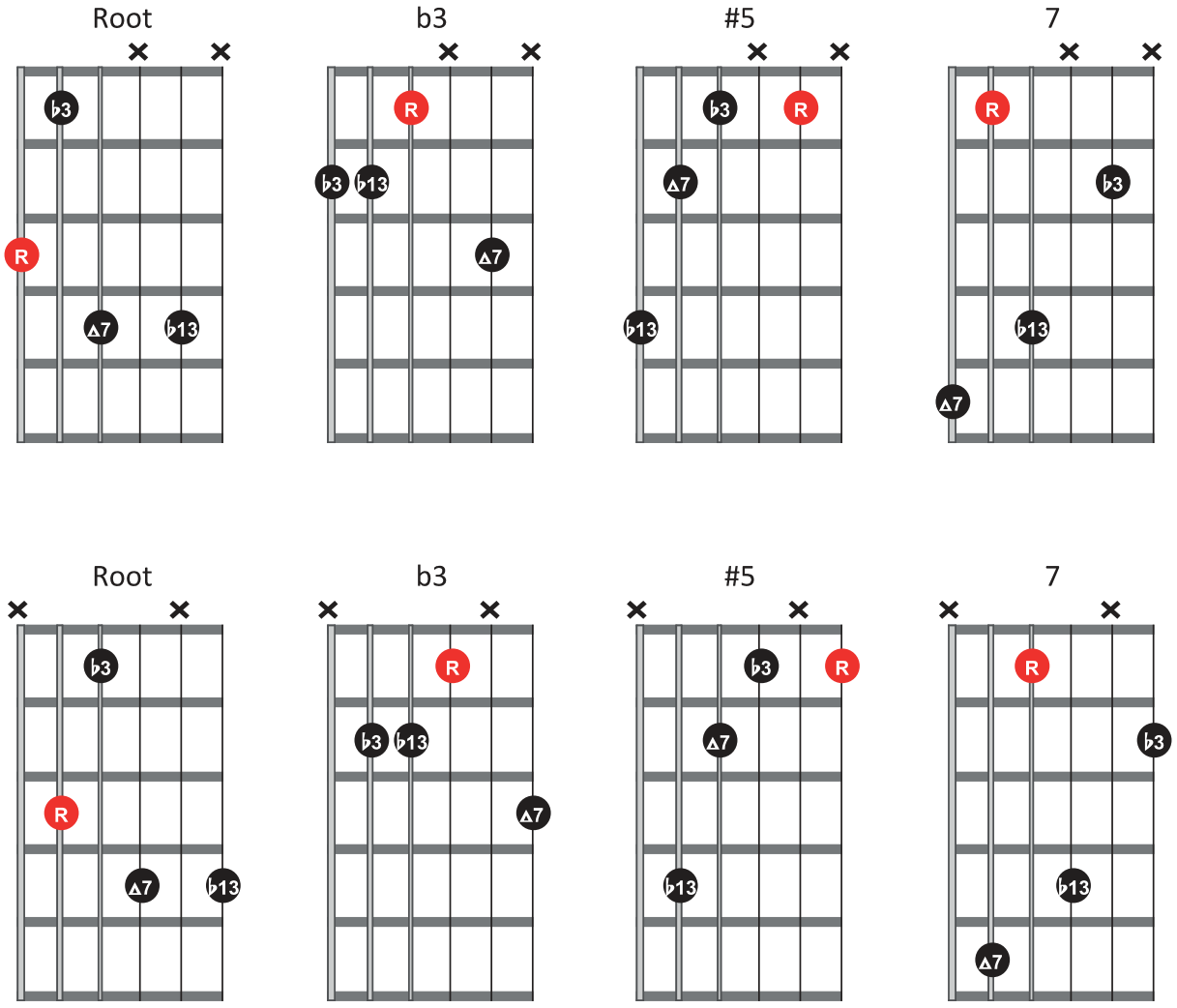


Lydian^{Dom7}



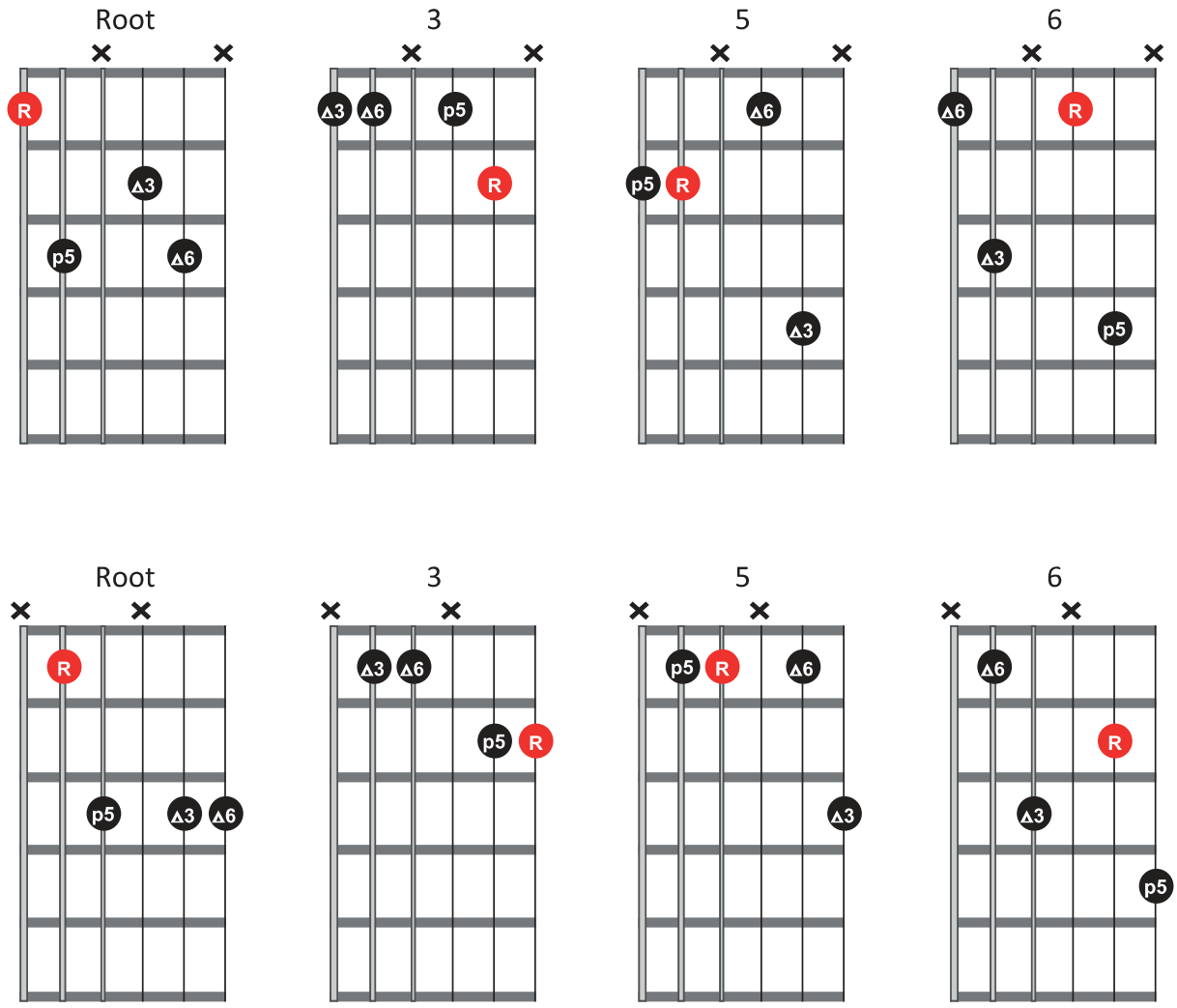


Dim^{maj7b13}

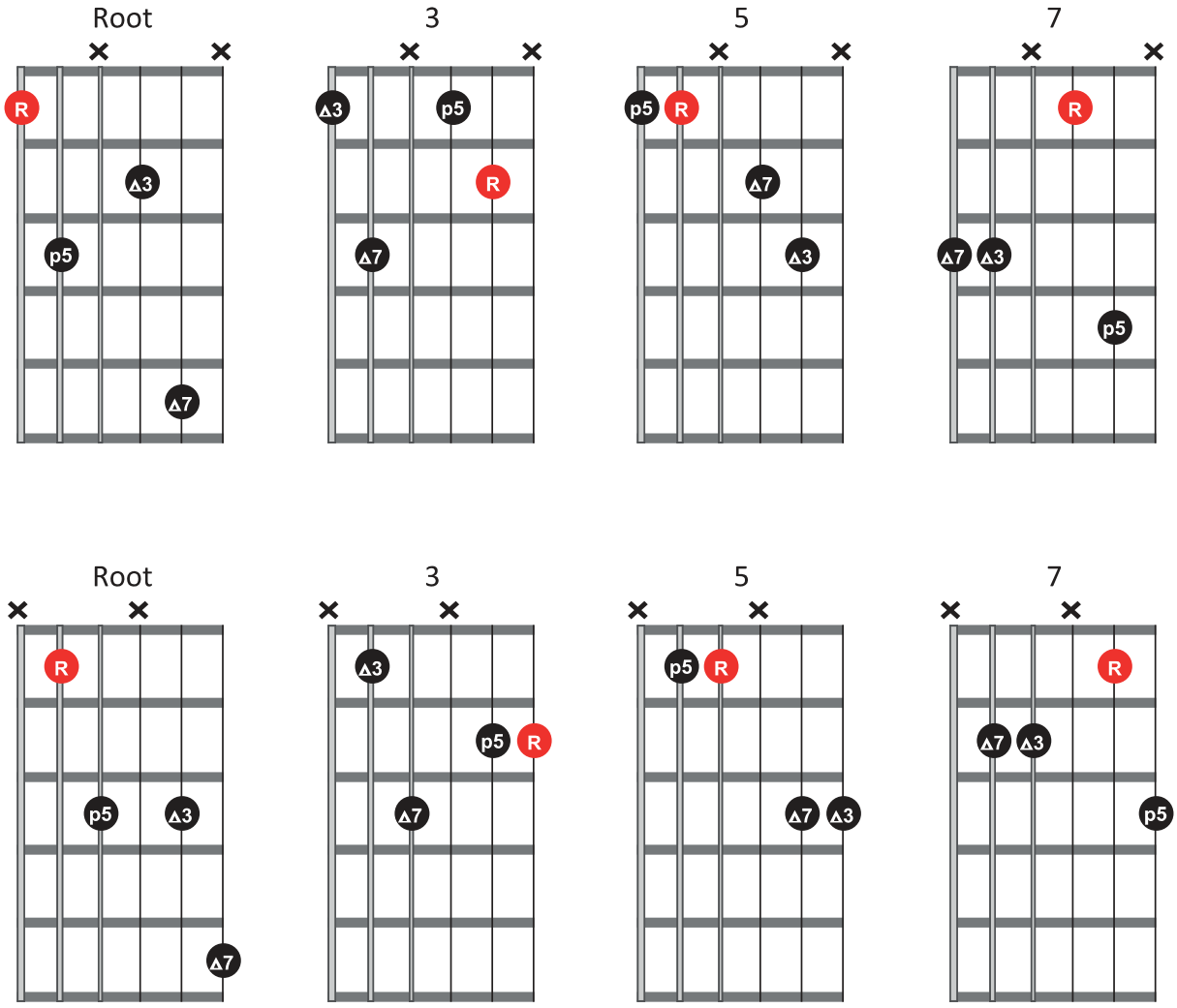


Drop 2 + 4

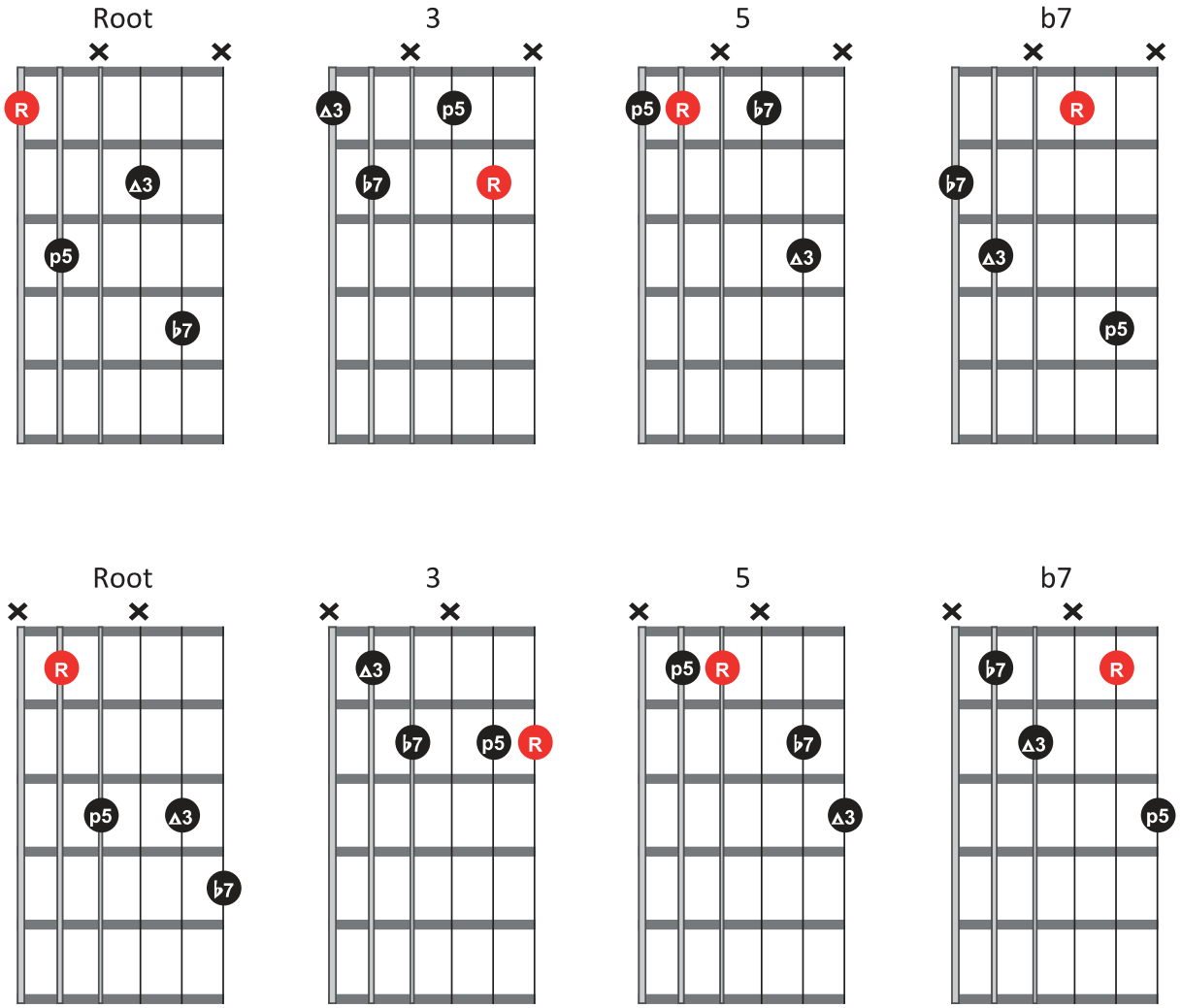
***Maj*^{6}**



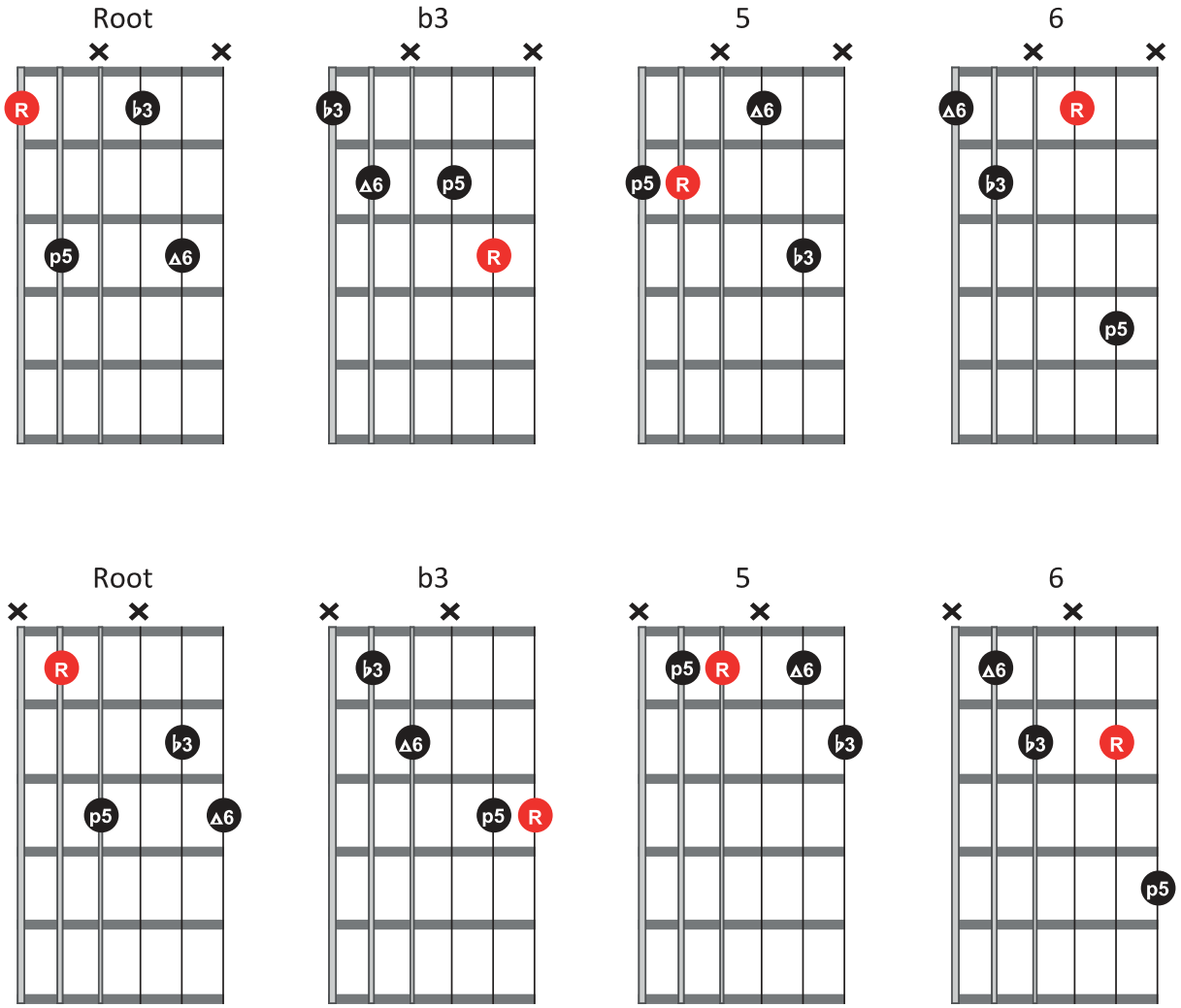
Maj^{7}



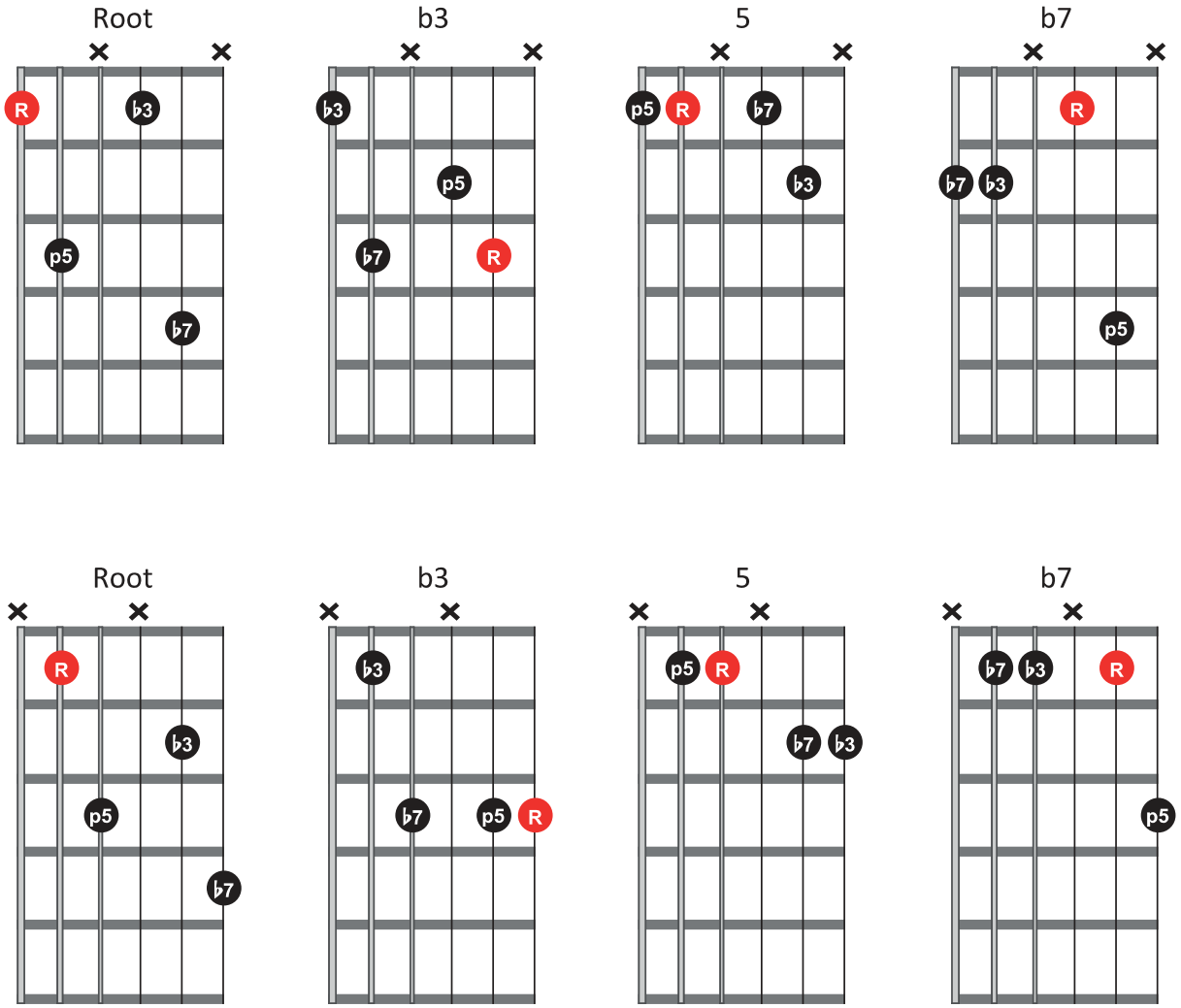
Dom^{7}



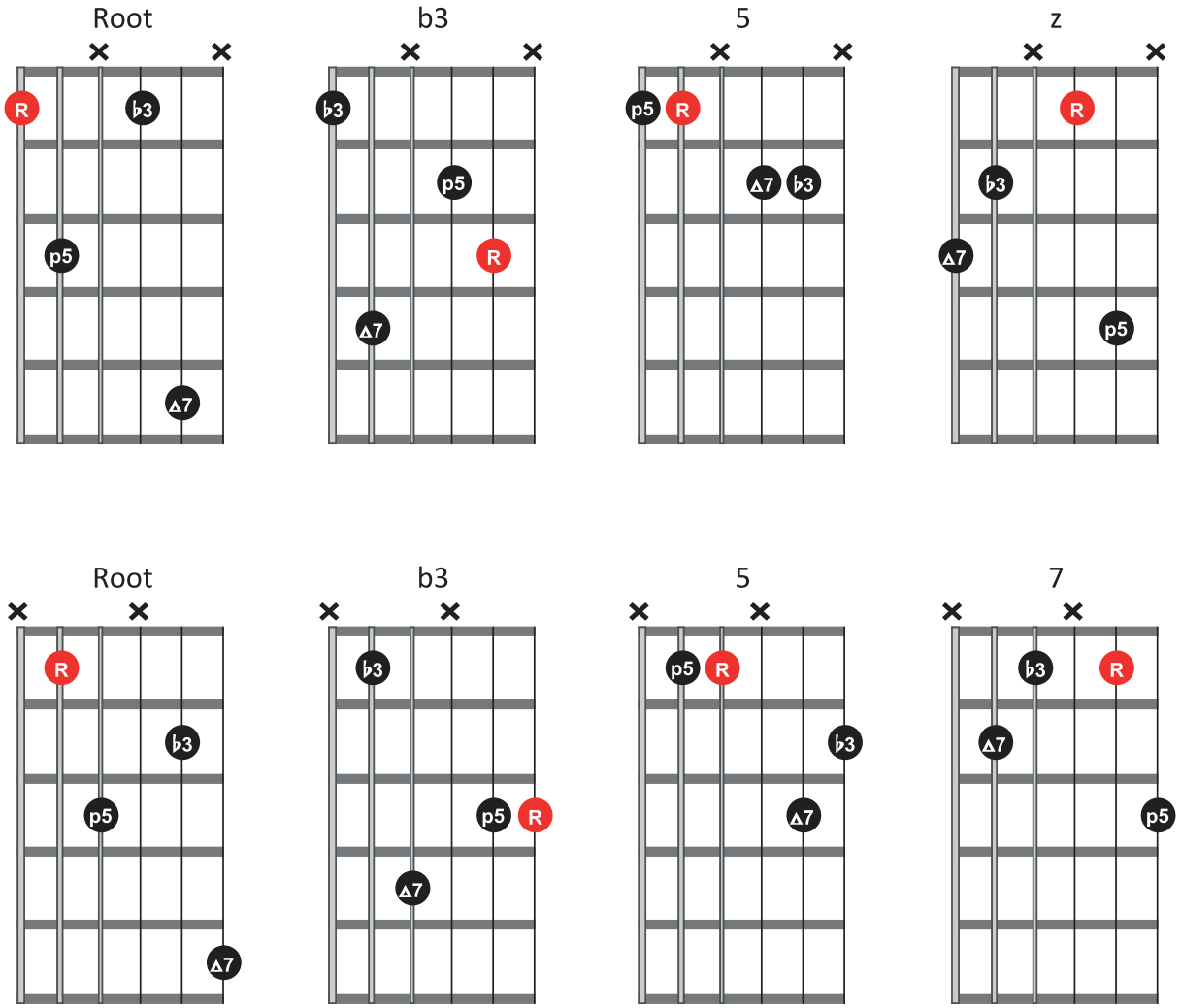
Min^{6}



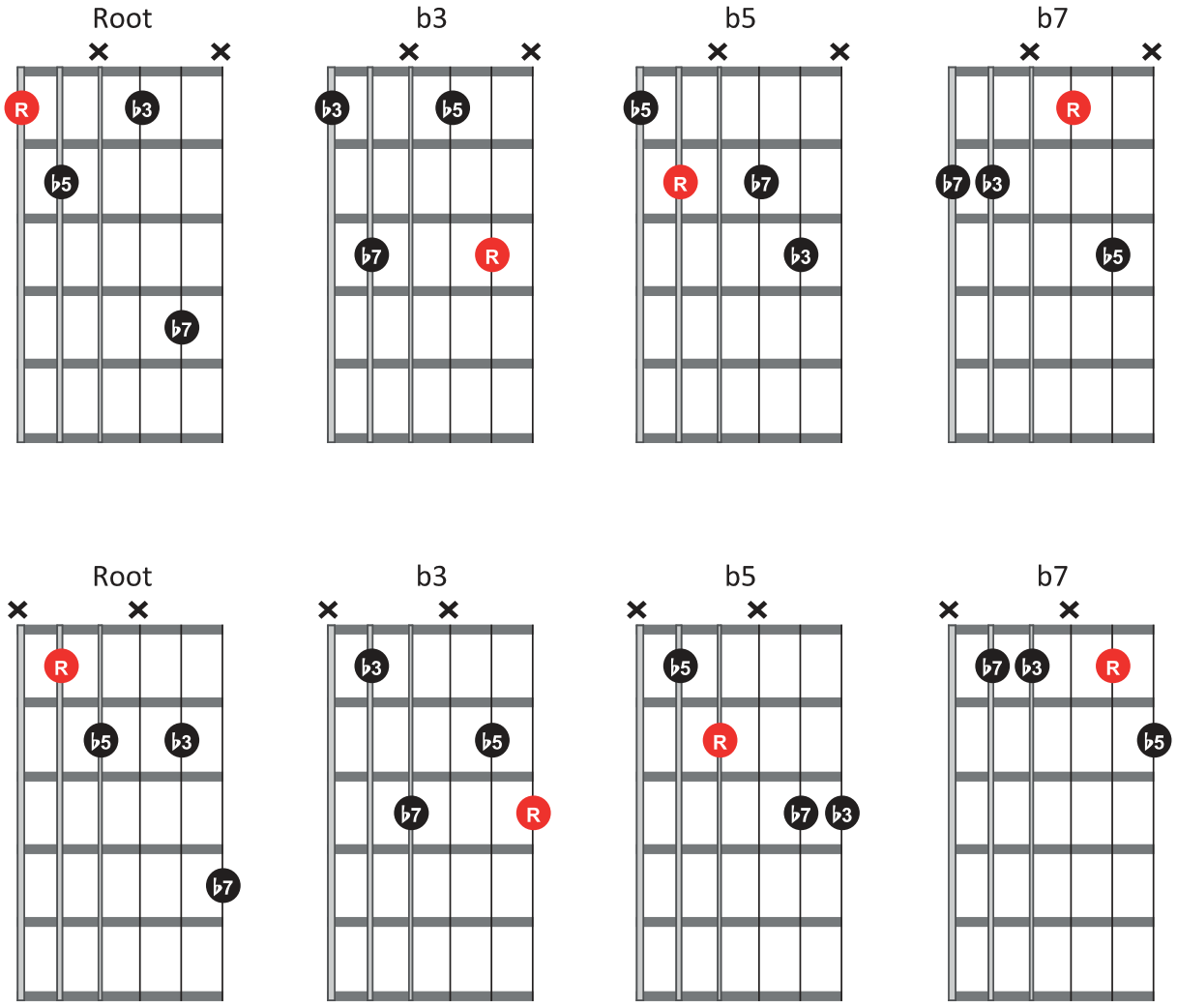
Min^{7}



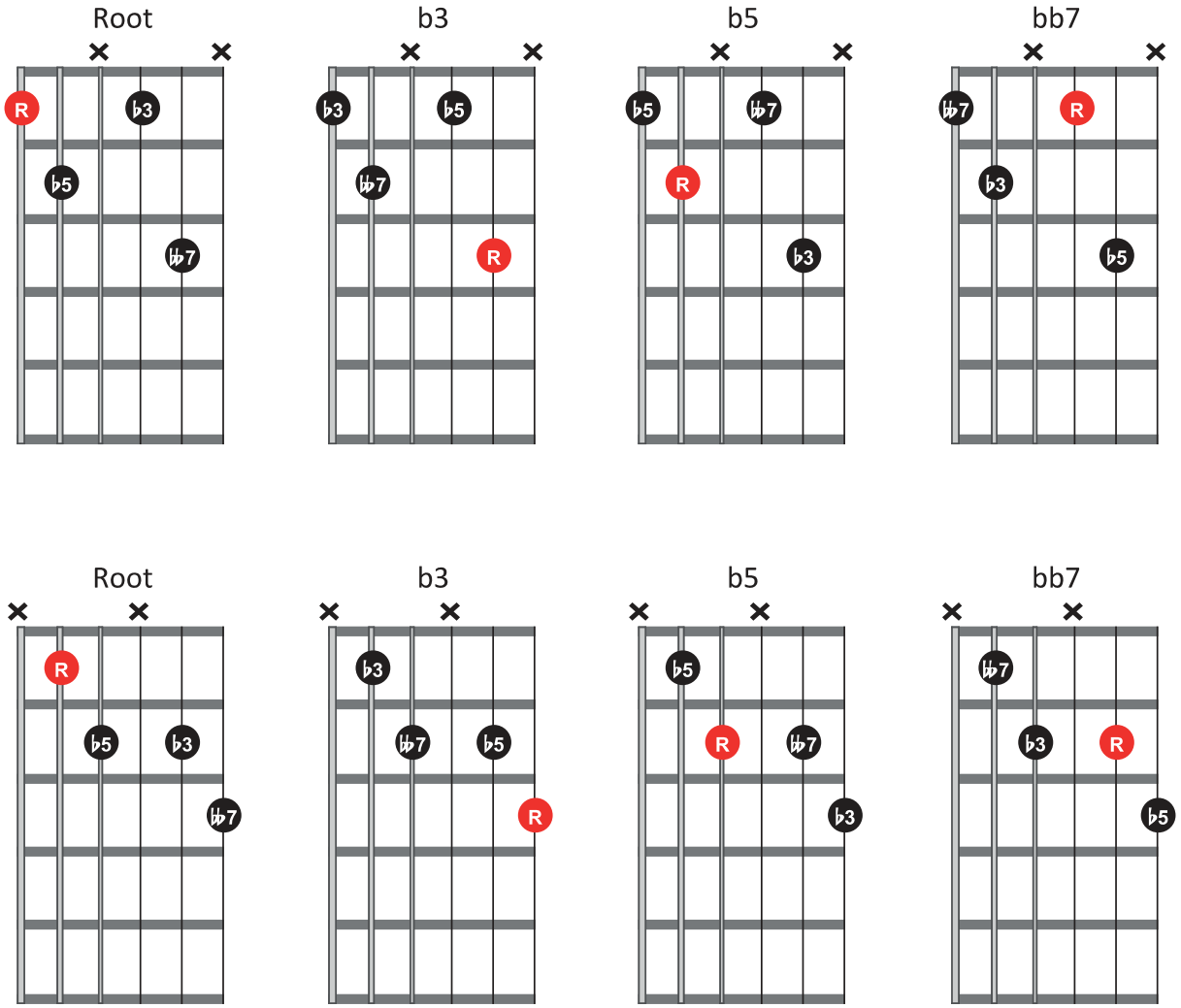
Min^{maj7}


Root
R
♭3
p5
Δ7
b3
♭3
p5
R
Δ7
5
p5
R
Δ7
♭3
z
R
♭3
Δ7
p5
Root
R
♭3
p5
Δ7
b3
♭3
p5
R
Δ7
5
p5
R
♭3
Δ7
7
♭3
R
Δ7
p5

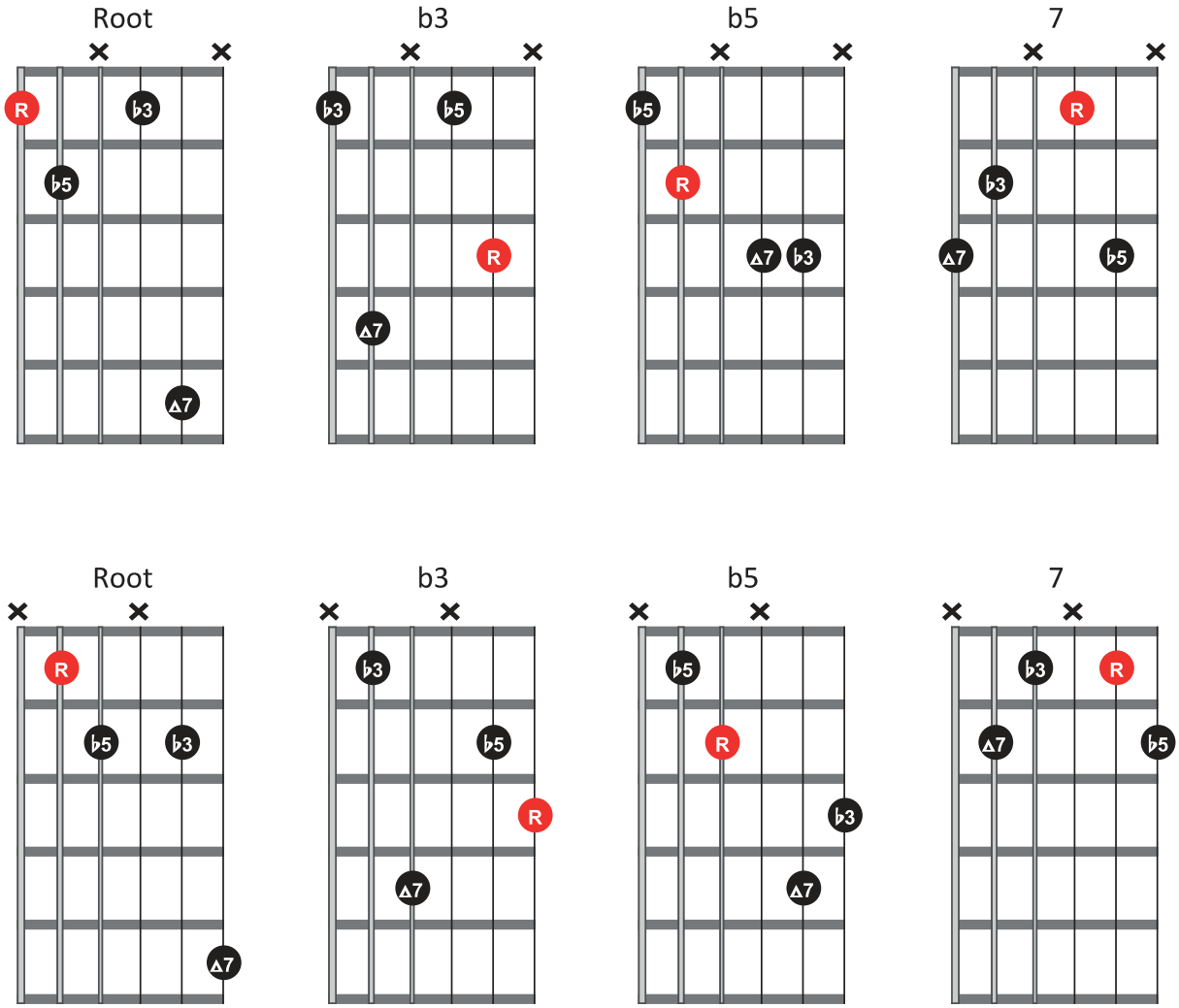
Min^{7b5}



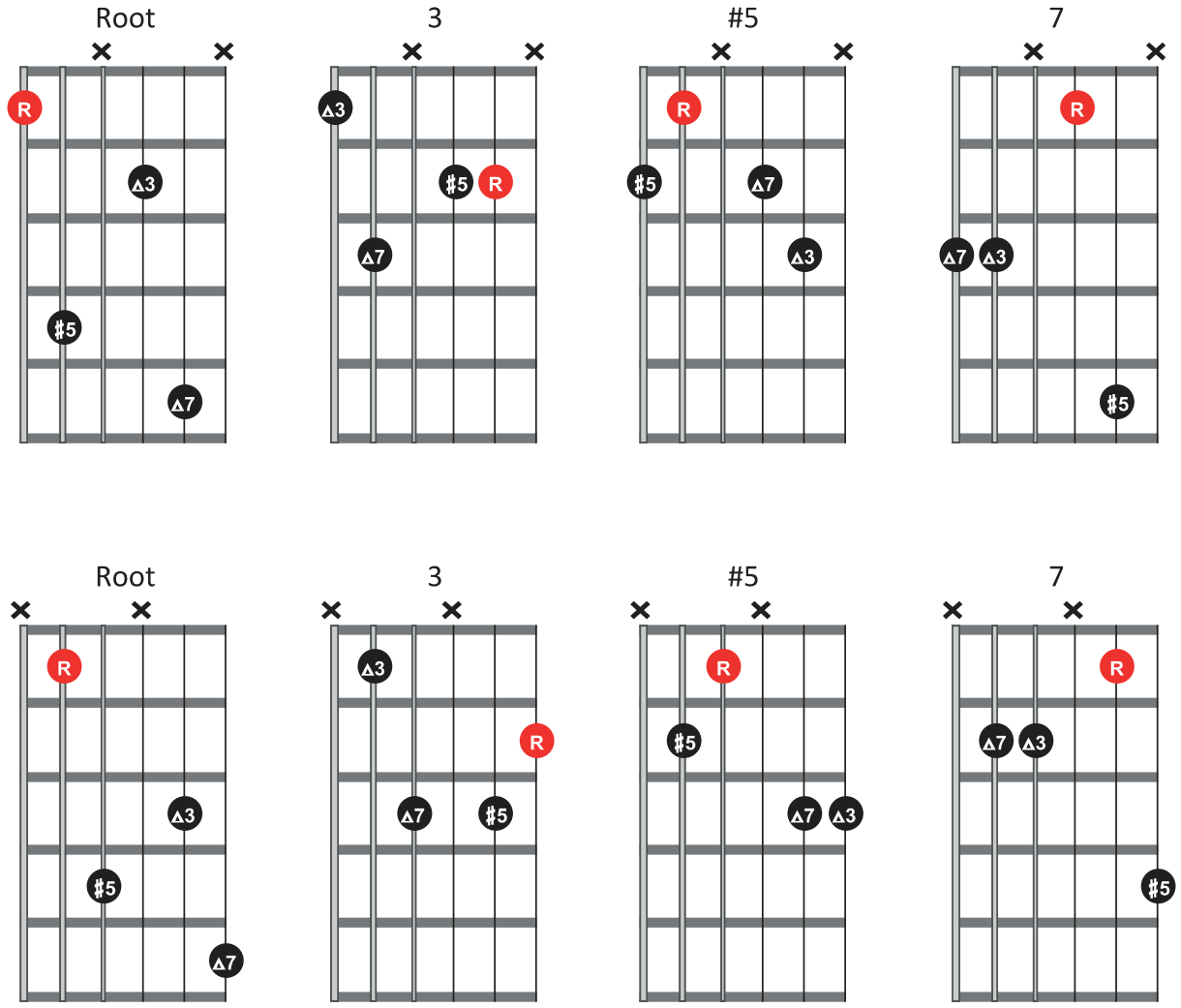
Dim^{7}



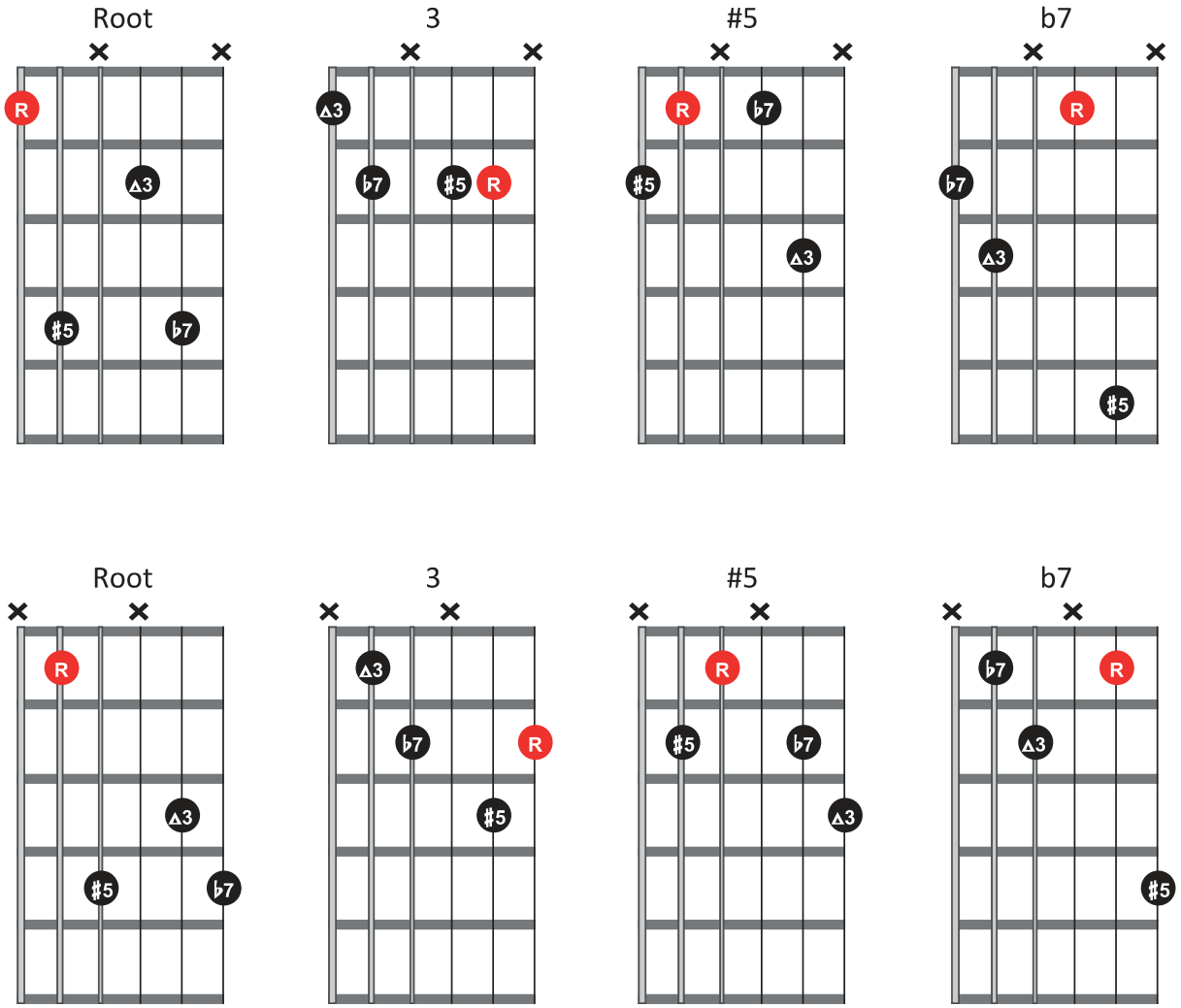
Dim^{maj7}



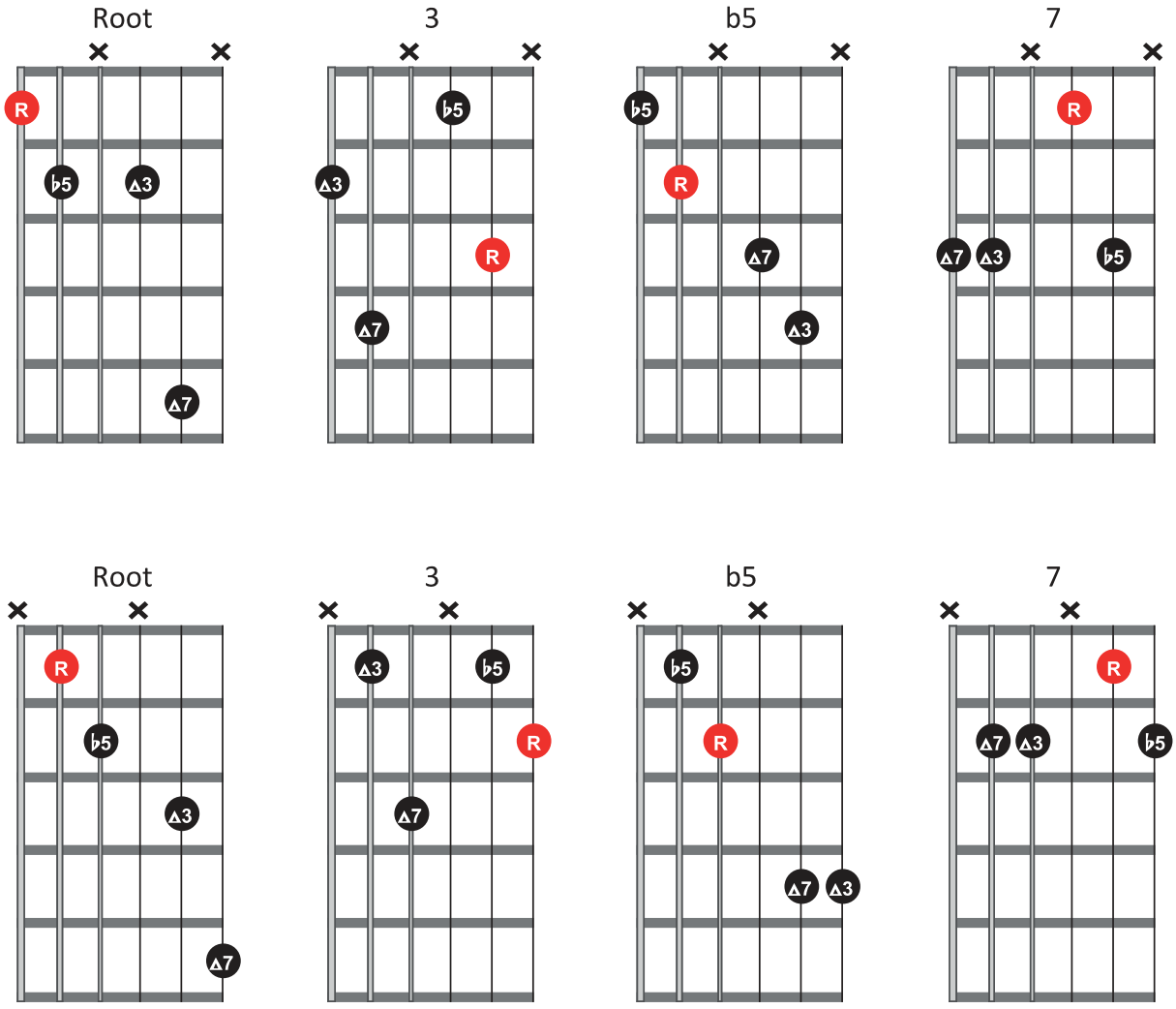
Aug^{maj7}



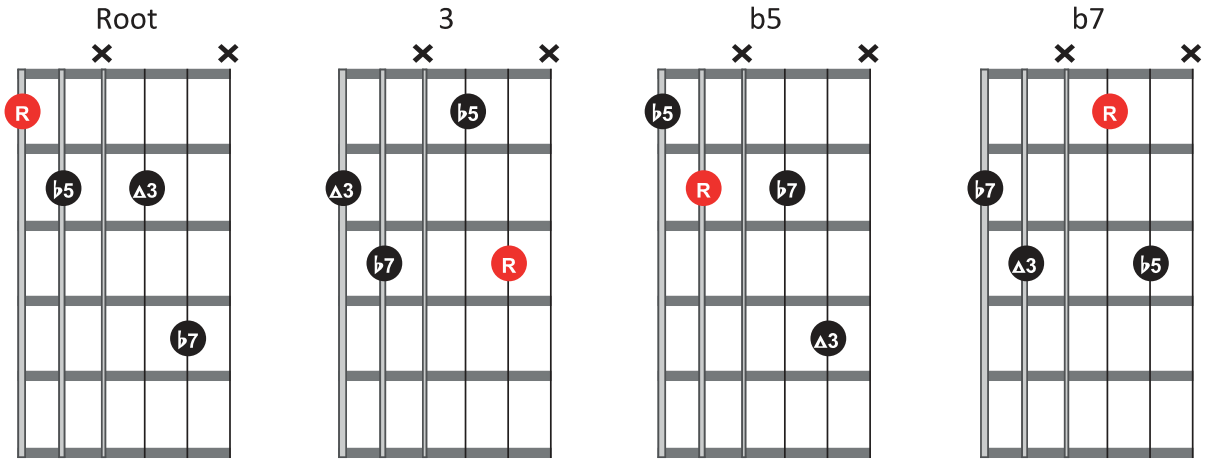
Aug^{7}



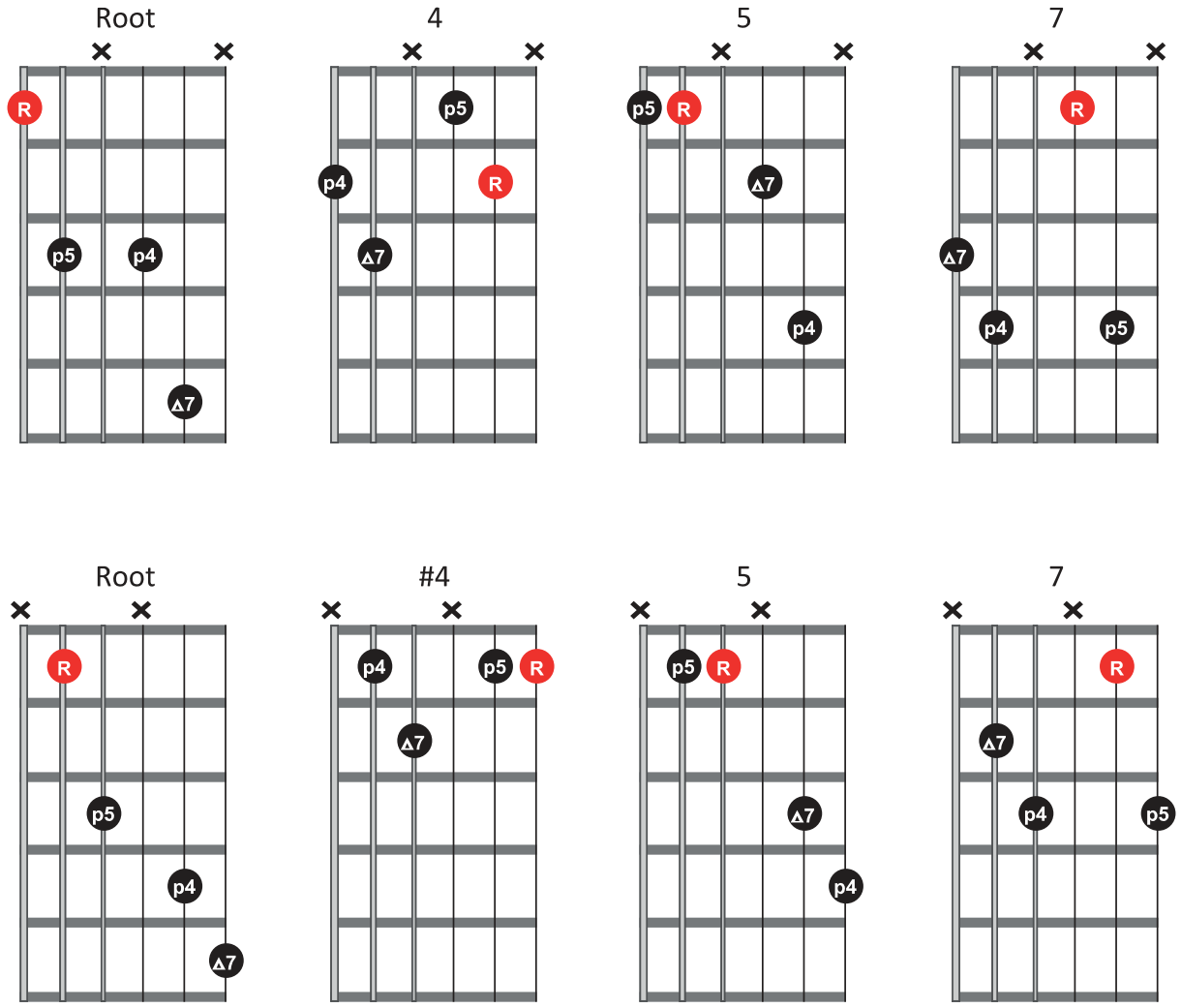
Maj^{7b5}



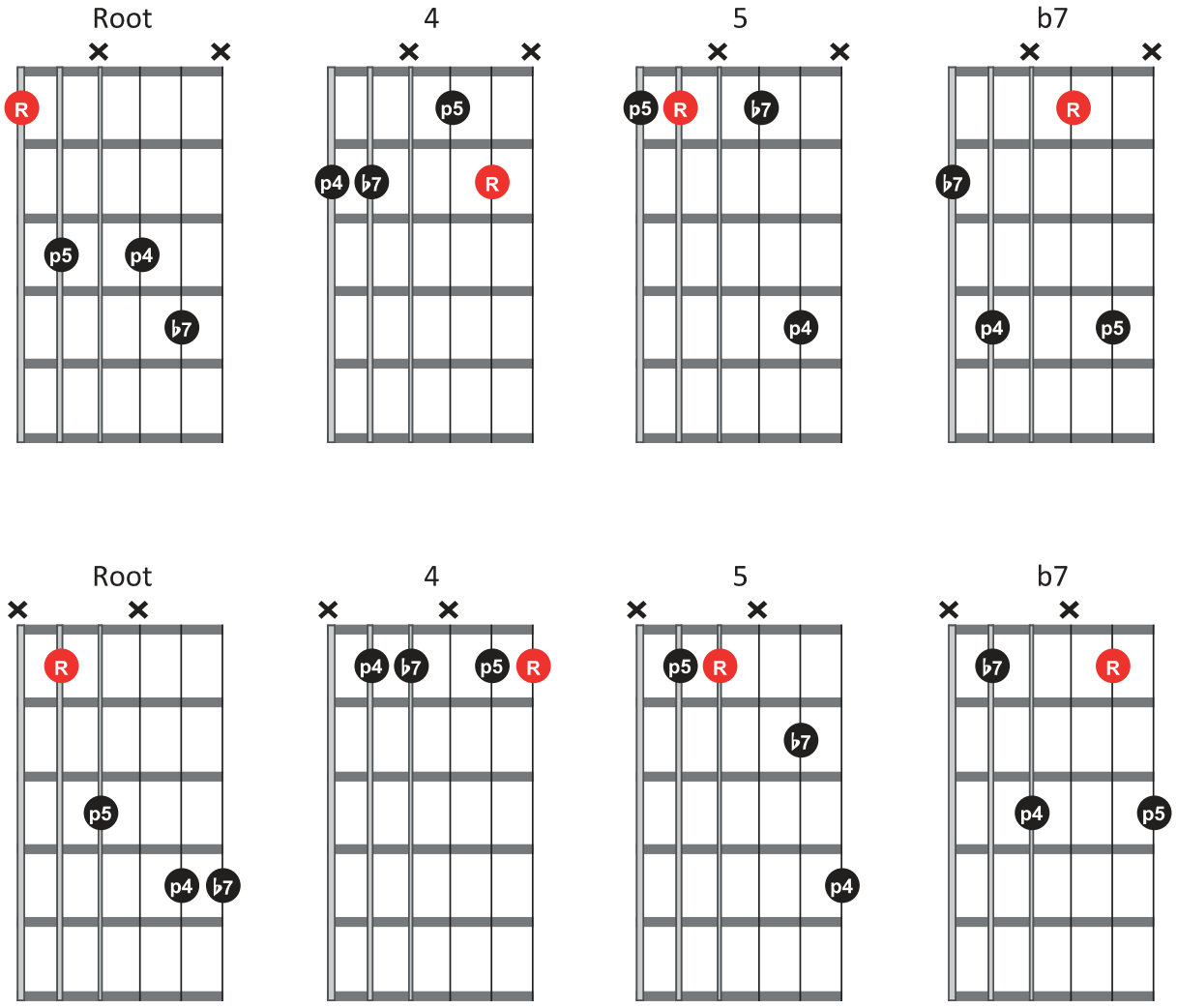
Dom^{7b5}


Root
3
b5
b7
R
♭5
♭7
♭5
♭7
♭5
♭7
♭7
♭5
♭7
♭5
♭7
♭5

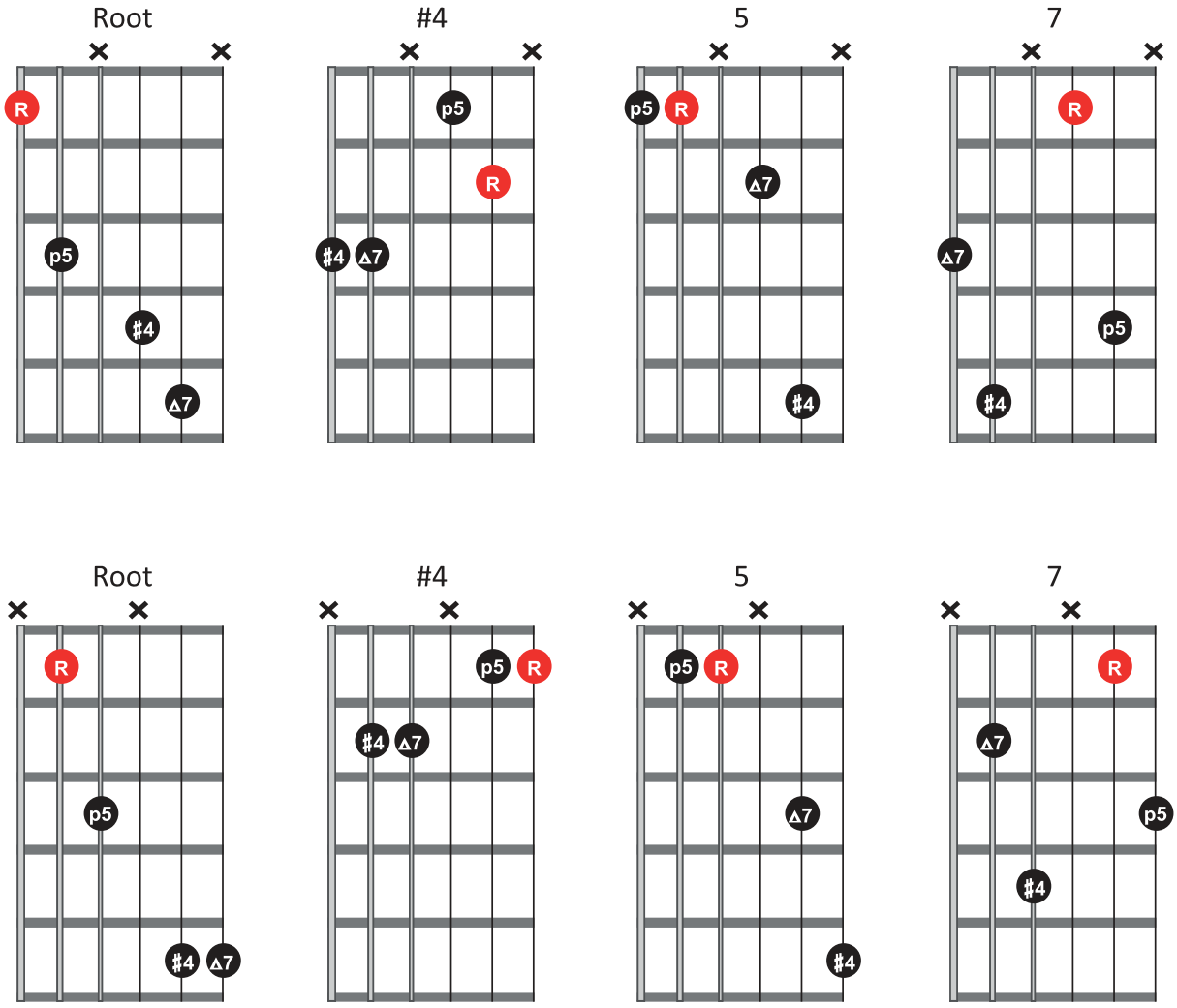
Maj^{7sus4}



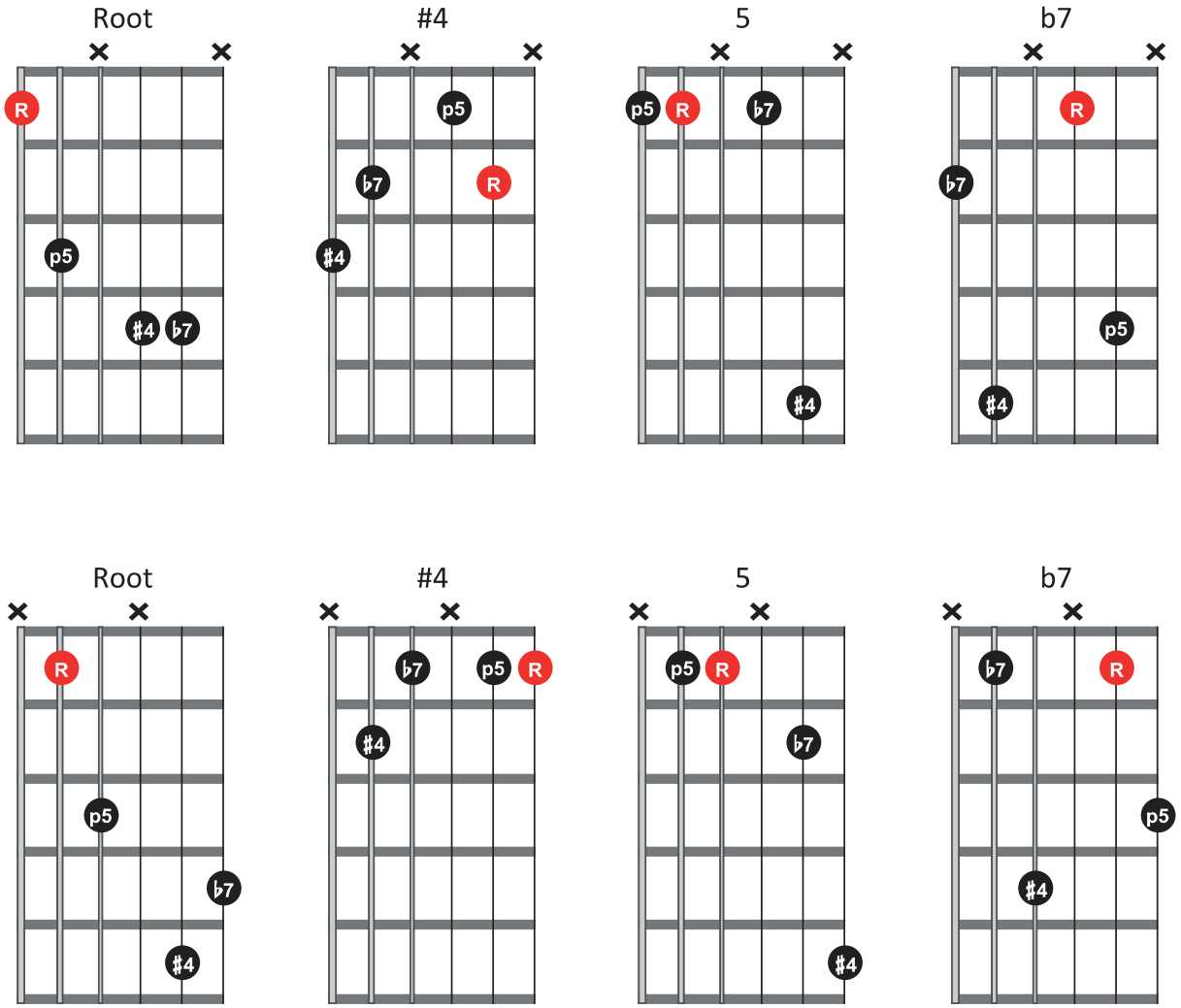
Dom^{7sus4}



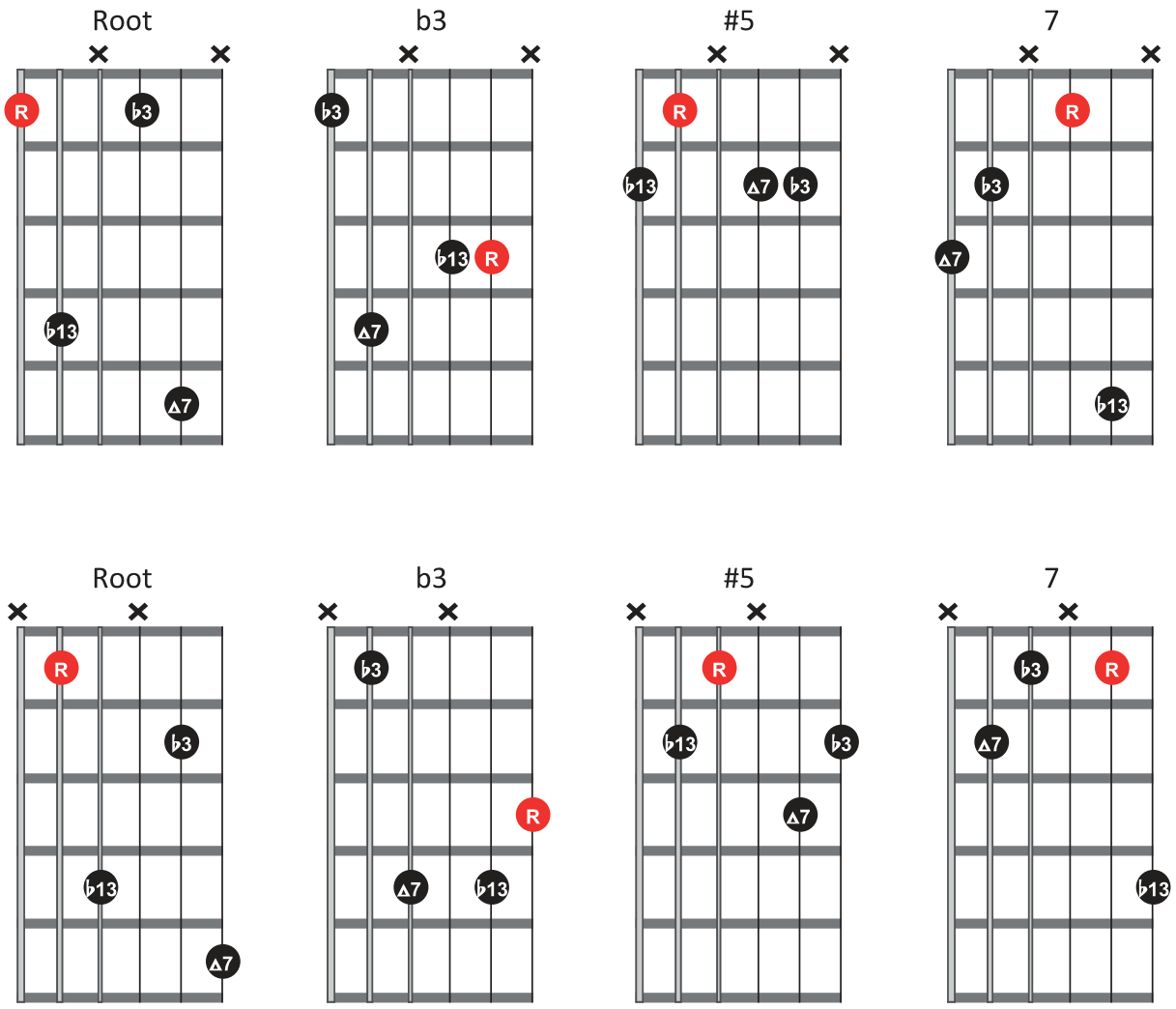
Lydian^{maj7}



Lydian^{Dom7}



***Dim**maj7b13*


Root
b3
#5
7
R
♭3
♭13
♭13
R
♭13
♭3
♭3
R
♭3
♭13
♭3
♭13
Root
b3
#5
7
R
♭3
♭13
R
♭3
♭13
♭3
♭13
R
♭13
♭3
♭3
R
♭13

Triads over Bass Notes

The subsequent chart contains six different forms of major, minor and suspended four triads over bass notes. I have included the bass note-to-triad relationship in addition to the common chord label.

Ex. 137

a)

$\frac{\text{MajΔ}}{4}$ ← Major triad

with its fourth in the bass

b)

$\frac{\text{C}}{\text{F}}$ ← C triad

Its 4^{th}, F, in the bass

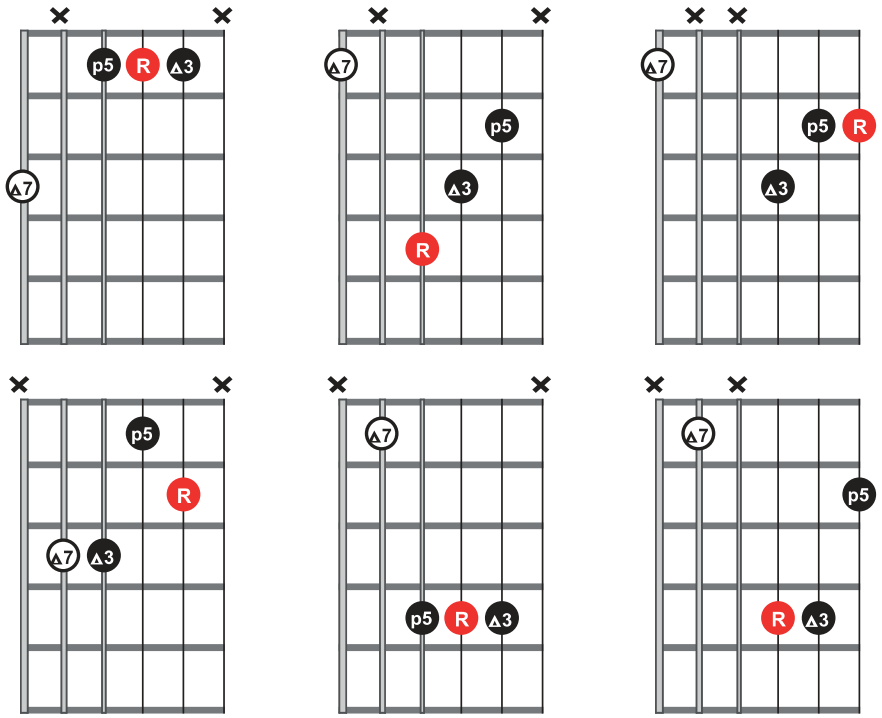
c) or

F^{MA9} no 3^{rd} =

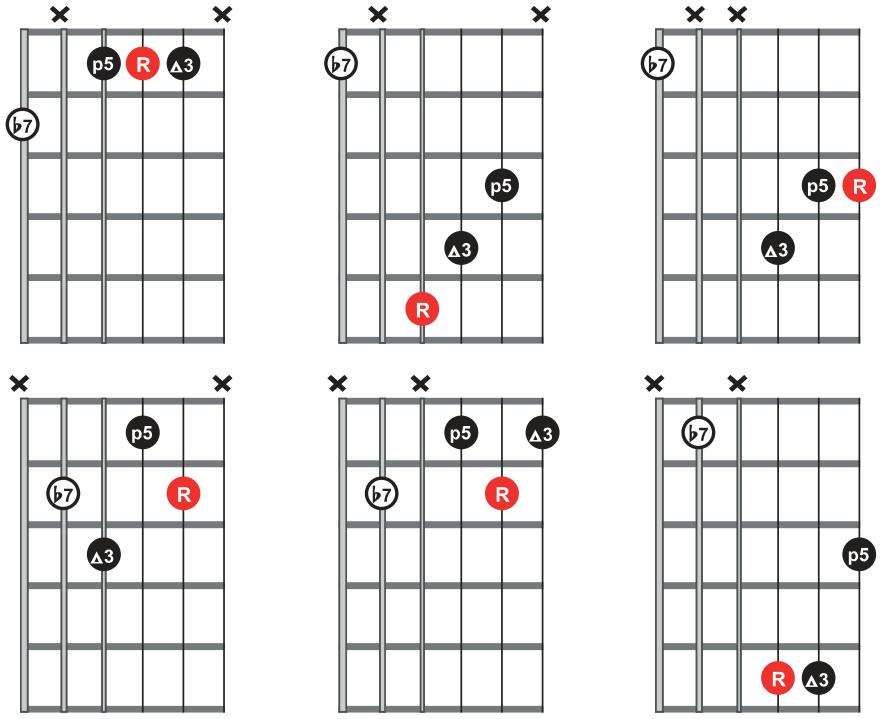
Bass	C triad		
F	C	E	G
1	5	7	9

Major Triads
Bass Notes

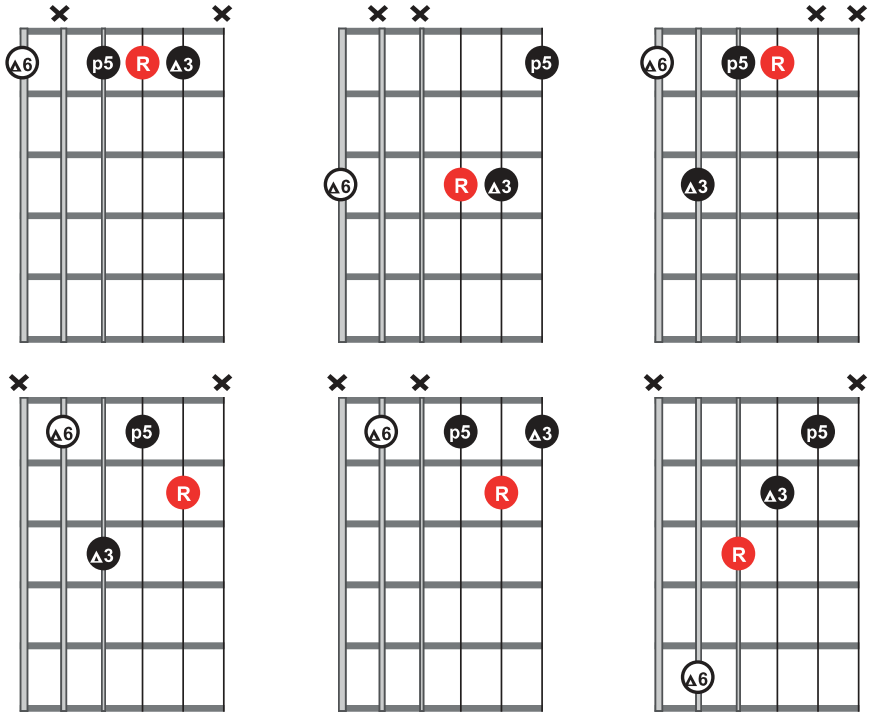
MajΔ / 7 = Phrygian



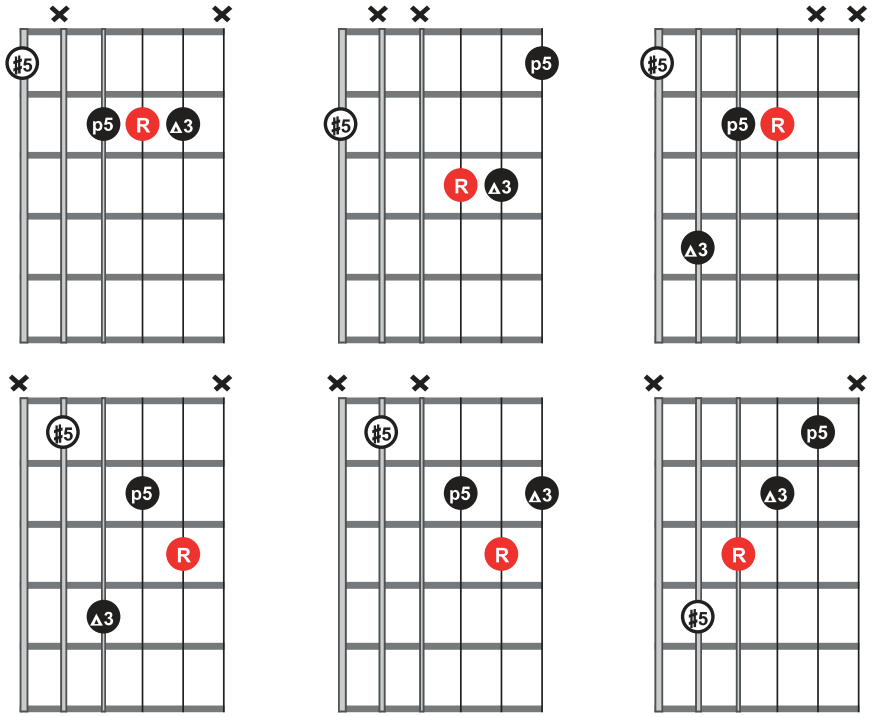
MajΔ / b7 = Dom 4 2



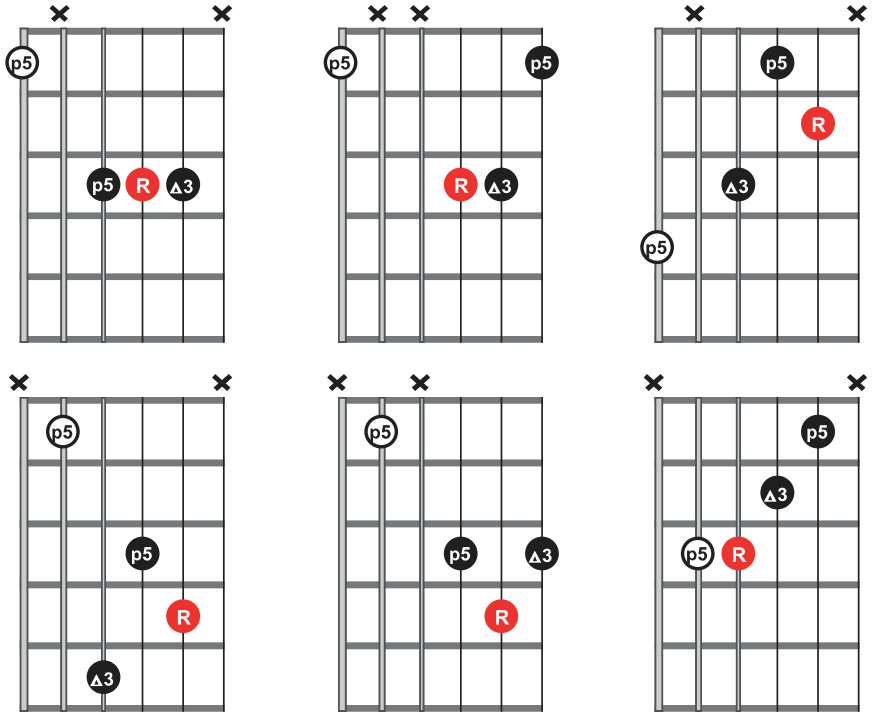
$$\frac{\text{Maj}\Delta}{6} = \text{Min}^{7}$$



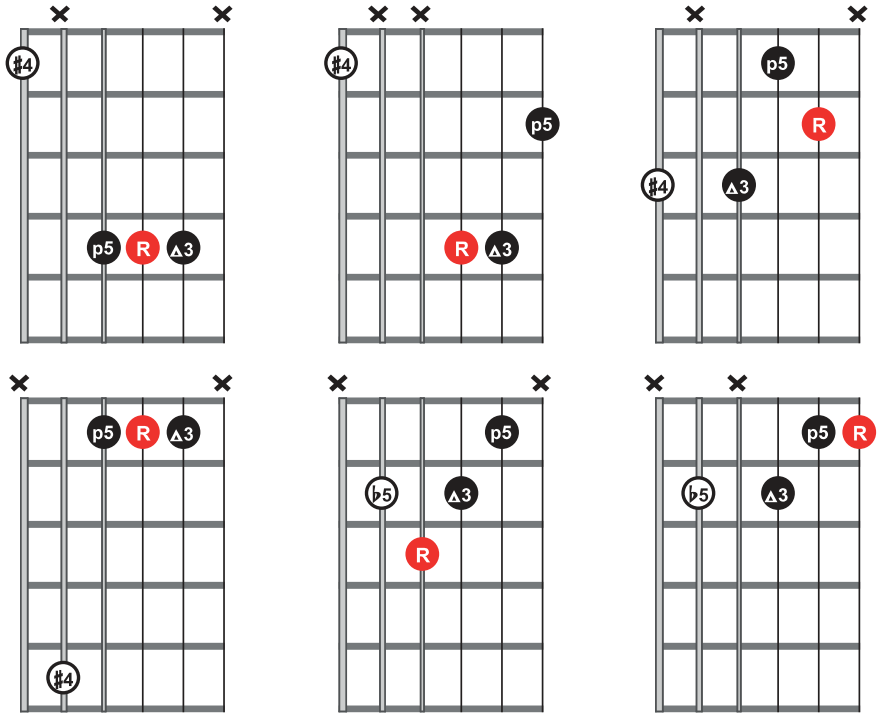
$$\frac{\text{Maj}\Delta}{\#5} = \text{Maj}^{7\#5}$$



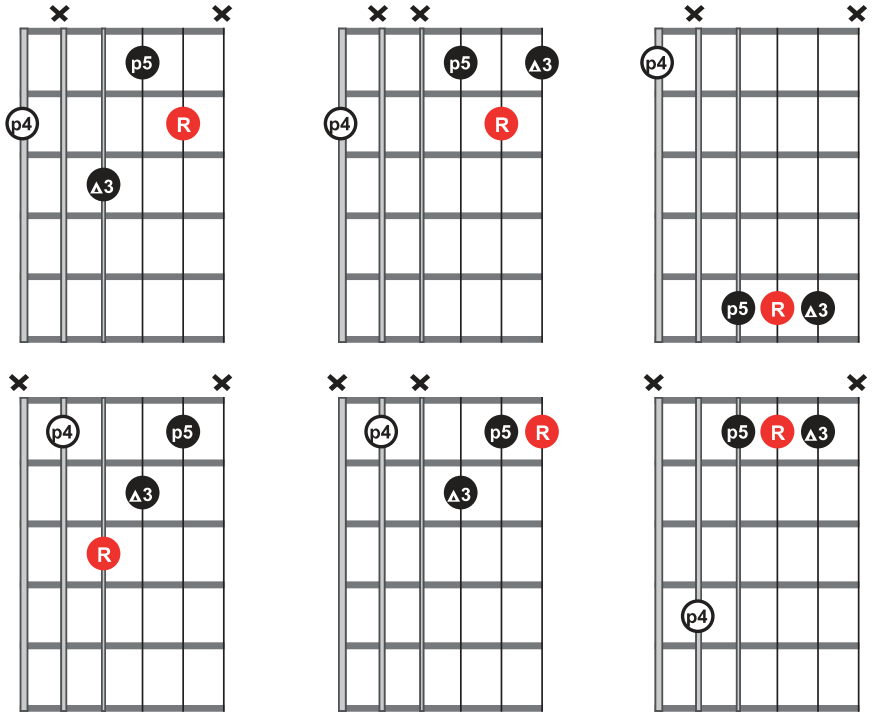
$$\frac{\text{MajΔ}}{5} = \text{Maj}\,\frac{6}{5}$$



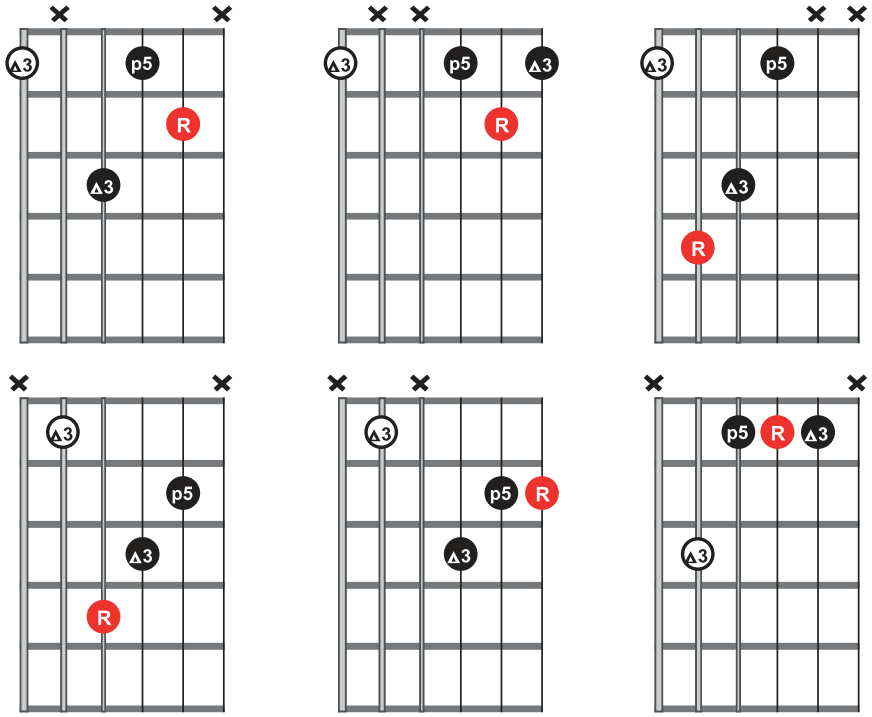
$$\frac{\text{MajΔ}}{\#4} = \text{Dom}^{7\flat9\flat5}$$



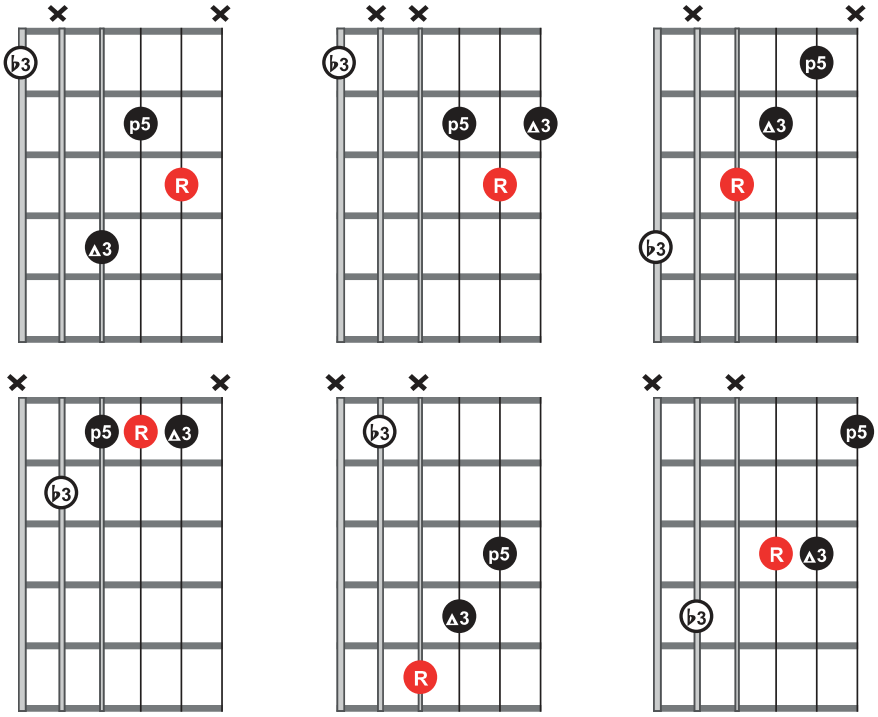
$\frac{\text{Maj}\Delta}{4} = \text{Maj}^{9\ \text{no}\ 3\underline{\text{rd}}}$



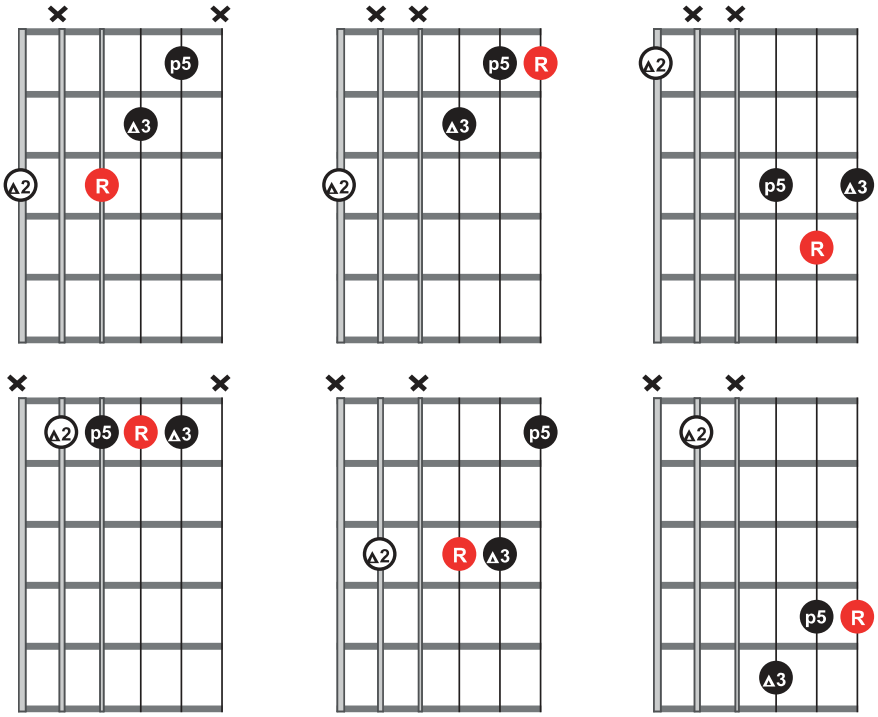
$\frac{\text{Maj}\Delta}{3} = \text{Maj 6}$



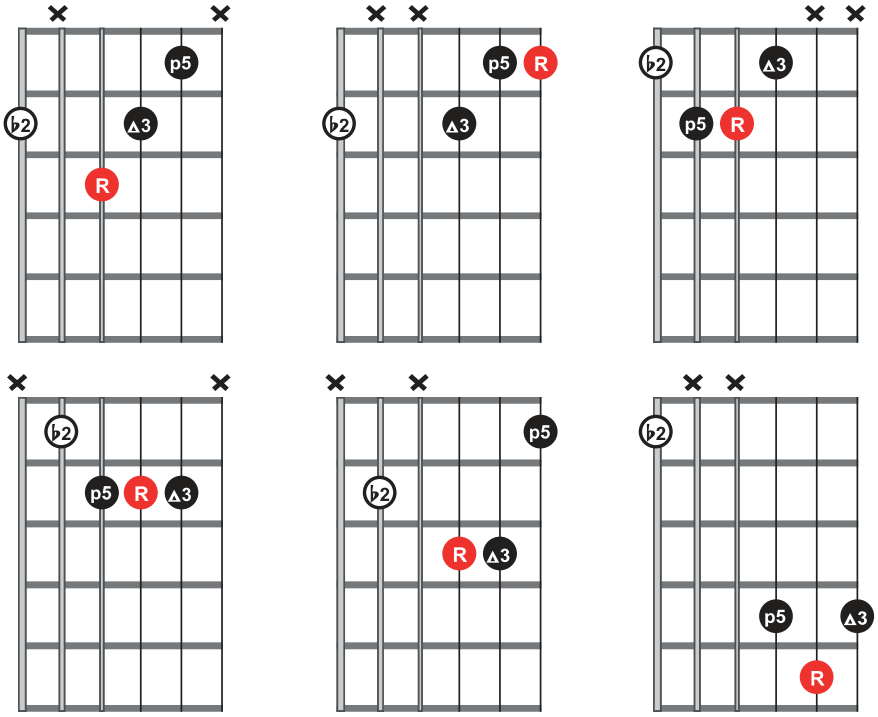
$$\frac{\text{Maj}\Delta}{\text{b3}} = \text{Dom}^{13\text{b9 no7th}}$$



$$\frac{\text{Maj}\Delta}{9} = \text{Dom}^{11}$$

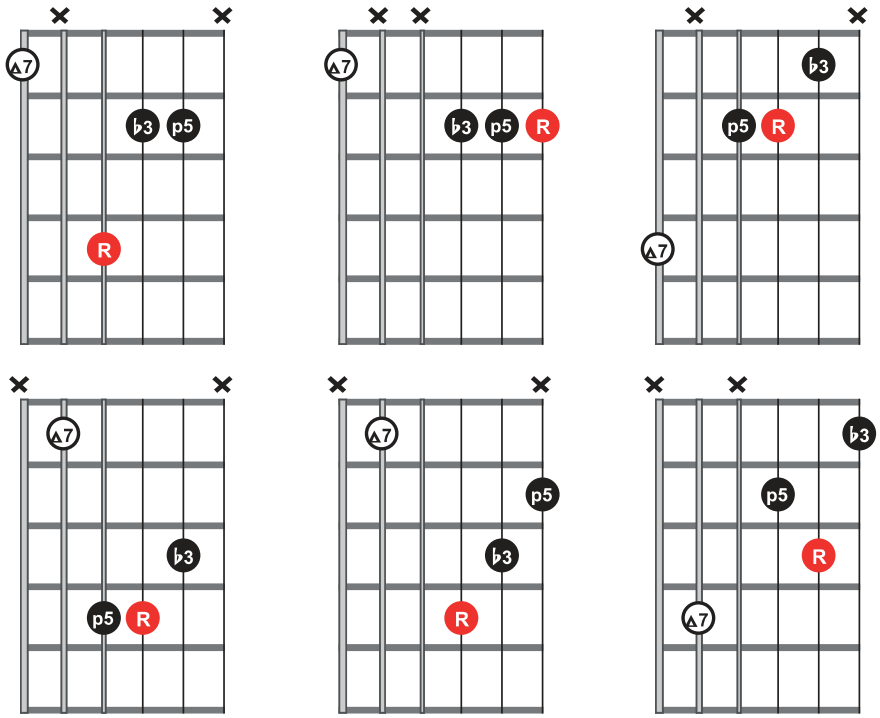


$$\frac{\text{Maj}\Delta}{\text{b9}} = \text{Dim}^{\text{maj7}}$$

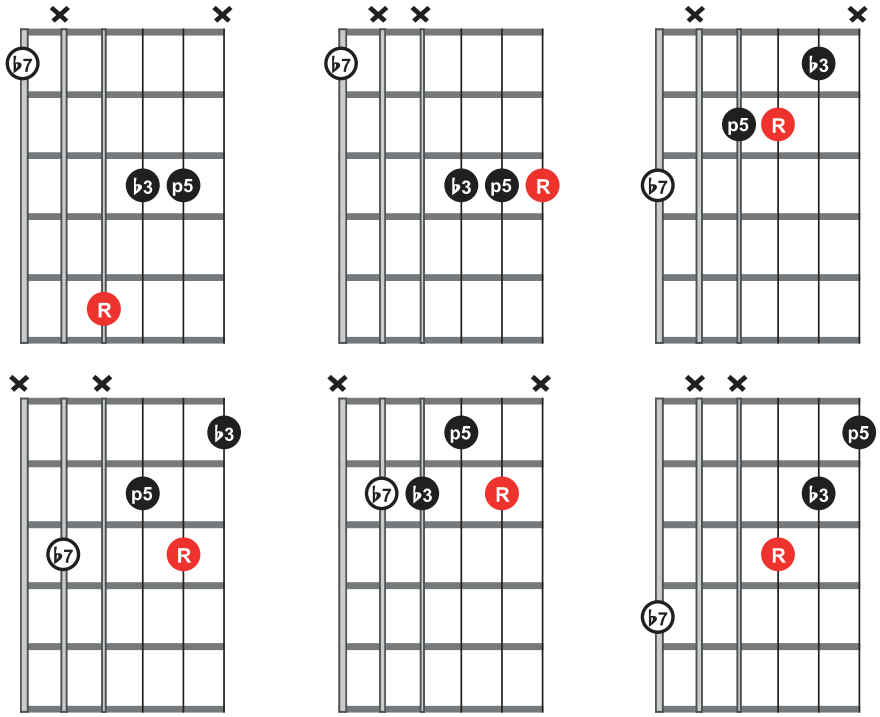


Minor Triads
Bass Notes

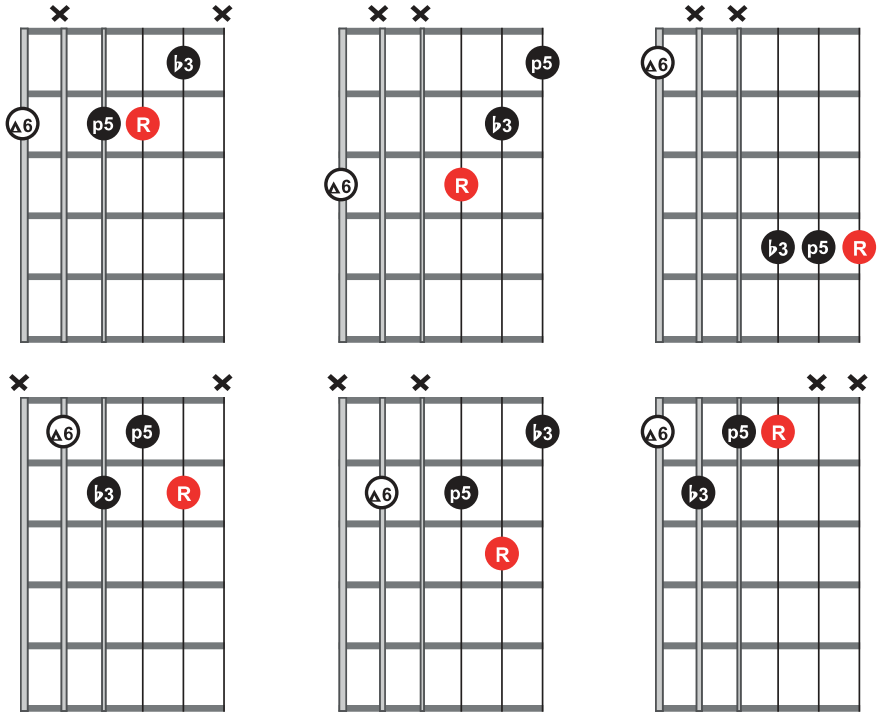
$\frac{\text{MinΔ}}{7} = \text{Dom}^{7/\#5/b9\ no7}$



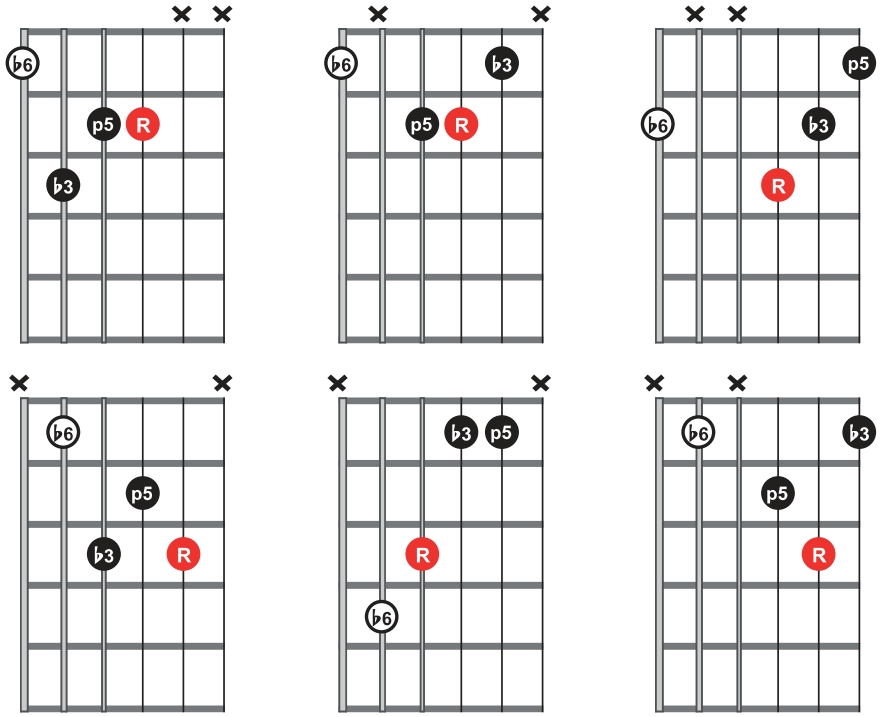
$\frac{\text{MinΔ}}{\text{b7}} = \frac{\text{Maj}^{6}}{5^{th}}$



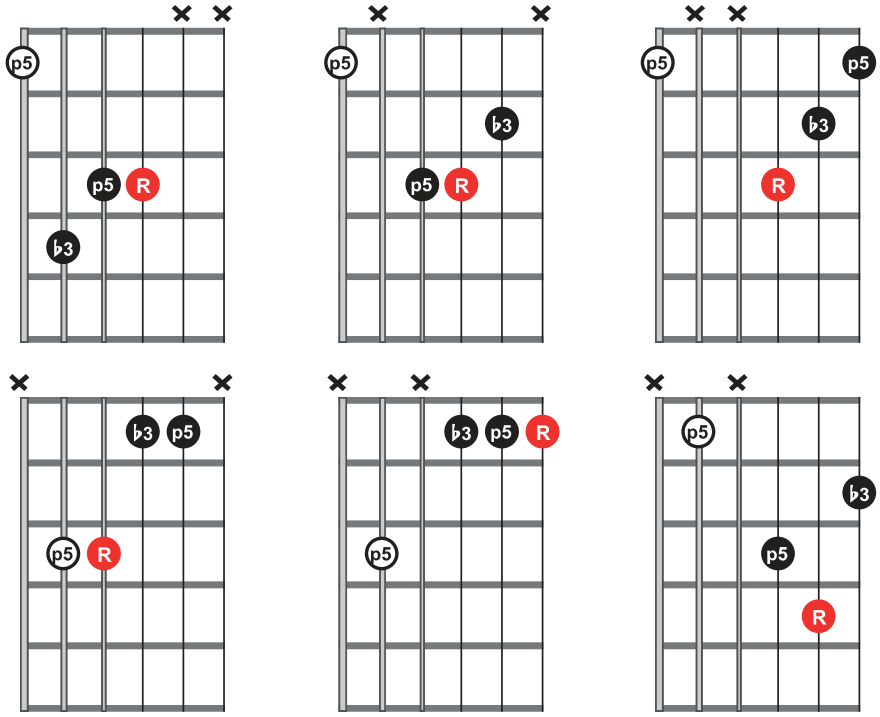
$$\frac{\text{Min}\Delta}{6} = \text{Min}^{7b5}$$



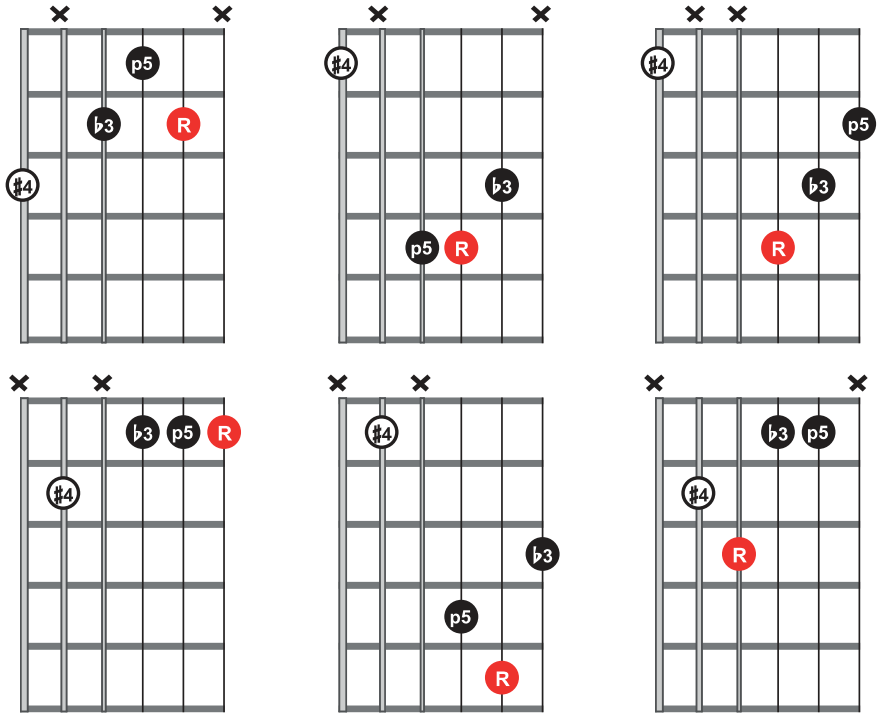
$$\frac{\text{Min}\Delta}{\text{b6}} = \text{Maj}^{7}$$



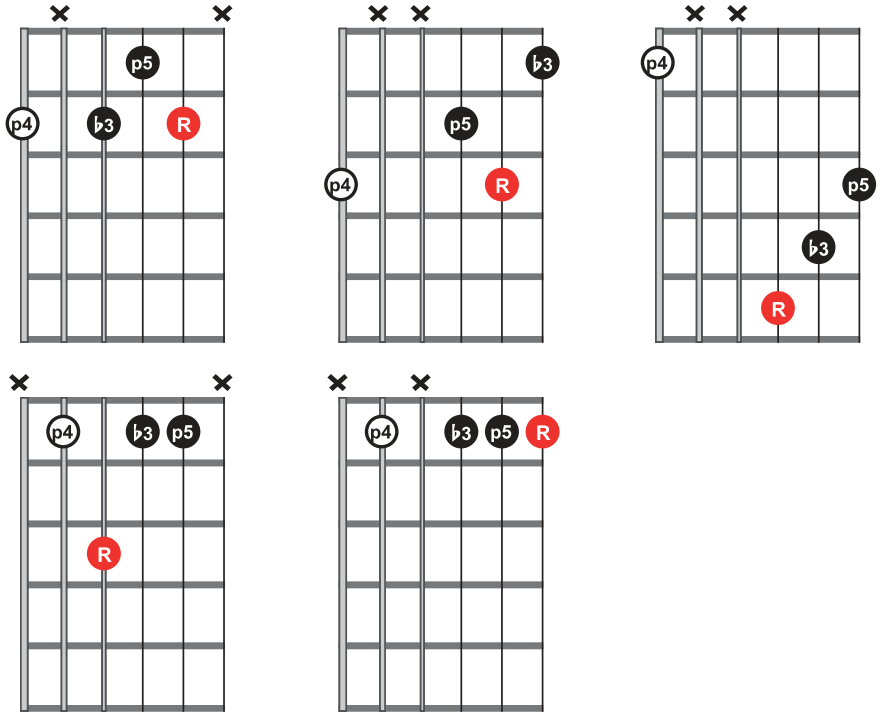
$$\frac{\text{Min}\Delta}{5} = \text{Min}\,\frac{6}{5}$$



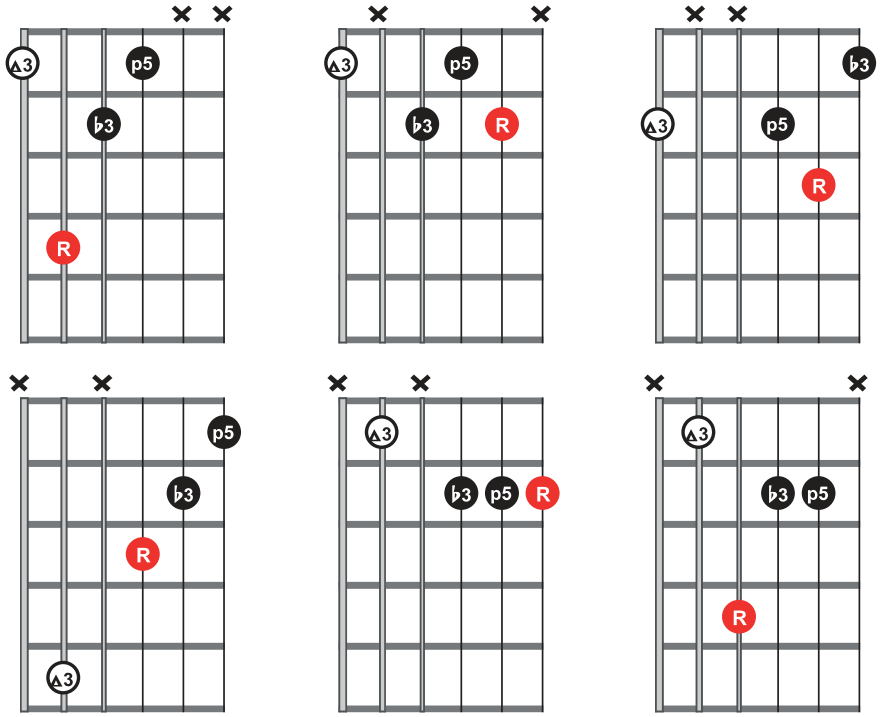
$$\frac{\text{Min}\Delta}{\#4} = \text{Dom}^{13/\flat9/\flat5\ \text{no}\ 3\underline{\text{rd}}}$$



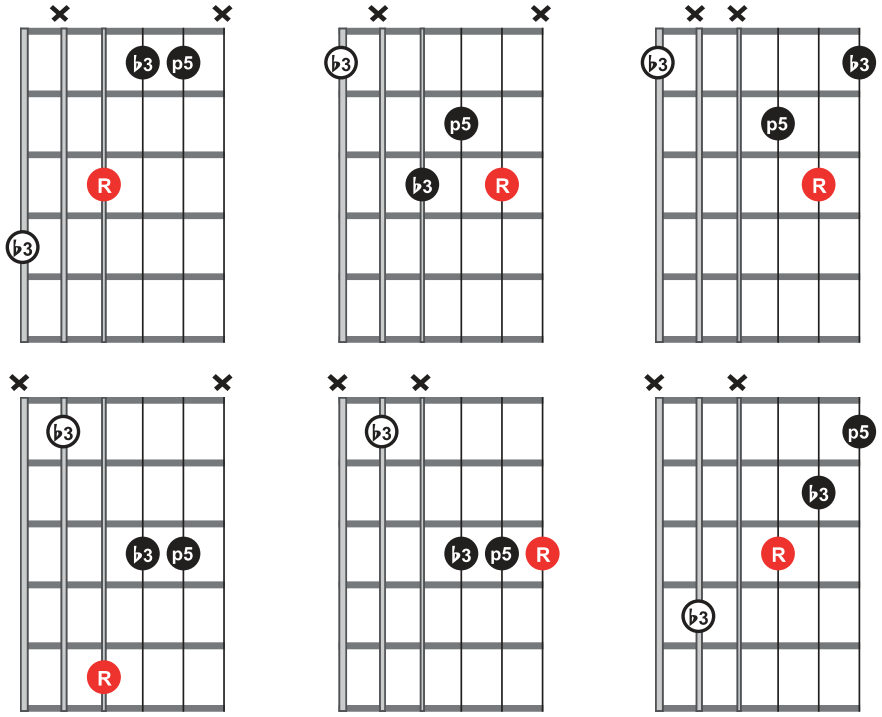
$$\frac{\text{Min}\Delta}{4} = \text{Dom}^{9\ \text{no}\ \underline{3\text{rd}}}$$



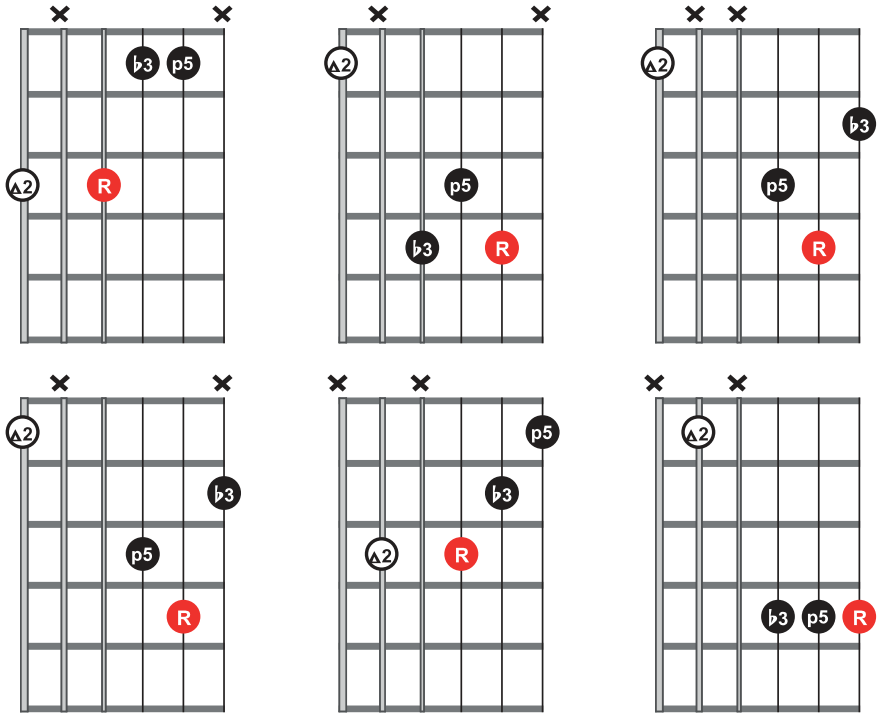
$$\frac{\text{Min}\Delta}{3} = \text{Dim}^{\text{maj7b13}}$$



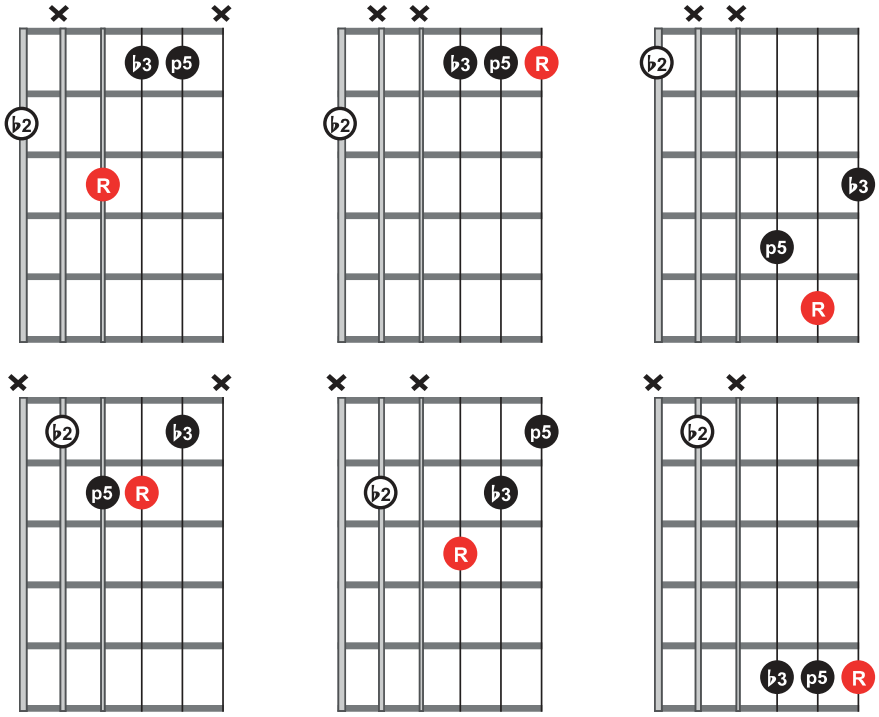
$$\frac{\text{Min}\Delta}{\text{b3}} = \text{Min 6}$$



$$\frac{\text{Min}\Delta}{9} = \text{Dom}^{11b9}$$

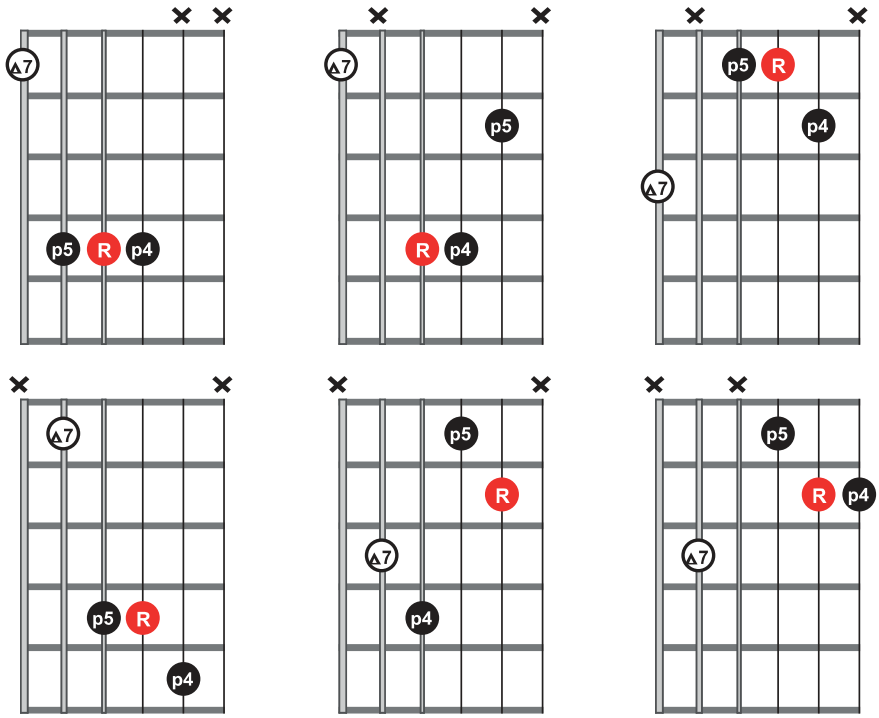


$$\frac{\text{Min}\Delta}{\text{b9}} = \text{Dim}^{\text{maj7}}$$

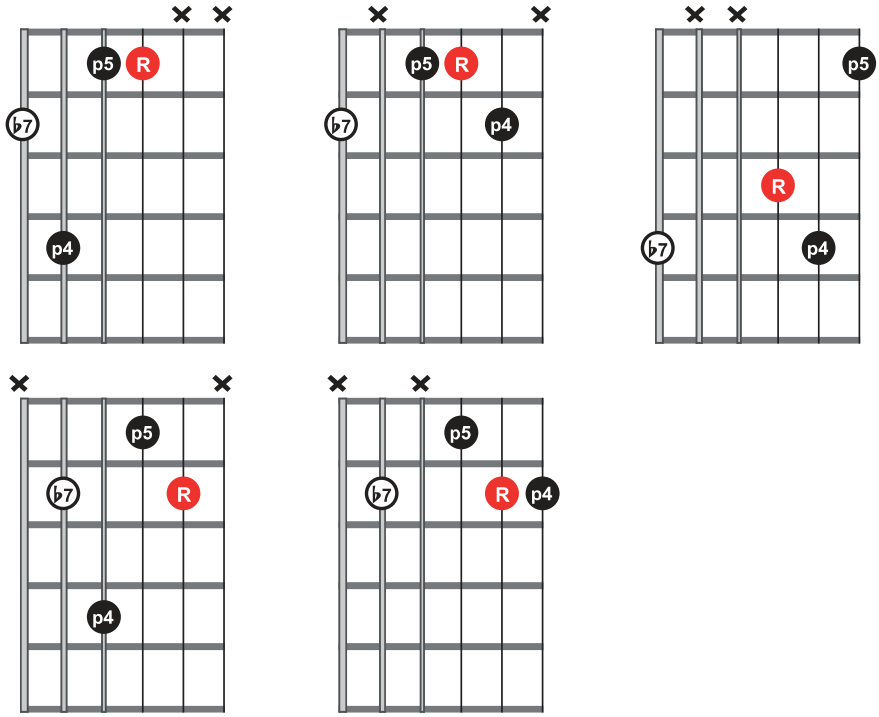


Sus4 Triads
Bass Notes

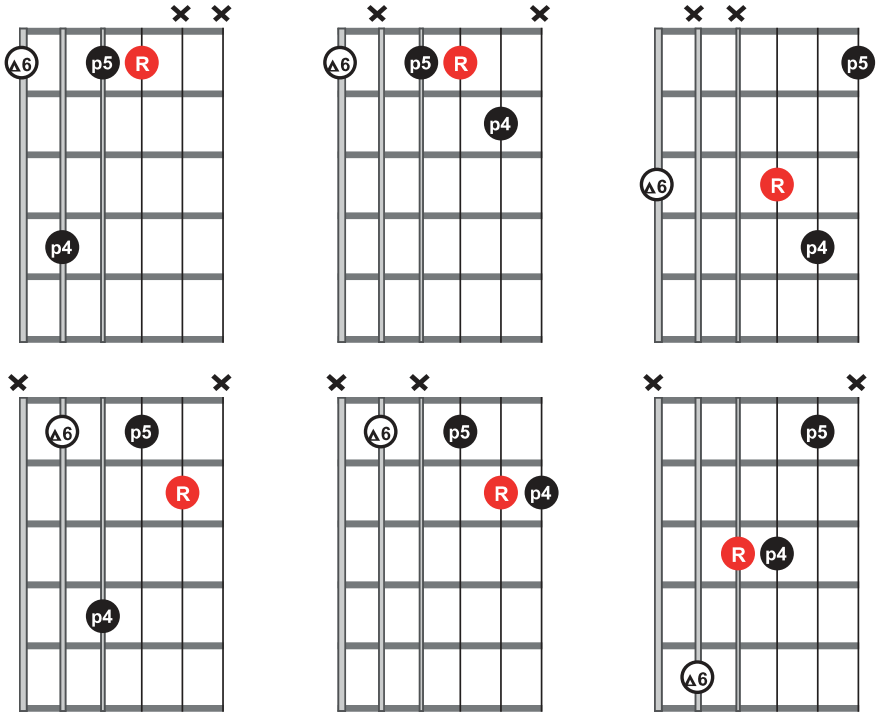
$\frac{\text{Sus4}\Delta}{7} = \text{Dom}^{7/b9/b5/\#5}$



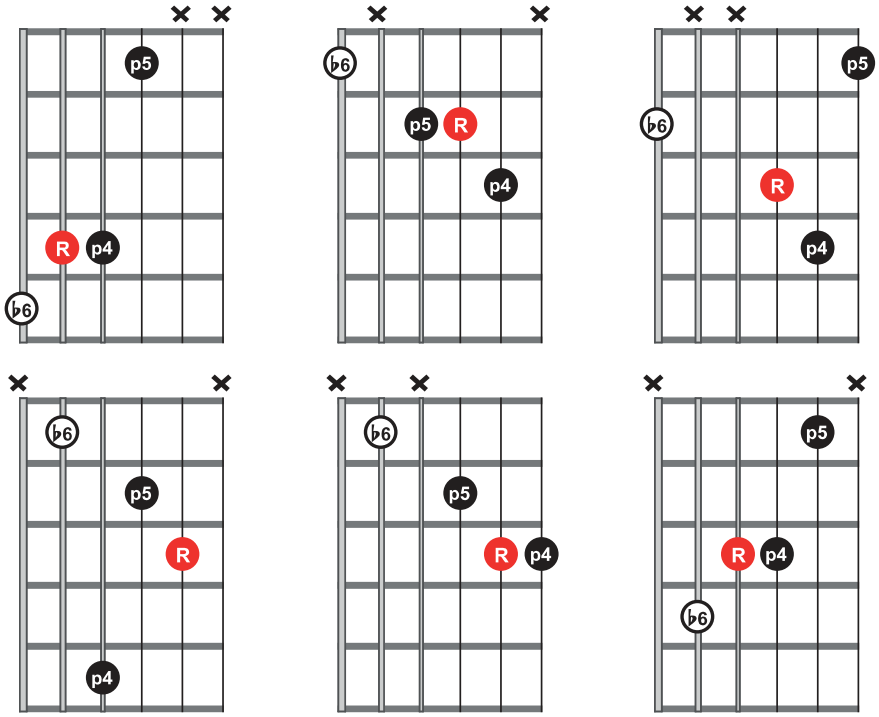
$\frac{\text{Sus4}\Delta}{\text{b7}} = \text{Maj}^{69}$



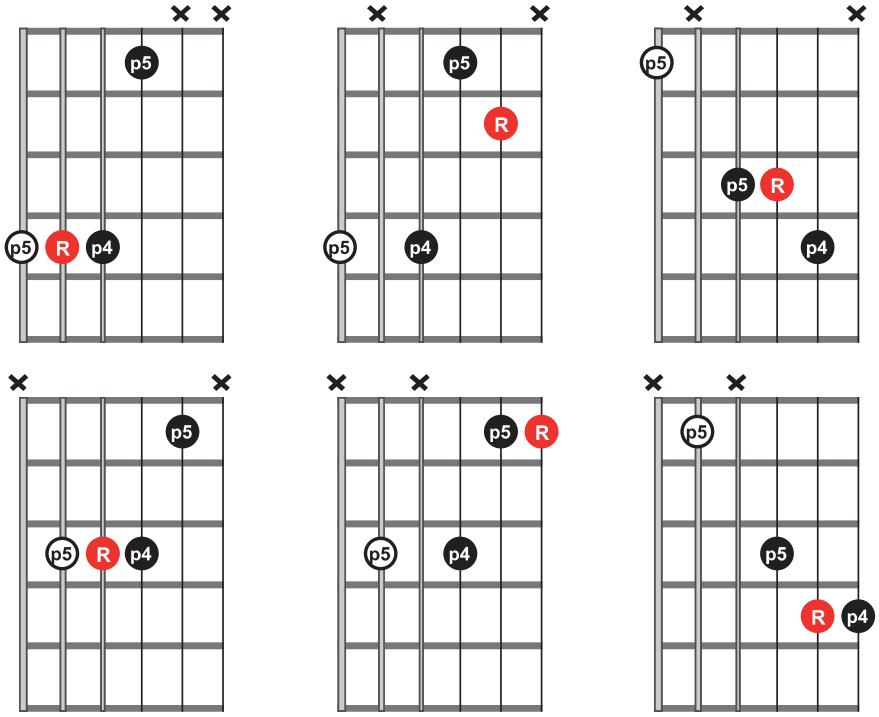
$$\frac{\text{Sus4}\Delta}{6} = \text{Min}^{7b6}$$



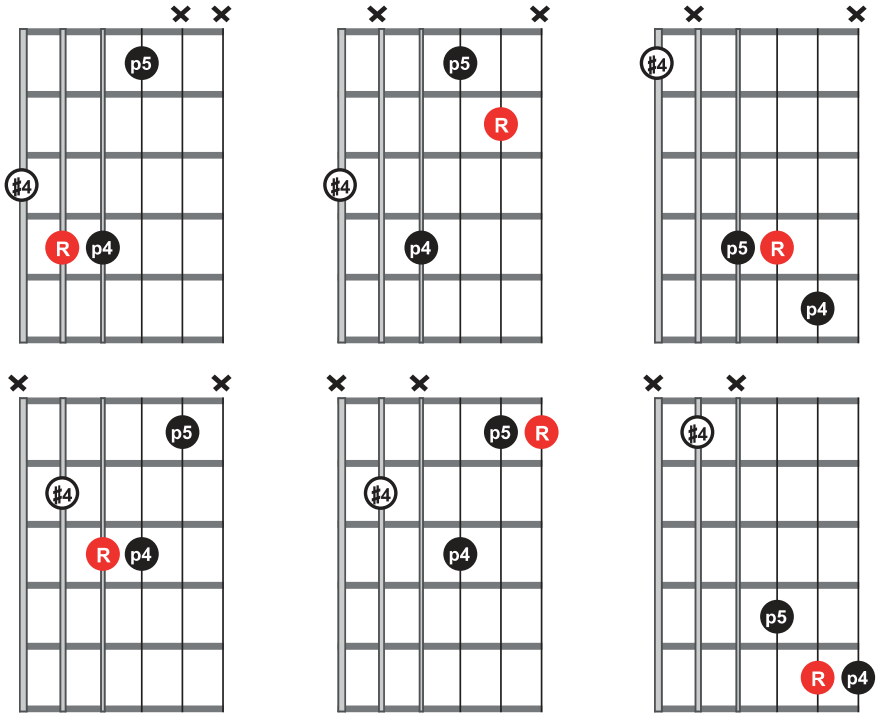
$$\frac{\text{Sus4}\Delta}{\text{b6}} = \text{Maj}^{13}$$



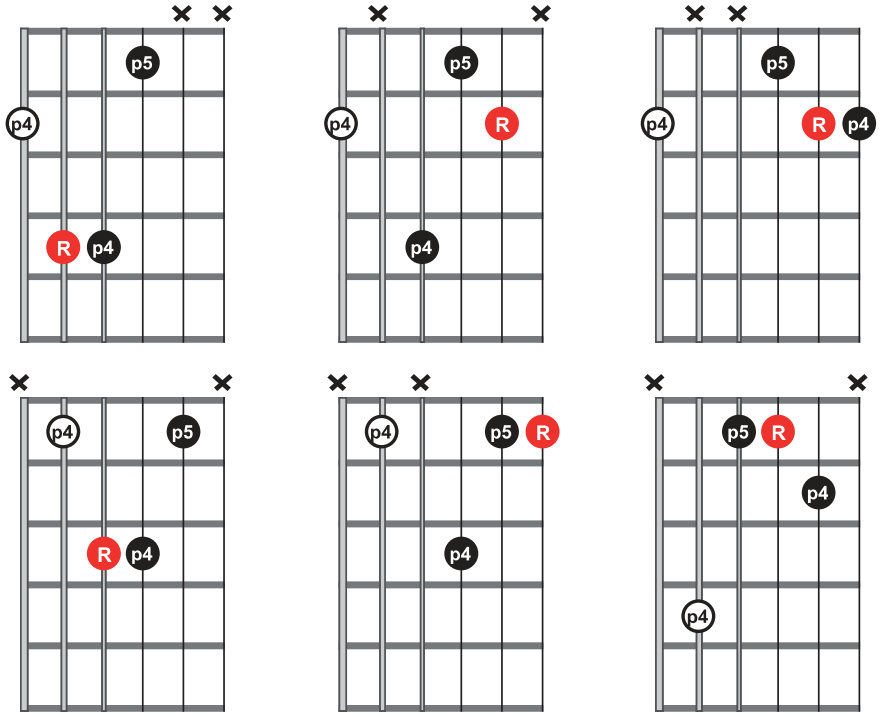
$\frac{\text{Sus4}\Delta}{5} = \text{Dom}^{7\text{sus4}}$



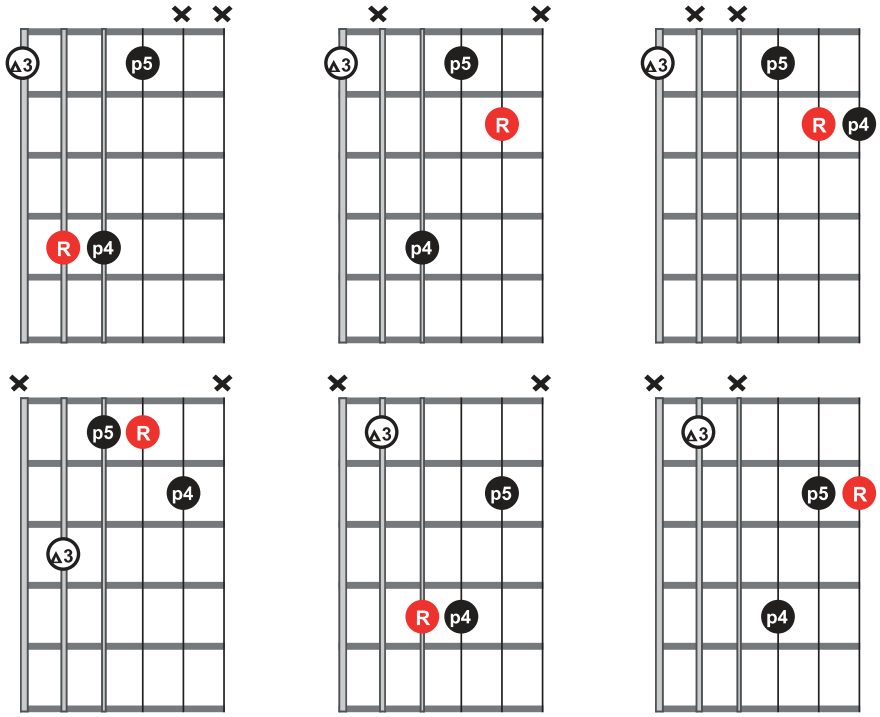
$\frac{\text{Sus4}\Delta}{\#4} = \text{Twelve Tone}$



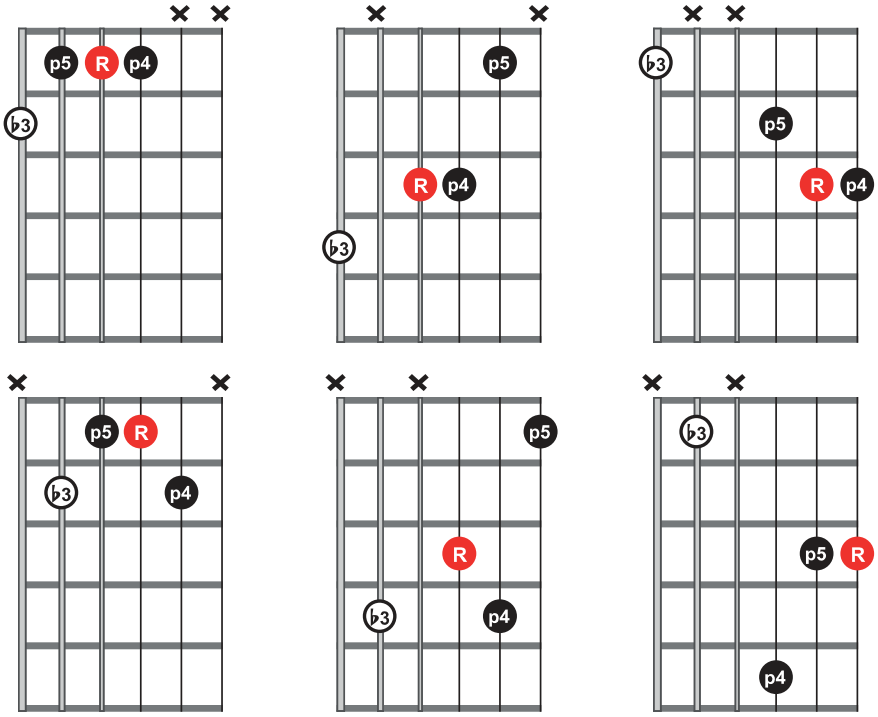
$\frac{\text{Sus4}\Delta}{4}$ = Sus2



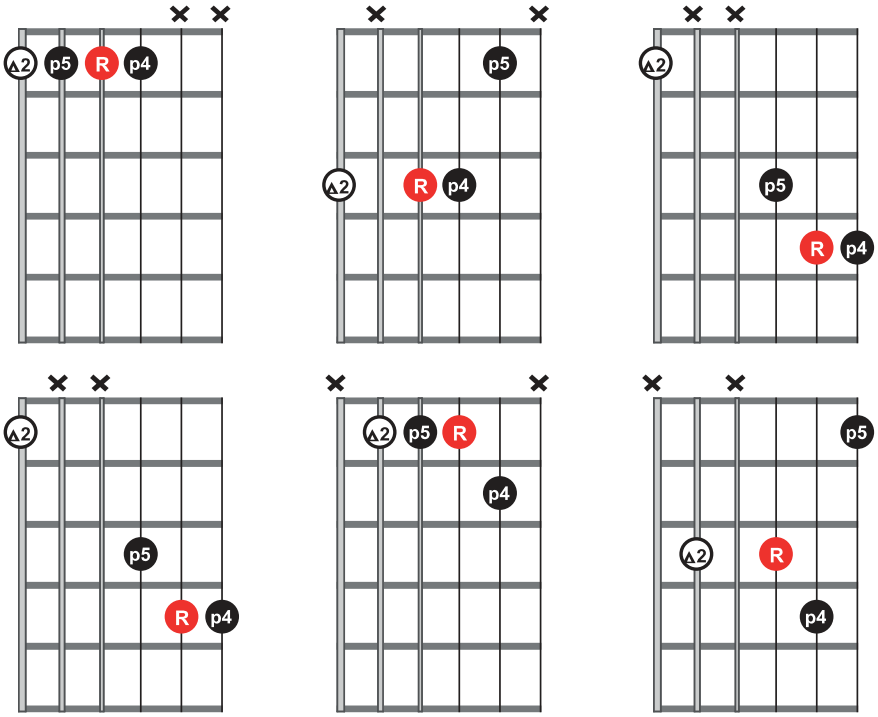
$\frac{\text{Sus4}\Delta}{3}$ = Dom$^{7/b9/b5/\#9}$ Min$^{7/b9b6}$



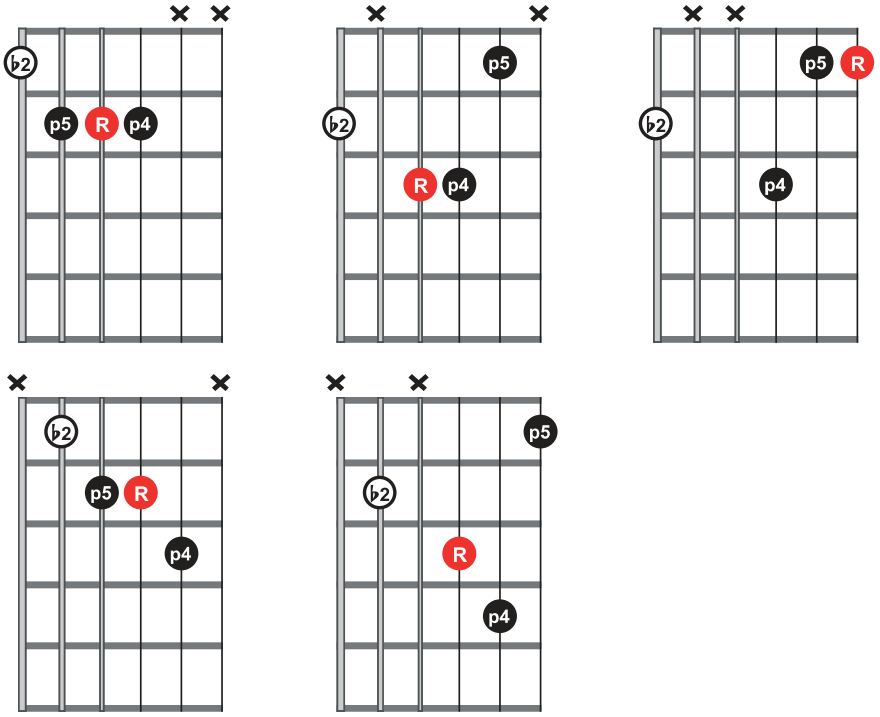
$\frac{\text{Sus4}\Delta}{\text{b3}} = \text{Maj}^{69}$



$\frac{\text{Sus4}\Delta}{9} = \text{Min}^{11}$



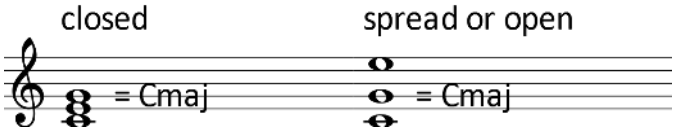
$$\frac{\text{Sus4}\Delta}{\text{b9}} = \text{Maj}^{7\text{b5}}$$



Spread triads over bass notes

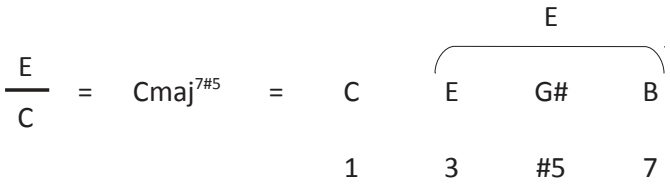
The term spread triad over bass note implies a triad in which the middle note has been raised an octave.

Ex. 138

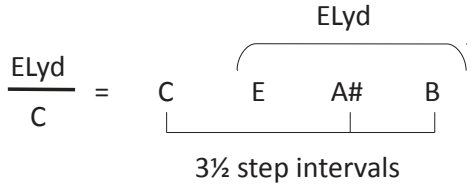


Whenever possible, the spread triads over bass notes have been reduced to seventh chord structures.

Ex. 139

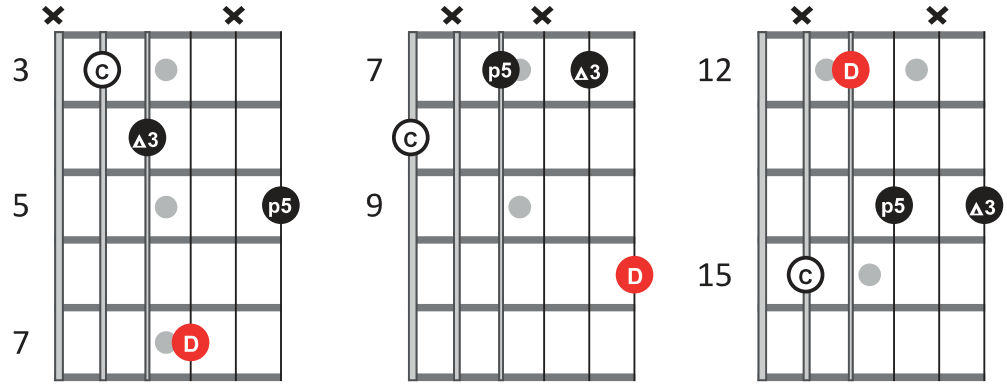


Because $\frac{C}{E}$ is a type of Cmajor chord it will be listed in the major chord category. The voicings labeled "Twelve Tone type" are derived from the chromatic scale. These only include chords with three consecutive chromatic tones as in:

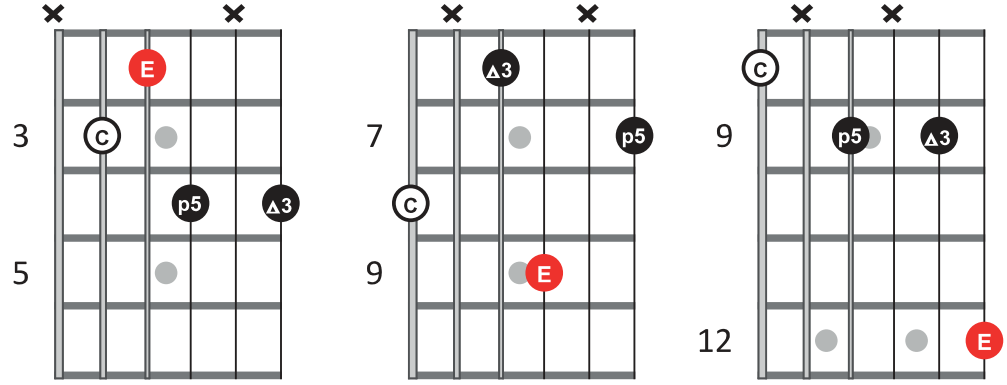


Major Type Spread Triads
Bass Notes (C)

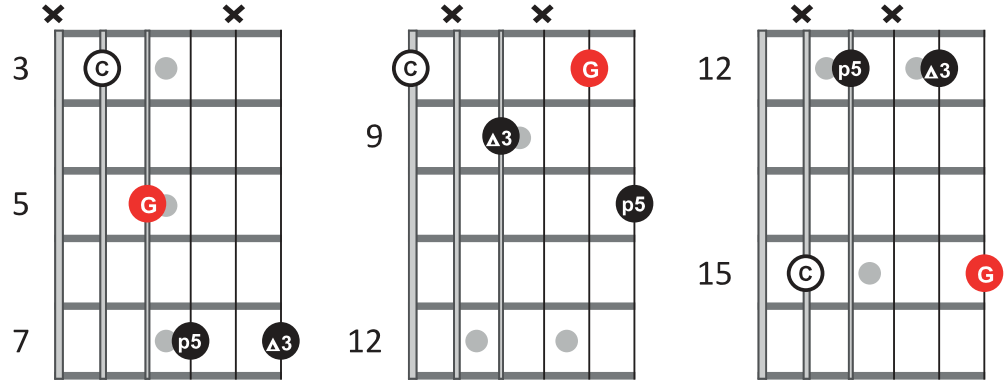
D/C



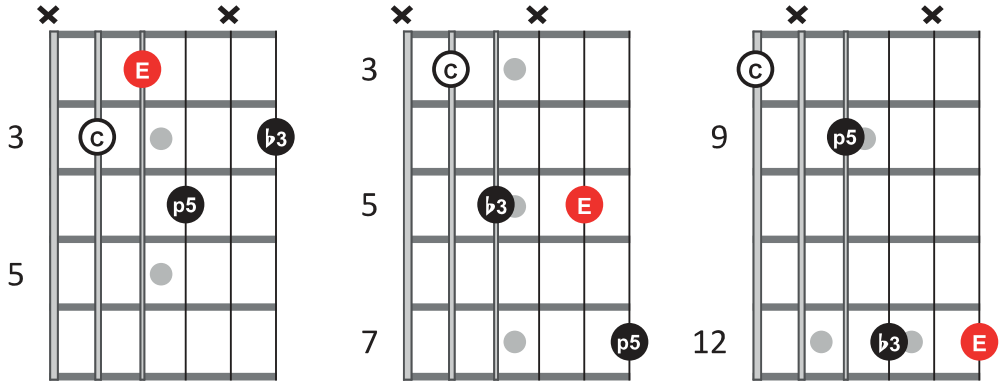
E/C



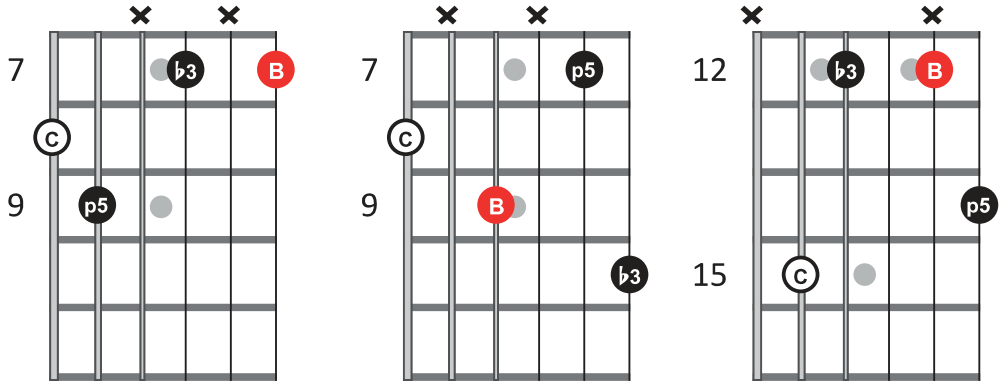
G/C



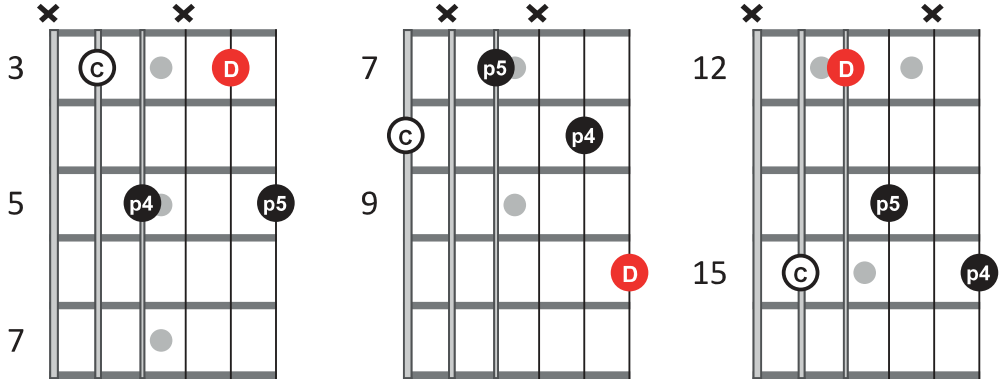
E-
C


3
5
E
C
♭3
p5
3
7
C
♭3
E
p5
9
12
C
p5
♭3
E

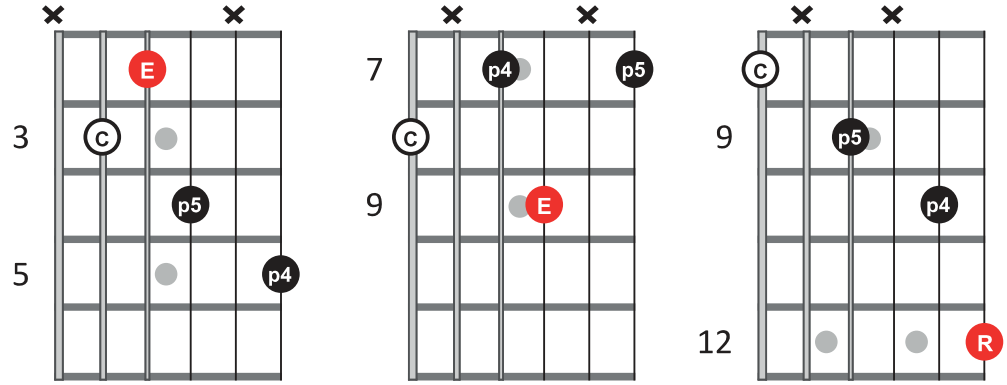
B-
C


7
9
♭3
B
C
p5
7
9
p5
B
C
♭3
12
15
♭3
B
p5
C

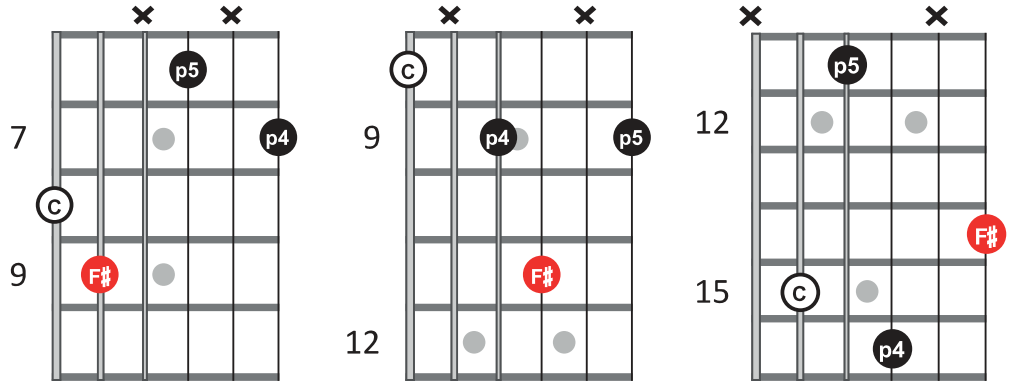
D^{sus4}
C


3
5
7
D
C
p4
p5
7
9
p5
p4
C
D
12
15
D
p5
C
p4

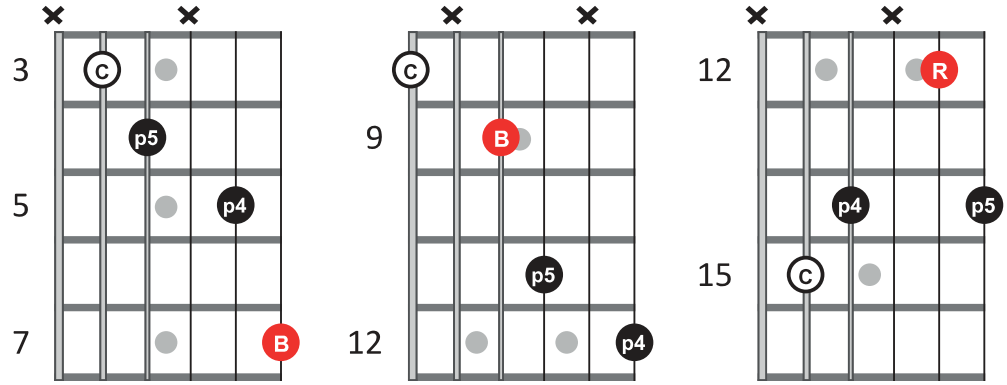
E^{sus4} / C



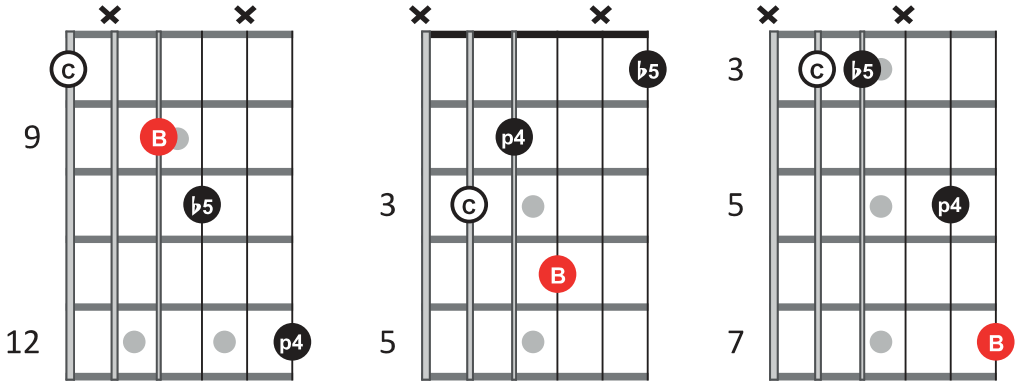
$F\#^{sus4}$ / C



B^{sus4} / C

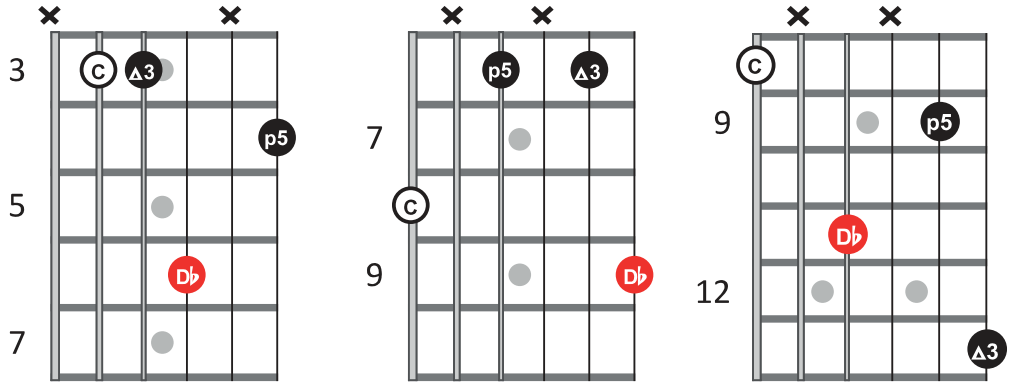


$\frac{B^{loc}}{C}$

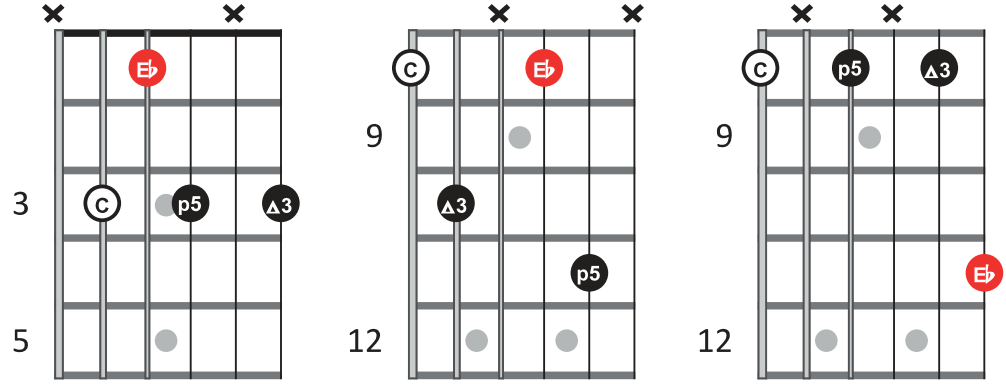


Minor Type Spread Triads
Bass Notes (C)

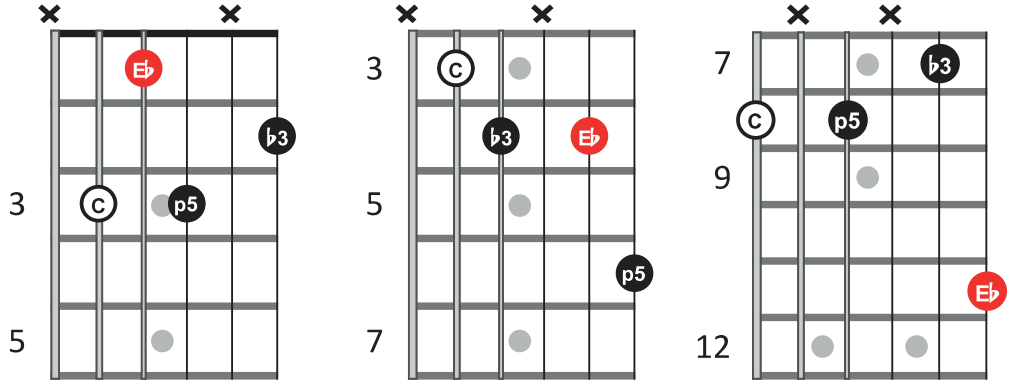
Db / C



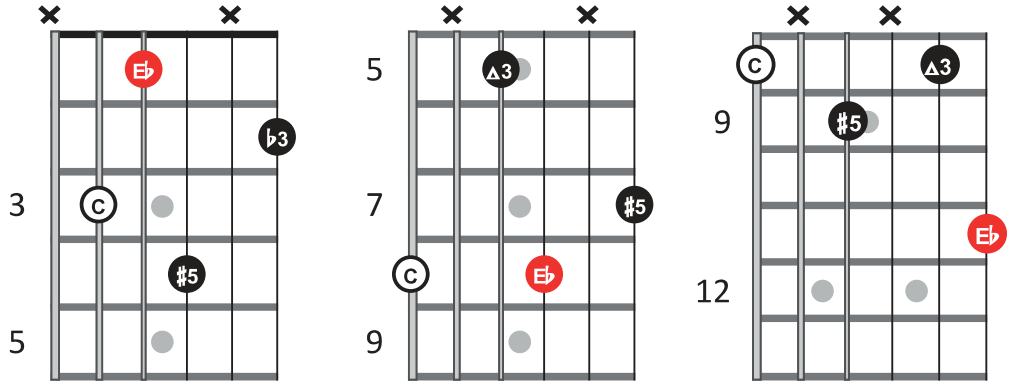
Eb / C



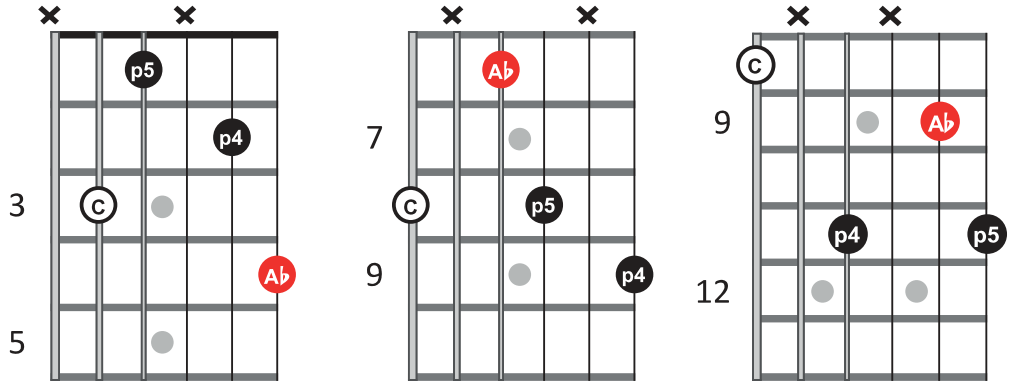
Eb- / C



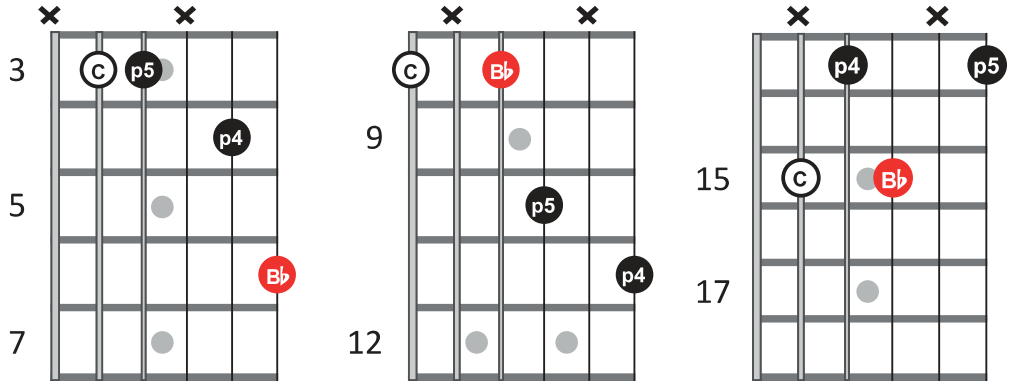
Eb+
C



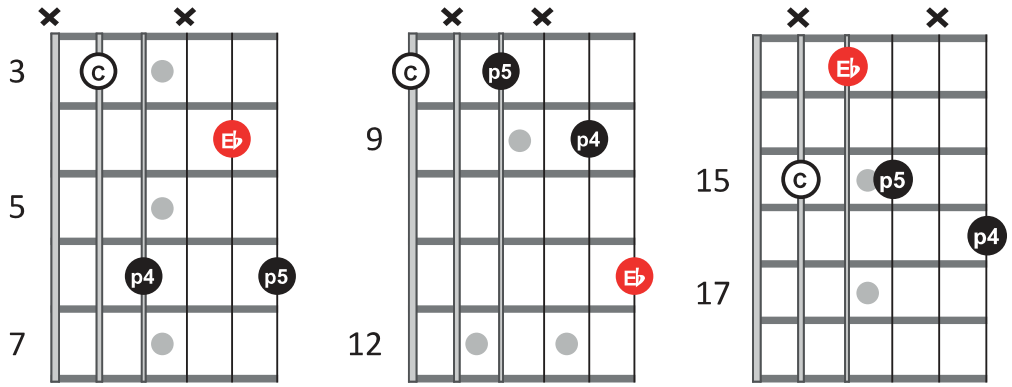
Ab^{sus4}
C



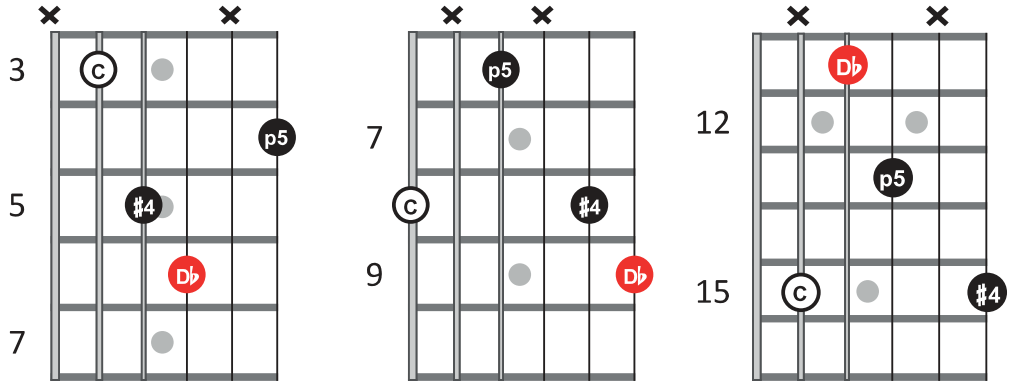
Bb^{sus4}
C



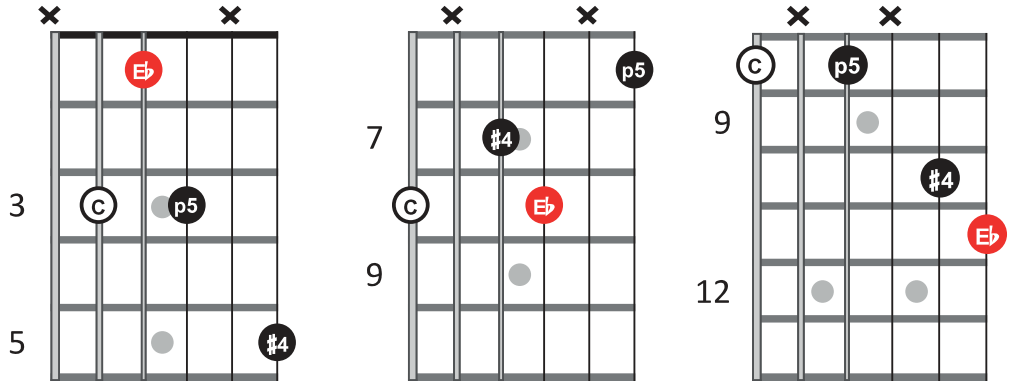
$\frac{Eb^{sus4}}{C}$



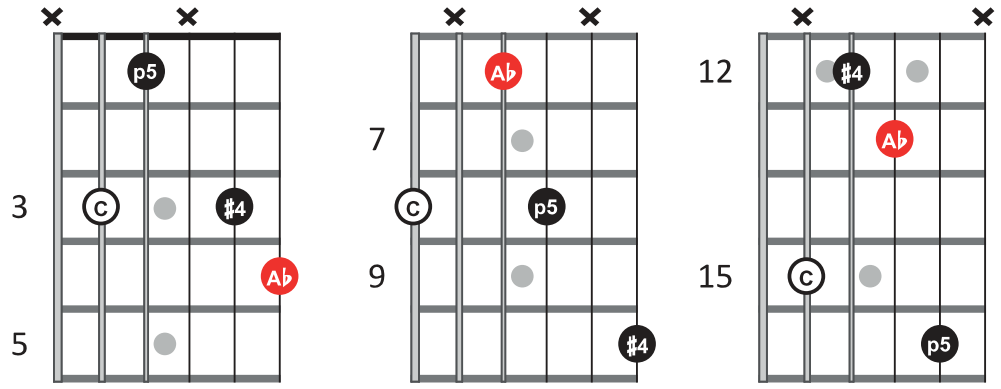
$\frac{Db^{lyd}}{C}$



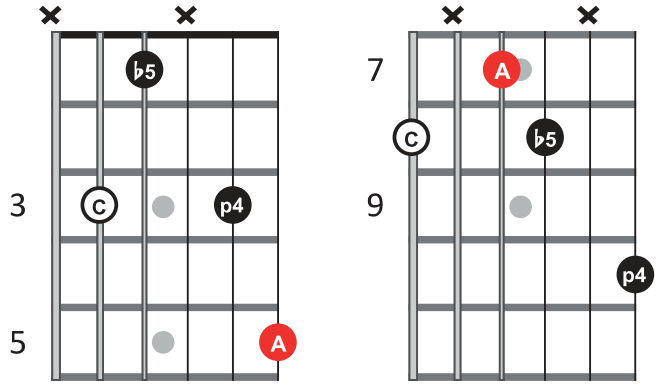
$\frac{Eb^{lyd}}{C}$



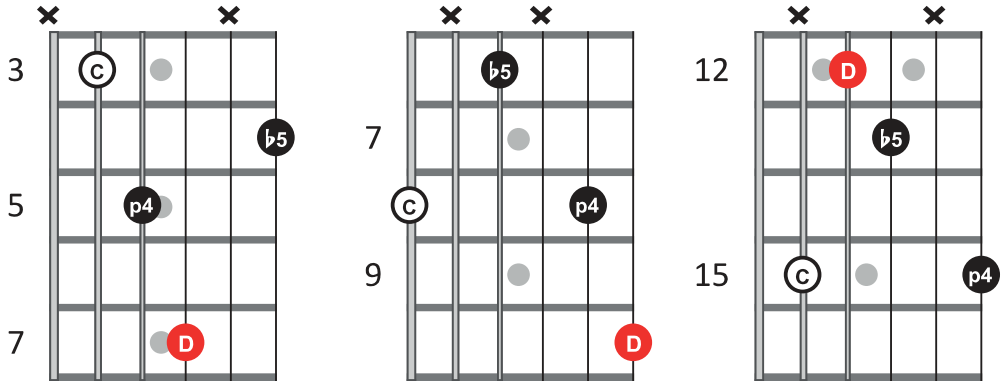
$\frac{Ab^{lyd}}{C}$



$\frac{A^{loc}}{C}$

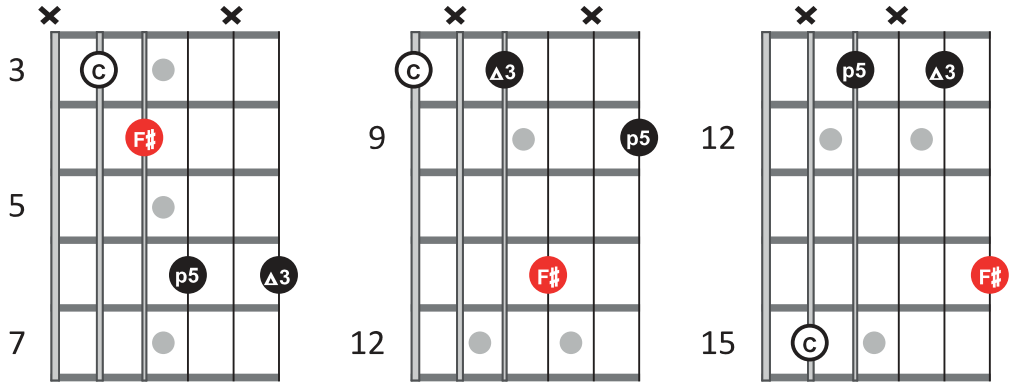


$\frac{D^{loc}}{C}$

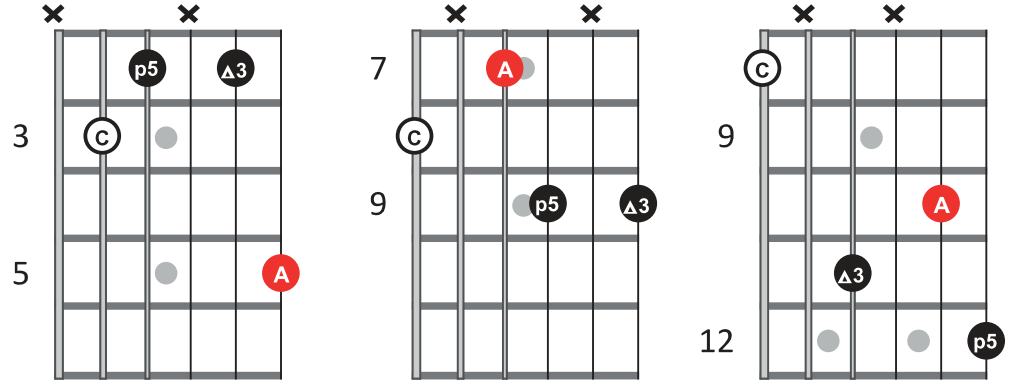


Dom^7 Type Spread Triads
Bass Notes (C)

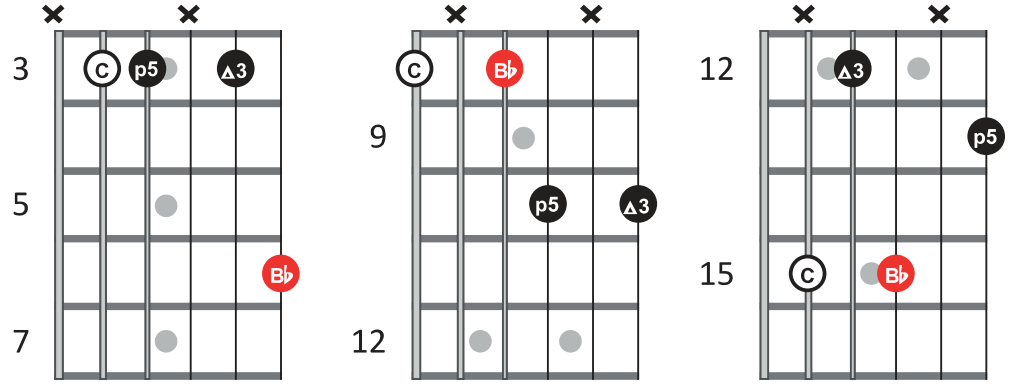
F#
C



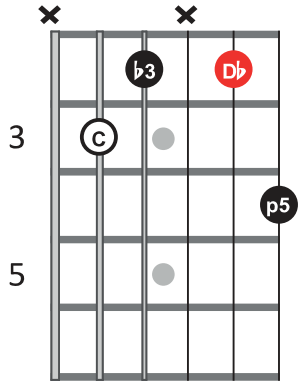
A
C

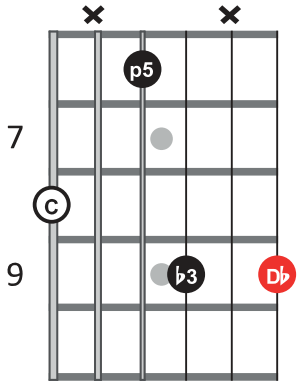


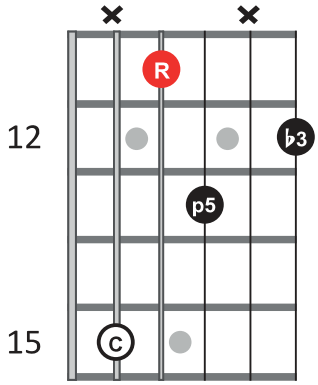
Bb
C



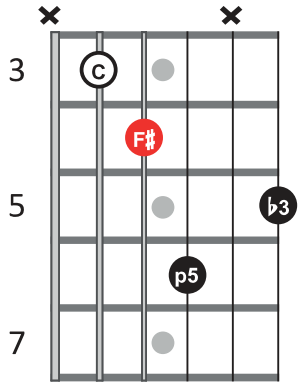
Db-
C

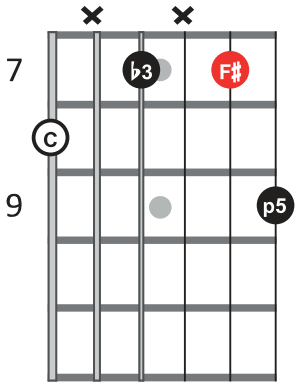

3
5
C
♭3
D♭
p5

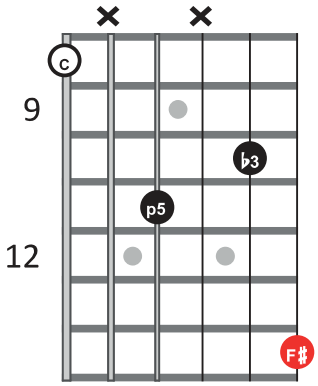

7
9
C
p5
♭3
D♭


12
15
C
R
♭3
p5

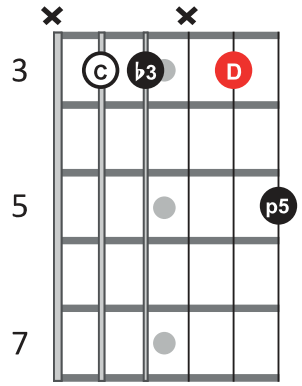
F#-
C

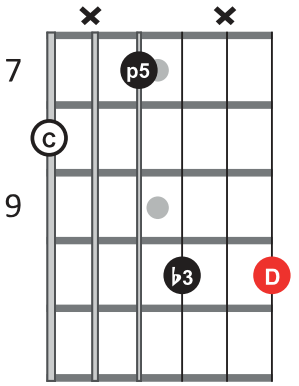

3
5
7
C
F♯
♭3
p5

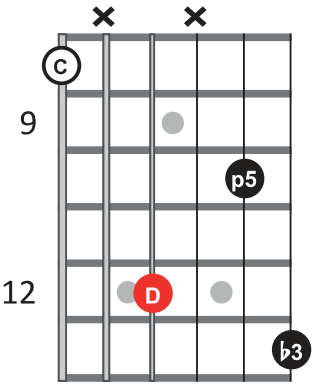

7
9
C
♭3
F♯
p5


9
12
C
♭3
p5
F♯

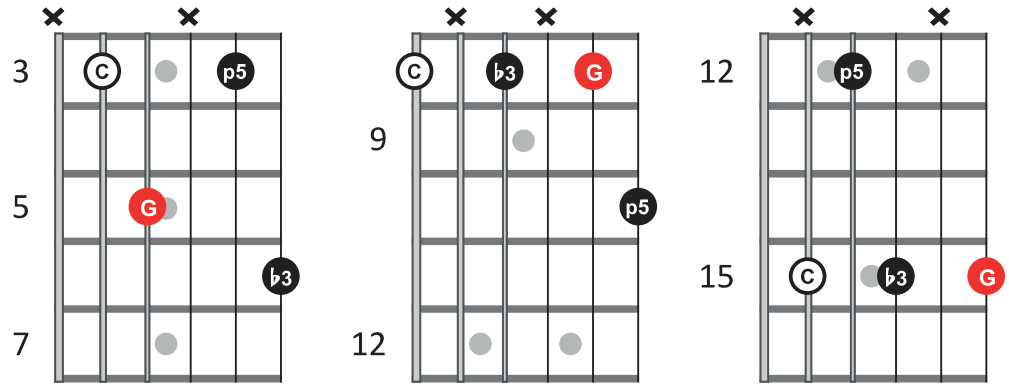
D-
C


3
5
7
C
♭3
D
p5

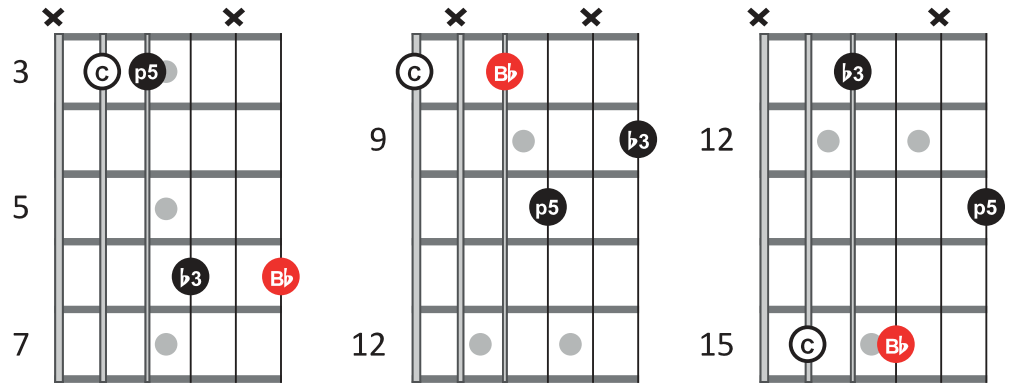

7
9
C
p5
♭3
D


9
12
C
p5
D
♭3

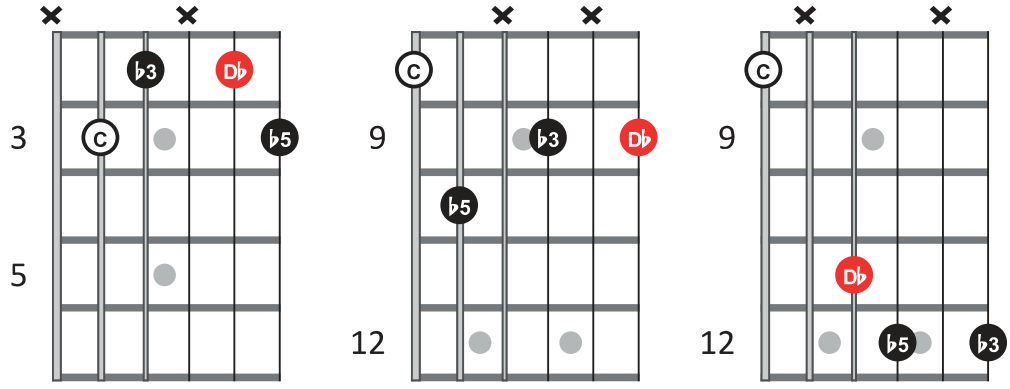
G-
C


3
5
7
C
p5
G
♭3
9
12
C
♭3
G
p5
12
15
C
p5
C
♭3
G

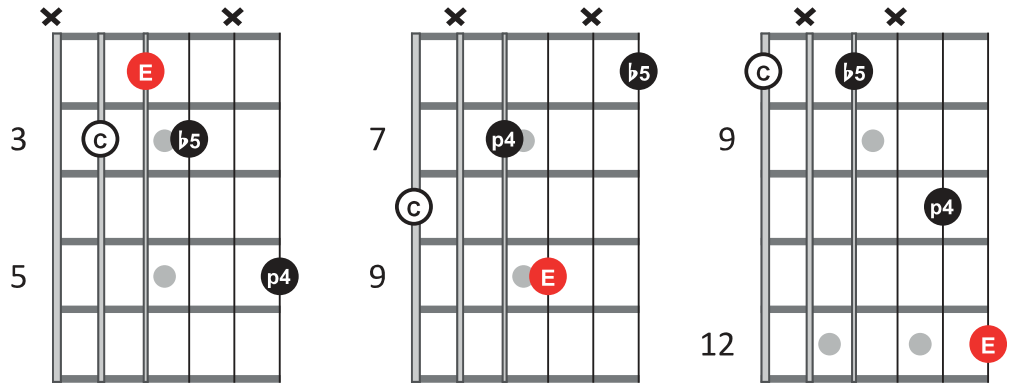
Bb-
C


3
5
7
C
p5
♭3
B♭
9
12
C
B♭
♭3
p5
12
15
C
♭3
p5
C
B♭

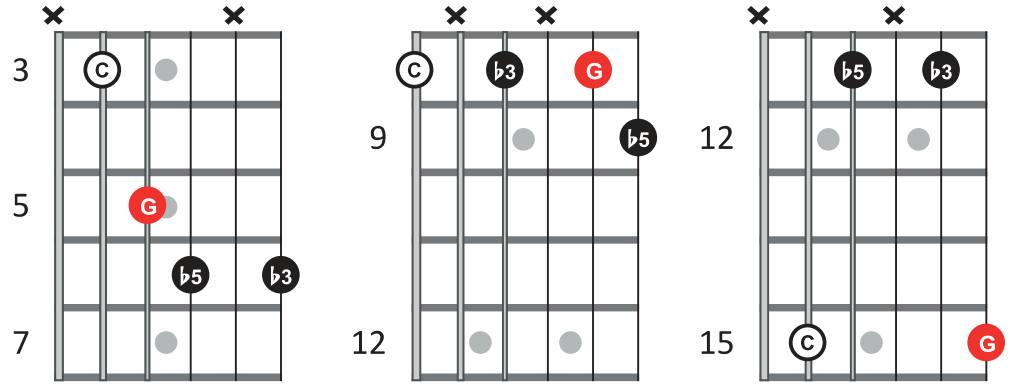
Db°
C


3
5
C
♭3
D♭
♭5
9
12
C
♭3
D♭
♭5
9
12
C
D♭
♭5
♭3

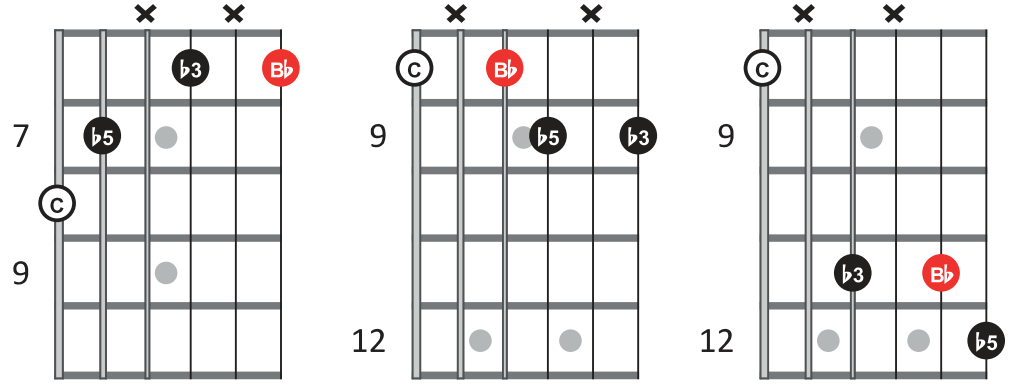
E°/C


3
E
C
♭5
5
p4
7
♭5
p4
C
9
E
9
C
♭5
p4
12
E

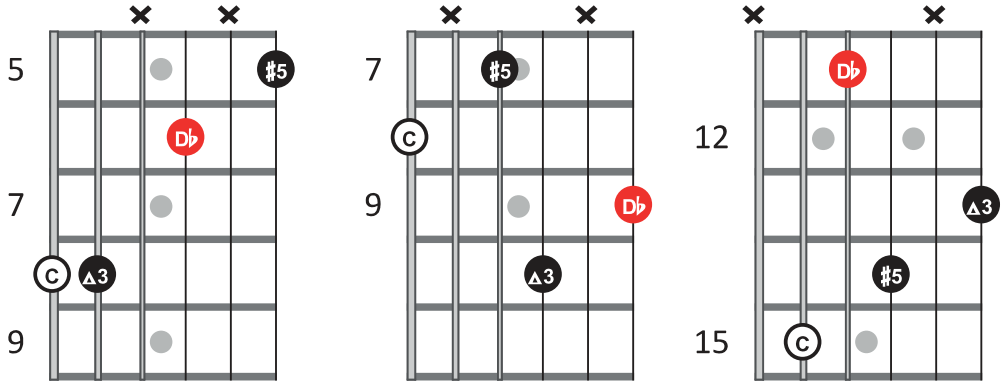
G°/C


3
C
5
G
♭5
♭3
7
9
C
♭3
G
♭5
12
12
♭5
♭3
15
C
G

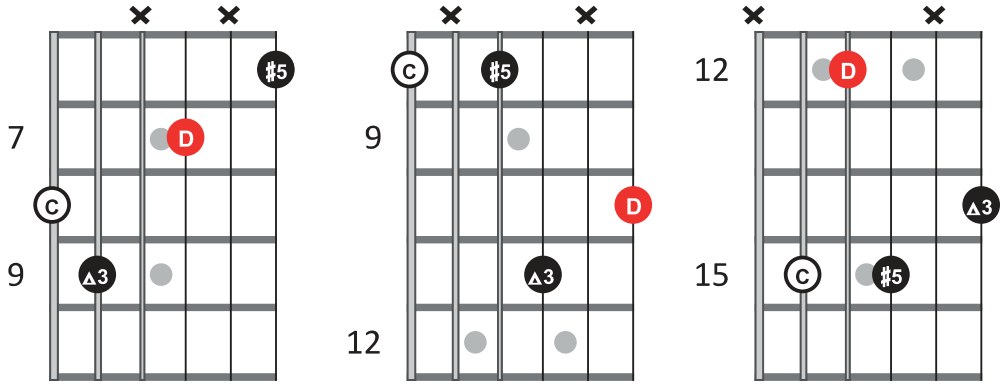
B♭°/C


7
♭3
B♭
♭5
C
9
9
B♭
C
♭5
♭3
12
9
C
♭3
B♭
12
♭5

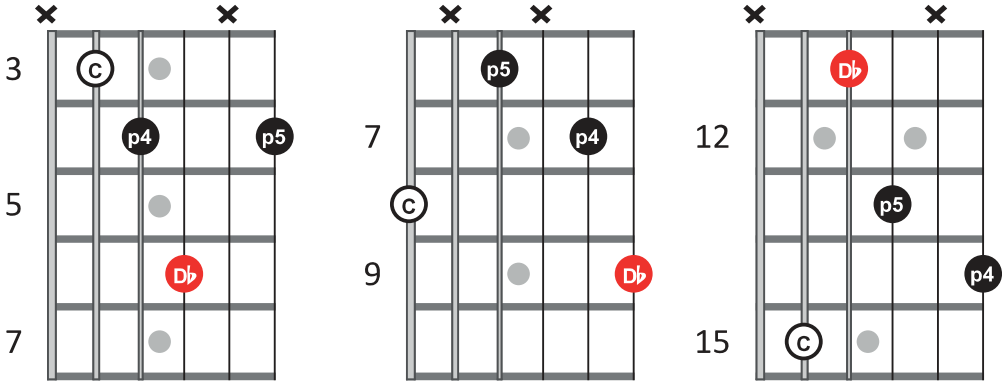
Db+
C



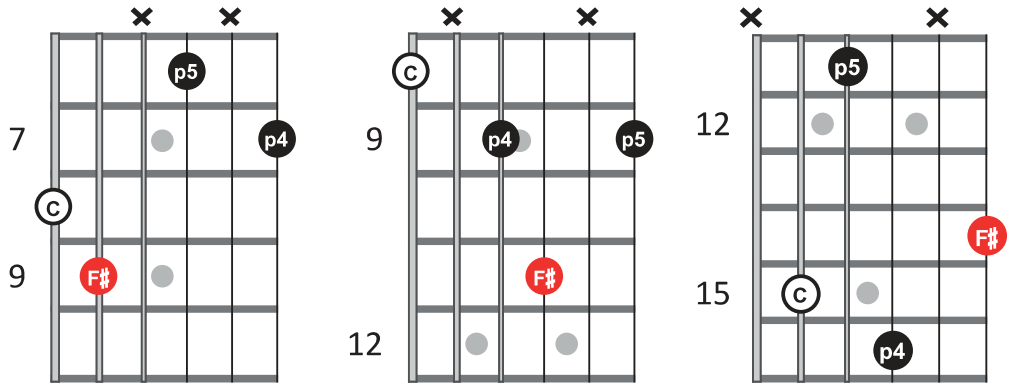
D+
C



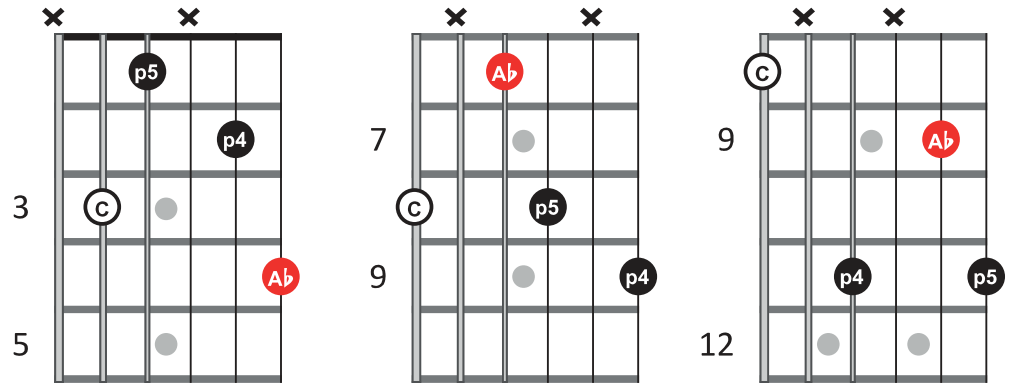
Db^{sus4}
C



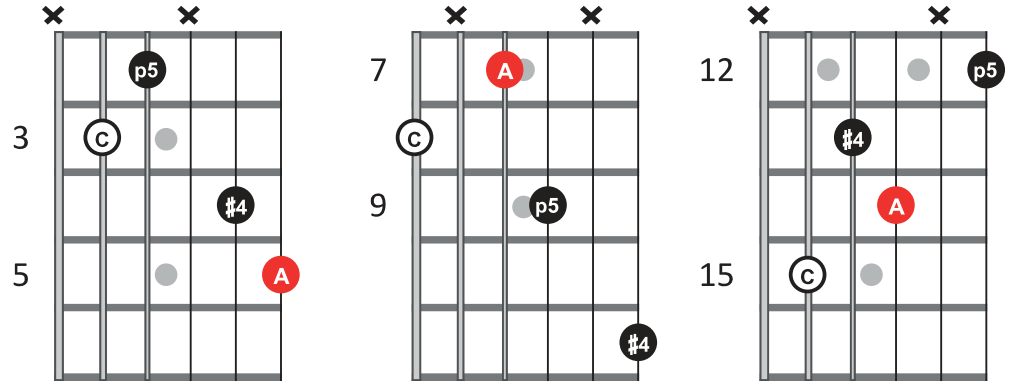
F#^{sus4} / C



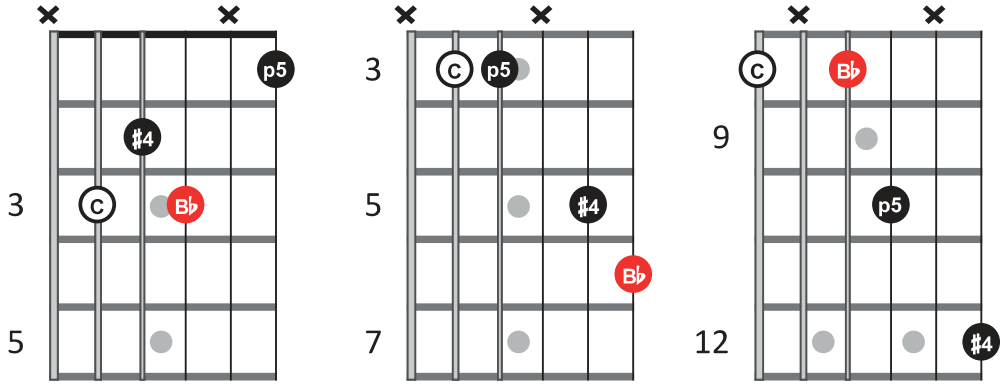
Ab^{sus4} / C



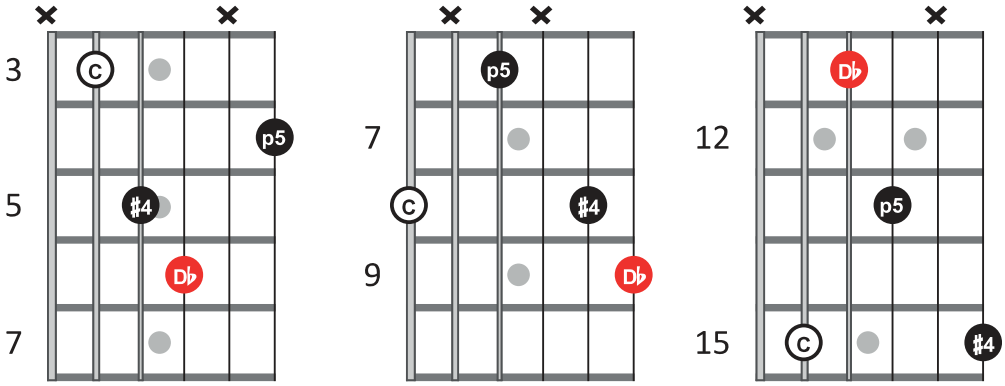
A^{lyd} / C



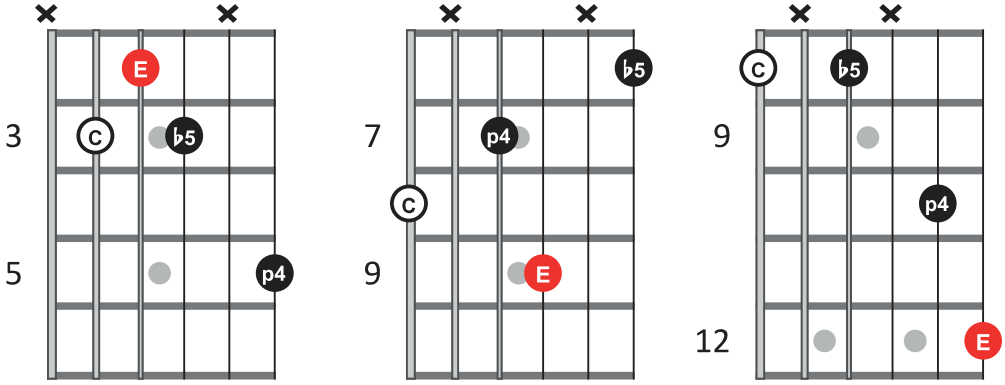
B♭lyd / C



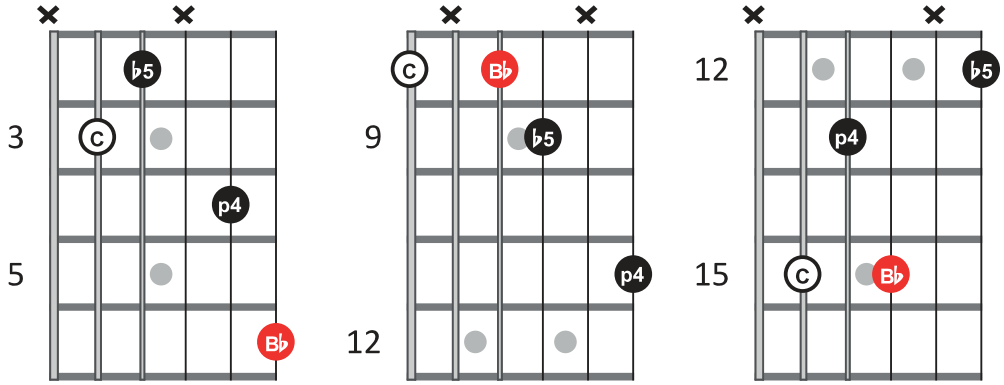
D♭lyd / C



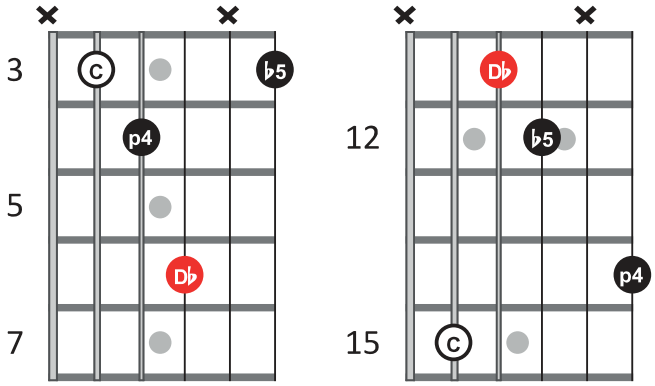
E♭loc / C



$\frac{Bb^{loc}}{C}$

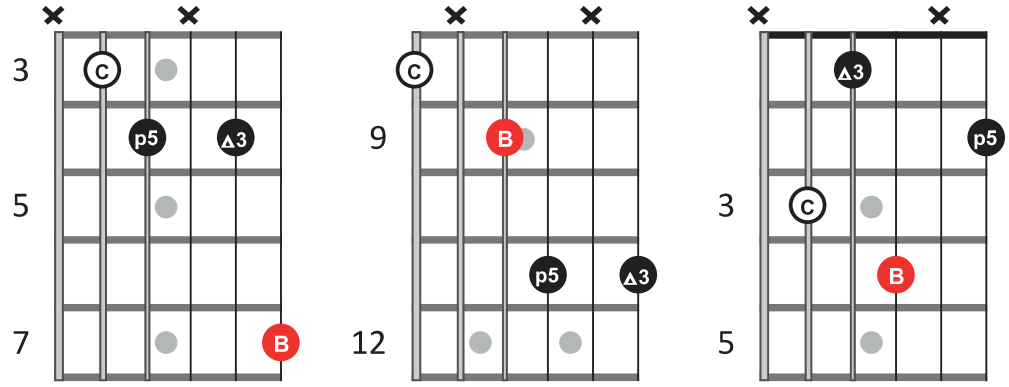


$\frac{Db^{loc}}{C}$

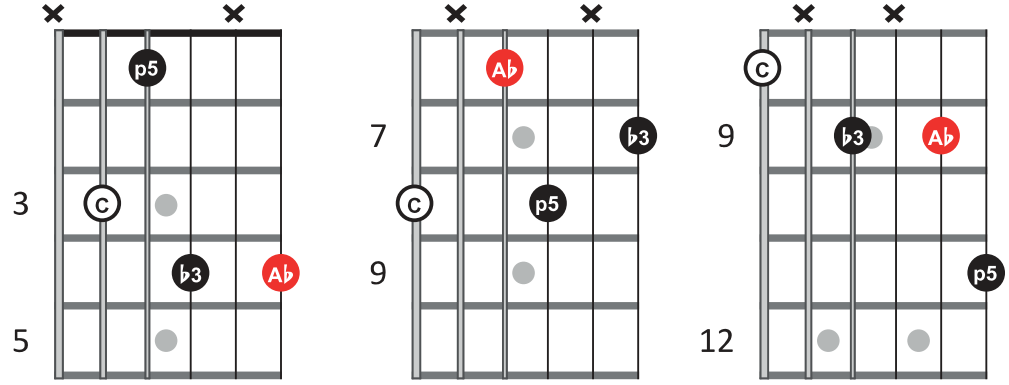


Diminished Type Spread Triads
Bass Notes (C)

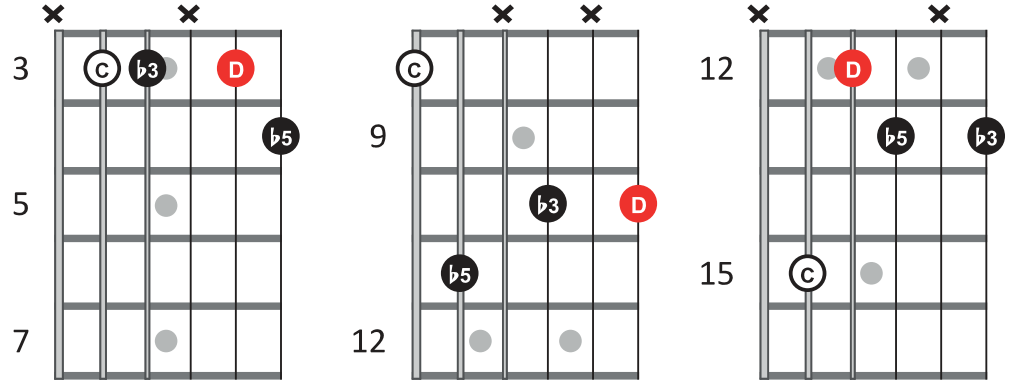
B
C



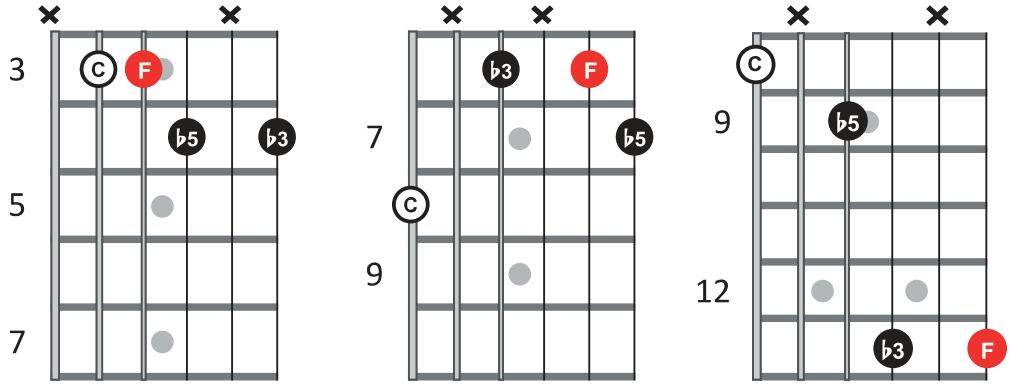
Ab-
C



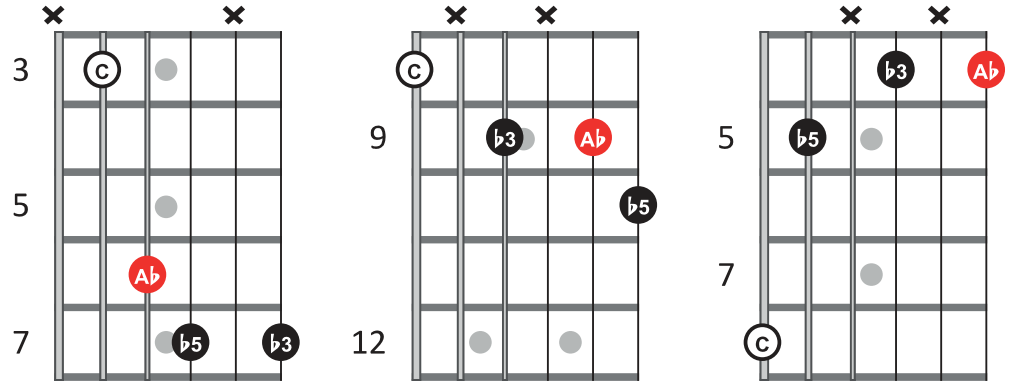
D°
C



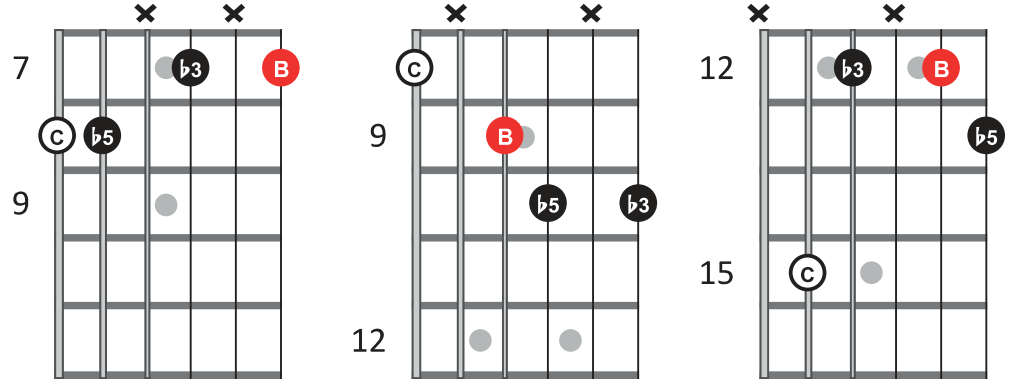
F°/C


3
C
F
♭5
♭3
5
7
7
♭3
F
♭5
9
C
9
9
♭5
C
12
♭3
F

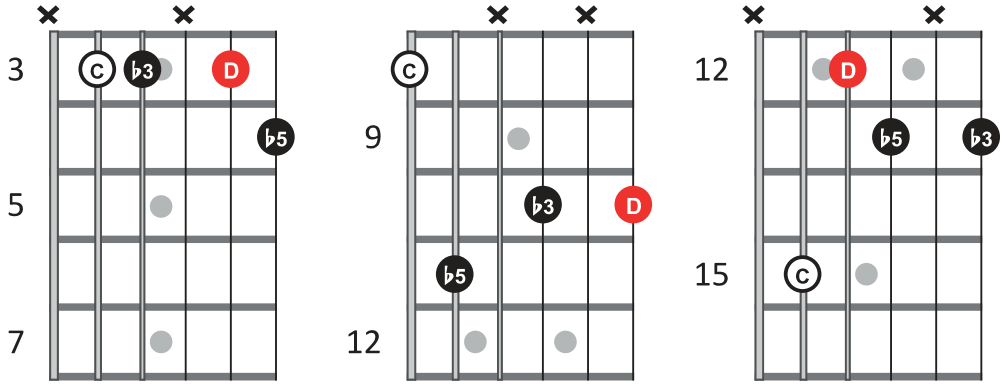
A♭°/C


3
C
5
A♭
7
♭5
♭3
9
C
♭3
A♭
♭5
12
5
♭5
♭3
A♭
7
C

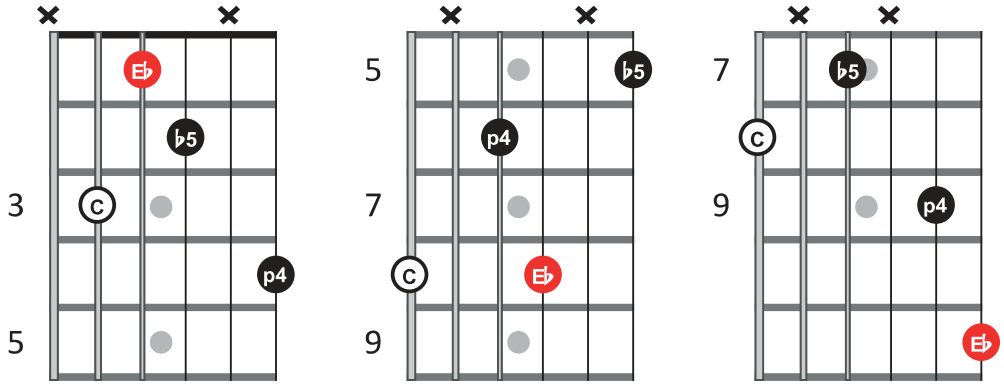
B°/C


7
♭3
B
C
♭5
9
9
B
♭5
♭3
12
12
♭3
B
♭5
15
C

$\frac{D^{o}}{C}$

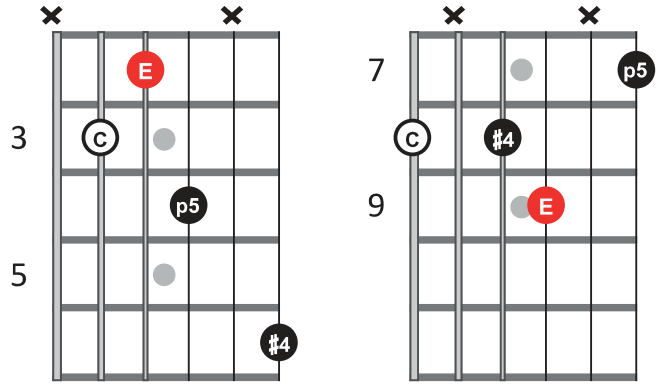


$\frac{Eb^{loc}}{C}$

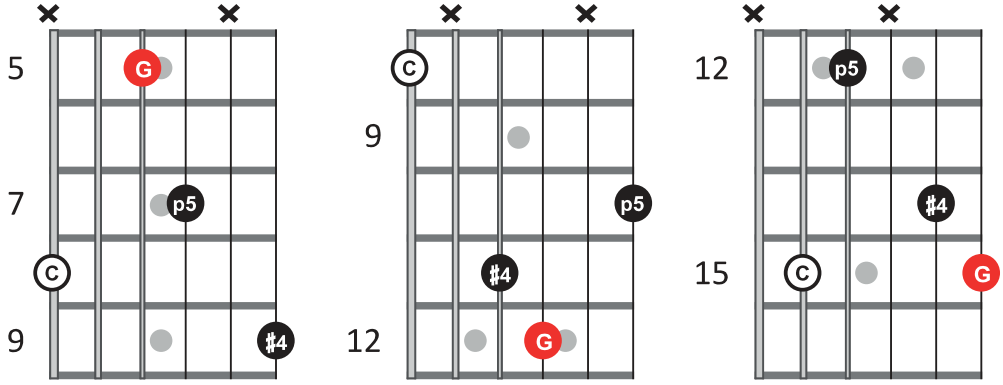


Twelve Tone Type Spread Triads
Bass Notes (C)

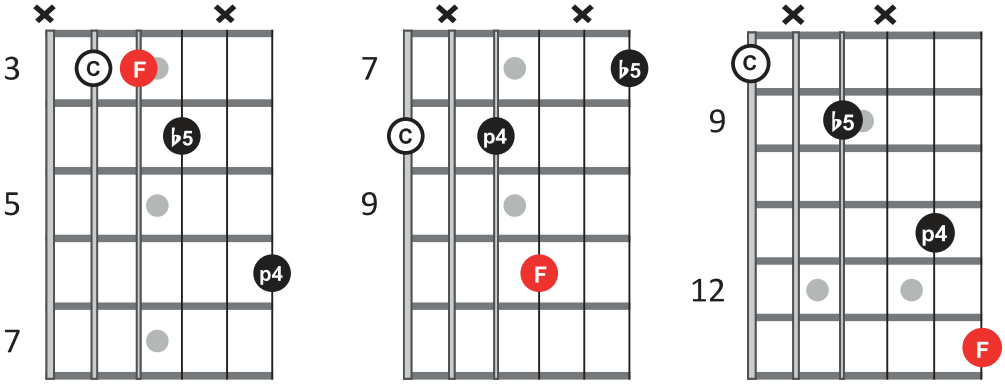
E^{lyd} / C



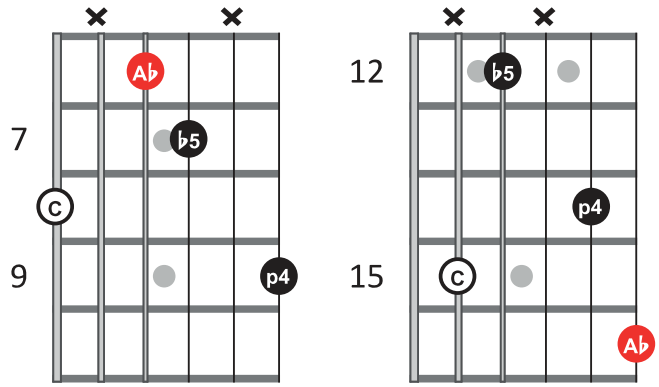
G^{lyd} / C



F^{loc} / C



$\frac{Ab^{loc}}{C}$

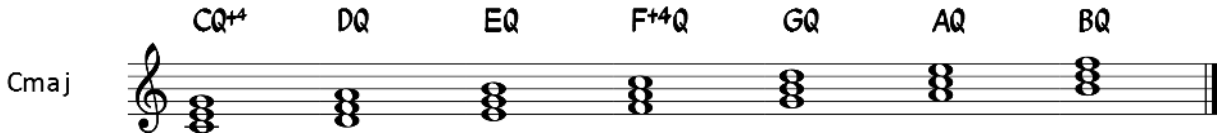


Quartal Voicings

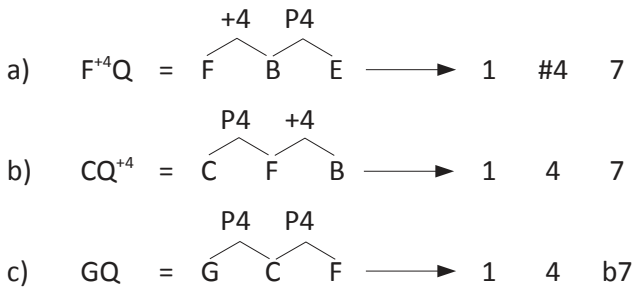
Quartal voicings are constructed by the superimposition of diatonic fourth intervals. The three part quartal voicing is most commonly used because it functions nicely in an upper-structure capacity. For this reason the (Q) nomenclature has been devised.

Three Part Quartal Voicings

Ex. 140



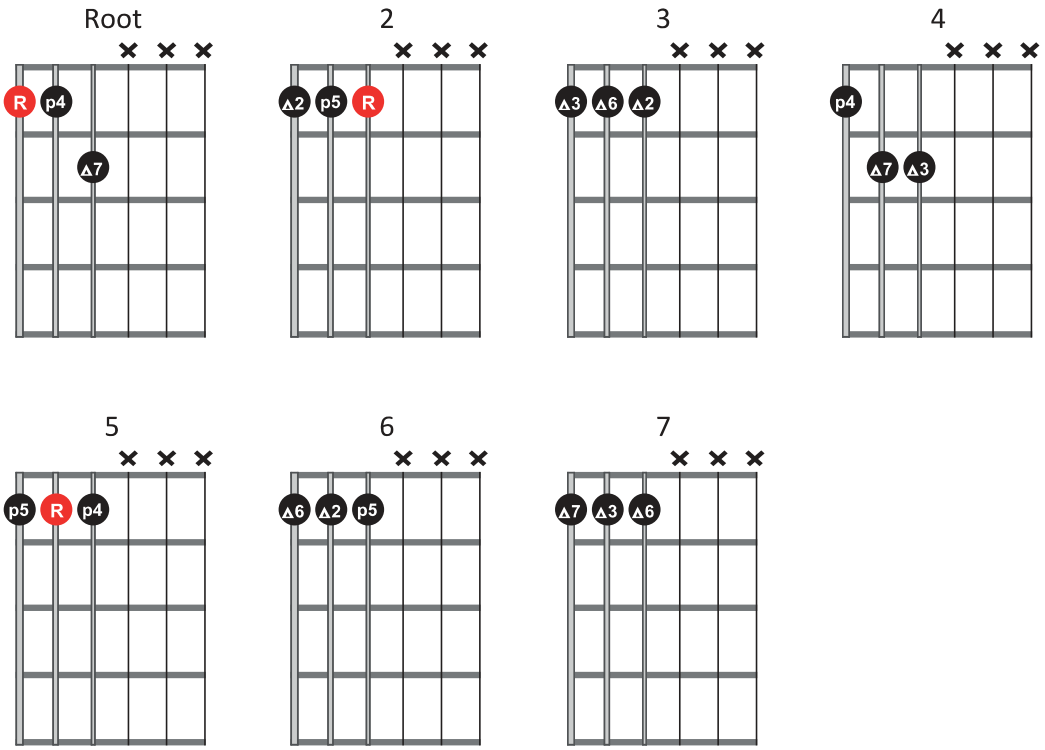
Ex. 141



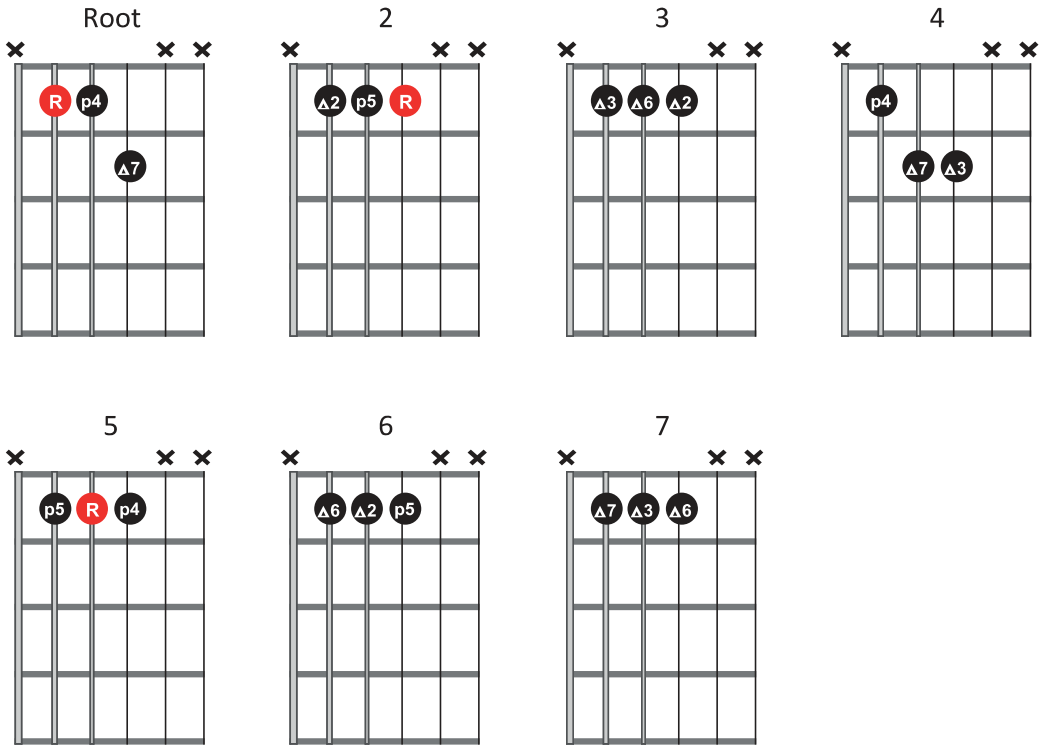
Four, Five, and Six-Part Quartal Voicings have not been specifically labeled.

3part-Quartal Voicings – Major Scale

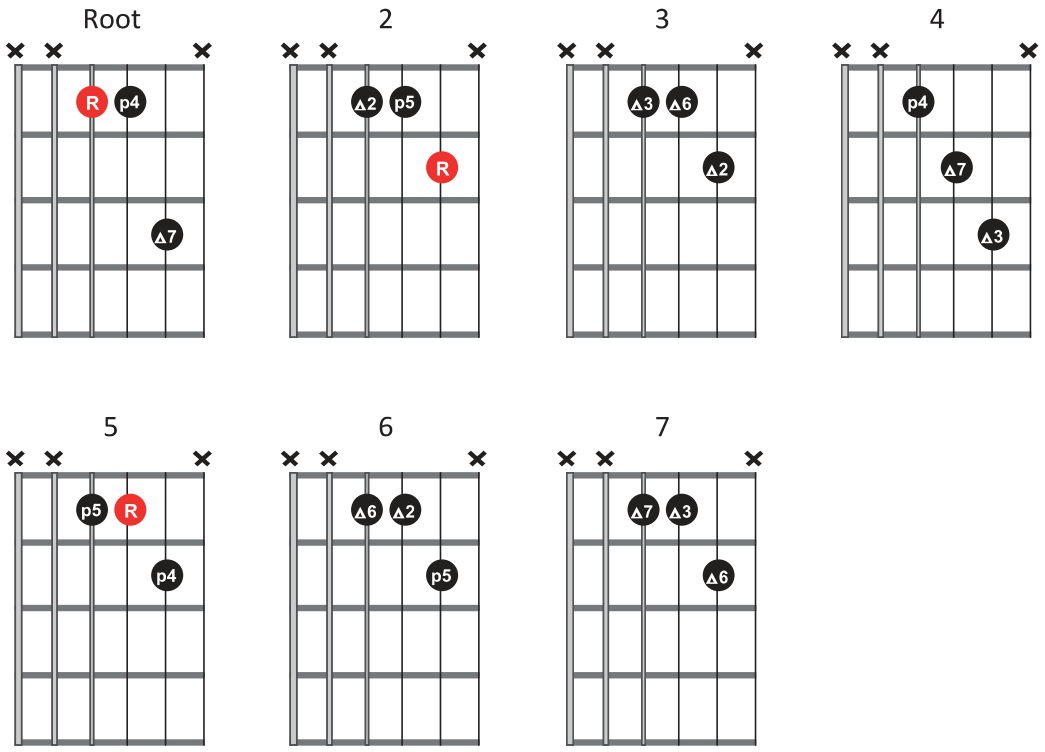
Stringset E-A-D



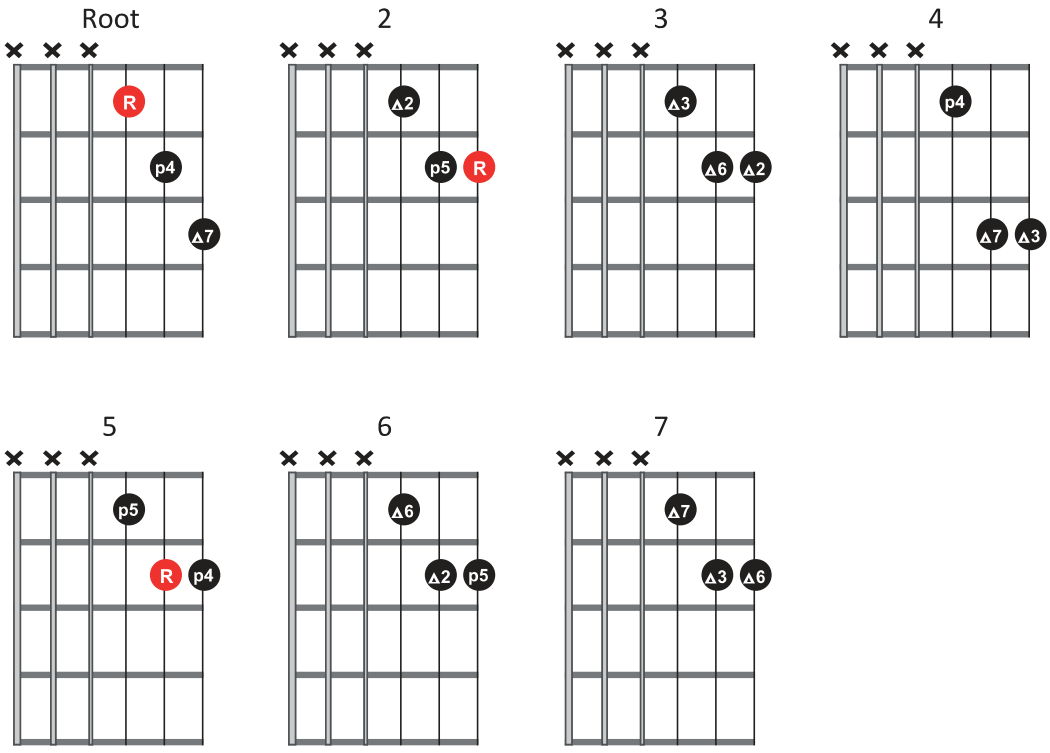
Stringset A-D-G



Stringset D-G-B

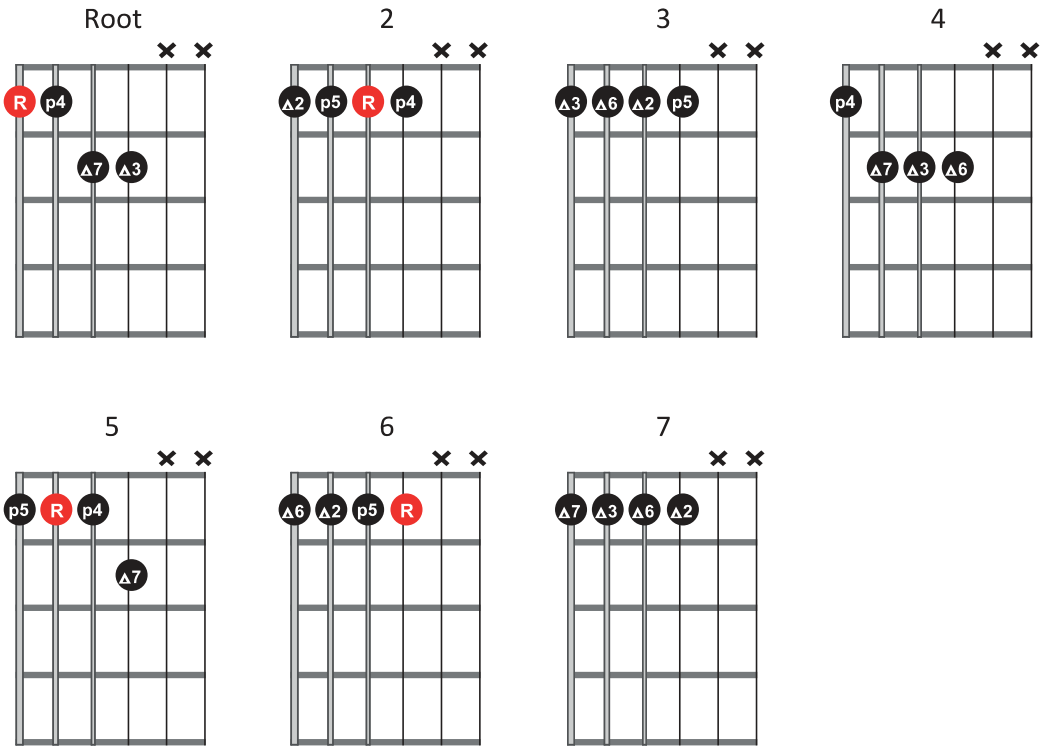


Stringset G-B-E

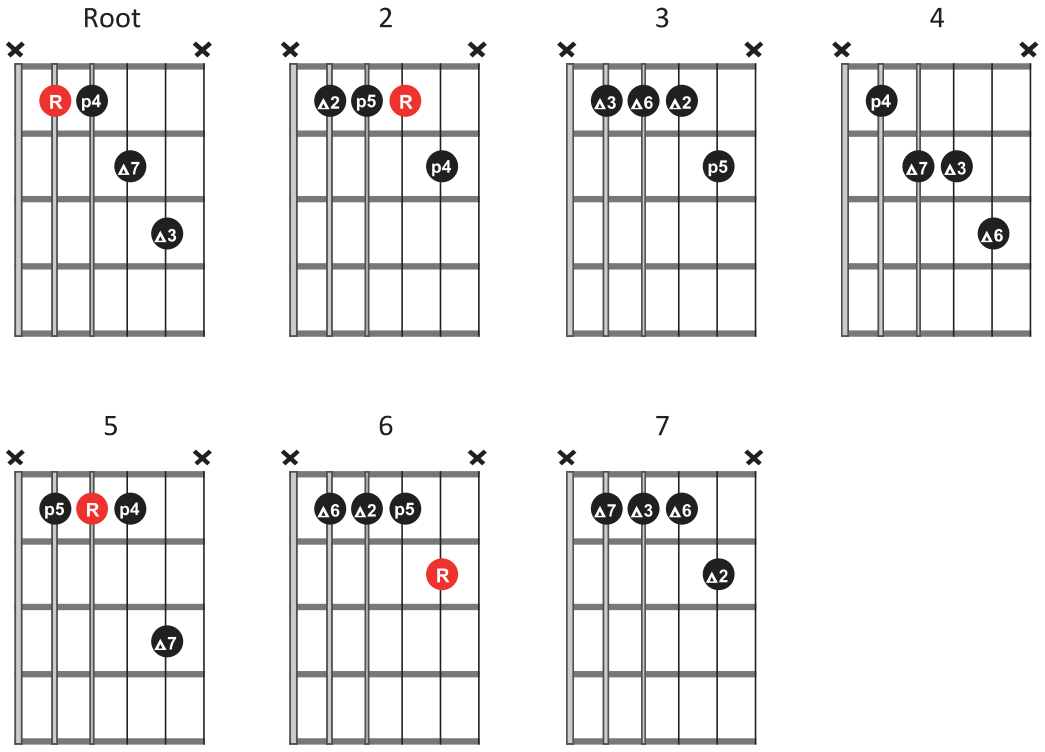


4part-Quartal Voicings – Major Scale

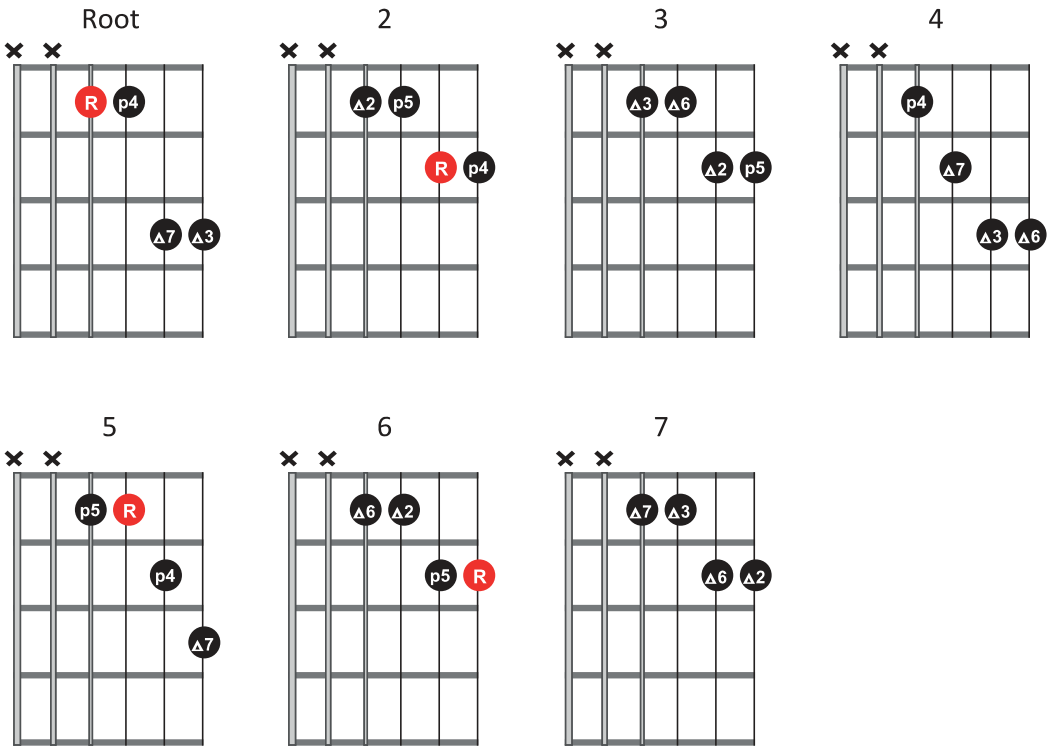
Stringset E-A-D-G



Stringset A-D-G-B

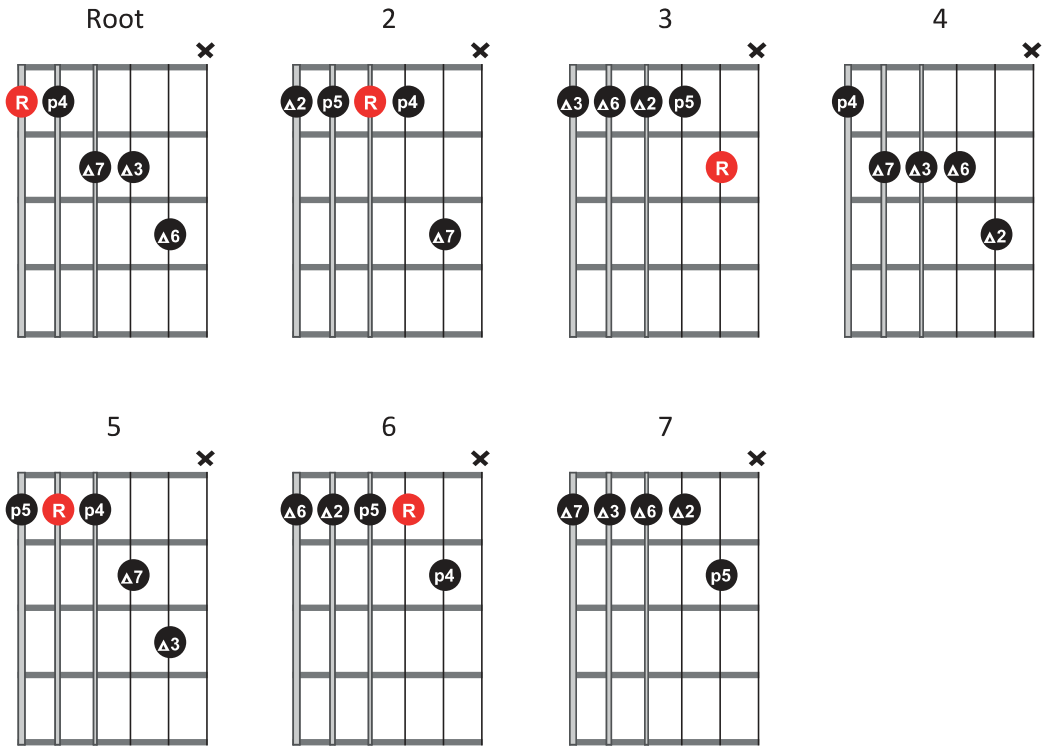


Stringset D-G-B-E

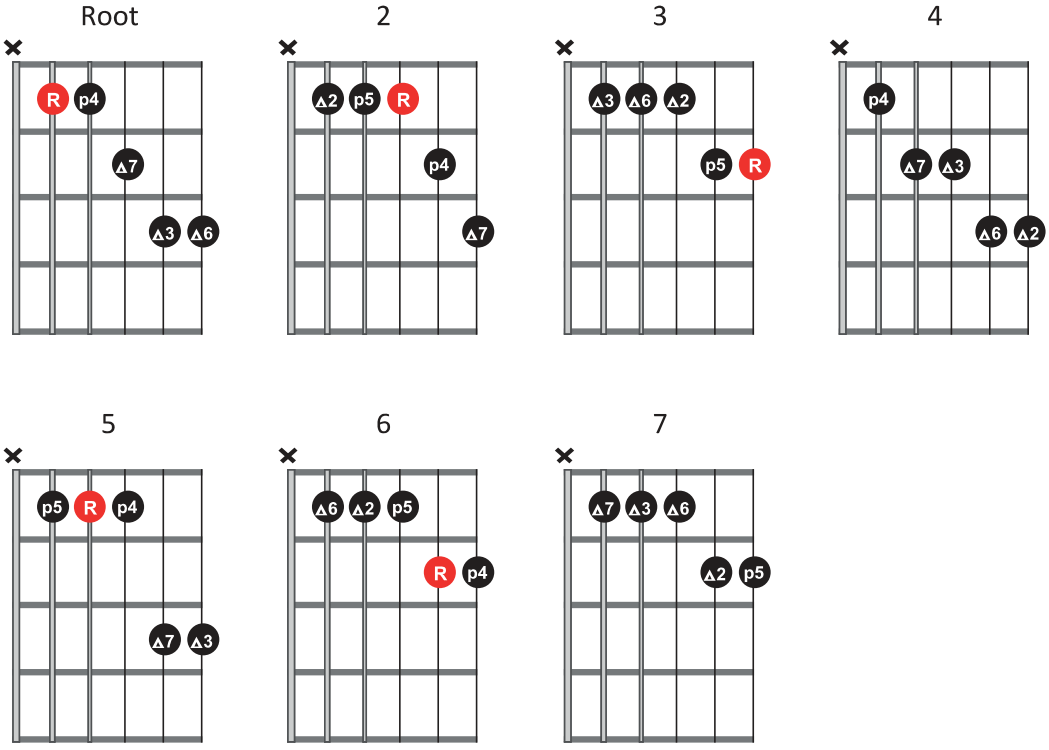


5part-Quartal Voicings – Major Scale

Stringset E-A-D-G-B



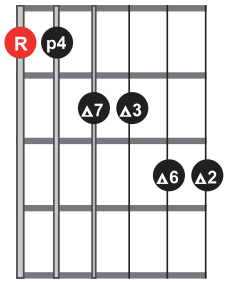
Stringset A-D-G-B-E



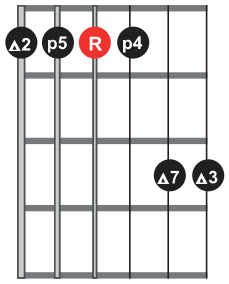
6part-Quartal Voicings – Major Scale

Stringset E-A-D-G-B-E

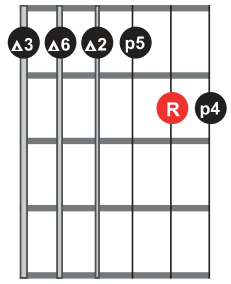
Root



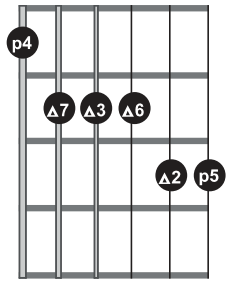
2



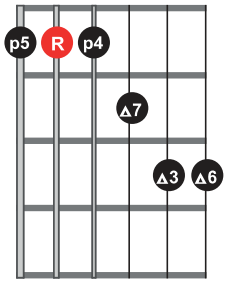
3



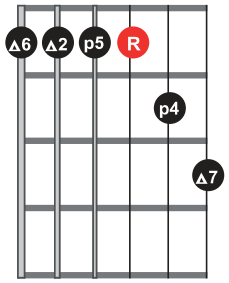
4



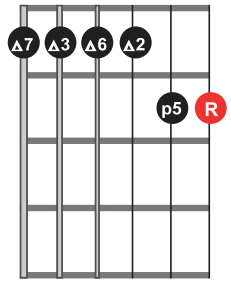
5



6

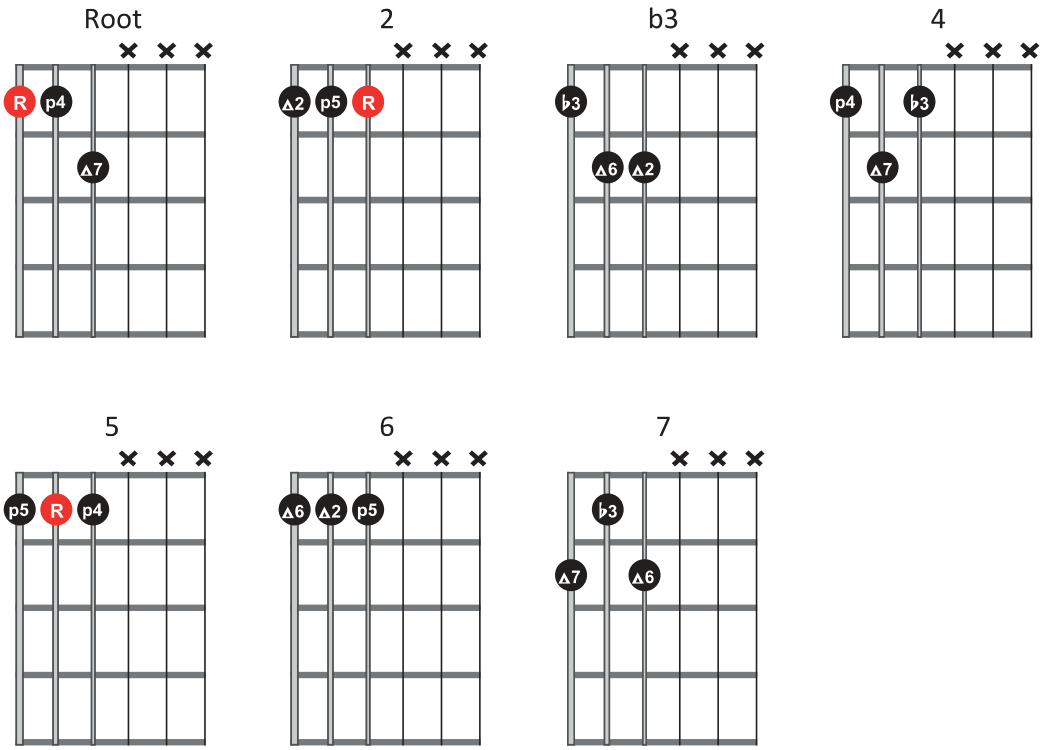


7

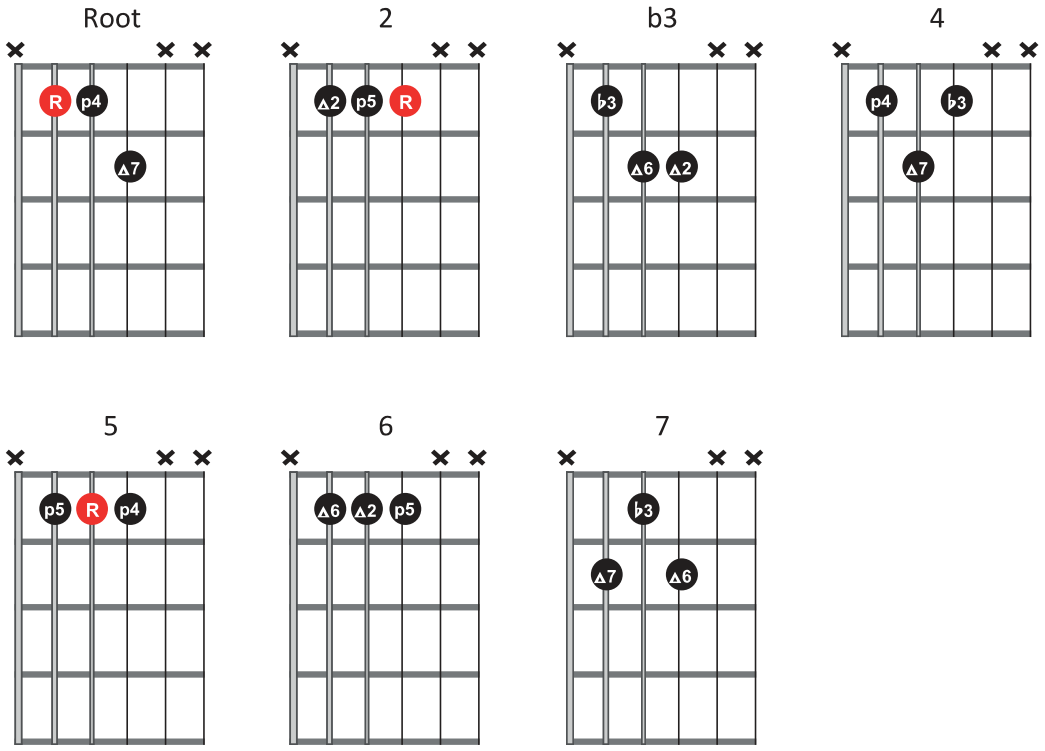


3part-Quartal Voicings – Melodic Minor

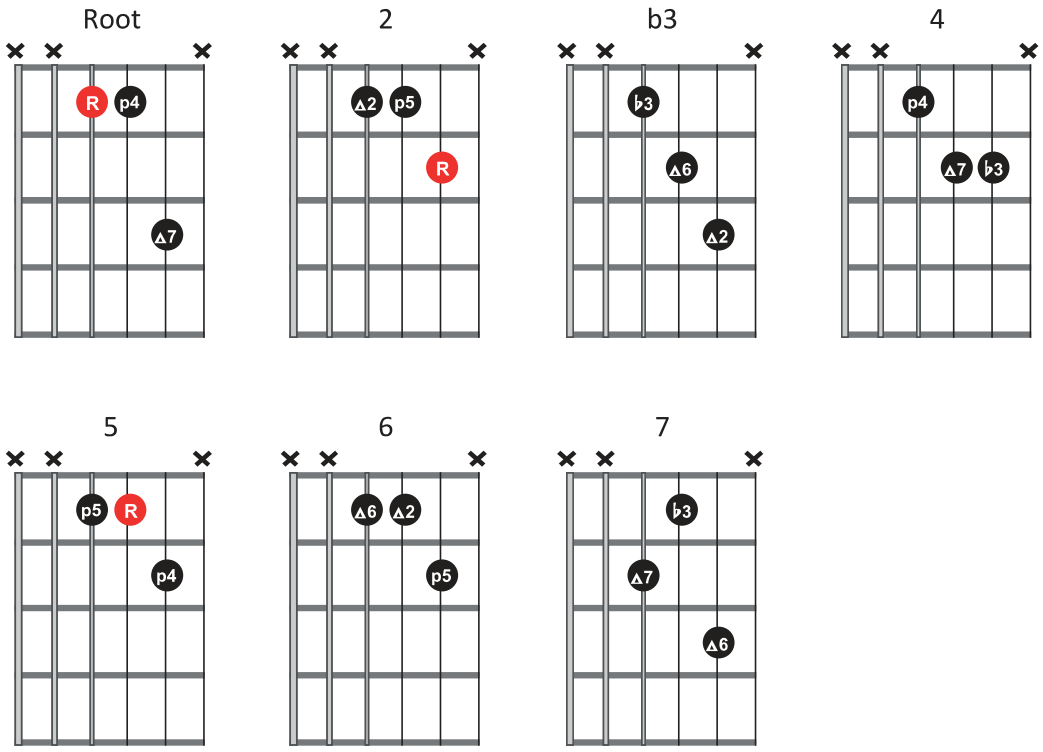
Stringset E-A-D



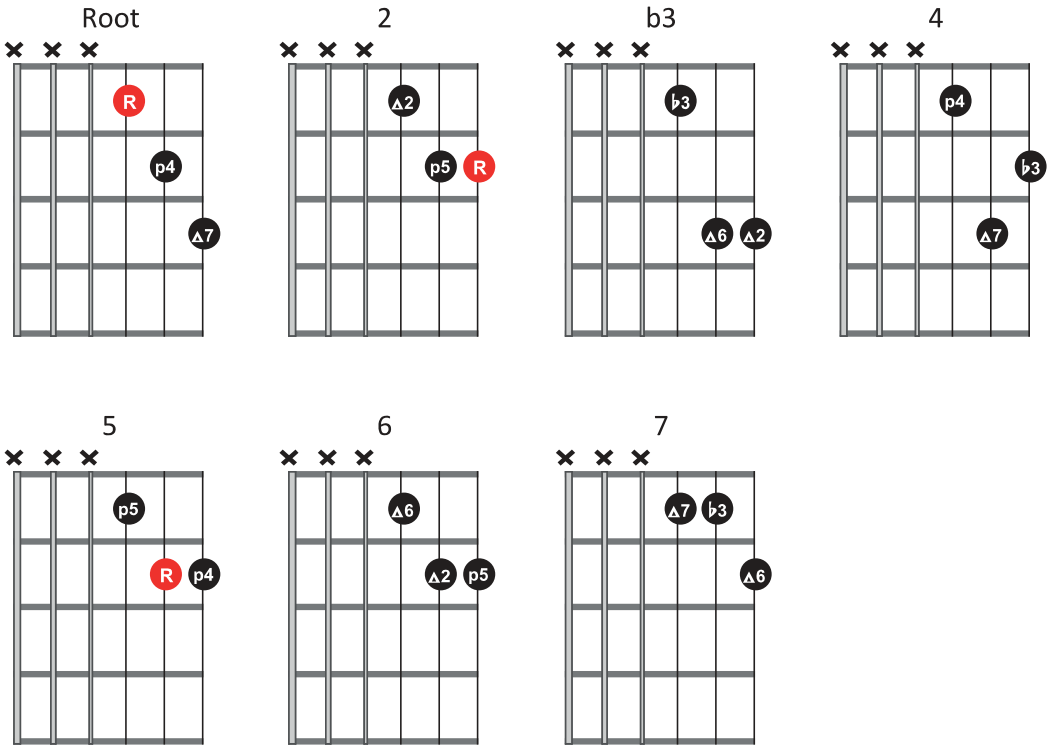
Stringset A-D-G



Stringset D-G-B

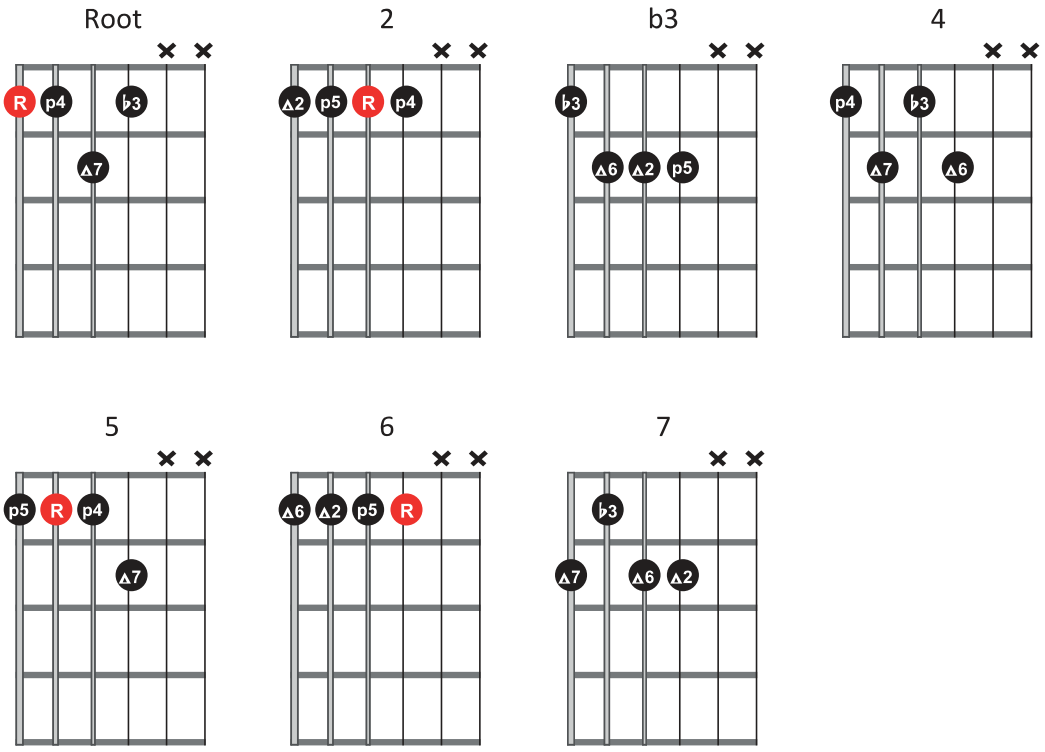


Stringset G-B-E

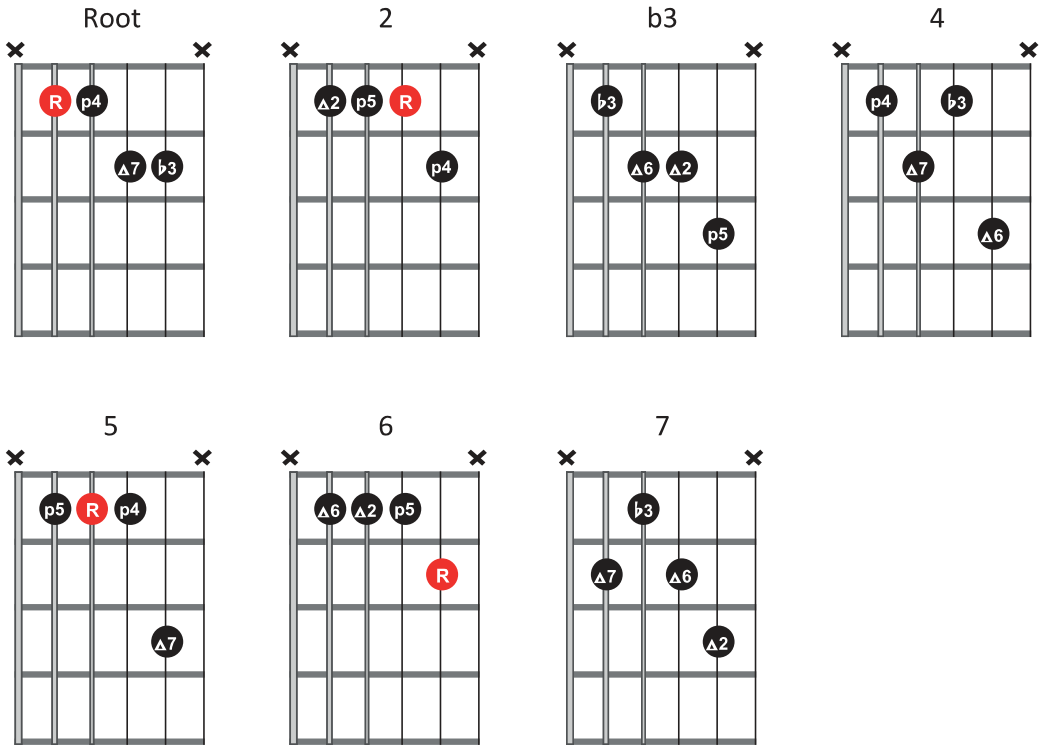


4part-Quartal Voicings – Melodic Minor

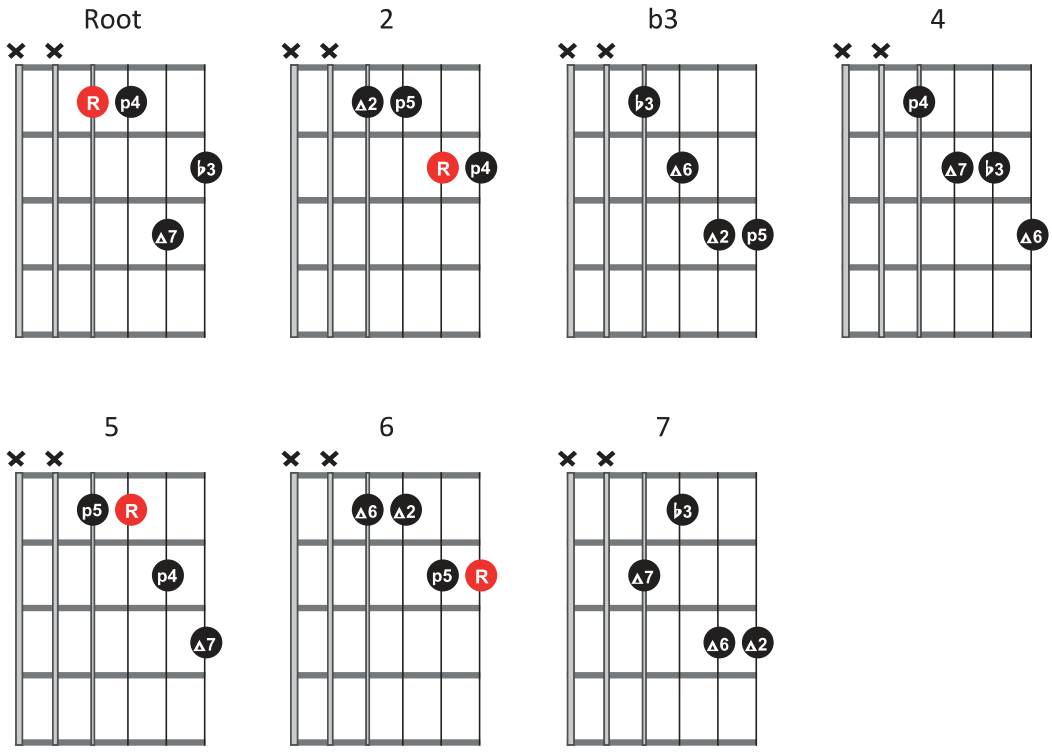
Stringset E-A-D-G



Stringset A-D-G-B

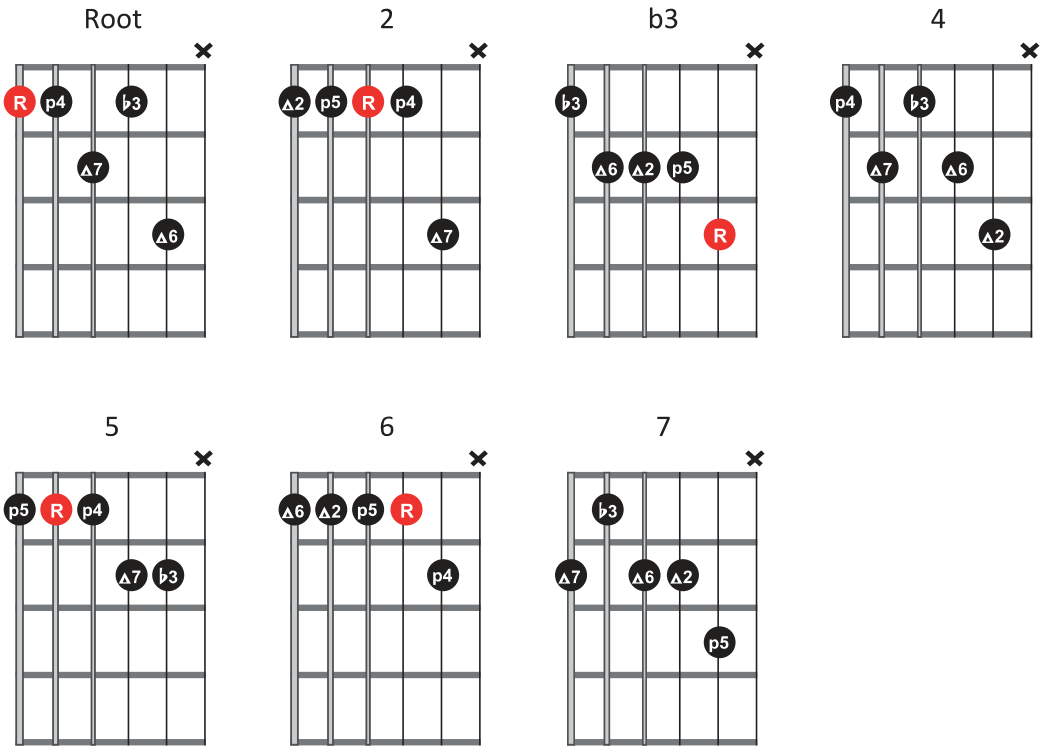


Stringset D-G-B-E

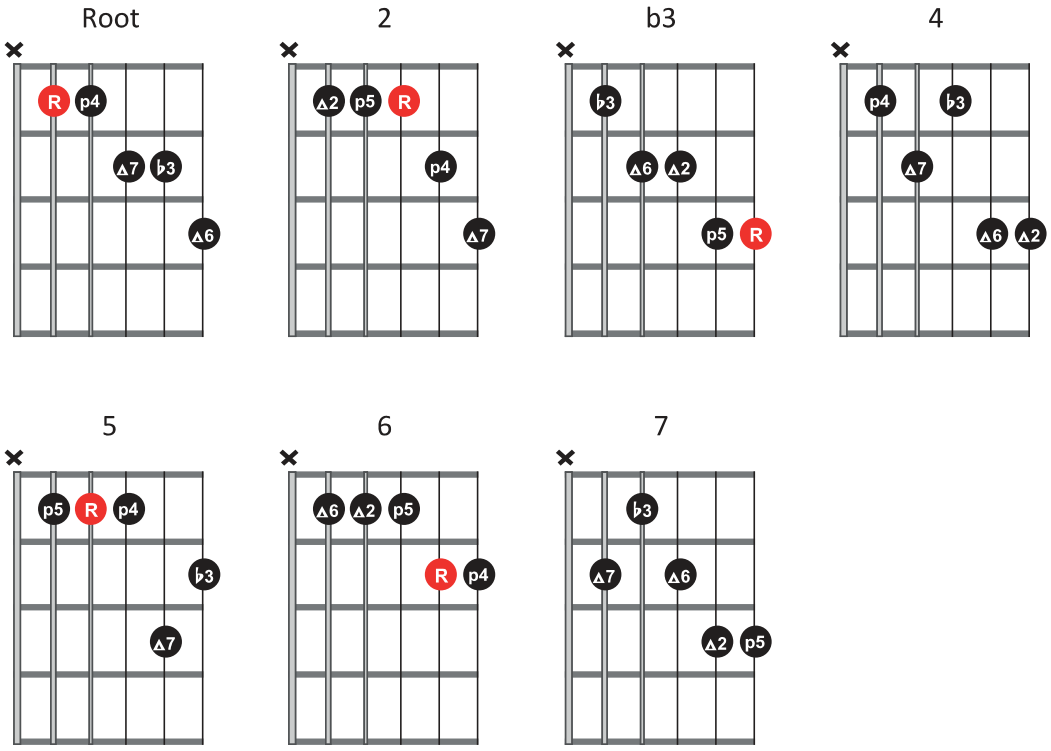


5part-Quartal Voicings – Melodic Minor

Stringset E-A-D-G-B



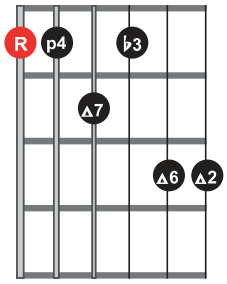
Stringset A-D-G-B-E



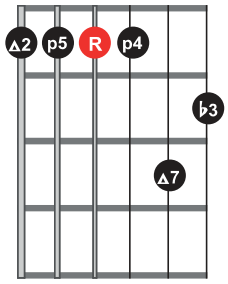
6part-Quartal Voicings – Melodic Minor

Stringset E-A-D-G-B-E

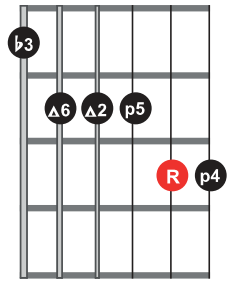
Root



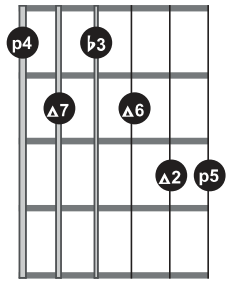
2



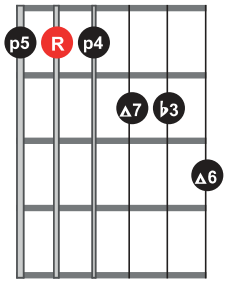
b3



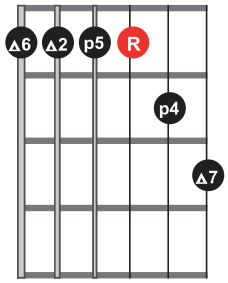
4



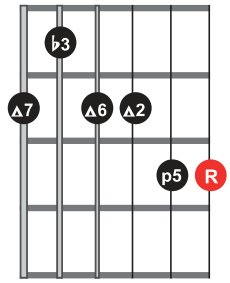
5



6

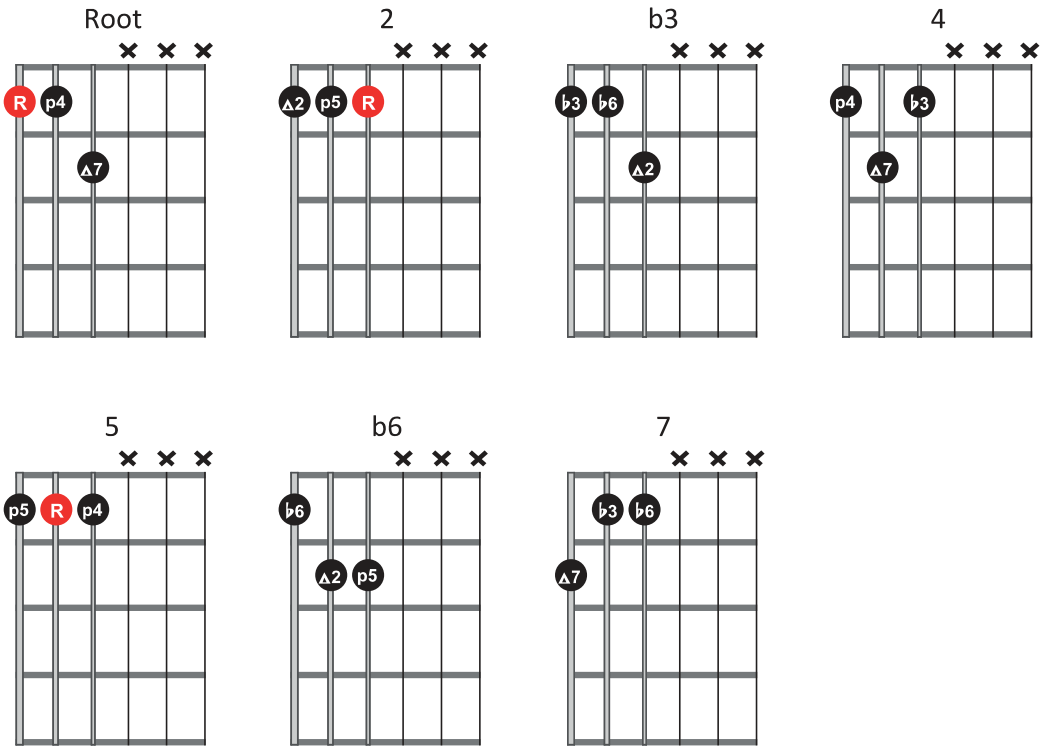


7

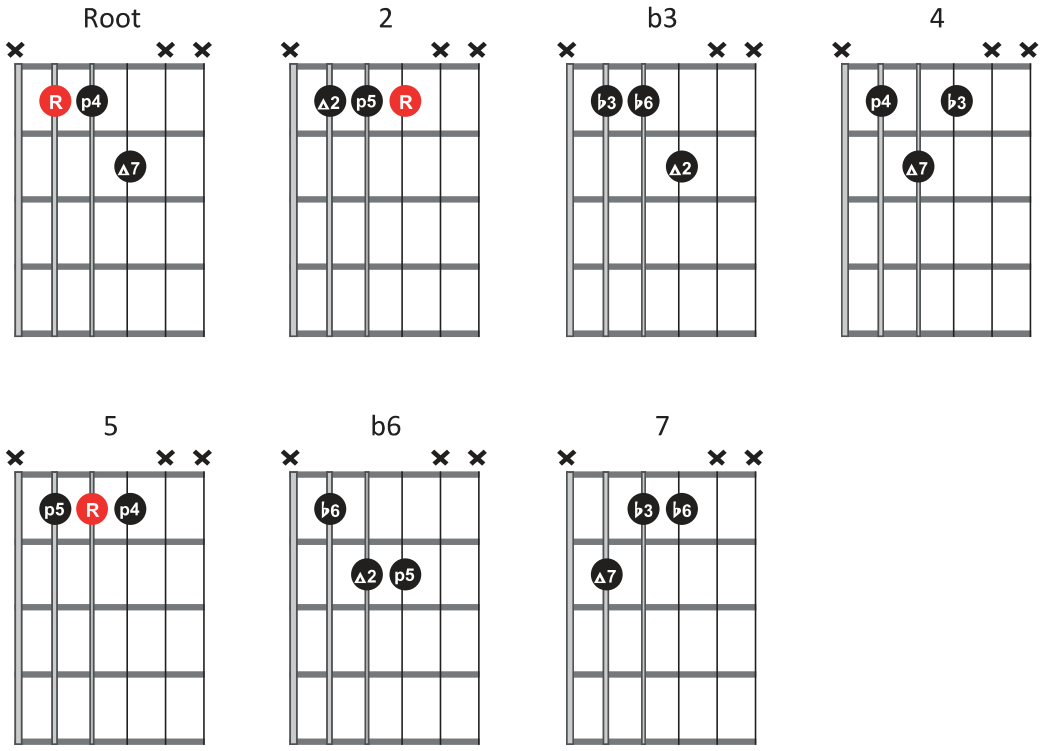


3part-Quartal Voicings – Harmonic Minor

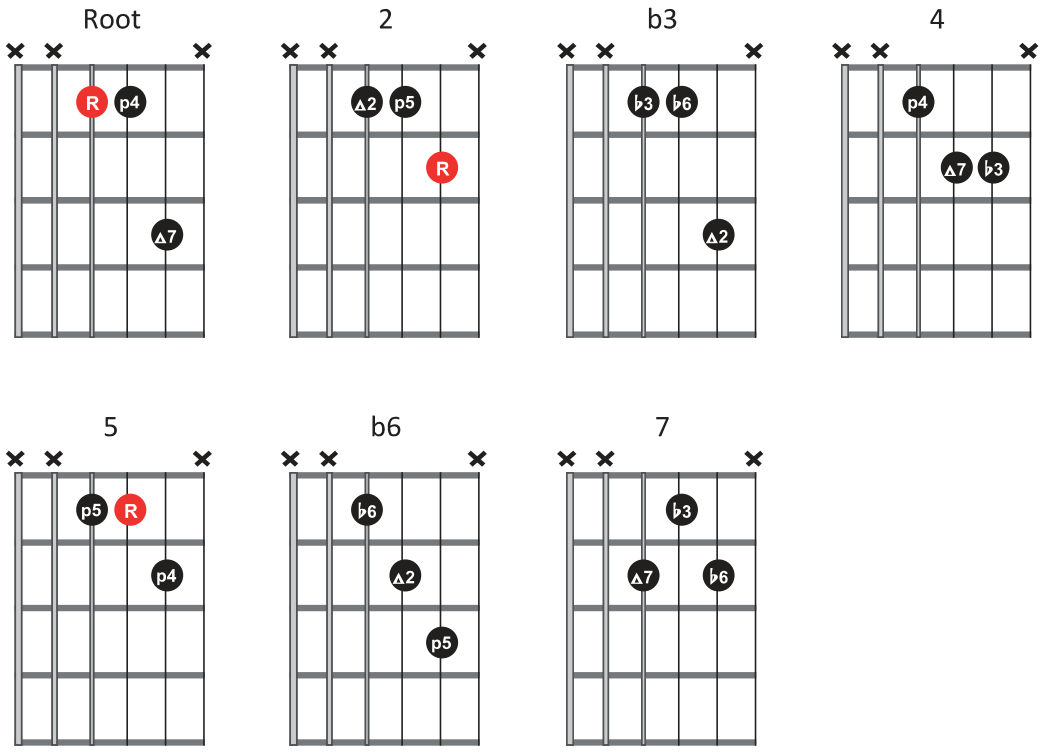
Stringset E-A-D



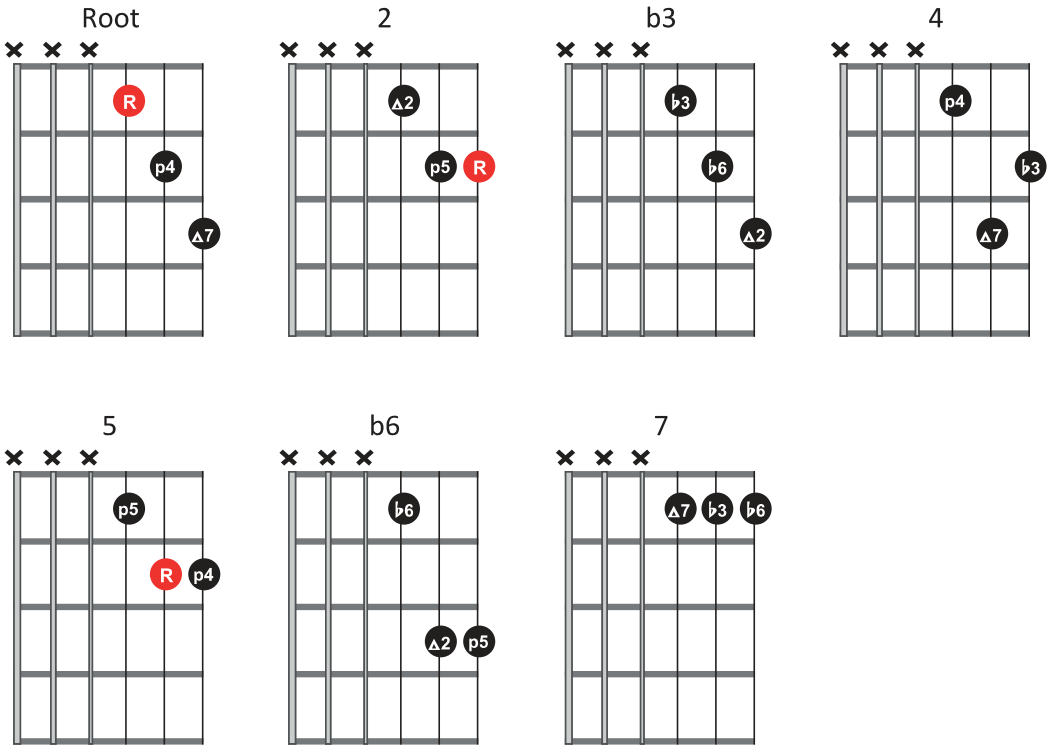
Stringset A-D-G



Stringset D-G-B

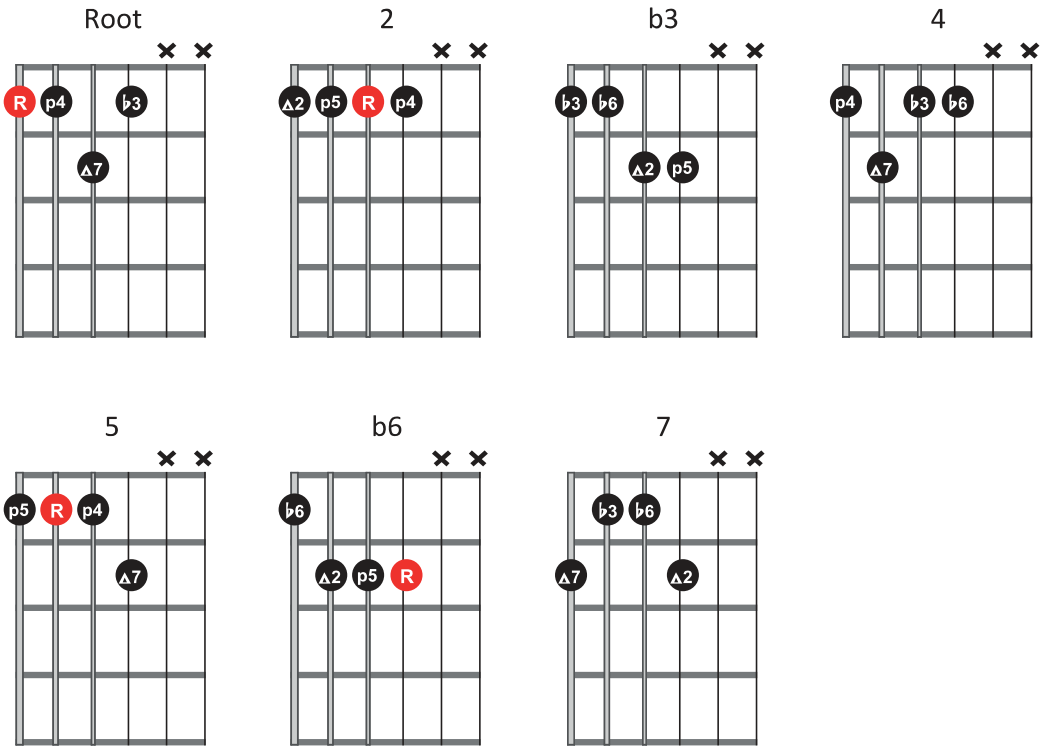


Stringset G-B-E

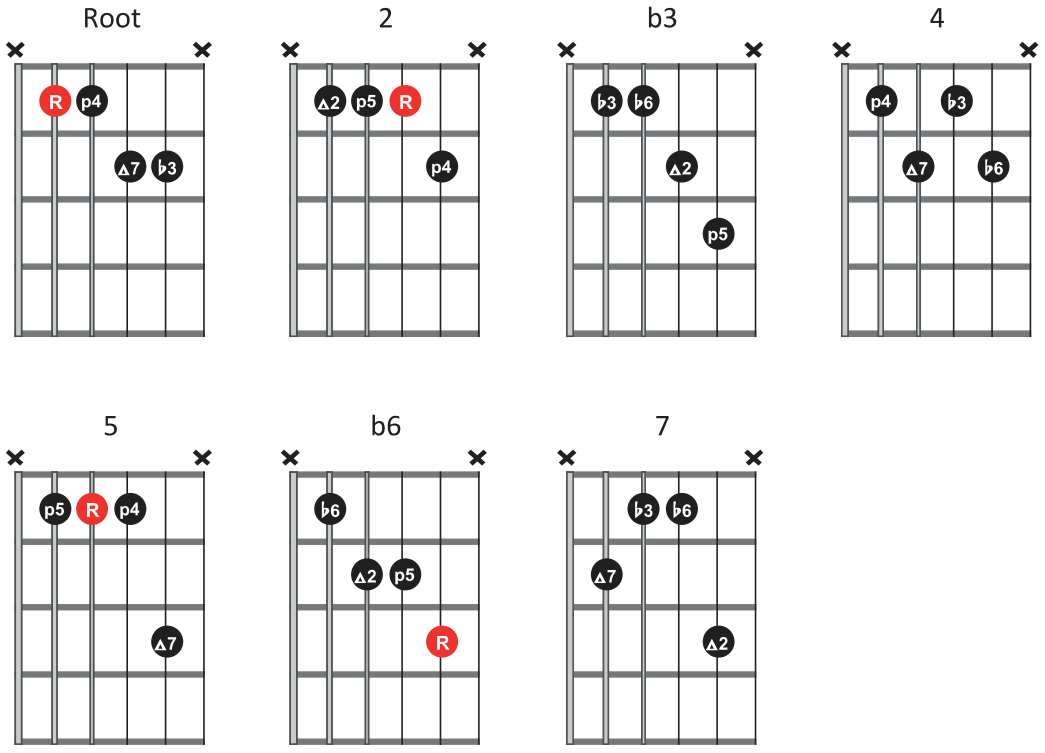


4part-Quartal Voicings – Harmonic Minor

Stringset E-A-D-G

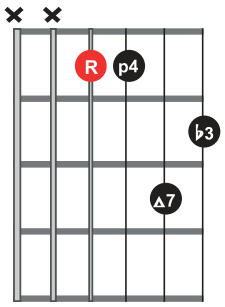


Stringset A-D-G-B

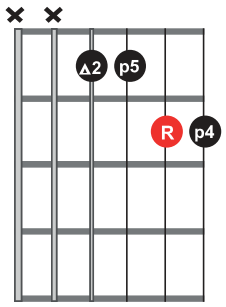


Stringset D-G-B-E

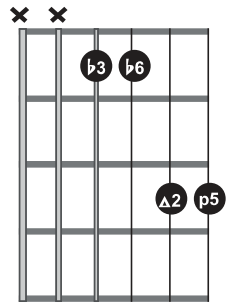
Root



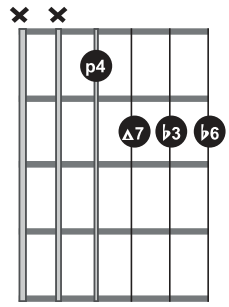
2



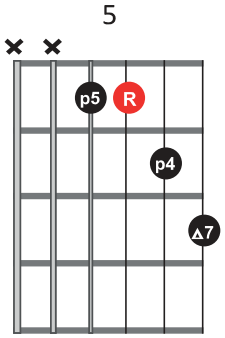
b3



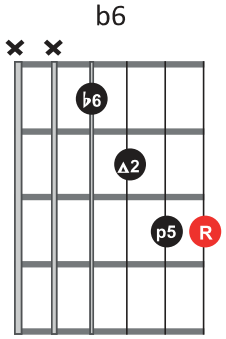
4



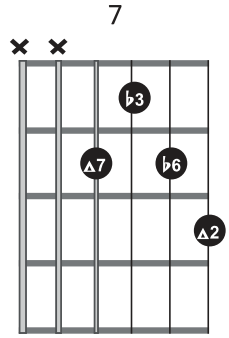
5



b6

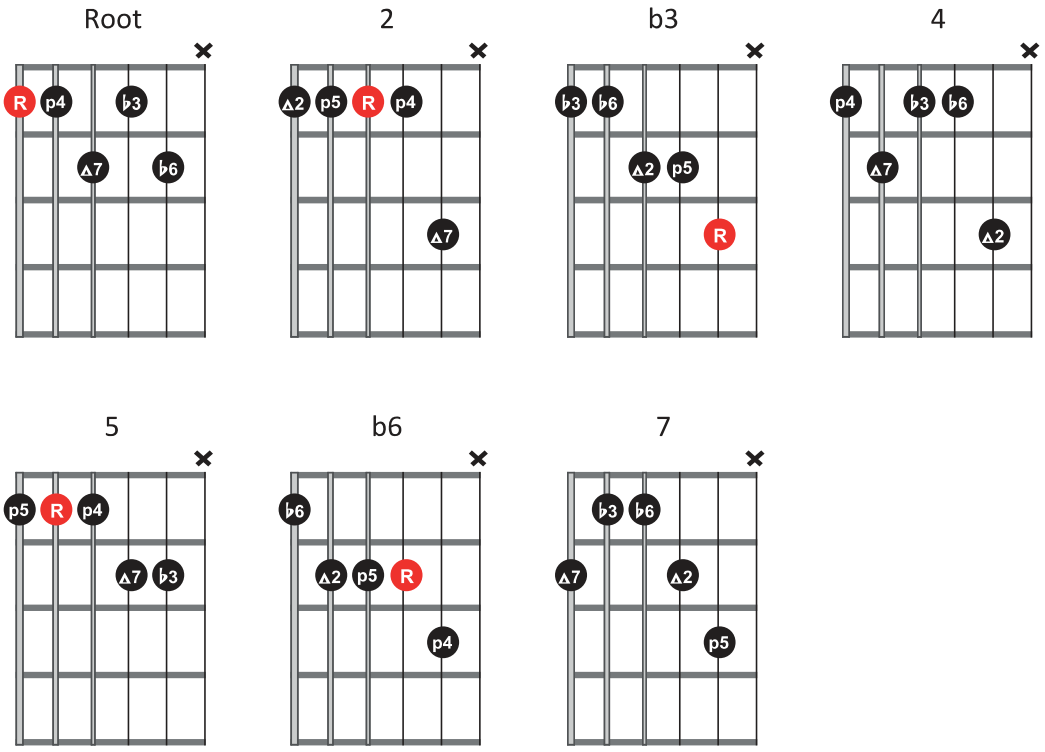


7

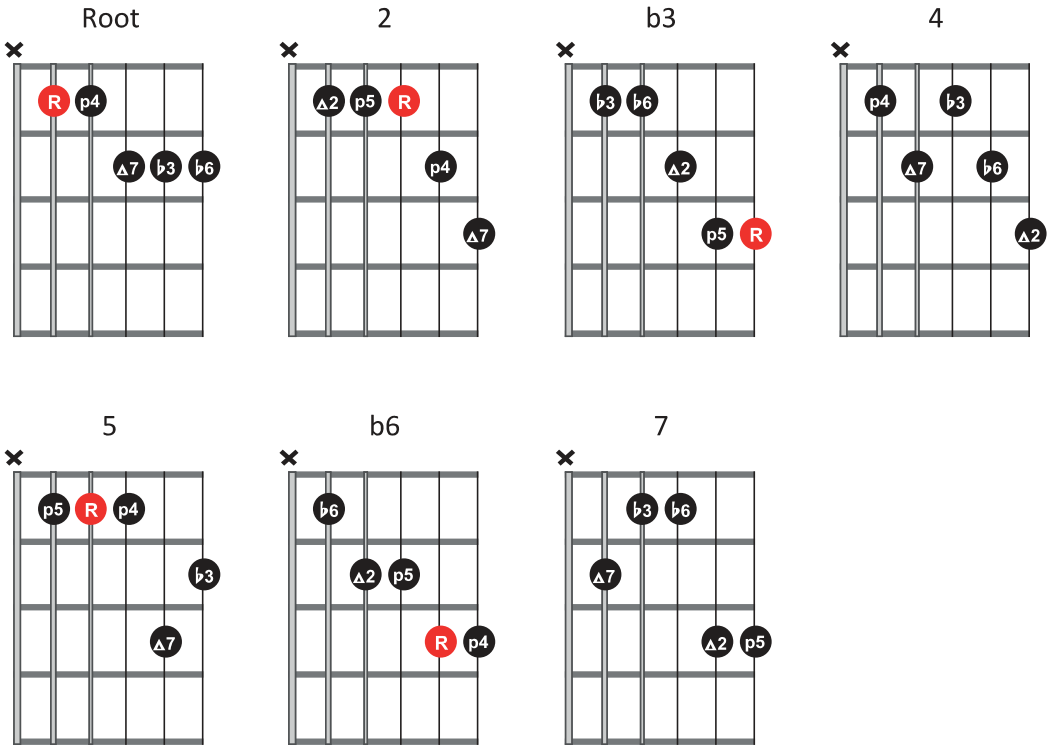


5part-Quartal Voicings – Harmonic Minor

Stringset E-A-D-G-B



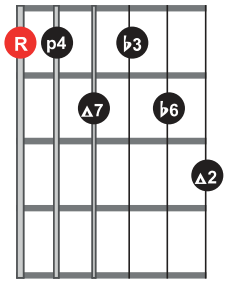
Stringset A-D-G-B-E



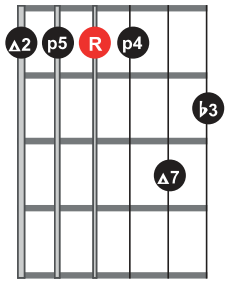
6part-Quartal Voicings – Harmonic Minor

Stringset E-A-D-G-B-E

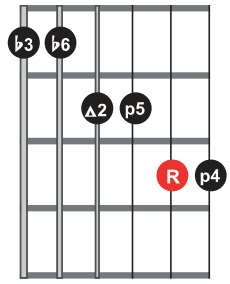
Root



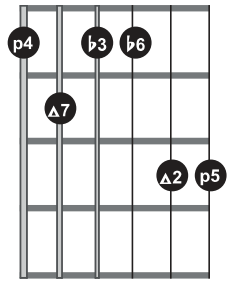
2



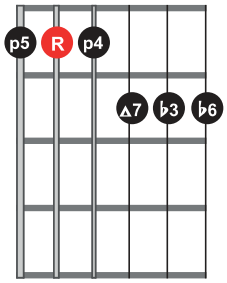
b3



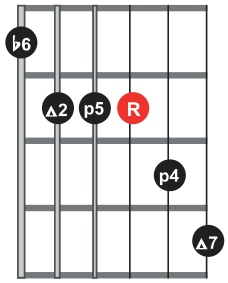
4



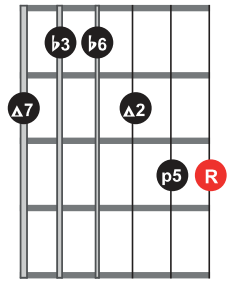
5



b6



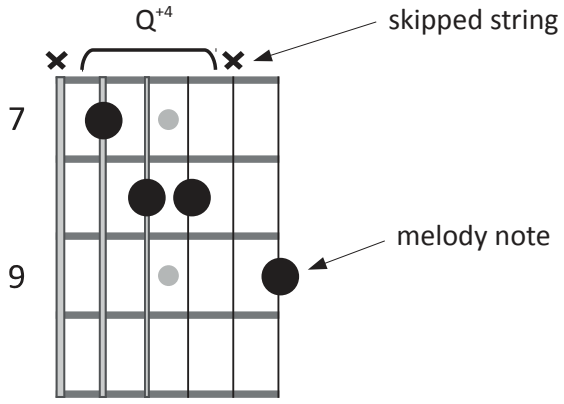
7



Skipped String Voicings

Skipped string voicings are particularly useful on the guitar. They consist of a quartal type voicing on the E,A,D or A,D,G strings and a melody note two strings above.

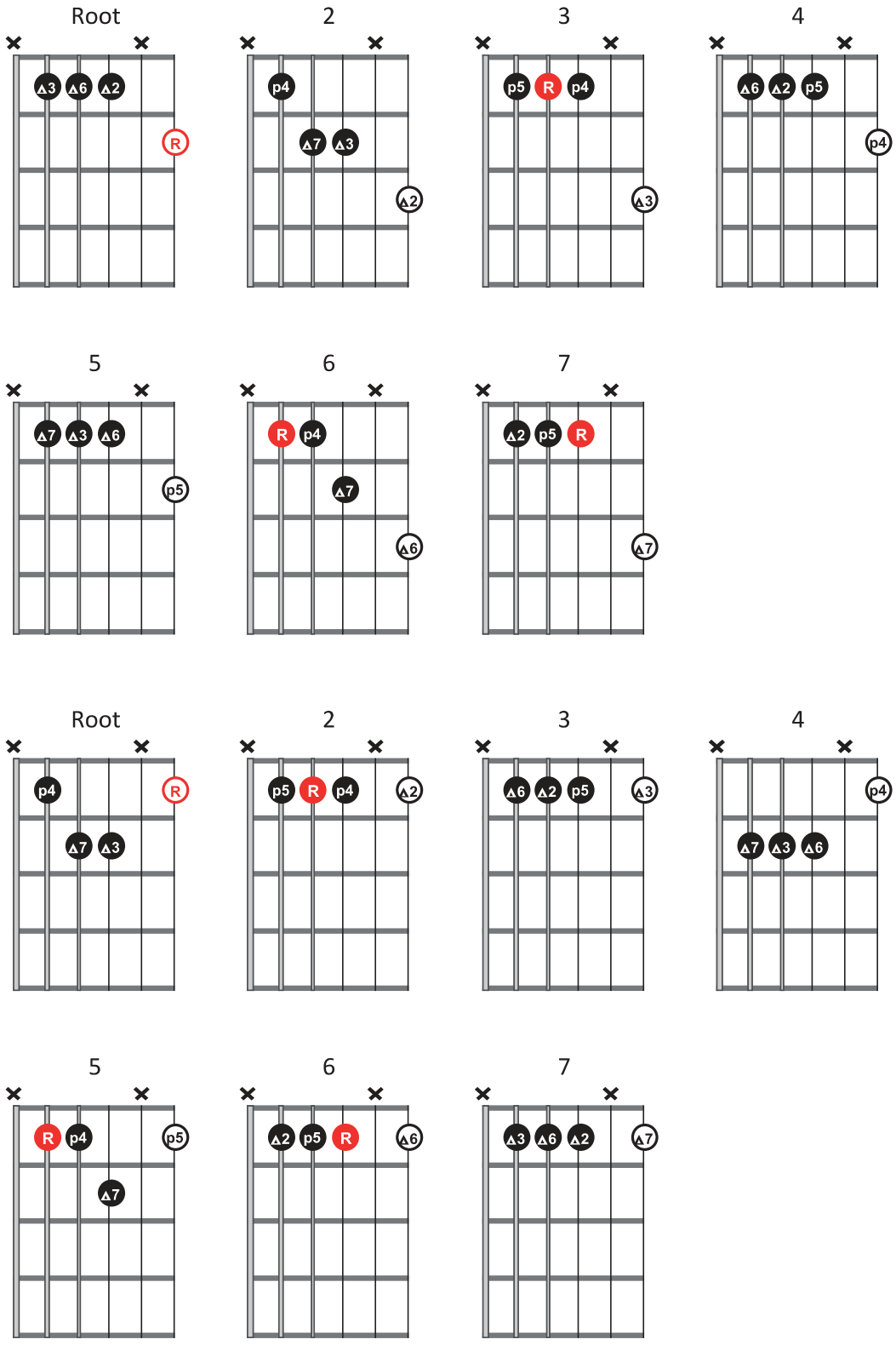
Ex. 142



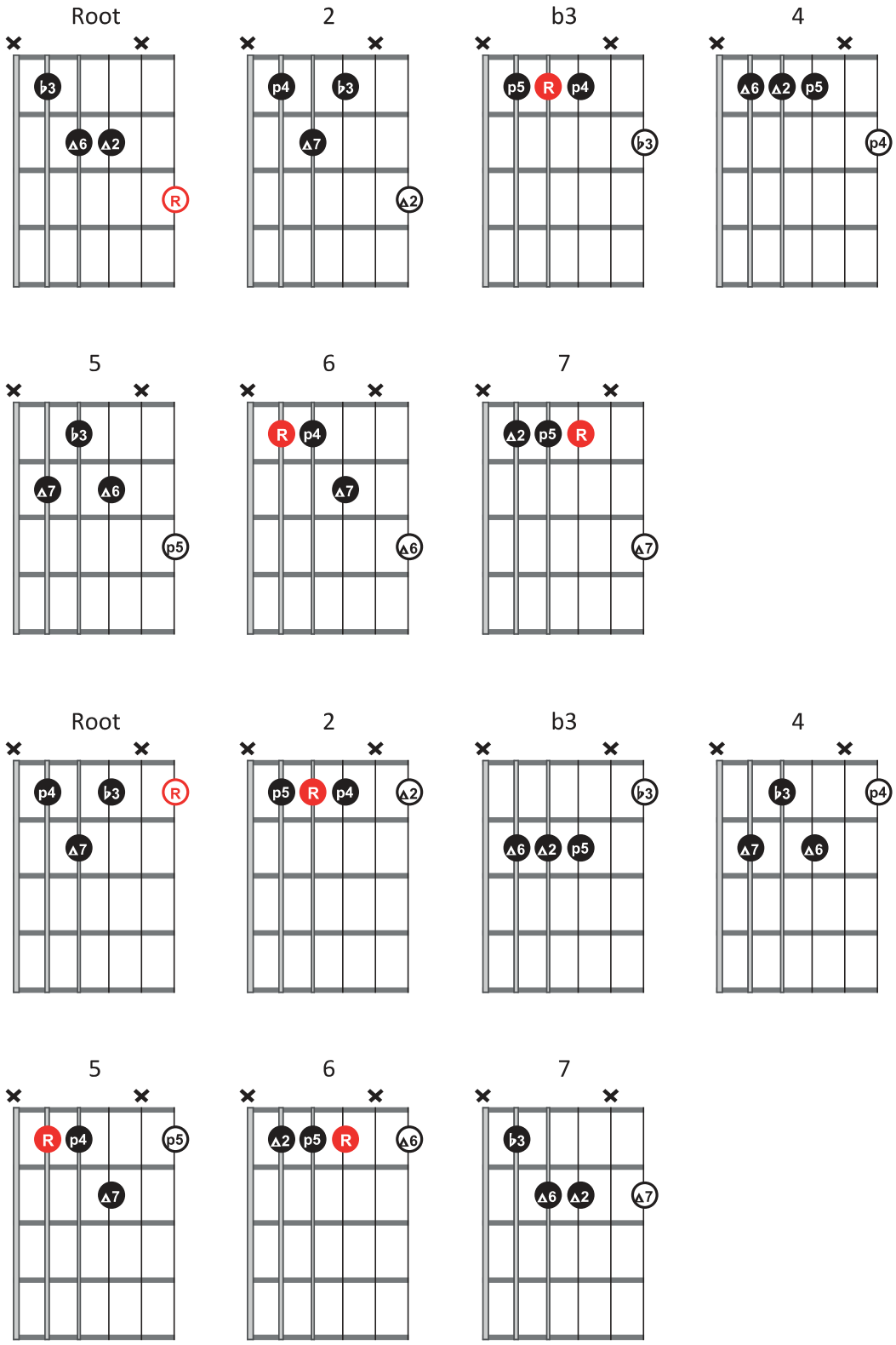
The voicing above is comprised of the notes E, Bb, Eb, Db. On its own this chord is incomplete but if played over certain bass notes it produces such chords as: $F\#^{13}$, $C^{7\#9b9}$, E^{b7b9}, A^{7b5b9}, $E^{MA\#11\#5,}$, G^{o7b13}.

Try to discover all the substitution posibilities of each form.

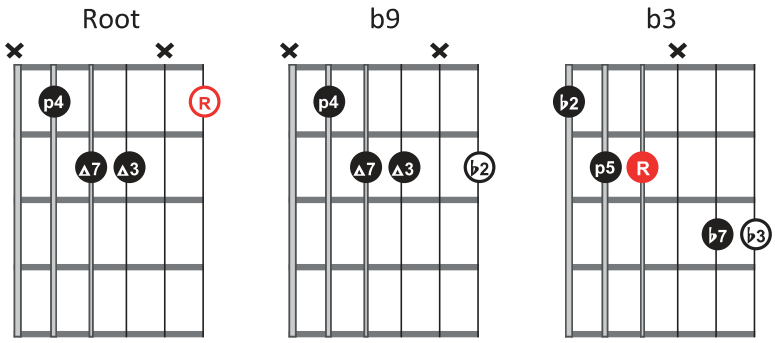
Major Scale Skipped String Voicings



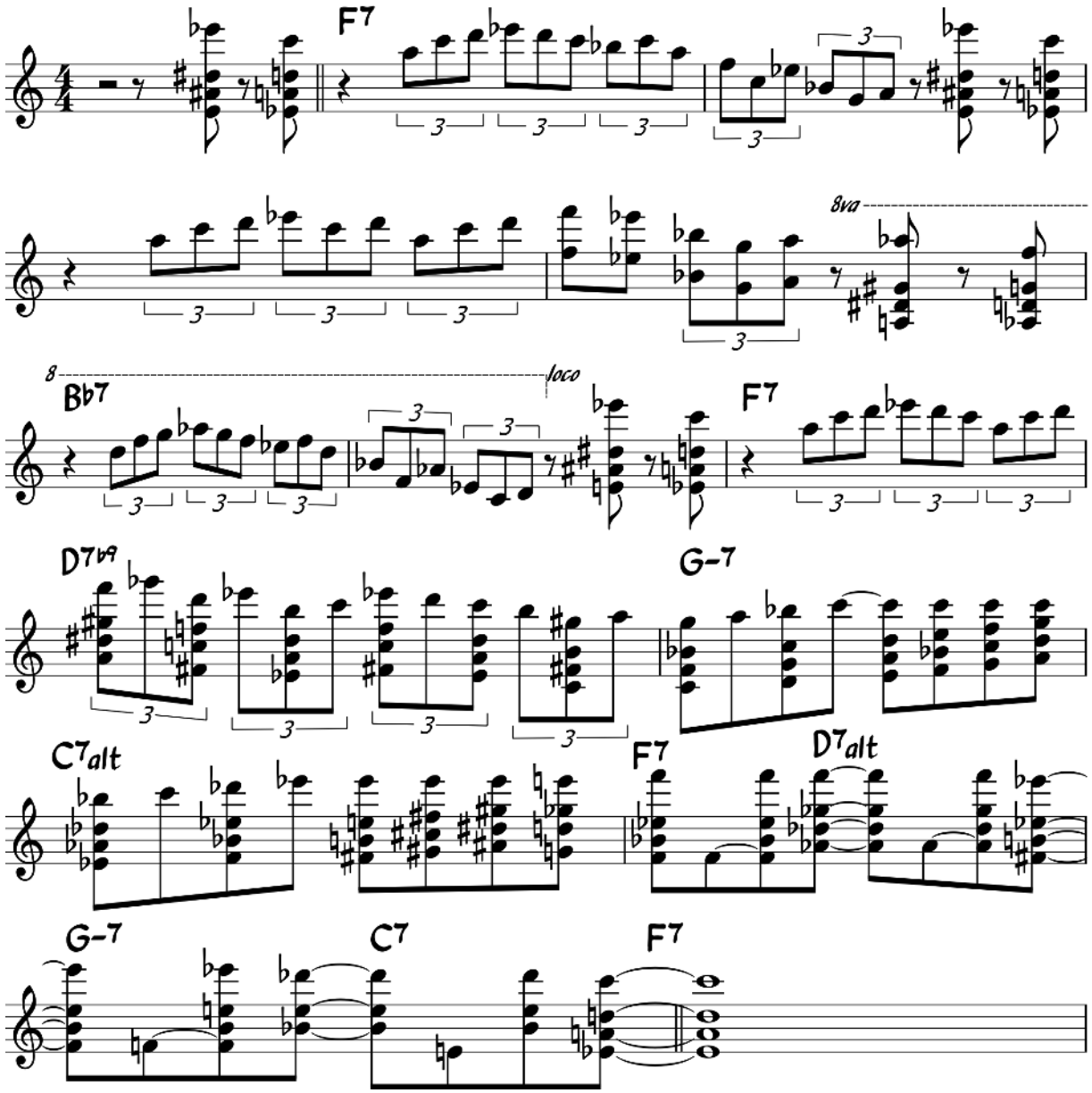
Melodic Minor Scale Skipped String Voicings



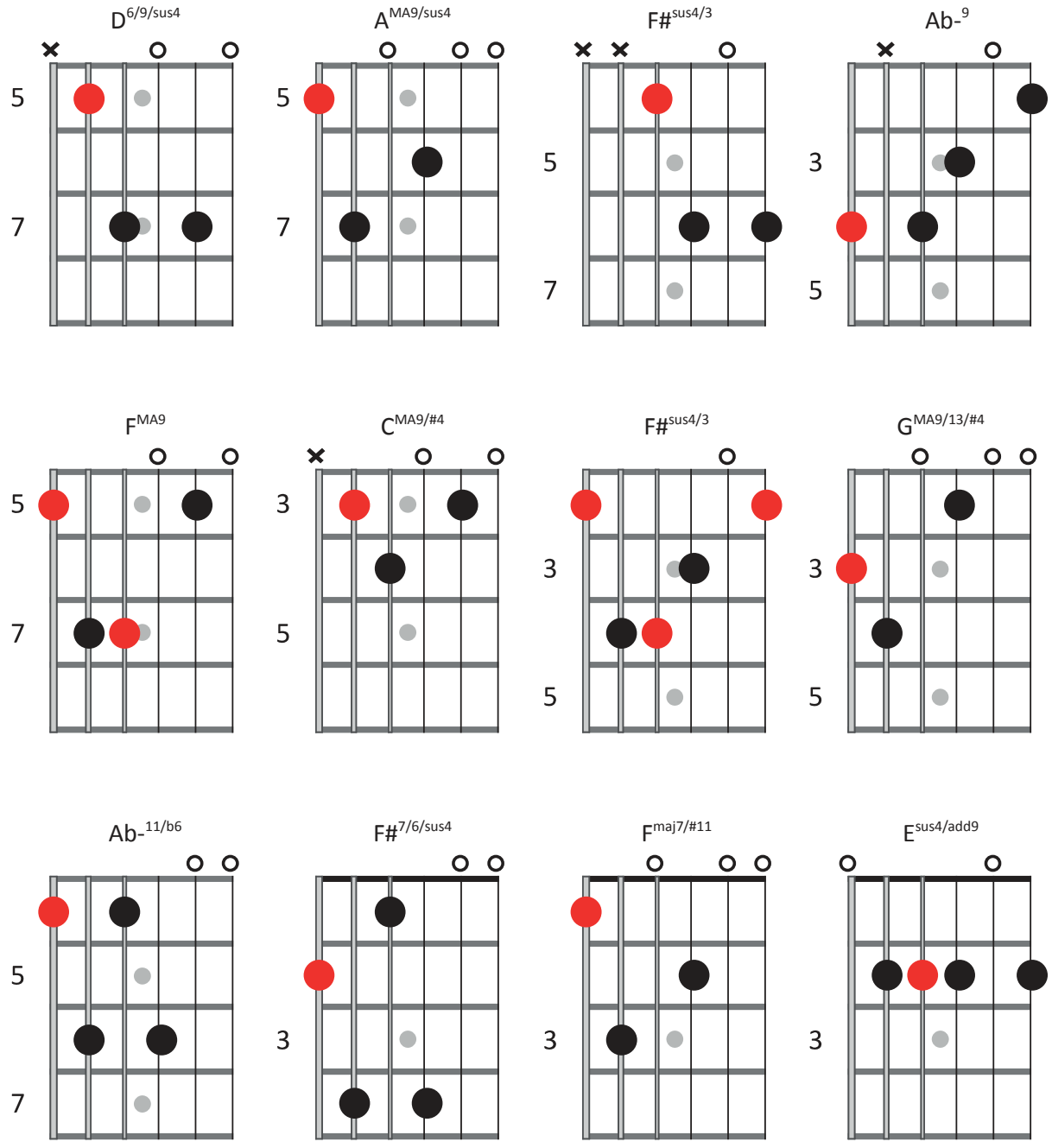
Dominant Diminished Skipped String Voicings

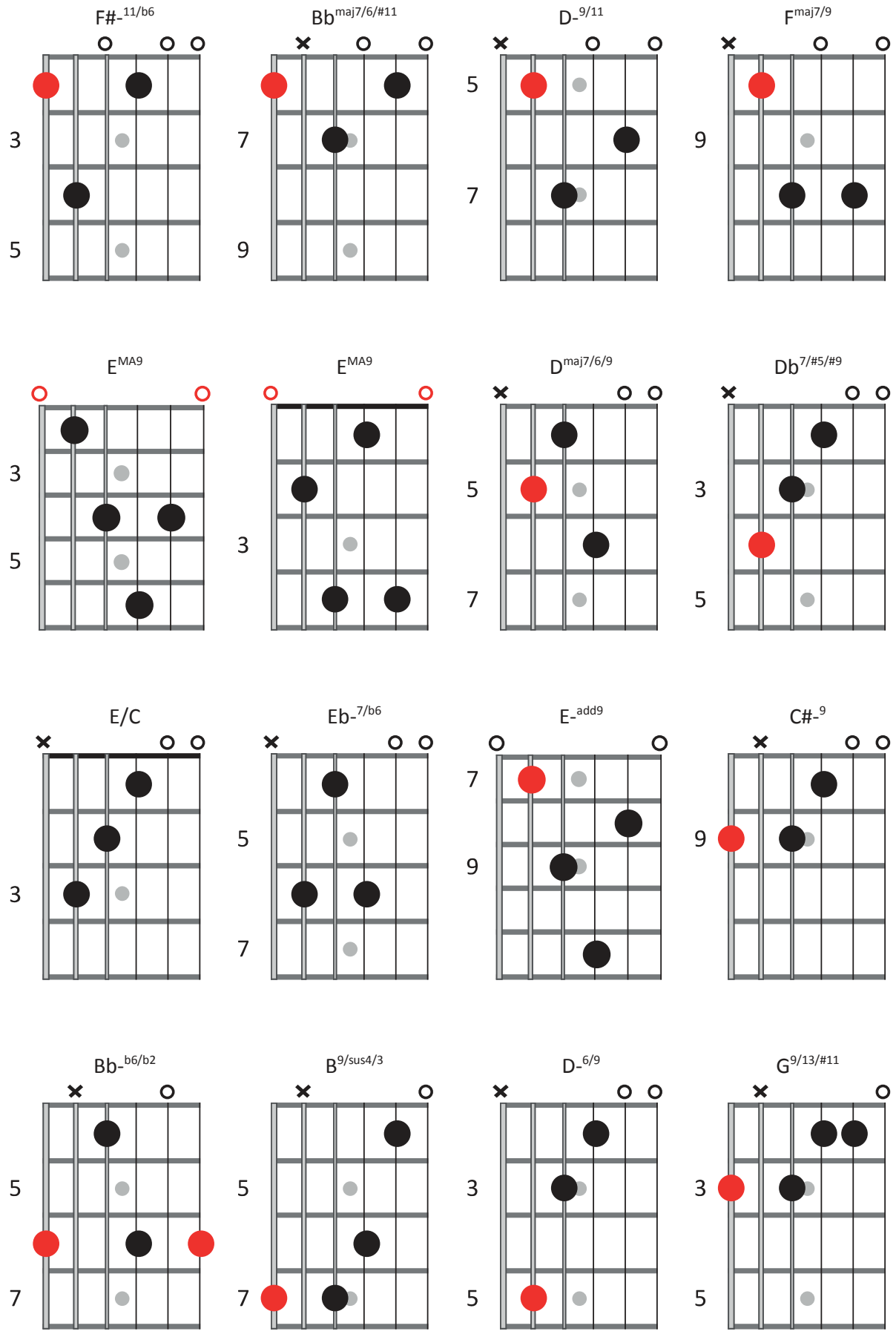


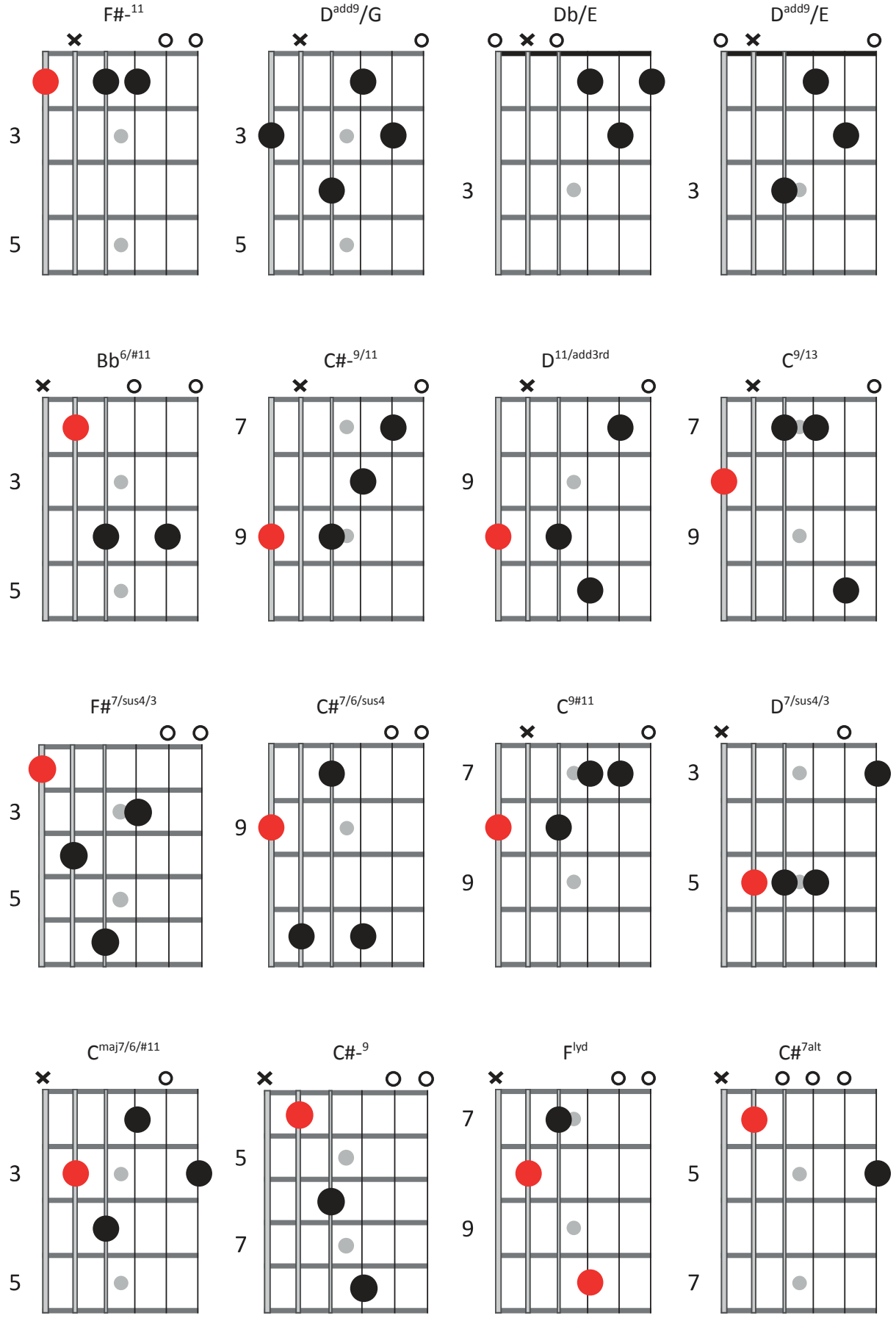
Here is an example of using skipped string quartal voicings in an F blues.

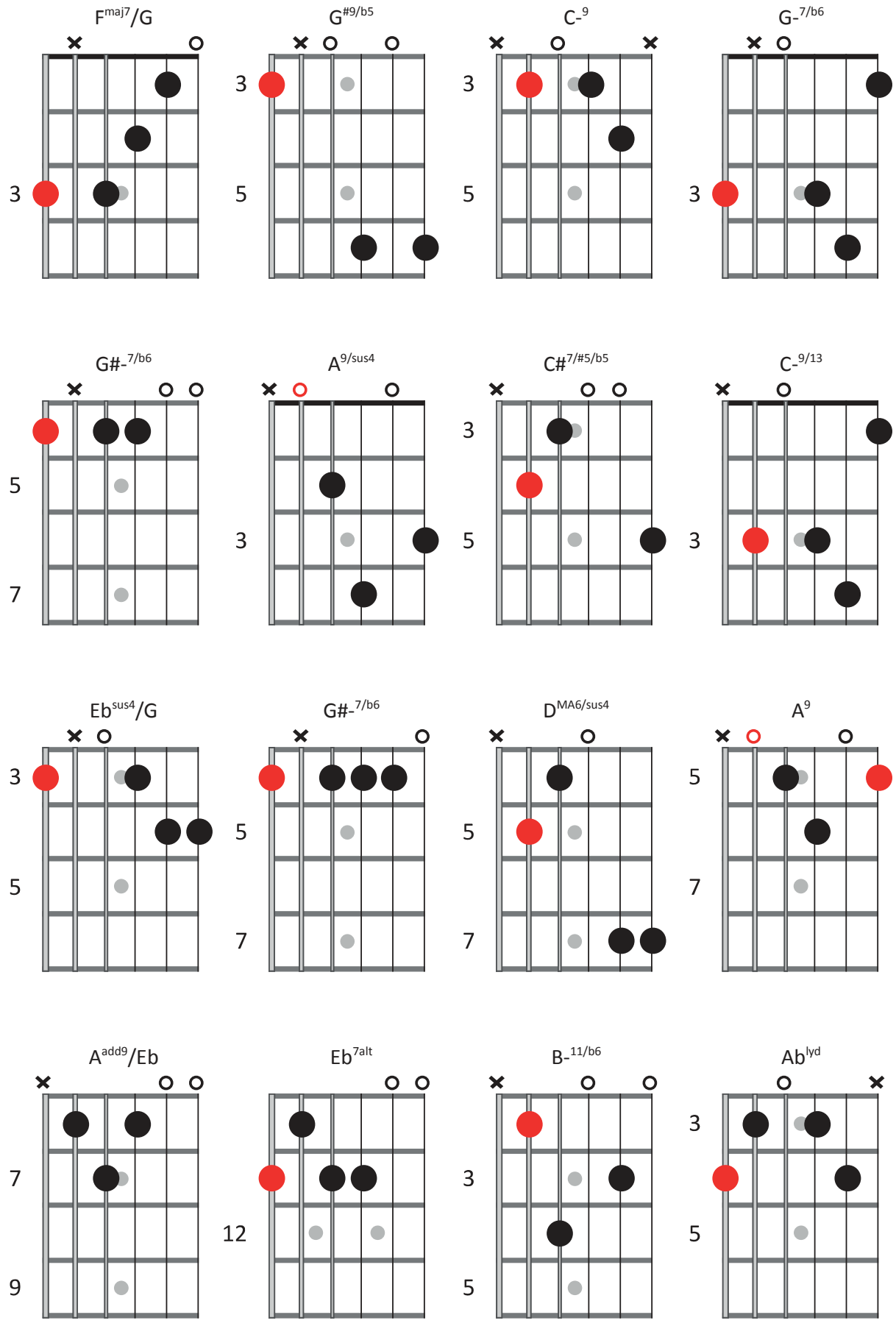
Ex. 143

Open String Voicings

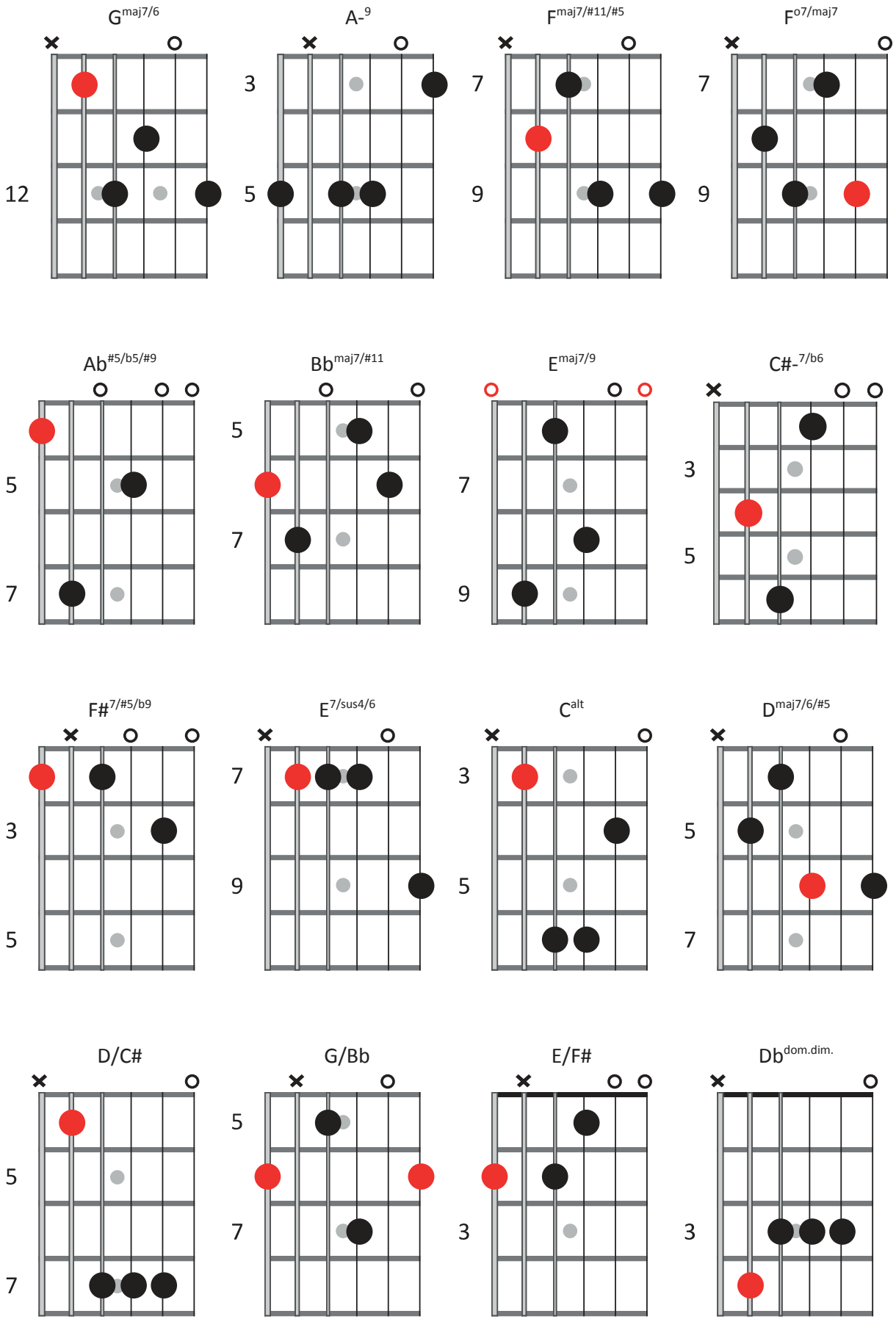
This section includes some of my favorite open string voicings. The open string(s) can occur as any member of the chord. Play through all the chords putting a check next to your favorites. It is also a good idea to organize the chords into progressions in order to remember them.

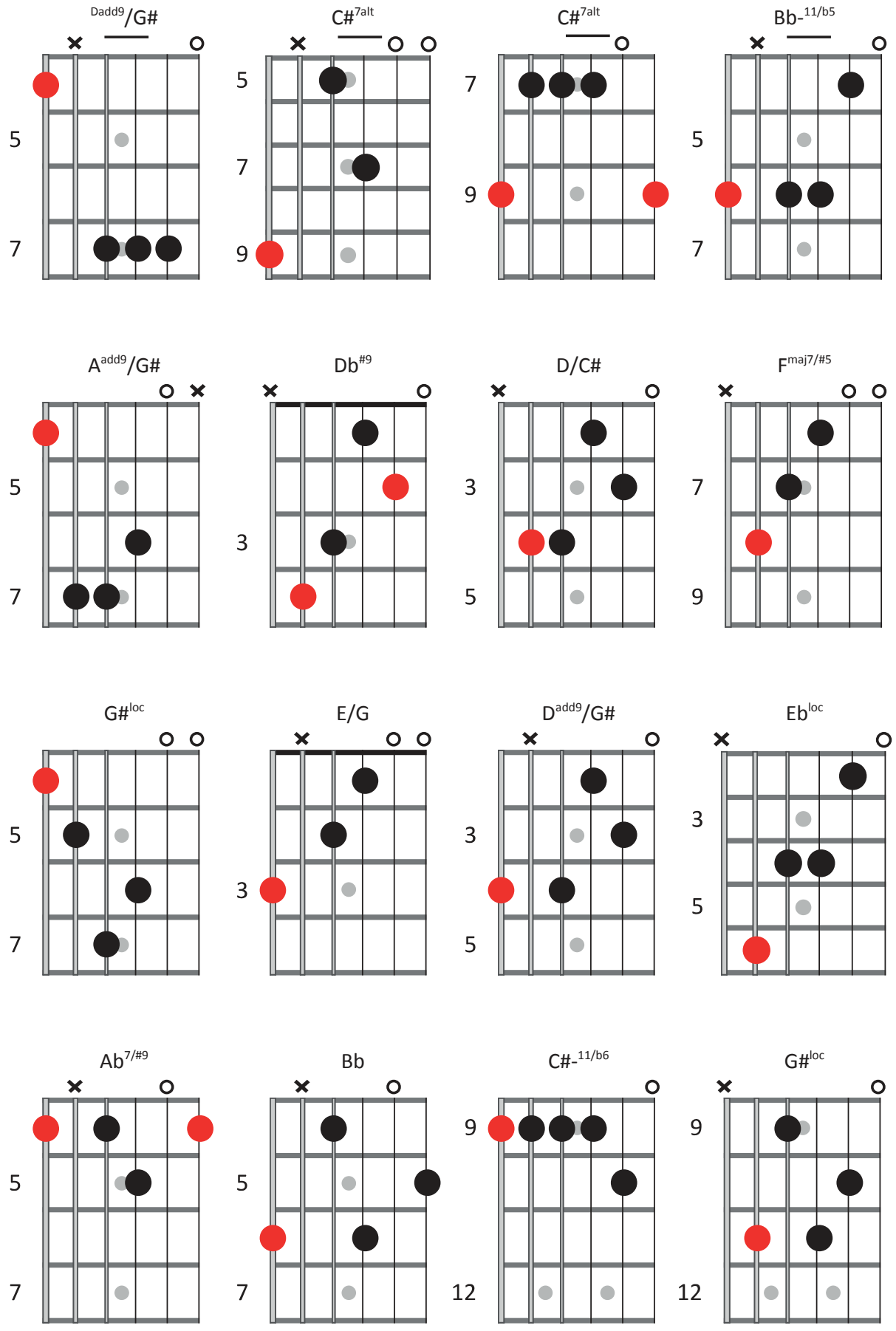


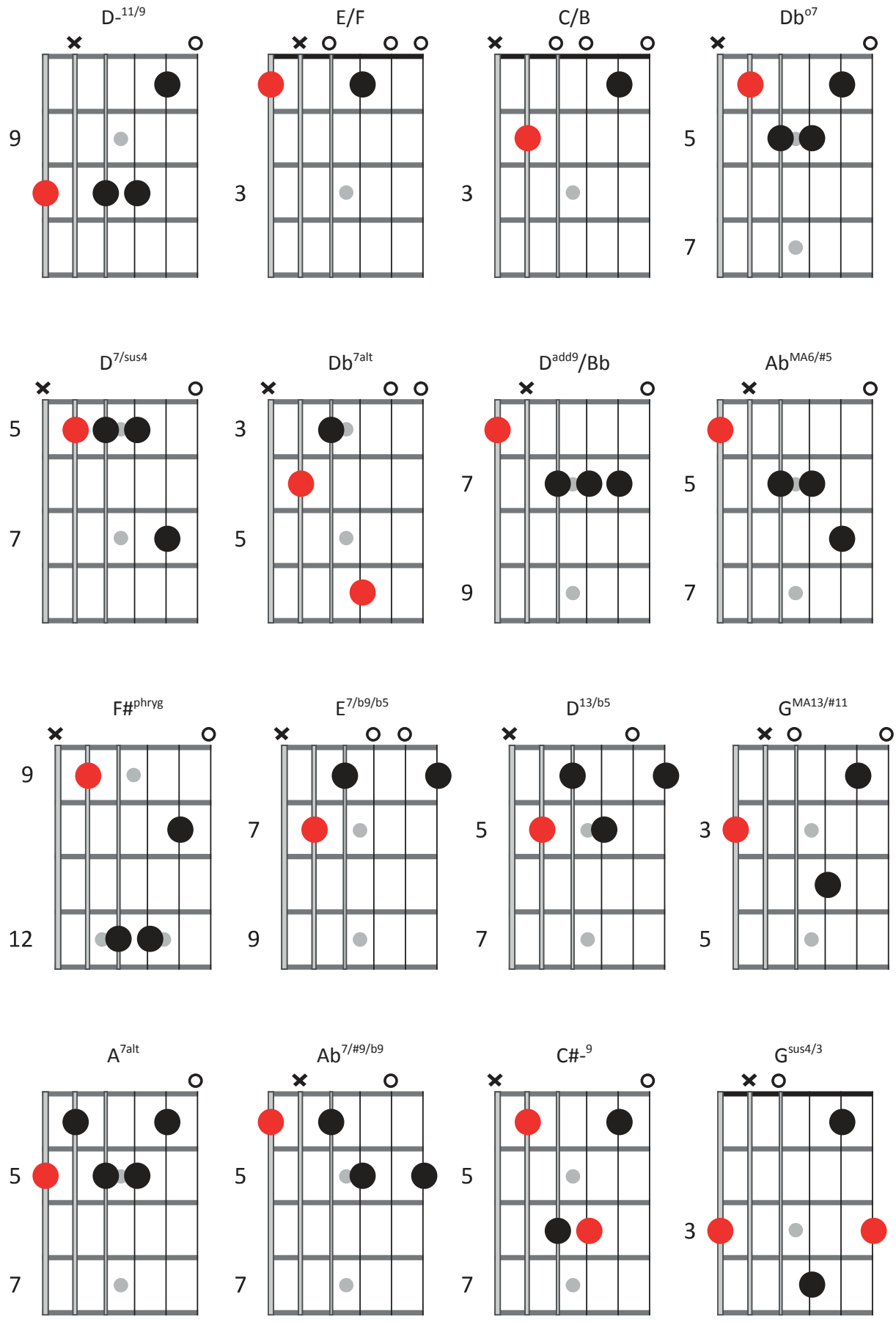

F#-11/b6
3
5
Bbmaj7/6/#11
7
9
D-9/11
5
7
Fmaj7/9
9
EMA9
3
5
EMA9
3
Dmaj7/6/9
5
7
Db7/#5/#9
3
5
E/C
3
Eb-7/b6
5
7
E-add9
7
9
C#-9
9
Bb-b6/b2
5
7
B9/sus4/3
5
7
D-6/9
3
5
G9/13/#11
3
5

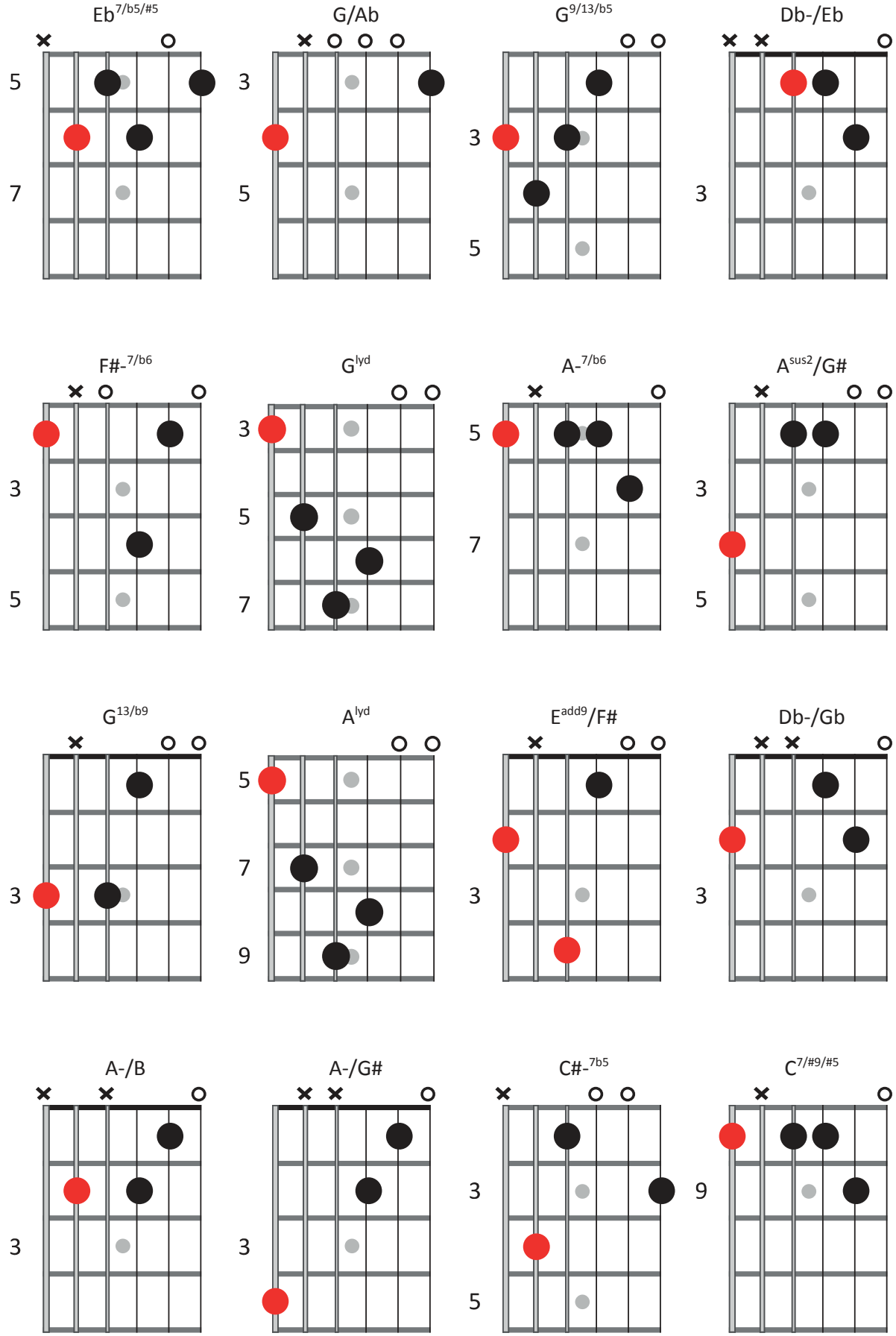

F#-11
3
5
Dadd9/G
3
5
Db/E
3
Dadd9/E
3
Bb6/#11
3
5
C#-9/11
7
9
D11/add3rd
9
C9/13
7
9
F#7/sus4/3
3
5
C#7/6/sus4
9
C9#11
7
9
D7/sus4/3
3
5
Cmaj7/6/#11
3
5
C#-9
5
7
Flyd
7
9
C#7alt
5
7

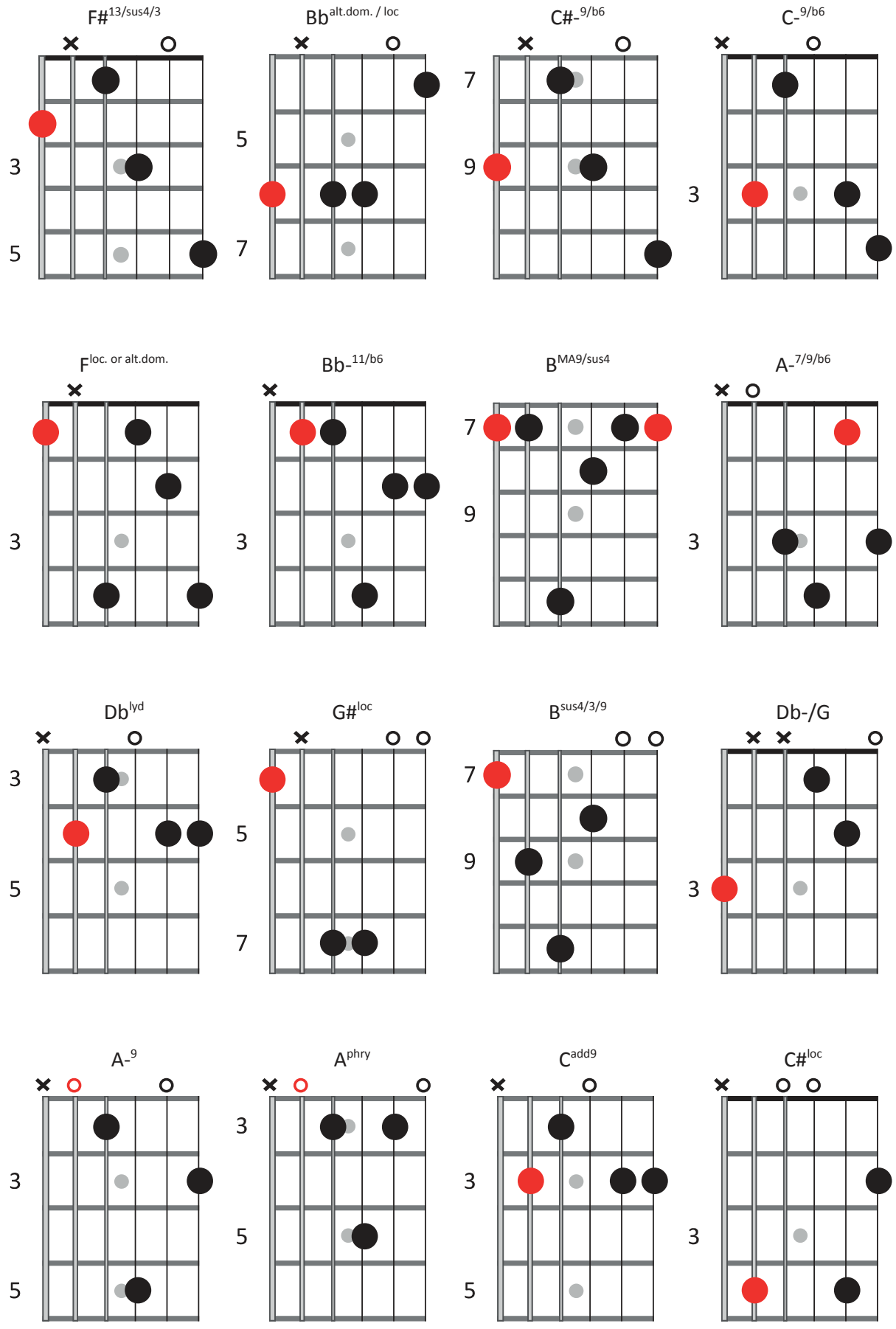

F maj7/G
G #9/b5
C -9
G -7/b6
G# -7/b6
A 9/sus4
C# 7/#5/b5
C -9/13
Eb sus4/G
G# -7/b6
D MA6/sus4
A 9
A add9/Eb
Eb 7alt
B -11/b6
Ab lyd

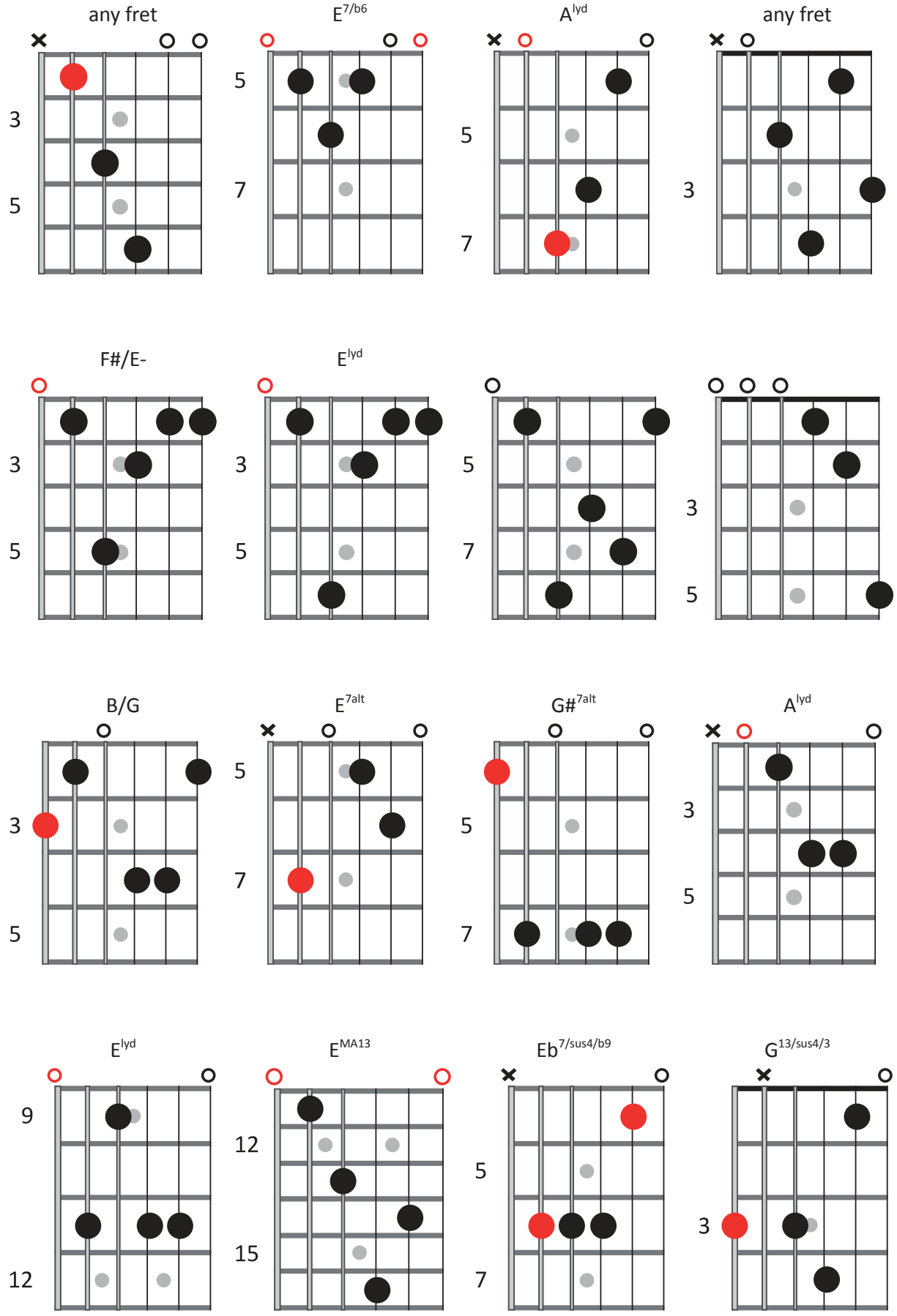

Gmaj7/6
12
A-9
3
5
Fmaj7/#11/#5
7
9
Fo7/maj7
7
9
Ab#5/b5/#9
5
7
Bbmaj7/#11
5
7
Emaj7/9
7
9
C#-7/b6
3
5
F#7/#5/b9
3
5
E7/sus4/6
7
9
Calt
3
5
Dmaj7/6/#5
5
7
D/C#
5
7
G/Bb
5
7
E/F#
3
Dbdom.dim.
3

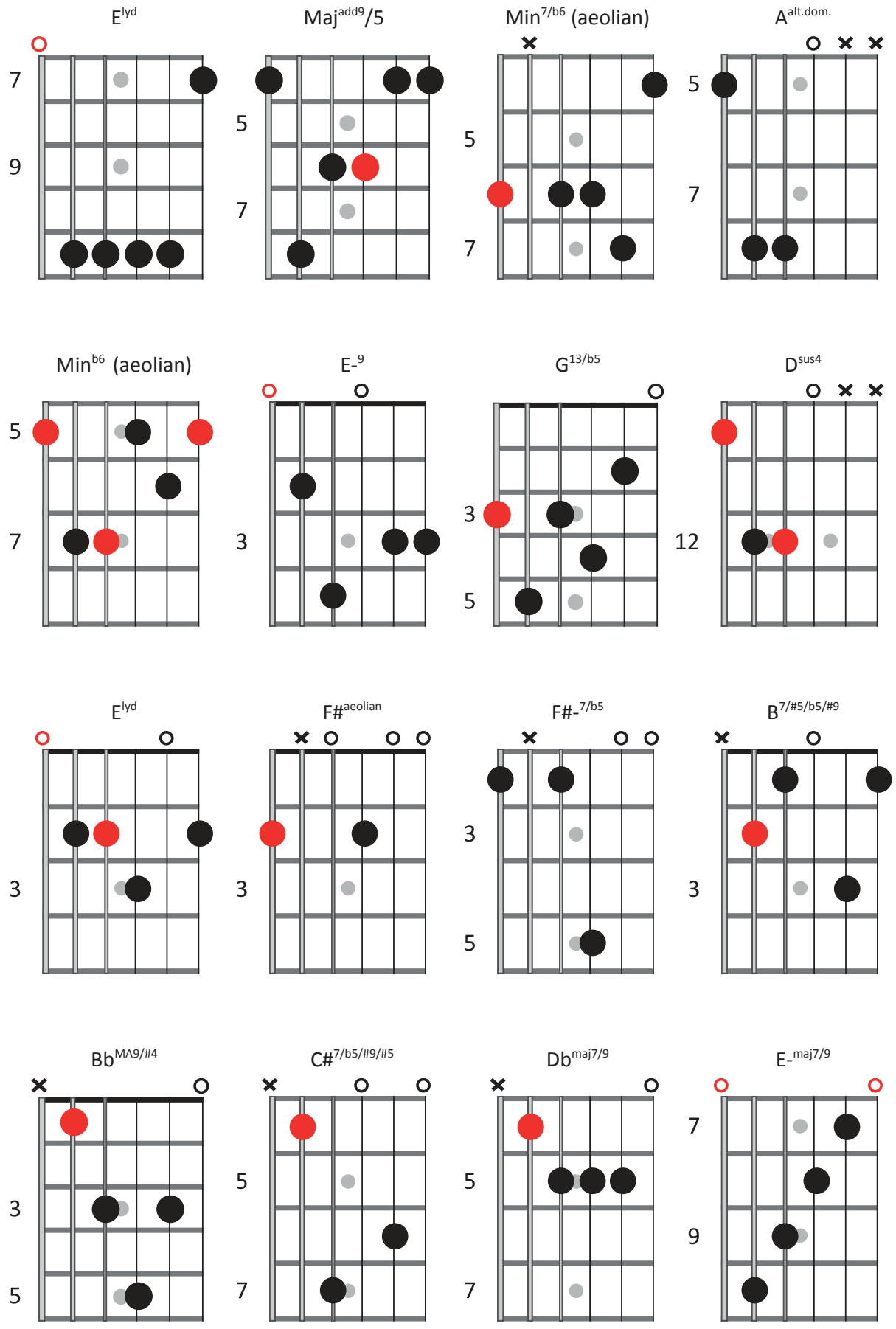

Dadd9/G#
5
7
C#7alt
5
7
9
C#7alt
7
9
Bb-11/b5
5
7
Aadd9/G#
5
7
Db#9
3
D/C#
3
5
Fmaj7/#5
7
9
G#loc
5
7
E/G
3
Dadd9/G#
3
5
Ebloc
3
5
Ab7/#9
5
7
Bb
5
7
C#-11/b6
9
12
G#loc
9
12


D-11/9
9
E/F
3
C/B
3
Db°7
5
7
D7/sus4
5
7
Db7alt
3
5
Dadd9/Bb
7
9
AbMA6/#5
5
7
F#phryg
9
12
E7/b9/b5
7
9
D13/b5
5
7
GMA13/#11
3
5
A7alt
5
7
Ab7/#9/b9
5
7
C#-9
5
7
Gsus4/3
3


Eb7/b5/#5
5
7
G/Ab
3
5
G9/13/b5
3
5
Db-/Eb
3
F#-7/b6
3
5
Glyd
3
5
7
A-7/b6
5
7
Asus2/G#
3
5
G13/b9
3
Alyd
5
7
9
Eadd9/F#
3
Db-/Gb
3
A-/B
3
A-/G#
3
C#-7b5
3
5
C7/#9/#5
9

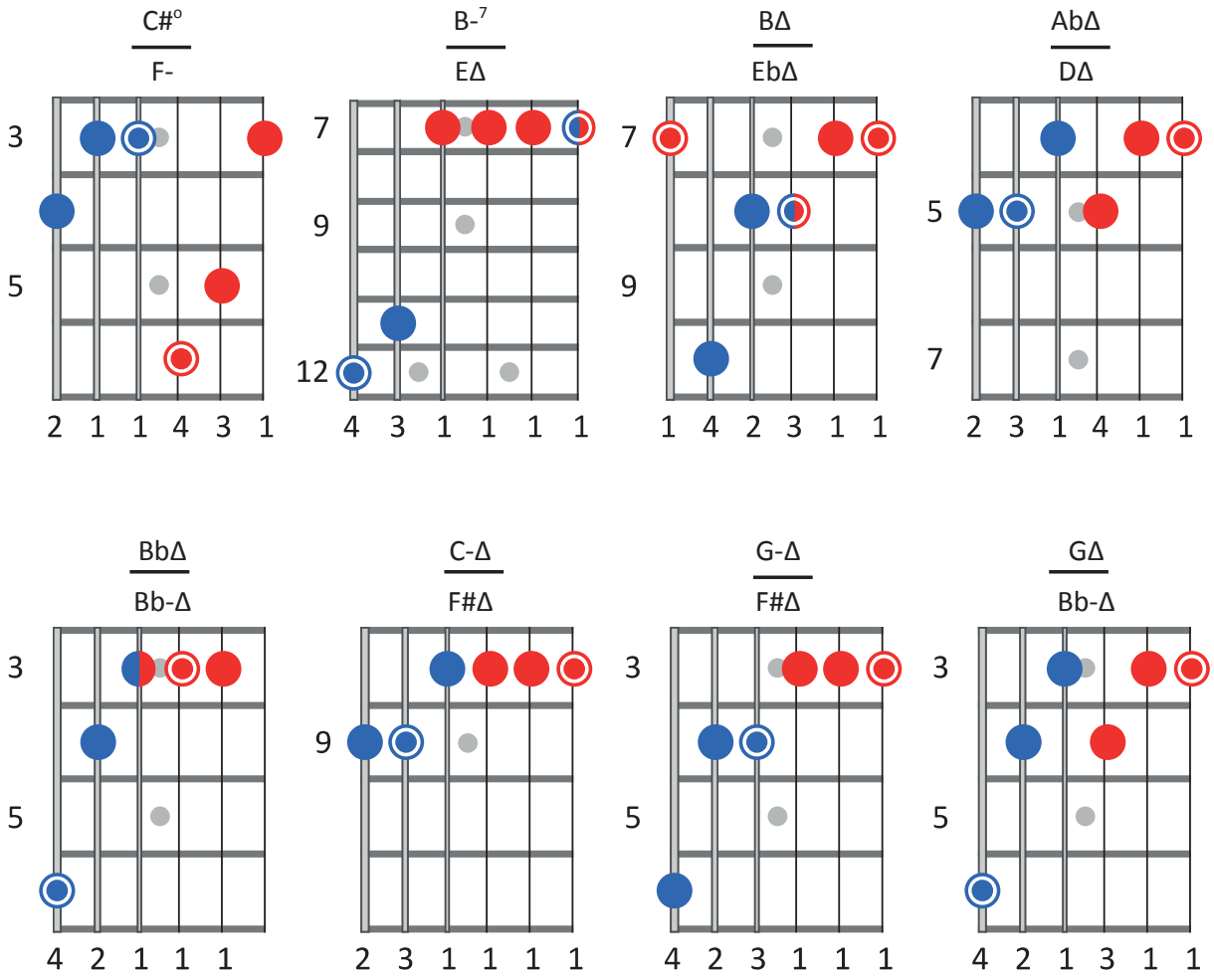

F#13/sus4/3
3
5
Bbalt.dom. / loc
5
7
C#-9/b6
7
9
C-9/b6
3
Floc. or alt.dom.
3
Bb-11/b6
3
BMA9/sus4
7
9
A-7/9/b6
3
Dblyd
3
5
G#loc
5
7
Bsus4/3/9
7
9
Db-/G
3
A-9
3
5
Aphry
3
5
Cadd9
3
5
C#loc
3


any fret
3
5
E7/b6
5
7
Alyd
5
7
any fret
3
F#/E-
3
5
Elyd
3
5
5
7
3
5
B/G
3
5
E7alt
5
7
G#7alt
5
7
Alyd
3
5
Elyd
9
12
EMA13
12
15
Eb7/sus4/b9
5
7
G13/sus4/3
3

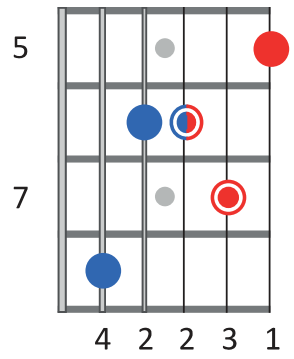

E lyd
7
9
Maj add9 /5
5
7
Min 7/b6 (aeolian)
5
7
A alt.dom.
5
7
Min b6 (aeolian)
5
7
E- 9
3
G 13/b5
3
5
D sus4
12
E lyd
3
F# aeolian
3
F#- 7/b5
3
5
B 7/#5/b5/#9
3
Bb MA9/#4
3
5
C# 7/b5/#9/#5
5
7
Db maj7/9
5
7
E- maj7/9
7
9

Polychords (Triads over Triads)

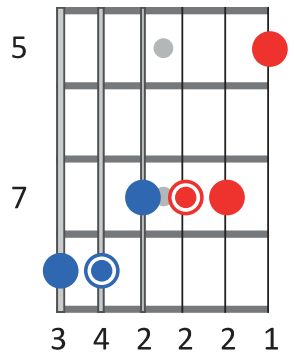
The polychords in this section are moveable but have been specifically labeled for easier understanding. These kind of voicings work well for intros and endings where there is more harmonic freedom.



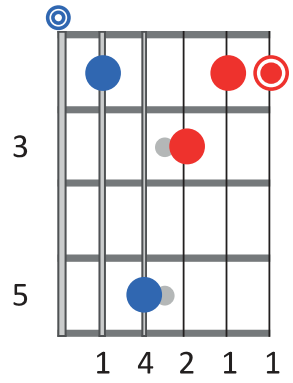
F#-Δ
DbΔ



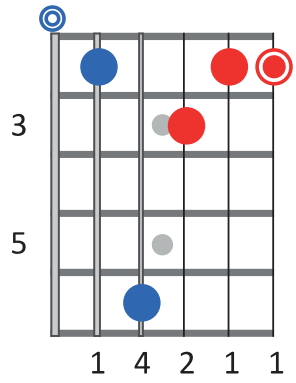
DΔ
FΔ



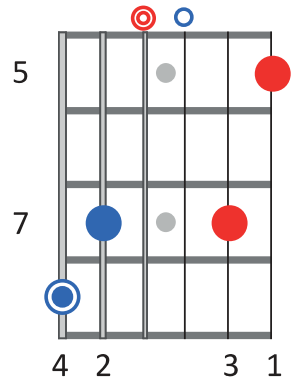
F#Δ
E-Δ



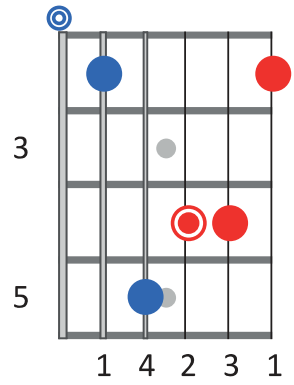
F#Δ
EΔ



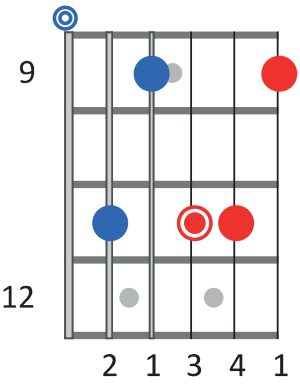
DΔ
CΔ



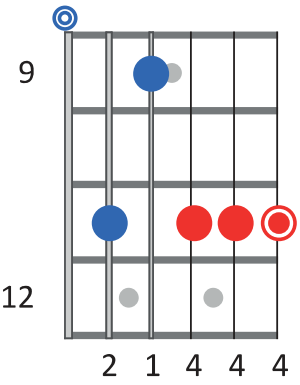
BΔ
E-Δ



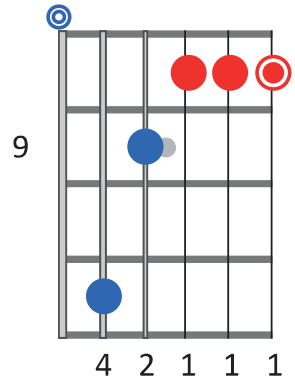
F#Δ
EΔ



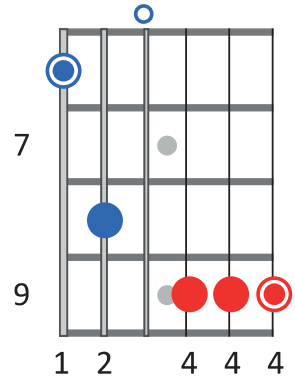
E-
EΔ



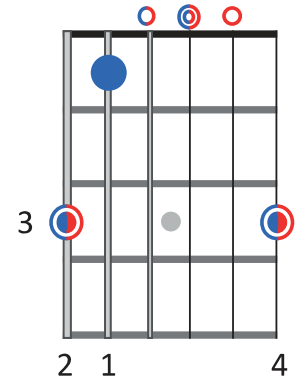
C-
EΔ



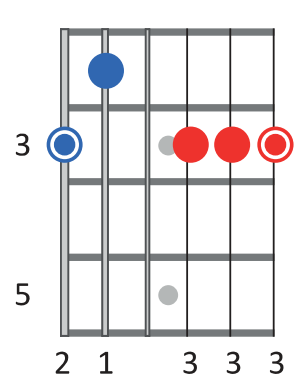
Db-Δ
BbΔ

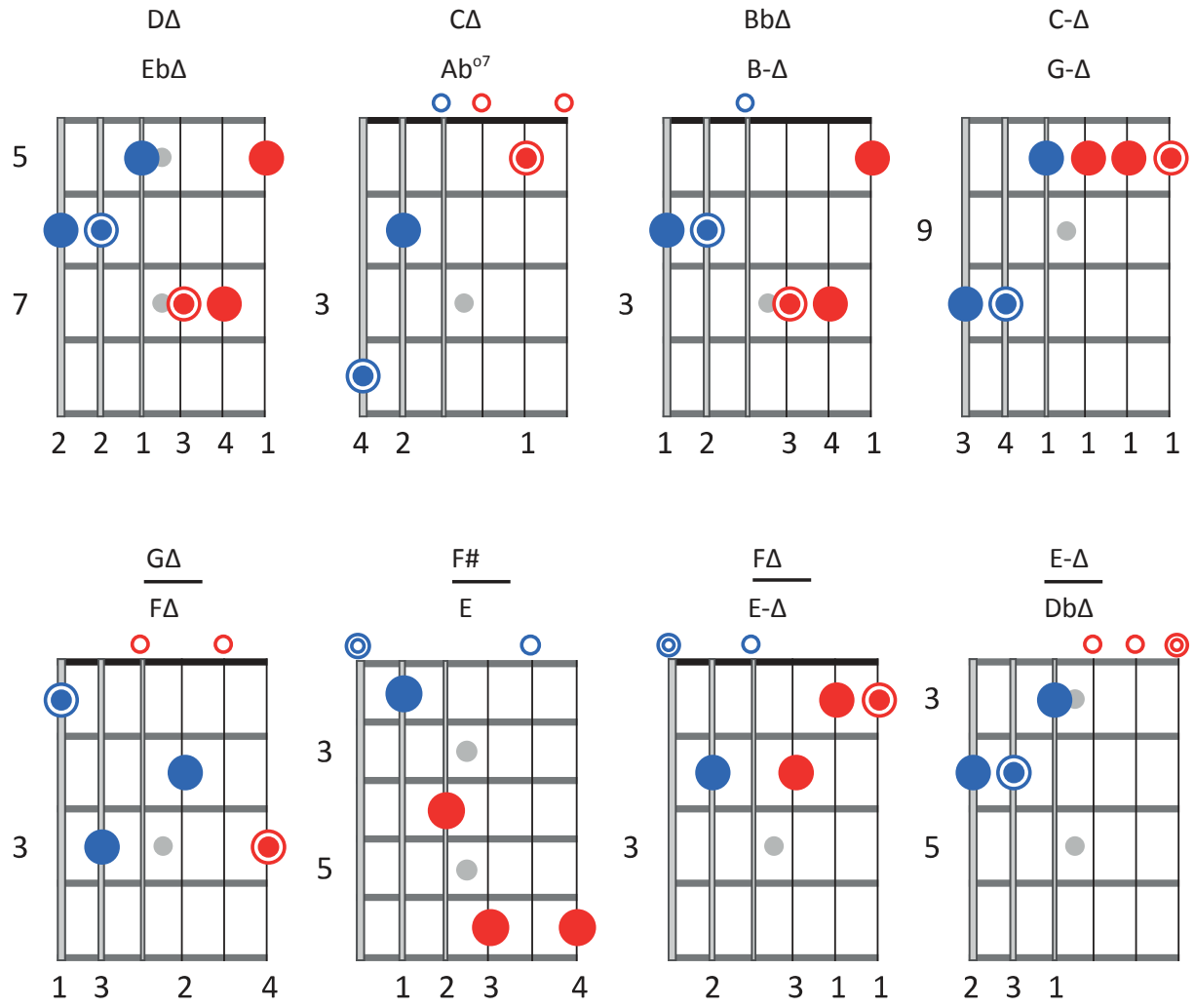


GΔ
G-Δ



G-Δ
GΔ




DΔ
EbΔ
5
7
2 2 1 3 4 1
CΔ
Ab°7
3
4 2 1
BbΔ
B-Δ
3
1 2 3 4 1
C-Δ
G-Δ
9
3 4 1 1 1 1
GΔ
FΔ
3
1 3 2 4
F#
E
3
5
1 2 3 4
FΔ
E-Δ
3
2 3 1 1
E-Δ
DbΔ
3
5
2 3 1

Using Comping Voicings in New Ways

The comping voicings on the following page may be used in a variety of ways. To demonstrate, let´s look at some possible substitutions for Eb^{7alt}.

Ex. 144

a)	$Eb^{7/\#5/\#9}$	=	3	#5	b7	#9
			G	**B**	**C#**	**F#**
sub.	G^{maj7b5}	=	1	3	b5	7
b)	$Eb^{7/b9/\#5}$	=	b9	3	#5	R
			E	**G**	**B**	**D#**
sub.			1	b3	5	7
c)	$Eb^{7/b9/\#5}$	=	b7	b9	3	#5
			Db	**Fb**	**Abb**	**Cb**
sub.	$Db\text{-}^{7b5}$	=	1	b3	b5	b7

Any of these substitutions (and their inversions) will work in place of Eb^{7alt}, because they all share the same parent melodic minor scale (E melodic minor).

Consult the substitution section for further applications.

Constructing Chord Scales

Chord scales for comping and soloing may be constructed from upper-string voicings. For demonstration purposes let´s use an F7 chord.

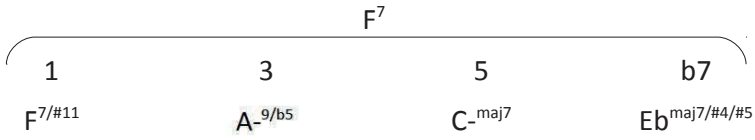
What I would first do is decide on an appropriate scale, in this case we´ll use Fmix#11. Then, using the parent melodic minor scale, list all of the chords generated by the scale.

Ex. 145

C-^{maj7} D$^{7/sus4/b13}$ Eb$^{maj7/\#4/\#5}$ F$^{7/\#11}$ G$^{7/b13}$ A-$^{9/b5}$ B$^{7/\#9/b9/\#5/b5}$

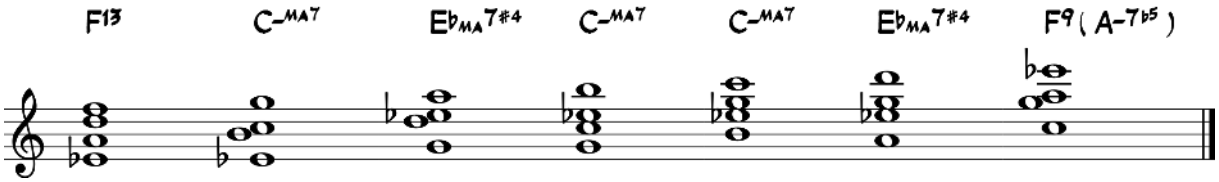
Next I would isolate all of the chords whose root was a basic chord tone of F^7.

Ex. 146



Out of these chords I would pick two, in this case C-^{maj7} and Eb$^{maj7/\#5/\#4}$, on which the scale would be based. By alternating these chords and their inversions an Fmix#11 scale may be built in the top voice. This may be done by alternating voicings as seen here in Ex. 147.

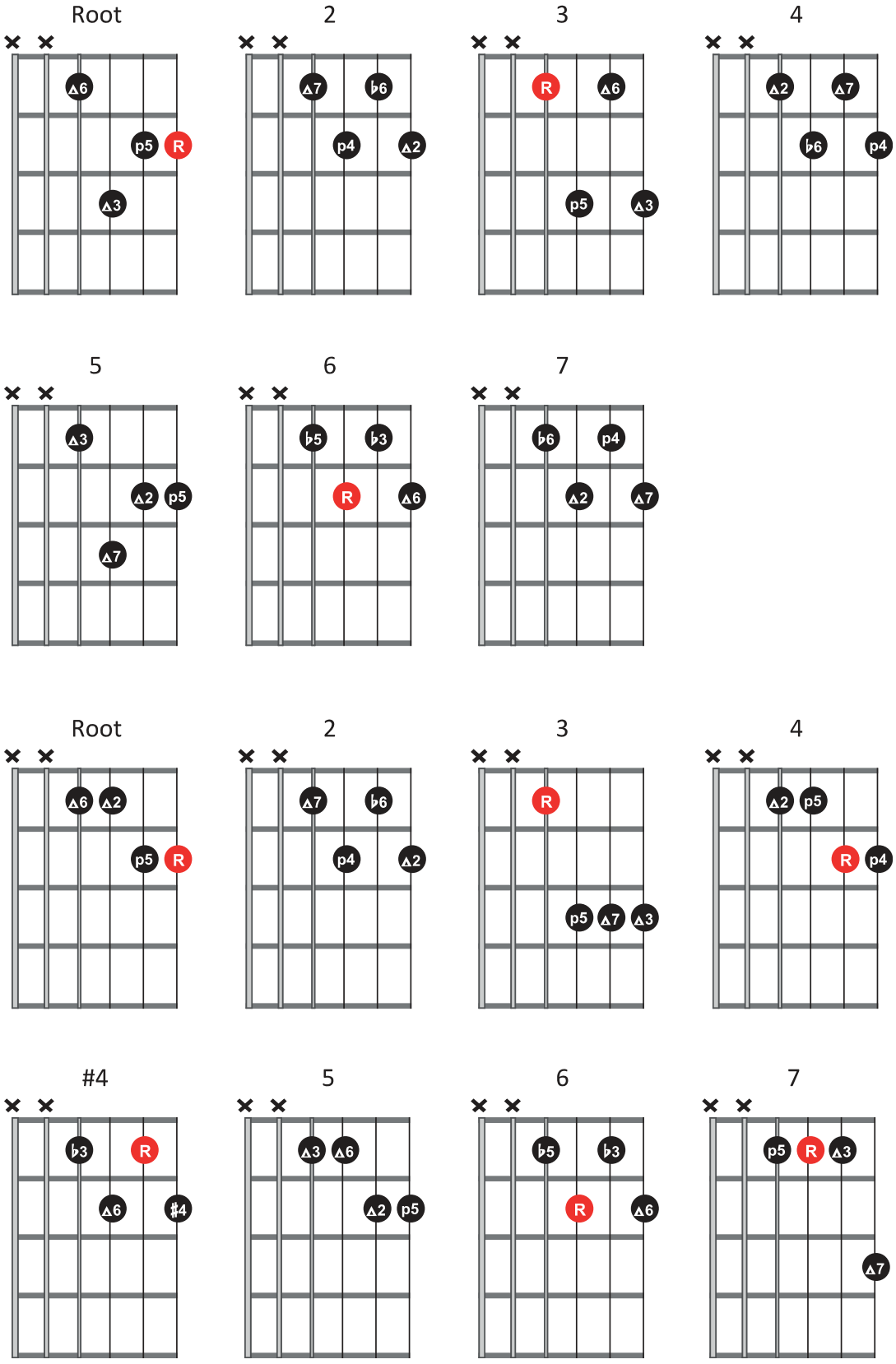
Ex. 147



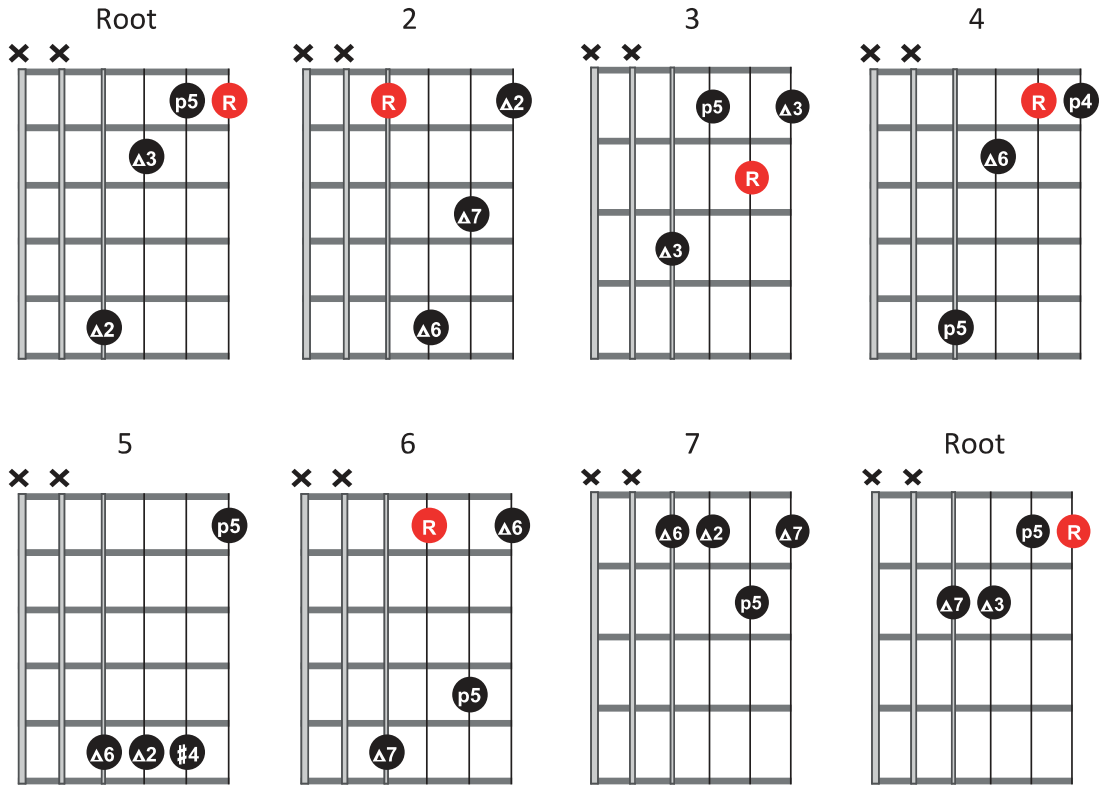
Remember, you may mix together any of the drop voicing groups for these purposes.

Chordal Scales

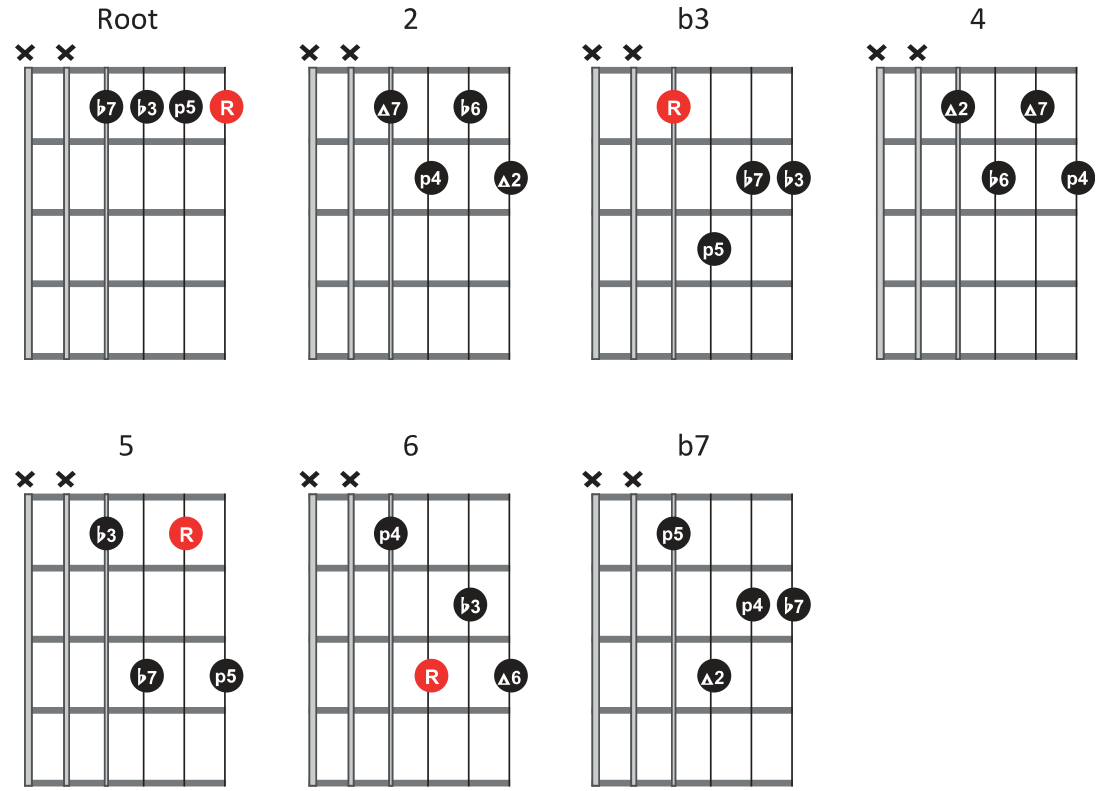
Major Scale (Bebop)



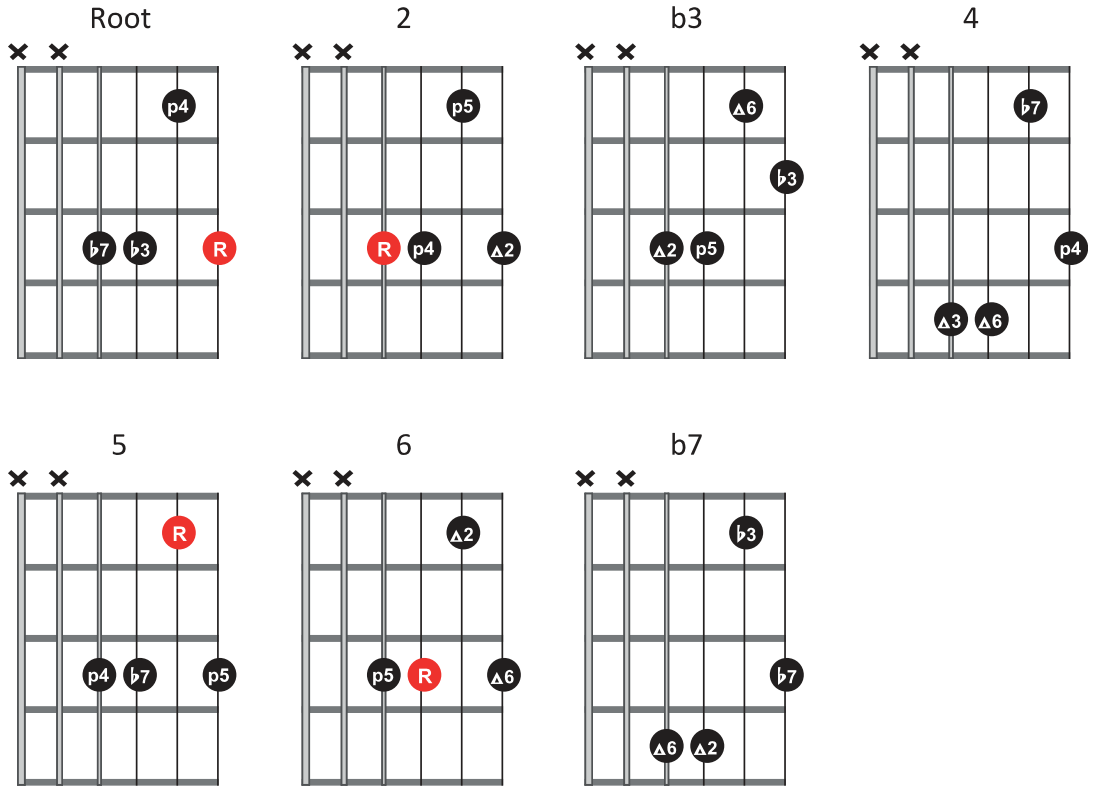
Major Scale (modern)

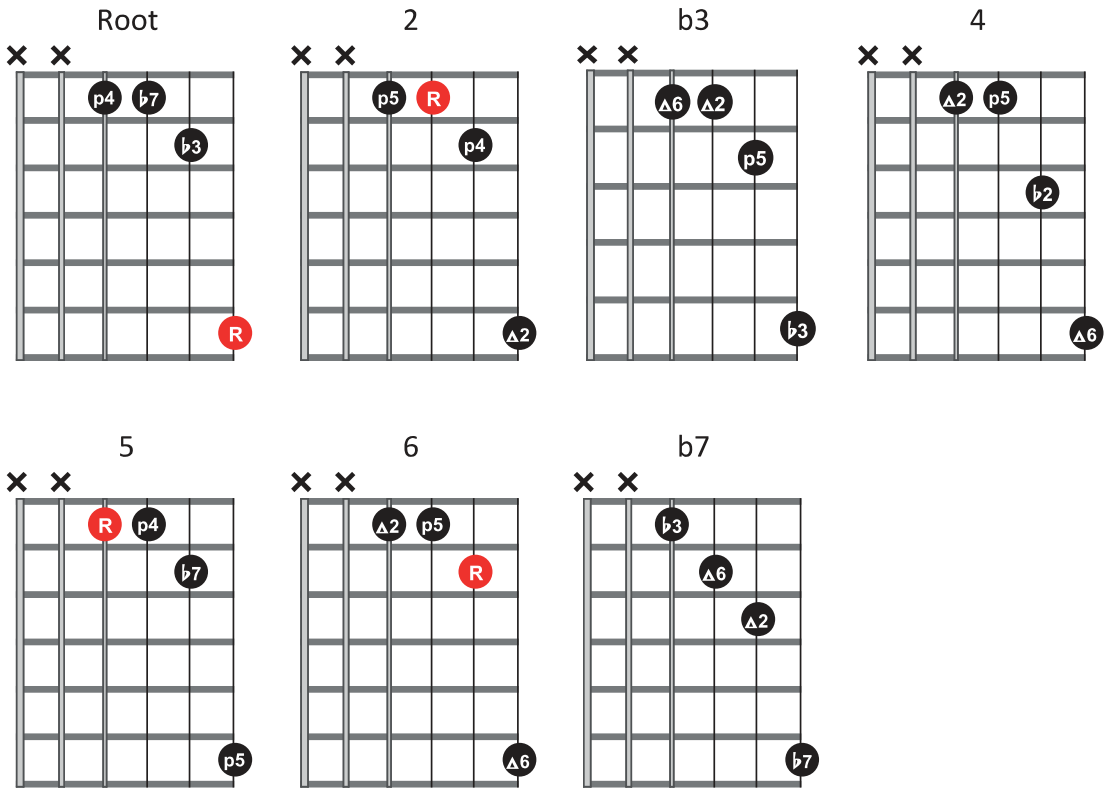


Dorian/Minor (Bebop)

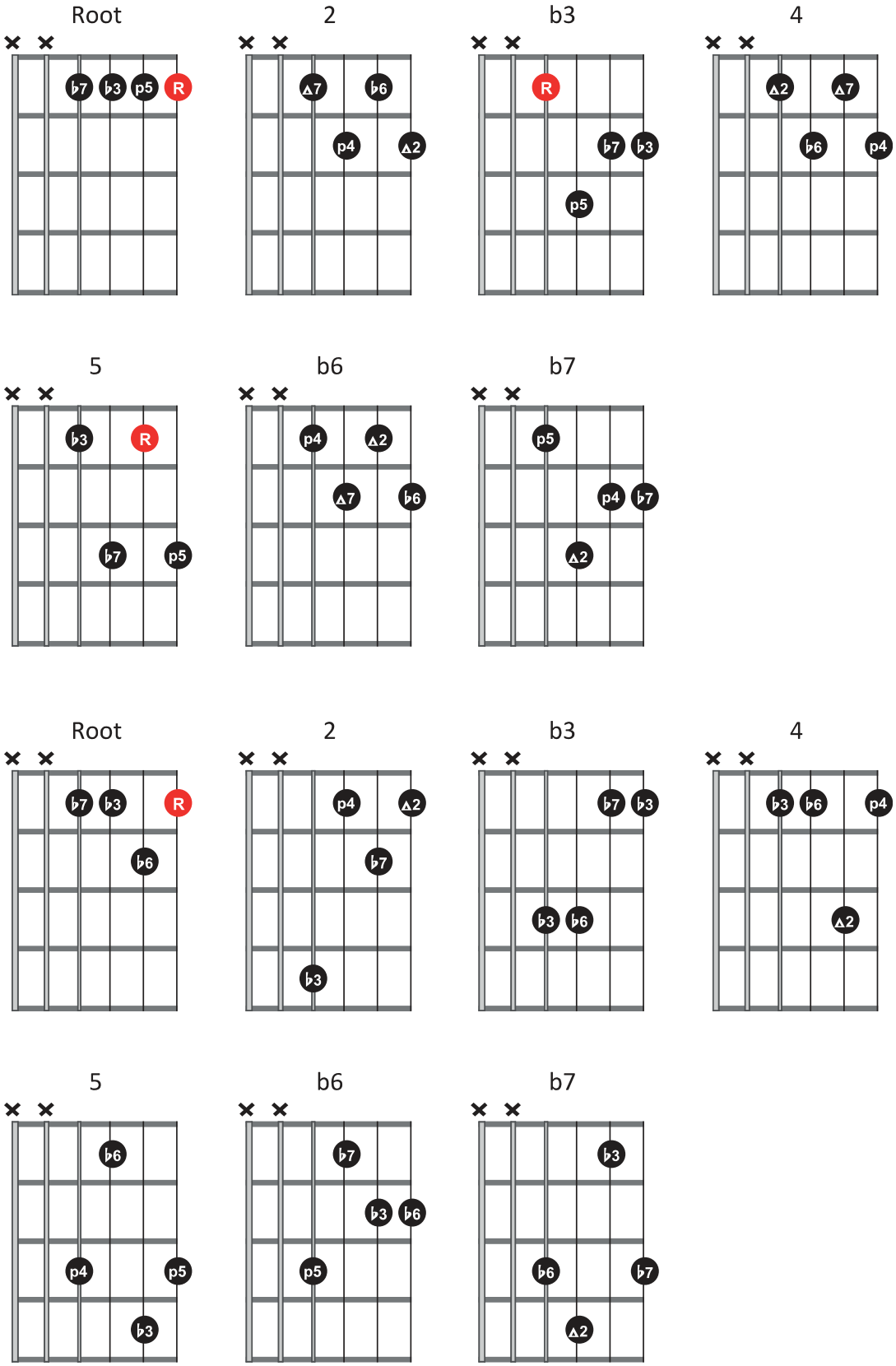


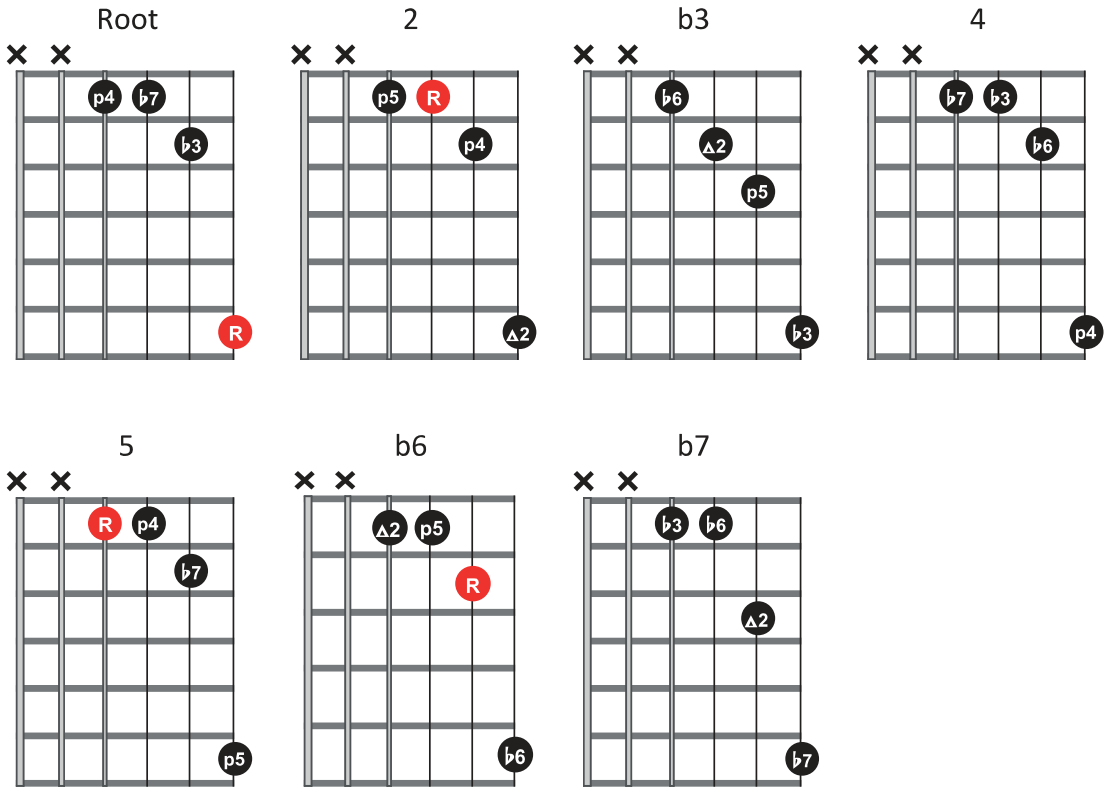
Dorian/Minor (modern)



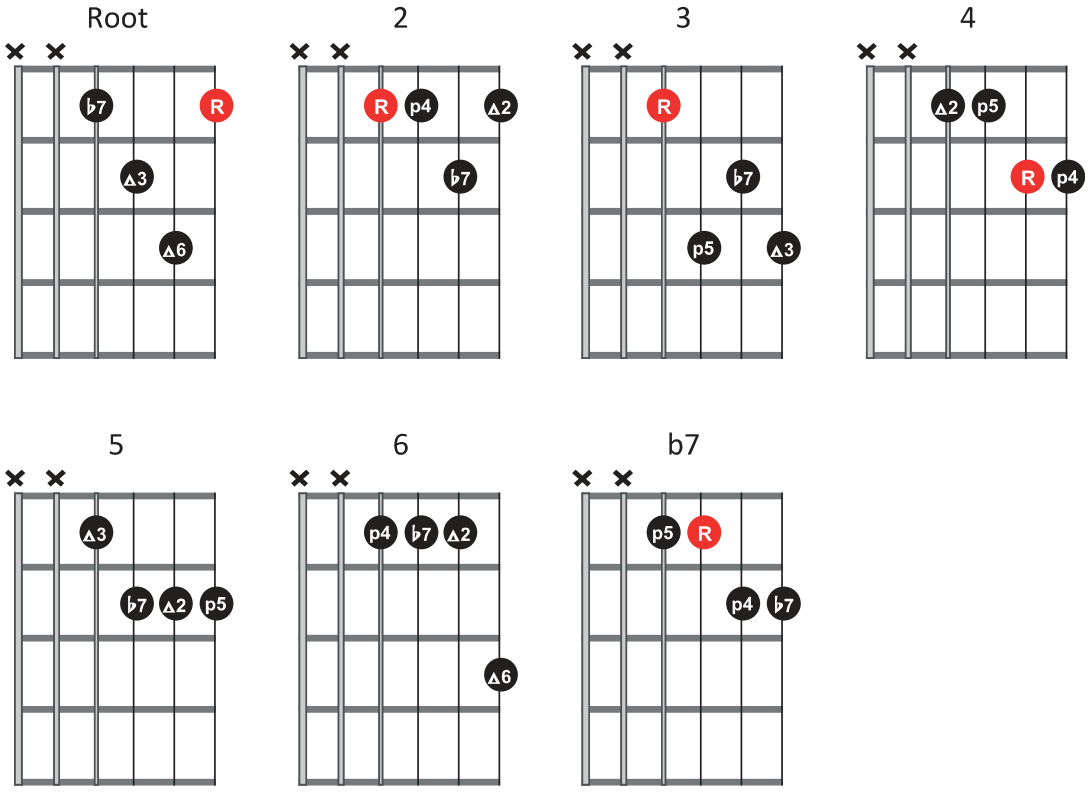


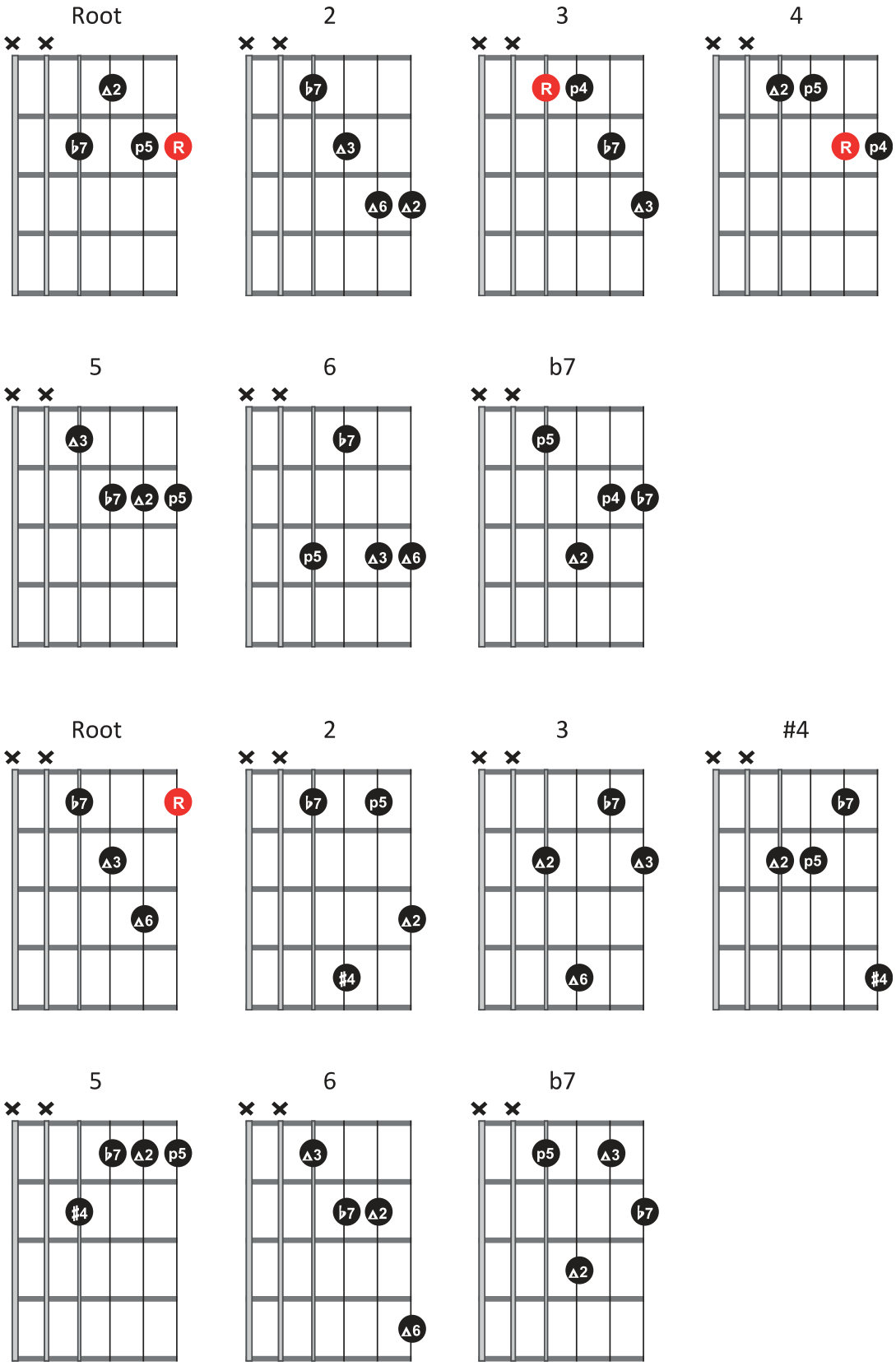
Aeolian/Minor





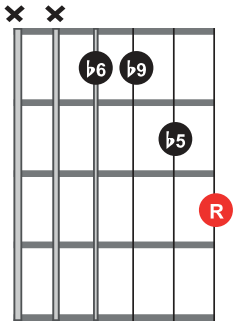
Dom^7 (unaltered)



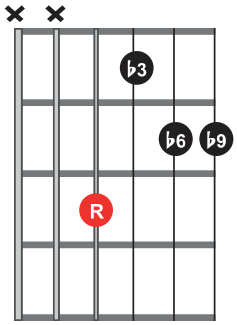

Root
2
3
4
5
6
b7
Root
2
3
#4
5
6
b7

Dom^{7} (altered Dom)

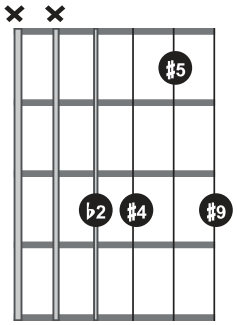
Root



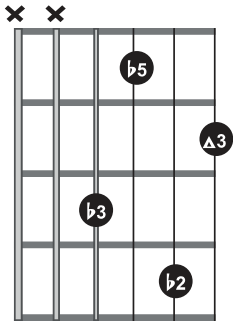
b9



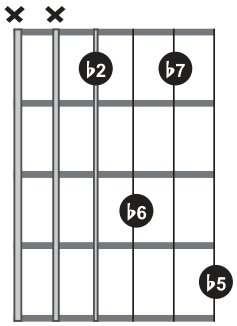
#9



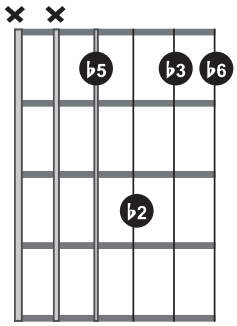
3



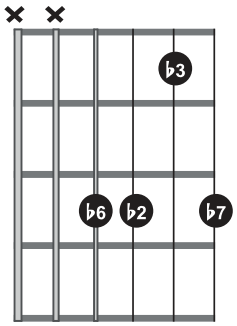
#4



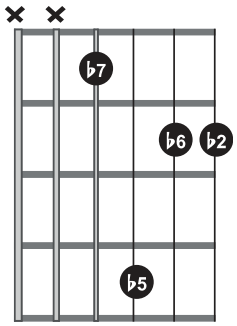
#5



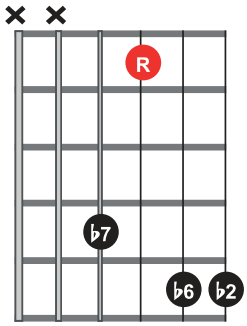
b7

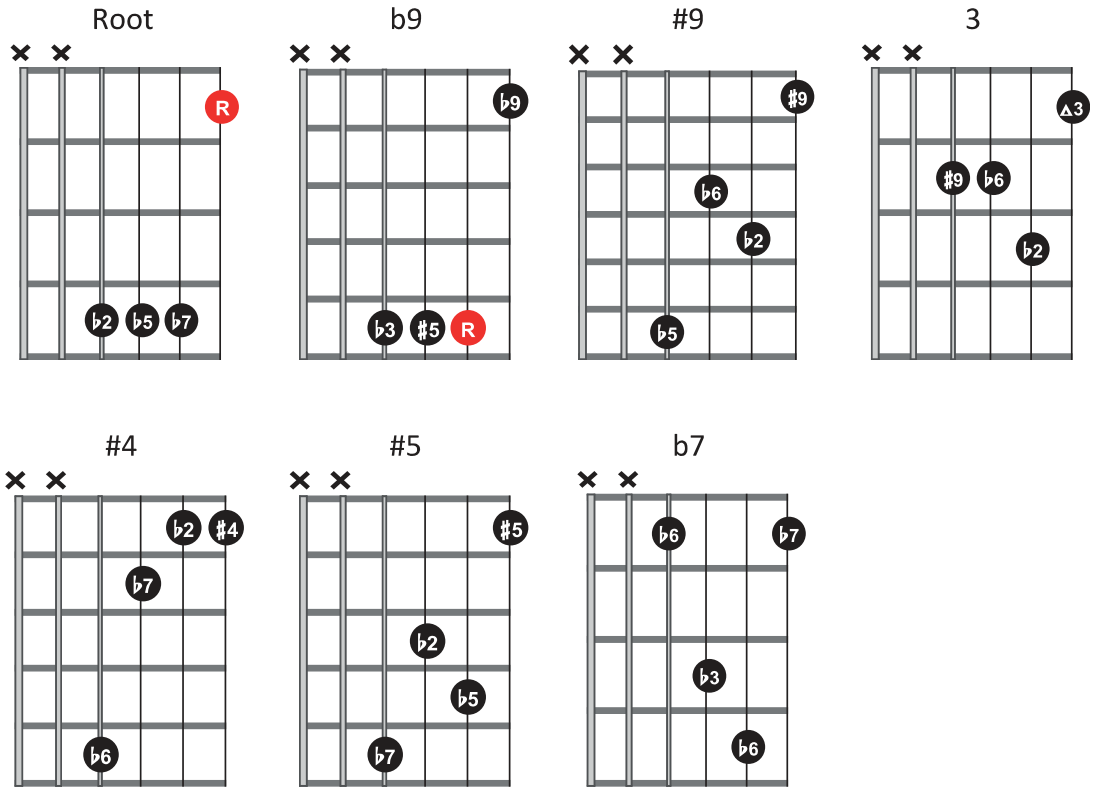


b9

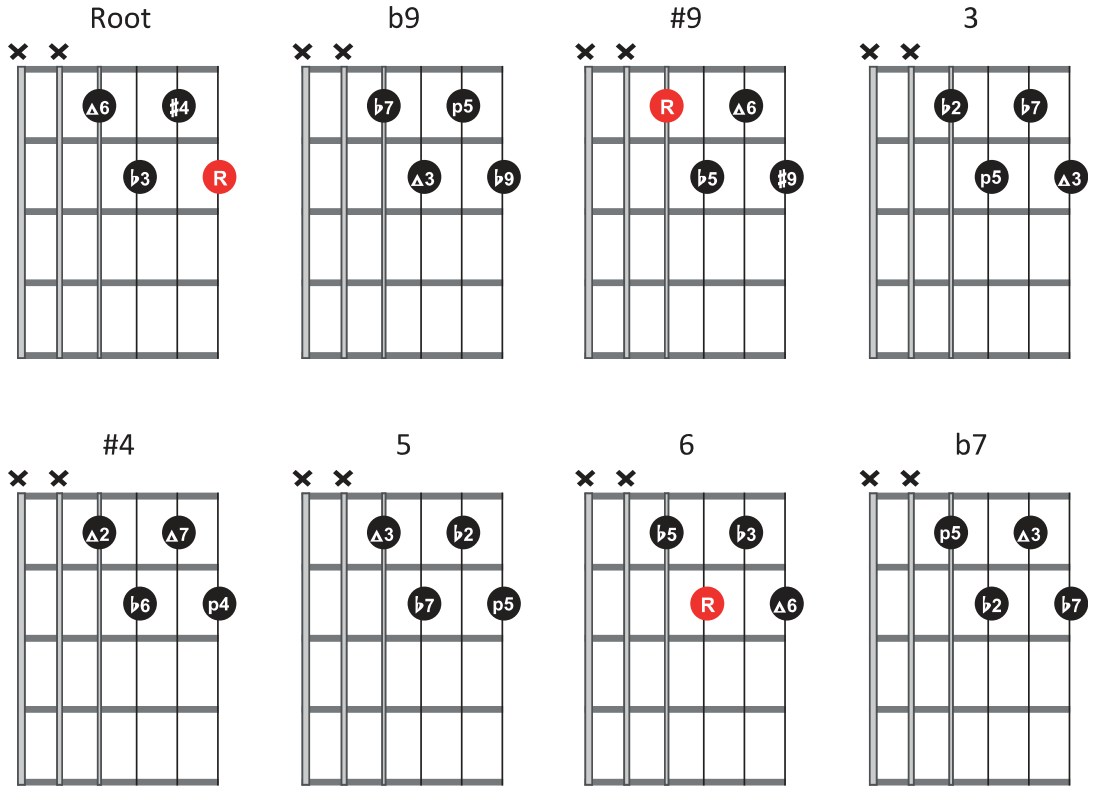


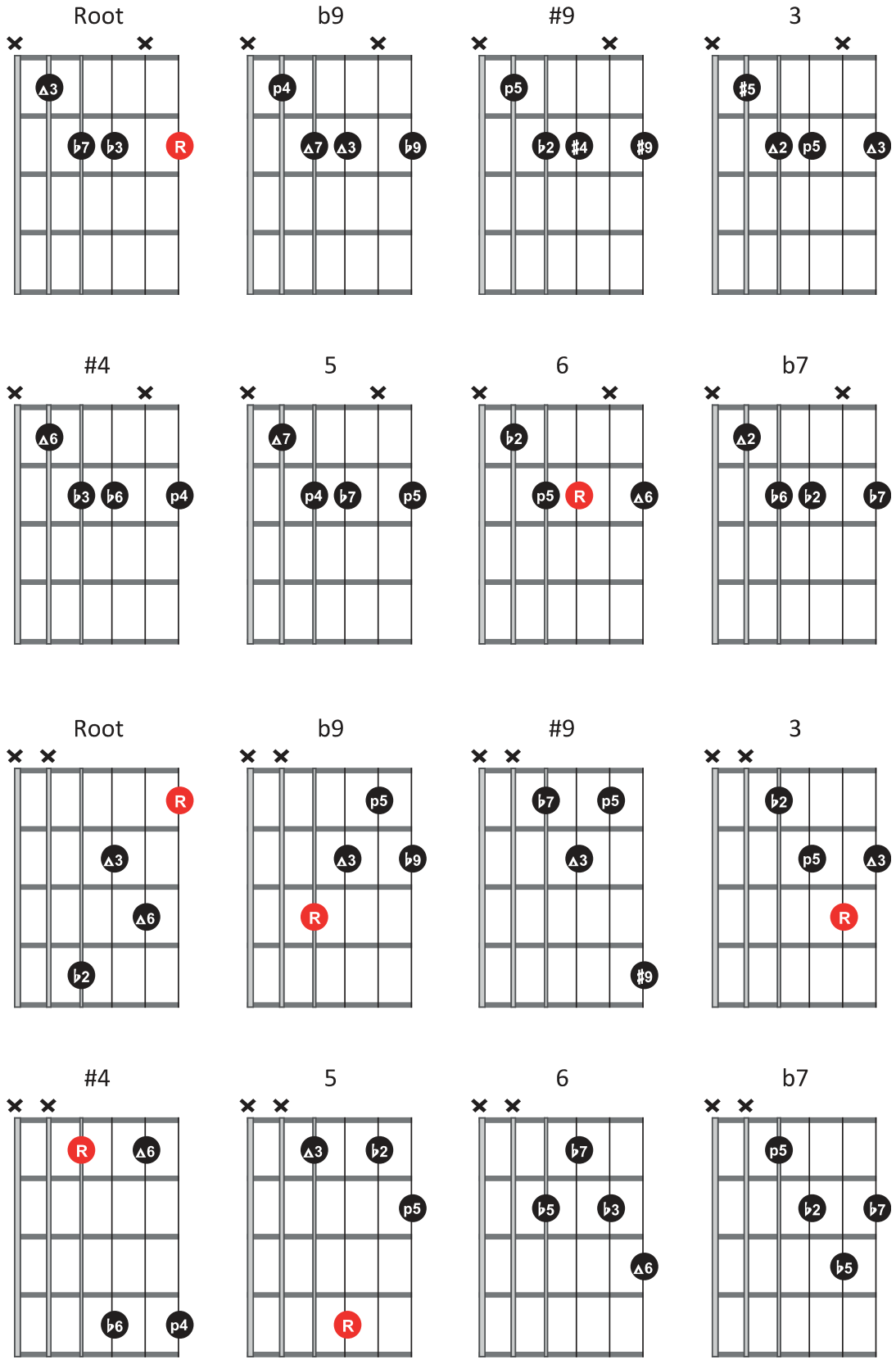
b9





Dom^{7} (Dominant Dim)

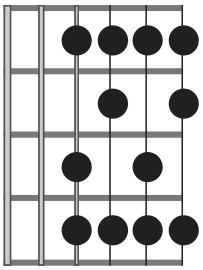



Root
b9
#9
3
#4
5
6
b7
Root
b9
#9
3
#4
5
6
b7

Diminished Scale Voicings

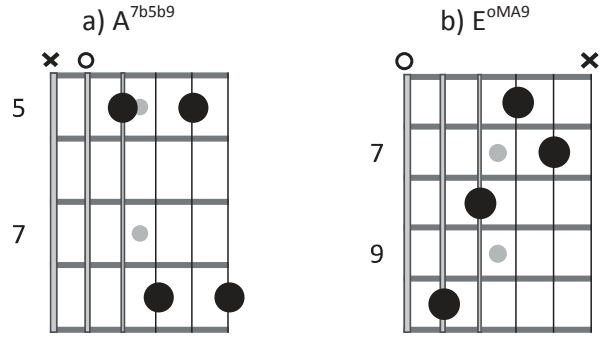
The accompanying diminished voicings can be used in tonic or dominant situations, depending on fret position. Most of the voicings are derived from this pattern.

Ex. 148



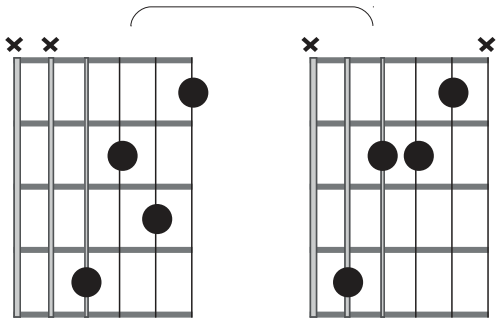
The open E and A strings work nicely as bass notes for these type voicings.

Ex. 149



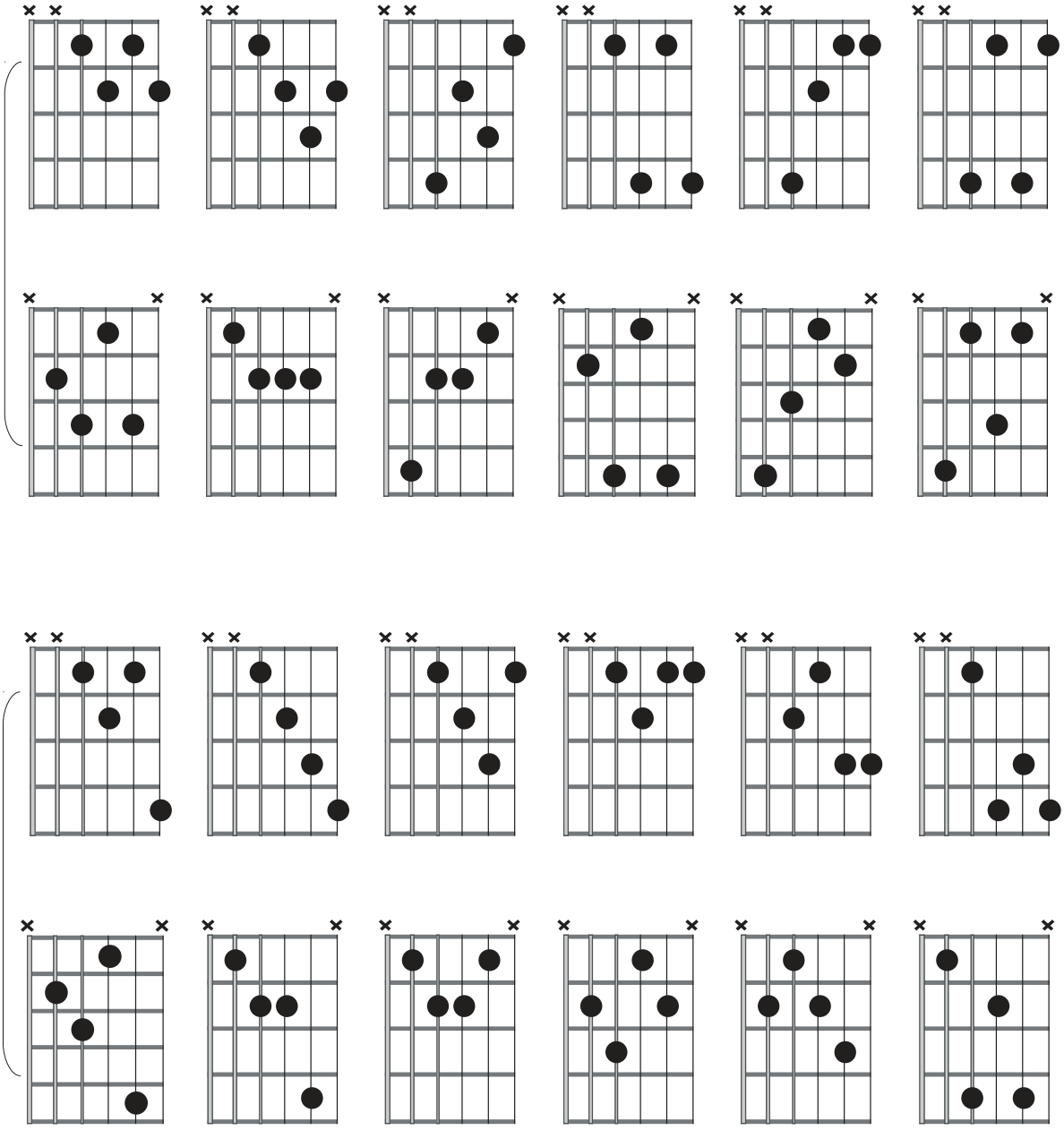
The voicings are grouped in pairs because they are the same shapes transposed.

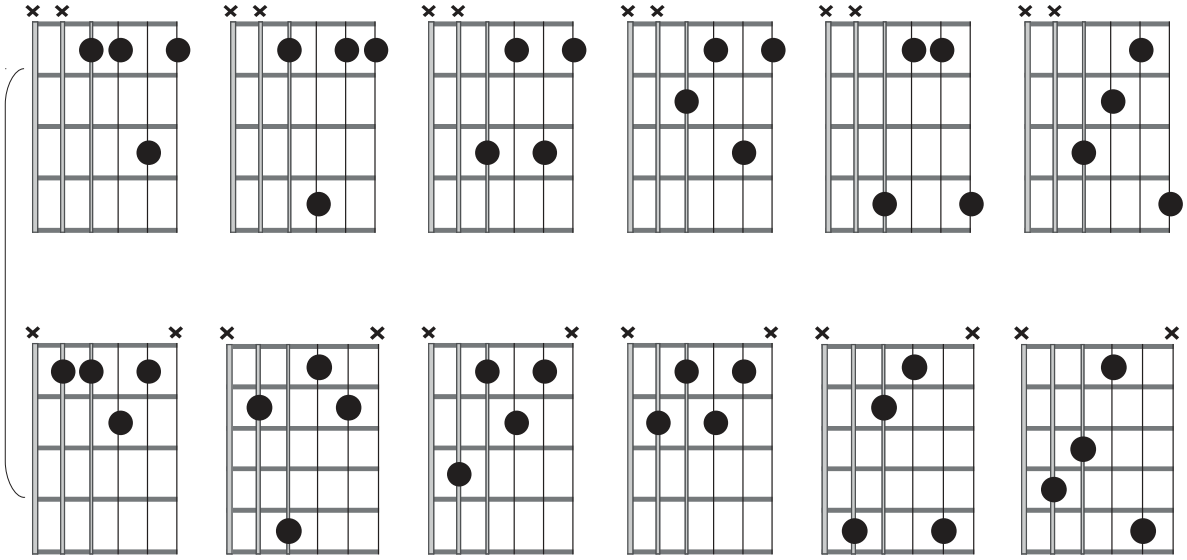
Ex. 150

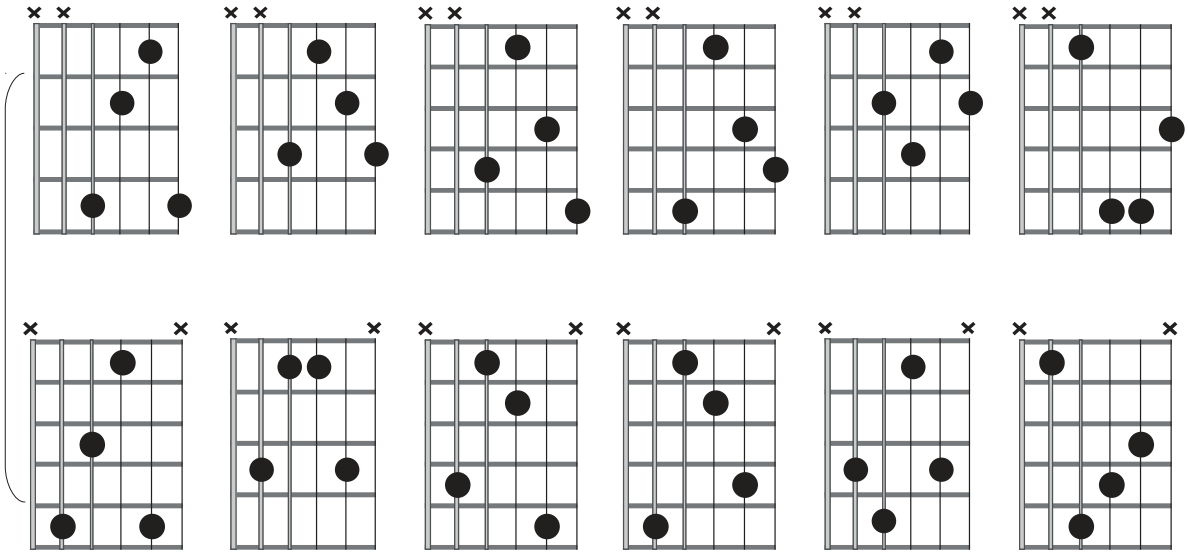


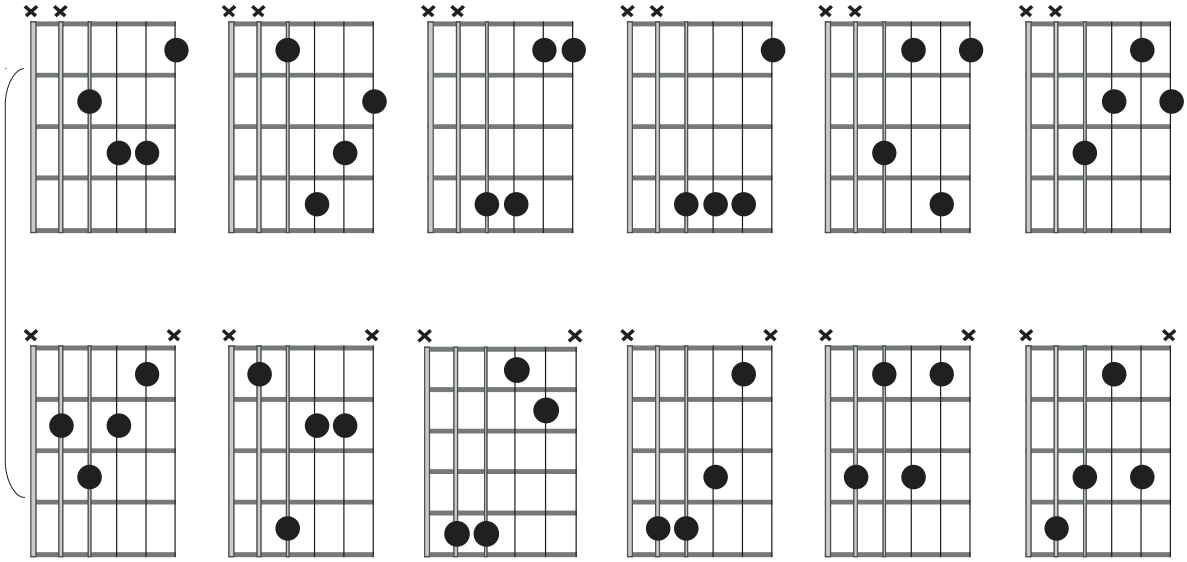
Remember, all of the voicings are moveable by minor thirds.

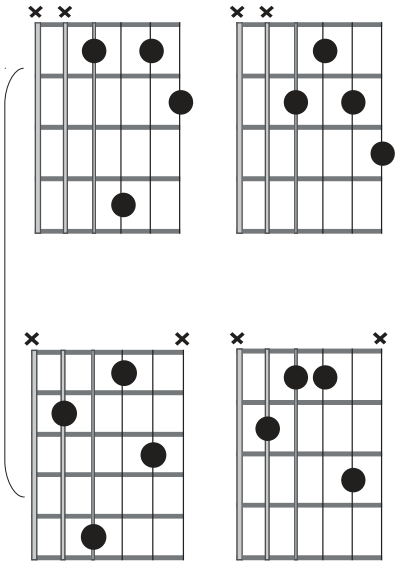
Dim^7 Type Comping Voicings











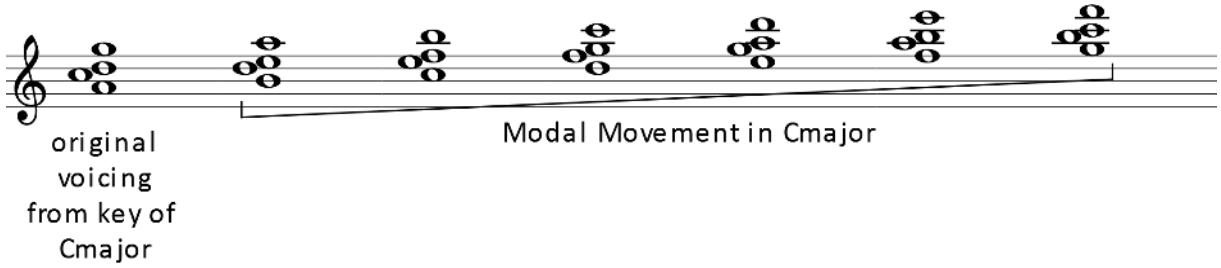
Stretch Voicings

For lack of a better term, the following harmonies are called stretch voicings. I recommend that you practice these voicings for no more than ten minutes at a time. This will prevent any left hand discomfort caused by the reaches involved. If you experience some pain it may simply be related to the stretching of the left hand muscles.

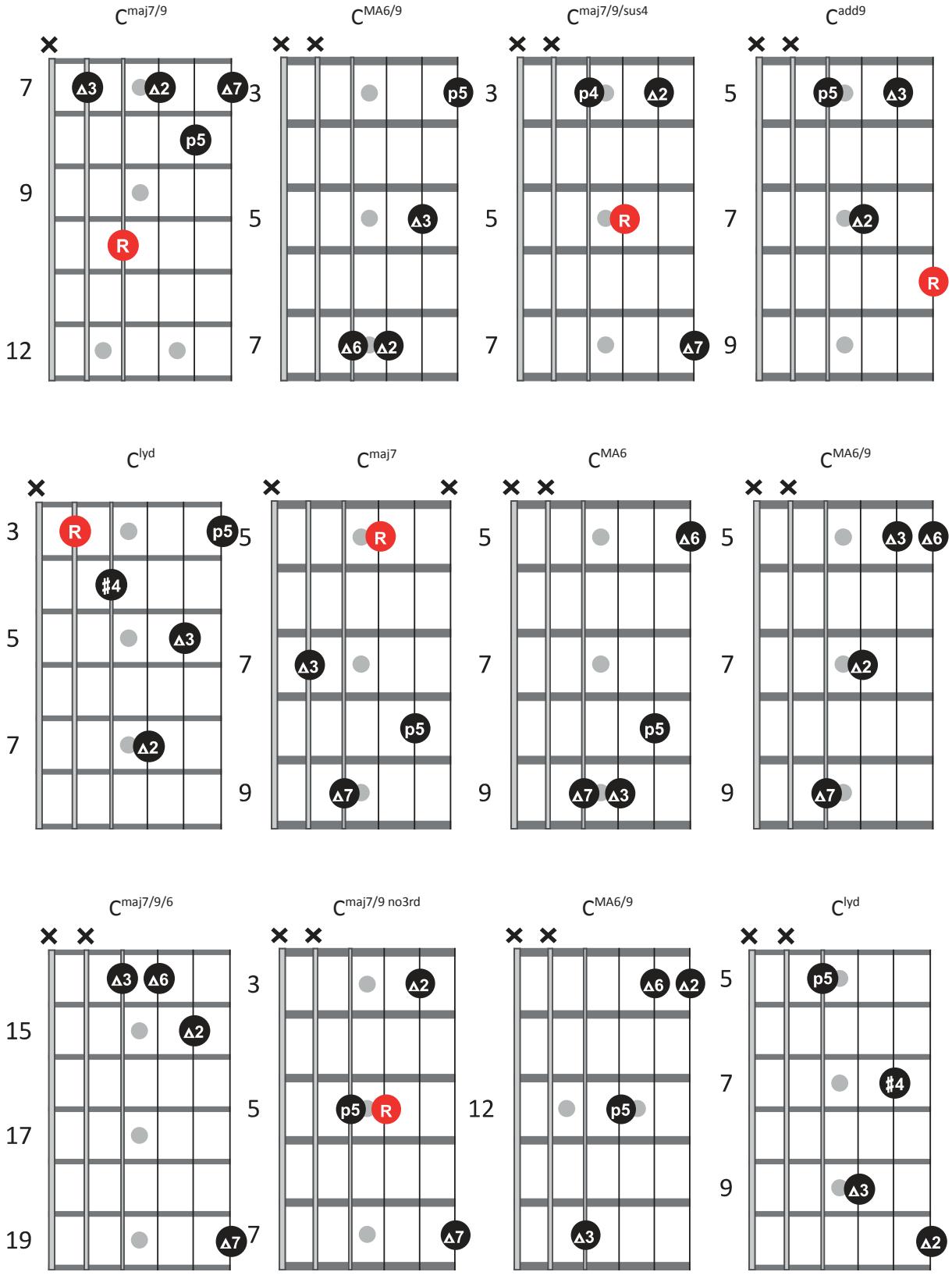
This pain, however, should diminish as your left hand muscles become more accustomed to the stretches.

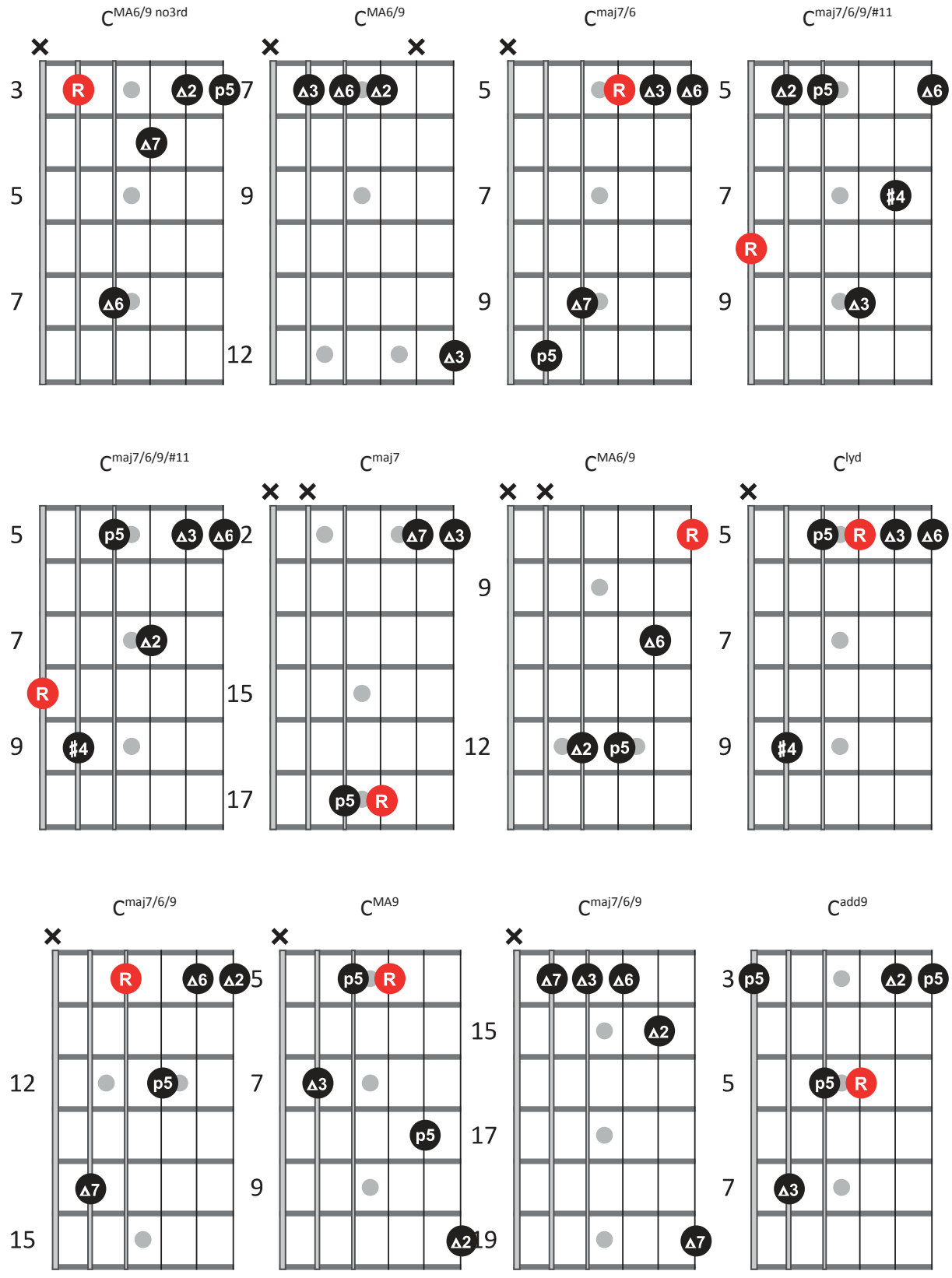
I would also like to remind you that all of these voicings can be moved modally on the fingerboard.

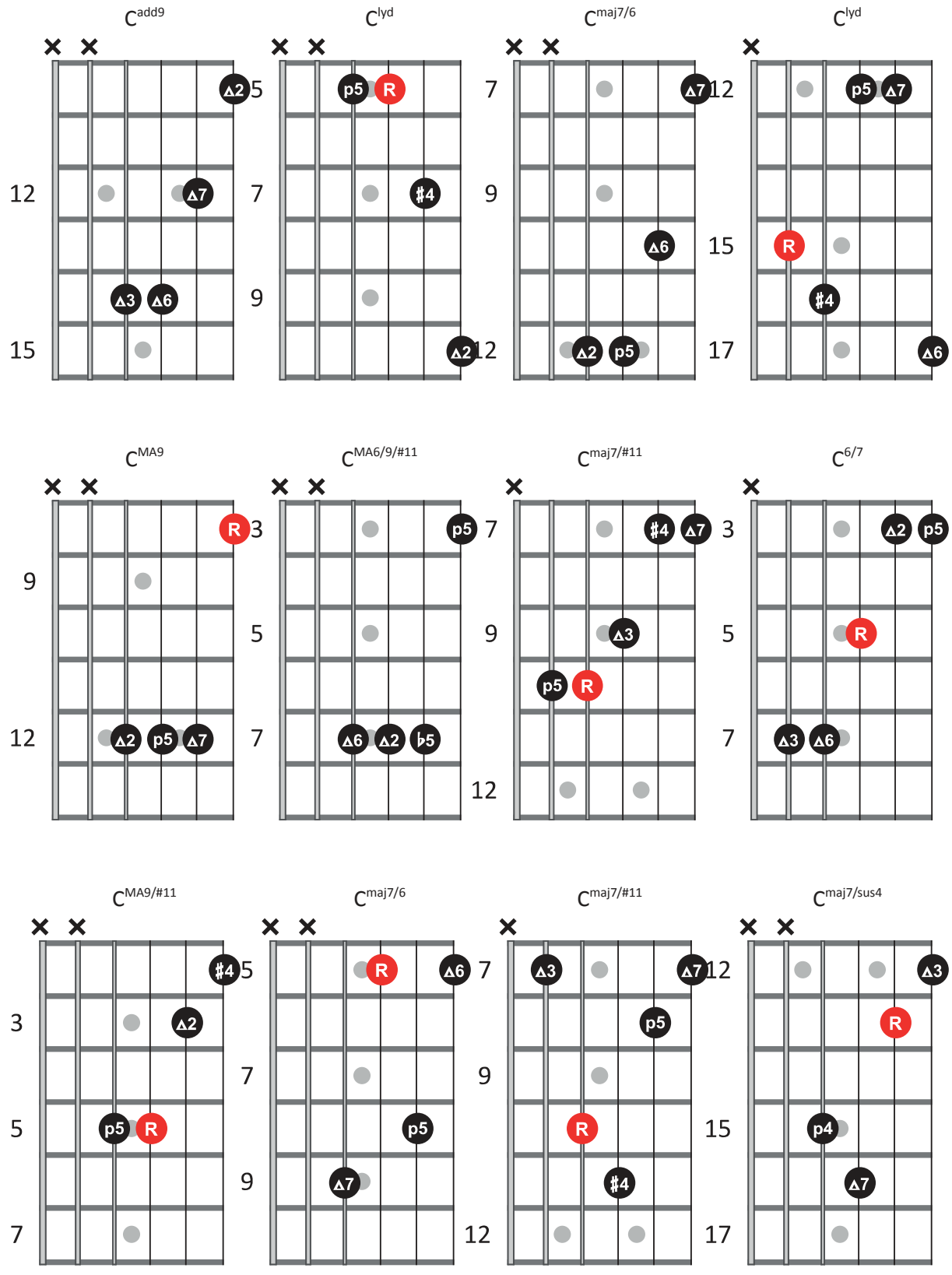
Ex. 151



Stretch Voicings




C MA6/9 no3rd
3
R
Δ2
p5
Δ7
5
7
Δ6
C MA6/9
7
Δ3
Δ6
Δ2
9
12
Δ3
C maj7/6
5
R
Δ3
Δ6
7
9
Δ7
p5
C maj7/6/9/#11
5
Δ2
p5
Δ6
7
♯4
R
9
Δ3
C maj7/6/9/#11
5
p5
Δ3
Δ6
7
Δ2
R
9
♯4
C maj7
2
Δ7
Δ3
15
17
p5
R
C MA6/9
R
9
Δ6
12
Δ2
p5
C lyd
5
p5
R
Δ3
Δ6
7
9
♯4
C maj7/6/9
R
Δ6
Δ2
5
12
p5
7
Δ7
9
15
C MA9
p5
R
7
Δ3
p5
9
Δ2
C maj7/6/9
Δ7
Δ3
Δ6
15
Δ2
17
19
Δ7
C add9
3
p5
Δ2
p5
5
p5
R
7
Δ3

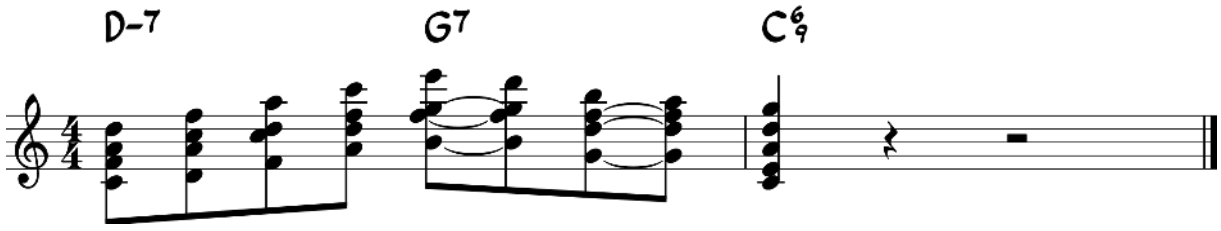

Cadd9
Clyd
Cmaj7/6
Clyd
CMA9
CMA6/9/#11
Cmaj7/#11
C6/7
CMA9/#11
Cmaj7/6
Cmaj7/#11
Cmaj7/sus4

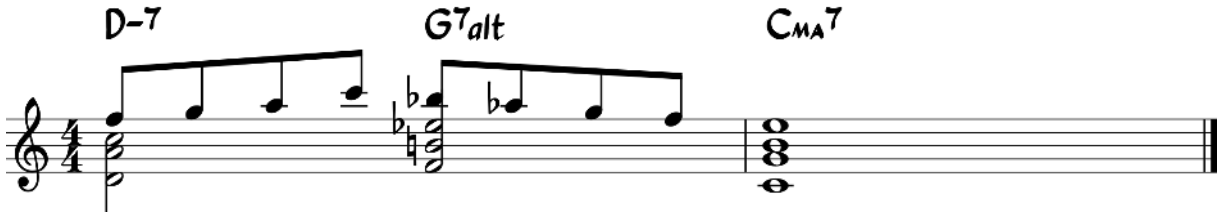
Chordal Lines (ii^7 V^7 I^{maj7})

Here are some chordal lines for ii^7 V^7 I^{maj7} progressions. I have used some single notes for variety.

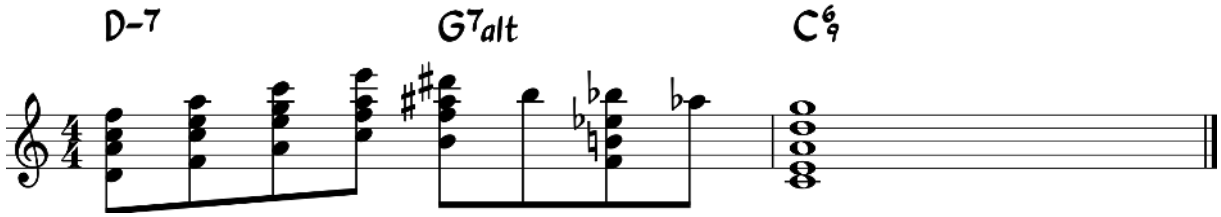
Try transposing these to all twelve keys. Transposing helps you to better understand the variety of chord fingerings on different string groups.

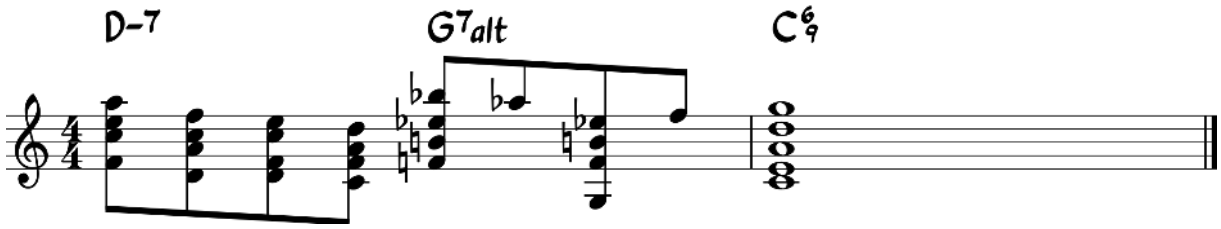
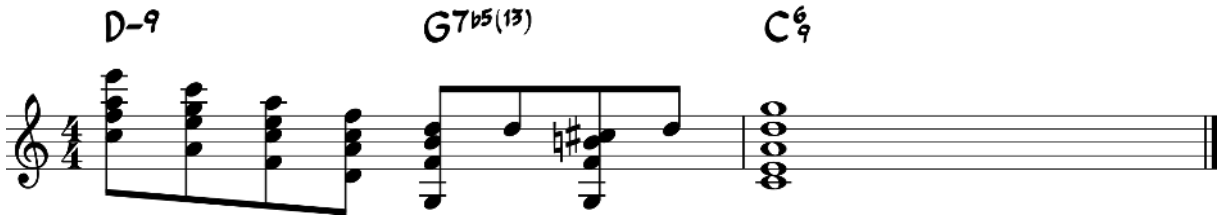
After mastering these, work out some of your own for minor ii V´s.

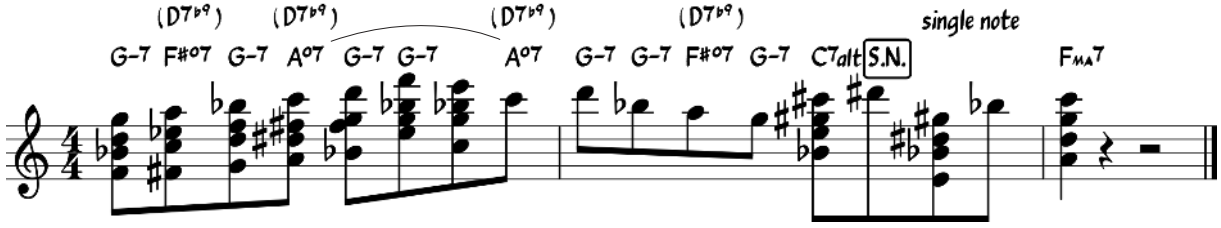
Ex. 152a

Ex. 152b

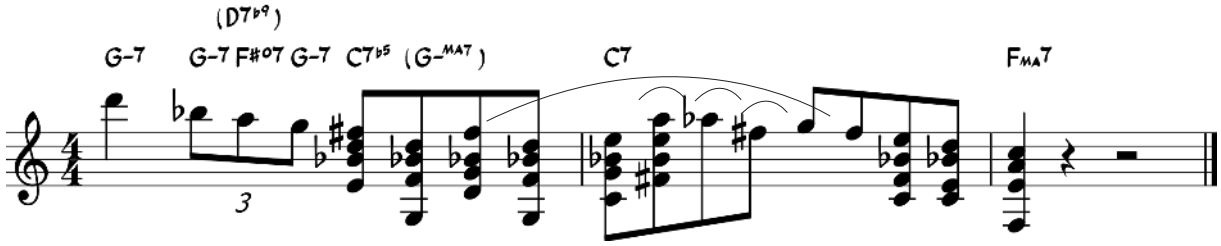


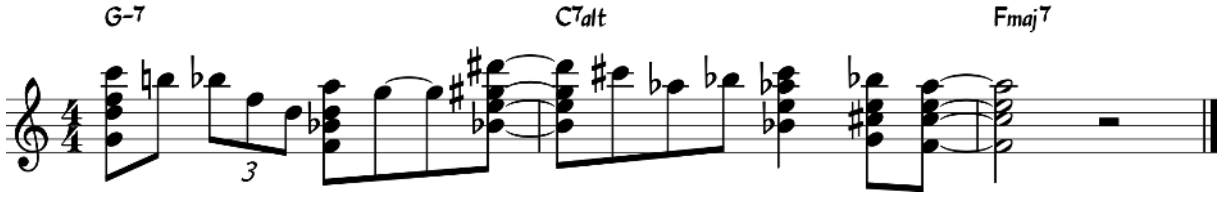
Ex. 152c

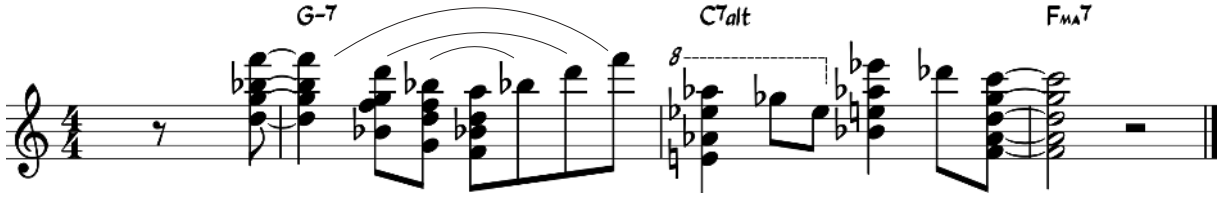


Ex. 152d

Ex. 152e

Here are some examples of ii^{7} V^{7} I^{maj7} chord lines in the key of Fmajor. The arrows are used to designate the use of the same voicing.

Ex. 153a

Ex. 153b

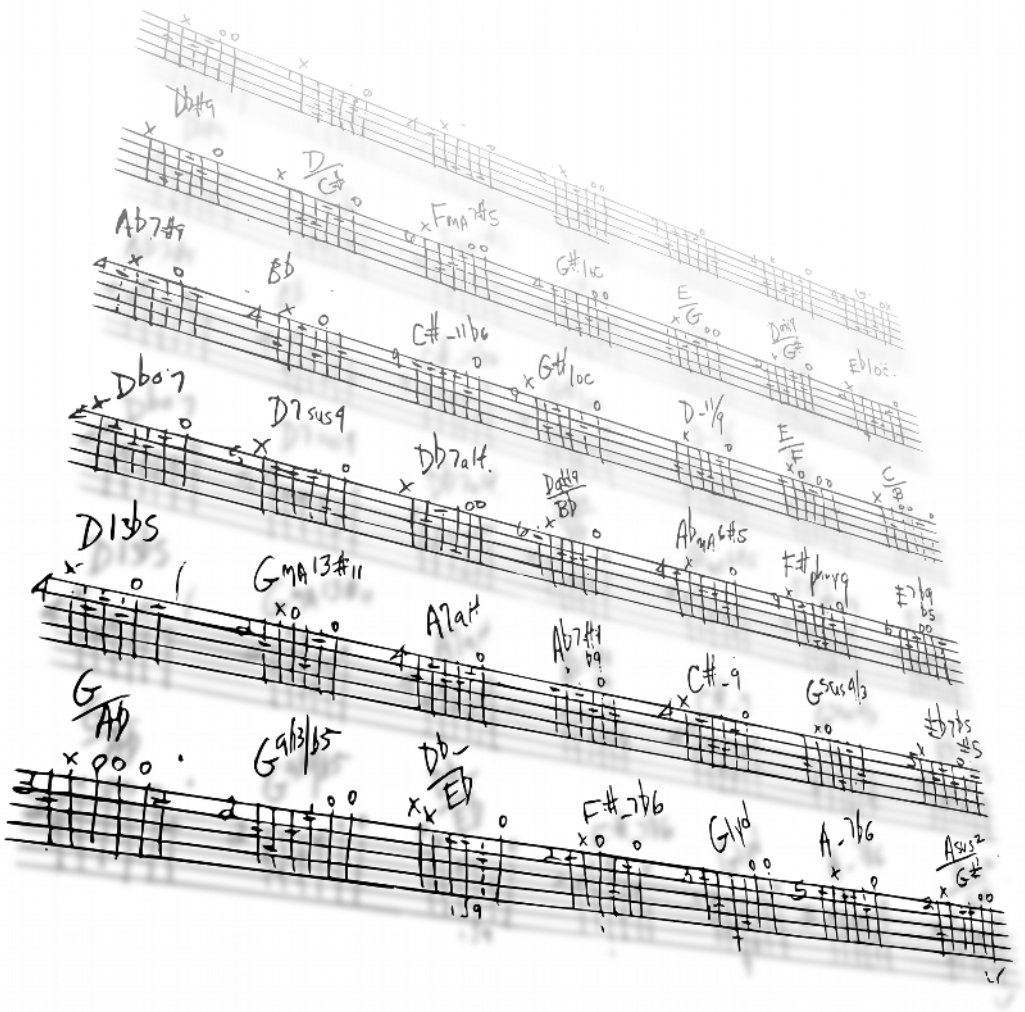


Ex. 153c

Ex. 153d



Chapter 3

Scales + Arpeggios



Chapter 3

Mode (Scale) Practice

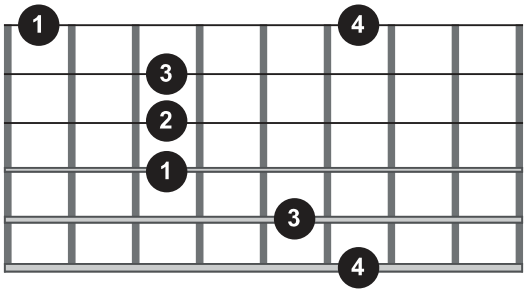
Modes should be practiced every day to warm-up the hands and develop fingerboard understanding. They should be practiced to a metronome in a variety of rhythms (eighths, quarters, eigth-note triplets, sixteenths, sixteenth-note triplets and dotted rhythms) paying particular attention to sound clarity.

Thorough knowledge of scale fingerings (two and three notes per string) will enable you to execute even the most difficult passages.

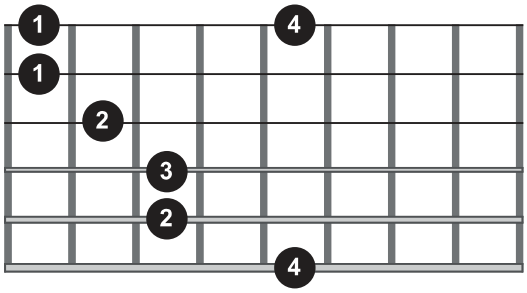
I would recommend practicing one scale group (major, melodic minor or harmonic minor) per day, isolating one mode for the purpose of interval studies. Practicing too many interval or arpeggio studies in one sitting will only add to your confusion.

Triads

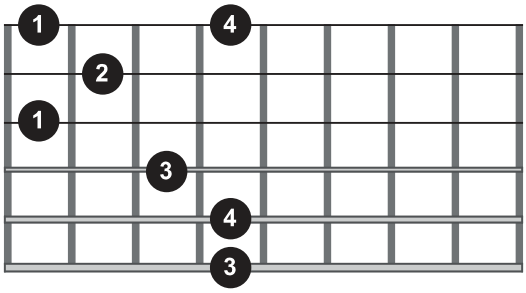
Major (Root Position)



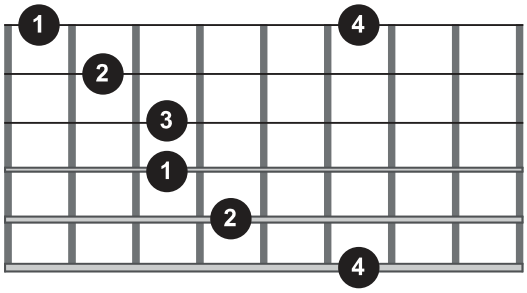
Major (3rd in Bass)



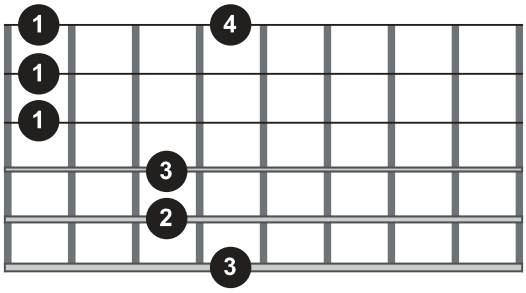
Major (5th in Bass)



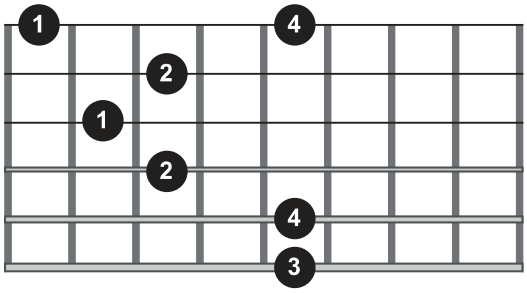
Minor (Root Position)



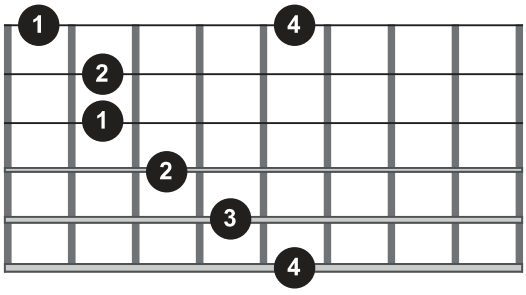
Minor (3rd in Bass)



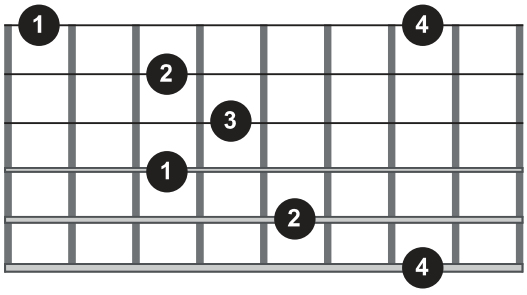
Minor (5th in Bass)



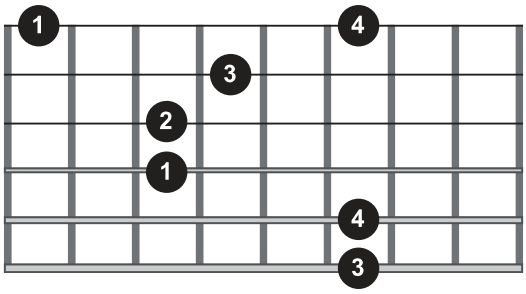
Augmented



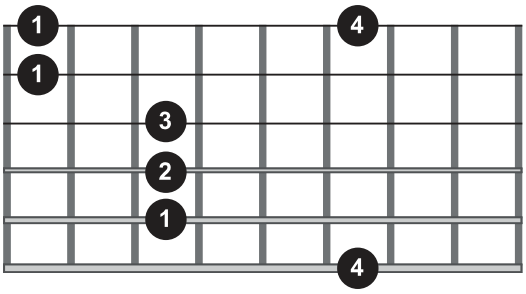
Diminished



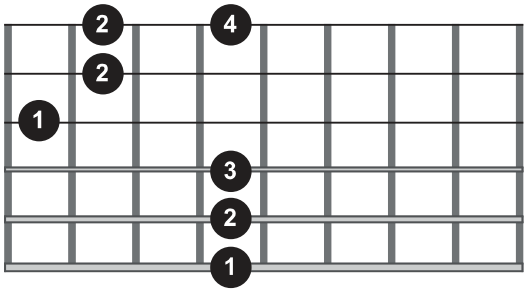
Sus^4



Sus^2

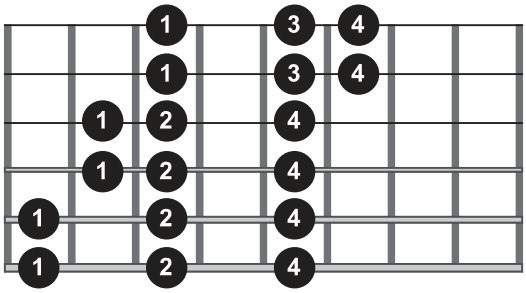


Quartal

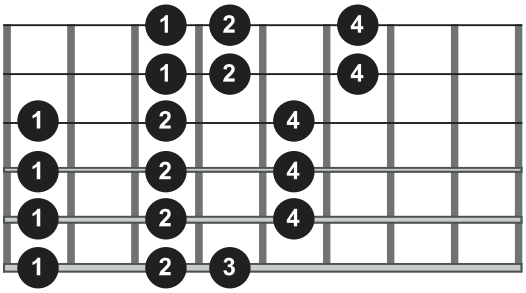


Modes of the Major Scale

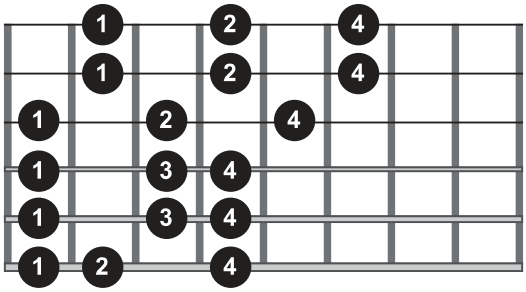
Ionian



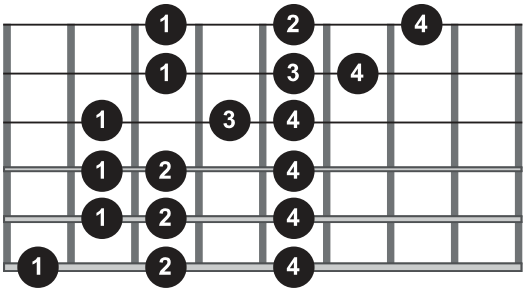
Dorian



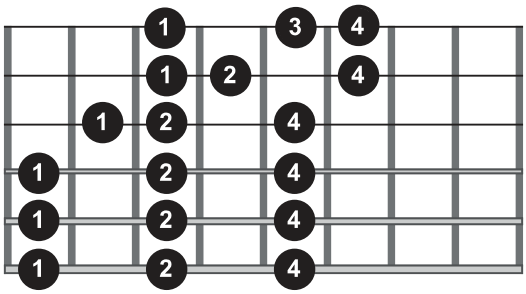
Phrygian



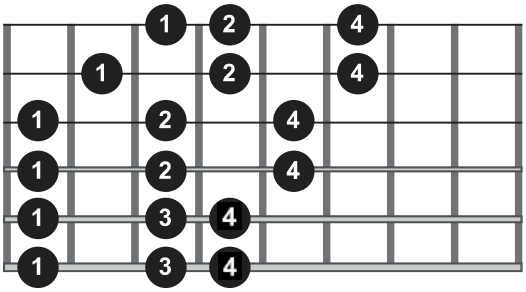
Lydian



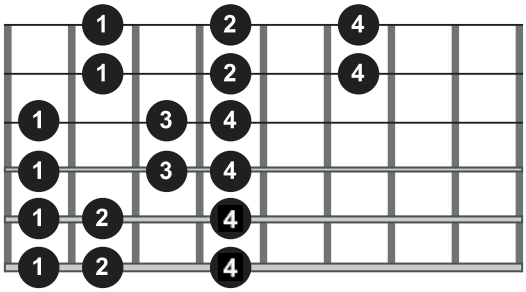
Mixolydian



Aeolian

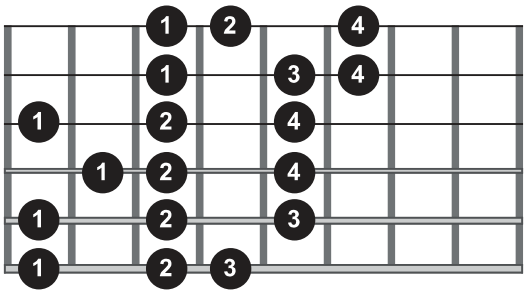


Locrian

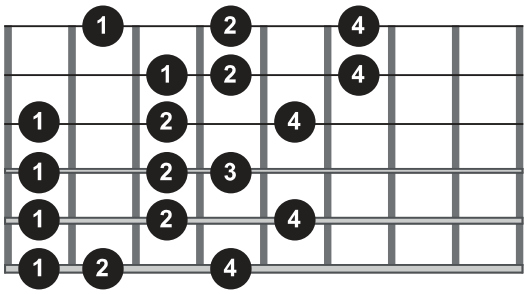


Modes of the Melodic Minor Scale

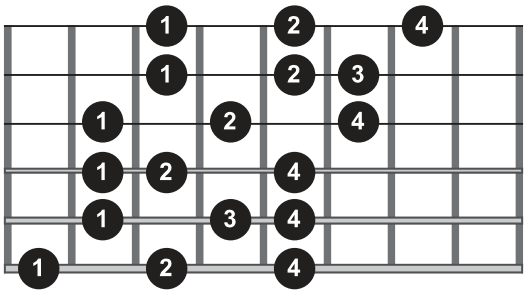
Melodic Minor



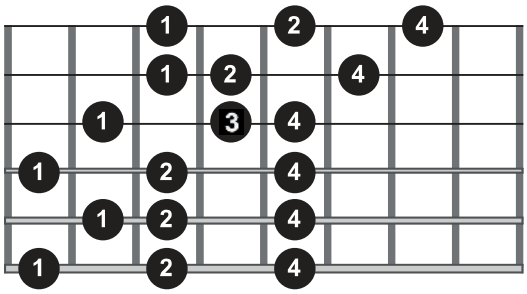
Dorian b2



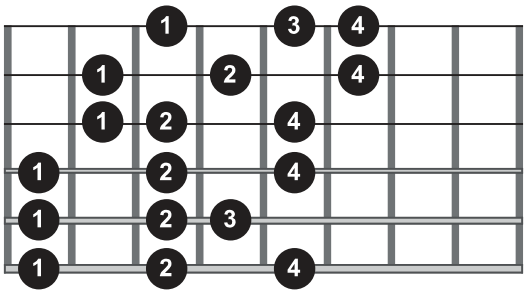
Lydian Augmented



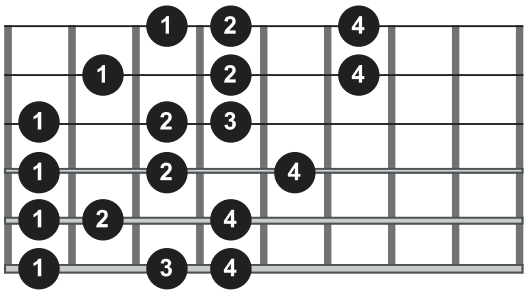
Mixolydian #11



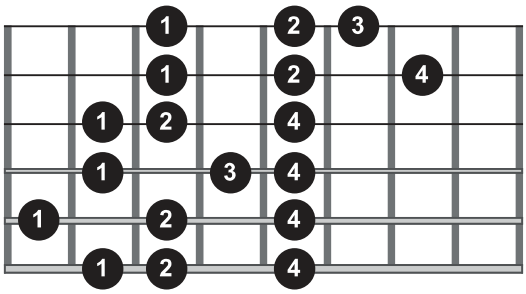
Mixolydian b6



Locrian nat.2

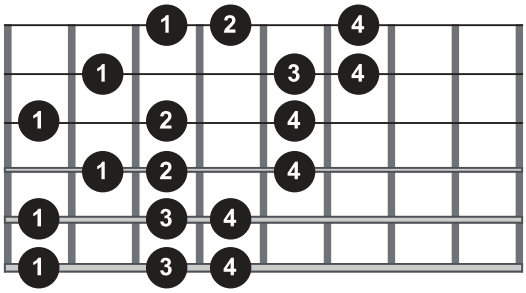


Altered Dominat

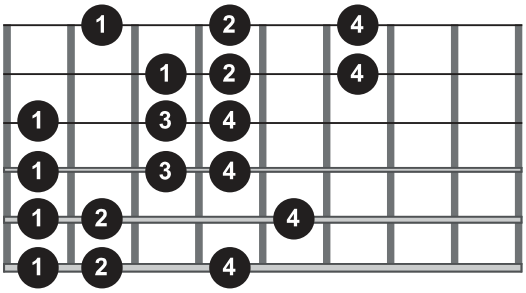


Modes of the Harmonic Minor Scale

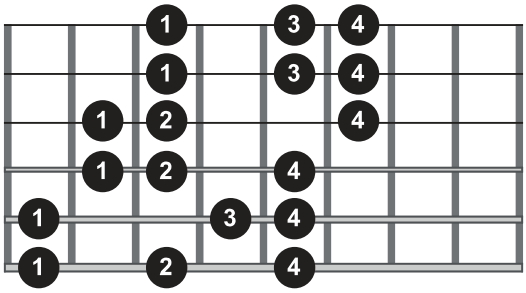
Harmonic Minor



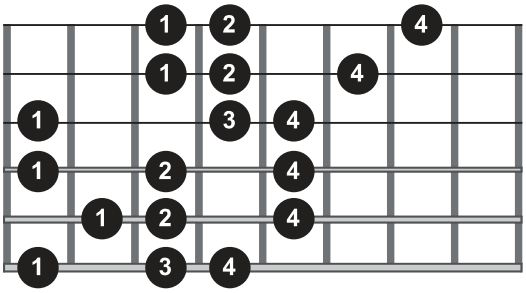
Locrian nat. 6



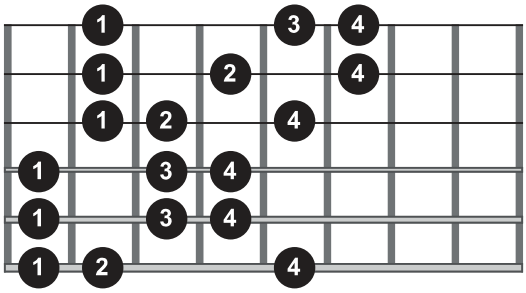
Ionian Augmented



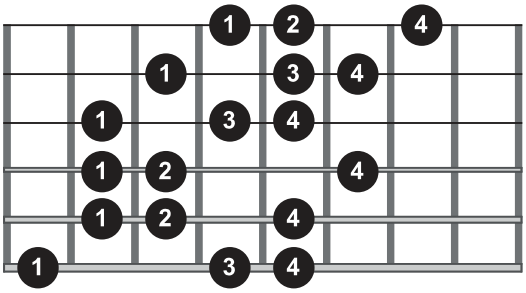
Dorian #4



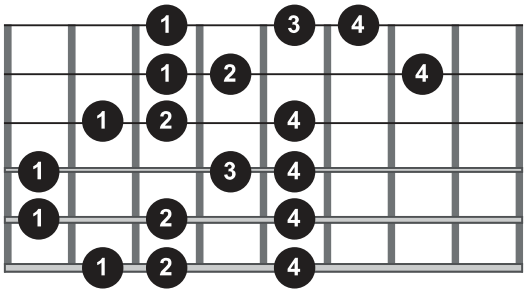
Phrygian Major



Lydian #9

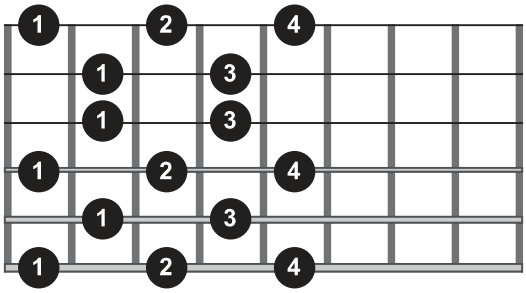


Altered Dominant bb7

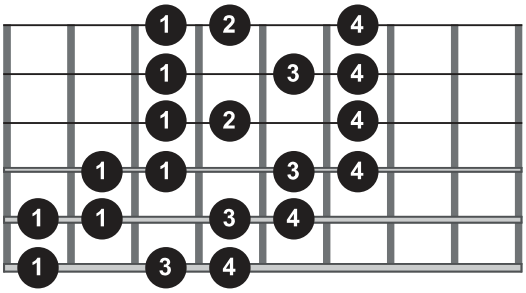


Miscellaneous Scales

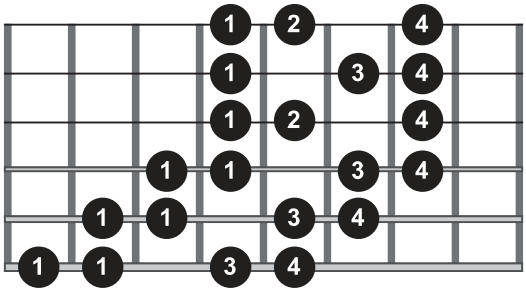
Whole Tone



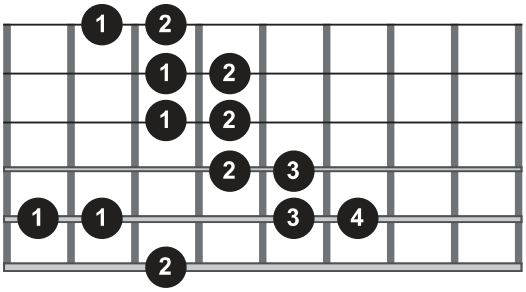
Tonic Diminished



Dominant Diminished



Augmented

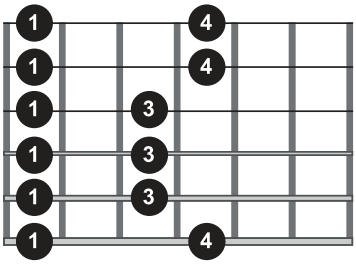


Blues and Pentatonic Scales

Pentatonic Scales:

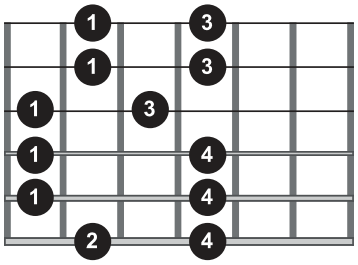
Major:	1	2	3	5	6
Minor:	1	b3	4	5	b7

Maj / Min (6^{th}/Root)



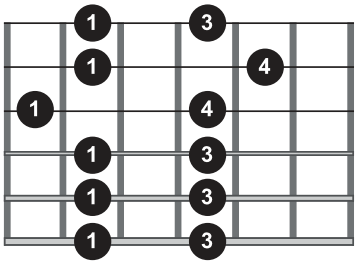
Position I

R/$b3^{rd}$



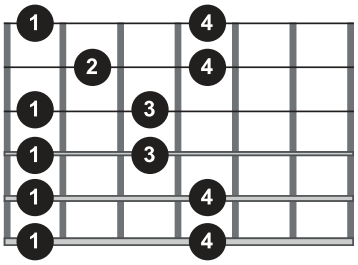
Position II

2^{nd}/4^{th}



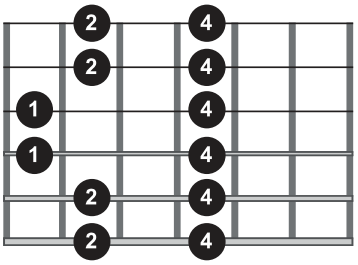
Position III

3^{rd}/5^{th}



Position IV

5^{th}/$b7^{th}$

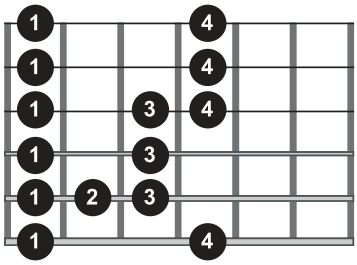


Position V

Blues Scales:

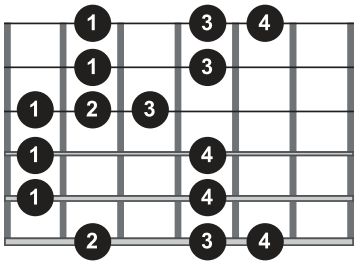
Major:	1	2	b3	3	5	6
Minor:	1	b3	4	#4	5	b7

Maj / Min (6^{th}/Root)



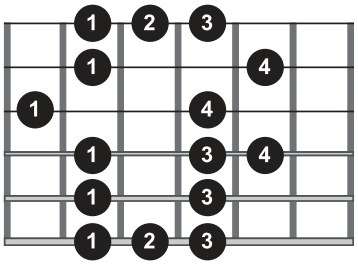
Position I

R/$b3^{rd}$



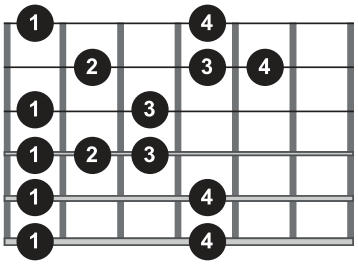
Position II

2^{nd}/4^{th}



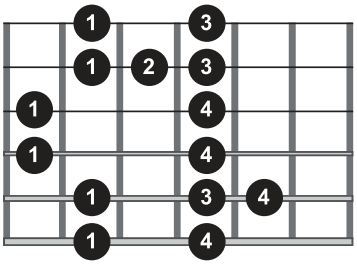
Position III

3^{rd}/5^{th}



Position IV

5^{th}/$b7^{th}$



Position V

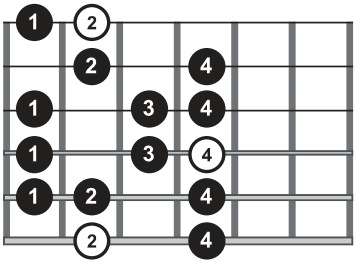
Close Position Fingerings

Close position fingerings work on a one finger per fret principle. This allows the left hand to be in a compact position giving the fingers greater striking power.

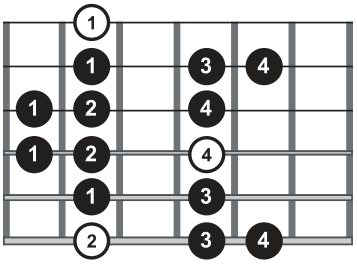
The added striking power creates a percussive attack aurally resembling a picked note. If you are already familiar with these fingerings move on to the two note per string scales.

Major Scale

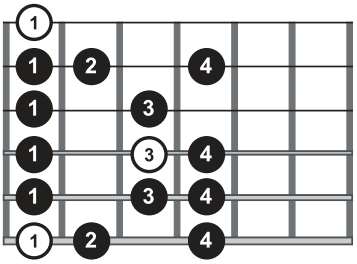
Ionian



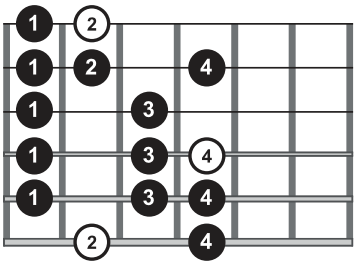
Dorian



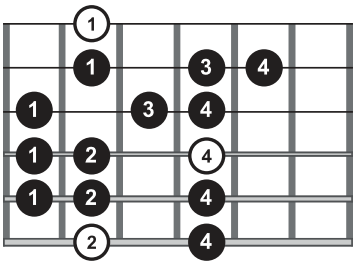
Phrygian



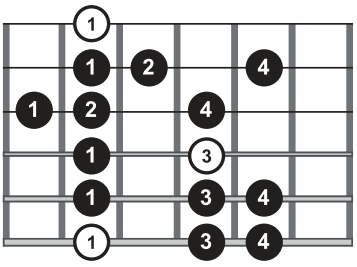
Lydian



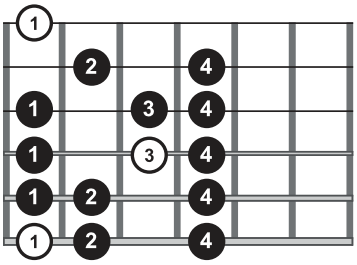
Mixolydian



Aeolian

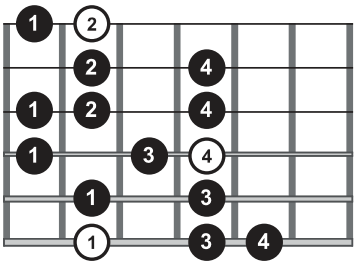


Locrian

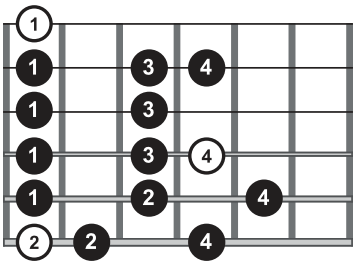


Melodic Minor Scale

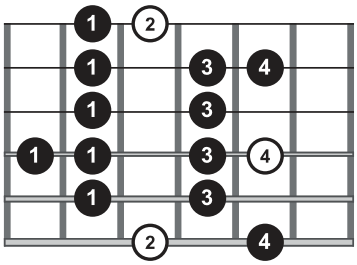
Melodic Minor



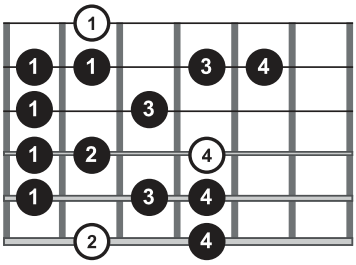
Dorian b2



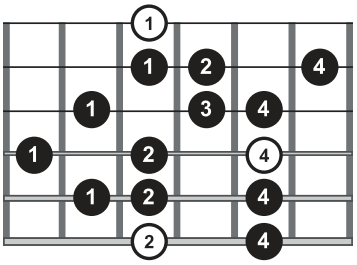
Lydian Augmented



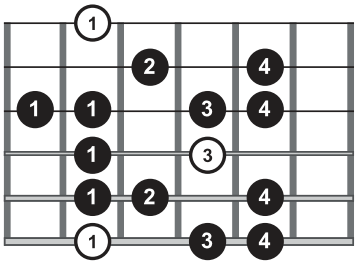
Mixolydian #11



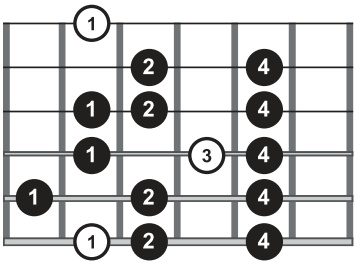
Mixolydian b6



Locrian nat.2

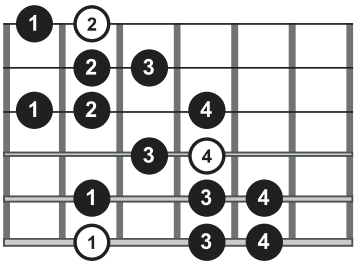


Altered Dominant

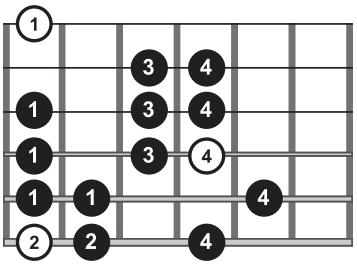


Harmonic Minor Scale

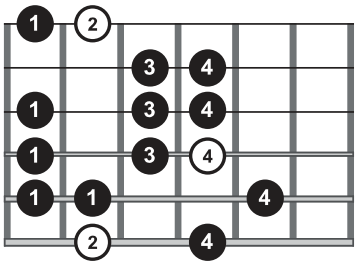
Harmonic Minor



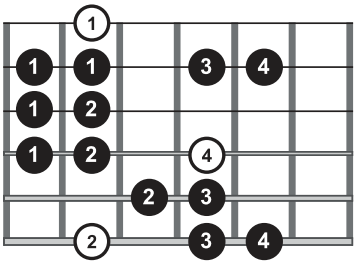
Locrian nat.6



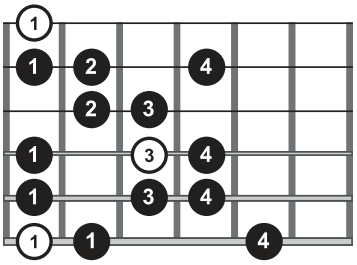
Ionian Augmented



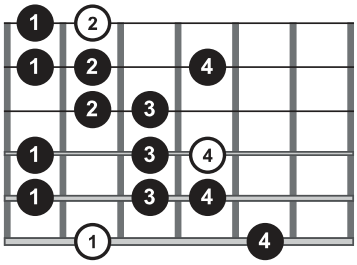
Dorian #4



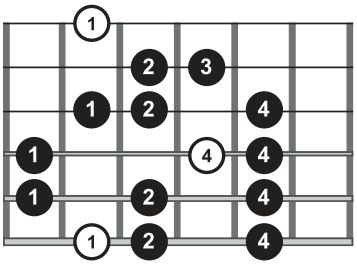
Phrygian Major



Lydian #9



Altered Dominant bb7



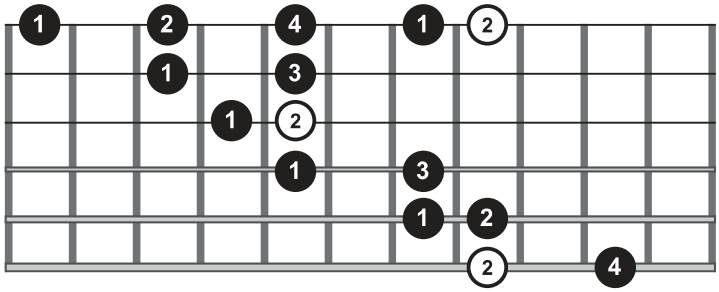
Scales with two notes per string

When playing through the following scale examples you will discover that unlike most scales, these ascend in pitch while the left hand moves towards the nut (this is generally associated with a descent in pitch).

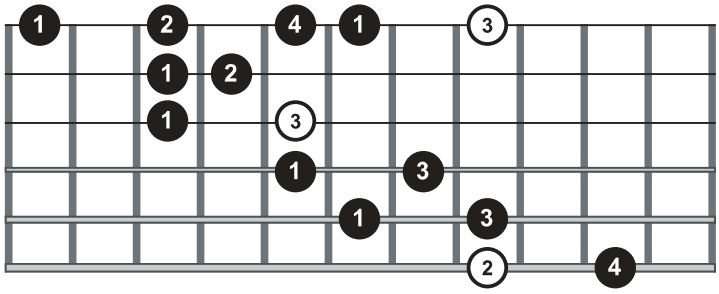
This unusual movement will open up your fingering possibilities and lead you in different directions while improvising. These fingerings work nicely if you slur on each string.

Modes of the Major Scale

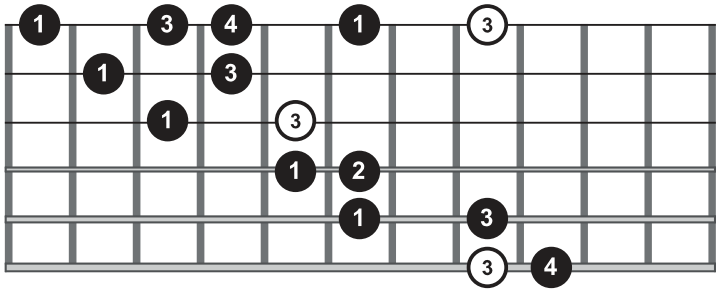
Ionian:



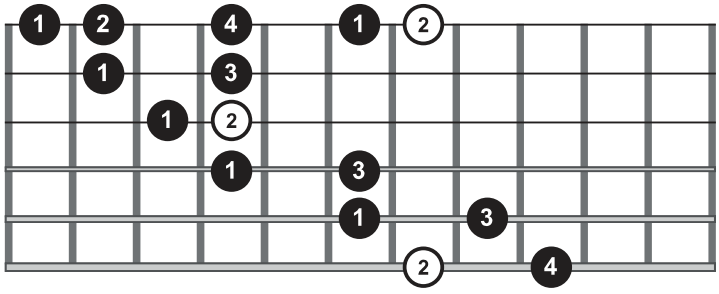
Dorian:



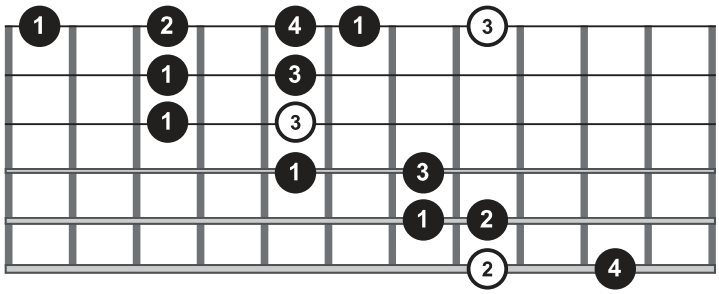
Phrygian:



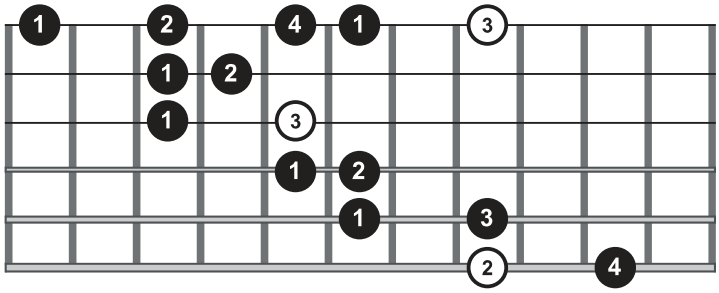
Lydian:



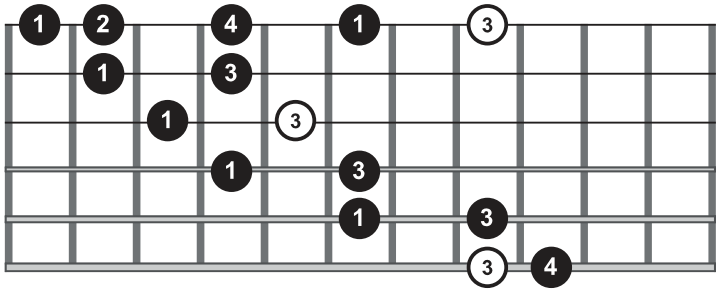
Mixolydian:



Aeolian:

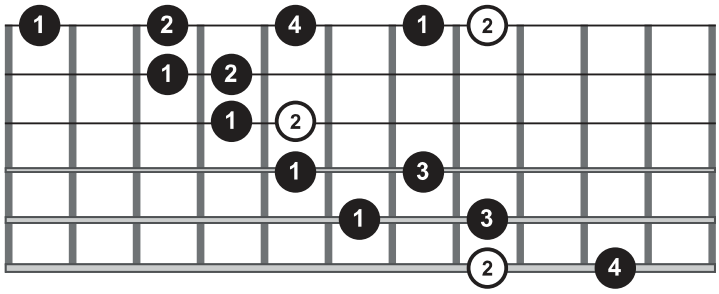


Locrian:

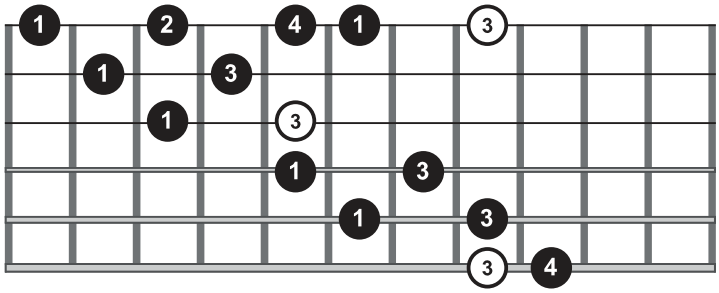


Modes of the Melodic Minor Scale

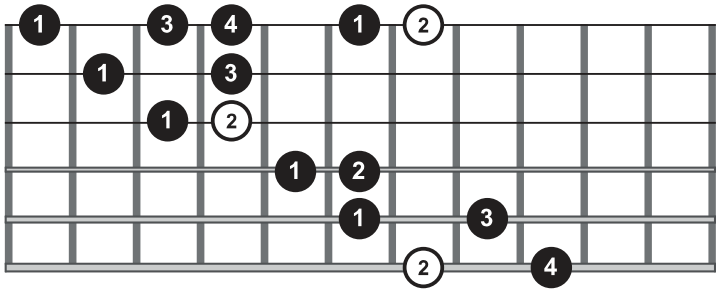
Melodic Minor:



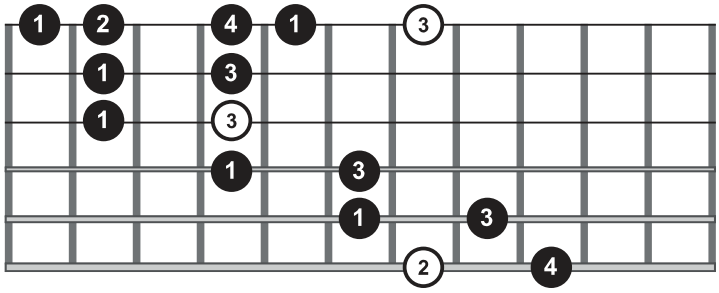
Dorian b2



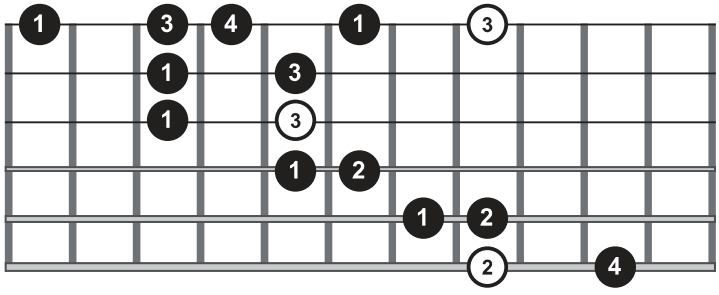
Lydian Augmented



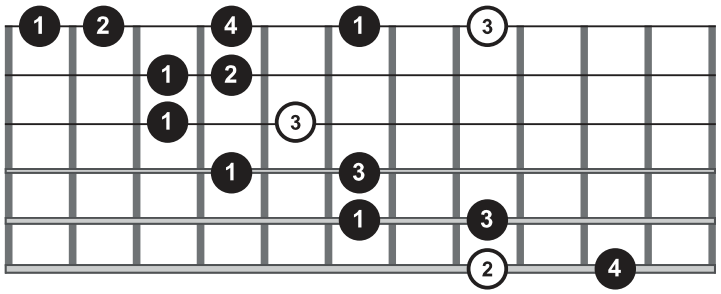
Mixolydian #11



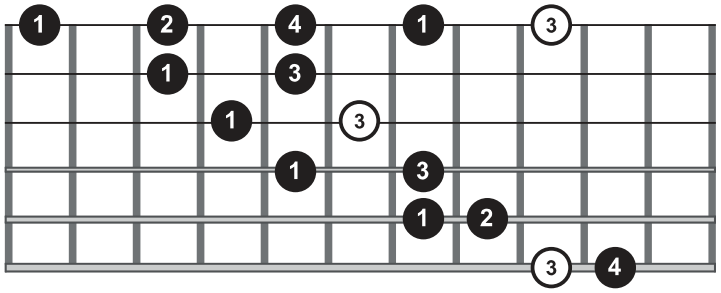
Mixolydian b6



Locrian nat.2

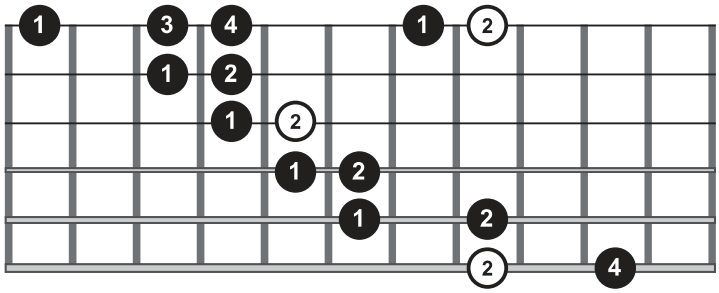


Altered Dominant

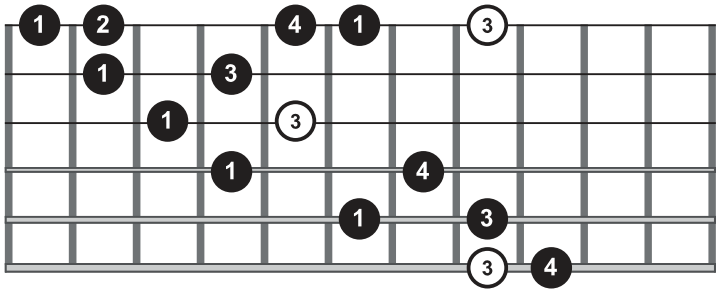


Modes of the Harmonic Minor Scale

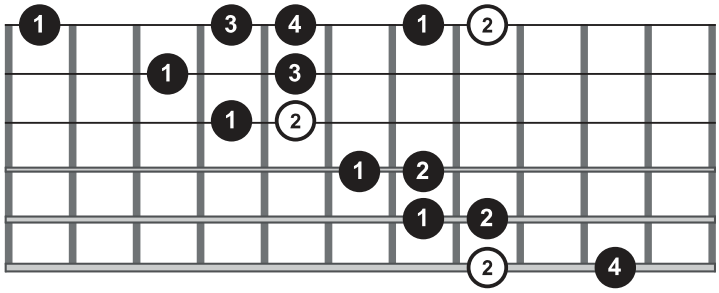
Harmonic Minor:



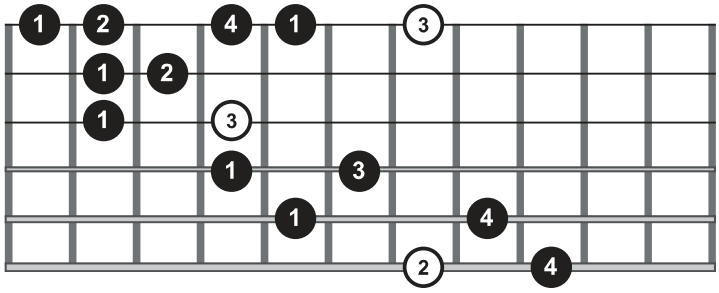
Locrian nat.6



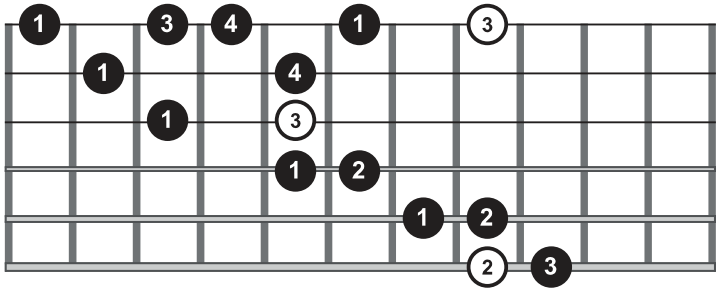
Ionian Augmented



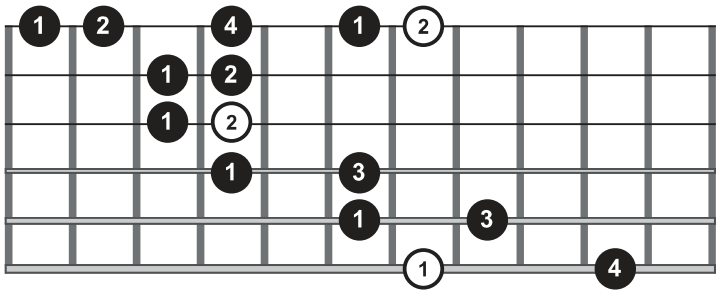
Dorian #4



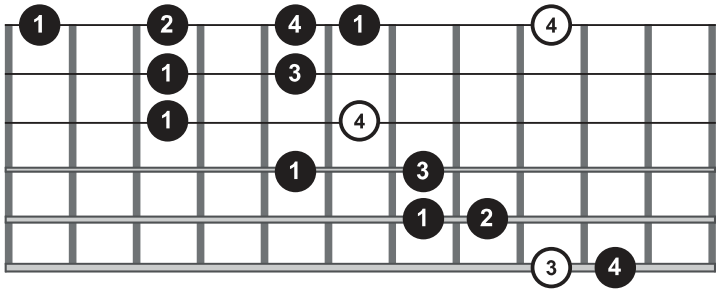
Phrygian Major



Lydian #9

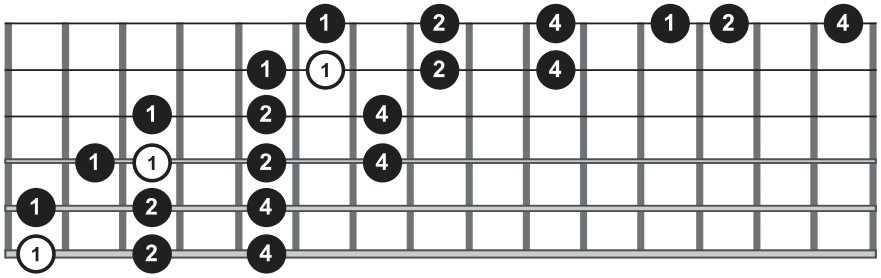


Altered Dominant bb7

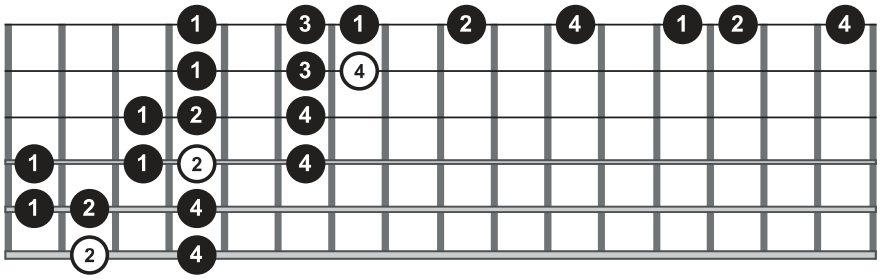


Examples of Extended Range Scale Fingerings

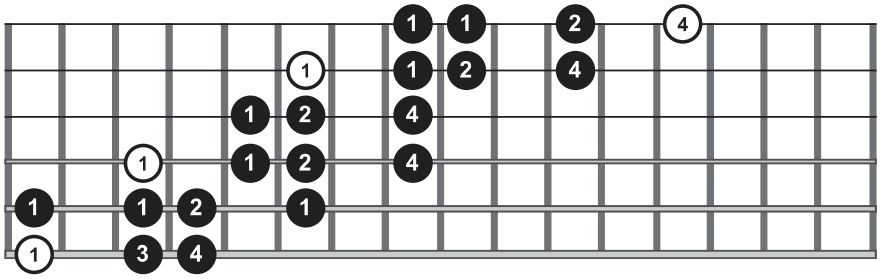
Major:



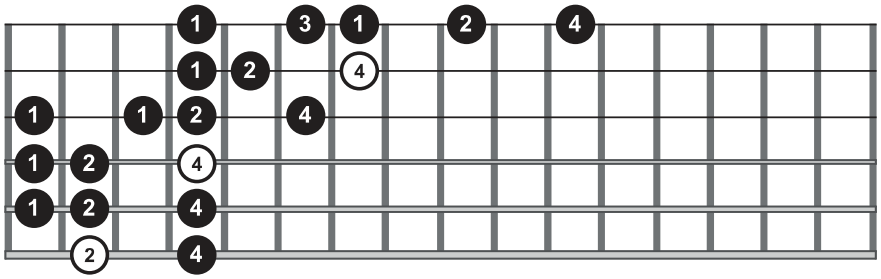
Major:



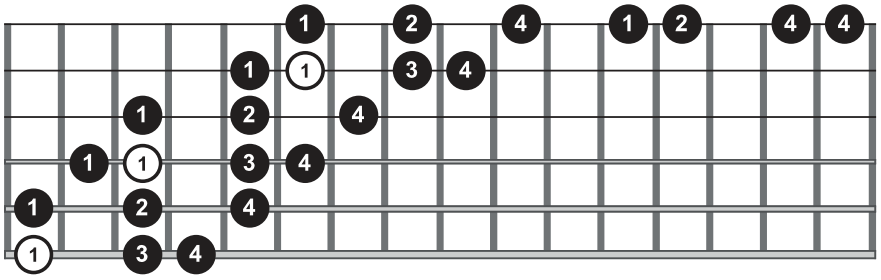
Aeolian:



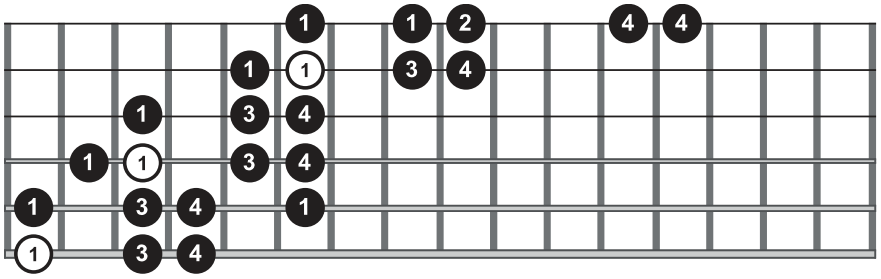
Mixolydian:



Melodic Minor:



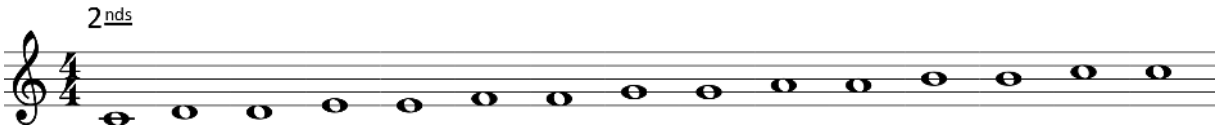
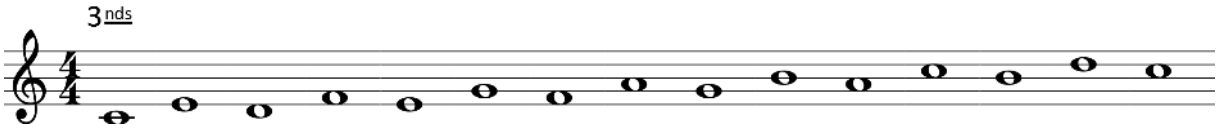
Harmonic Minor:

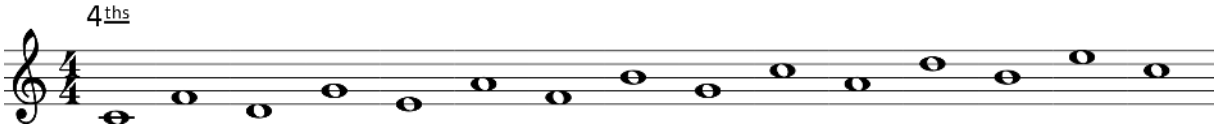


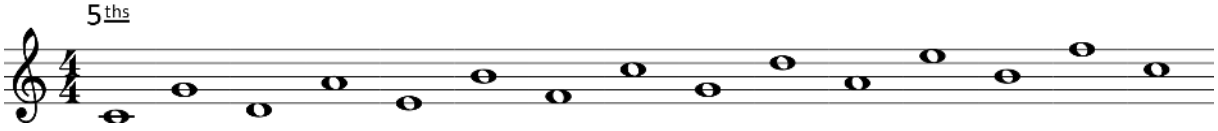
Diatonic Intervals

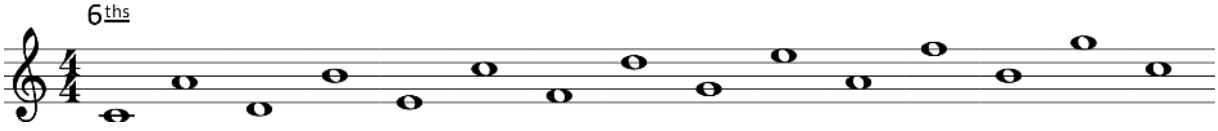
Practicing scales in diatonic interval combinations should be part of your daily practicing. It will help you develop both technical and improvisational skills as well as improving left and right hand coordination.

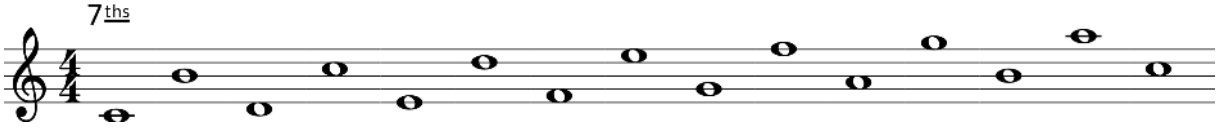
Here is a example of the diatonic intervals of a (C) major scale.

Ex. 154a

Ex. 154b

Ex. 154c



Ex. 154d

Ex. 154e



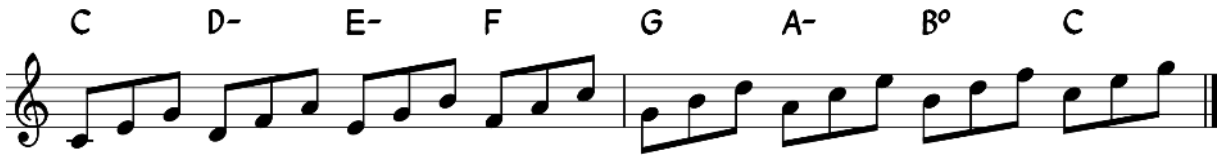
Ex. 154f

Also practice descending. Then practice all scales and modes in intervals including all auxiliary scales such as the diminished, augmented, whole-tone, pentatonics and Blues.

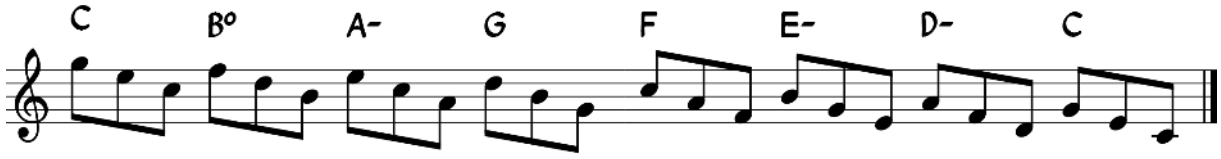
Diatonic Arpeggios

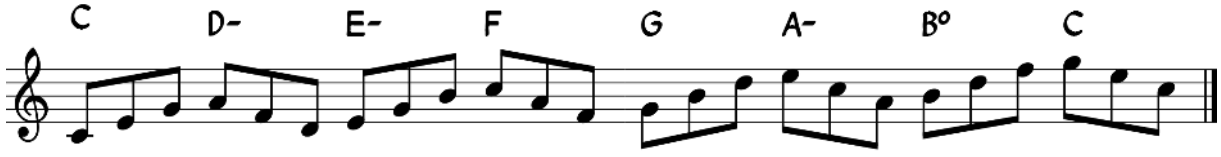
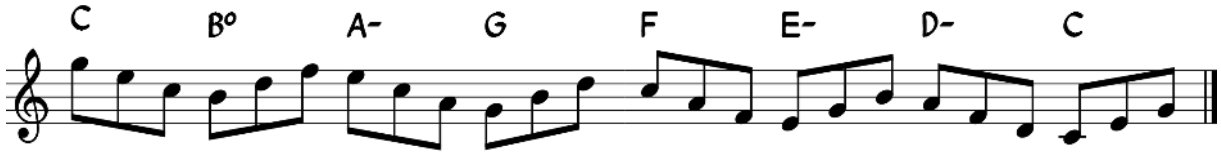
In addition to intervals, all scales should be broken into both diatonic triads and seventh chords. I suggest practicing these within the close (2 note per string) and open (3 note per string) fingering systems.

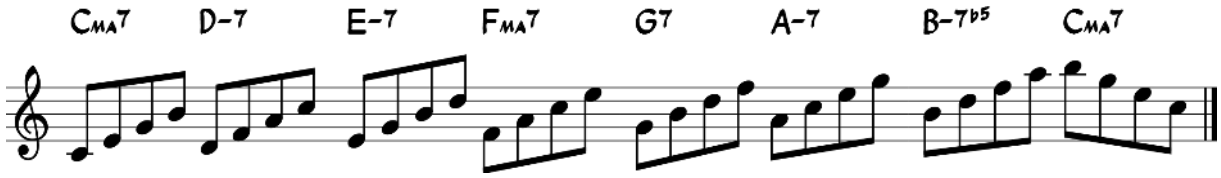
Here are a few practice examples.

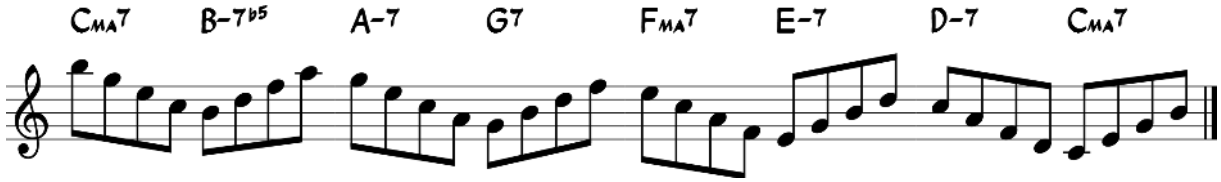
Triads (in C Major)

Ex. 155a

Ex. 155b

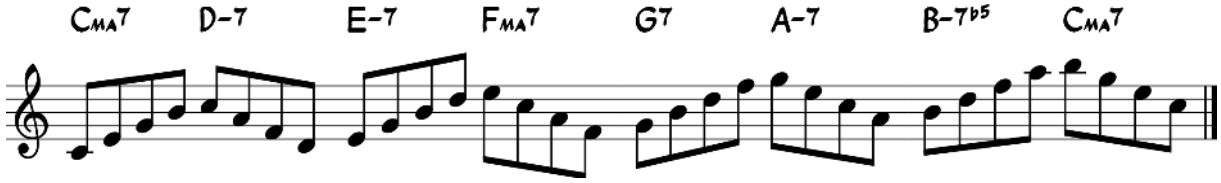


Ex. 155c

Ex. 155d

Seventh Chords

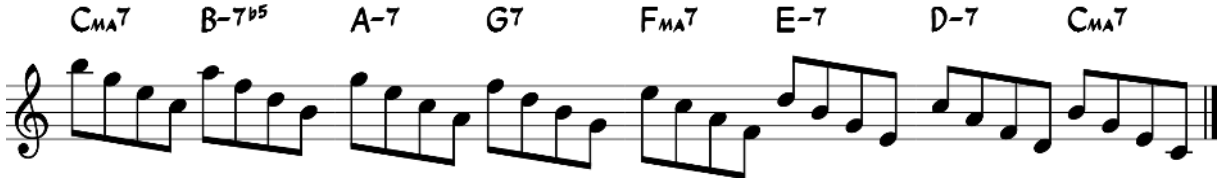
Ex. 156a

Ex. 156b



Ex. 156c

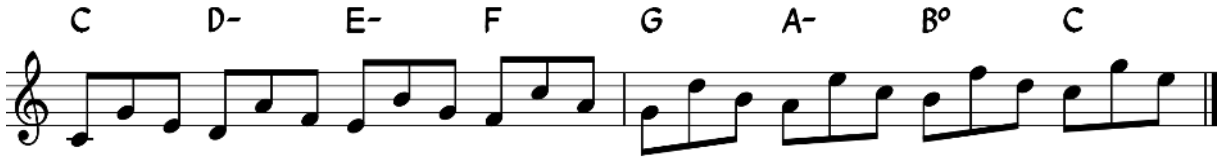


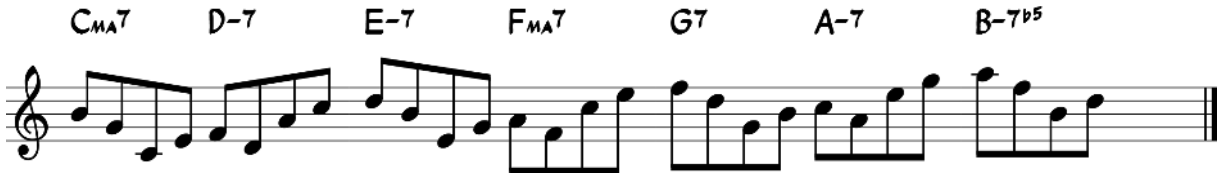
Ex. 156d



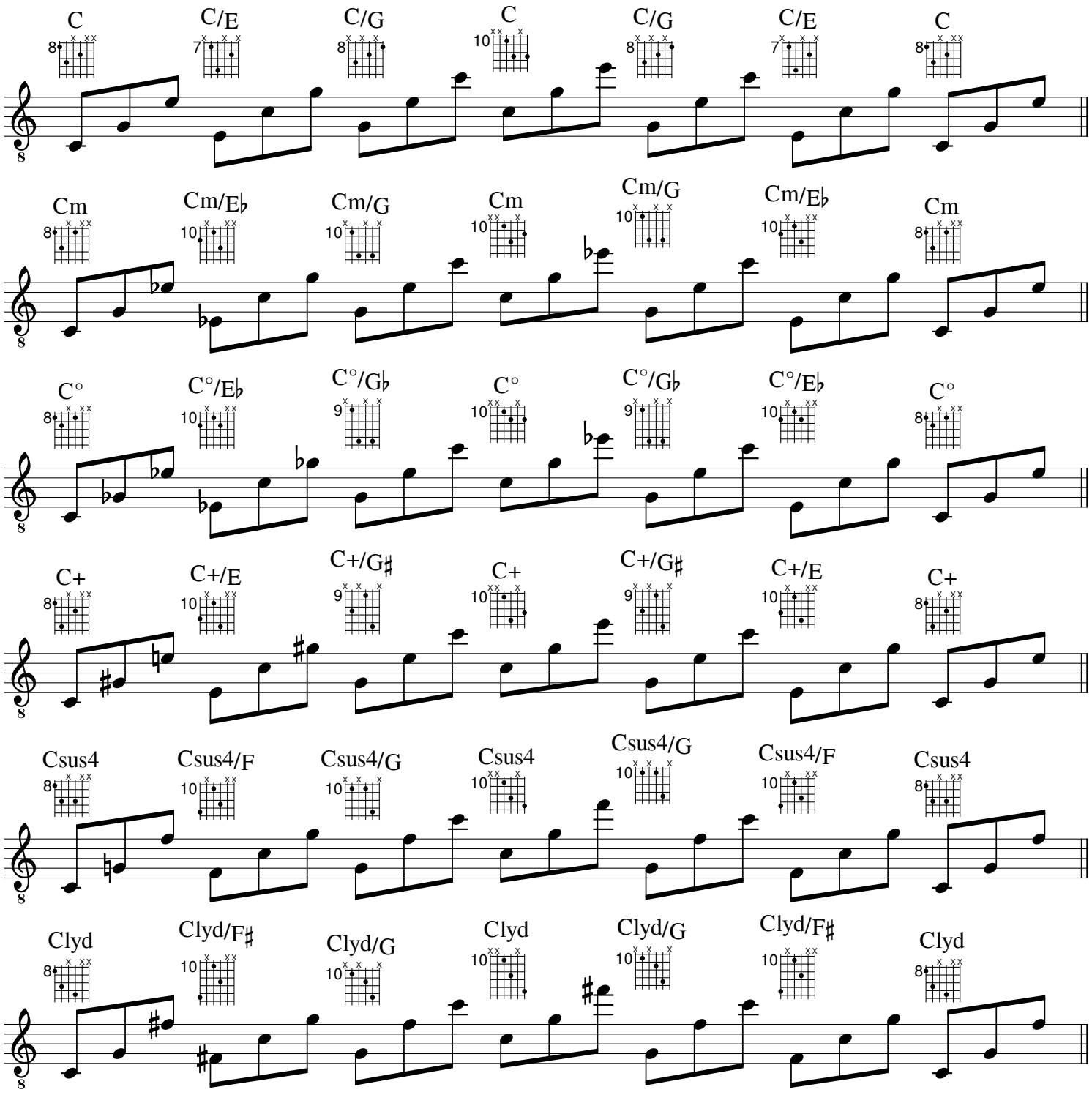
Try to experiment with as many variations of note and chord order as possible.

Triad Variation

Ex. 157

Seventh Chord Variation

Ex. 158

Spread Triad Arpeggios

Ascending & Descending



Seventh Chord Arpeggios

I have compiled this chart of seventh chord arpeggios by combining all of the possible chromatic alterations of the basic chord tones. the naming of arpeggios relates to the intervals above the root (see seventh chord formula chart).

Seventh Chord Formulas

Maj^{7}	1	3	5	7	$\frac{Maj\Delta^{add9}}{5}$	1	4	5	bb7
Dom^{7}	1	3	5	b7	$Maj^{7/b5/sus4}$	1	4	b5	7
Maj^{6}	1	3	5	6 (bb7)	$Dom^{7/b5/sus4}$	1	4	b5	b7
Aug^{maj7}	1	3	#5	7	$\frac{Maj\Delta^{b9}}{5}$	1	4	b5	bb7
Aug^{7}	1	3	#5	b7	$Maj^{7/sus4/\#5}$	1	4	#5	7
Aug^{MA6}	1	3	#5	6 (bb7)	$\frac{Min\Delta^{add9}}{5}$	1	4	#5	b7
Min^{maj7}	1	b3	5	7	$\frac{Maj\Delta^{\#9}}{5}$	1	4	#5	bb7
Min^{7}	1	b3	5	b7	$Maj^{7/sus2}$	1	2	5	7
Min^{6}	1	b3	5	6	$Dom^{7/sus2}$	1	2	5	b7
Tonic Dim	1	b3	#5	7	$Maj^{6/sus2}$	1	2	5	6 (bb7)
$\frac{Maj\Delta^{add9}}{3}$	1	b3	#5	b7	$Maj^{7/sus2/\#5}$	1	2	#5	7
$Min^{6/\#5}$	1	b3	#5	6 (bb7)	$Dom^{7/sus2/\#5}$	1	2	#5	b7
$Maj^{7/b5}$	1	3	b5	7	$Dim^{MA9/b13}$	1	2	#5	bb7
$Dom^{7/b5}$	1	3	b5	b7	$Maj^{7/b5/sus2}$	1	2	b5	7
$ø^{4}{}_{3}$	1	3	b5	bb7	$Dom^{7/b5/sus2}$	1	2	b5	b7
Dim^{maj7}	1	b3	b5	7	$Dom^{4}{}_{2}$	1	2	b5	bb7 (6)
$Min^{7/b5}$	1	b3	b5	b7	$Maj^{7/bb5}$	1	3	4 (bb5)	7
Dim^{7}	1	b3	b5	bb7	$Dom^{7/bb5}$	1	3	4	b7
$Maj^{7/sus4}$	1	4	5	7	$Maj^{6/bb5}$	1	3	4	bb7
$Dom^{7/sus4}$	1	4	5	b7	$Min^{maj7/bb5}$	1	b3	4	7
$Min^{7/bb5}$	1	b3	4	b7	$Min^{6/bb5}$	1	b3	4	6 (bb7)
$Maj^{7/sus2/bb5}$	1	2	4	7	$Dom^{7/sus2/bb5}$	1	2	4	b7
$Maj^{6/sus2/bb5}$	1	2	4	6 (bb7)					

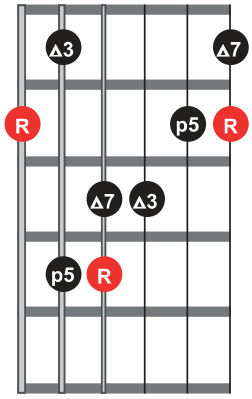
Triads and Suspensions

Maj	1	3	5	#7 (8)
Min	1	b3	5	8
Dim	1	b3	b5	8
Aug	1	3	#5	8
ΔSus$^{4\ (no\ 5th)}$	1	3	4	8
-ΔSus$^{4(no\ 5th)}$	1	b3	4	8
Sus^{2}	1	2	5	8
Sus^{2b5}	1	2	b5	8
Sus$^{2/4\ (no\ 5th)}$	1	2	4	8
Sus$^{2\#5}$	1	2	#5	8
Sus^{4}	1	4	5	8
Sus^{4b5}	1	4	b5	8
Sus$^{4\#5}$	1	4	#5	8

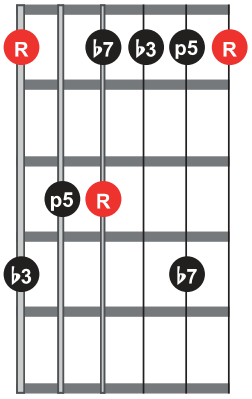
Phryg.Δ	1	b2	5	8
Loc.Δ	1	b2	b5	8
Lyd.Δ	1	#4	5	8
MajΔ^{b5}	1	3	#4	8
Q^{+4}	1	4	7	8
Q	1	4	b7	8
+4Q	1	#4	7	8
+4^{d4}	1	#4	b7	8

Seventh Chord Arpeggios Root 6

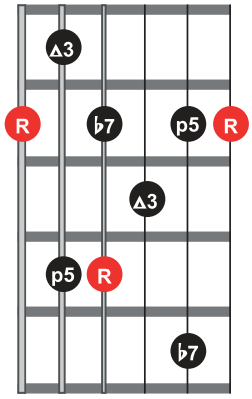
Maj^{7}



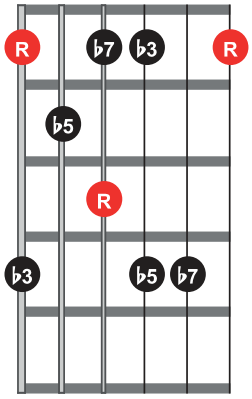
Min^{7}



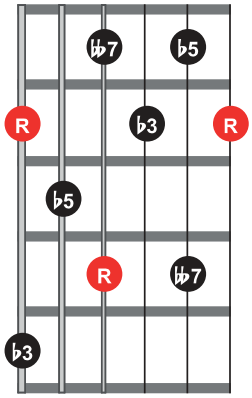
Dom^{7}



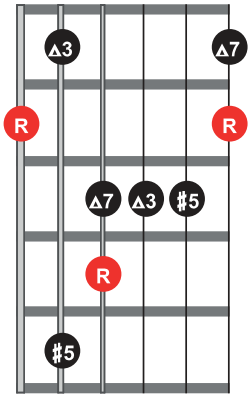
Min^{7b5}



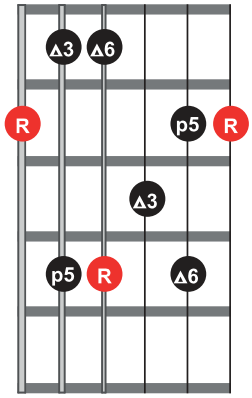
Dim^{o7}



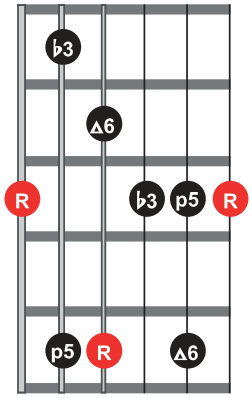
Aug^{maj7}



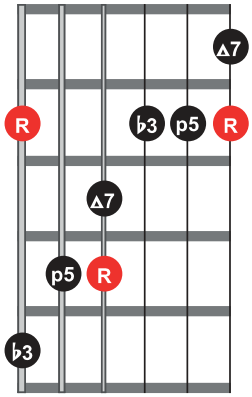
Maj^{6}



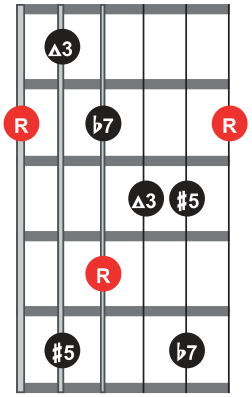
Min^{6}



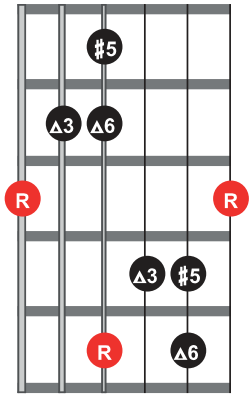
Min^{maj7}



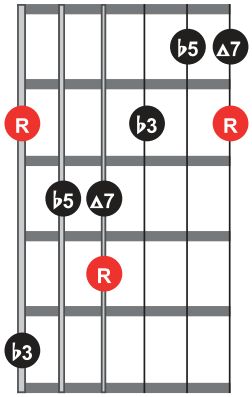
Aug^{7}



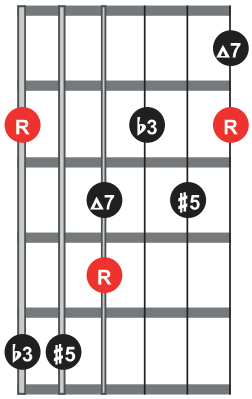
Aug^{MA6}



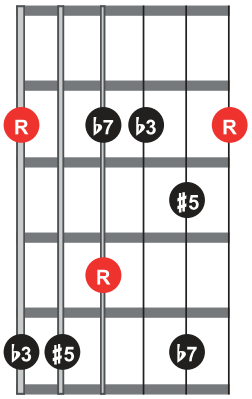
Dim^{maj7}



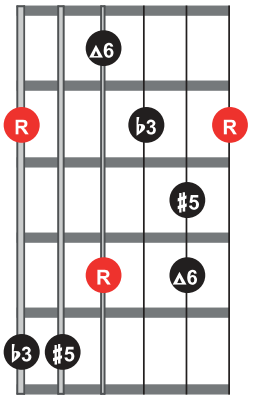
Tonic Dim



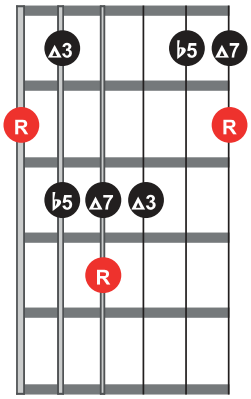
MajΔ^{add9}/3rd



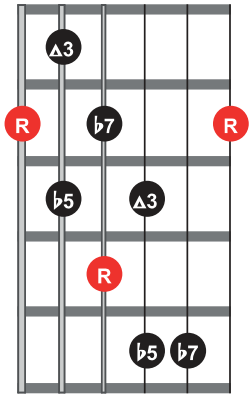
Min$^{6/\#5}$



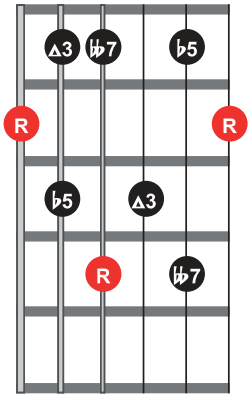
Maj$^{7/b5}$



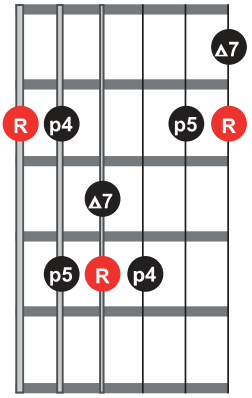
Dom$^{7/b5}$



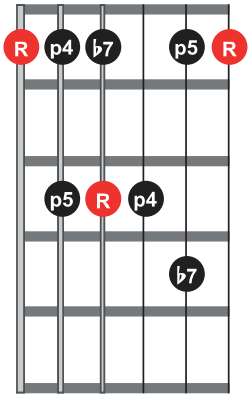
ø$^{4}_{3}$



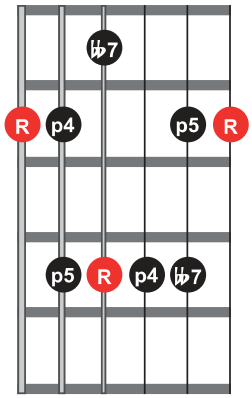
Maj^{7sus4}



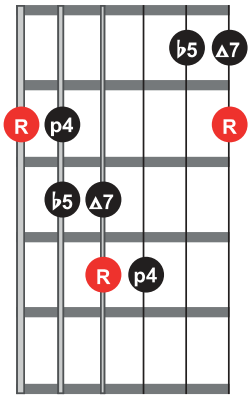
Dom^{7sus4}



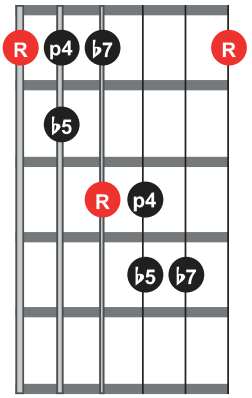
MajΔ^{add9}/5th



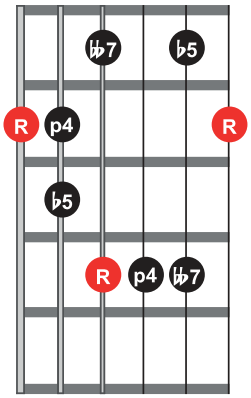
Maj$^{7/b5/sus4}$



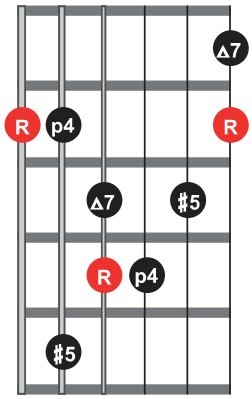
Dom$^{7/b5/sus4}$



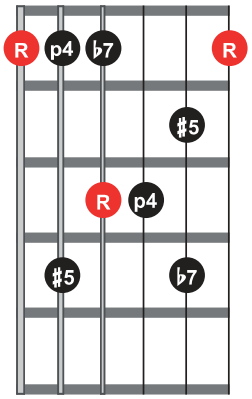
MajΔ^{b9}/5th



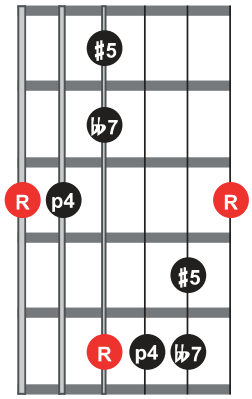
Maj$^{7/sus4/\#5}$



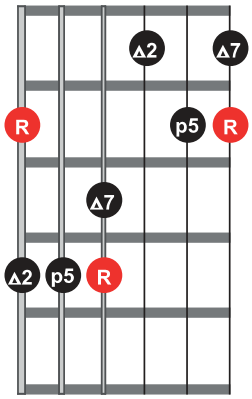
Min$\Delta^{add9}/5^{th}$



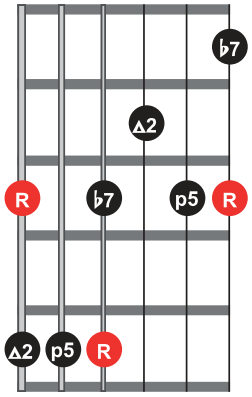
Maj$\Delta^{add9}/5^{th}$



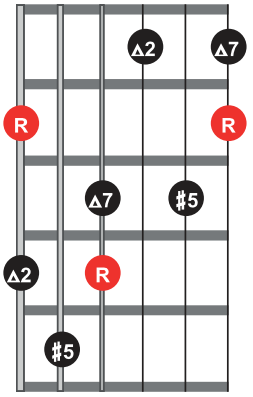
Maj^{7sus2}



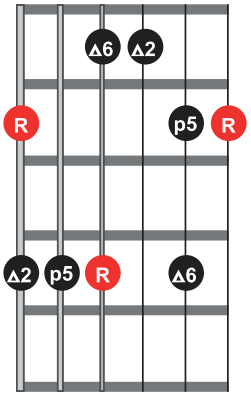
Dom^{7sus2}



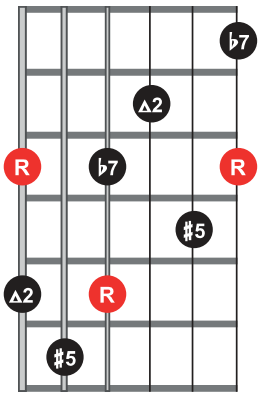
Maj$^{7/sus2/\#5}$



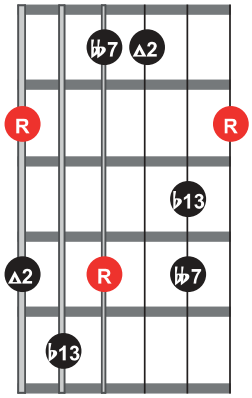
Maj$^{6/sus2}$



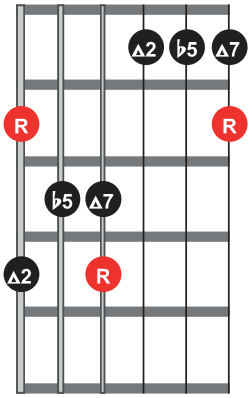
Dom$^{7/sus2/\#5}$



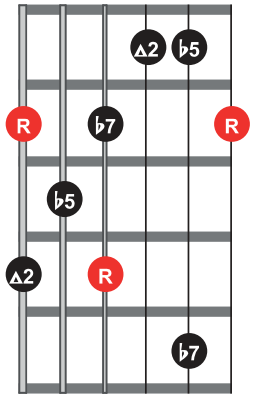
Dim$^{o7/MA9/b13}$



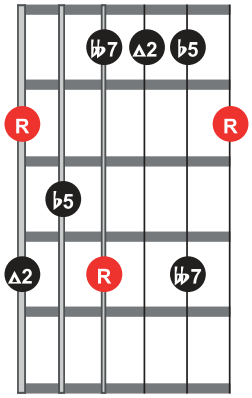
Maj$^{7/b5/sus2}$



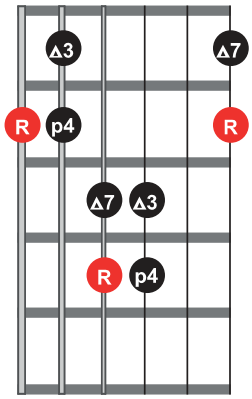
Dom$^{7/b5/sus2}$



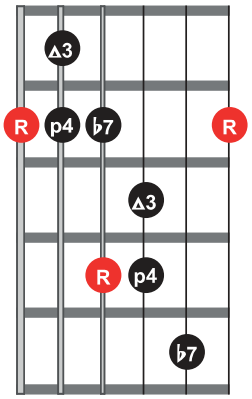
Dom$^{4}{}_{2}$



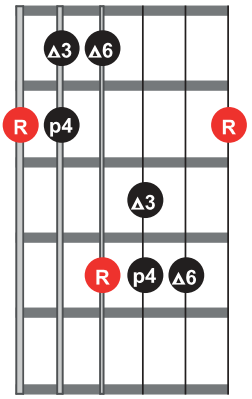
Maj$^{7/bb5}$



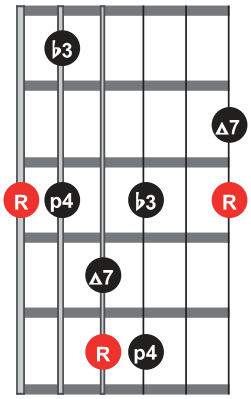
Dom$^{7/bb5}$



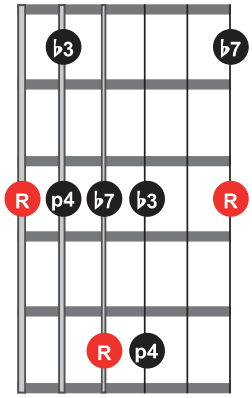
Maj$^{6/bb5}$



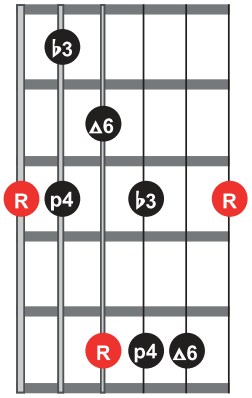
Min$^{maj7/bb5}$



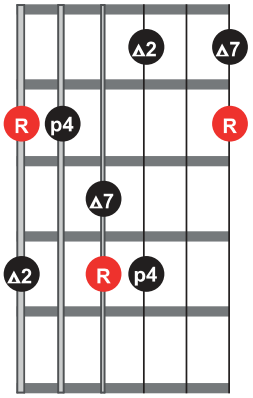
Min$^{7/bb5}$



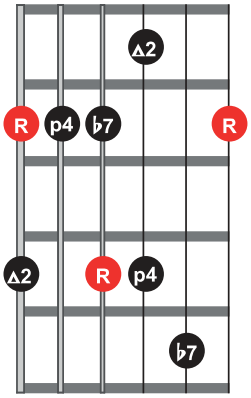
Min$^{6/bb5}$



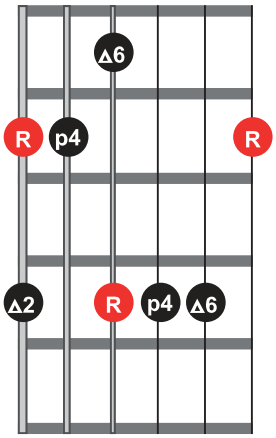
Maj$^{7/sus2/bb5}$



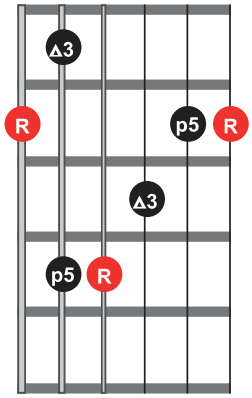
Dom$^{7/sus2/bb5}$



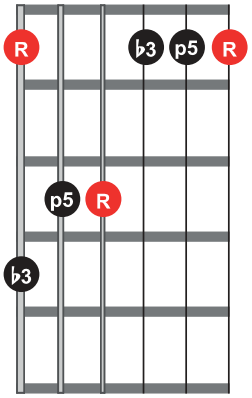
Maj$^{6/sus2/bb5}$



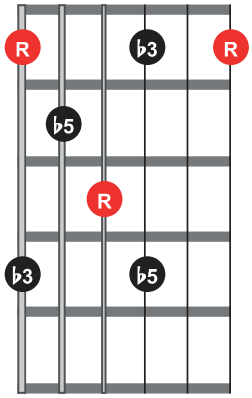
Maj



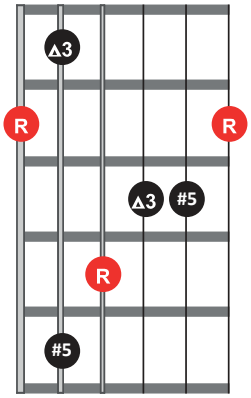
Min



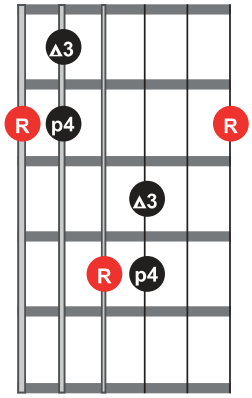
Dim



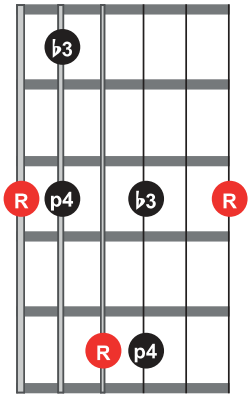
Aug



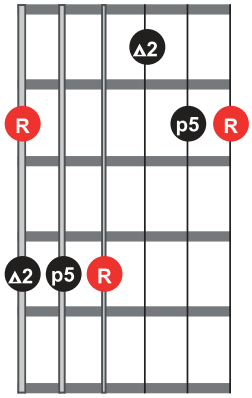
Δsus $^{4\text{ no }5\underline{\text{th}}}$



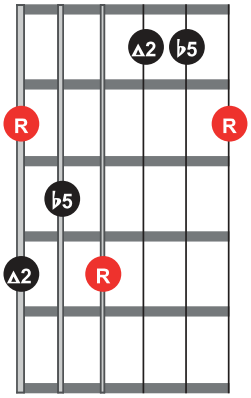
-Δsus $^{4\text{ no }5\underline{\text{th}}}$



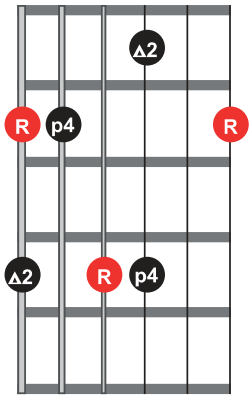
sus^{2}



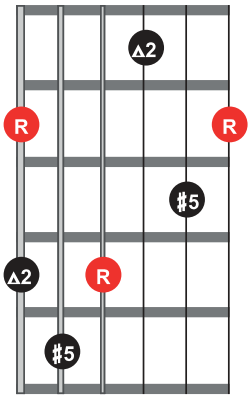
sus$^{2/b5}$



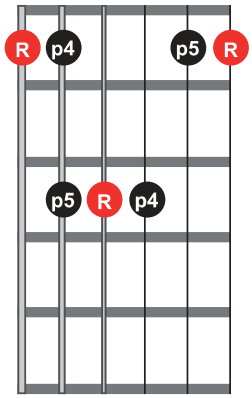
sus$^{2/4\text{ no }5\underline{\text{th}}}$



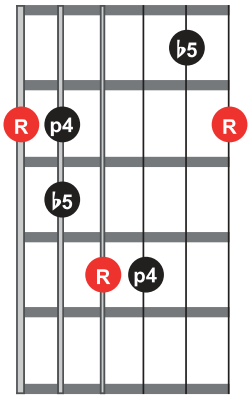
sus$^{2/\#5}$



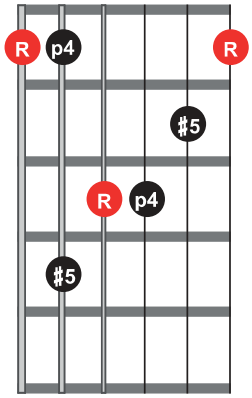
sus^{4}



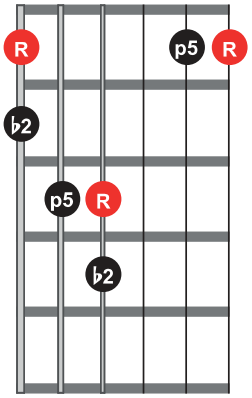
sus$^{4/b5}$



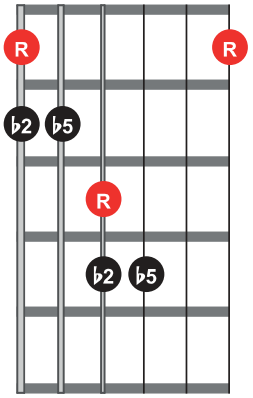
sus$^{4/\#5}$



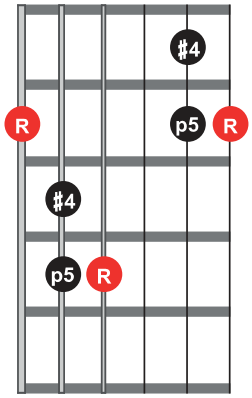
phrygΔ



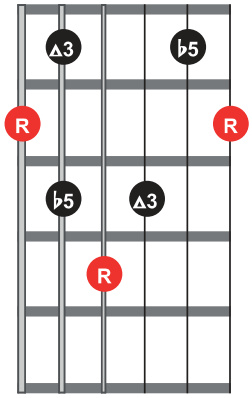
locΔ



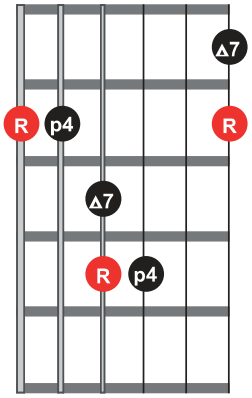
lydΔ



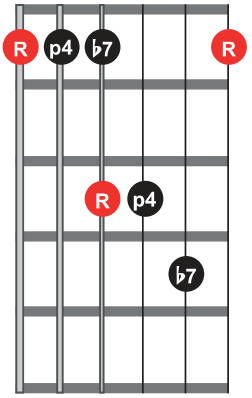
MajΔ^{b5}



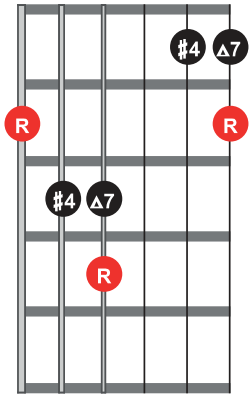
Q^{+4}



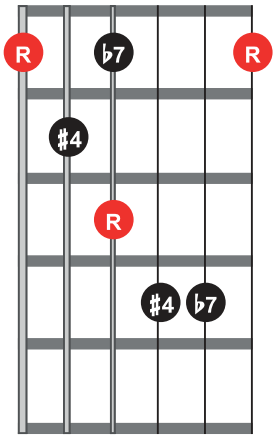
Q



+4Q

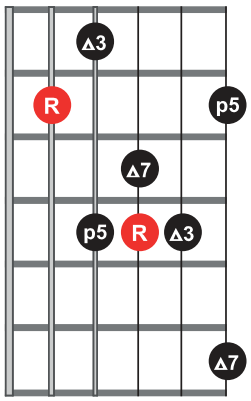


+4^{d4}

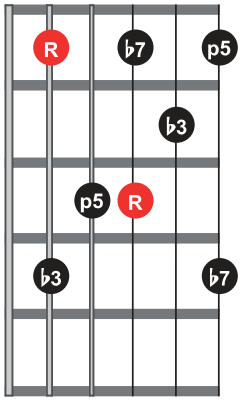


Seventh Chord Arpeggios Root 5

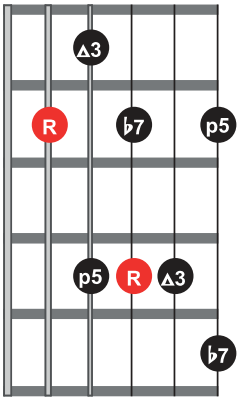
Maj^{7}



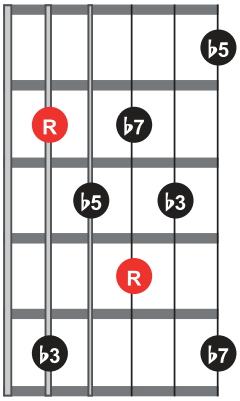
Min^{7}



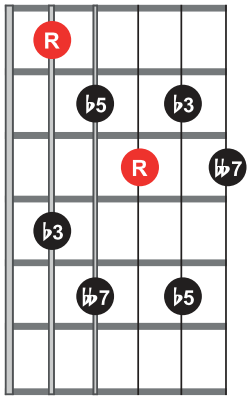
Dom^{7}



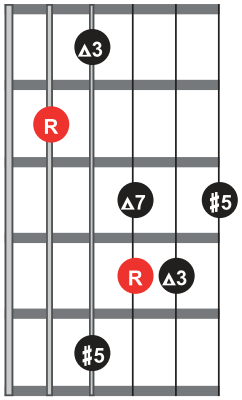
Min^{7b5}



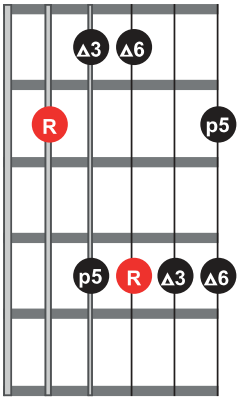
Dim^{o7}



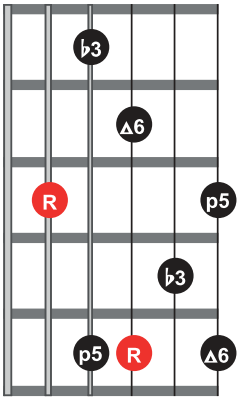
Aug^{maj7}



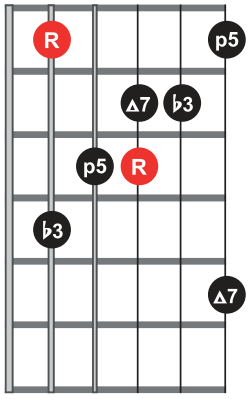
Maj^{6}



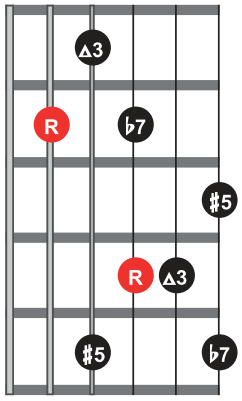
Min^{6}



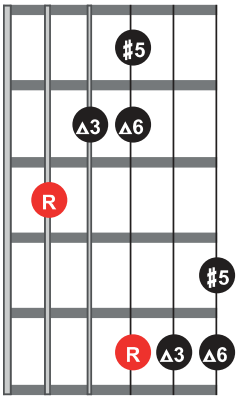
Min^{maj7}



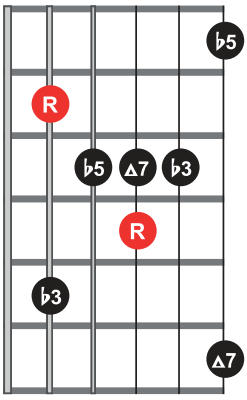
Aug^{7}



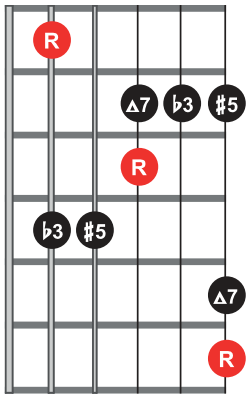
Aug^{MA6}



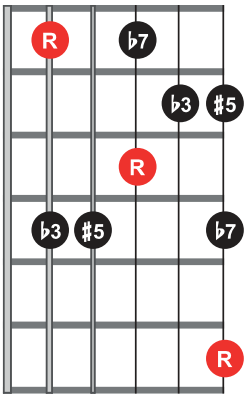
Dim^{maj7}



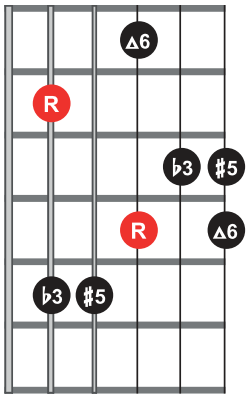
Tonic Dim



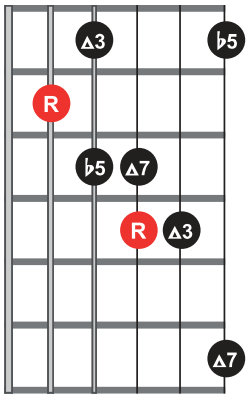
MajΔ^{add9}/3rd



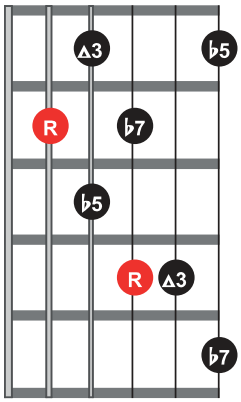
Min$^{6/\#5}$



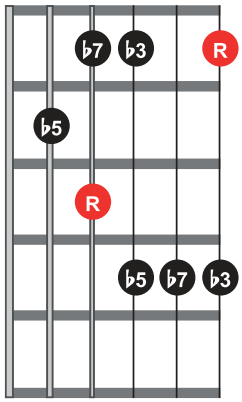
Maj$^{7/b5}$



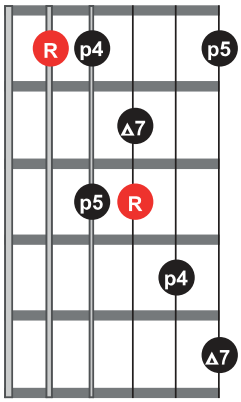
Dom$^{7/b5}$



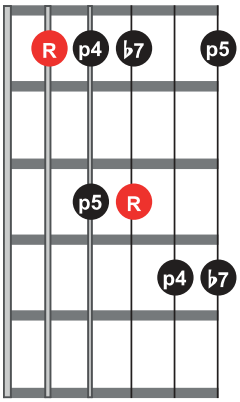
ø$^{4}_{3}$



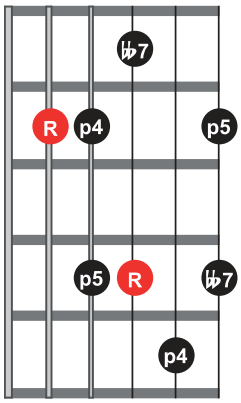
Maj^{7sus4}



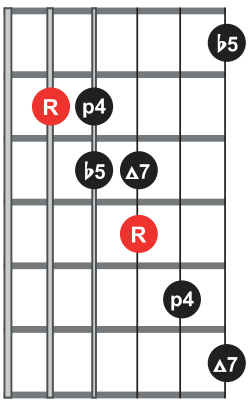
Dom^{7sus4}



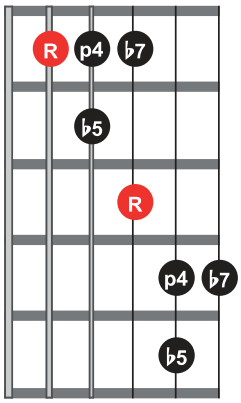
MajΔ^{add9}/5th



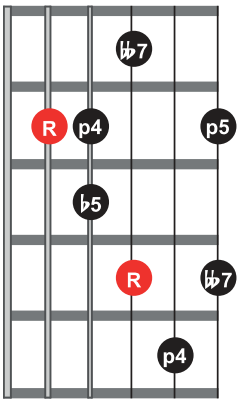
Maj$^{7/b5/sus4}$



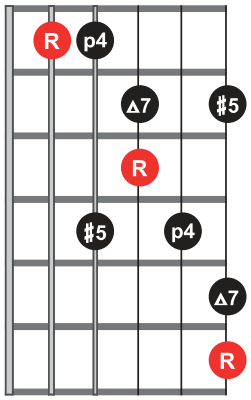
Dom$^{7/b5/sus4}$



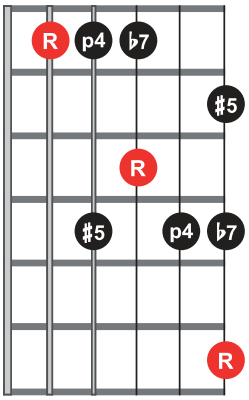
MajΔ^{b9}/5th



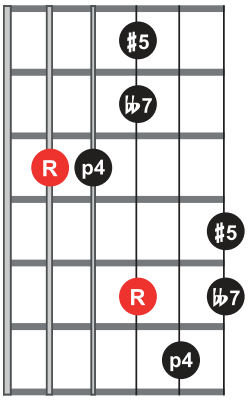
Maj$^{9/sus4/\#5}$



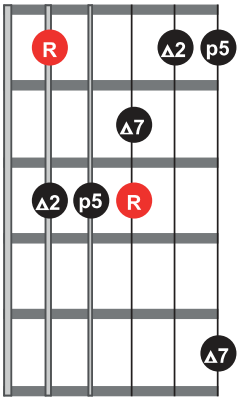
MinΔ^{add9}/5th



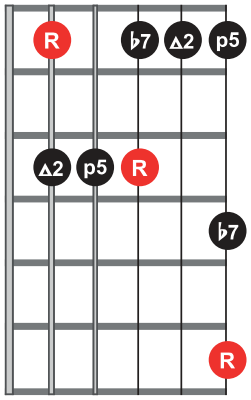
MajΔ$^{\#9}$/5th



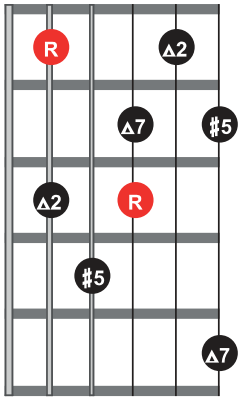
Maj^{7sus2}



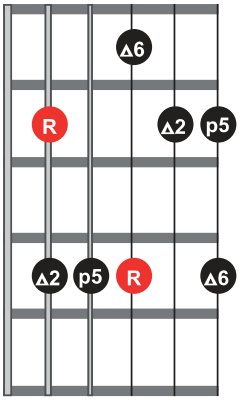
Dom^{7sus2}



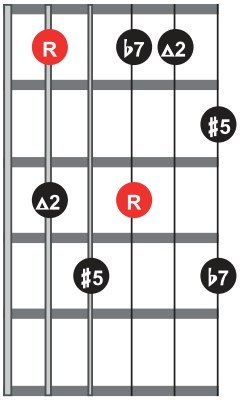
Maj$^{7/sus2/\#5}$



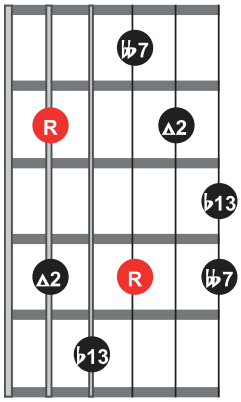
Maj$^{6/sus2}$



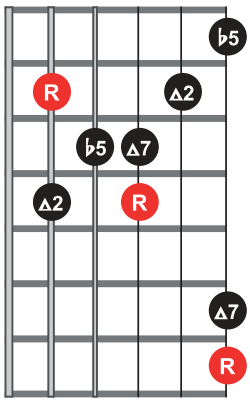
Dom$^{7/sus2/\#5}$



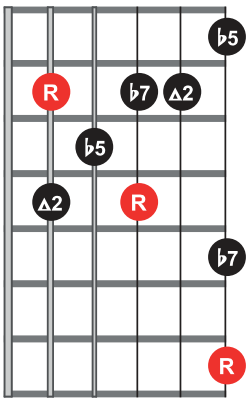
Dim$^{o7/MA9/b13}$



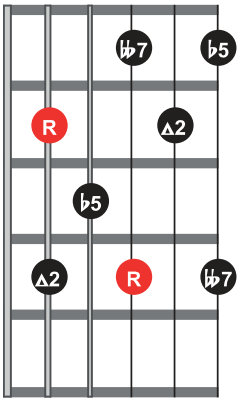
Maj$^{7/b5/sus2}$



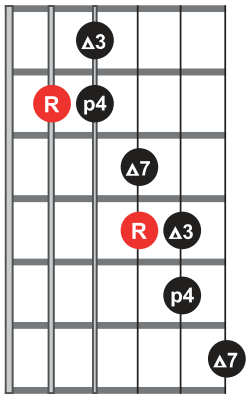
Dom$^{7/b5/sus2}$



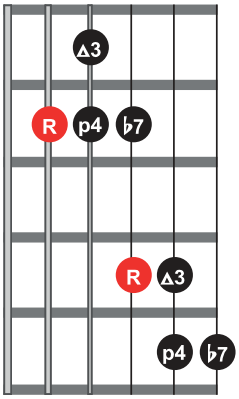
Dom$^{4}_{2}$



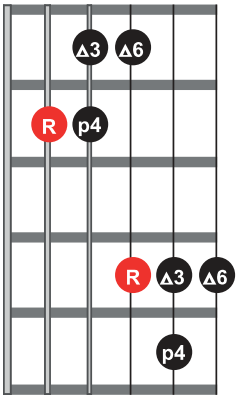
Maj$^{7/bb5}$



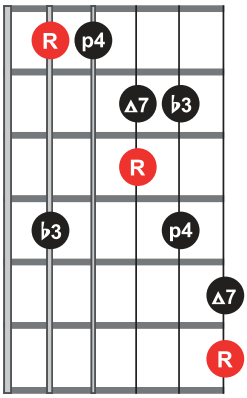
Dom$^{7/bb5}$



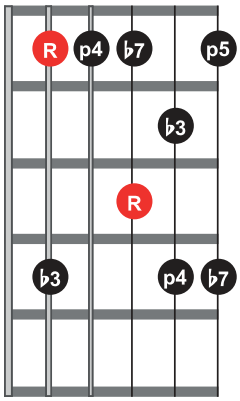
Maj$^{6/bb5}$



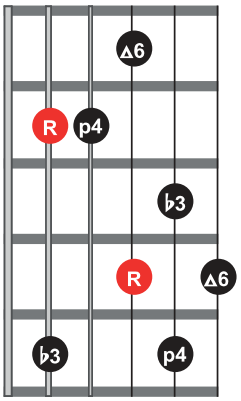
Min$^{maj7/bb5}$



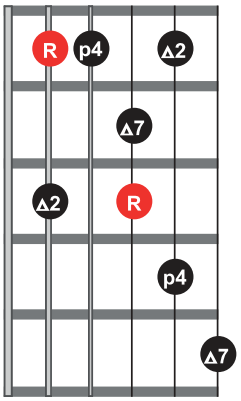
Min$^{7/bb5}$



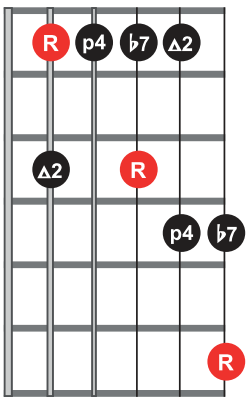
Min$^{6/bb5}$



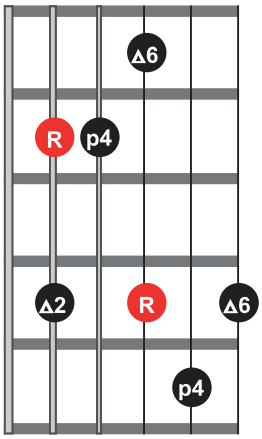
Maj$^{7/sus2/bb5}$



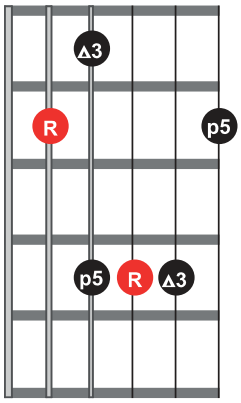
Dom$^{7/sus2/bb5}$



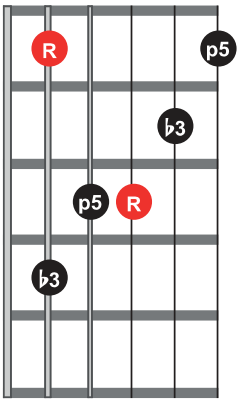
Maj$^{6/sus2/bb5}$



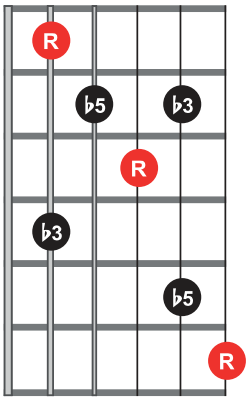
Maj



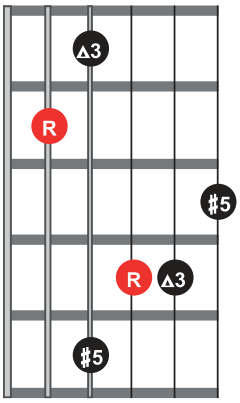
Min



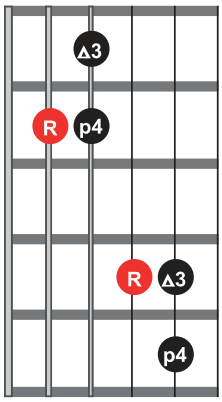
Dim



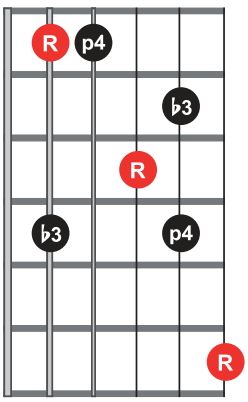
Aug



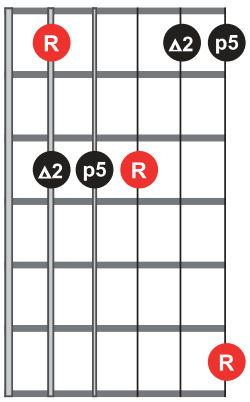
Δsus $^{4 \text{ no } 5\underline{th}}$



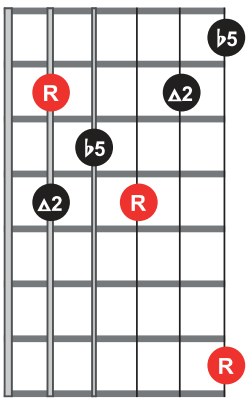
-Δsus $^{4 \text{ no } 5\underline{th}}$



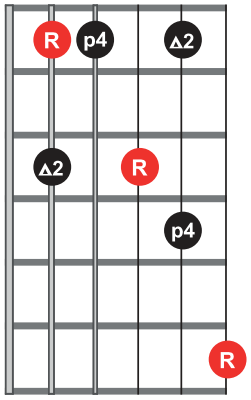
sus^{2}



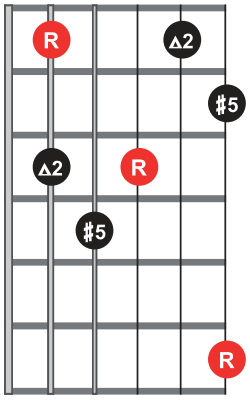
sus$^{2/b5}$



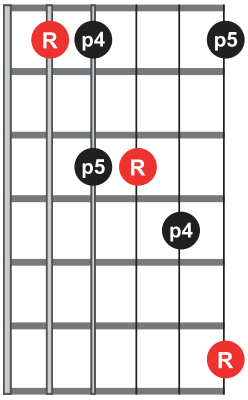
sus$^{2/4 \text{ no } 5\underline{th}}$



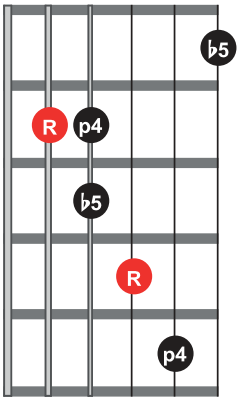
sus$^{2/\#5}$



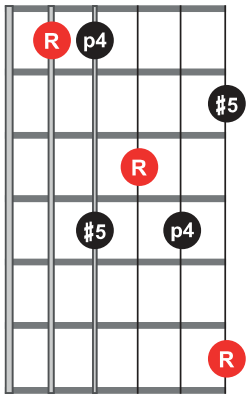
sus^{4}



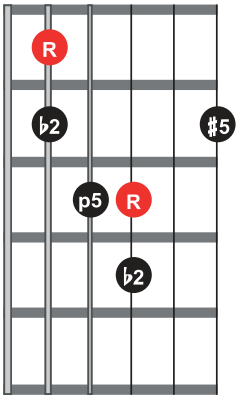
sus$^{4/b5}$



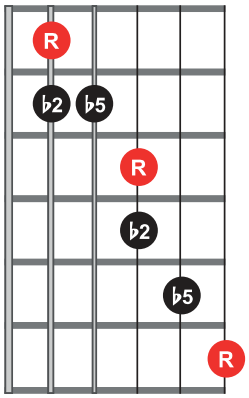
sus$^{4/\#5}$



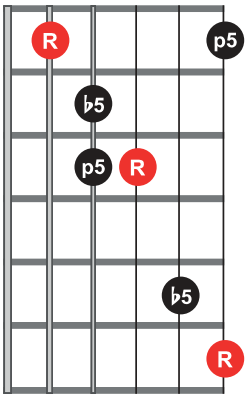
PhrygΔ



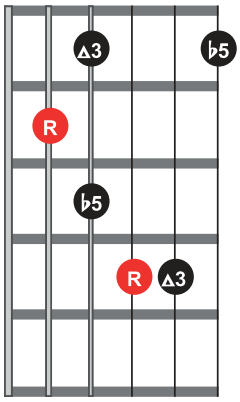
LocΔ



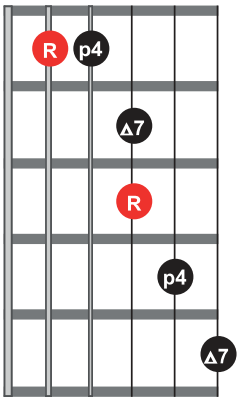
LydΔ



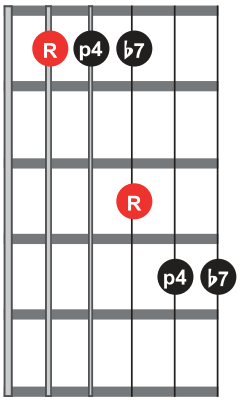
MajΔ^{b5}



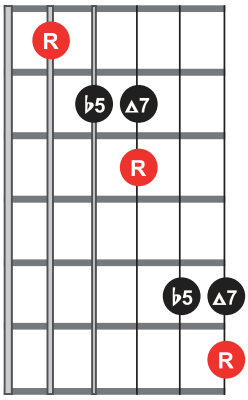
Q^{+4}



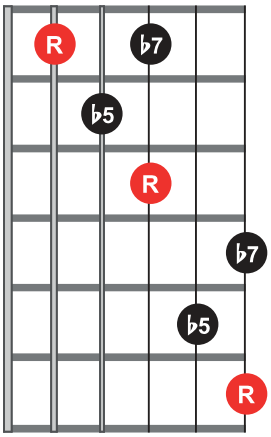
Q



+4Q



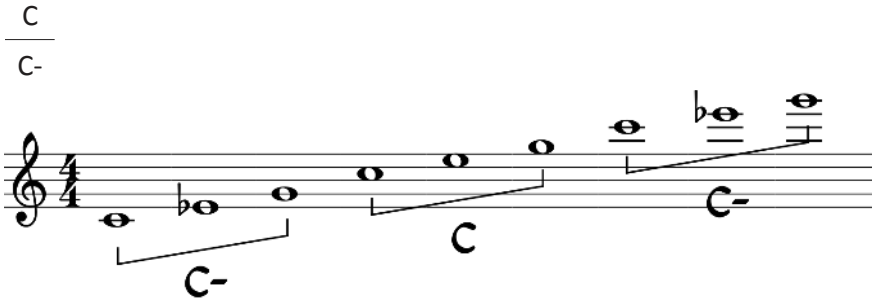
+4^{d4}



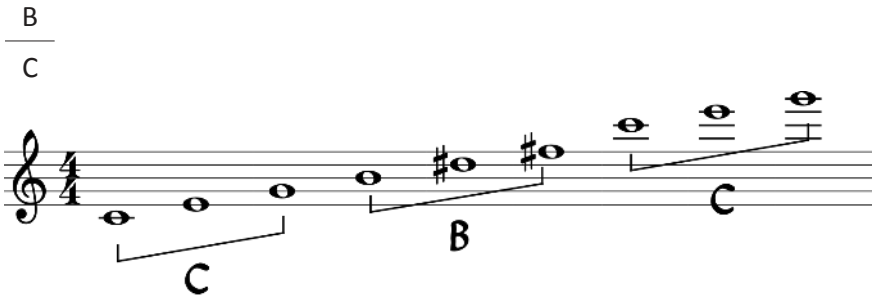
Bitonal Arpeggios

These particular bitonal arpeggios alternate between the bottom and top triads.

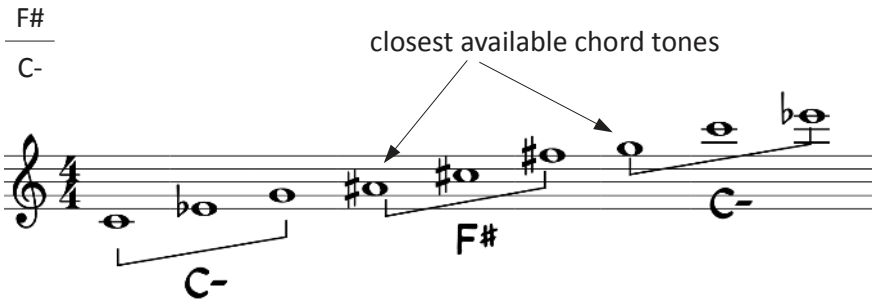
Ex. 159a



Ex. 159b



Ex. 159c



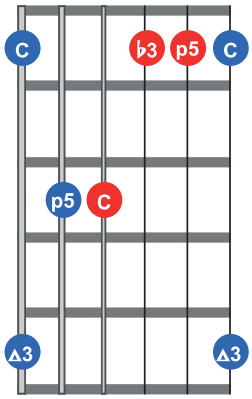
Major and Minor Bitonal Arpeggios

The charts include only major and minor triads over major, minor, diminished and augmented triads.

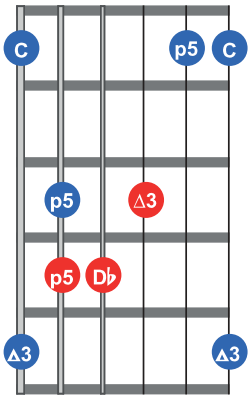
C- / C	=	1	3	5	#9		
Db / C	=	1	3	5	b6	b9	11
Db- / C	=	1	3	5	b6	b9	
D / C	=	1	3	5	13	9	#11
D- / C	=	1	3	5	6	9	11
Eb / C	=	1	3	5	b7	#9	
Eb- / C	=	1	3	5	b7	#9	#11
E / C	=	1	3	5	#5	7	
E- / C	=	1	3	5	7		
F / C	=	1	3	5	6	11	
F- / C	=	1	3	5	b6	11	
F# / C	=	1	3	5	b7	b9	#11
F#- / C	=	1	3	5	13	b9	#11
G / C	=	1	3	5	7	9	
G- / C	=	1	3	5	b7	9	
Ab / C	=	1	3	5	#5	#9	
Ab- / C	=	1	3	5	#5	7	#9
A / C	=	1	3	5	6	b9	
A- / C	=	1	3	5	6		
Bb / C	=	1	3	5	b7	9	11
Bb- / C	=	1	3	5	b7	b9	sus4
B / C	=	1	33	5	7	#9	#11
B- / C	=	1	3	5	7	9	#11

Major Bitonal Arpeggios (over C at the 8th fret)

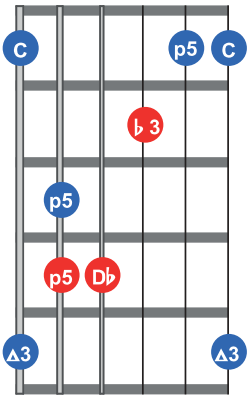
C-
C



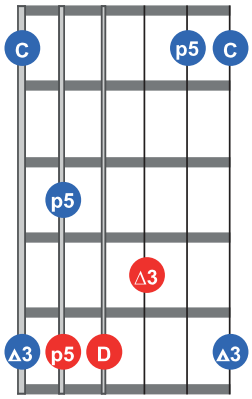
Db
C



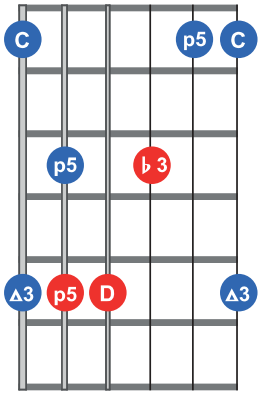
Db-
C



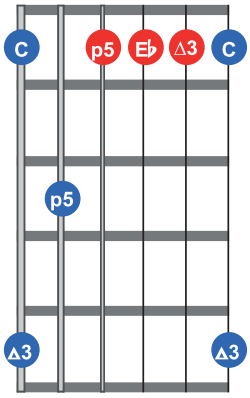
D
C



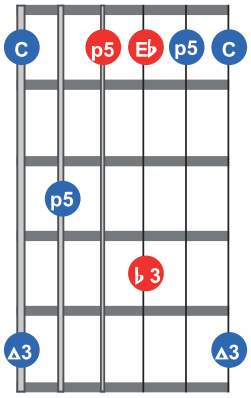
D-
C



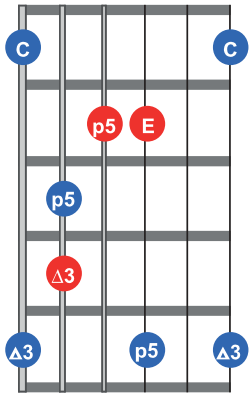
Eb
C



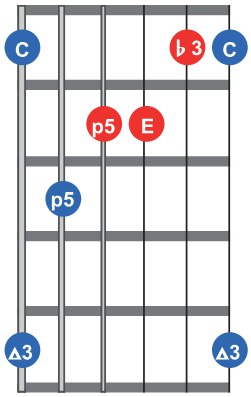
Eb-
C



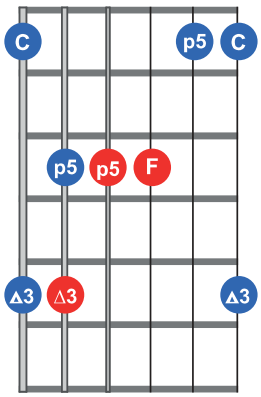
E
C



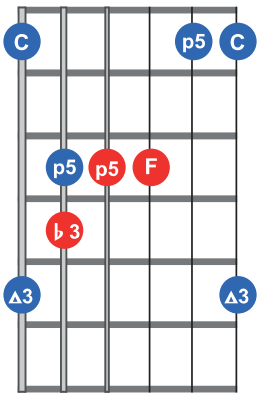
E-
C



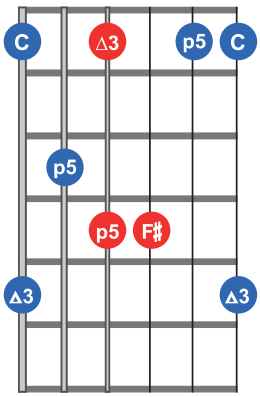
F
C



F-
C

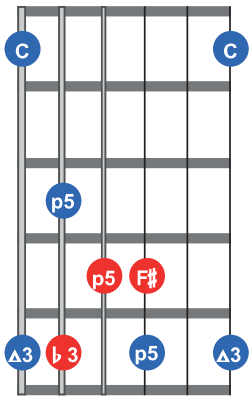


F#
C



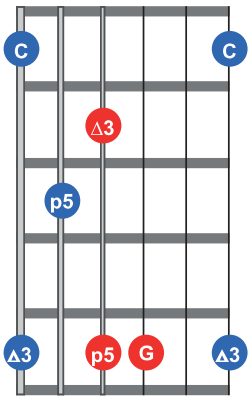
F#-

C



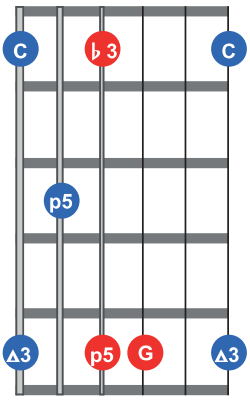
G

C



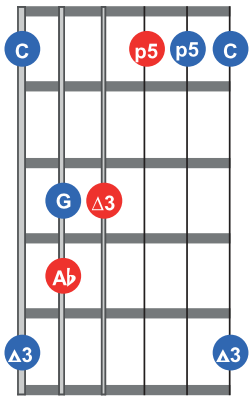
G-

C



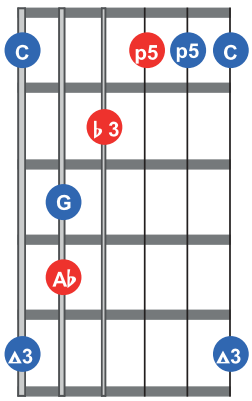
Ab

C



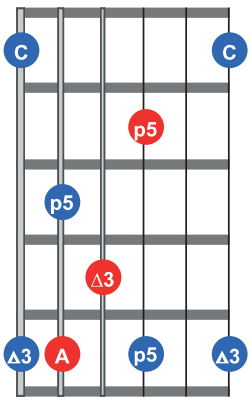
Ab-

C



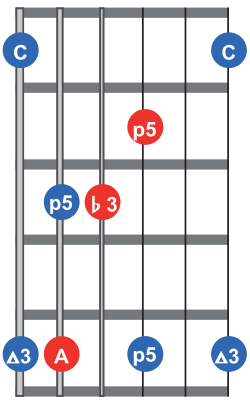
A

C



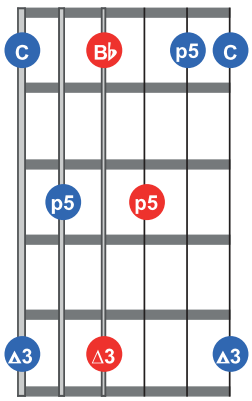
A-

C



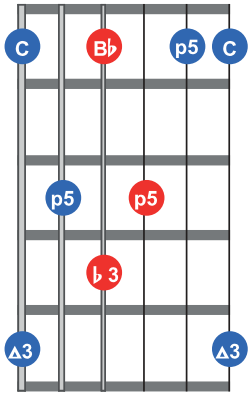
Bb

C



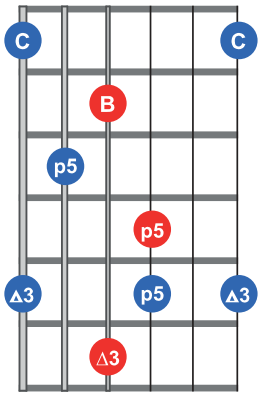
Bb-

C



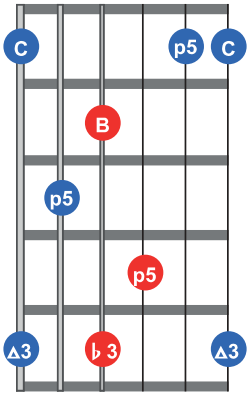
B

C



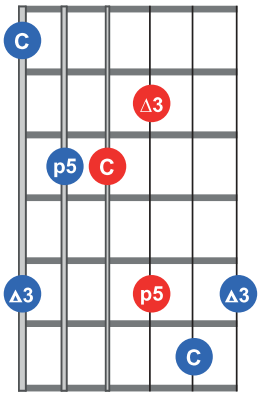
B-

C



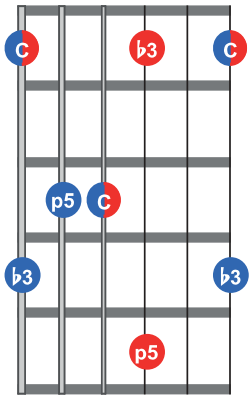
C

C

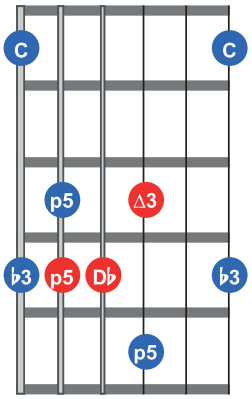


Minor Bitonal Arpeggios (over C- at the 8th fret)

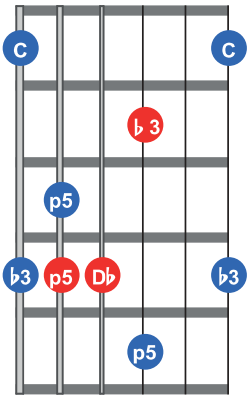
C
C-



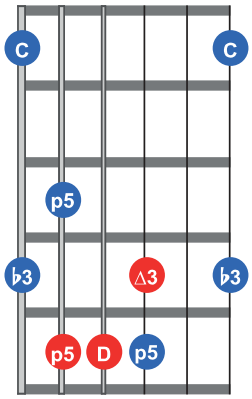
Db
C-



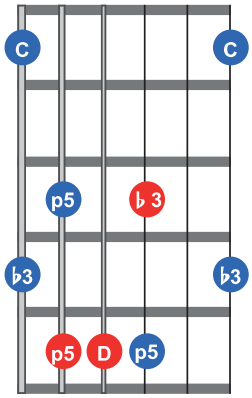
Db-
C-



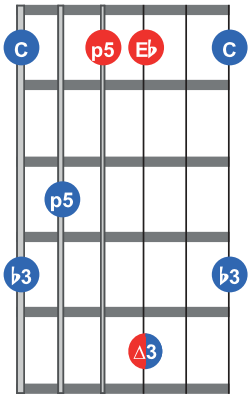
D
C-



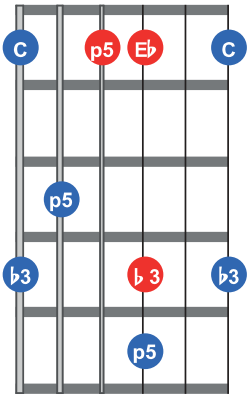
D-
C-



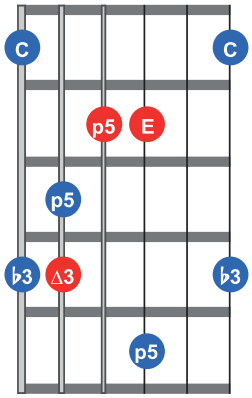
Eb
C-



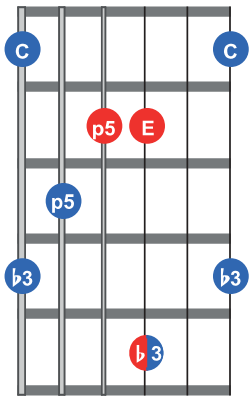
Eb-
C-



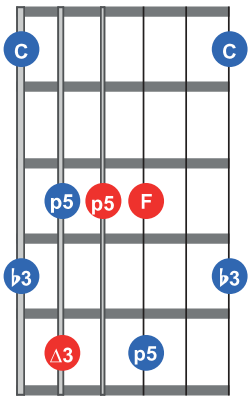
E
C-



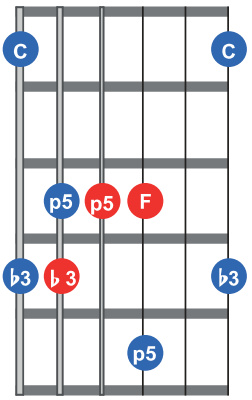
E-
C-



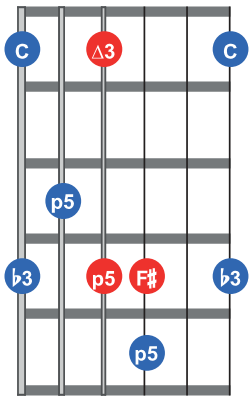
F
C-



F-
C-

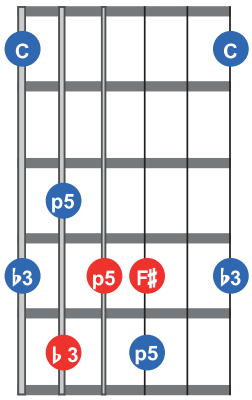


F#
C-



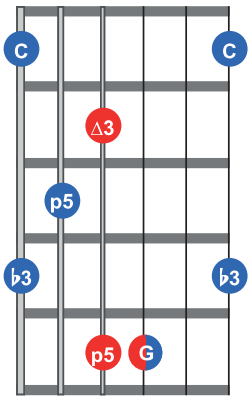
F#-

C-



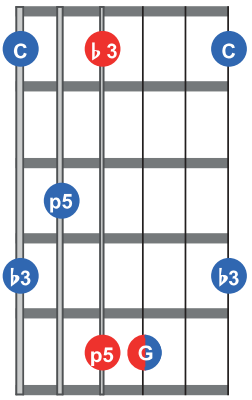
G

C-



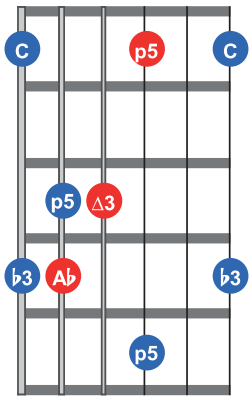
G-

C-



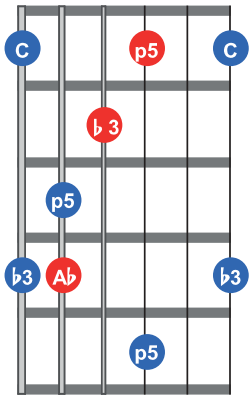
Ab

C-



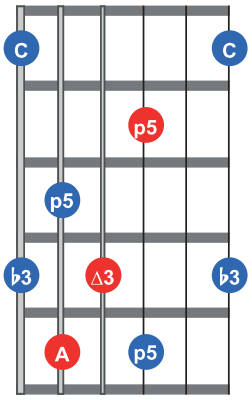
Ab-

C-



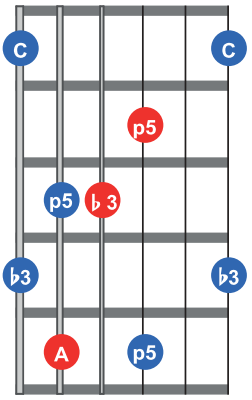
A

C-



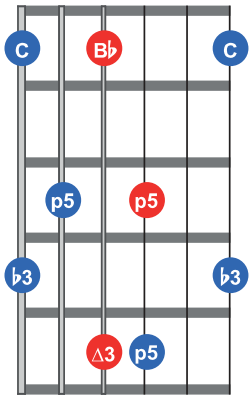
A-

C-



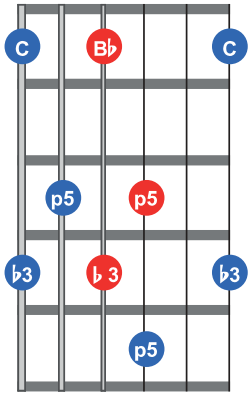
Bb

C-



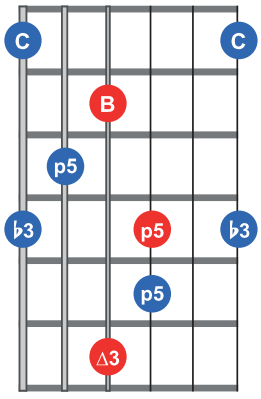
Bb-

C-



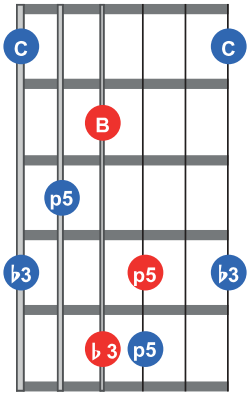
B

C-



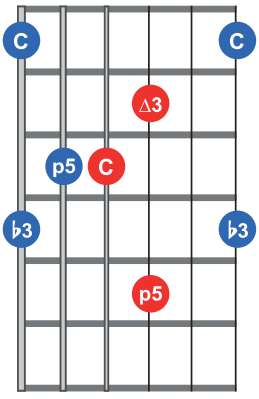
B-

C-



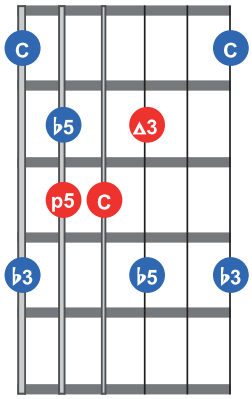
C

C-

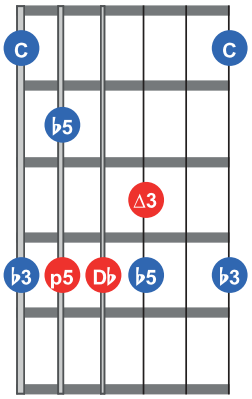


Diminished Bitonal Arpeggios (over C° at the 8th fret)

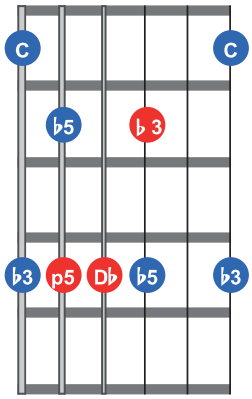
C-
C°



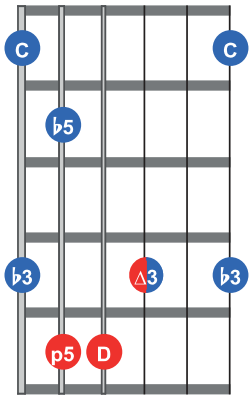
Db
C°



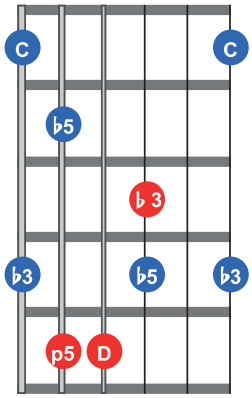
Db-
C°



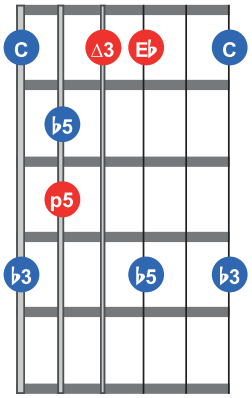
D
C°



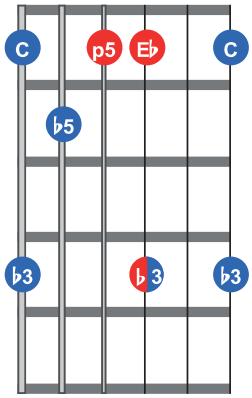
D-
C° C°



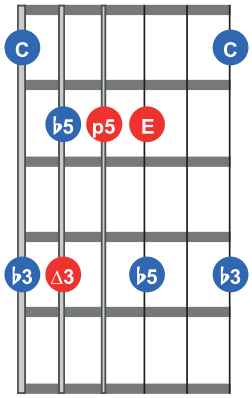
Eb
C°



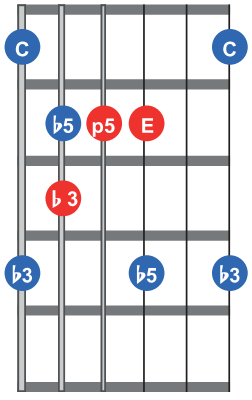
Eb-
C°



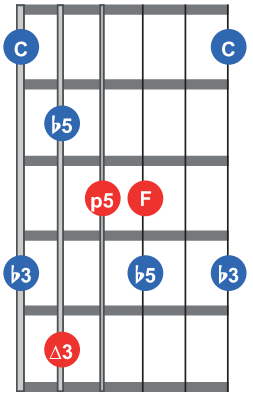
E
C°



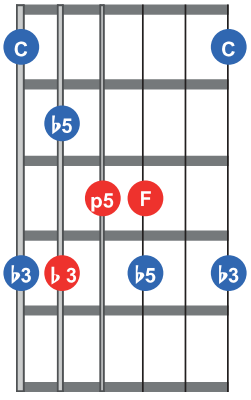
E-
C°



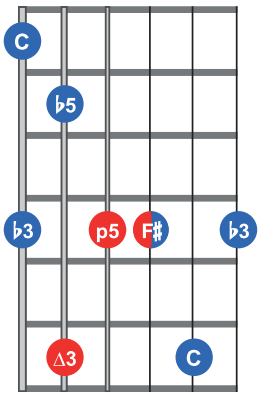
F
C°



F-
C°

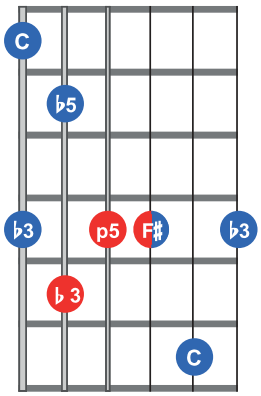


F#
C°



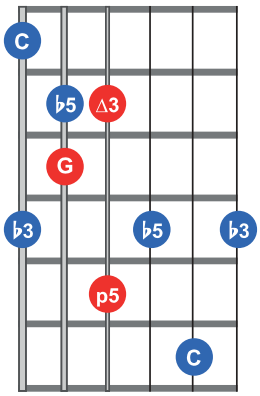
F#-

C°



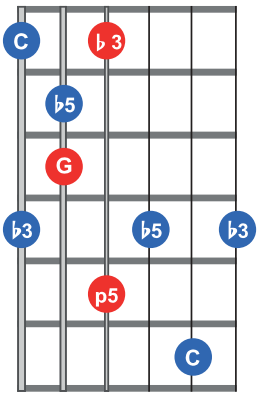
G

C°



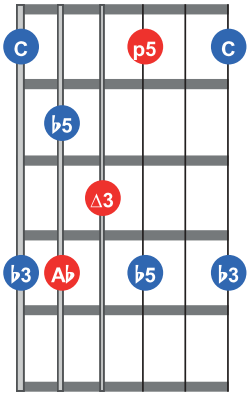
G-

C°



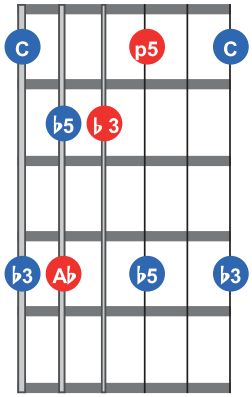
Ab

C°



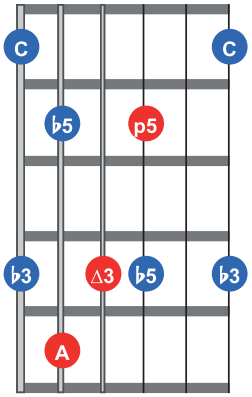
Ab-

C°



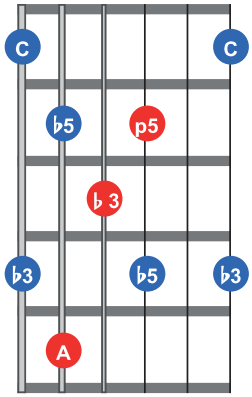
A

C°



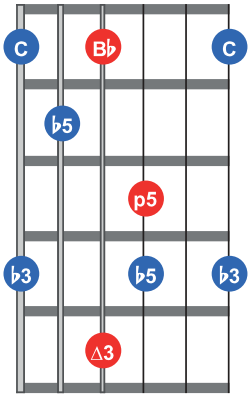
A-

C°



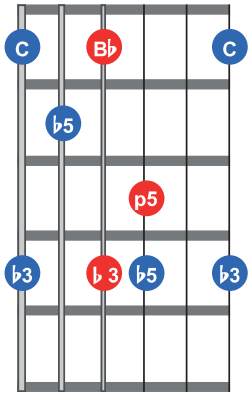
Bb

C°



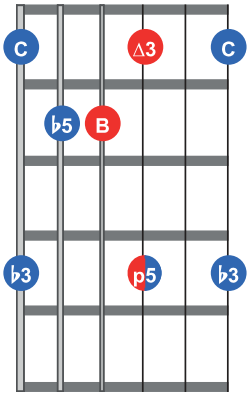
Bb-

C°



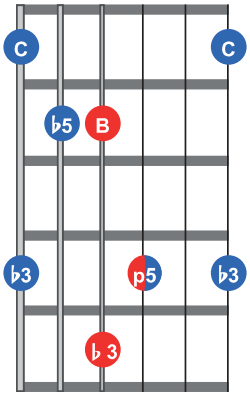
B

C°



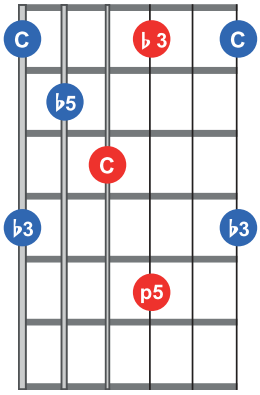
B-

C°



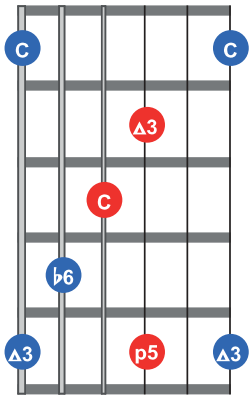
C

C°

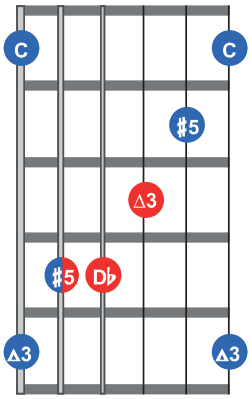


Augmented Bitonal Arpeggios (over C+ at the 8th fret)

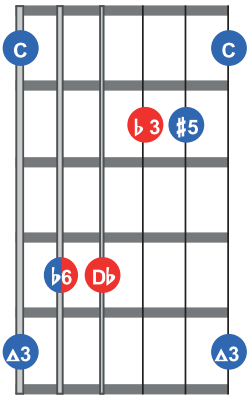
C-
C+



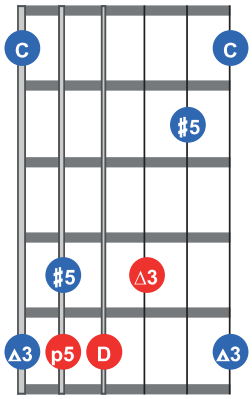
Db
C+



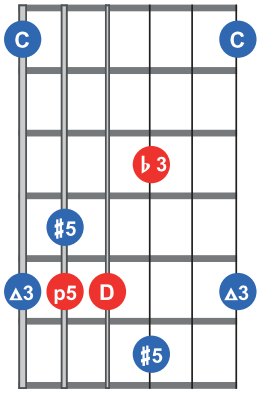
Db-
C+



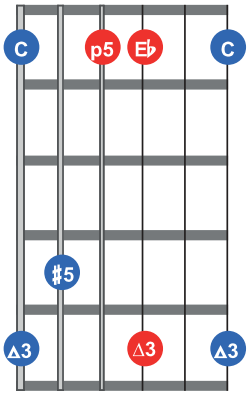
D
C+



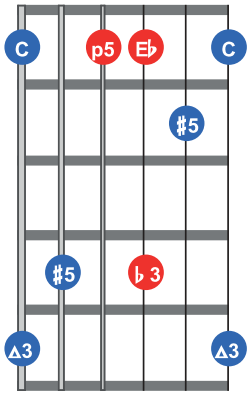
D-
C+



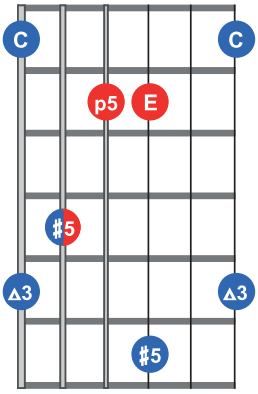
Eb
C+



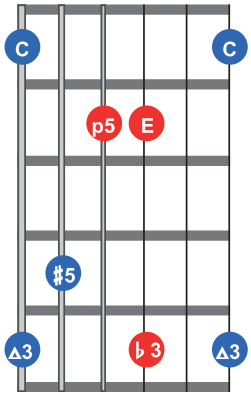
Eb-
C+



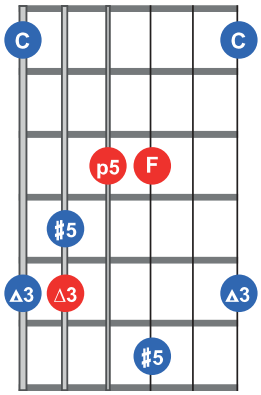
E
C+



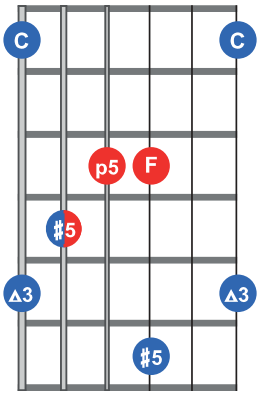
E-
C+



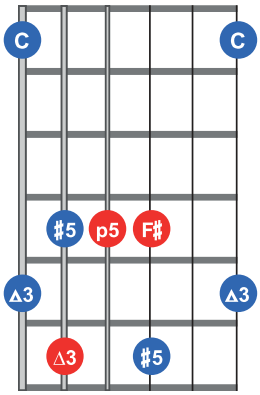
F
C+



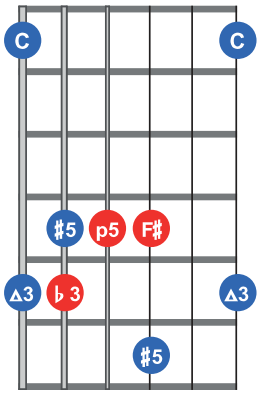
F-
C+



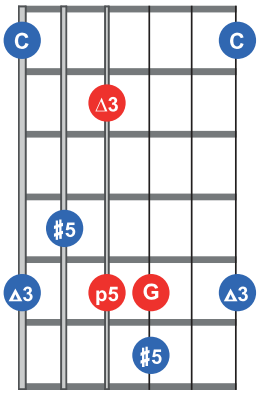
F#
C+



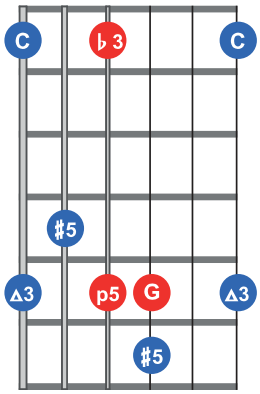
F#-
C+



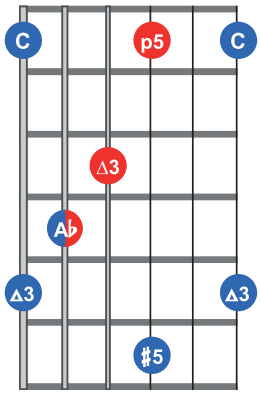
G
C+



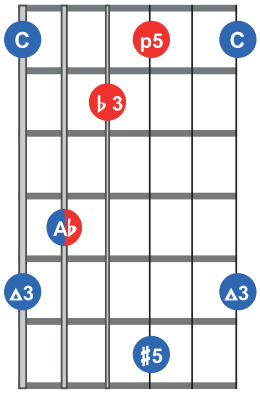
G-
C+



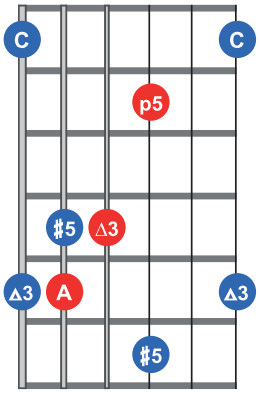
Ab
C+



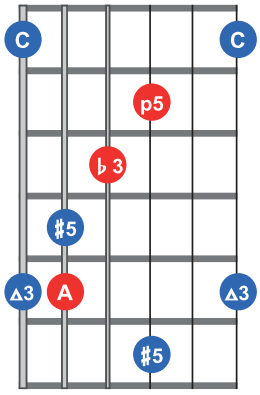
Ab-
C+



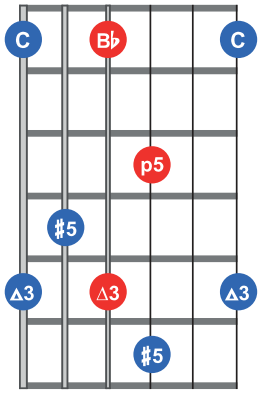
A
C+



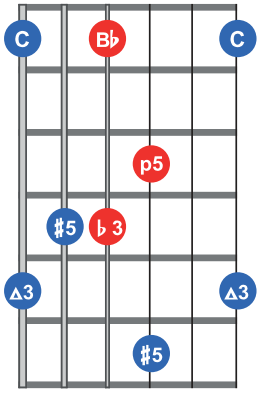
A-
C+



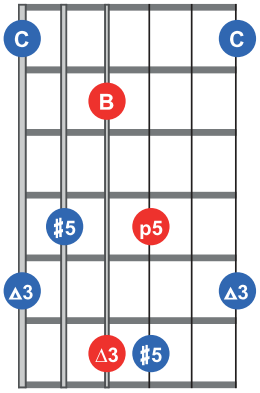
Bb
C+



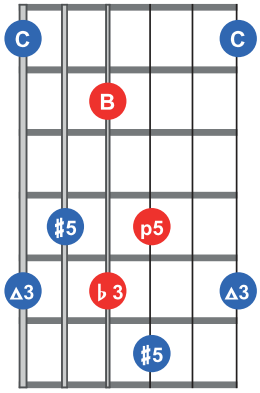
Bb-
C+



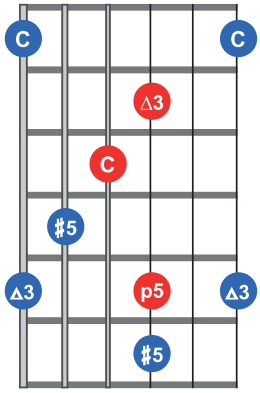
B
C+



B-
C+

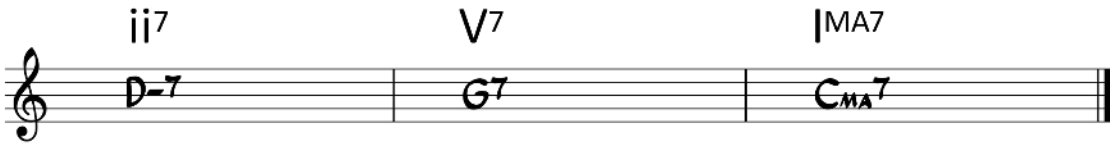


C
C+



More About Practicing Scales and Arpeggios

In order to maximize your practicing time, scales should be practiced in the combinations in which they fall in a particular chord progression.

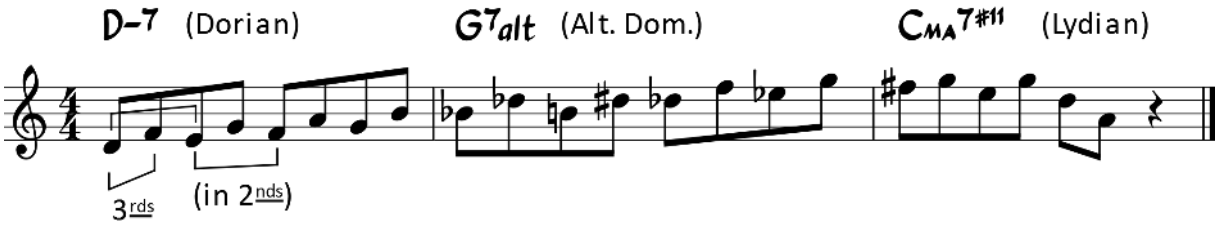
Ex. 160

Comb. 1 Parent Scale	D-Dorian (C-Major)	G-Mixolydian (C-Major)	C-Ionian (C-Major)
Comb. 2 Parent Scale	D-Dorian (C-Major)	G-Alt.Dom. (Ab-Mel. Min.)	C-Lydian (G-Major)

There are many combinations to choose from if you consider all the possible scale choices for each chord.

Ex. 161

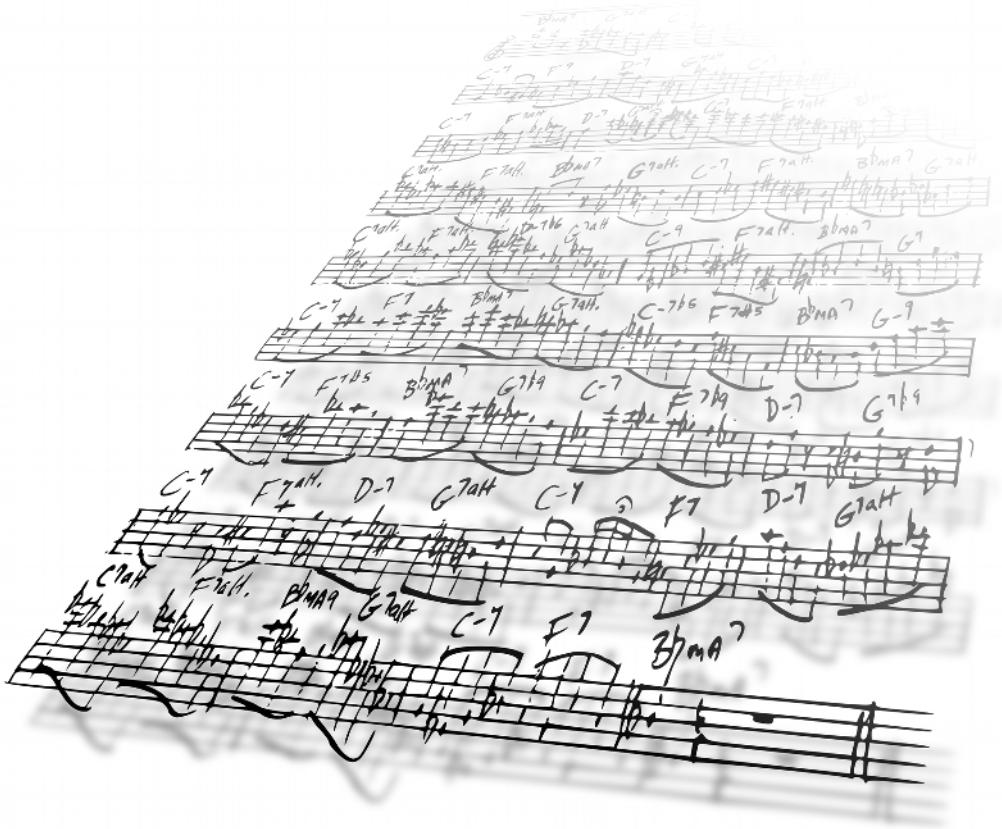
ii^7	V^7	I^{maj7}
Dorian	Mixolydian	Ionian
Melodic Minor	Mixolydian #11	Lydian
Aeolian	Whole Tone	Lydian Augmented
Phrygian	Dominant Diminished	Major Pentatonic
Minor Pentatonic	Altered Dominant	Major Pentatonic P5 ↑
Major Pentatonic M2↓	Major Pentatonic Tritone ↑↓	Major Pentatonic M2 ↑
Harmonic Minor	Phrygian Major	

After getting comfortable with the mode changes you should begin connecting them by diatonic interval patterns.

Ex. 162

As you can see, the interval pattern continues on the closest available note of the next mode.

Chapter 4

Linear Studies



Chapter 4: Linear Studies

ii V I Progressions

The major (key) ii V I progression can fall into a number of different catagories.

Ex. 163

a) ii V^7 I^{maj7}

$Dmin^7$ $G^{7(13)}$ C^{maj7}

unaltered (basic)

b) ii^7 V^{7alt} I^{maj7}

$Dmin^7$ $G^{7/\#5/b9}$ C^{maj7}

with altered Dom. Chord

c) ii^7 V^7 I^7

$D\text{-}^7$ $G^{7(9)}$ $C^{7(13)}$

Dom.^{7} tonic chord with unaltered V^7

d) ii^7 V^{7alt} I^{7alt}

$D\text{-}^7$ $G^{7/\#9/\#5}$ $C^{7/b9/b5}$

altered Dom.^{7} tonic chord with altered V^7

e) $ii^{7/b6}$ V^{7alt} $I^{maj7/\#5}$

$D\text{-}^{7/b6}$ $\frac{Ab^{sus4}}{G}$ $C^{maj7/\#4/\#5}$

modal ii V I

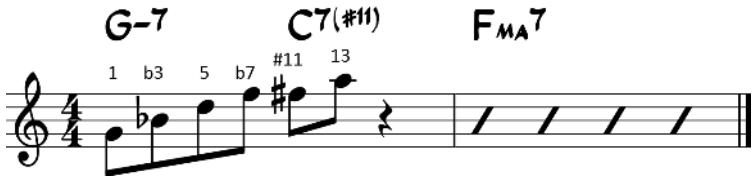
The minor (key) ii V i Progression contains a half-diminished supertonic chord, an altered dominant seventh chord and a minor or minor-major seventh tonic chord.

Ex. 164

ii^{7b5}	V^{7alt}	I$^{7\ (maj7)}$
D-^{7b5}	G$^{7/\#5/b9}$	C-$^{7\ (maj7)}$

The following examples are in two or four bar phrase lengths. The interval relationship of the melody notes to the chords should be analyzed as follows.

Ex. 165

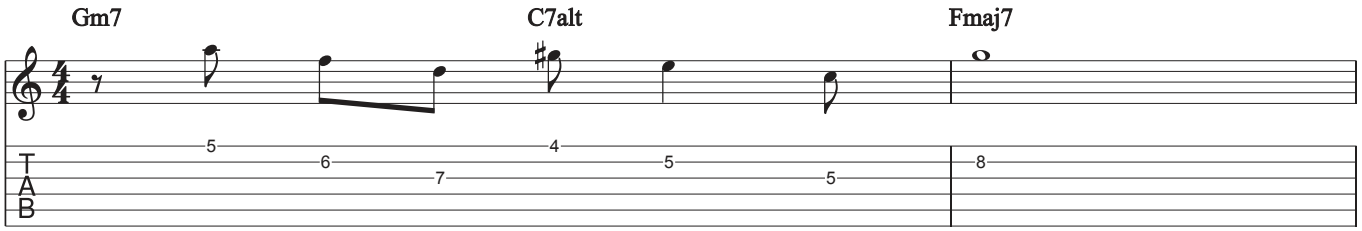


ii V's in One Measure

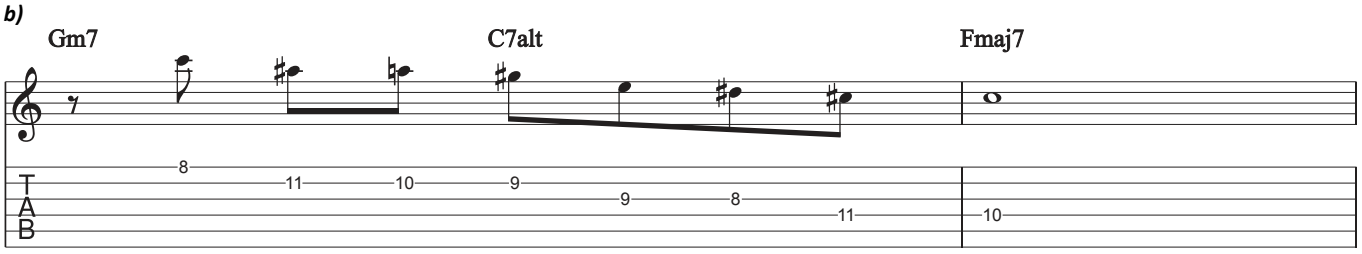
Ex. 166

Major keys

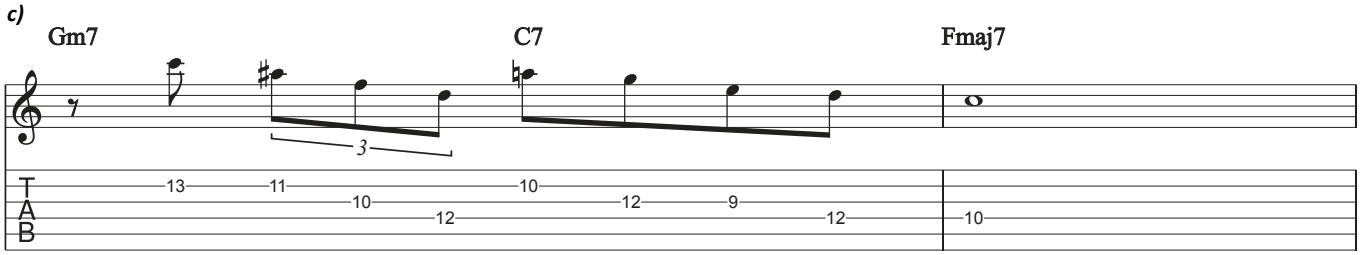
a)



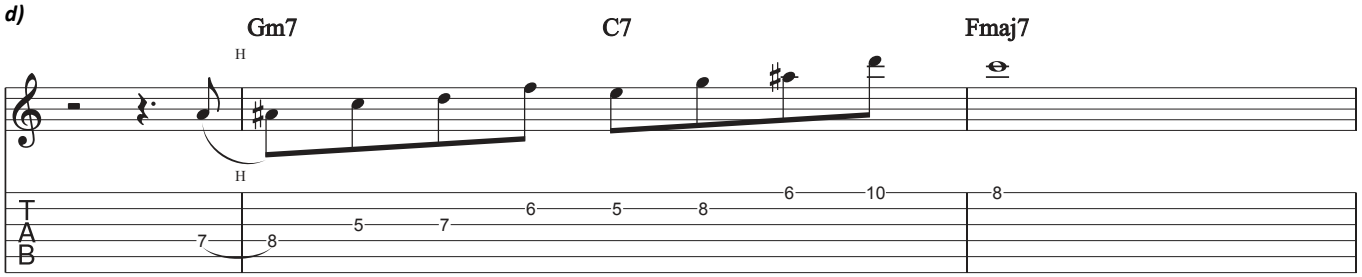
b)



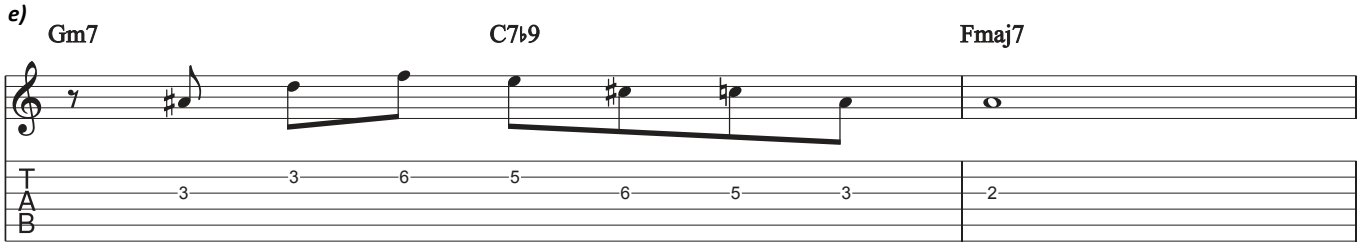
c)



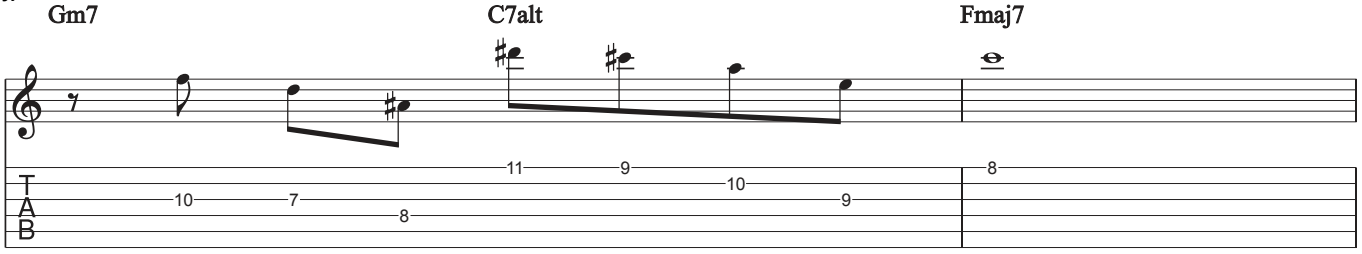
d)



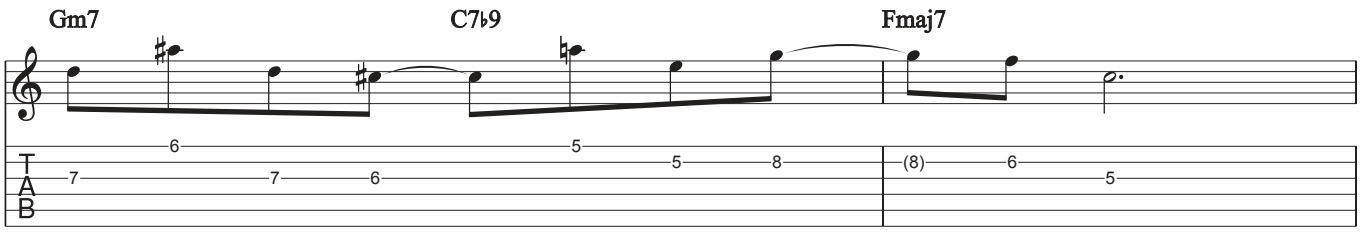
e)



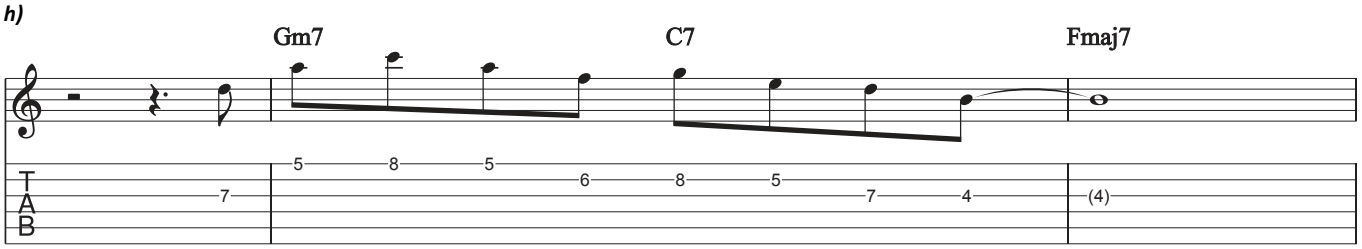
f)



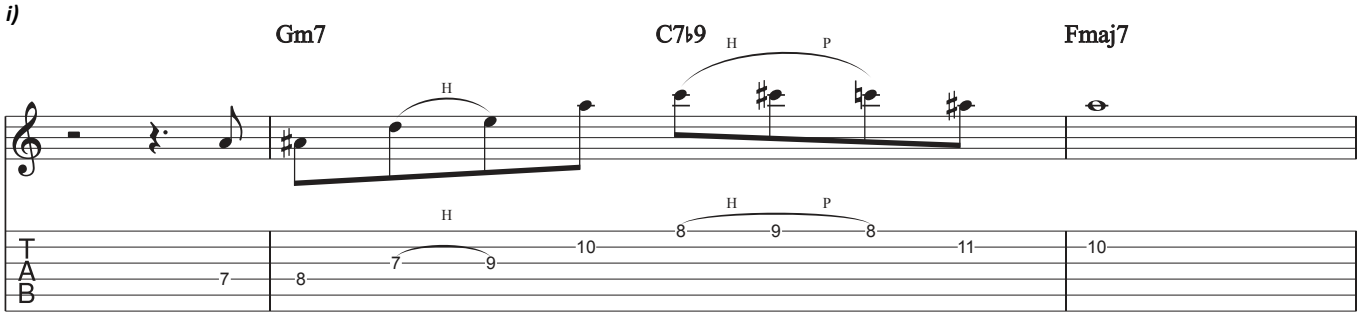
g)



h)

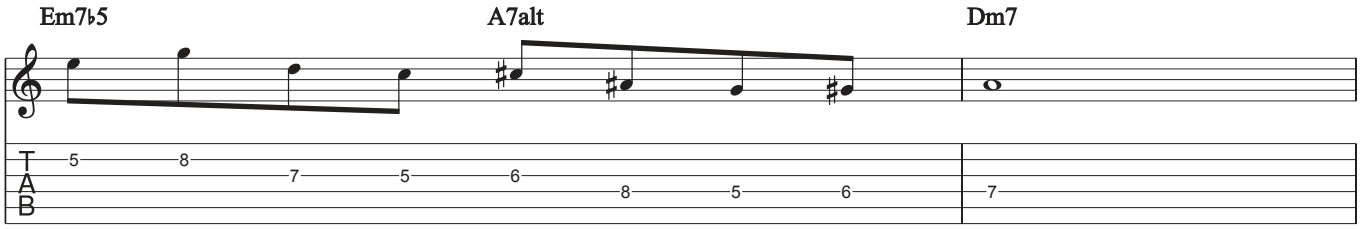


i)

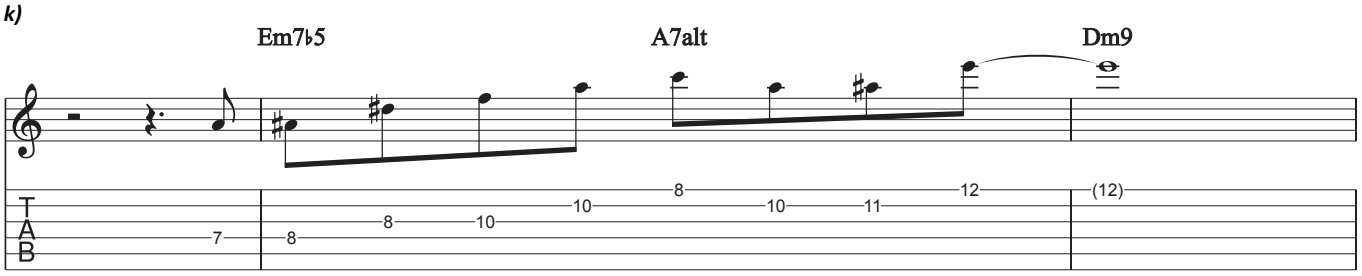


Minor keys

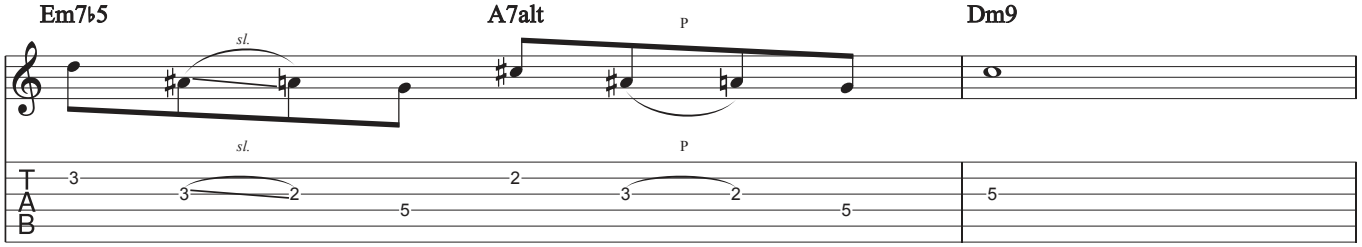
j)



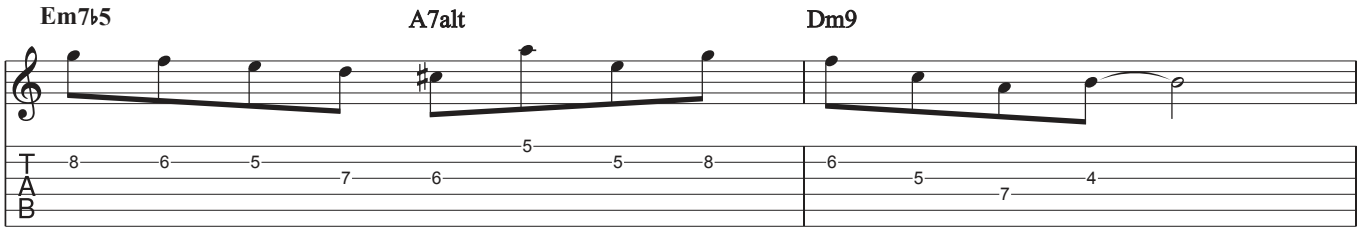
k)



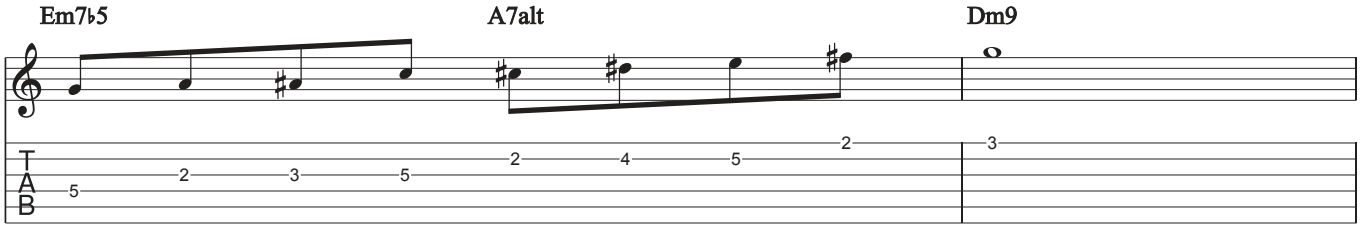
l)



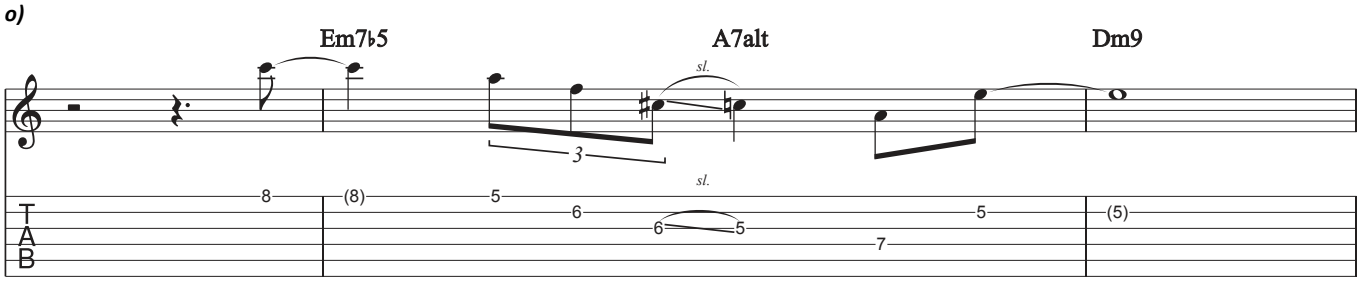
m)


Em7♭5
A7alt
Dm9
T
A
B

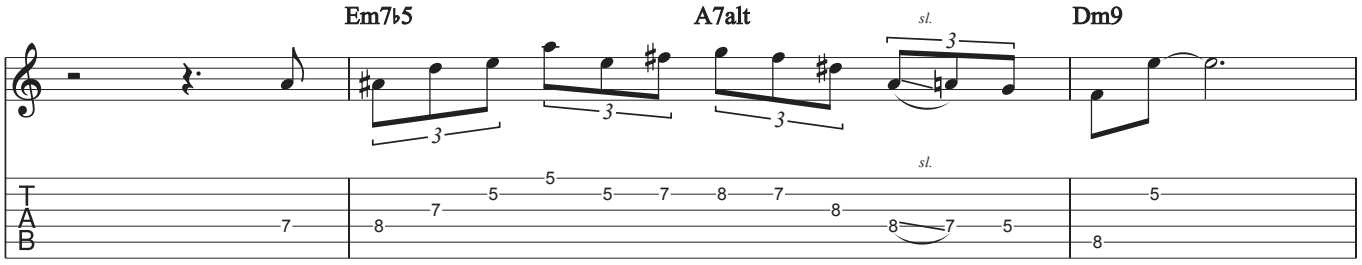
n)


Em7♭5
A7alt
Dm9
T
A
B

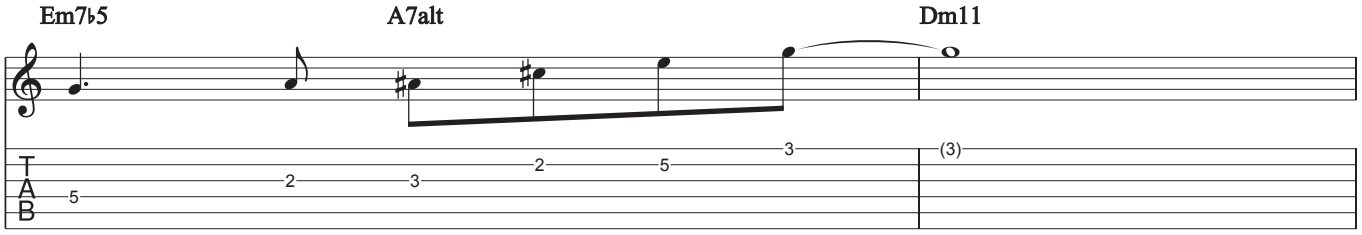
o)


Em7♭5
A7alt
Dm9
sl.
sl.
T
A
B

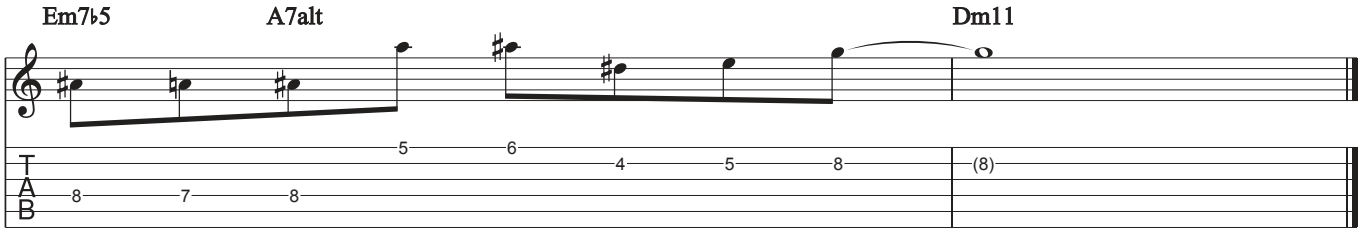
p)


Em7♭5
A7alt
Dm9
sl.
sl.
T
A
B

q)


Em7♭5
A7alt
Dm11
T
A
B

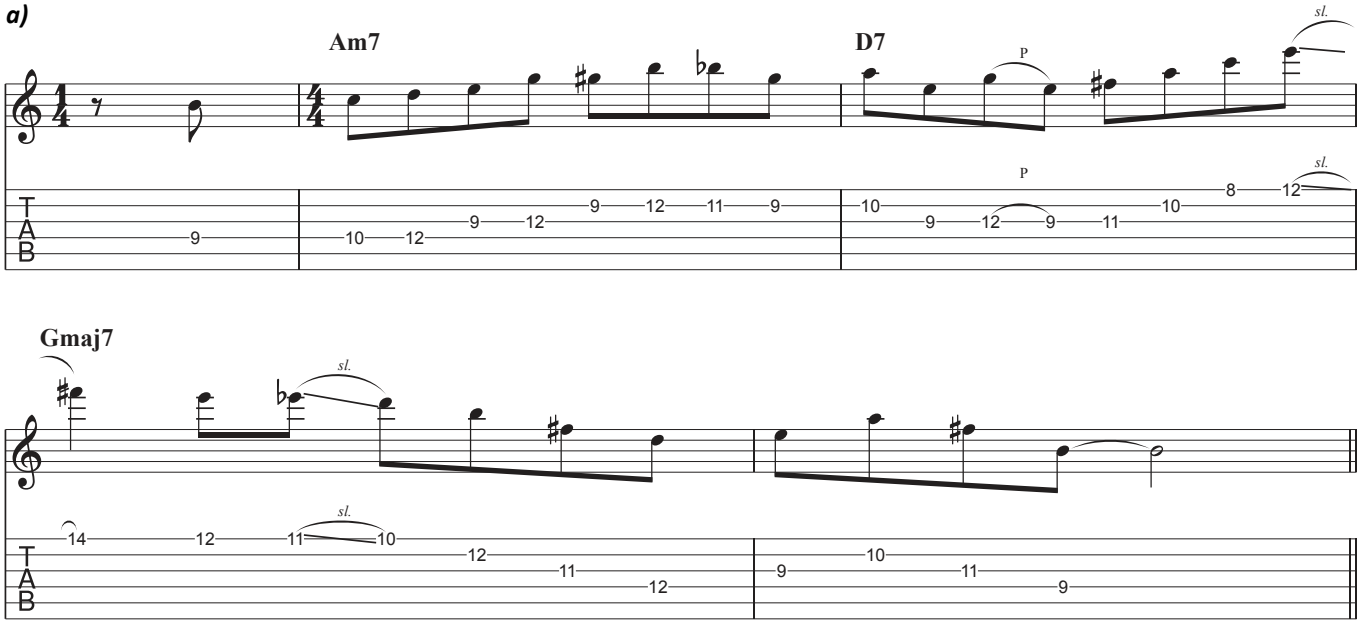
r)


Em7♭5
A7alt
Dm11
T
A
B

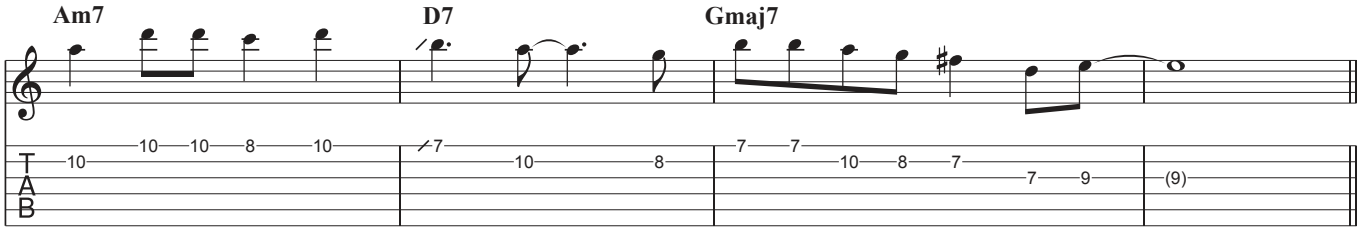
One Measure ii V I's in Major

Ex. 167

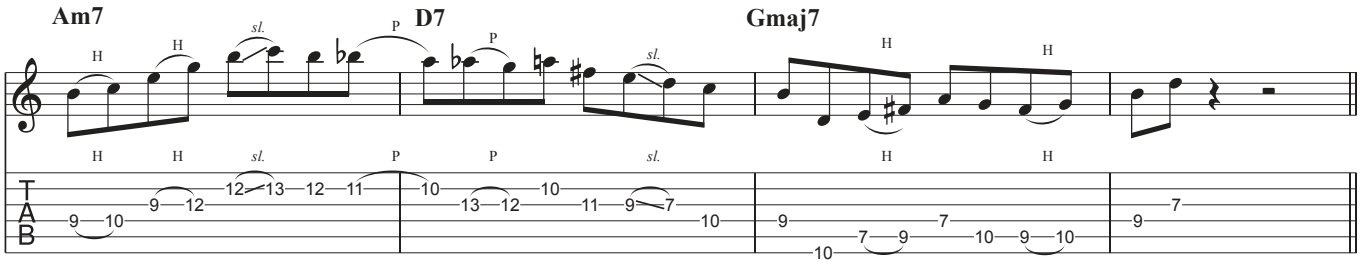
a)



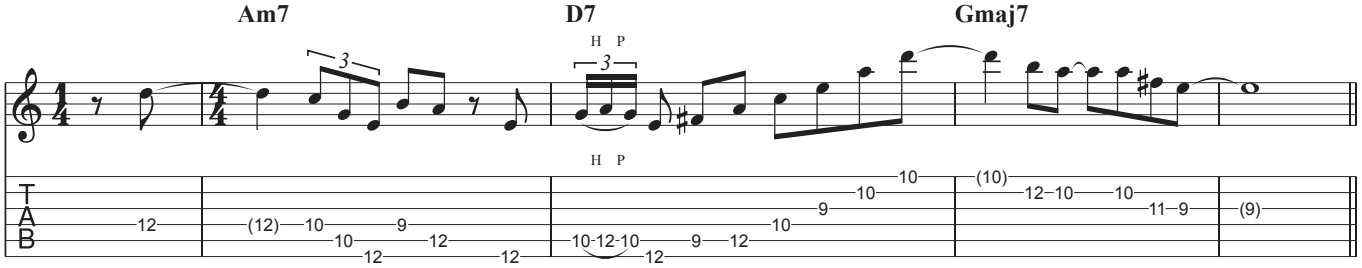
b)



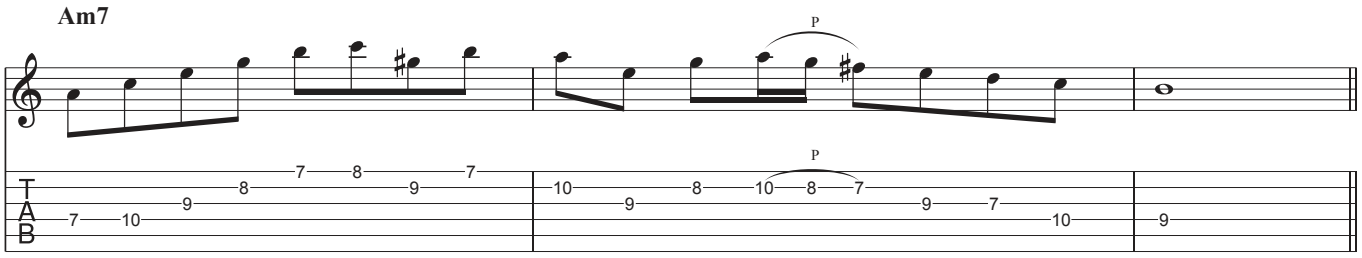
c)



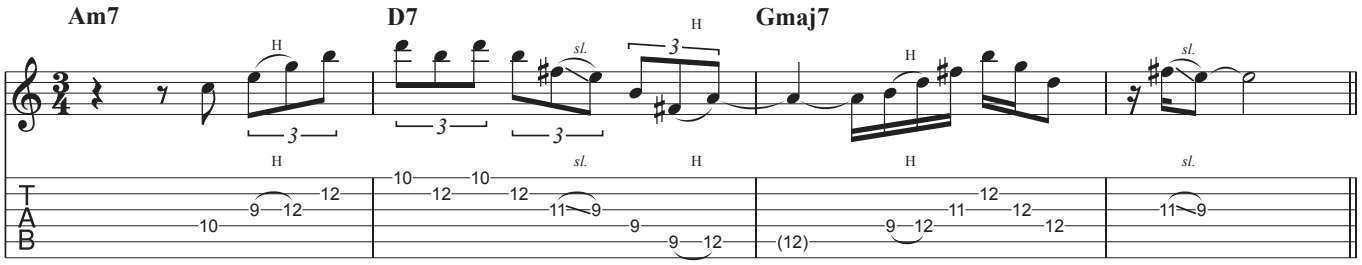
d)



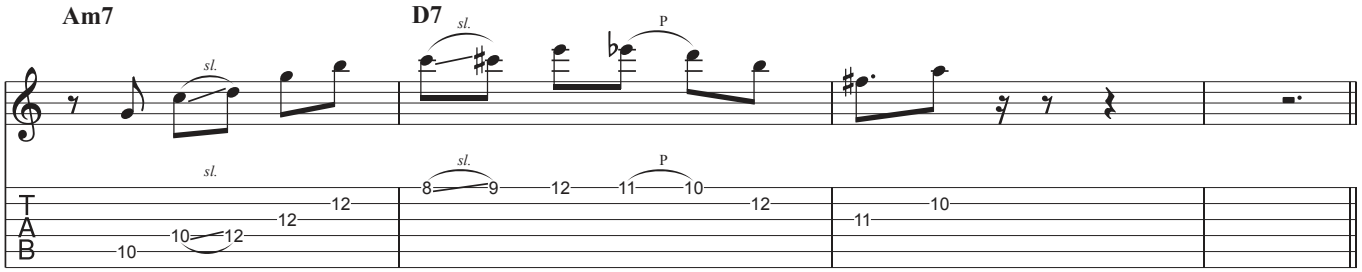
e)


Am7
P
P

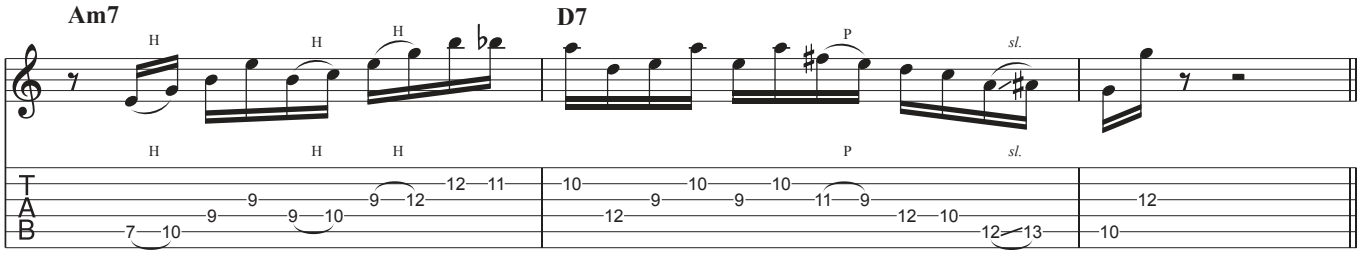
f)


Am7
D7
Gmaj7
H
sl.
3
(12)

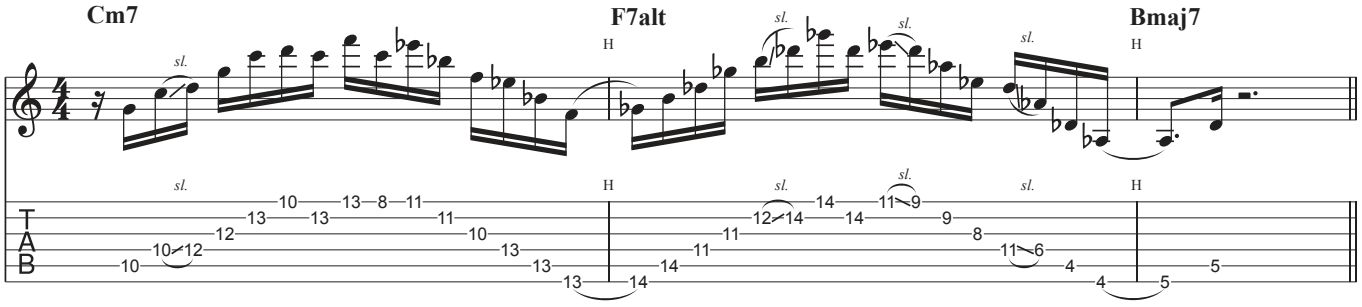
g)


Am7
D7
sl.
P

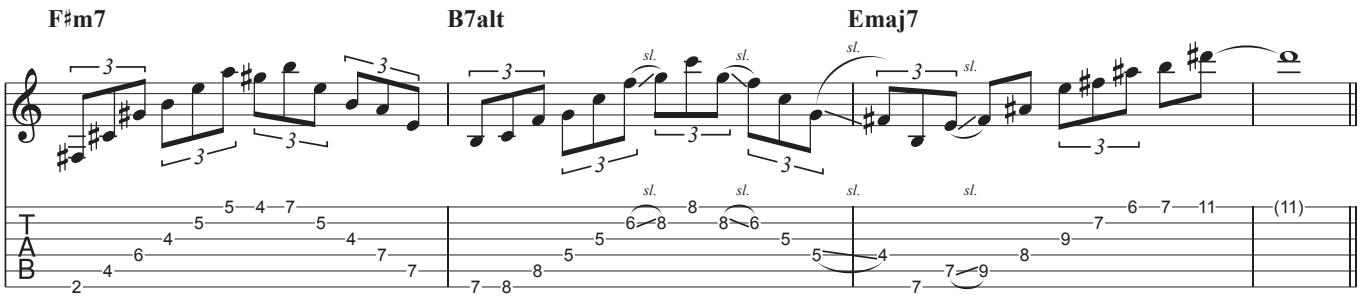
h)


Am7
D7
H
P
sl.

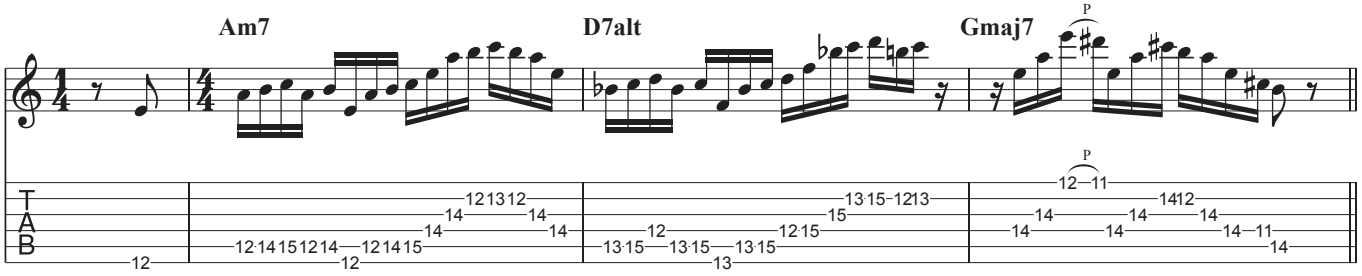
i)


Cm7
F7alt
Bmaj7
sl.
H

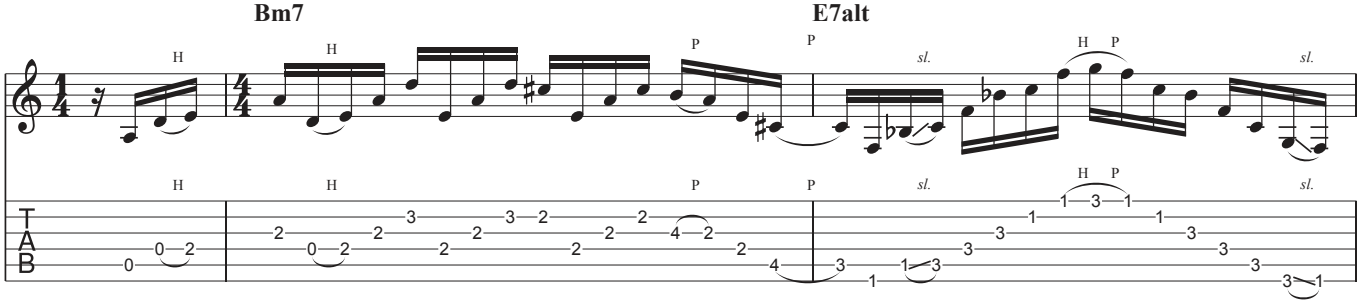
j)

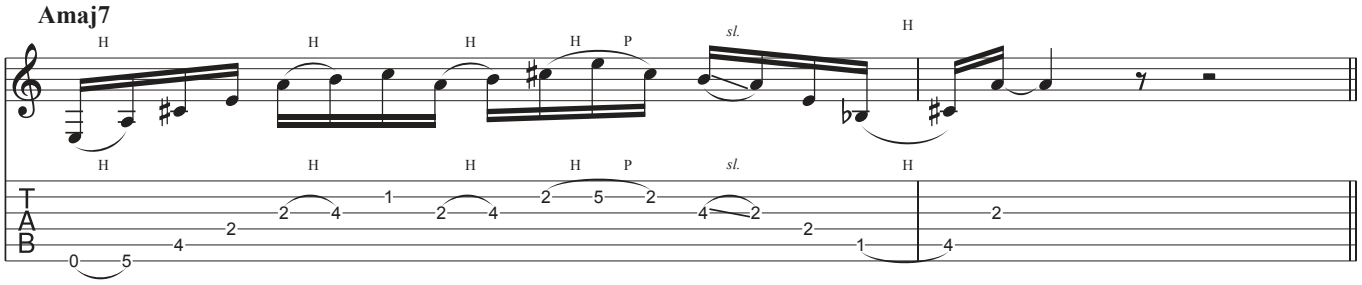

F♯m7
B7alt
Emaj7

k)

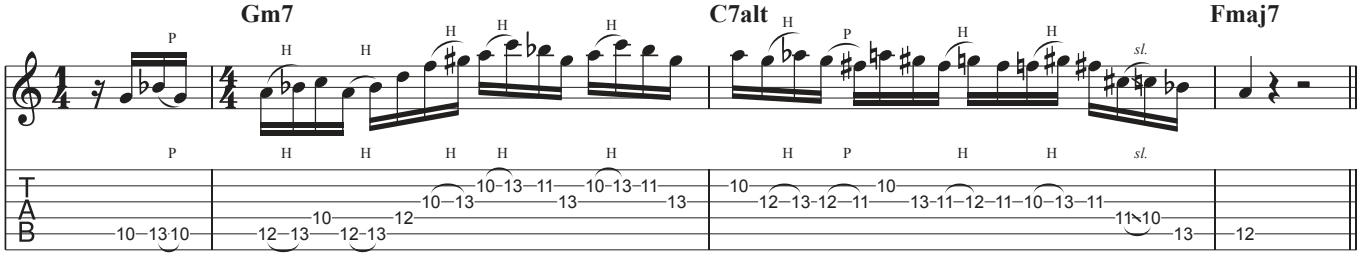

Am7
D7alt
Gmaj7

l)

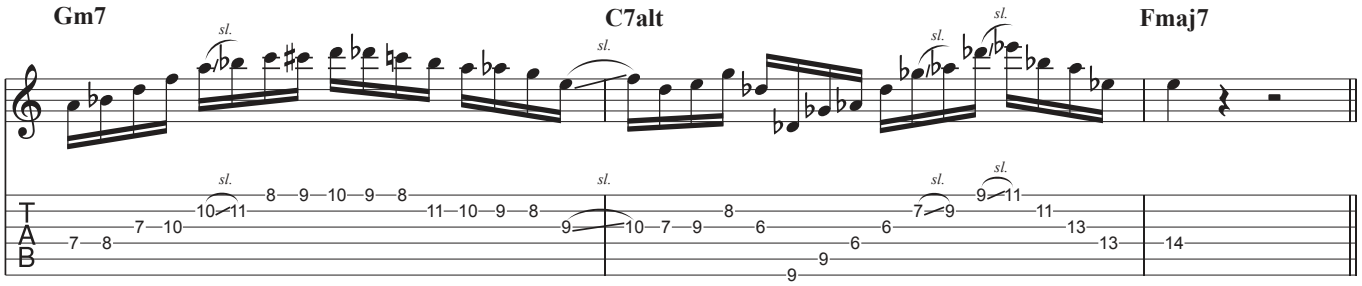

Bm7
E7alt


Amaj7

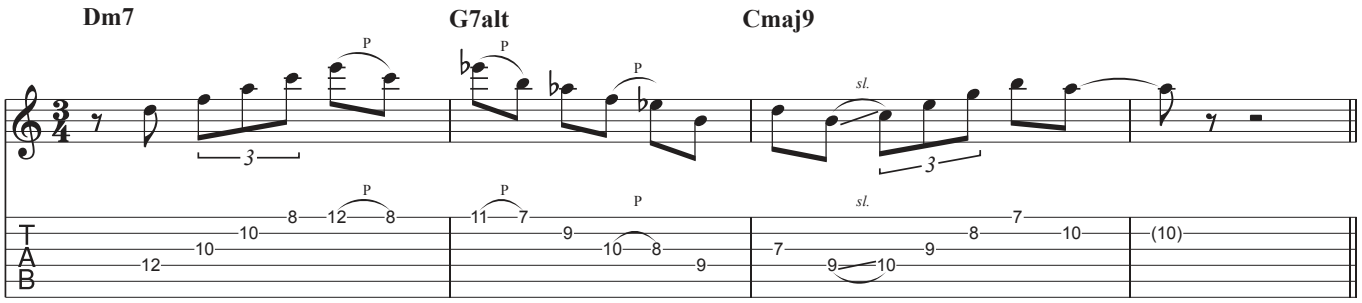
m)


Gm7
C7alt
Fmaj7

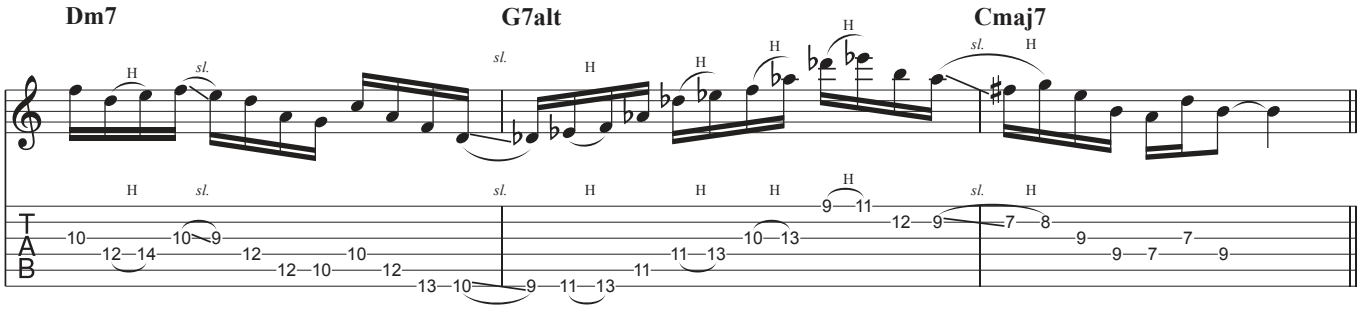
n)


Gm7
C7alt
Fmaj7
sl.
T
A
B

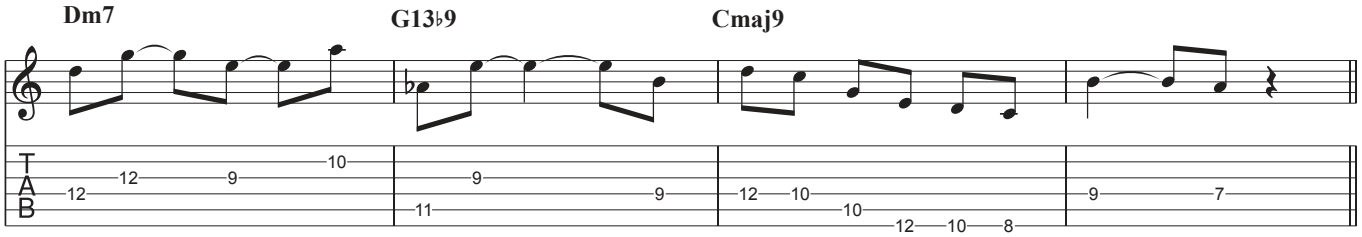
o)


Dm7
G7alt
Cmaj9
P
sl.
T
A
B

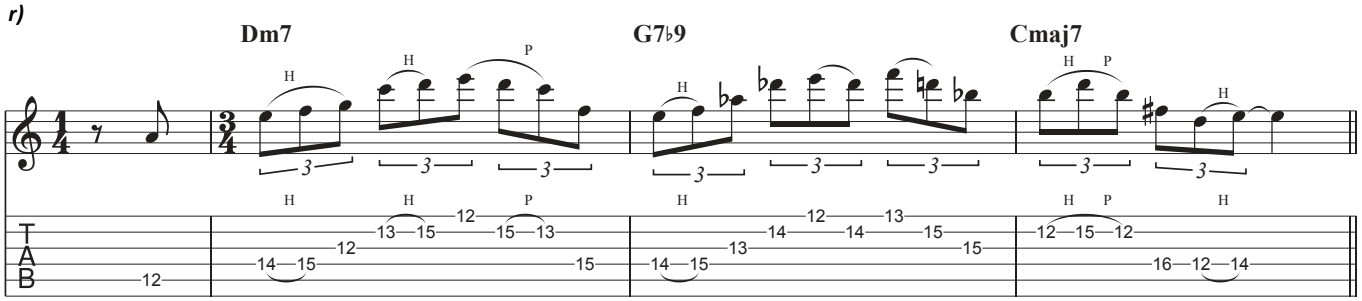
p)


Dm7
G7alt
Cmaj7
H
sl.
T
A
B

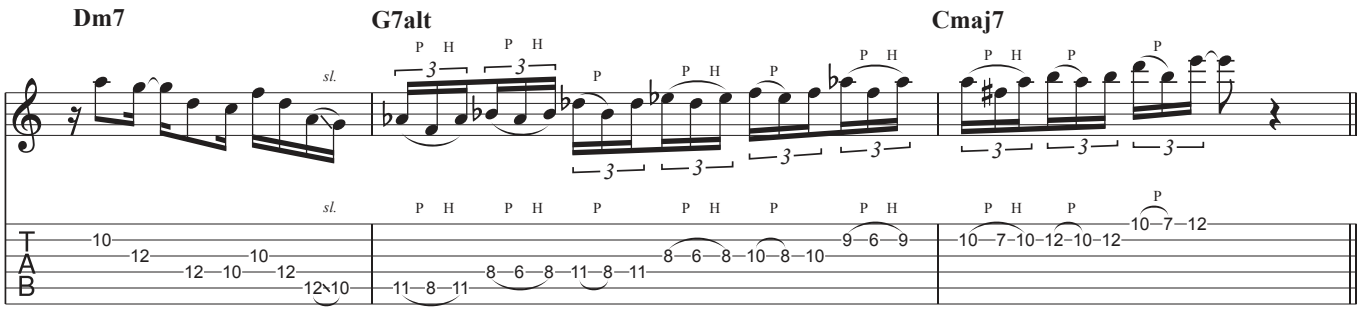
q)


Dm7
G13♭9
Cmaj9
T
A
B

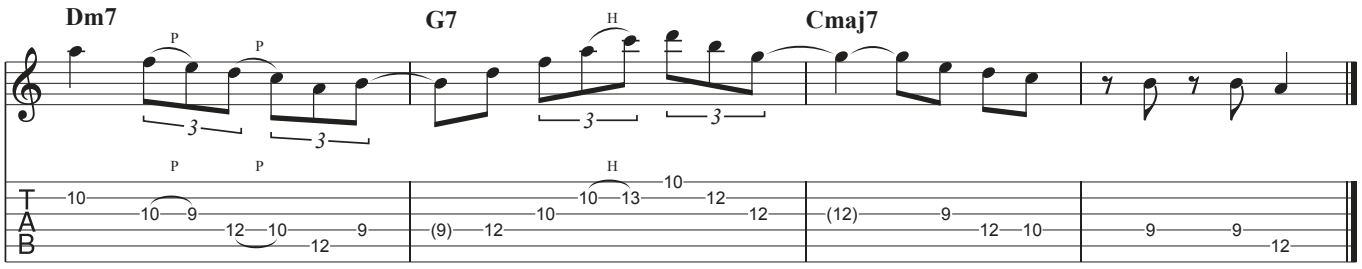
r)


Dm7
G7♭9
Cmaj7
H
P
T
A
B

s)



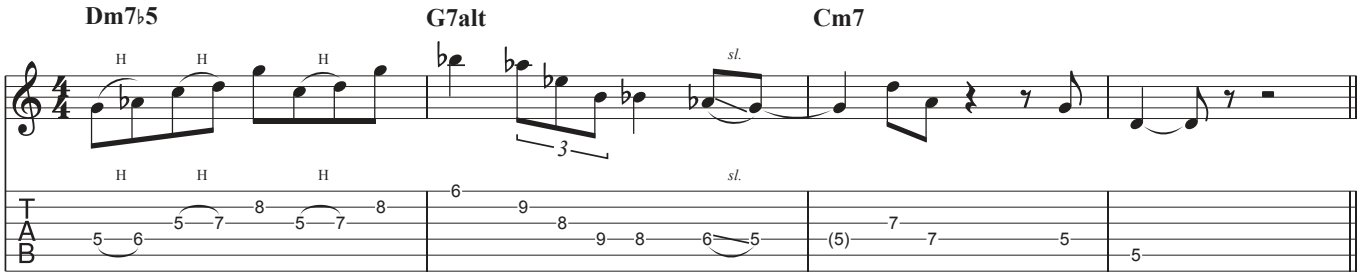
t)



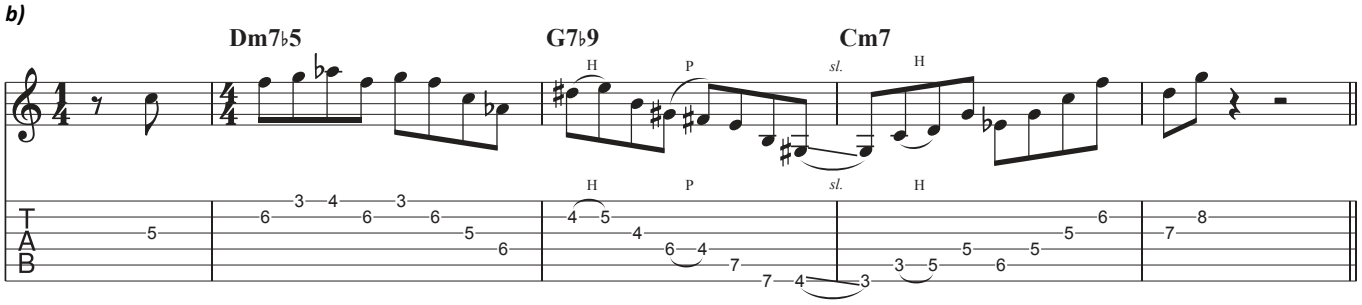
One Measure ii V I's in Minor

Ex. 168

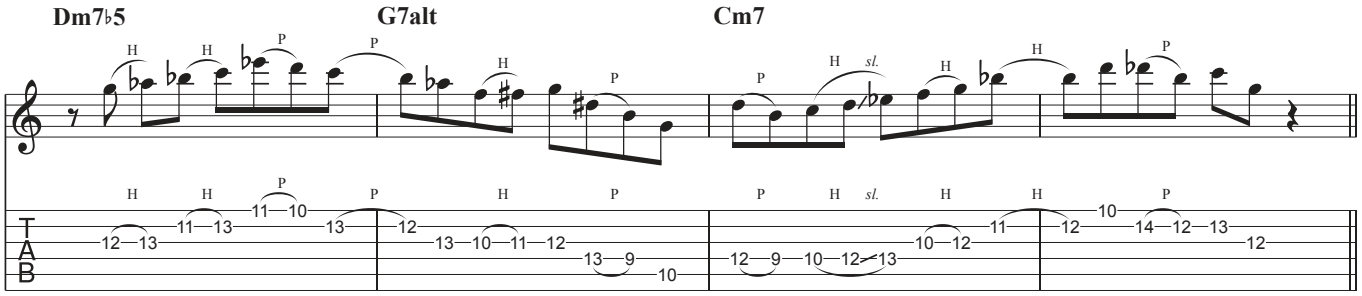
a)



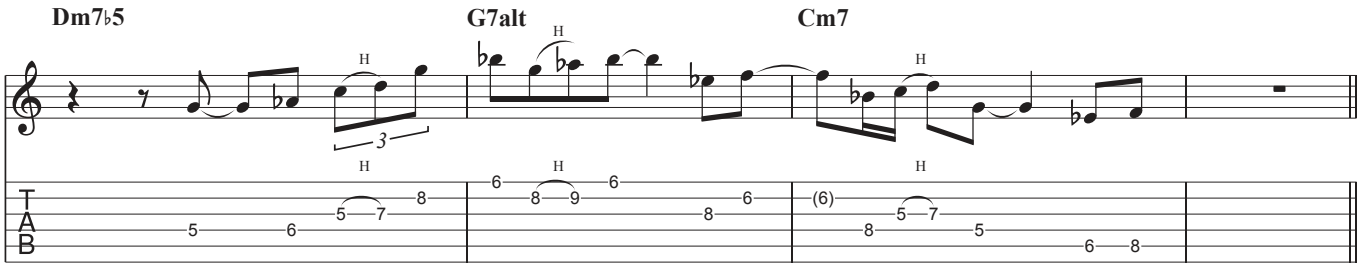
b)



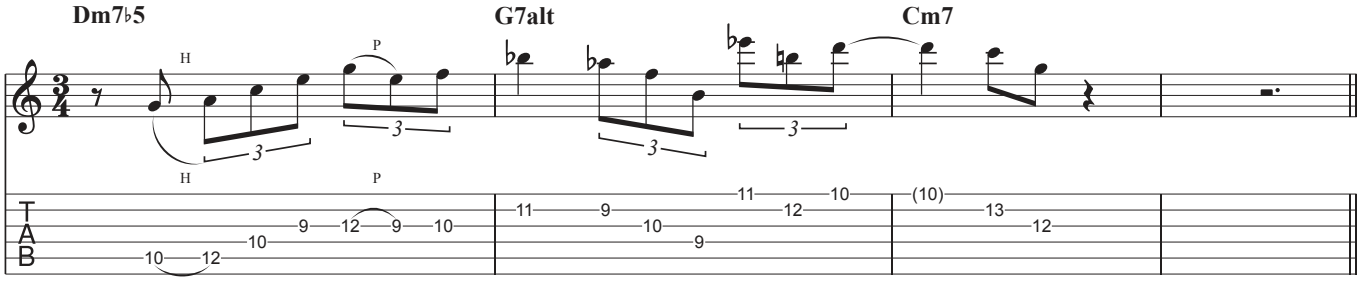
c)



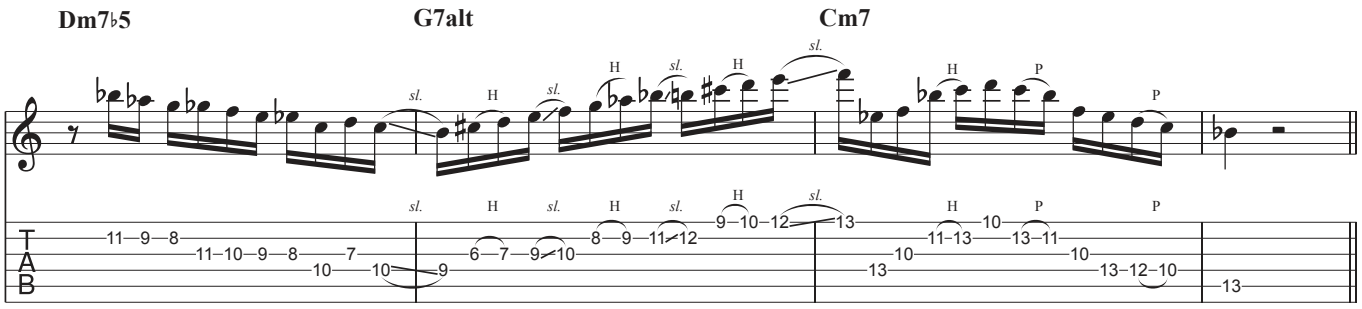
d)



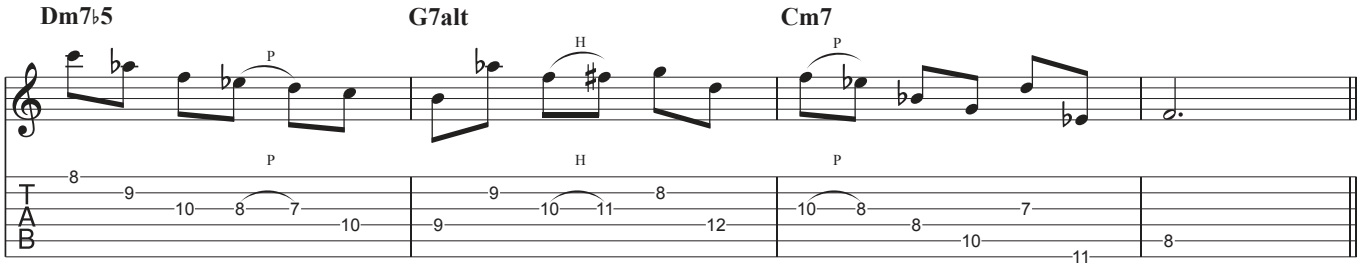
e)



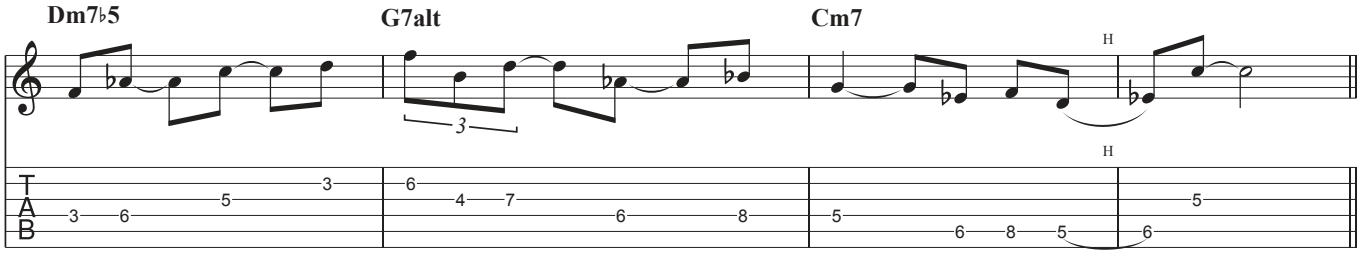
f)


Dm7♭5
G7alt
Cm7

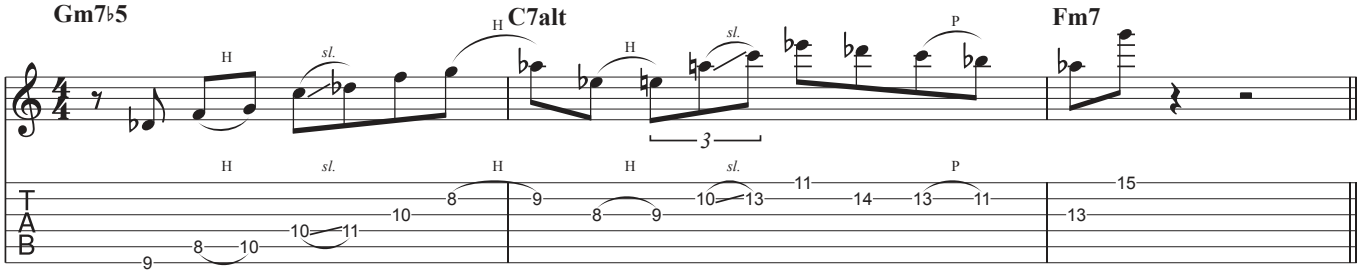
g)


Dm7♭5
G7alt
Cm7

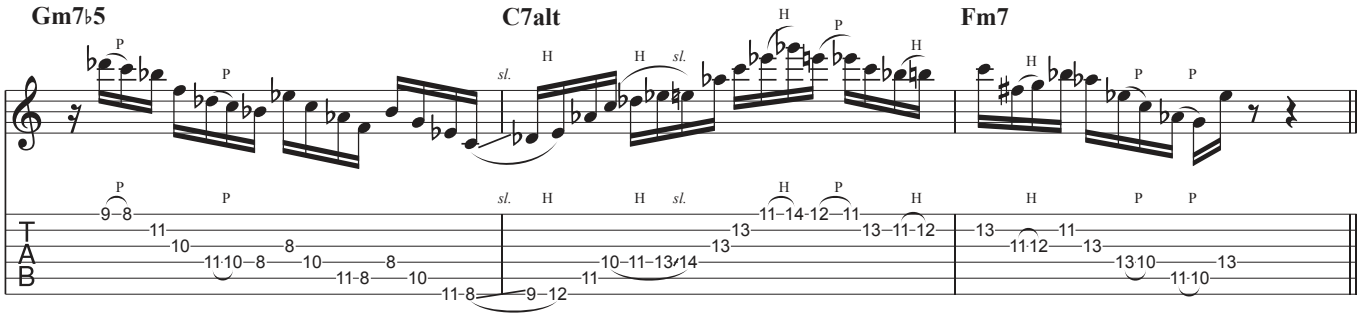
h)


Dm7♭5
G7alt
Cm7

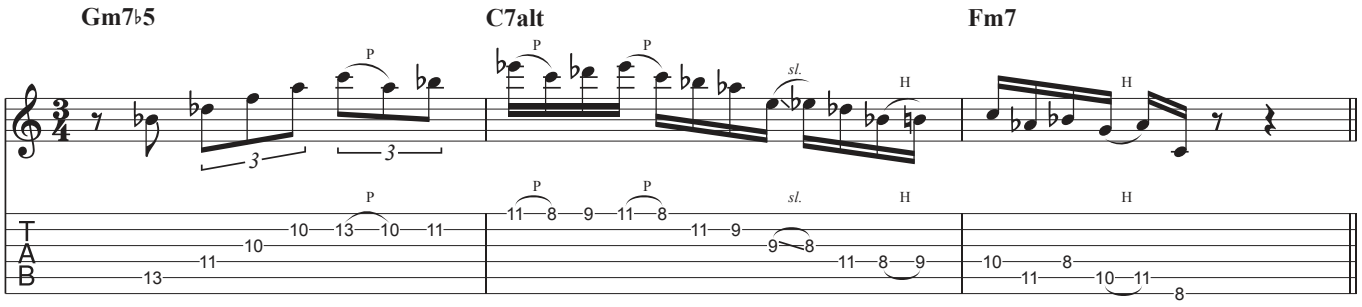
i)


Gm7♭5
C7alt
Fm7

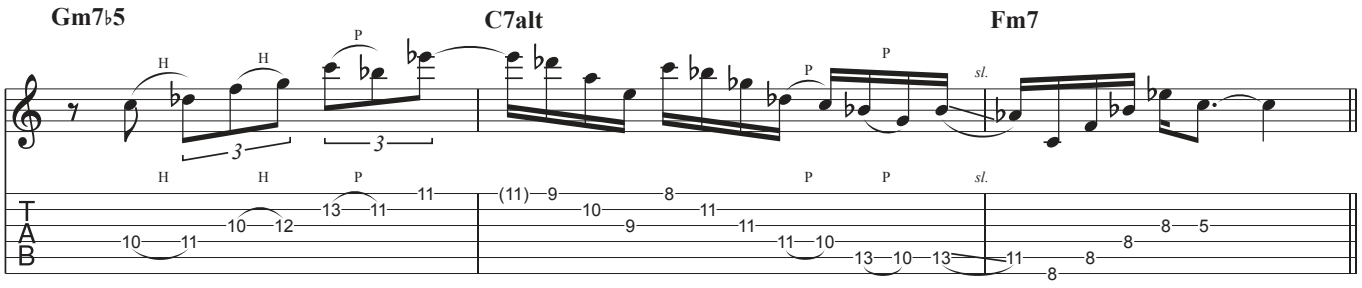
j)


Gm7♭5
C7alt
Fm7

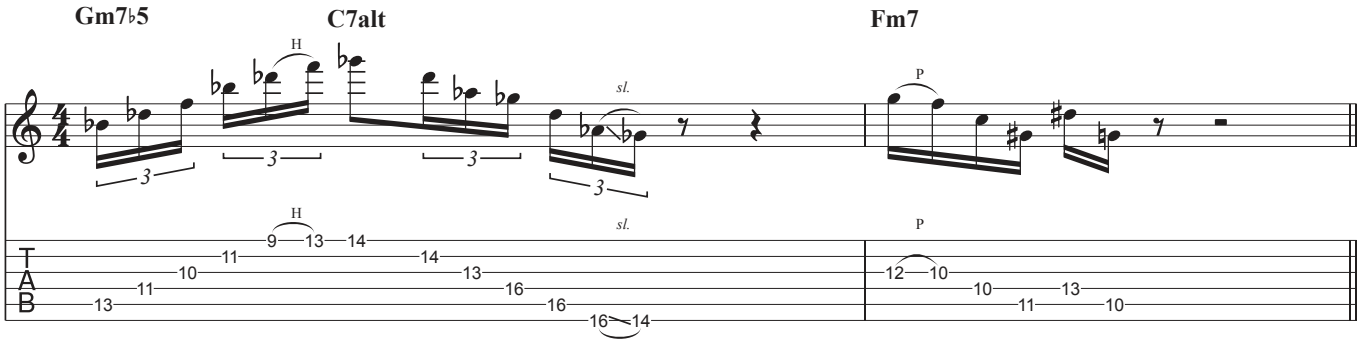
k)


Gm7♭5
C7alt
Fm7

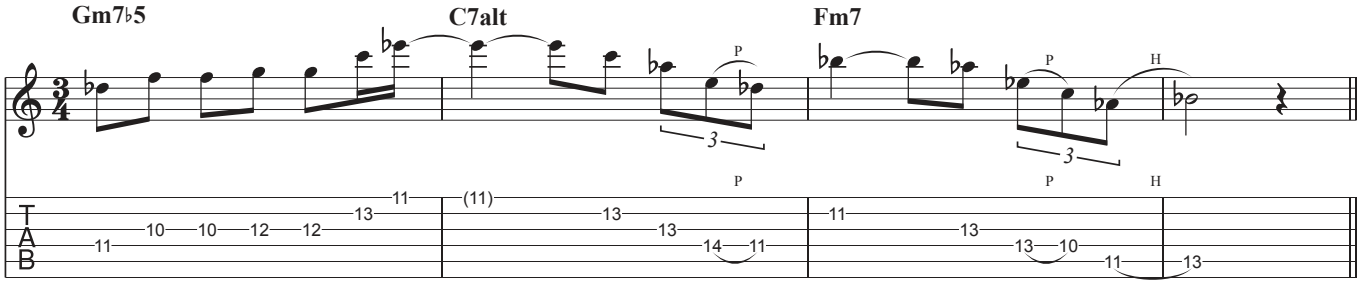
l)


Gm7♭5
C7alt
Fm7

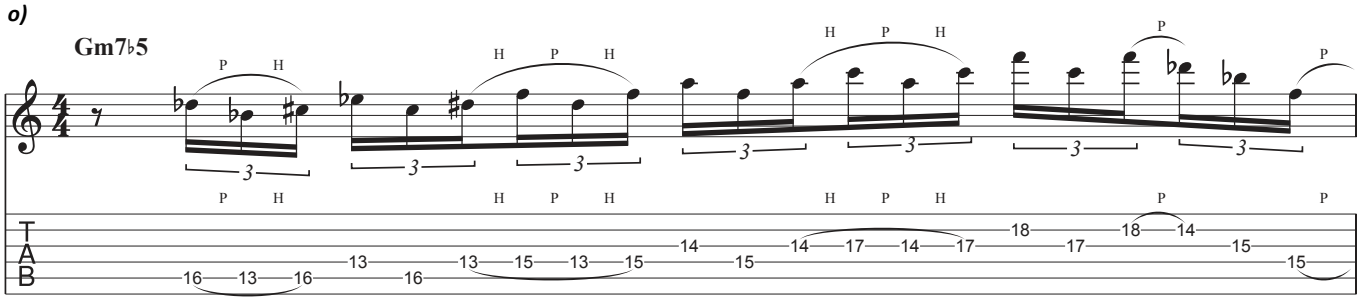
m)

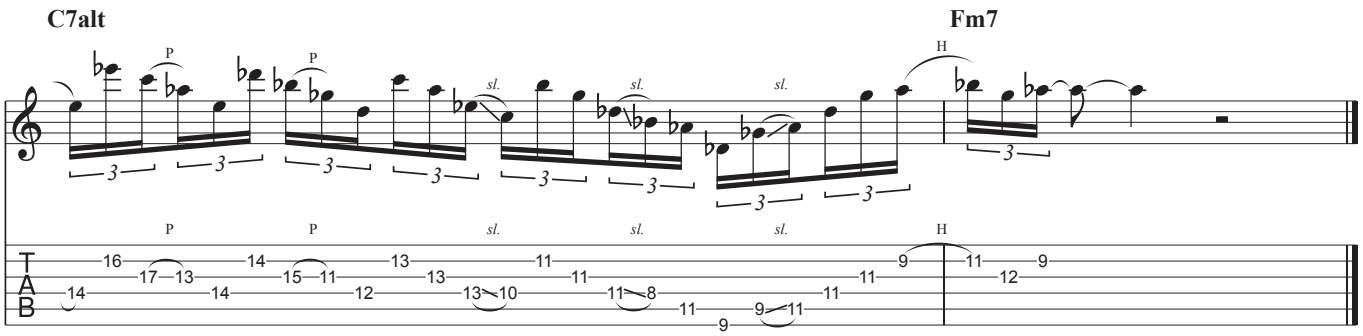

Gm7♭5
C7alt
Fm7

n)


Gm7♭5
C7alt
Fm7

o)


Gm7♭5


C7alt
Fm7
P
P
sl.
sl.
sl.
H
3
T
A
B
16 17 13 14 14 15 11 12 13 13 10 11 8 11 9 11 9 11 9 11 12

ii^{7} V^{7} I^{maj7} substitute patterns

The substitutions presented in this section may be used over the major ii V^{7} I^{maj7} progression. I have indicated the root relationship of the first substitute chord to the supertonic chord to better facilitate the transposition of these progressions.

Ex. 169

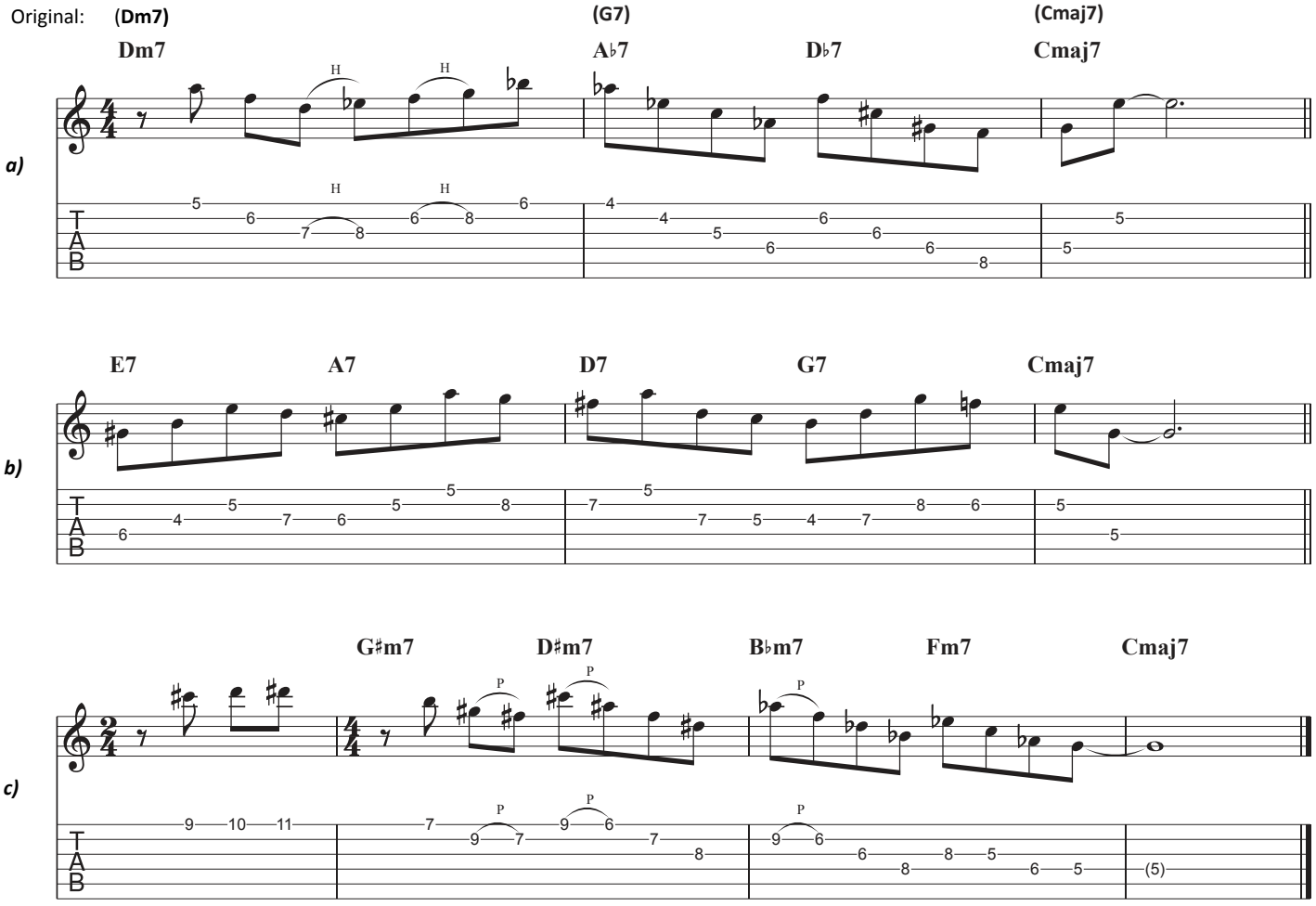
ii^{7} V^{7} I^{maj7} Substitues

Interval Relation	D-^{7}		G^{7}		C^{maj7}
R	D^{maj7}	F^{7}	Ab^{7}	B^{7}	C^{maj7}
R	D^{maj7}	F^{7}	Bb^{maj7}	Db^{7}	C^{maj7}
TT ↕	Ab^{maj7}	B^{7}	E^{maj7}	G^{7}	C^{maj7}
R	D-^{7}	G^{7}	Ab-^{7}	Db^{7}	C^{maj7}
R	D-^{7}		F-^{7}		C^{maj7} (E-^{7})
R	D-^{7}		Db-^{7}		C^{maj7}
M2 ↑	E^{7}	A^{7}	D^{7}	G^{7}	C^{maj7}
M3 ↓	Bb^{7}	Eb^{7}	Ab^{7}	Db^{7}	C^{maj7}
TT ↕	Ab-^{7}	Eb-^{7}	Bb-^{7}	F-^{7}	C^{maj7}
M2 ↑	E-^{7}	B-^{7}	Gb-^{7}	Db-^{7}	C^{maj7}
R	D-^{7}	Eb^{7}	Ab^{7}	Db^{7}	C^{maj7}
M3 ↓	Bb^{7}	Eb^{7}	Ab^{7}	G^{7}	C^{maj7}
m3 ↓	B-^{7}	Gb^{7}	Db-^{7}	Ab-^{7}	C^{maj7}
M2 ↑	E-^{7}	B-^{7}	F#-^{7}	C#-^{7}	C^{maj7}

Examples off ii^{7}-V^{7}-I^{maj7} Substitutions

Ex. 170

Original: (**Dm7**) (**G7**) (**Cmaj7**)

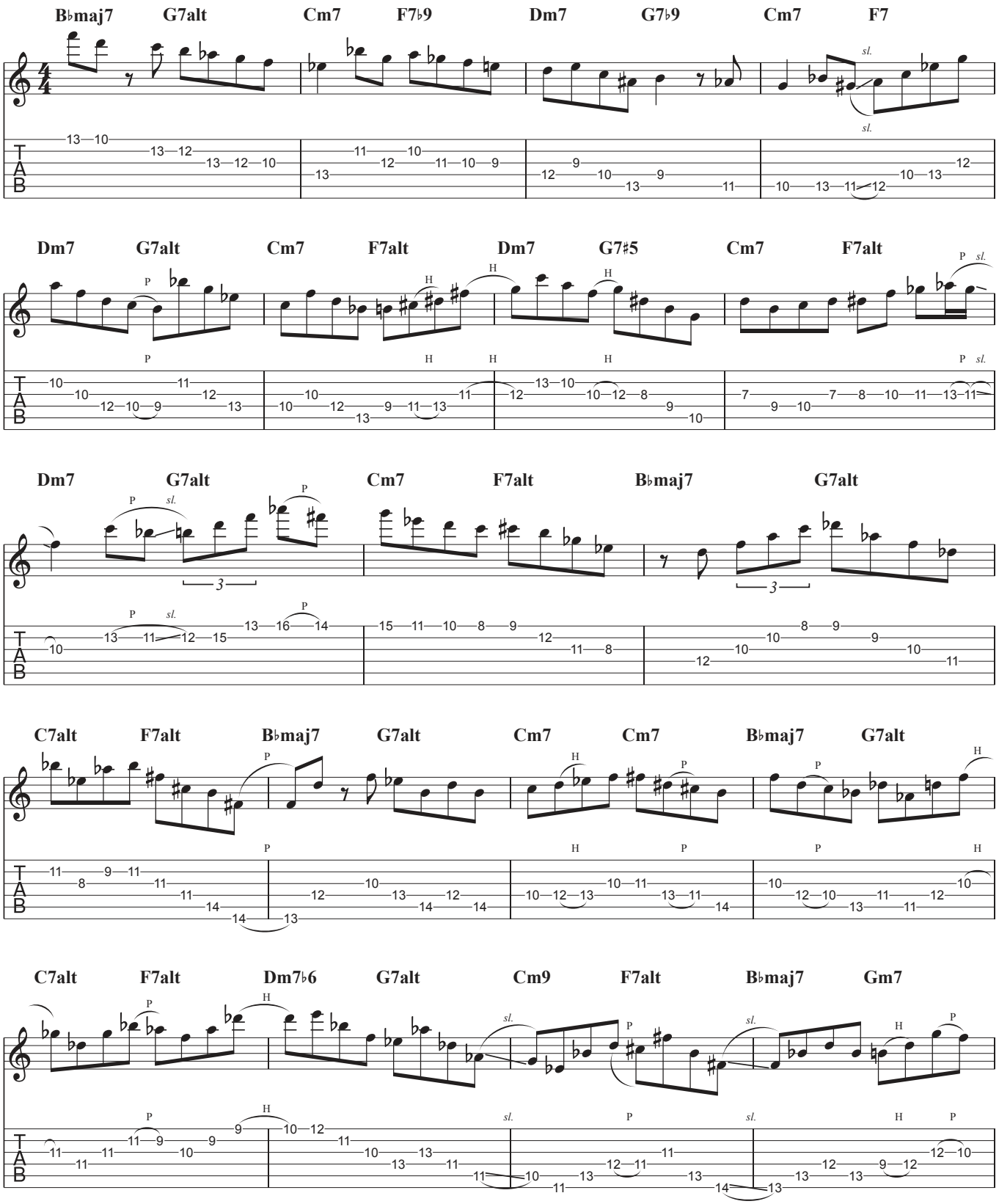
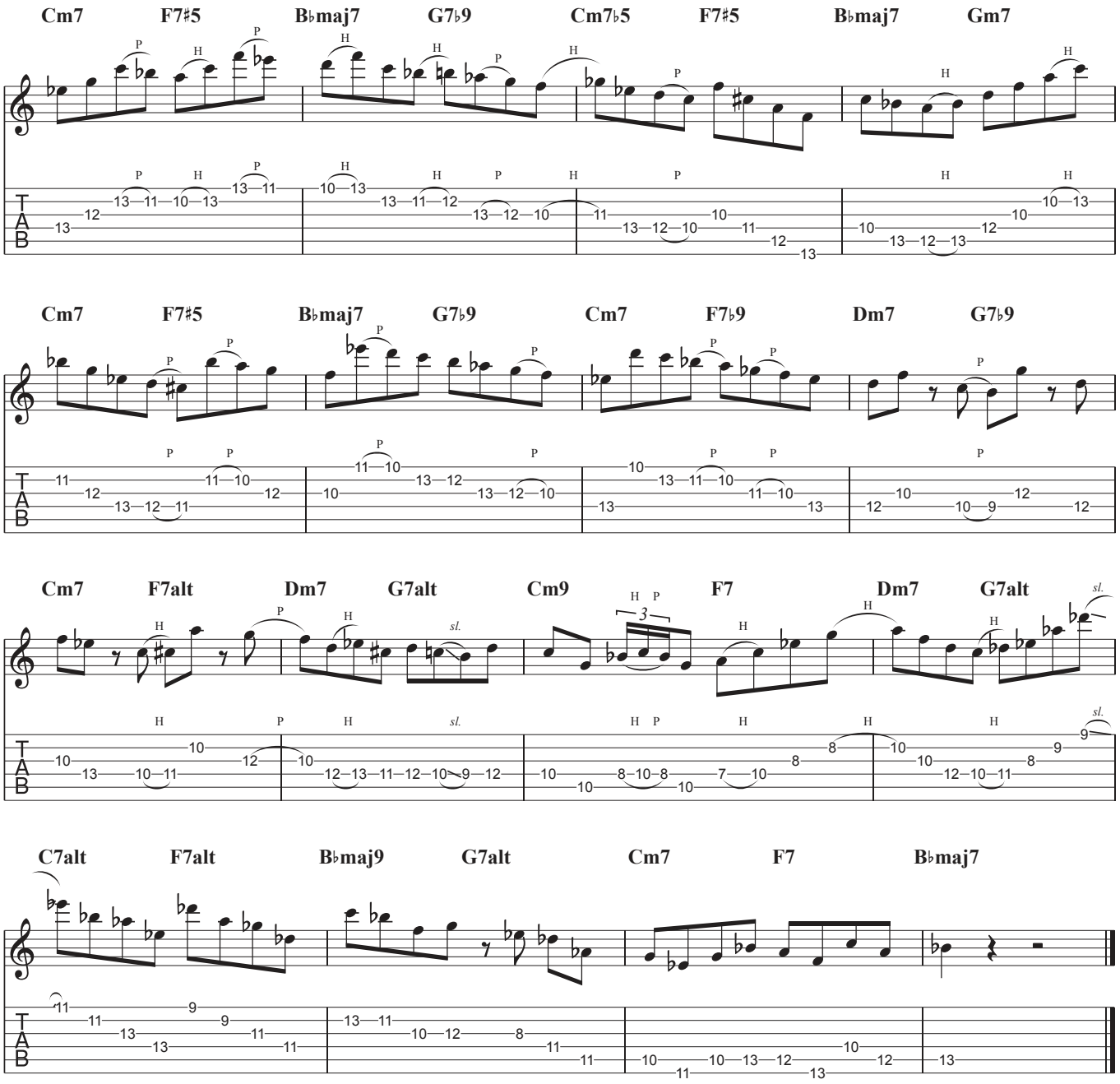


Turnarounds

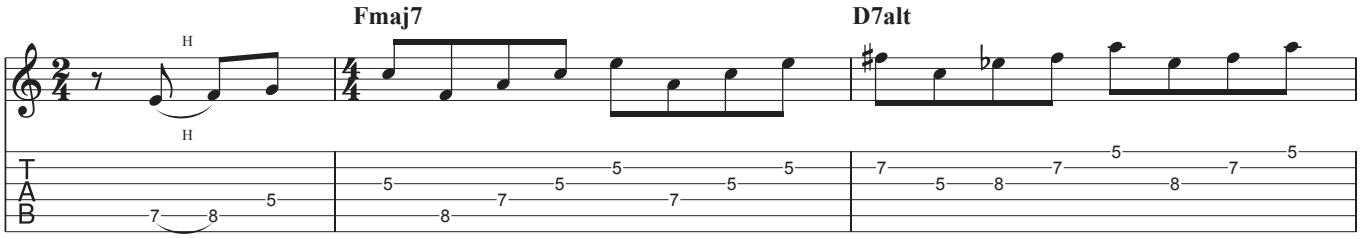
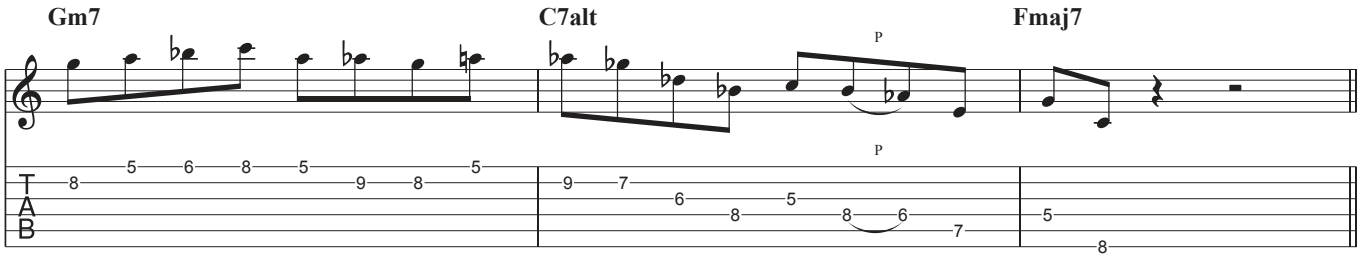
The term turnaround commonly refers to a four chord progression found in the last two measures of most AABA and blues form tunes.

Its purpose is to relieve monotony and help to aurally define the form of a tune by creating strong harmonic motion which ultimately resolves to the tonic chord. The frequent use of turnarounds in the jazz idiom requires the improviser to have a large variety of harmonic and melodic variations from which to draw upon.

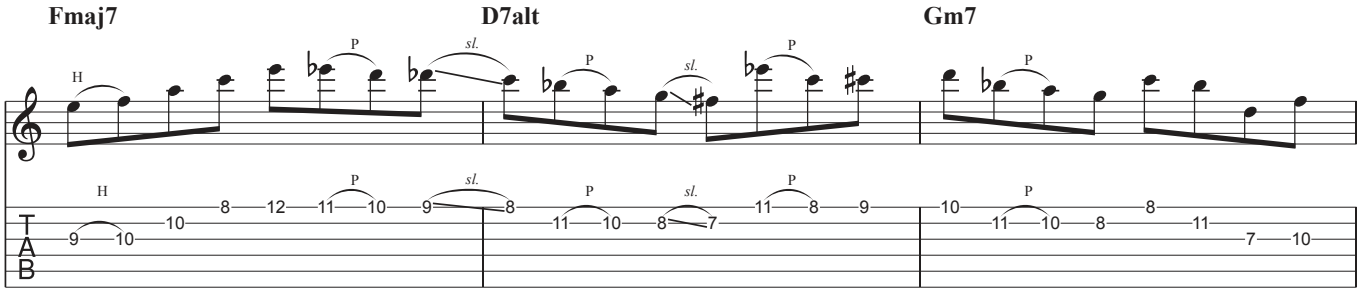
$I^{VI7}/_{ii7}$ V7 Exercise

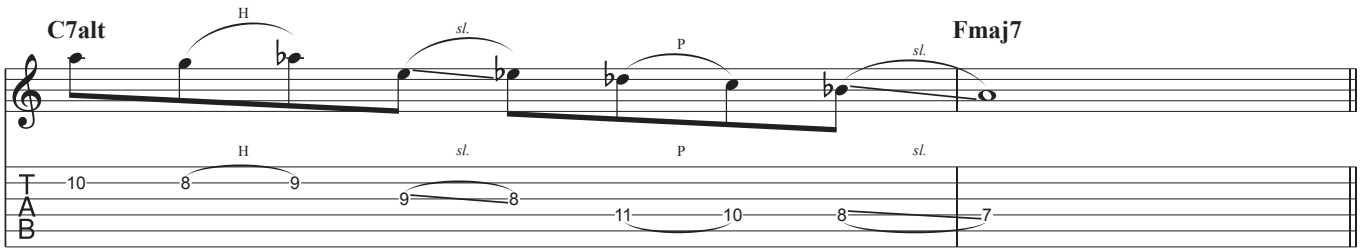
Ex. 171


Cm7 F7♯5 B♭maj7 G7♭9 Cm7♭5 F7♯5 B♭maj7 Gm7
Cm7 F7♯5 B♭maj7 G7♭9 Cm7 F7♭9 Dm7 G7♭9
Cm7 F7alt Dm7 G7alt Cm9 F7 Dm7 G7alt
C7alt F7alt B♭maj9 G7alt Cm7 F7 B♭maj7

Turnarounds Over Four Measures

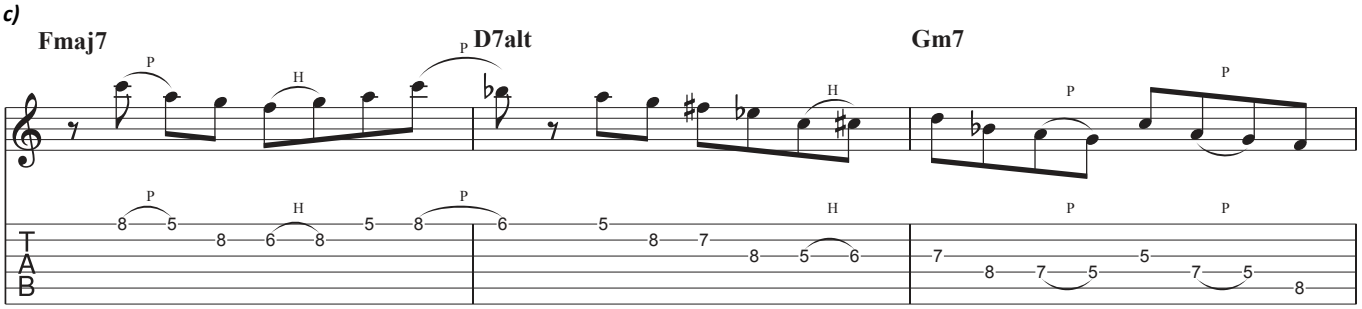
Ex. 172a



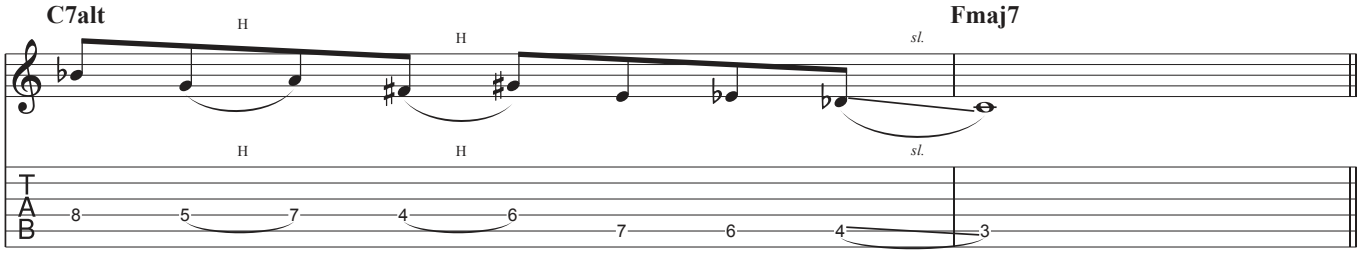
b)



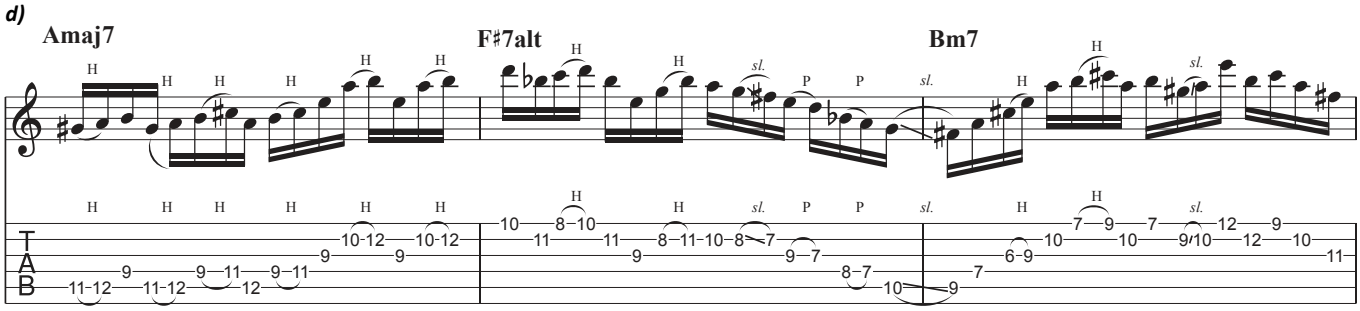


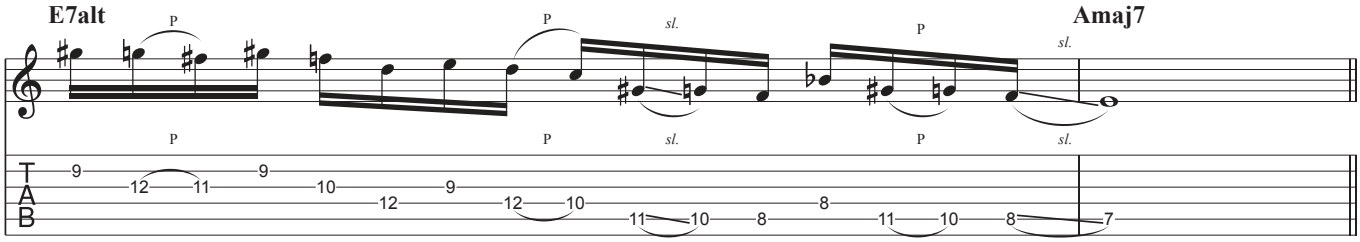
c)



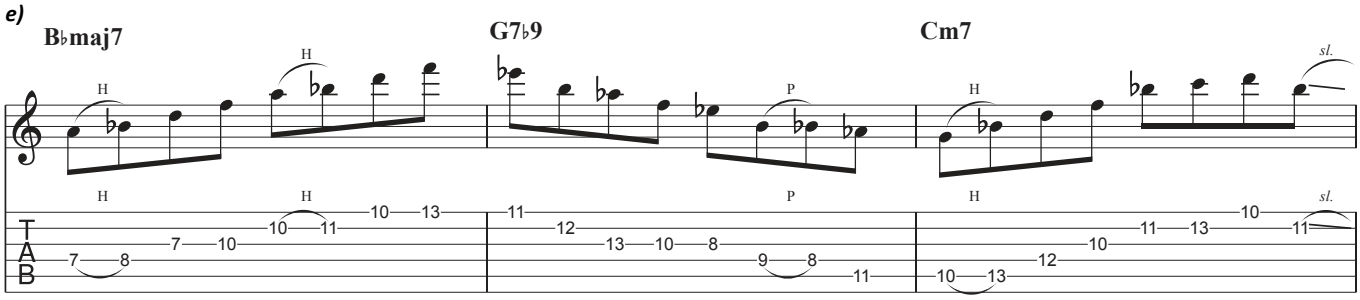

C7alt
Fmaj7

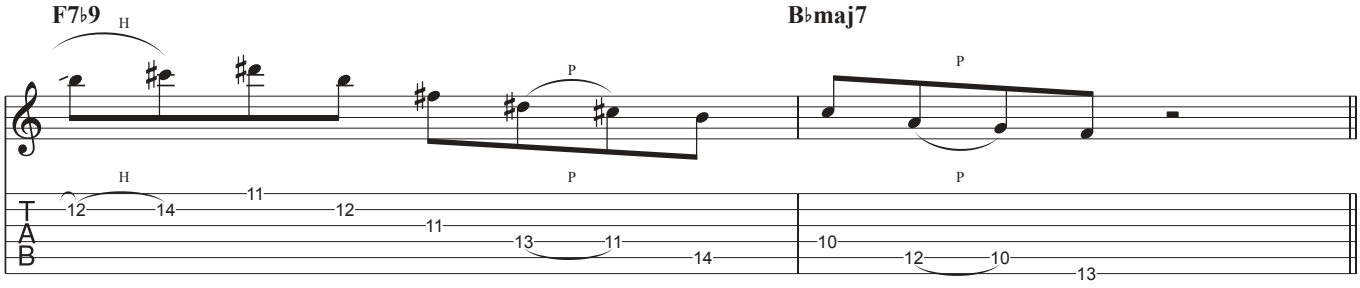
d)


Amaj7
F♯7alt
Bm7

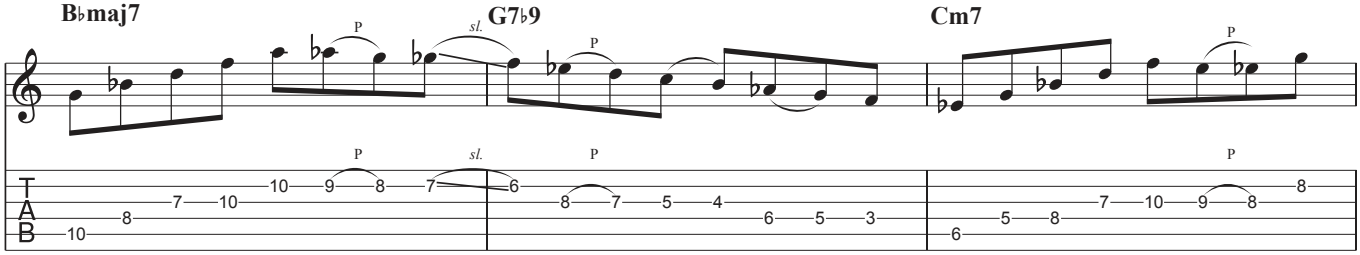
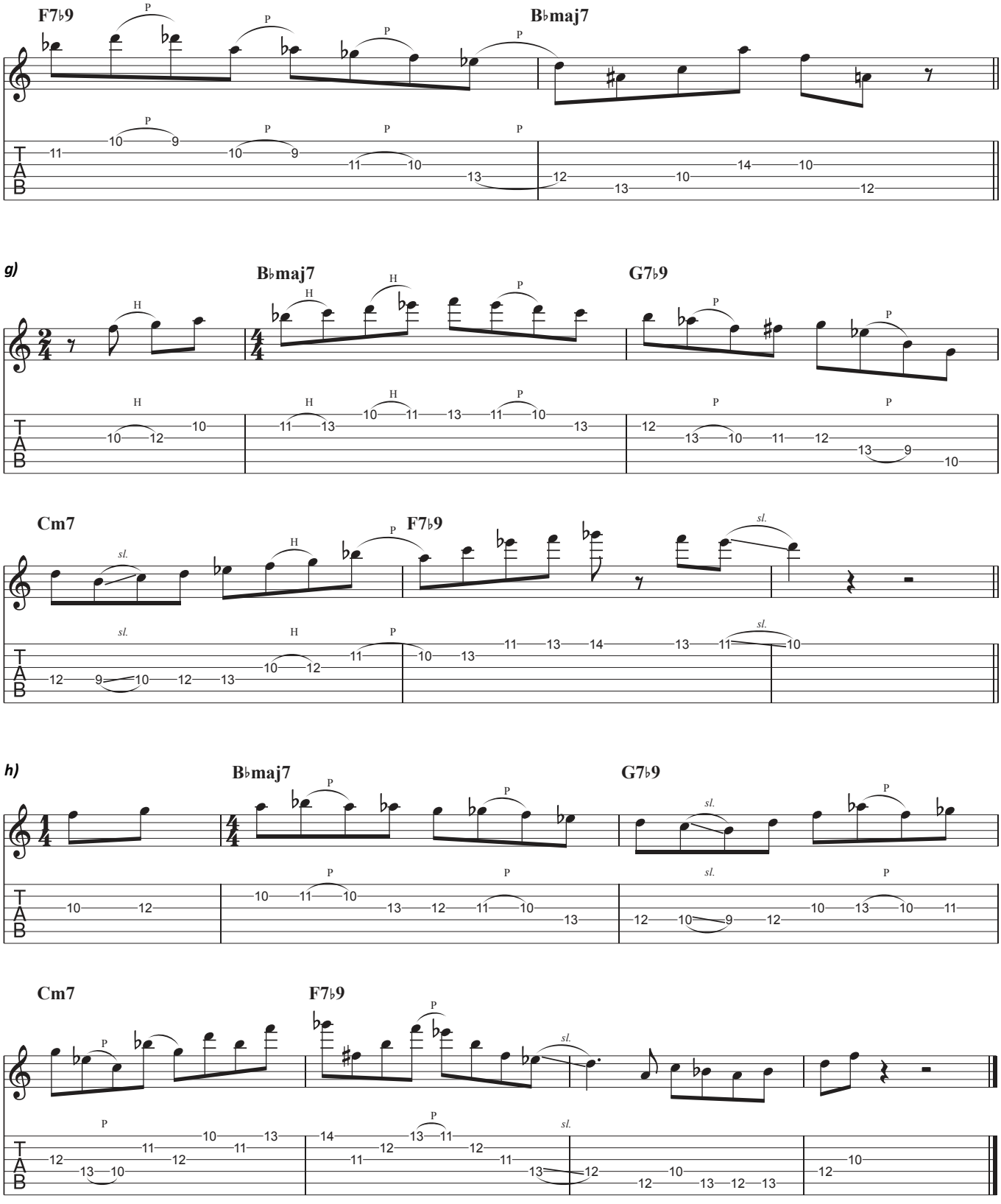

E7alt
Amaj7

e)


B♭maj7
G7♭9
Cm7


F7♭9
B♭maj7

f)

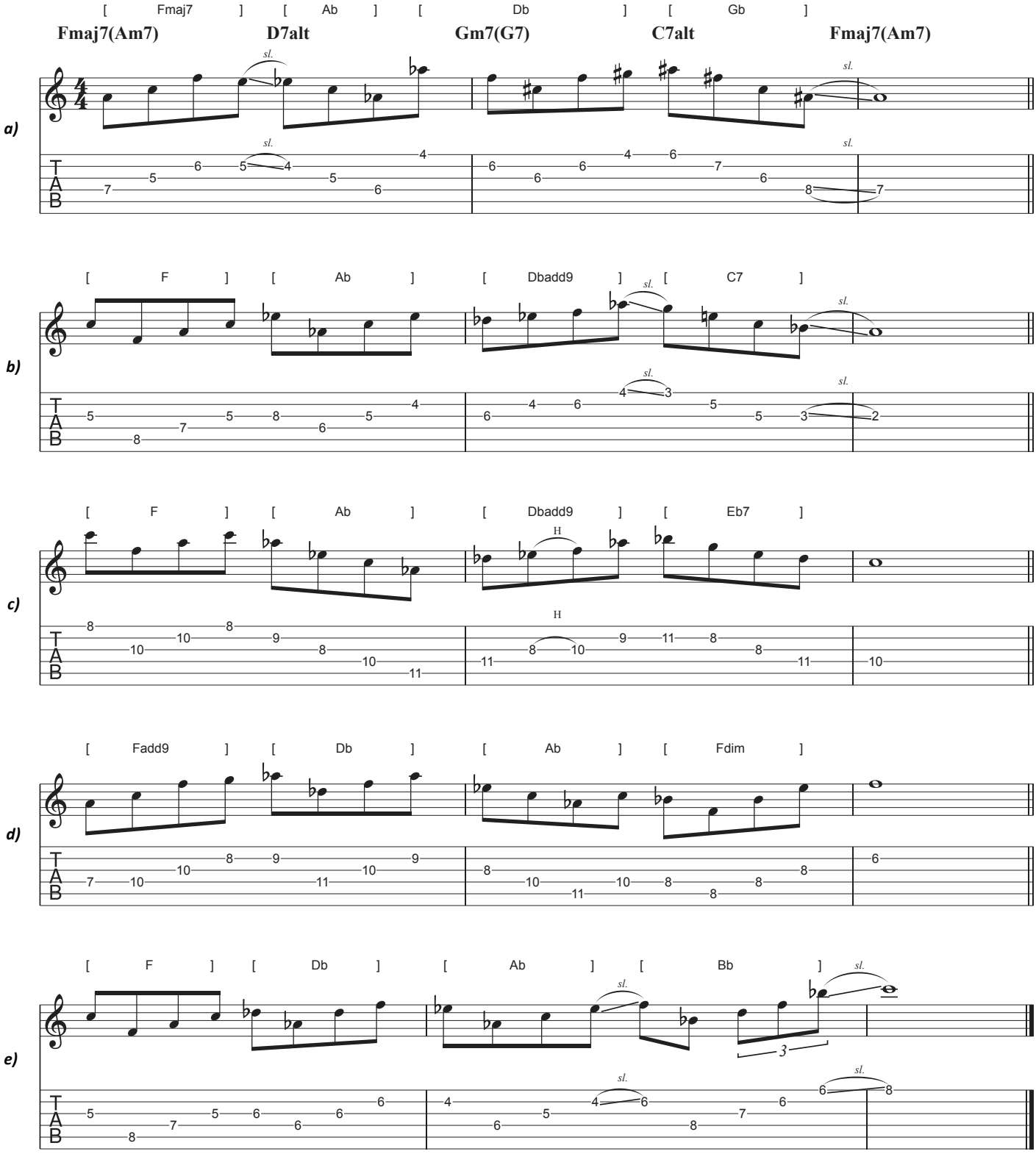

B♭maj7
G7♭9
Cm7

F7♭9
B♭maj7
g)
B♭maj7
G7♭9
Cm7
F7♭9
h)
B♭maj7
G7♭9
Cm7
F7♭9

Turnaround Substitutions

The following examples demonstrate the use of triadic substitutions over to turnaround progression.
Try to come up with your own lines based on these principles.

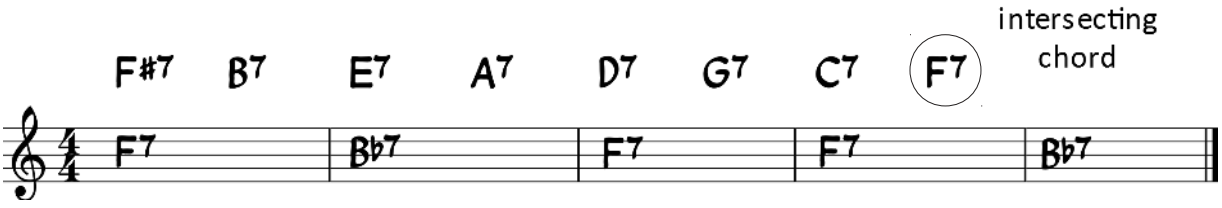
Additional substitutions can be found in the triadic superimposition chart later in this chapter.

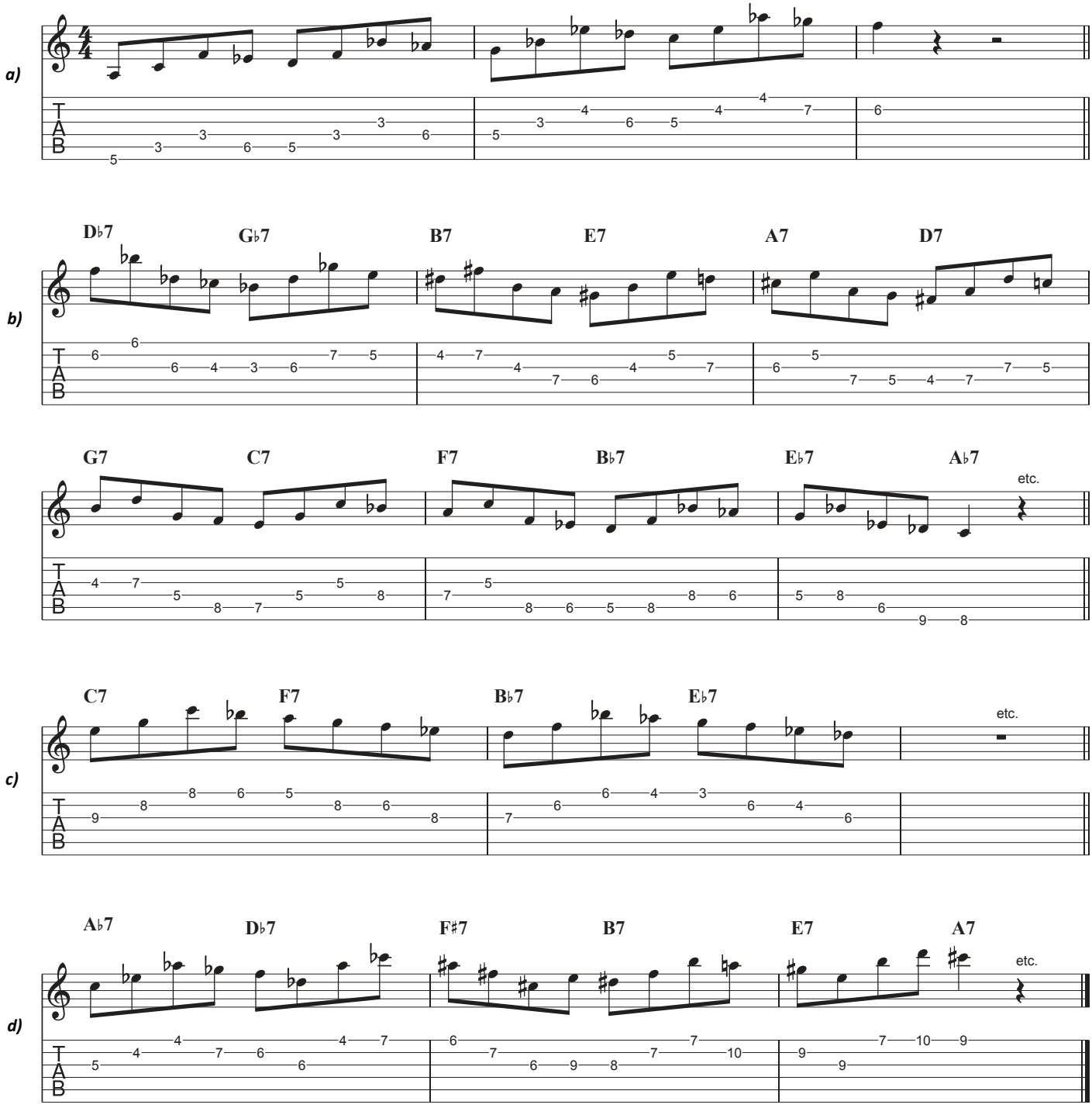
Ex. 173

The Triads are labeled for analysis

Cycle of Fifths

The term cycle of fifths or simply cycle describes the movement of dominant seventh chords with a descending fifth or ascending fourth root pattern. The cycle may be superimposed over any progression as long as the intersecting chord's root is a P5 or m2 above its resolution chord.

Ex. 174

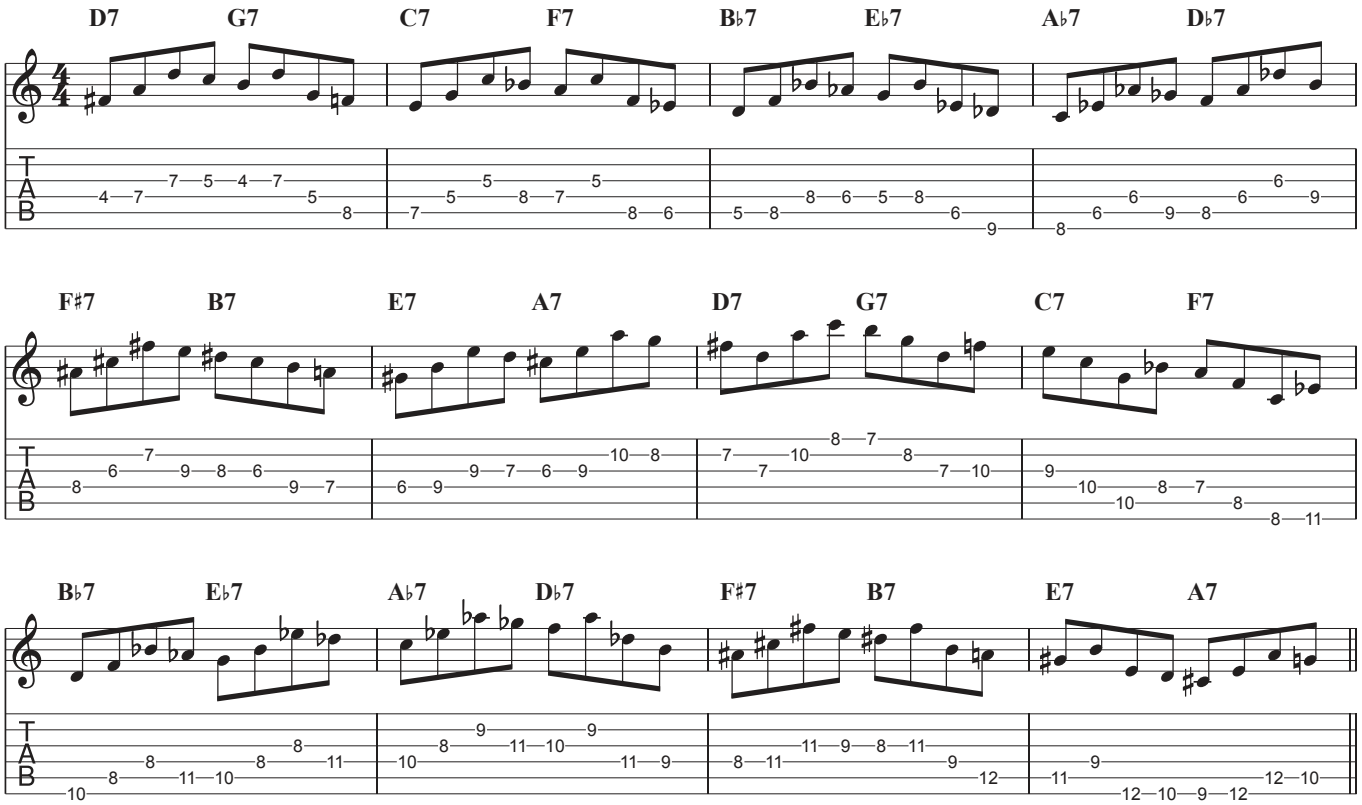


Ex. 175

a)
b)
D♭7
G♭7
B7
E7
A7
D7
G7
C7
F7
B♭7
E♭7
A♭7
etc.
c)
C7
F7
B♭7
E♭7
etc.
d)
A♭7
D♭7
F♯7
B7
E7
A7
etc.

Cycle of Fifths Exercise

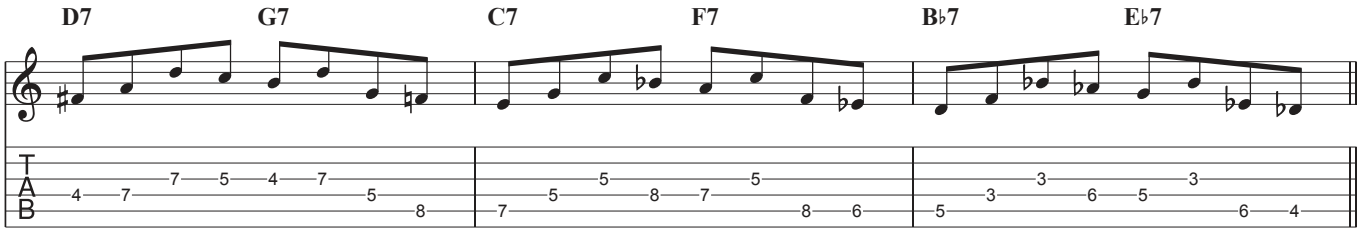
Here is an extended cycle of fifths exercise
based on the shapes from the previous page.

Ex. 176

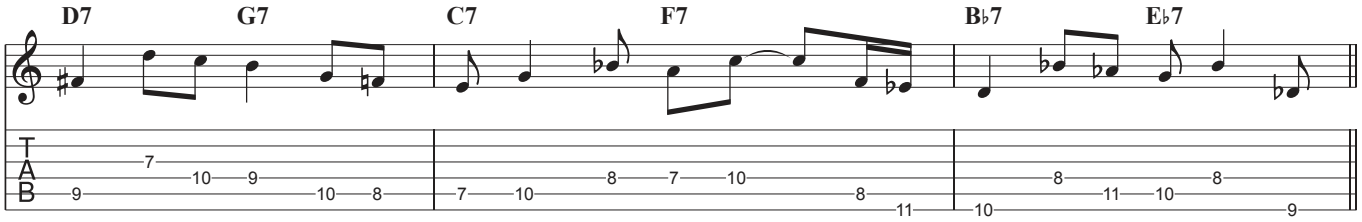


Ex. 177 Try to break up the rhythm in different ways.

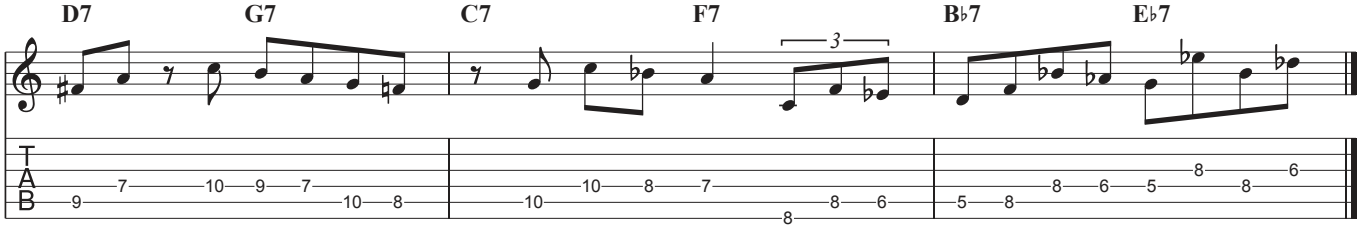
a)



b)



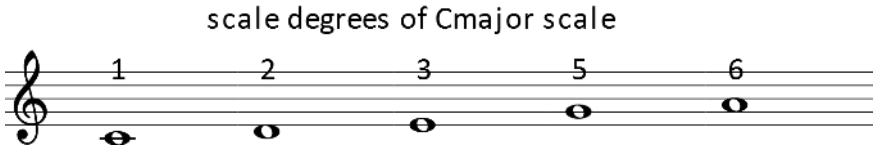
c)



Pentatonic Scales

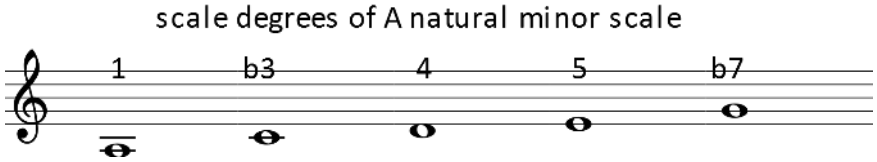
A pentatonic scale is simply a scale consisting of five notes. The two most common pentatonic scales are:

C major pentatonic:



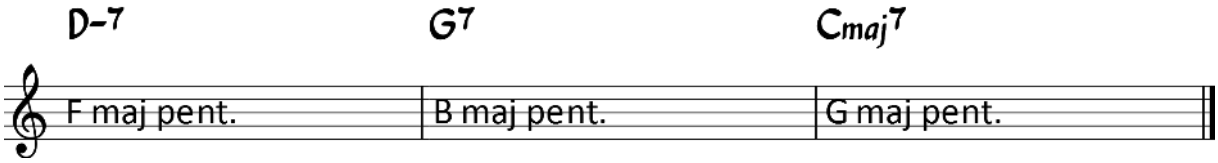
and

A minor pentatonic:

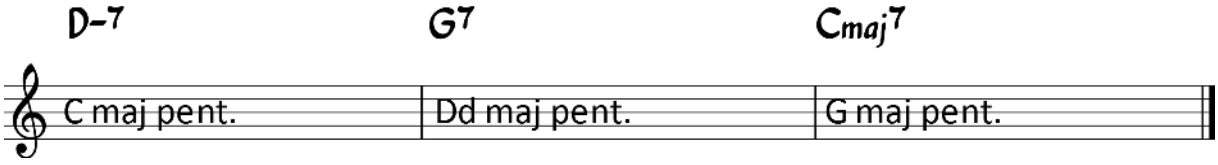


The following chart is of pentatonic scale usage over major^7, min^7, and Dom^7 chords. They should also be grouped into progressions and used over chord changes.

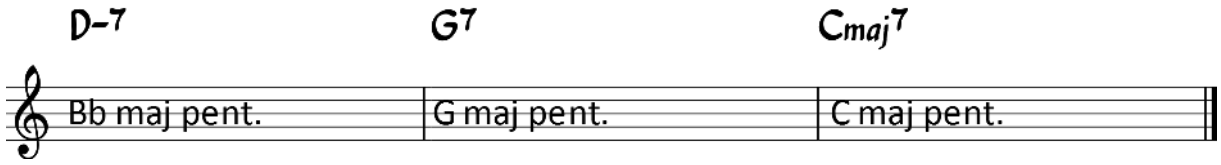
Ex. 178a



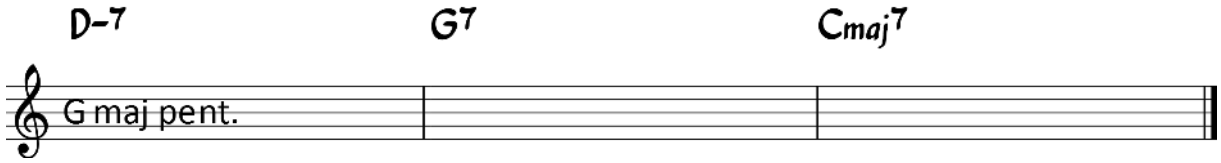
Ex. 178b



Ex. 178c

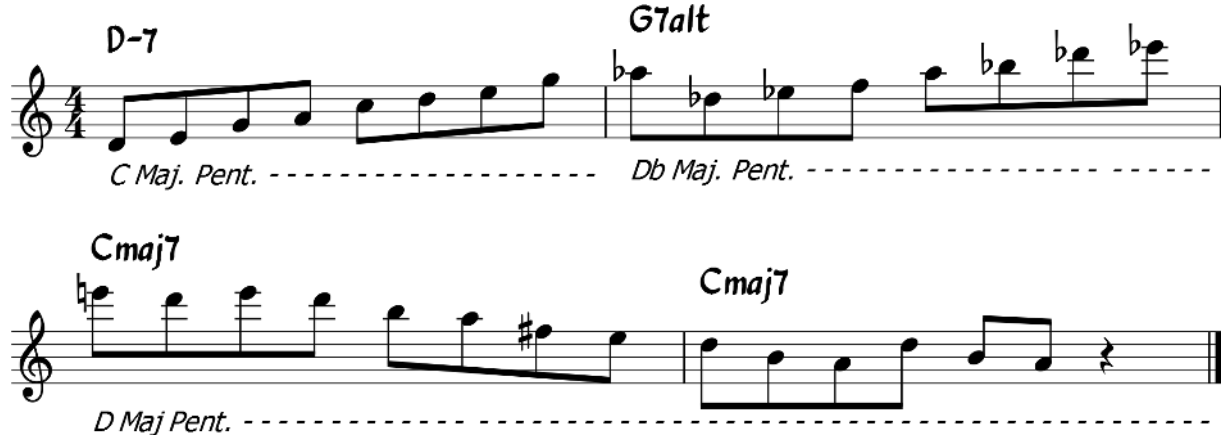


Ex. 178d



Pentatonic Usage over a ii V^7 I^{maj7} progression

Ex. 179



Pentatonic Scales Over Maj$^{(7)}$ Chords

	start major pentatonic on	chord tones					start minor pentatonic on	Maj / Min Ex. C^{maj7}
1)	Root of chord	1	2	3	5	6	6th of chord	C/A-
2)	5th of chord	5	6	7	9	3	3rd of chord	G/E-
3)	9th of chord	9	3	#11	6	7	7th of chord	D/B-

Pentatonic Scales Over Min$^{(7)}$ Chords

	start major pentatonic on	chord tones					start minor pentatonic on	Maj / Min Ex. C-^{7}
1)	3rd of chord	b3	4	5	b7	1	Root of chord	Eb/C-
2)	4th of chord	4	5	6	1	9	9th of chord	F/D-
3)	b6th of chord	b6	b7	1	b3	4	4th of chord	Ab/F-
4)	b7th of chord	b7	1	9	11	5	5th of chord	Bb/G-

Pentatonic Scales Over Dom^{7th} Chords

	start major pentatonic on	chord tones					start minor pentatonic on	Maj / Min Ex. C^{7}
1)	Root of chord	1	2	3	5	6	6th of chord	C/A-
2)	b3rd of chord	1	#9	11	5	b7	Root of chord	Eb/C-
3)	3rd of chord	b9	3	#11	#5	7	b9th of chord	E/Db-
4)	#4th of chord	#11	#5	b7	b9	#9	#9th of chord	F#/D#-
5)	b7th of chord	b7	1	9	11	5	5th of chord	Bb/G-
6)	6th of chord	13	7	b9	3	#11	#4th of chord	A/F#-

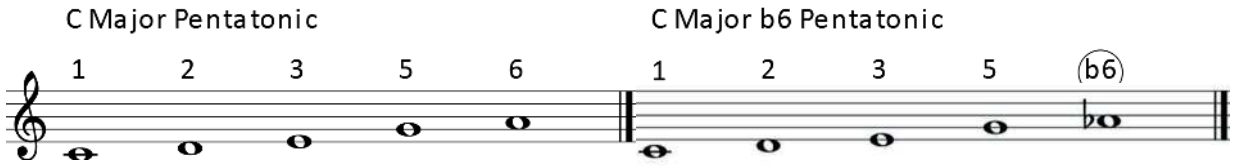
Or In Relation to One Pentatonic

C major Pentatonic		C	D	E	G	A	A minor pentatonic
Maj^{7}	C^{maj7}	1	2	3	5	6	
	F^{maj7}	5	6	7	9	3	
	Bb^{maj7}	9	3	#11	6	7	
Min^{7}	$A-^{7}$	b3	11	5	b7	1	
	$D-^{7}$	b7	1	9	11	5	
	$E-^{7b6}$	b6	b7	1	3	11	
	$G-^{7}$	11	5	6	1	9	
Dom^{7}	C^{7}	1	9	3	5	13	
	A^{7}	#9	11	5	b7	1	
	D^{7}	b7	1	9	11	5	
	$F\#^{7\ alt}$	#11	#5	b7	b9	#9	
	$Ab^{7\ alt}$	3	#11	#5	(7)	b9	
	$Eb^{7\ alt}$	13	7	b9	3	#11	

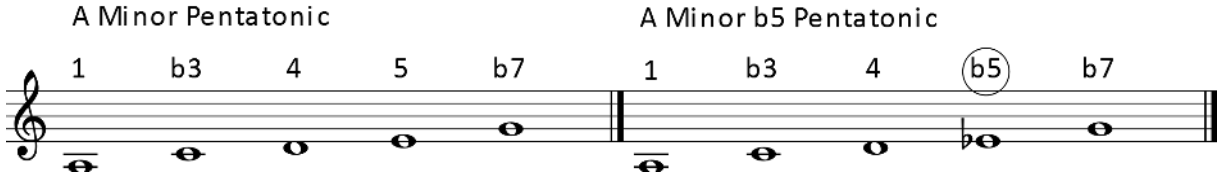
Melodic Minor Pentatonics

By making a simple adjustment to the Major and Minor Pentatonics, we can create pentatonics from the Melodic Minor.

Ex. 180a

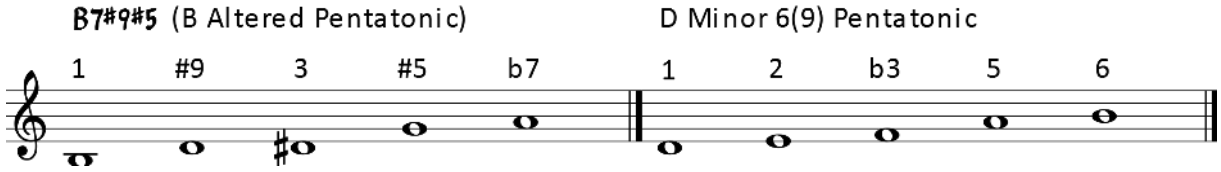


Ex. 180b

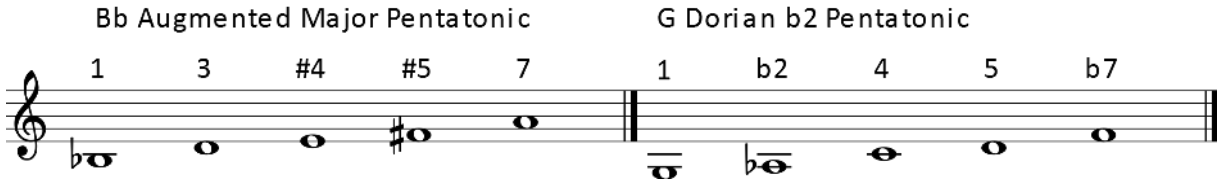


Other Melodic Minor Pentatonics can also be created with different combinations of 5 note patterns.

Ex. 181a



Ex. 181b

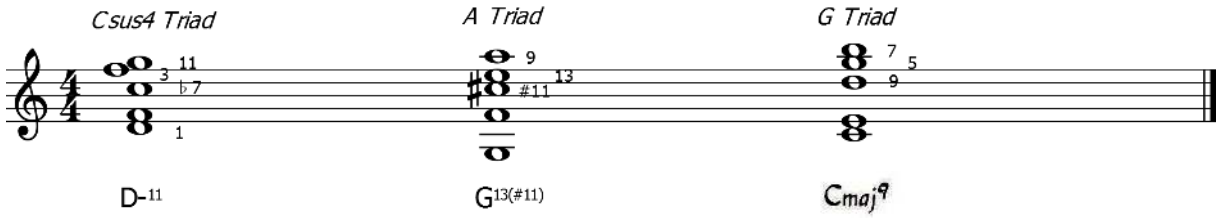


Triadic Superimposition

Triadic Superimposition is the melodic or harmonic stacking of triads over existing harmonies.

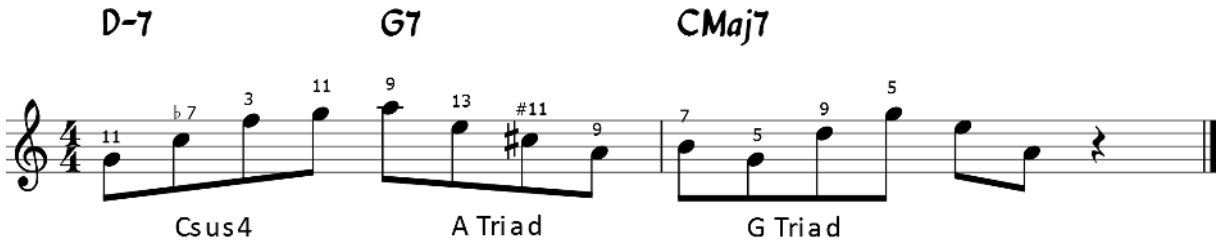
Ex. 182a

Harmonic superimposition.



Ex. 182b

Melodic superimposition.



The addition of lower chord members (3rds, 5ths, 7ths) in melodic and harmonic usage will keep the superimpositions from sounding detached from the harmony (though this may be desirable in a more modern context).

The triadic superimposition chart is broken down into the basic chord types maj^7, min^7 and dom^7, all with C as the tonic. I would suggest transposing these to different keys in order to create harmonic and melodic lines over common chord progressions.

Make a list of a few superimpositions over ii V I progressions and then try to improvise over one or more of the chords using triads.

Ex. 182c

	$F\text{-}^{7}$	Bb^{7}	Eb^{maj7}
①	Ab triad	E triad	$Csus^{4}$
②	F-	$Bsus^{4}$	$Dsus^{4}$
③	$Absus^{4}$ (aeolian)	G Triad	F triad

The Ab^{sus4} triad used on the Fm^{7} chord produces an aeolian sound where a dorian sound is normally heard. This type of interchange is called modal mixture (which will be discussed later in this chapter).

Check the scale derivation column to see if the chord function matches with the scale or not.

Triadic Superimposition Chart (Maj^7)

Chord	Superimposed Triad	Location from the root of the chord	Note functions of superimposition			Modal sound (scale derivation)
C^{maj7}	C	Root	1	3	5	Ionian, Lydian
	C+	Root	1	3	#5	Lydian Aug. (Ionian Aug.)
	C^{sus4}	Root	1	sus4	5	Ionian
	D	M2↑	9	#11	13	Lydian (Lydian Aug.)
	D^{o}	M2↑	9	#11	#5	Lydian Aug.
	D^{sus4}	M2↑	9	5	13	Ionian, Lydian
	E	M3↑	3	#5	7	Lydian Aug.
	E-	M3↑	3	5	7	Ionian, Lydian
	E^{+}	M3↑	3	#5	1	Lydian Aug. (Ionian Aug.)
	E^{sus4}	M3↑	3	13	7	Ionian, Lydian
	F	P4↑	sus4	13	1	Ionian
	F-	P4↑	sus4	#5	1	Ionian Aug.
	F^{o}	P4↑	sus4	#5	7	Ionian Aug.
	F#-	+4↕	#11	13	b9	
	$F\#^{o}$	+4↕	#11	13	1	Lydian (Aug.)
	$F\#^{sus4}$	+4↕	#11	7	b9	
	G	P4↓ P5↑	5	7	9	Ionian, Lydian
	G^{sus4}	P4↓ P5↑	5	1	9	Ionian, Lydian
	G#-	M3↓ m6↑	#5	7	#9	
	$G\#^{o}$	M3↓ m6↑	#5	7	9	Lydian Aug.
	A-	m3↓ M6↑	13	1	3	Ionian, Lydian
	A^{sus4}	m3↓ M6↑	6	9	3	Ionian, Lydian
	B	m2↓ M7↑	7	#9	#11	Lydian #9
	B-	m2↓ M7↑	7	9	#11	Lydian
	B^{o}	m2↓ M7↑	7	9	sus4	Ionian
	B^{loc}	m2↓ M7↑	7	3	sus4	Ionian
	B^{sus4}	m2↓ M7↑	7	3	#11	Lydian (Aug.)

Triadic Superimposition Chart ($Min^{7\,(b5)}$)

Chord	Superimposed Triad	Location from the root of the chord	Note functions of superimposition			Modal sound (scale derivation)
$C-^{7(b5)}$	C-	Root	1	b3	5	Minor (all)
	C^{sus4}	Root	1	4	#5	Minor (all except Locrian type)
	Db+	m2↑	b9	11	13	Dorian b2 (Dom^7 function)
	Db^{lyd}	m2↑	b9	5	b6	Phrygian
	D-	M2↑	9	11	13	Dorian
	D^{+}	M2↑	9	#11	b7	Dorian #4 (Dom^7 also)
	D^{o}	M2↑	9	11	b6	Aeolian
	D^{sus4}	M2↑	9	5	13	Dorian
	D^{loc}	M2↑	9	5	b6	Aeolian
	Eb	m3↑	b3	5	b7	Minor (all without b5)
	Eb-	m3↑	b3	b5	b7	Locrian (all), Dorian #4
	Eb^{o}	m3↑	b3	b5	13	Locrian nat.6, Dorian #4
	Eb^{+}	m3↑	b3	5	7	Melodic + Harmonic Minor
	Eb^{sus4}	m3↑	b3	b6	b7	Aeolian
	Eb^{lyd}	m3↑	b3	13	b7	Dorian
	F	P4↑	11	13	1	Dorian
	F-	P4↑	11	b6	1	Aeolian, Phrygian
	F^{+}	P4↑	11	13	b9	Dorian b2
	F^{o}	P4↑	11	b6	7	Harmonic Minor
	F^{sus4}	P4↑	11	b7	1	Minor (all without 7)
	$F\#^{o}$	+4↕	#4	13	1	Dorian #4, Locrian nat.6
	$F\#^{+}$	+4↕	b5	b7	9	Dorian #4, Locrian nat.2
	G	P4↓ P5↑	5	7	9	Melodic + Harmonic Minor
	G-	P4↓ P5↑	5	b7	9	Dorian, Aeolian
	G^{o}	P4↓ P5↑	5	b7	b9	Phrygian
	G^{+}	P4↓ P5↑	5	7	b3	Melodic + Harmonic Minor
	G^{sus4}	P4↓ P5↑	5	1	9	Dorian, Aeolian
	An	M3↓ m6↑	b6	1	b3	Aeolian, (Phrygian)
	Ab-	M3↓ m6↑	b6	7	b3	Harmonic Minor
	Ab^{sus4}	M3↓ m6↑	b6	b9	b3	Phrygian, Locrian
	Ab^{lyd}	M3↓ m6↑	b6	9	b3	Aeolian

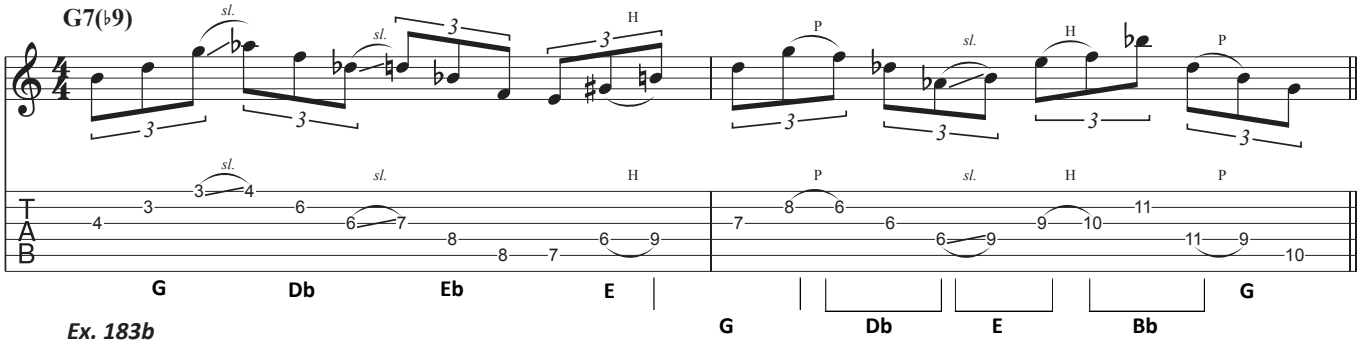
Chord	Superimposed Triad	Location from the root of the chord	Note functions of superimposition			Modal sound (scale derivation)
	A^{o}	m3↓ M6↑	13	1	b3	Dorian
	A^{+}	m3↓ M6↑	13	b9	11	Dorian b2
	A^{loc}	m3↓ M6↑	13	9	b3	Dorian (#4)
	Bb	M2↓ m7↑	b7	9	11	Dorian, Aeolian
	Bb^{sus4}	M2↓ m7↑	b7	b3	11	Dorian, Phrygian, Aeolian

Triadic Superimposition Chart (Dom^7)

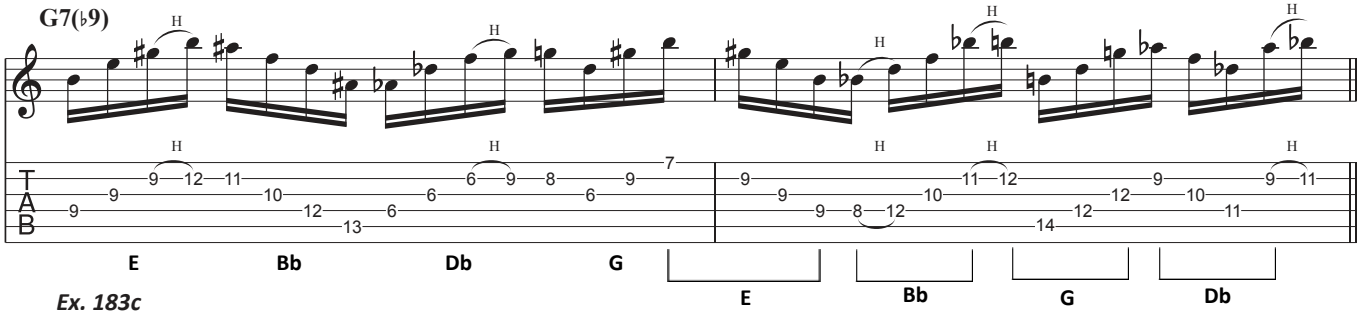
Chord	Superimposed Triad	Location from the root of the chord	Note functions of superimposition			Modal Sound (scale derivation)
C^7	C-	Root	1	3	5	Mixolydian
	C-	Root	1	#9	5	Dom.Dim.
	C^o	Root	1	#9	#11	Dom.Dim., Alt.Dom.
	C^+	Root	1	3	#5	Whole Tone, Alt.Dom.
	C^{sus4}	Root	1	4	5	Mixolydian
	C^{lyd}	Root	1	#4	5	Mixolydian #11, Dom.Dim.
	Db	m2↑	b9	11	#5	Phrygian (major)
	Db-	m2↑	b9	3	#5	Alt.Dom., Phrygian Major
	Db^o	m2↑	b9	3	5	Dom.Dim.
	Db^+	m2↑	b9	11	6	Dorian b2
	Db^{sus4}	m2↑	b9	#11	#5	Alt.Dom.
	Db^{lyd}	m2↑	b9	5	#5	Phrygian Major
	Db^{loc}	m2↑	b9	#11	5	Dom.Dim.
	D	M2↑	9	#11	6	Mixolydian #11
	D-	M2↑	9	11	6	Mixolydian
	D^o	M2↑	9	11	#5	Mixolydian b6
	D^+	M2↑	9	#11	b7	Mixolydian #11, Whole Tone
	D^{sus4}	M2↑	9	5	13	Mixolydian
	Eb	m3↑	#9	5	b7	Dom.Dim.
	Eb-	m3↑	#9	#11	b7	Dom.Dim., Alt.Dom.
	Eb^o	m3↑	#9	#11	13	Dom.Dim.
	Eb^{sus4}	m3↑	#9	#5	b7	Alt.Dom.
	E^o	M3↑	3	5	b7	Mixolydian (#11)
	E^+	M3↑	3	#5	1	Whole Tone, Alt.Dom.
	E^{loc}	M3↑	3	13	b7	Mixolydian (#11)
	F	P4↑	11	6	1	Mixolydian
	F-	P4↑	11	#5	1	Phrygian major
	F^+	P4↑	11	6	b9	Dorian b2 (Dom^7 function)
	F^{sus4}	P4↑	11	b7	1	Mixolydian
	F#	+4↕	#11	b7	b9	Alt.Dom., Dom.Dim.
	$F\#^o$	+4↕	#11	6	1	Mixolydian #11

Chord	Superimposed Triad	Location from the root of the chord	Note functions of superimposition			Modal sound (scale derivation)
	F#$^{+}$	+4↕	#11	b7	9	Mixolydian #11
	F#^{sus4}	+4↕	#11	(7)	b9	Upper Extension
	G-	P4↓ P5↑	5	b7	9	Mixolydian
	G^{o}	P4↓ P5↑	5	b7	b9	Dom.Dim., Phrygian Major
	Ab	M3↓ m6↑	#5	1	#9	Alt.Dom.
	Ab^{+}	M3↓ m6↑	#5	1	3	Whole Tone, Alt.Dom.
	Ab^{sus4}	M3↓ m6↑	#5	b9	#9	Alt.Dom.
	A	m3↓ M6↑	13	b9	3	Dom.Dim., Dorian b2
	A-	m3↓ M6↑	13	1	3	Mixolydian
	A^{o}	m3↓ M6↑	13	1	#9	Dom.Dim.
	A^{+}	m3↓ M6↑	13	b9	11	Dorian b2 (Dom^{7} function)
	A^{sus4}	m3↓ M6↑	13	9	3	Mixolydian
	A^{lyd}	m3↓ M6↑	13	#9	3	Dom.Dim.
	Bb	M2↓ m7↑	b7	9	11	Mixolydian
	Bb-	M2↓ m7↑	b7	b9	11	Dorian b2, Phrygian Major
	Bb^{o}	M2↓ m7↑	b7	b9	3	Dom.Dim., Alt.Dom.
	Bb+	M2↓ m7↑	b7	b9	#11	Dom.Dim., Alt.Dom.
	Bb^{lyd}	M2↓ m7↑	b7	3	11	Mixolydian
	Bb^{loc}	M2↓ m7↑	b7	#9	3	Dom.Dim., Alt.Dom.

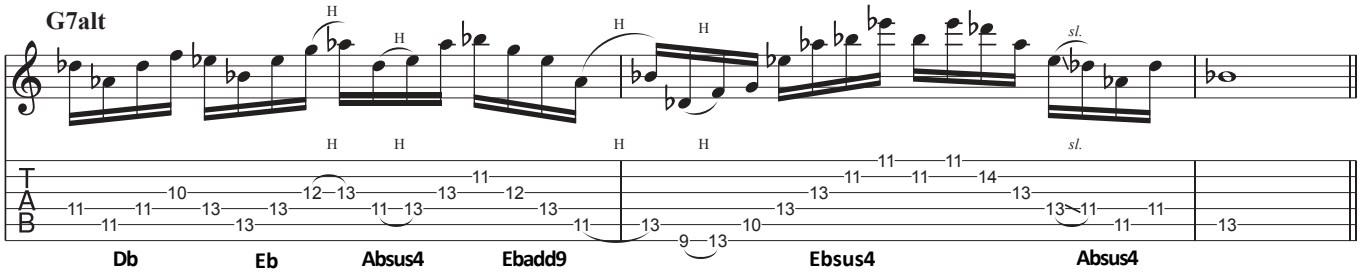
Ex. 183a



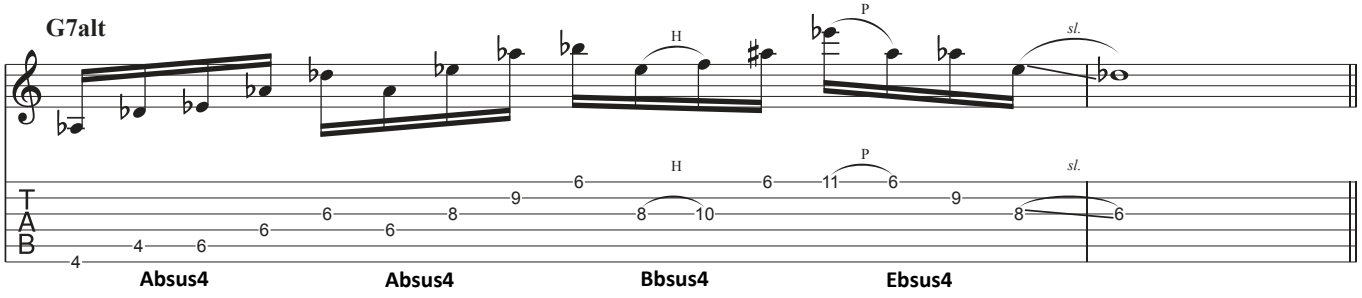
Ex. 183b



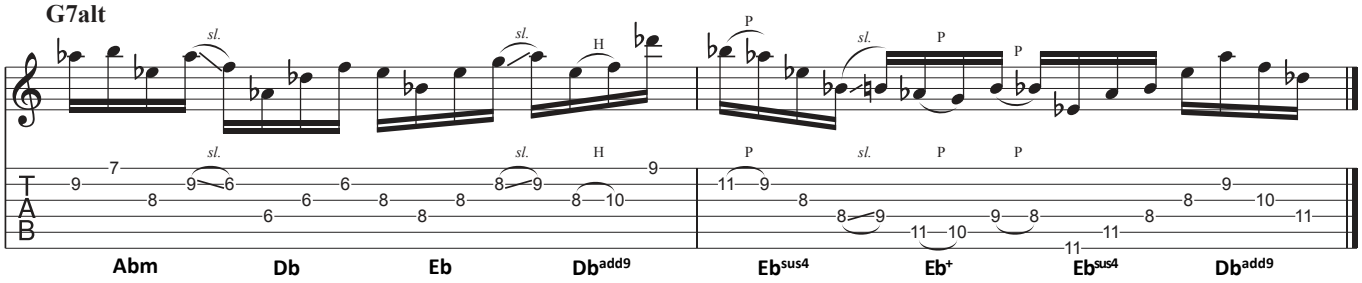
Ex. 183c



Ex. 183d



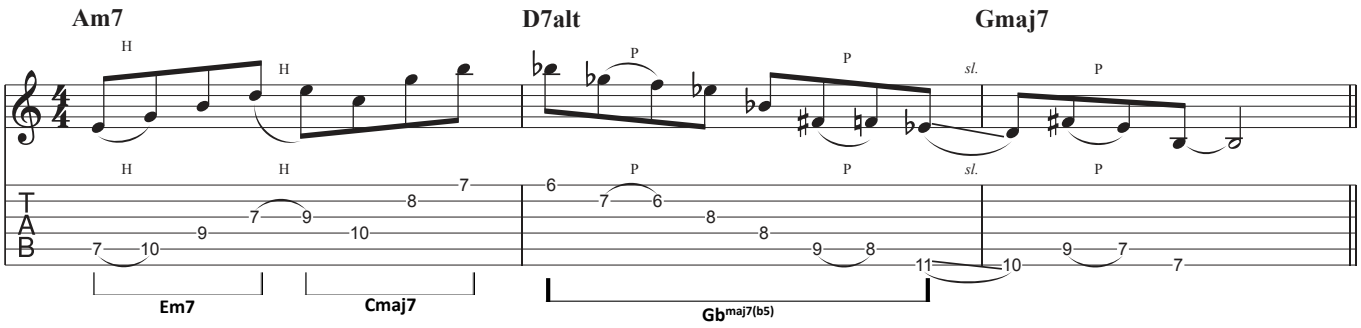
Ex. 183e This one uses some added note arpeggios.



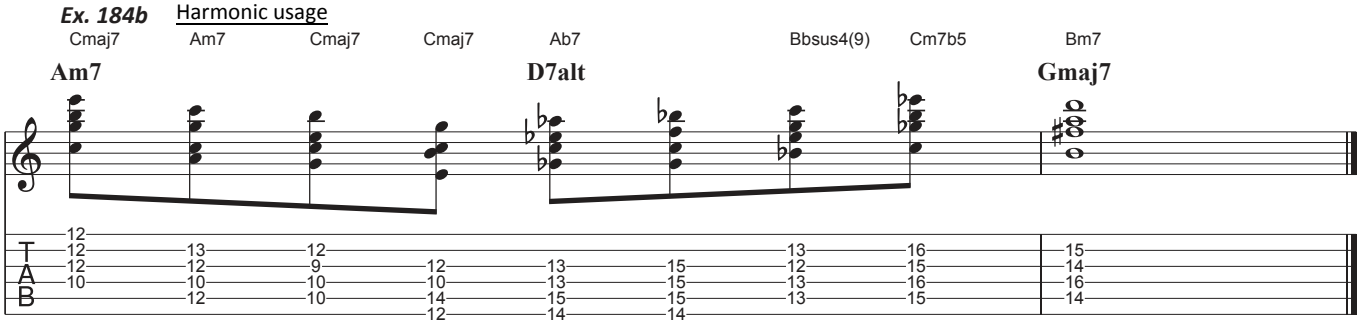
Seventh Chord Superimposition

Seventh chord superimposition works the same way as triadic superimposition.
We are just stacking a seventh chord instead of triads.

Ex. 184a Melodic usage



Ex. 184b Harmonic usage



Once again, try to group these together in progressions to facilitate the contextual integration of this concept.

Seventh Chord Superimposition Chart (Maj^7)

Chord	Superimposed seventh chord	Location from the root of the chord	Note functions of superimposition				Modal sound (scale derivation)
C^{maj7}	D^6	M2↑	9	#11	13	7	Lydian
	D^7	M2↑	9	#11	13	1	Lydian
	$D^{7/sus4}$	M2↑	9	5	13	1	Ionian
	$E\text{-}^7$	M3↑	3	5	7	9	Ionian
	$E^{7/\#5}$	M3↑	3	#5	1	9	Lydian
	$E^{7/sus4}$	M3↑	3	13	7	9	Ionian
	$F\#\text{-}^{7b5}$	+4↕	#11	13	1	3	Lydian
	G^6	P4↓ P5↑	5	7	9	3	Ionian
	G^{maj7}	P4↓ P5↑	5	7	9	#11	Lydian
	$G^{maj7/sus4}$	P4↓ P5↑	5	1	9	#11	Lydian
	$G\#\text{-}^{7b5}$	M3↓ m6↑	#5	7	9	#11	Lydian Augmented
	$A\text{-}^6$	m3↓ M6↑	13	1	3	#11	Lydian
	$A\text{-}^7$	m3↓ M6↑	13	1	3	5	Ionian
	$A\text{-}^{maj7}$	m3↓ M6↑	13	1	3	#5	Lydian Augmented
	$A^{maj7/sus4}$	m3↓ M6↑	13	9	3	#5	Lydian Augmented
	$A^{7/sus4}$	m3↓ M6↑	13	9	3	5	Ionian
	$B\text{-}^6$	m2↓ M7↑	7	9	#11	#5	Lydian Augmented
	$B\text{-}^7$	m2↓ M7↑	7	9	#11	13	Lydian
	$B^{7/sus4}$	m2↓ M7↑	7	3	#11	13	Lydian

Seventh Chord Superimposition Chart (Min$^{7\,(b5)(♮7)}$)

Chord	Superimposed seventh chord	Location from the root of the chord	Note functions of superimposition				Modal sound (scale derivation)
C-$^{7(b5)(♮7)}$	C-^{6}	Root	1	b3	5	6	Dorian Types
	C-^{maj7}	Root	1	b3	5	7	Melodic + Harmonic Minor
	C-^{7b5}	Root	1	b3	b5	b7	Locrian Types
	C$^{7/sus4}$	Root	1	4	5	b7	Dorian, Aeolian, Phrygian
	C-$^{maj7/\#5}$	Root	1	b3	#5	7	Harmonic Minor
	C-$^{7/\#5}$	Root	1	b3	#5	b7	Aeolian, Phrygian
	Db^{maj7}	m2↑	b2	11	b6	1	Phrygian
	Db$^{maj7/b5}$	m2↑	b2	11	5	1	Phrygian
	Db$^{maj7/\#5}$	m2↑	b2	11	6	1	Dorian b2
	Db$^{maj7/sus4}$	m2↑	b2	b5	b6	1	Locrian
	Db$^{lyd/maj7}$	m2↑	b2	5	b6	1	Phrygian
	D-^{7}	M2↑	9	11	6	1	Dorian
	D-$^{7/b5}$	M2↑	9	11	b6	1	Aeolian
	D$^{7/\#5}$	M2↑	9	b5	b7	1	Locrian
	D$^{7/b5}$	M2↑	9	b5	b6	1	Locrian nat.2
	Eb^{MA6}	m3↑	b3	5	b7	1	Dorian, Aeolian, Phrygian
	Eb^{maj7}	m3↑	b3	5	b7	9	Dorian, Aeolian
	Eb^{7}	m3↑	b3	5	b7	b2	Phrygian
	Eb-^{7}	m3↑	b3	b5	b7	b2	Locrian
	Eb$^{maj7/b5}$	m3↑	b3	5	6	9	Dorian
	Eb$^{maj7/\#5}$	m3↑	b3	5	7	9	Melodic + Harmonic Minor
	Eb$^{lyd/maj7}$	m3↑	b3	6	b7	9	Dorian
	F^{MA6}	P4↑	11	6	1	9	Dorian, Melodic Minor
	F-^{7}	P4↑	11	b6	1	b3	Aeolian, Phrygian
	F-^{6}	P4↑	11	b6	1	9	Aeolian
	F$^{7/sus4}$	P4↑	11	b7	1	b3	Dorian, Aeolian, Phrygian
	F$^{lyd/dom7}$	P4↑	11	7	1	b3	Melodic + Harmonic Minor
	F#^{MA6}	+4↕	b5	b7	b2	b3	Locrian
	F#^{maj7}	+4↕	b5	b7	b2	11	Locrian
	F#$^{maj7/b5}$	+4↕	b5	b7	1	11	Locrian
	F#$^{maj7/\#5}$	+4↕	b5	b7	2	11	Locrian nat. 2

Chord	Superimposed seventh chord	Location from the root of the chord	Note functions of superimposition				Modal sound (scale derivation)
	F#$^{lyd/maj7}$	+4↕	b5	1	b2	11	Locrian
	G^{7}	P4↓ P5↑	5	7	2	11	Melodic + Harmonic Minor
	G-^{7}	P4↓ P5↑	5	b7	2	11	Dorian, Aeolian
	G-^{7b5}	P4↓ P5↑	5	b7	b2	11	Phrygian, Dorian b2
	G$^{7/b5}$	P4↓ P5↑	5	7	2	11	Melodic + Harmonic Minor
	Ab^{NA6}	M3↓ m6↑	b6	1	b3	11	Aeolian, Phrygian
	Ab^{maj7}	M3↓ m6↑	b6	1	b3	5	Aeolian, Phrygian
	Ab^{7}	M3↓ m6↑	b6	1	b3	b5	Locrian
	Ab$^{7/b5}$	M3↓ m6↑	b6	1	2	b5	Locrian nat. 2
	Ab$^{maj7/sus4}$	M3↓ m6↑	b6	b2	b3	5	Phrygian
	Ab$^{lyd/maj7}$	M3↓ m6↑	b6	2	b3	5	Aeolian
	Ab$^{lyd/dom7}$	M3↓ m6↑	b6	2	b3	b5	Dorian #4
	A-$^{7/b5}$	m3↓ M6↑	6	1	b3	5	Dorian
	A$^{7/sus4/b5}$	m3↓ M6↑	6	2	b3	5	Dorian
	Bb^{MA6}	M2↓ m7↑	b7	2	11	5	Dorian, Aeolian
	Bb^{maj7}	M2↓ m7↑	b7	2	11	6	Dorian
	Bb-^{7}	M2↓ m7↑	b7	b2	11	b6	Phrygian
	Bb-^{6}	M2↓ m7↑	b7	b2	11	5	Phrygian
	Bb$^{7/\#5}$	M2↓ m7↑	b7	2	b5	b6	Locrian nat. 2
	Bb$^{maj7/sus4}$	M2↓ m7↑	b7	b3	11	6	Dorian
	Bb$^{7/sus4}$	M2↓ m7↑	b7	b3	11	b6	Aeolian, Phrygian
	B$^{7/\#5}$	m2↓ M7↑	7	b3	5	6	Melodic Minor

Seventh Chord Superimposition Chart (Dom^7)

Chord	Superimposed seventh chord	Location from the root of the chord	Note functions of superimposition				Modal sound (scale derivation)
C^7	C^6	Root	1	3	5	6	Mixolydian
	C^7	Root	1	3	5	b7	Mixolydian
	C^{+7}	Root	1	3	#5	b7	Whole Tone, Alt. Dom.
	$C^{7/b5}$	Root	1	3	b5	b7	Whole Tone, Alt. Dom., Mix.#11
	$C^{7/sus4}$	Root	1	4	5	b7	Mixolydian
	$C^{lyd/b7}$	Root	1	#4	5	b7	Mixolydian #11
	Db^6	m2↑	b9	11	#5	b7	Phrygian Major
	Db^{maj7}	m2↑	b9	11	#5	1	Phrygian Major
	$Db\text{-}^6$	m2↑	b9	3	#5	b7	Alt. Dom., Phrygian Major
	$Db\text{-}^{maj7}$	m2↑	b9	3	#5	1	Alt. Dom., Phrygian Major
	Db^{o7}	m2↑	b9	3	5	b7	Dom. Dim.
	Db^{omaj7}	m2↑	b9	3	5	1	Dom. Dim.
	$Db^{maj7/b5}$	m2↑	b9	11	5	1	Phrygian Major
	$Db^{maj7/sus4}$	m2↑	b9	b5	#5	1	Alt. Dom.,
	$Db^{lyd/maj7}$	m2↑	b9	5	#5	1	Phrygian Major
	$Db\text{-}^{maj7/\#5}$	m2↑	b9	3	13	1	Dom. Dim.
	D^7	M2↑	9	#11	13	1	Mixolydian #11
	$D\text{-}^7$	M2↑	9	11	13	1	Mixolydian
	$D^{7/\#5}$	M2↑	9	#11	13	1	Mixolydian #11
	$D^{7/b5}$	M2↑	9	#11	#5	1	Whole Tone
	$D^{7/sus4}$	M2↑	9	5	13	1	Mixolydian
	Eb^7	m3↑	#9	5	b7	b9	Dom. Dim.
	$Eb\text{-}^7$	m3↑	#9	#11	b7	b9	Dom. Dim., Alt. Dom.,
	$Eb\text{-}^6$	m3↑	#9	#11	b7	1	Dom. Dim., Alt. Dom.,
	$Eb\text{-}^{7/b5}$	m3↑	#9	#11	13	b9	Dom. Dim.
	$Eb^{7/b5}$	m3↑	#9	5	13	b9	Dom. Dim.
	$Eb^{7/sus4}$	m3↑	#9	#5	b7	b9	Alt. Dom.
	$Eb^{lyd/b7}$	m3↑	#9	13	b7	b9	Dom. Dim.
	$E\text{-}^{7b5}$	M3↑	3	5	b7	9	Mixolydian
	E^{o7}	M3↑	3	5	b7	b9	Dom. Dim

Chord	Superimposed seventh chord	Location from the root of the chord	Note functions of superimposition				Modal sound (scale derivation)
	E^{omaj7}	M3↑	3	5	b7	#9	Dom. Dim
	$E^{maj7/\#5}$	M3↑	3	#5	1	#9	Alt. Dom.
	$E^{maj7/b5}$	M3↑	3	#5	b7	#9	Alt. Dom.
	E^{7b5}	M3↑	3	#5	b7	9	Whole Tone
	$E\text{-}^{maj7/\#5}$	M3↑	3	5	1	#9	Dom. Dim
	$F^{maj7/sus4}$	P4↑	11	b7	1	3	Mixolydian
	$F\#^{6}$	+4↕	#11	b7	b9	#9	Dom. Dim, Alt. Dom.
	$F\#^{7}$	+4↕	#11	b7	b9	3	Dom. Dim, Alt. Dom.
	$F\#\text{-}^{7}$	+4↕	#11	13	b9	#9	Dom. Dim
	$F\#\text{-}^{6}$	+4↕	#11	13	b9	#9	Dom. Dim
	$F\#\text{-}^{7b5}$	+4↕	#11	13	1	3	Dom. Dim
	$F\#^{7\#5}$	+4↕	#11	b7	4	3	Mix. #11, Whole Tone
	$F\#^{7b5}$	+4↕	#11	b7	1	3	Mix. #11, Whole Tone
	$F\#^{lydb7}$	+4↕	#11	1	b9	3	Dom. Dim, Alt. Dom.
	$G\text{-}^{7}$	P4↓ P5↑	5	b7	9	11	Mixolydian
	$G\text{-}^{6}$	P4↓ P5↑	5	b7	9	3	Mixolydian
	$G\text{-}^{maj7}$	P4↓ P5↑	5	b7	9	#11	Mixolydian #11
	$G\text{-}^{7b5}$	P4↓ P5↑	5	b7	b9	11	Phrygian Major
	G^{o7}	P4↓ P5↑	5	b7	b9	3	Dom. Dim
	G^{omaj7}	P4↓ P5↑	5	b7	b9	#11	Dom. Dim
	$G^{maj7/sus4}$	P4↓ P5↑	5	1	9	#11	Mixolydian #11
	G^{7sus4}	P4↓ P5↑	5	1	9	11	Mixolydian
	$G\text{-}^{maj7/\#5}$	P4↓ P5↑	5	b7	#9	#11	Dom. Dim
	Ab^{7}	M3↓ m6↑	#5	1	#9	#11	Alt. Dom.
	$Ab^{7\#5}$	M3↓ m6↑	#5	1	3	#11	Whole Tone
	Ab^{7sus4}	M3↓ m6↑	#5	b9	#9	#11	Alt. Dom.
	A^{6}	m3↓ M6↑	13	b9	3	#11	Dom. Dim
	A^{7}	m3↓ M6↑	13	b9	3	5	Dom. Dim
	$A\text{-}^{7}$	m3↓ M6↑	13	1	3	5	Mixolydian
	$A\text{-}^{6}$	m3↓ M6↑	13	1	3	#11	Mixolydian #11
	$A\text{-}^{7b5}$	m3↓ M6↑	13	1	#9	5	Dom. Dim
	A^{7sus4}	m3↓ M6↑	13	9	3	5	Mixolydian
	$A^{lyd/b7}$	m3↓ M6↑	13	#9	3	5	Dom. Dim
	Bb^{6}	M2↓ m7↑	b7	9	11	5	Mixolydian

Chord	Superimposed seventh chord	Location from the root of the chord	Note functions of superimposition				Modal sound (scale derivation)
	Bb^{maj7}	M2↓ m7↑	b7	9	11	13	Mixolydian
	$Bb\text{-}^{7}$	M2↓ m7↑	b7	b9	11	#5	Phrygian Major
	$Bb\text{-}^{6}$	M2↓ m7↑	b7	b9	11	5	Phrygian Major
	$Bb\text{-}^{7b5}$	M2↓ m7↑	b7	b9	3	#5	Alt. Dom.
	Bb^{o7}	M2↓ m7↑	b7	b9	3	5	Dom. Dim
	Bb^{omaj7}	M2↓ m7↑	b7	b9	3	13	Dom. Dim
	$Bb^{maj7/\#5}$	M2↓ m7↑	b7	9	#11	13	Mixolydian #11
	$Bb^{7/\#5}$	M2↓ m7↑	b7	9	#11	#5	Whole Tone
	$Bb^{maj7/b5}$	M2↓ m7↑	b7	9	3	13	Mixolydian
	Bb^{7b5}	M2↓ m7↑	b7	9	3	#5	Whole Tone
	$Bb^{lyd/maj7}$	M2↓ m7↑	b7	3	11	13	Mixolydian
	$Bb\text{-}^{maj7/\#5}$	M2↓ m7↑	b7	b9	#11	13	Dom. Dim

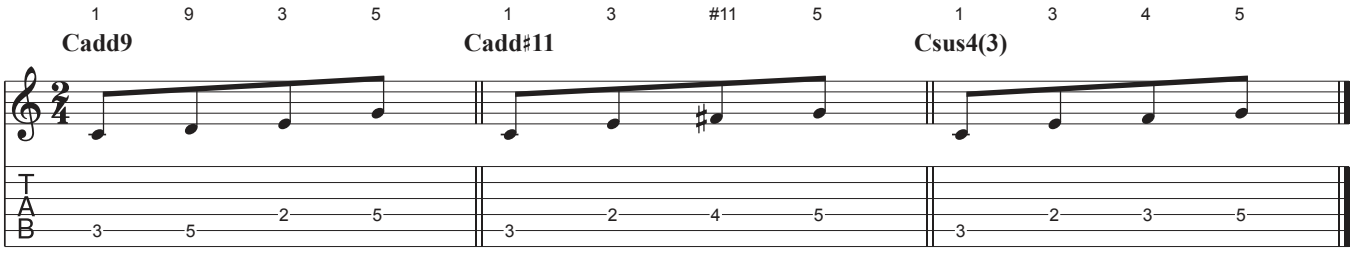
Also From The Dominant Diminished Scale

Chord	Superimposed seventh chord	Location from the root of the chord	Note functions of superimposition				Modal sound (scale derivation)
	C^{o7}	Root	1	#9	b5	13	Dom. Dim
	Eb^{o7}	m3↑	#9	b5	13	1	Dom. Dim
	$F\#^{o7}$	+4↕	b5	13	1	#9	Dom. Dim
	A^{o7}	m3↓ M6↑	13	1	#9	b5	Dom. Dim

Hybrid Arpeggios

Hybrid Arpeggios do not fall into triadic or seventh chord categories but contain pieces of each. Added note chords are types of hybrid arpeggios.

Ex. 185a-c



This next chart contains serveral possible hybrid arpeggios over each of the four chord types.

Hybrid Arpeggio Chart (Maj^7)

Chord	Hybrid Structure	Note functions of superimposition					Modal sound (scale derivation)
C^{maj7}	C^{add9}	1	9	3	5		Ionian, Lydian
	G^{add9}	5	6	7	9		Ionian, Lydian
	D^{add9}	9	3	#11	13		Lydian (Augmented)
	E^{add9}	3	#11	#5	7		Lydian Augmented
	C^{add11}	1	3	4	5		Ionian
	D^{add11}	9	#11	5	13		Lydian
	E^{add11}	3	#5	13	7		Lydian Augmented
	G^{add11}	5	7	1	9		Ionian, Lydian
	$C^{add\#11}$	1	3	#11	5		Lydian
	$D^{add\#11}$	9	#11	5	13		Lydian
	$C^{add9/11}$	1	9	3	11	5	Ionian
	$G^{add9/11}$	3	13	7	1	9	Ionian, Lydian
	$D^{add9/11}$	9	3	#11	5	13	Lydian
	$C^{add9/\#11}$	1	9	3	#11	5	Lydian
	$D^{add9/\#11}$	9	3	#11	5	13	Lydian
	$A\text{-}^{add9}$	6	7	1	3		Ionian, Lydian
	$B\text{-}^{add11}$	7	9	3	#11		Lydian
	$E\text{-}^{add9}$	3	#11	5	7		Lydian
	$E\text{-}^{add9/11}$	3	#11	5	13	7	Lydian
	$A\text{-}^{add9/11}$	6	7	1	9	3	Ionian, Lydian

Hybrid Arpeggio Chart (Minor^{7})

Chord	Hybrid Structure	Note functions of superimposition					Modal sound (scale derivation)
C-^{7}	Eb^{add9}	b3	11	5	b7		Dorian, Aeolian, Phrygian
	F^{add9}	11	5	13	1		Dorian
	G^{add9}	5	13	7	9		Melodic Minor
	Bb^{add9}	b7	1	9	11		Dorian, Aeolian
	Db^{add9}	b9	b3	11	b6		Phrygian
	Eb^{add11}	b3	5	b6	b7		Aeolian
	F^{add11}	11	13	b7	1		Dorian
	Bb^{add11}	b7	9	b3	11		Dorian, Aeolian
	G^{add11}	5	7	1	9		Melodic Minor
	Eb$^{add\#11}$	b3	5	13	b7		Dorian (b2)
	F$^{add\#11}$	11	13	7	1		Melodic Minor
	Db$^{add\#11}$	b9	11	5	b6		Phrygian
	Db$^{add9/\#11}$	b9	b3	11	5	b6	Phrygian
	Eb$^{add9/\#11}$	b3	11	5	13	b7	Dorian
	F$^{add9/\#11}$	11	5	13	7	1	Melodic Minor
	C-^{add9}	1	9	b3	5		Dorian, Aeolian
	G-^{add9}	5	13	b7	9		Dorian
	F-^{add9}	11	5	b6	1		Aeolian
	C-$^{add9/11}$	1	9	b3	11	5	Dorian, Aeolian
	F-$^{add9/11}$	11	5	b6	b7	1	Aeolian
	G-$^{add9/11}$	5	13	b7	1	9	Dorian

Hybrid Arpeggio Chart (Min^{7b5})

Chord	Hybrid Structure	Note functions of superimposition					Modal sound (scale derivation)
C-^{7b5}	Eb-^{add9}	b3	11	b5	b7		Locrian
	F#^{add9}	b5	b6	b7	b9		Locrian
	F#$^{add\#11}$	b5	b7	1	b9		Locrian
	F#$^{add9/\#11}$	b5	b6	b7	1	b9	Locrian
	Ab^{add11}	b6	1	b9	b3		Locrian
	Eb-$^{add9/11}$	b3	11	b5	b6	b7	Locrian (nat. 2)
	Ab$^{add9/11}$	b6	b7	1	b9	b3	Locrian

Hybrid Arpeggio Chart (Dom^{7})

hord	Hybrid Structure	Note functions of superimposition					Modal sound (scale derivation)
C^{7}	C^{add9}	1	9	3	5		Mixolydian
	C^{add11}	1	3	11	5		Mixolydian
	$C^{add\#11}$	1	3	#11	5		Mixolydian #11
	D^{add9}	9	3	#11	13		Mixolydian #11
	D^{add11}	9	3	#11	5	13	Mixolydian #11
	$D^{add\#11}$	9	#11	#5	13		Mixolydian b6
	$F\#^{add9}$	#11	#5	b7	b9		Altered Dominant
	$F\#^{add\#11}$	#11	b7	1	b9		Altered Dominant
	$G\#^{add9}$	#5	b7	1	#9		Altered Dominant
	$G\#^{add11}$	#5	b9	1	#9		Altered Dominant
	$A^{add\#11}$	13	b9	ä9	3		Dominant Diminished
	Bb^{add9}	b7	1	9	11		Mixolydian
	$Bb^{add\#11}$	b7	9	3	11		Mixolydian
	$Db\text{-}^{add9}$	b9	#9	3	#5		Altered Dominant
	$G\text{-}^{add9}$	5	13	b7	9		Mixolydian
	$C^{add9/11}$	1	9	3	11	5	Mixolydian
	$C^{add9/\#11}$	1	9	3	#11	5	Mixolydian #11
	$Db\text{-}^{add9/11}$	b9	#9	3	#11	#5	Altered Dominant
	$D^{add9/11}$	9	3	#11	5	13	Mixolydian #11
	$D^{add9/\#11}$	9	3	#11	#5	13	Mixolydian b6
	$F\#^{add9/\#11}$	#11	#5	b7	1	b9	Altered Dominant
	$G\text{-}^{add9/11}$	5	13	b7	1	9	Mixolydian
	$G\#^{add9/11}$	#5	b7	1	b9	#9	Altered Dominant
	$Bb^{add9/\#11}$	b7	1	9	3	11	Mixolydian

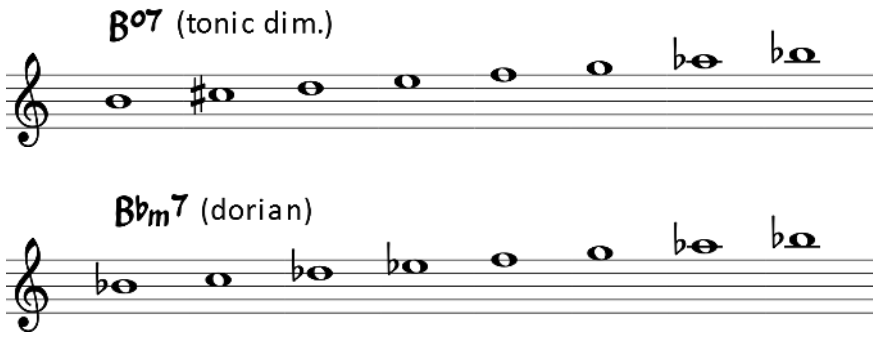
Playing Over Unusual Resolutions

① Dim7 chords resolving down by ½ step to minor

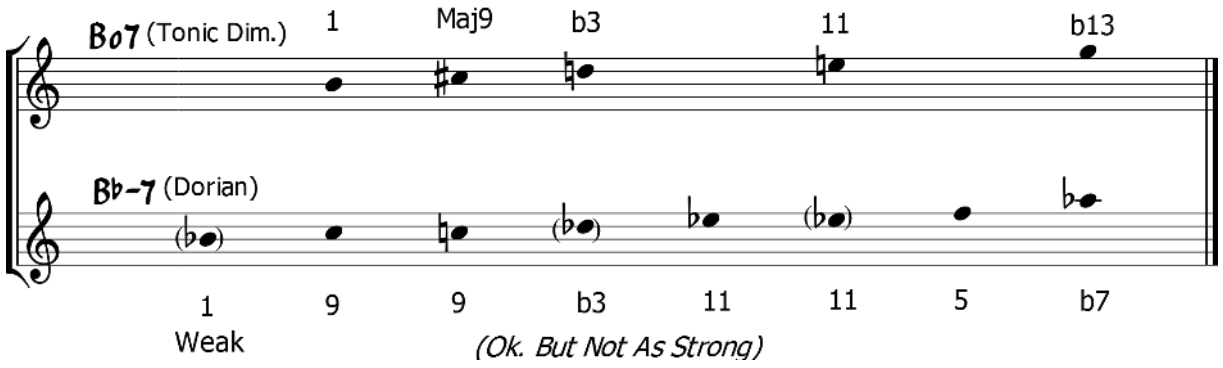
Tunes found in: All the Things You Are
Wave
Triste
Night and Day
Body and Soul
Here´s That Rainy Day

The resolution tendencies are as follows:

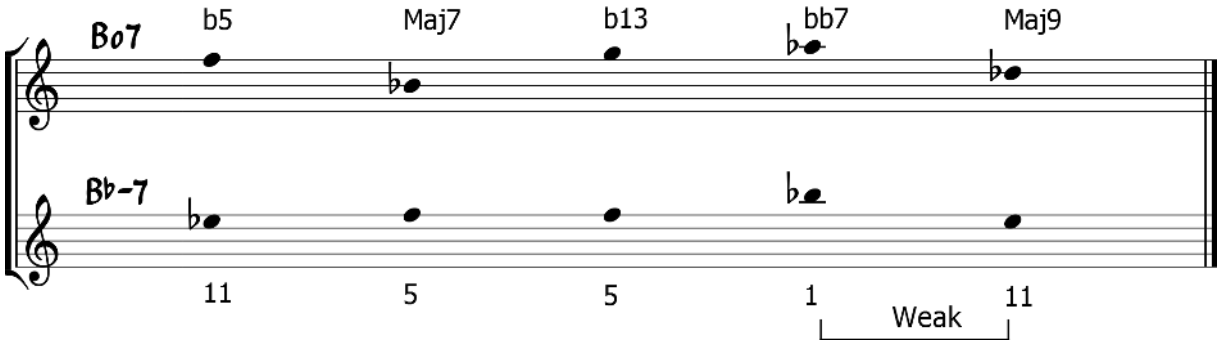
Scales



Ex. 186a

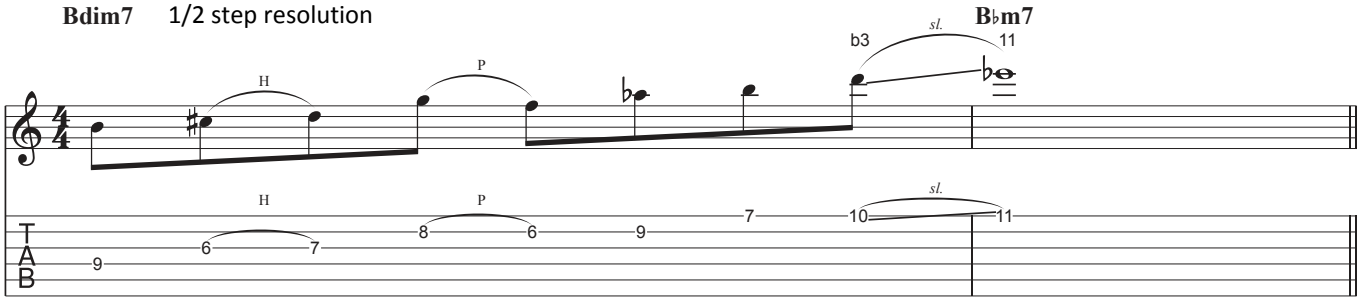
½ step resolutions

Ex. 186b

whole step resolution



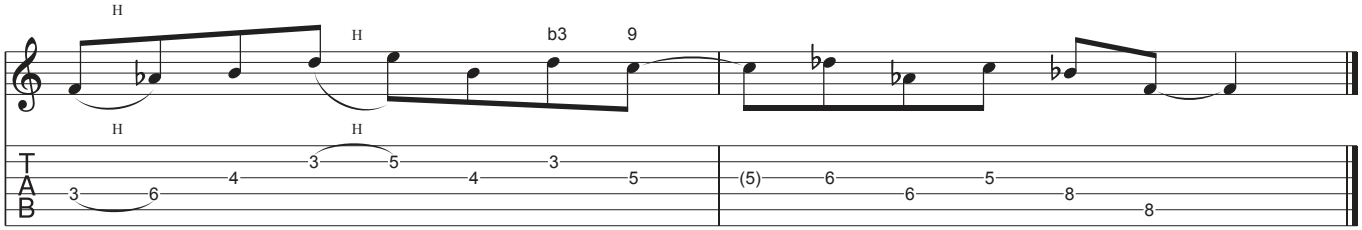
Ex. 186c

Bdim7 1/2 step resolution **B♭m7**



Ex. 186d

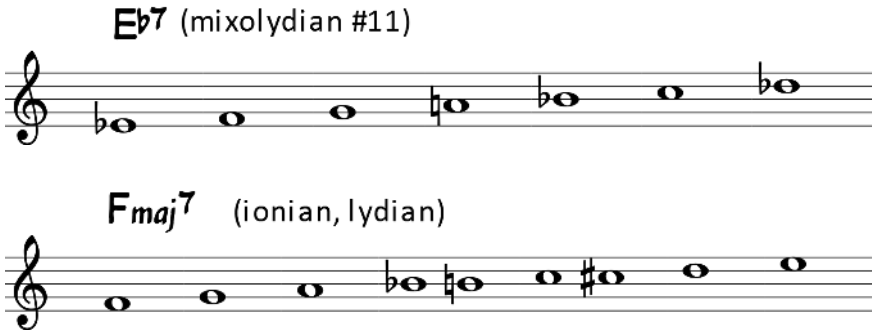
Bdim7 Whole step resolution **B♭m7**



② Dom7 chords resolving up by whole step

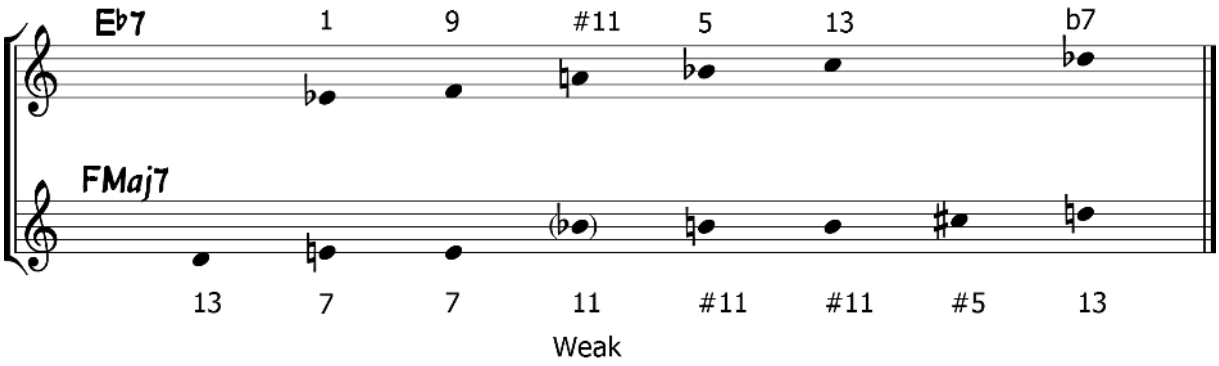
Tunes found in: Just Friends
Yardbird Suite
There Will Never Be Another You
Donna Lee
Cherokee
Stella By Starlight

Scales



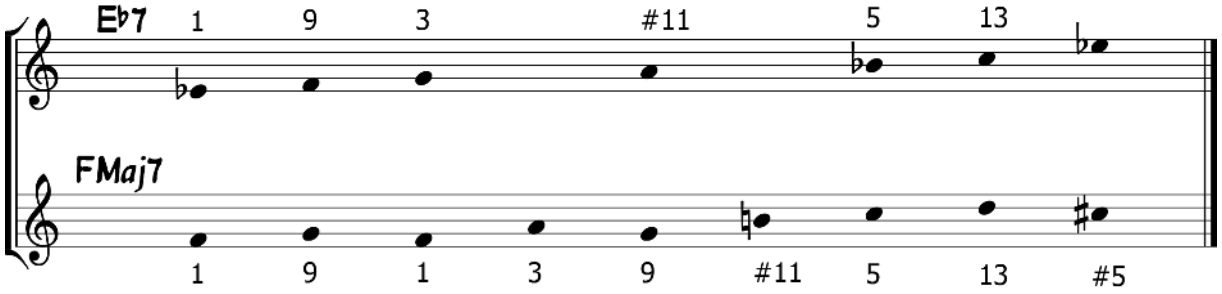
Ex. 187a

½ step resolutions

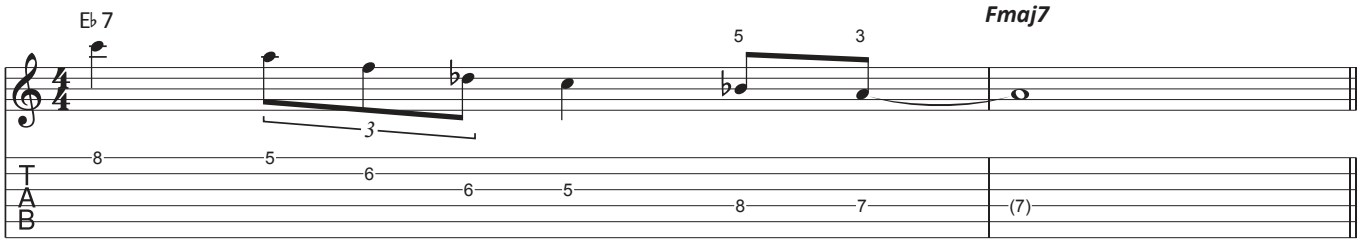


Ex. 187b

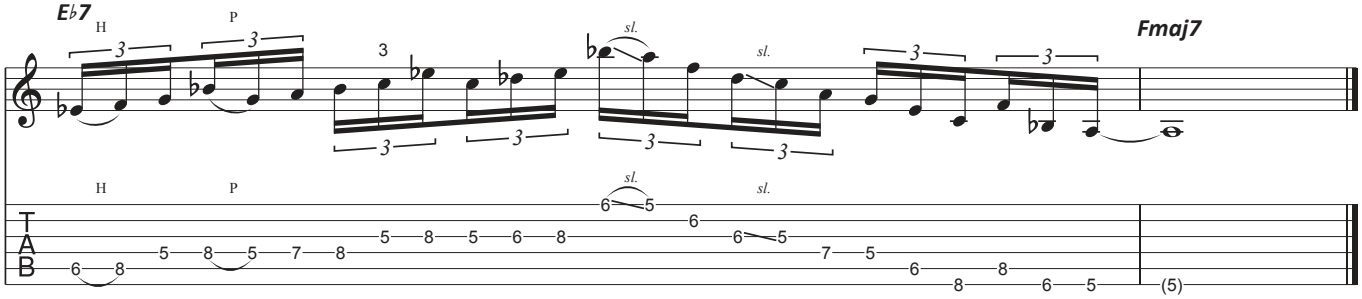
whole step resolutions



Ex. 187c



Ex. 187d



You will find the whole step resolutions are not as strong as half-step ones. The harmonic tendencies of Eb^{7} to F^{maj7} are the same for the chords.

Ex. 188

1 $E\flat^{7}$ → F^{MAJ7}

2 A^{7ALT} → Dm^{7}

3 A^{7ALT} → $B\flat^{MAJ7}$

4 $E\flat^{7}$ → $B\flat^{MAJ7}$

5 A^{7ALT} → F^{MAJ7} OR F^{7}

6 $E\flat^{7}$ → Am^{7}

7 A^{7ALT} → Am^{7}

Further explanation of this concept can be found in the harmony chapter.

Other Resolutions To Know

1. Third related chords (similar types)

Ex. 189

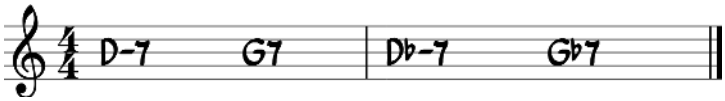
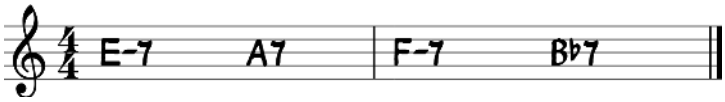
C^{maj7} → Eb^{maj7}

$F\text{-}^{7}$ → $D\text{-}^{7}$

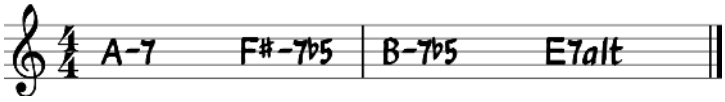
E^{maj7} → Ab^{maj7}

$B\text{-}^{7}$ → $G\text{-}^{7}$

2. ii V´s moving up and down by half steps

Ex. 190a

Ex. 190b

3. i^{7} vi^{7b5} ii^{7b5} V^{7alt}

Ex. 191



Many of the other resolution possibilities are covered in the harmony chapter.

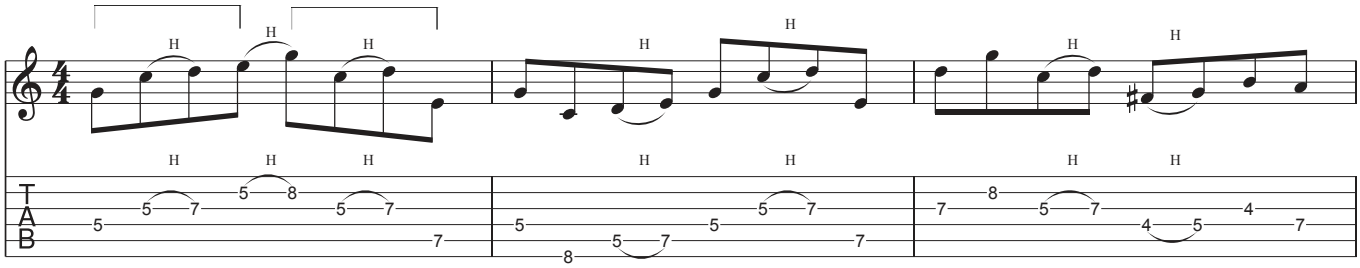
Remember: Try to take advantage of half step resolutions because they make the transitions much smoother.

Melodic Ideas

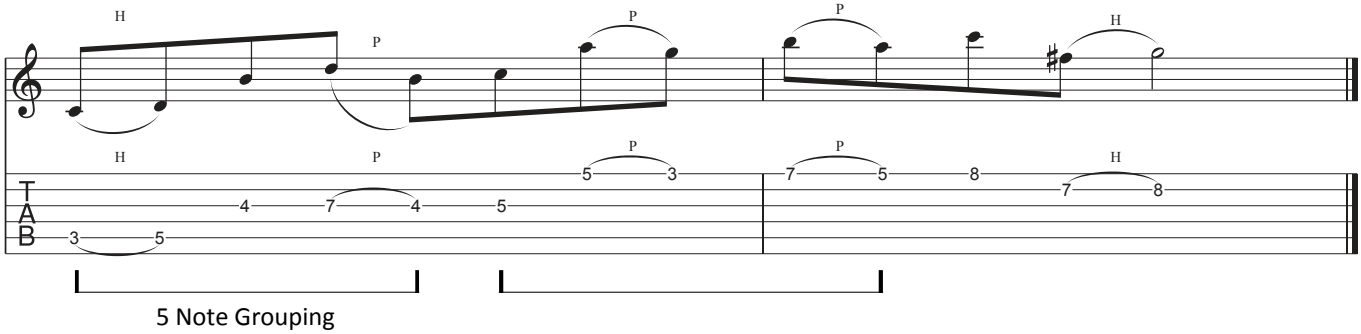
A small collection of scale tones can bear a large amount of melodic material through the use of octave displacement.

A four note order such as G C D E can become this.

Ex. 192 & 193



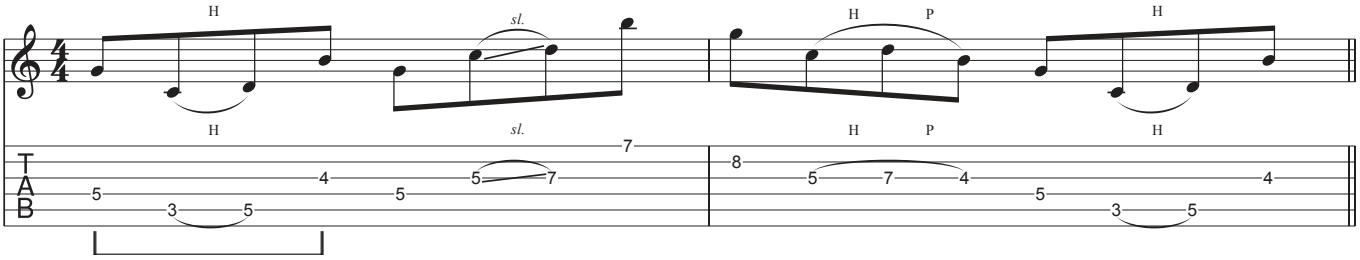
Try this over C, Csus4, Bbmaj7, F#7alt, Em7b6. Odd number note ideas metrically overlap in the measure.



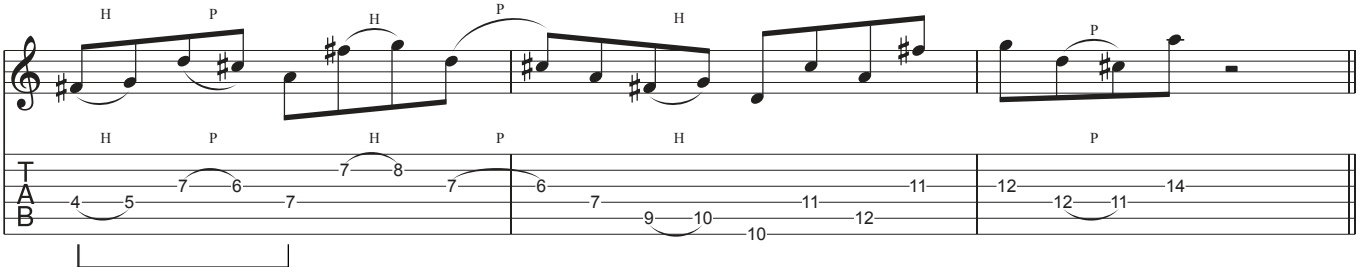
Try this over Em7b5, Cmaj7, F#7alt, D7#11

I have included some more examples like these from different scales.

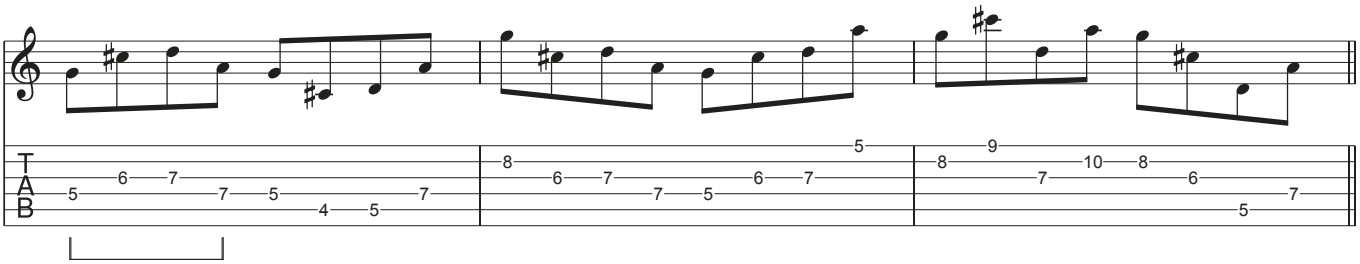
Ex. 194a



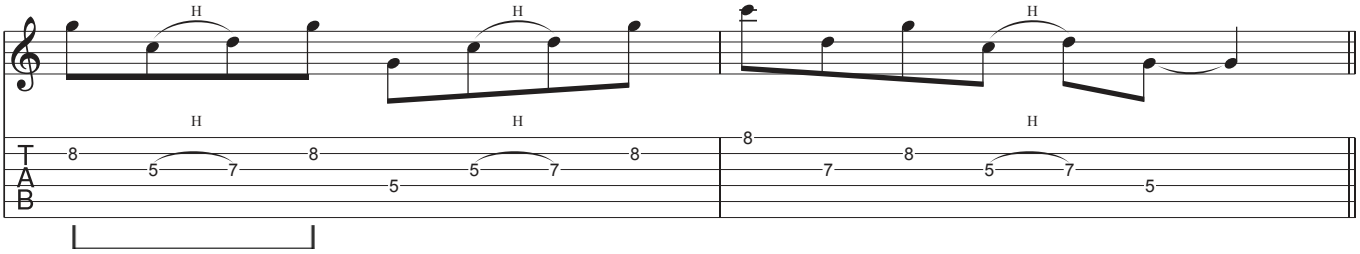
Ex. 194b



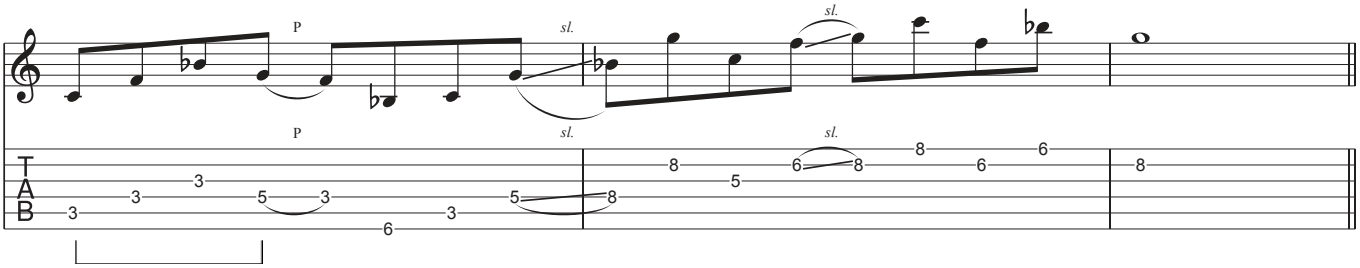
Ex. 194c



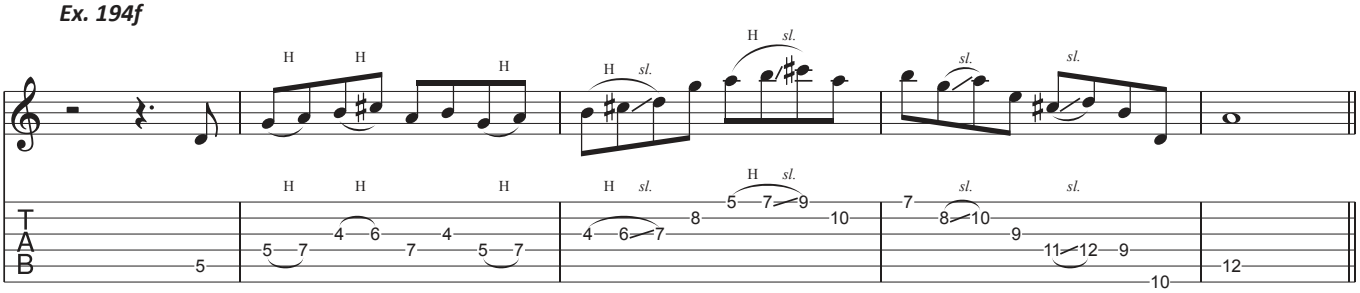
Ex. 194d



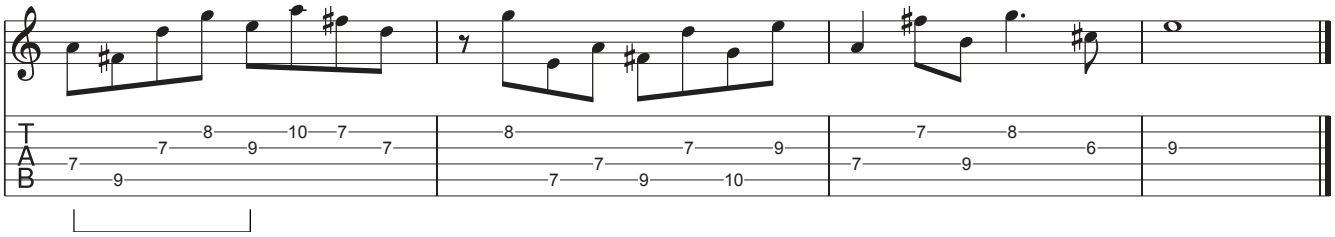
Ex. 194e



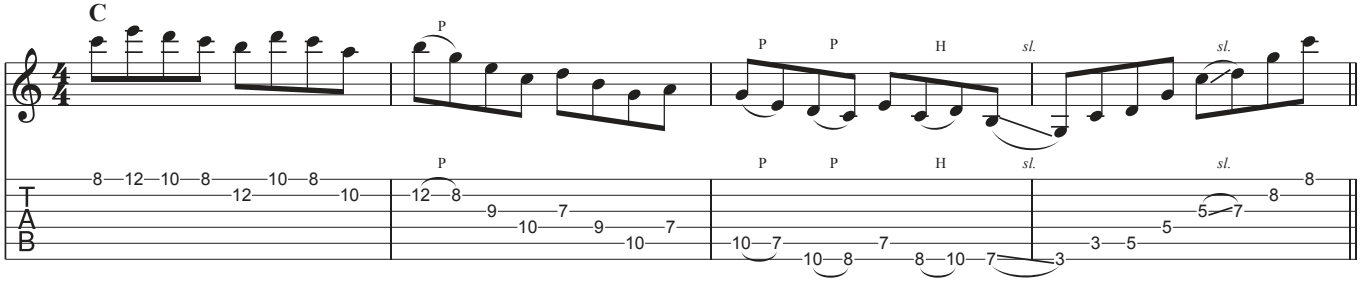
Ex. 194f



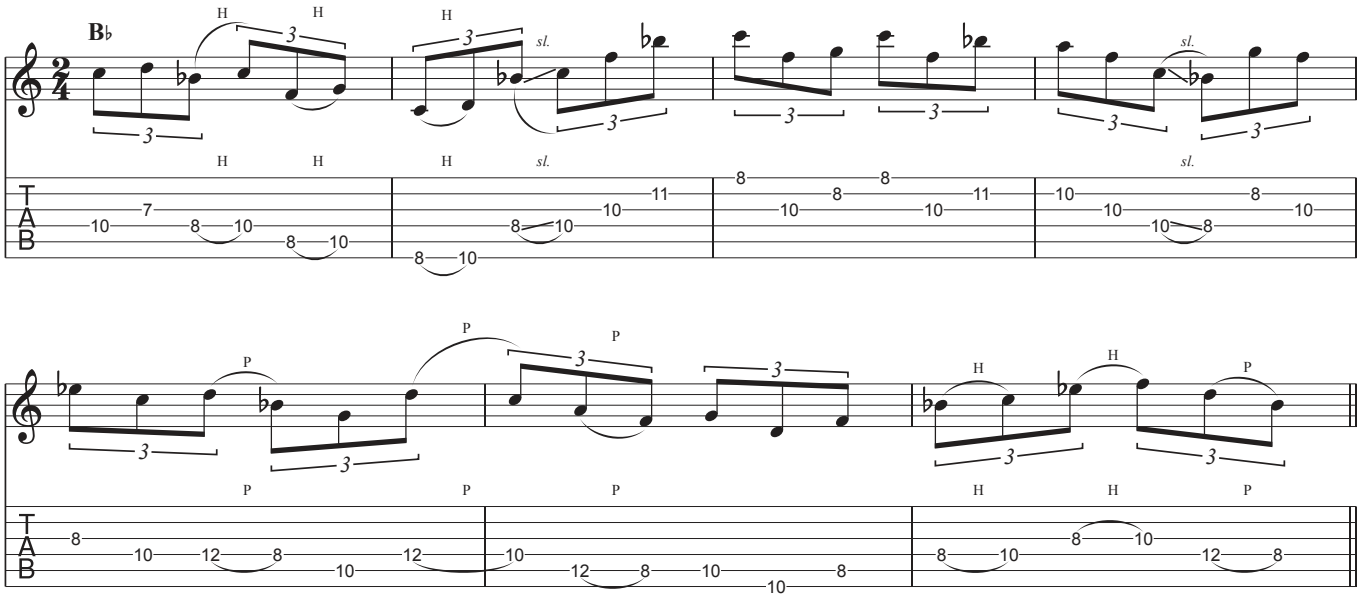
Ex. 194g

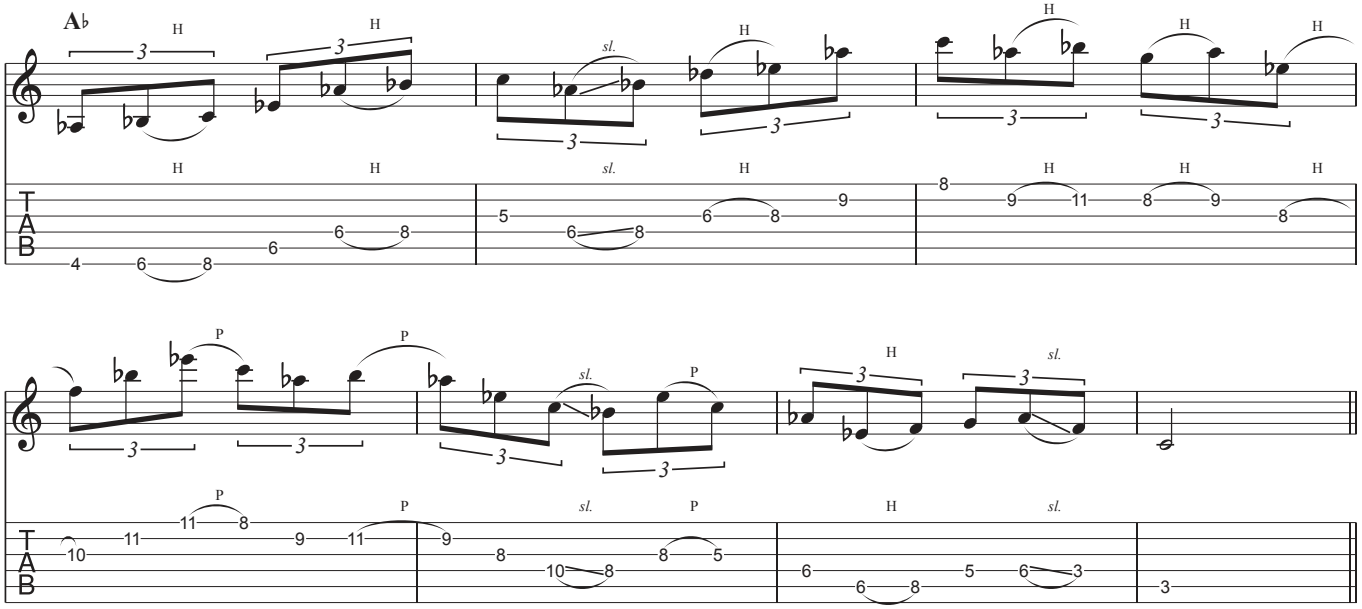


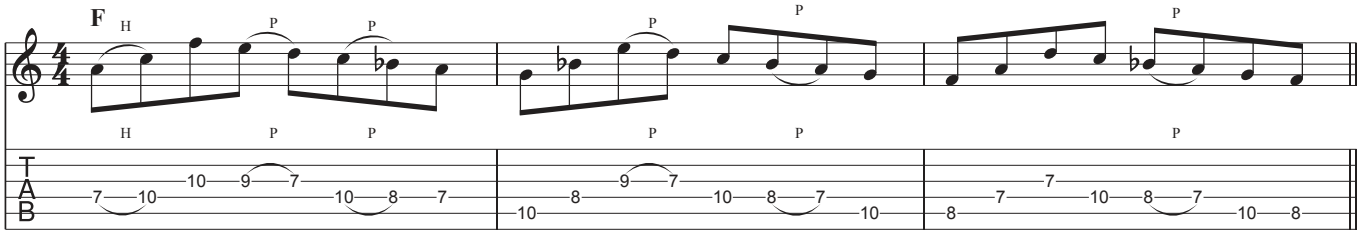
Ex. 195a



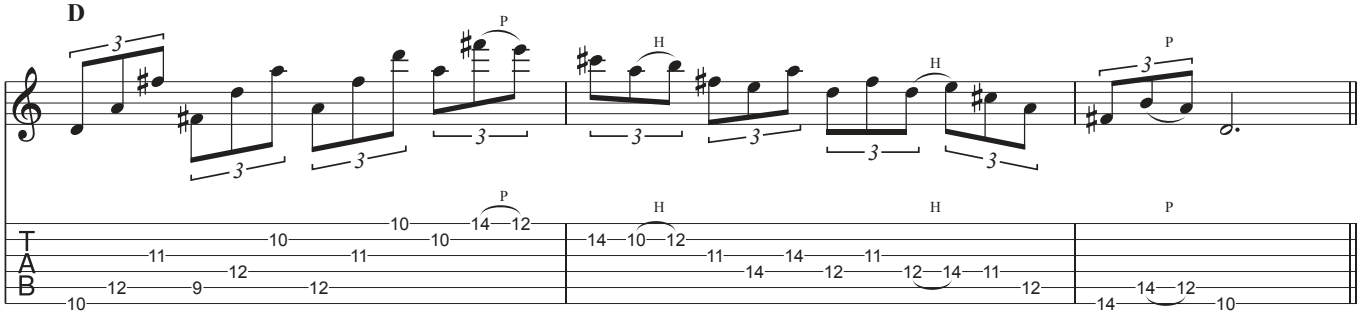
Ex. 195b



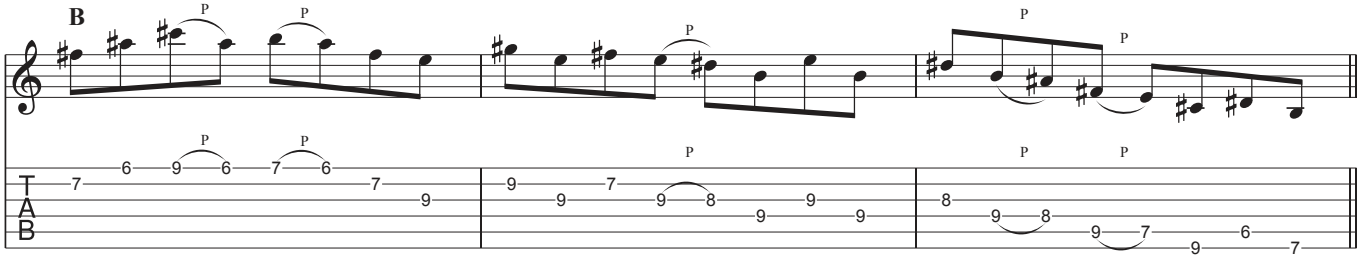
Ex. 195c

Ex. 195d

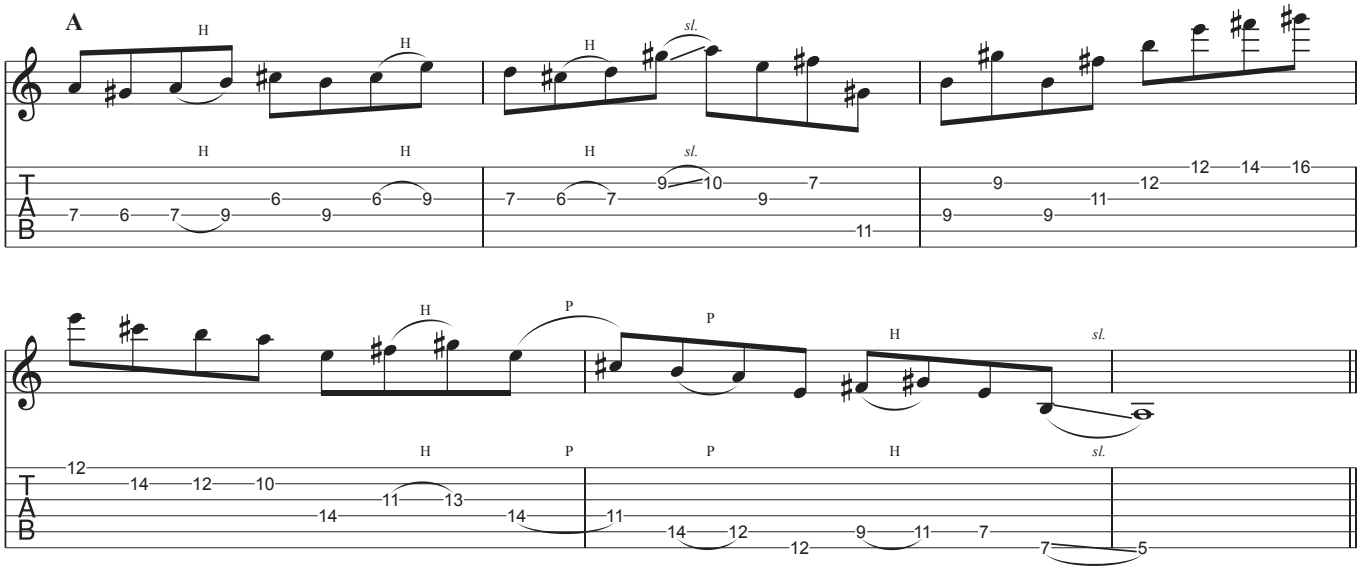


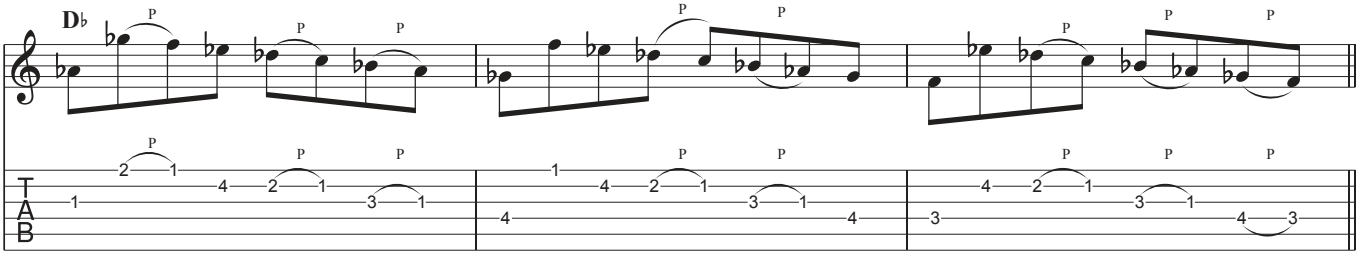
Ex. 195e



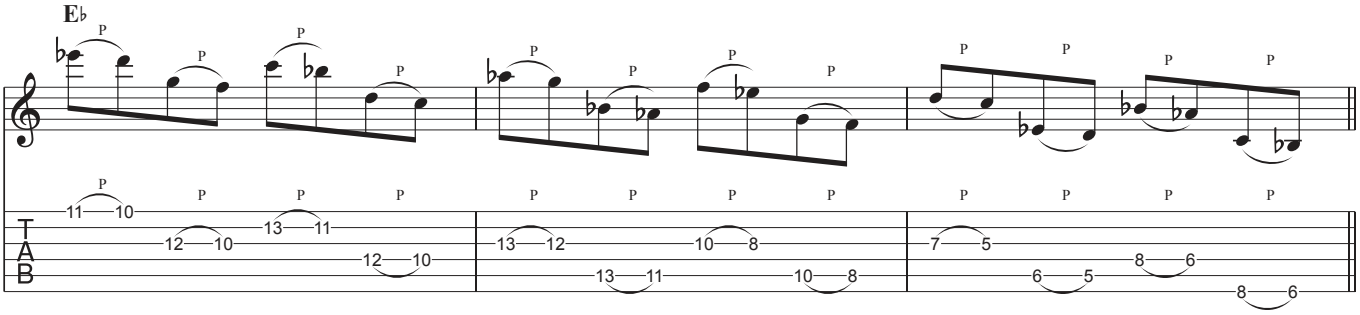
Ex. 195f



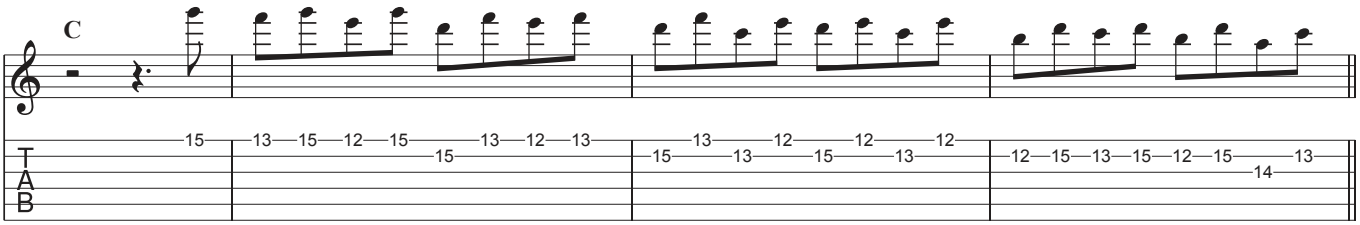
Ex. 195g

Ex. 195h



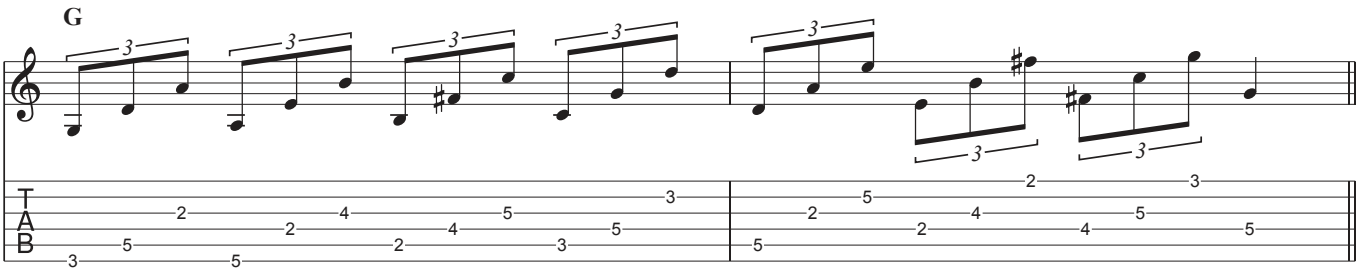
Ex. 195i



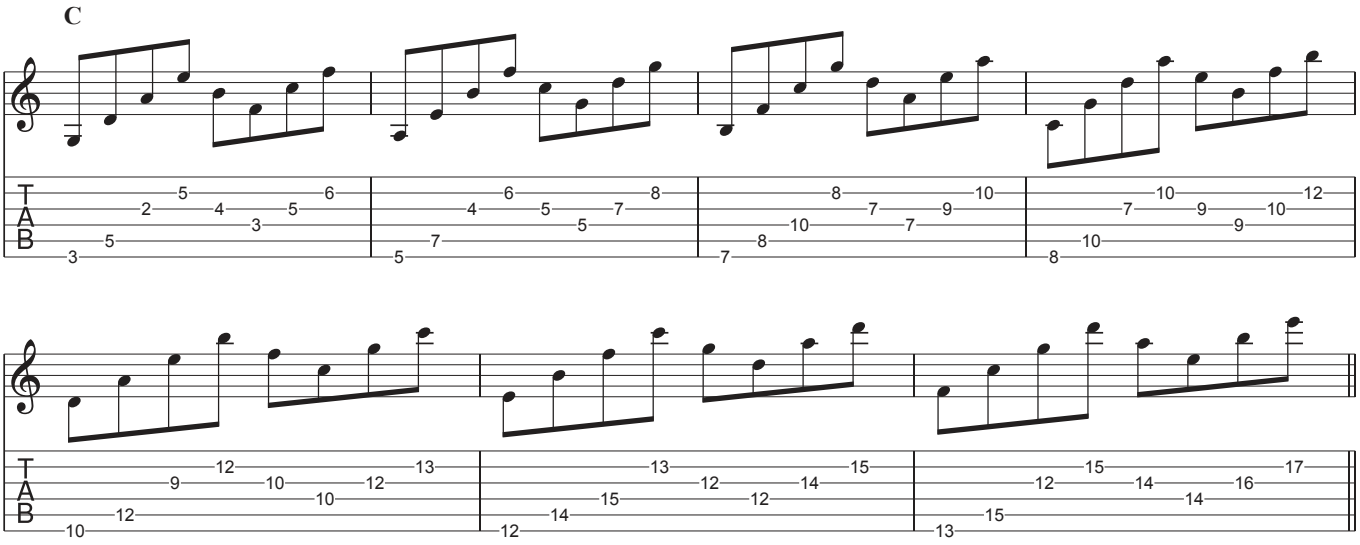
Ex. 195j



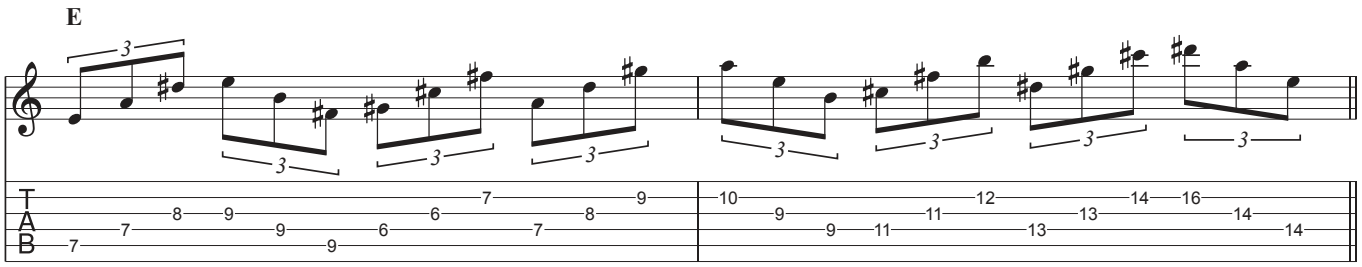
Ex. 195k



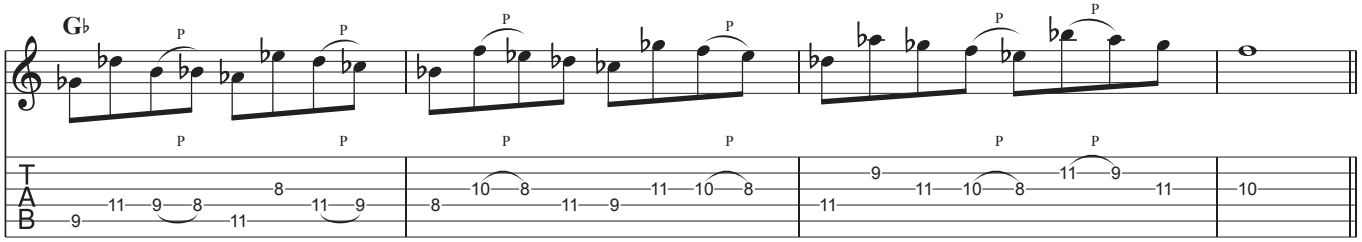
Ex. 195l



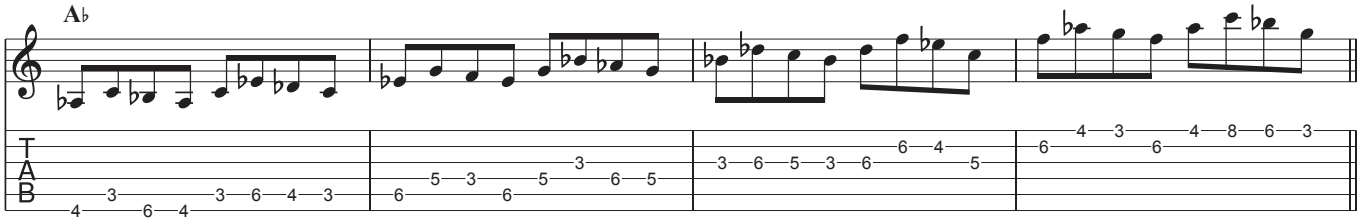
Ex. 195m



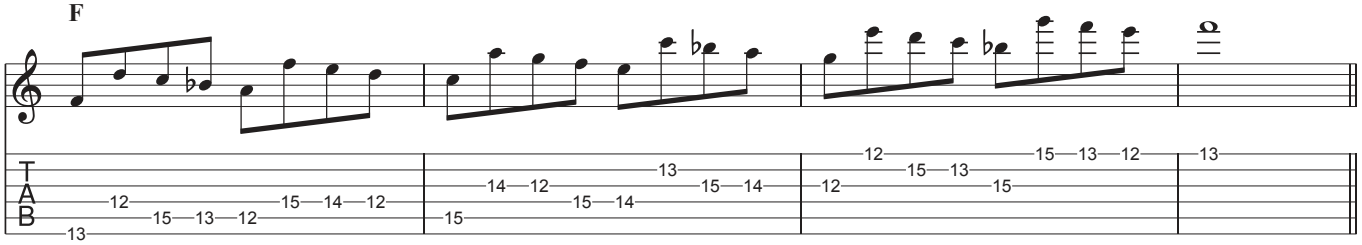
Ex. 195n



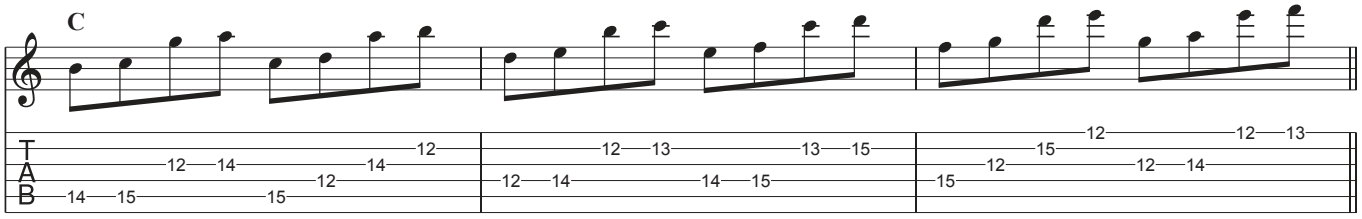
Ex. 195o



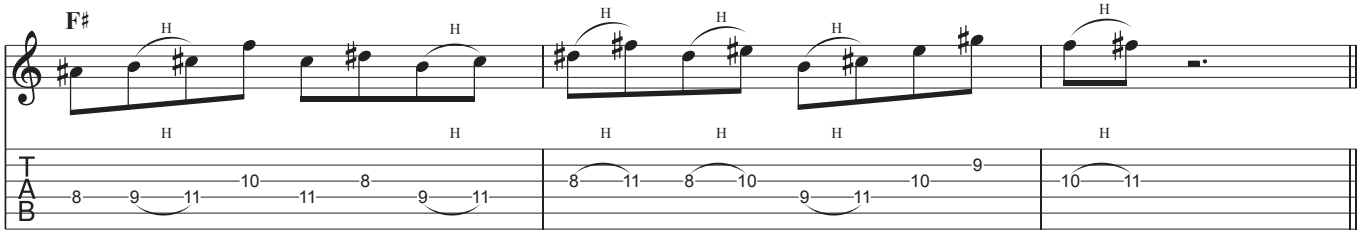
Ex. 195p



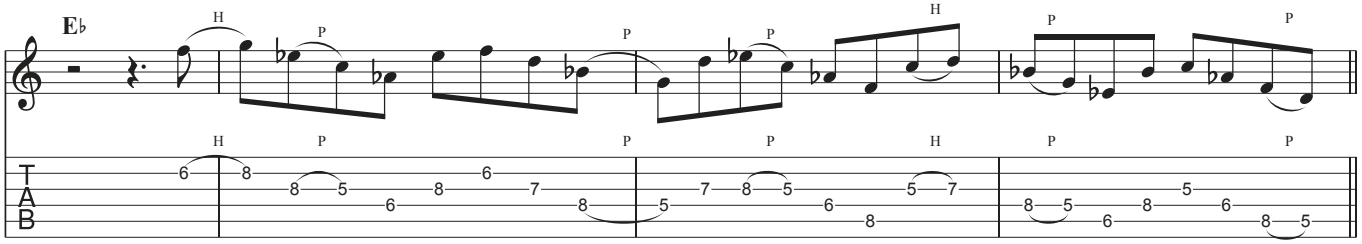
Ex. 195q



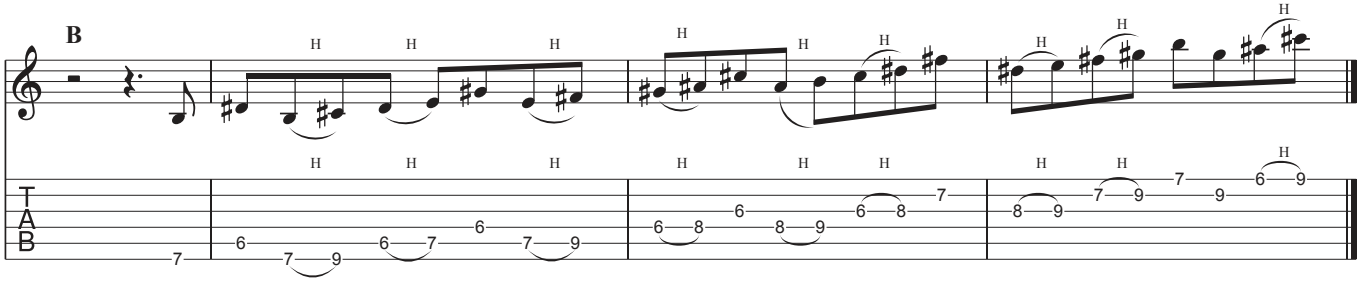
Ex. 195r



Ex. 195s



Ex. 195t



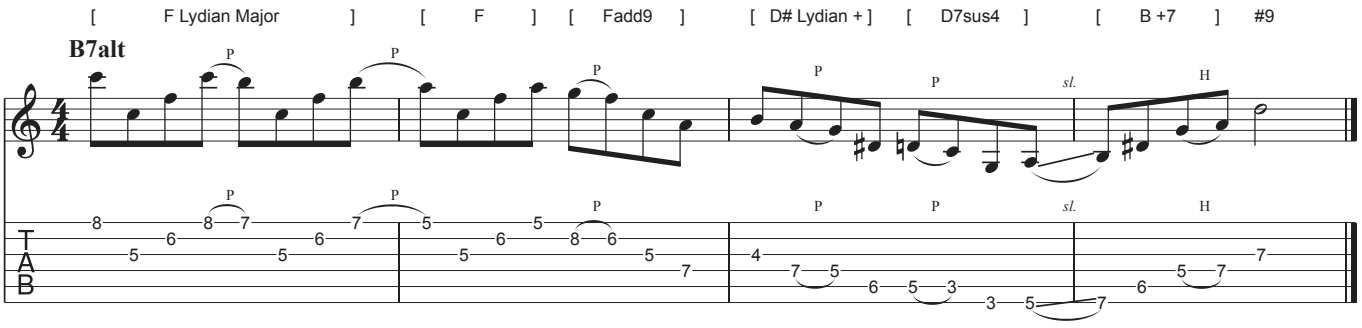
Modes: Linear Approach

The following line studies should be practiced in all keys and positions on the guitar. Each study concentrates on a particular mode and strives to bring out its own distinctive flavor.

This is done through the use of target notes and intervallic structures.

Since no harmonic analysis has been provided, it will be your job to look for harmonic structures such as broken intervals, diatonic triads and seventh chords.

Ex. 196



This can also be used over:

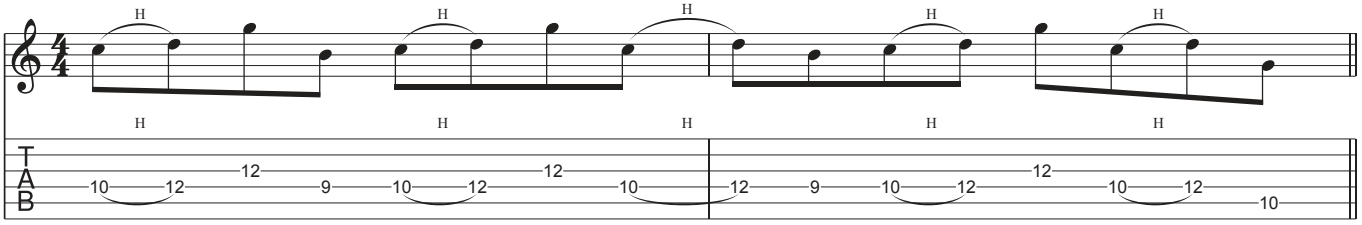
F7#11, Ebmaj7#5, Am9b5, Cmmaj7, D7/sus4/b9 (C melodic minor chords)

Before we go on I would like to make an important point regarding passing tones.

Because of the abundance of linear possibilities using only scale tones, I would encourage developing lines without passing tones so as to not cluster your playing with superfluous notes. This is why most of the examples contain only scale tones (a great starting place for developing these kinds of lines are the diatonic triads and seventh chords we covered in both chapters 1+3).

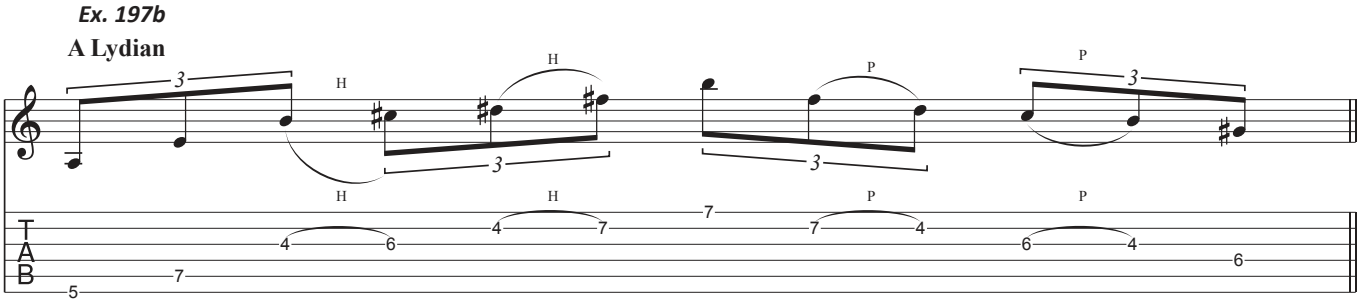
Ex. 197a

C Ionian



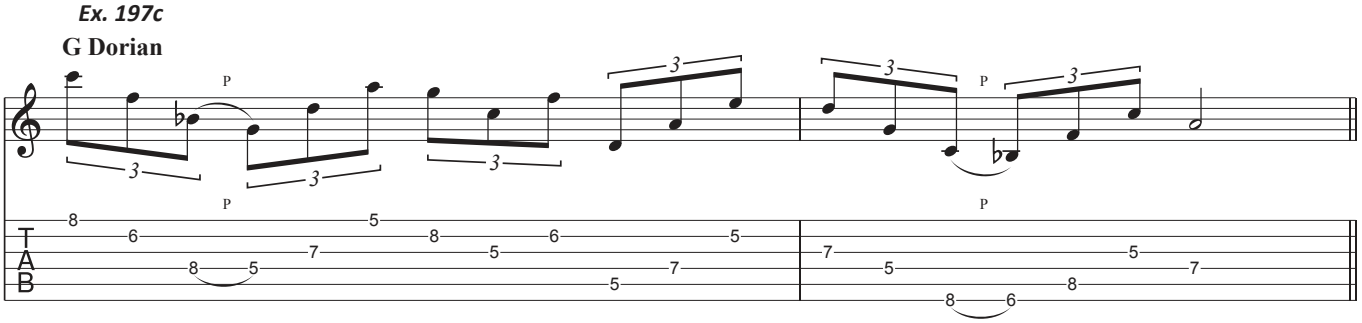
Ex. 197b

A Lydian



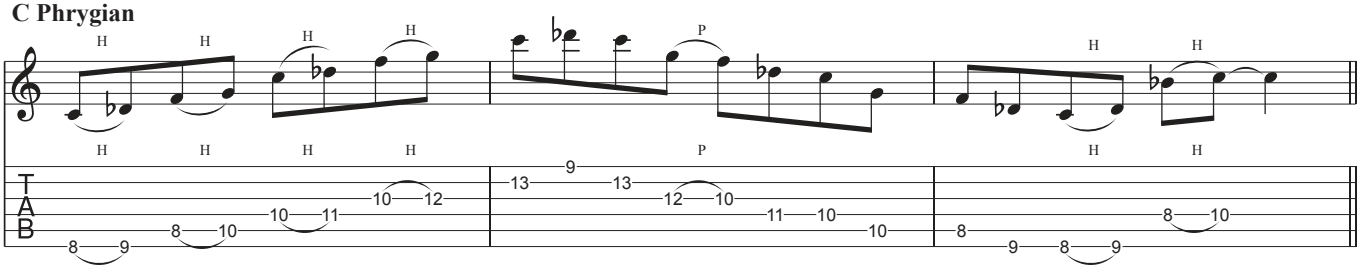
Ex. 197c

G Dorian



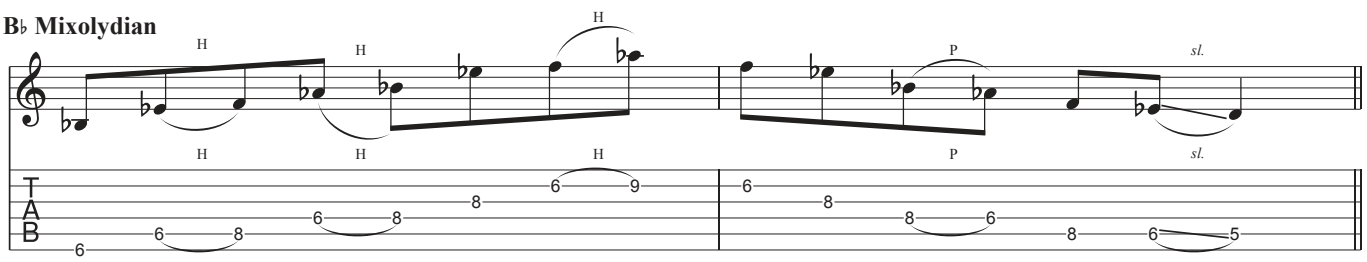
Ex. 197d

C Phrygian

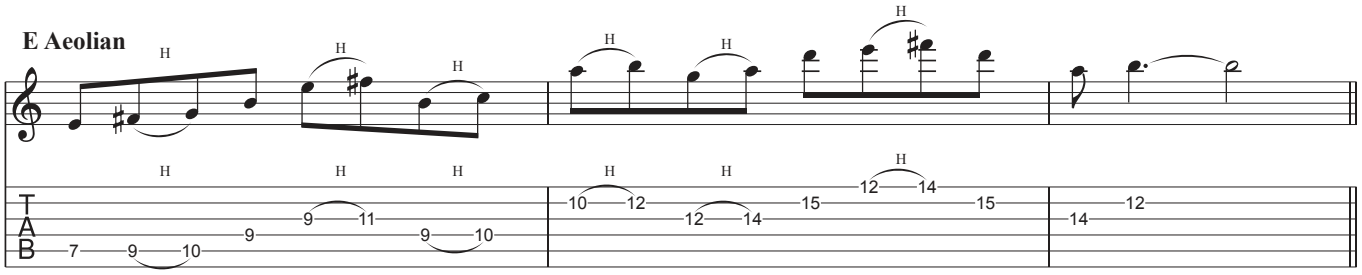


Ex. 197e

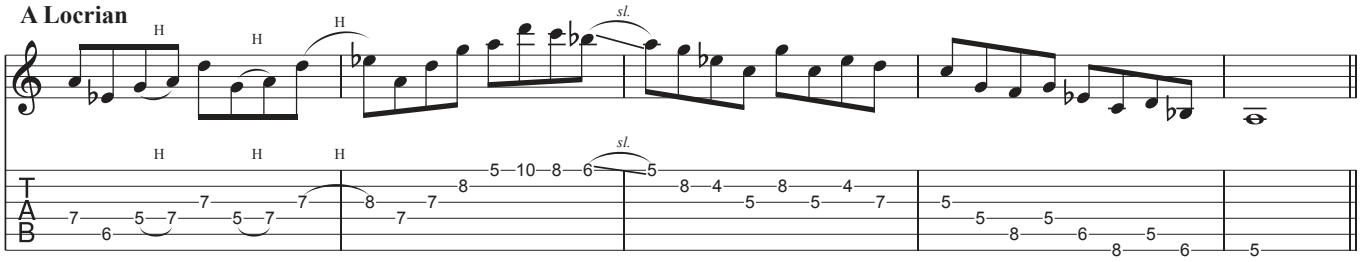
B♭ Mixolydian



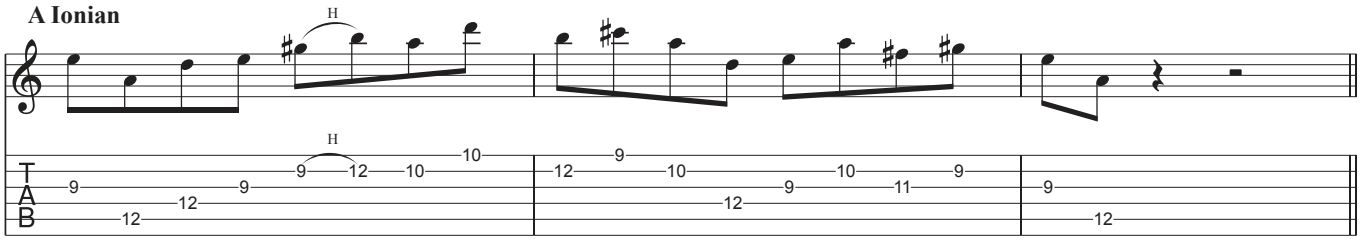
Ex. 197f



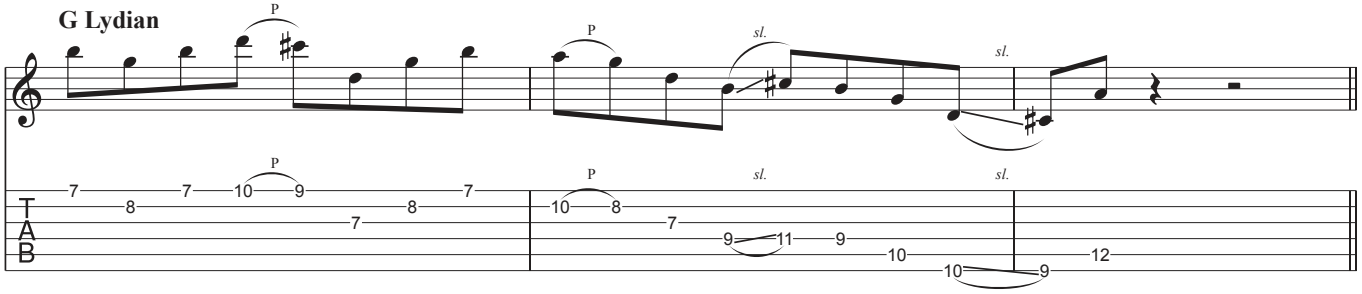
Ex. 197g



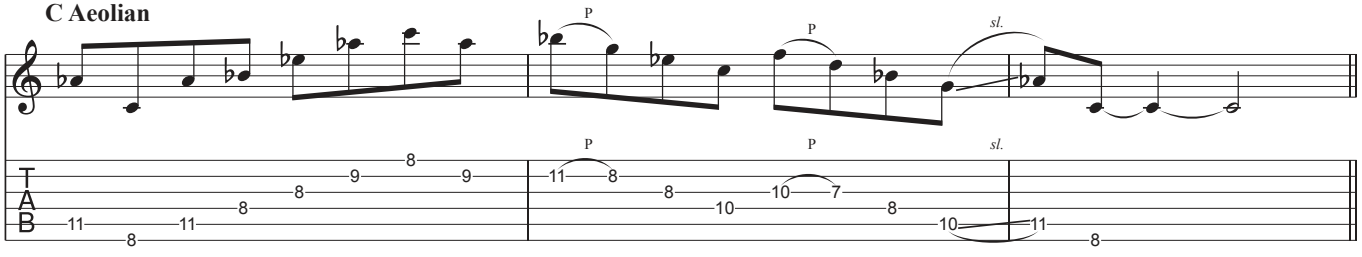
Ex. 197h



Ex. 197i

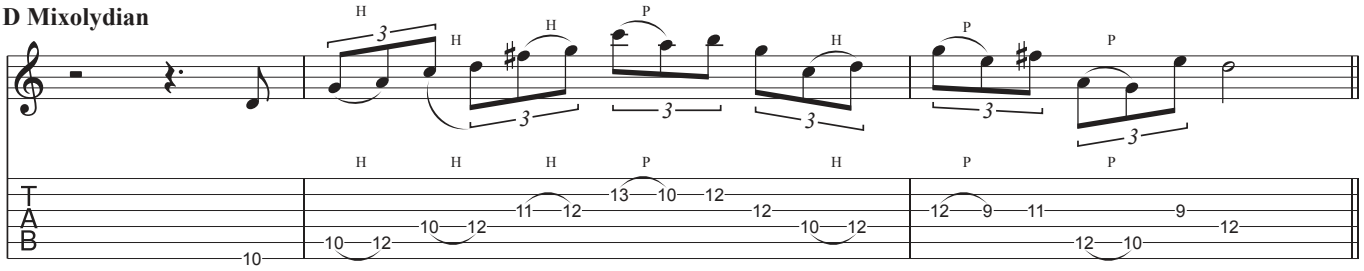


Ex. 197j



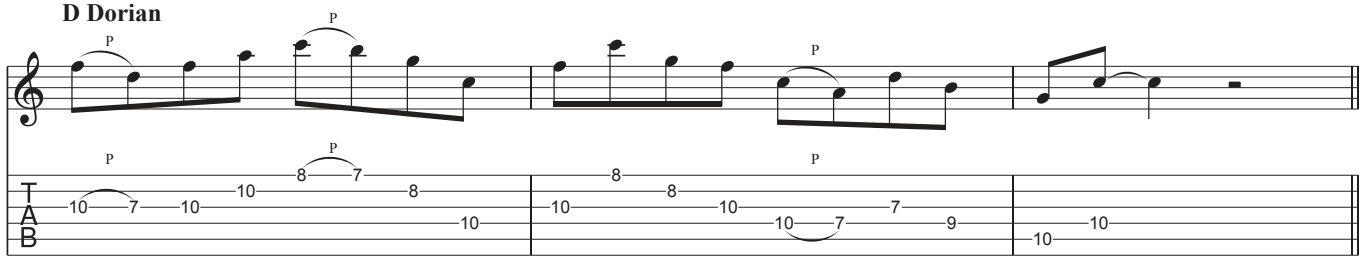
Ex. 197k

D Mixolydian



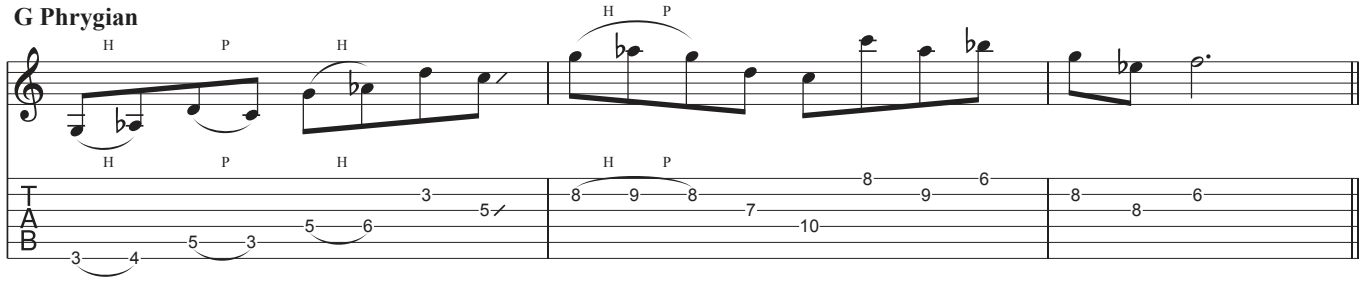
Ex. 197l

D Dorian



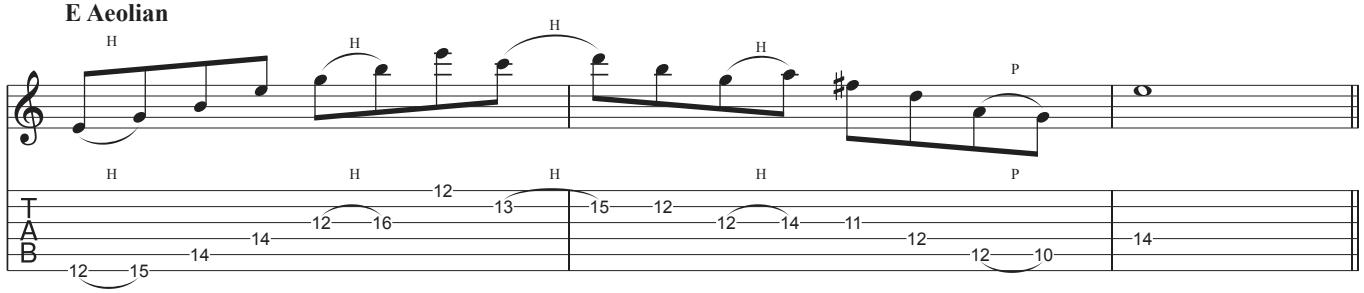
Ex. 197m

G Phrygian



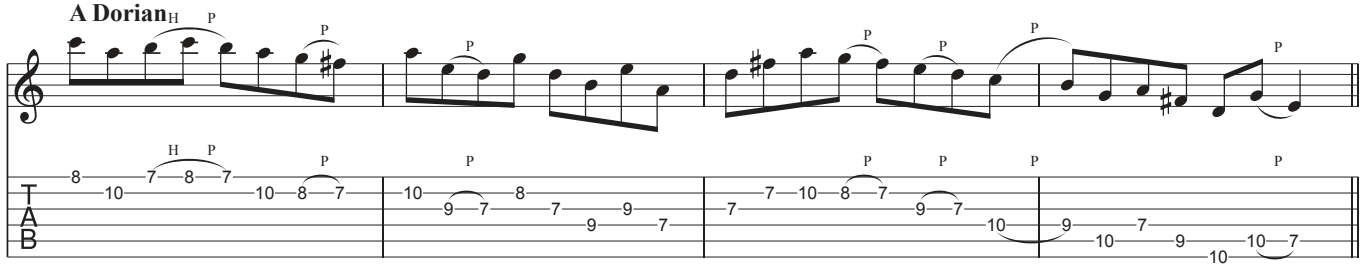
Ex. 197n

E Aeolian



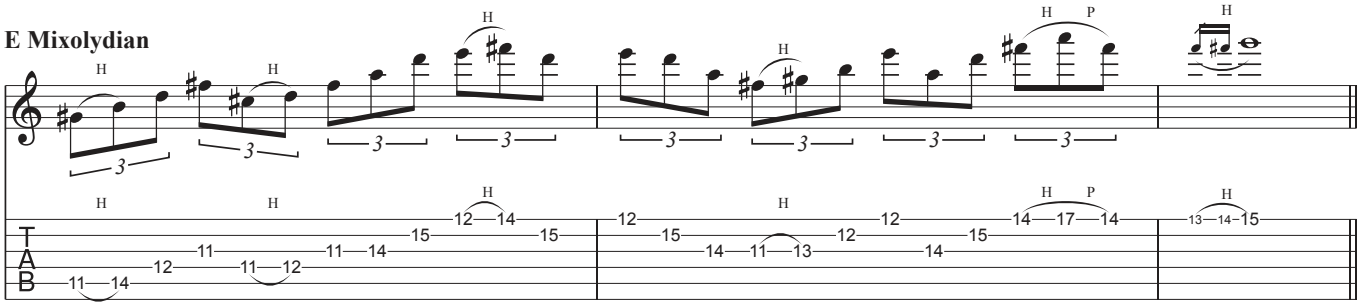
Ex. 197o

A Dorian



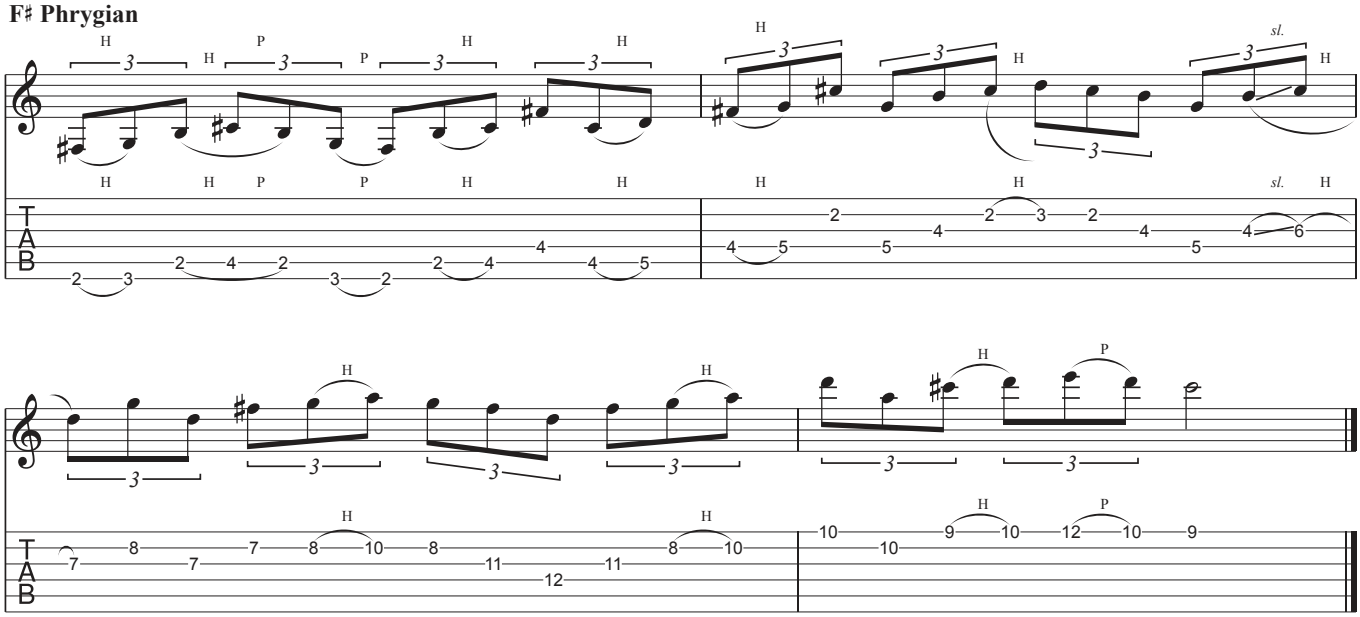
Ex. 197p

E Mixolydian



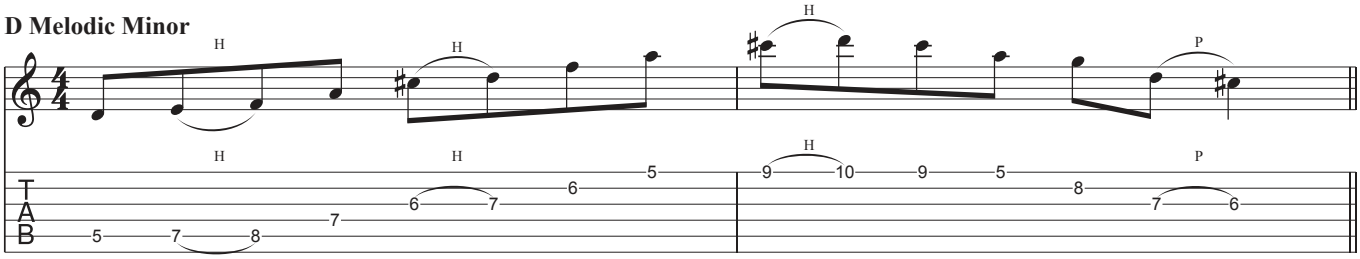
Ex. 197q

F♯ Phrygian



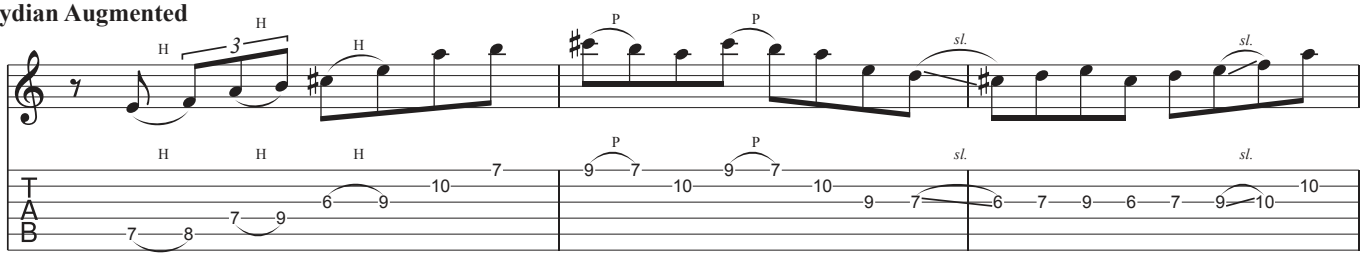
Ex. 198a

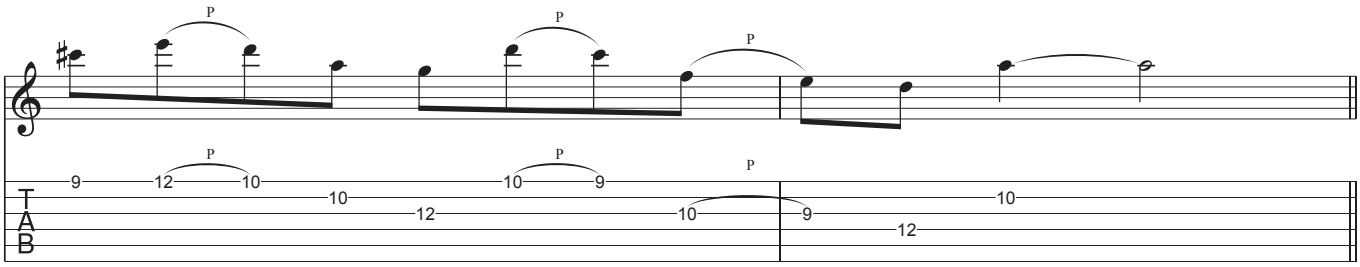
D Melodic Minor



Ex. 198b

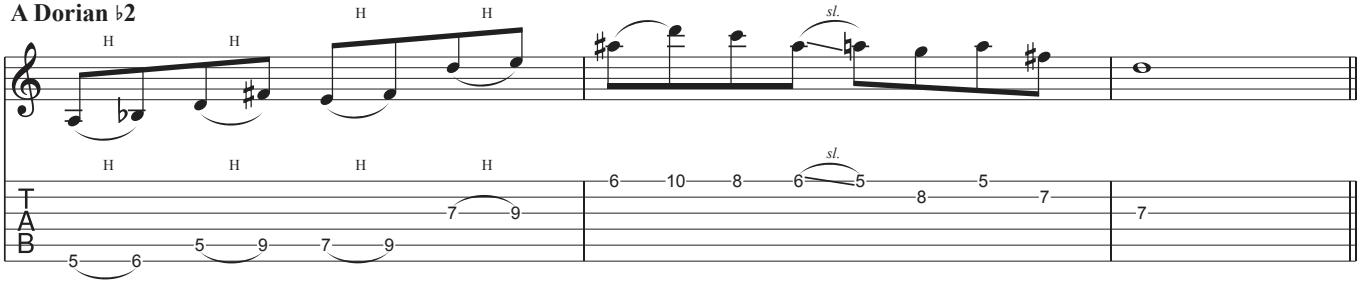
F Lydian Augmented





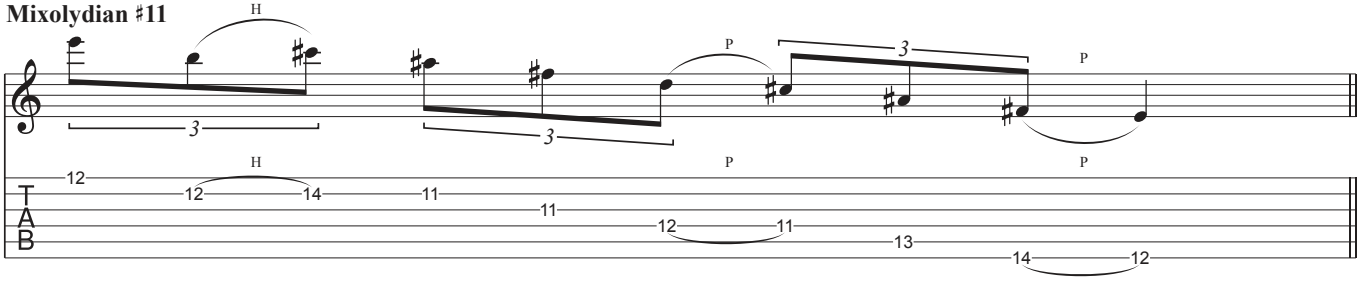
Ex. 198c

A Dorian ♭2



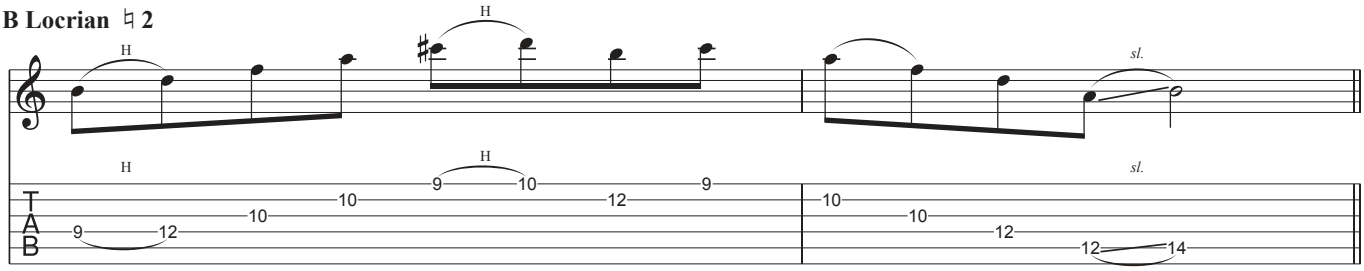
Ex. 198d

E Mixolydian ♯11

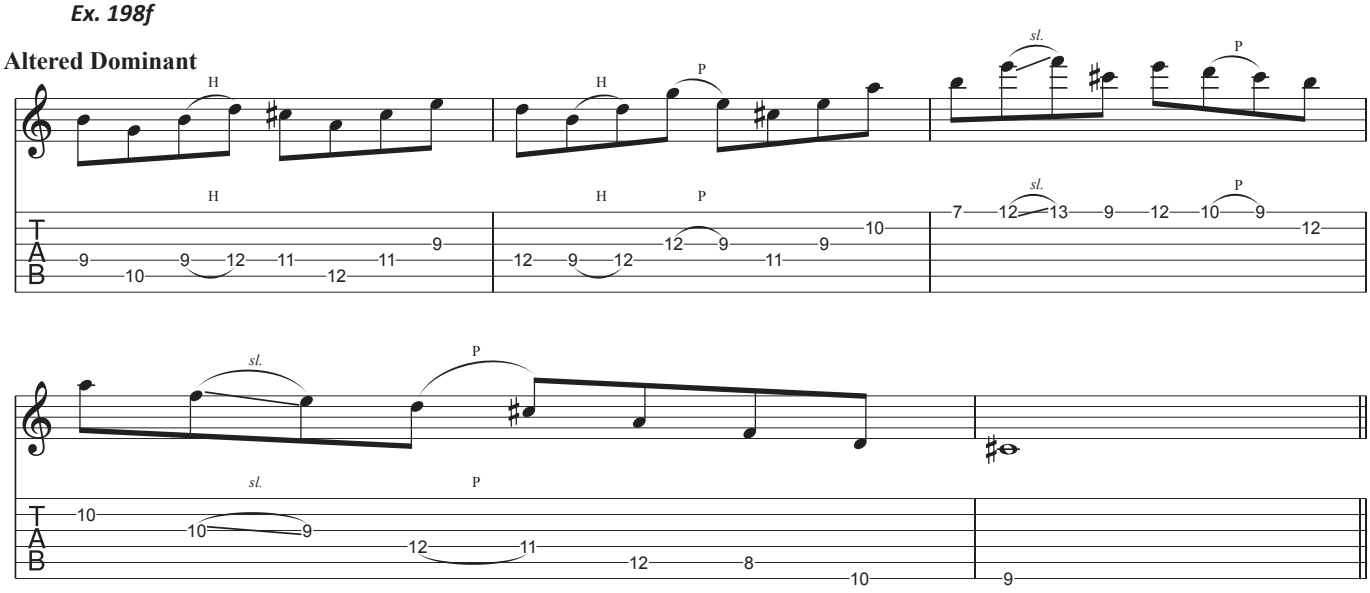


Ex. 198e

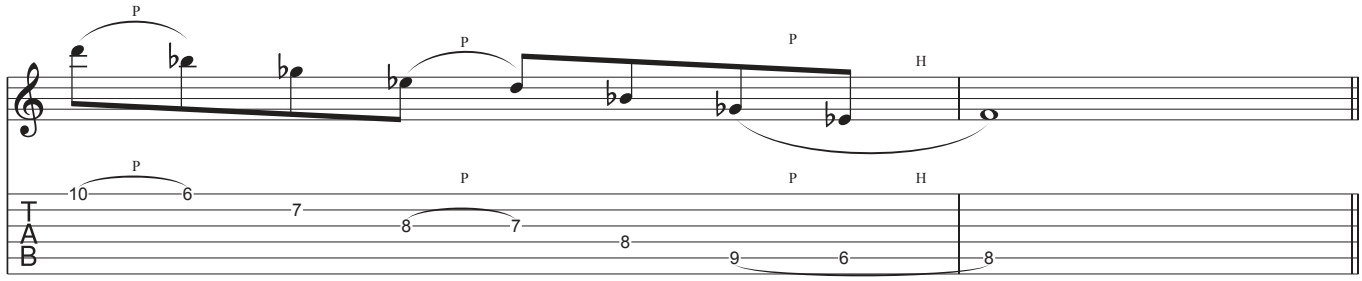
B Locrian ♮2



Ex. 198f

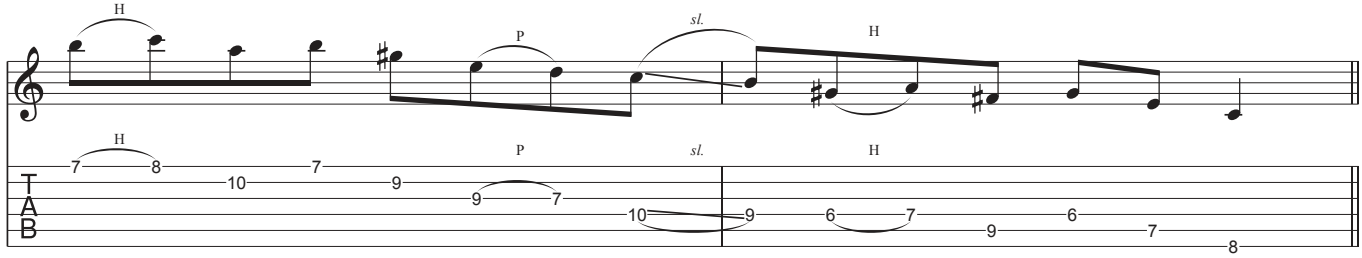
C♯ Altered Dominant

Ex. 198g

E♭ Melodic Minor



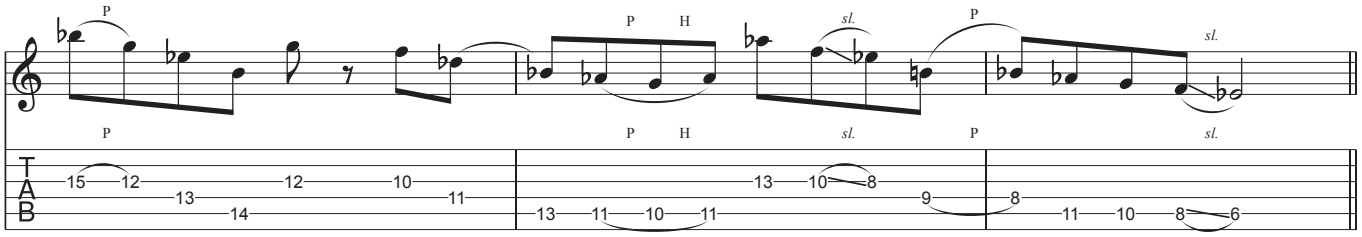
Ex. 198h

C Lydian Augmented



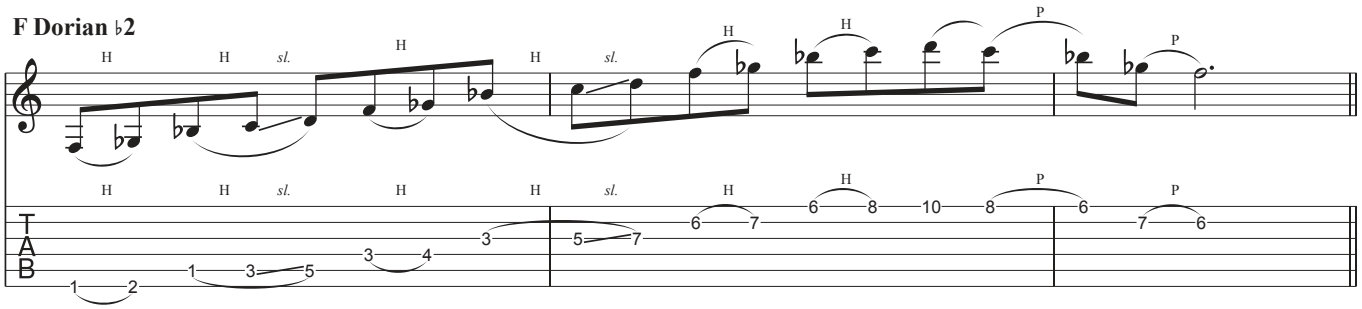
Ex. 198i

G Altered Dominant



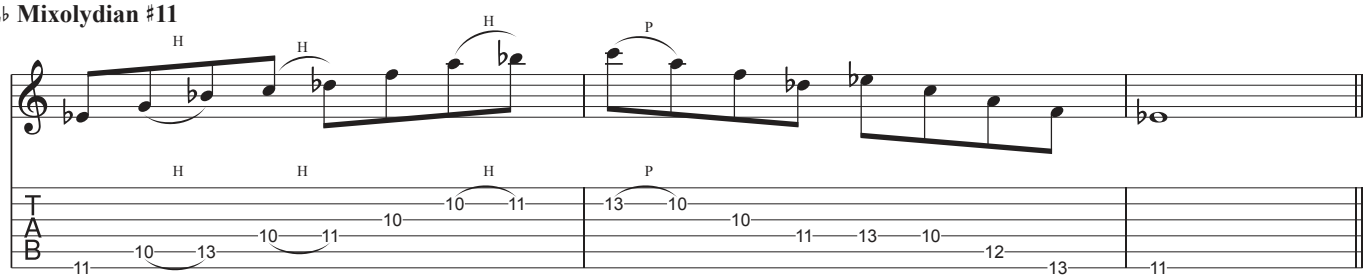
Ex. 198j

F Dorian ♭2



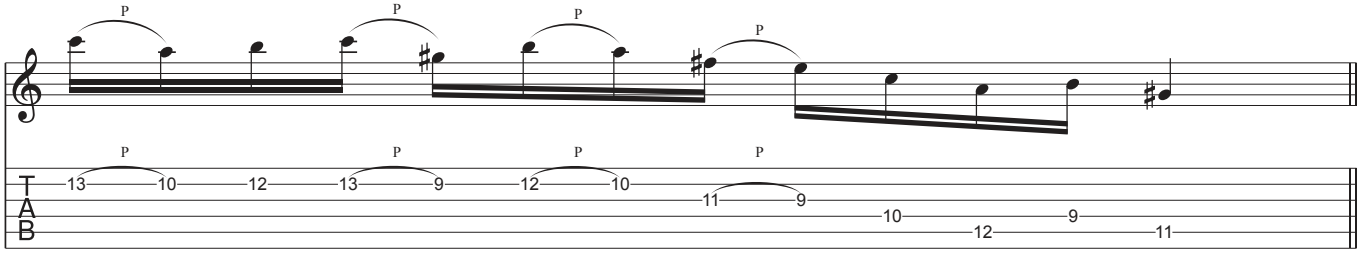
Ex. 198k

E♭ Mixolydian ♯11



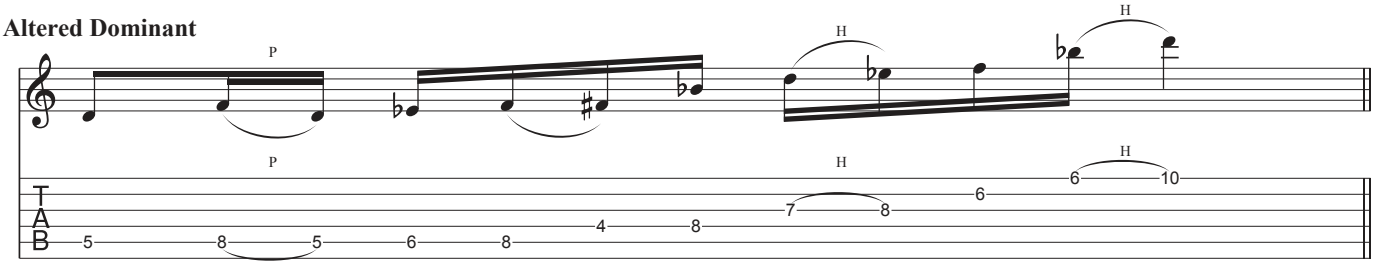
Ex. 198l

F♯ Locrian 𝄫2



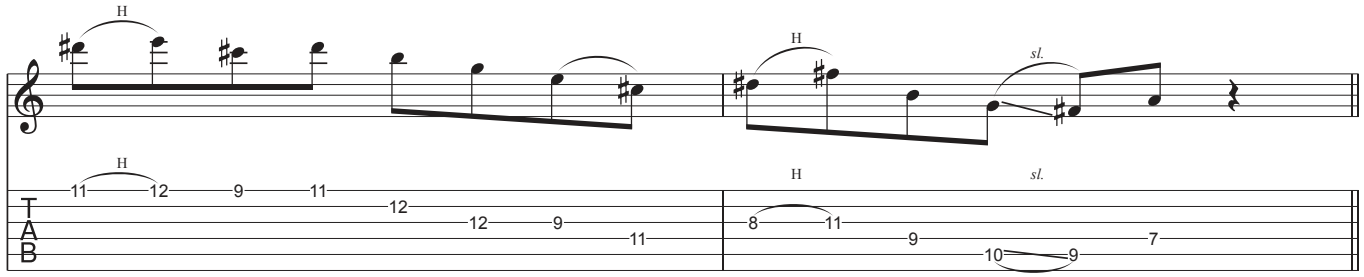
Ex. 198m

D Altered Dominant



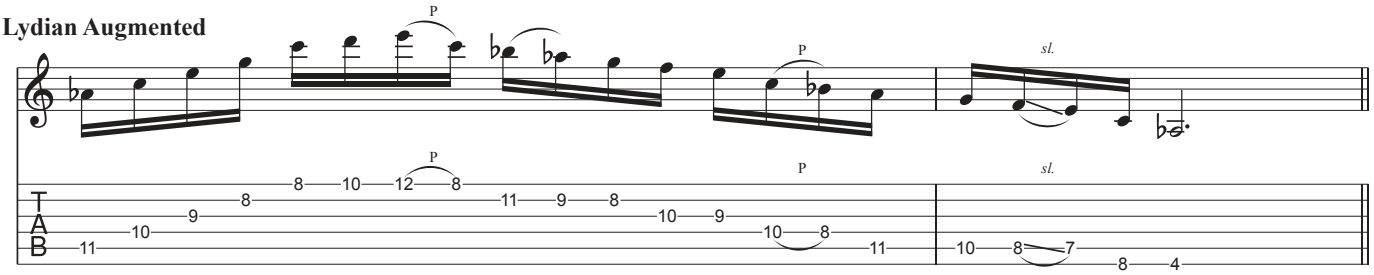
Ex. 198n

E Melodic Minor



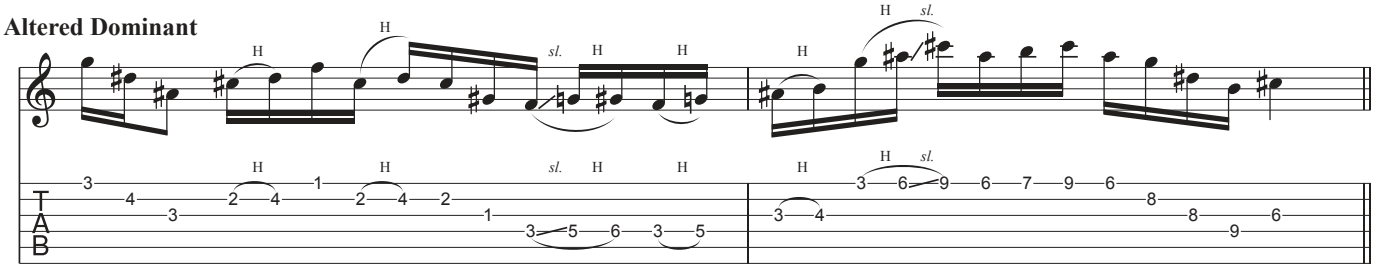
Ex. 198o

A♭ Lydian Augmented



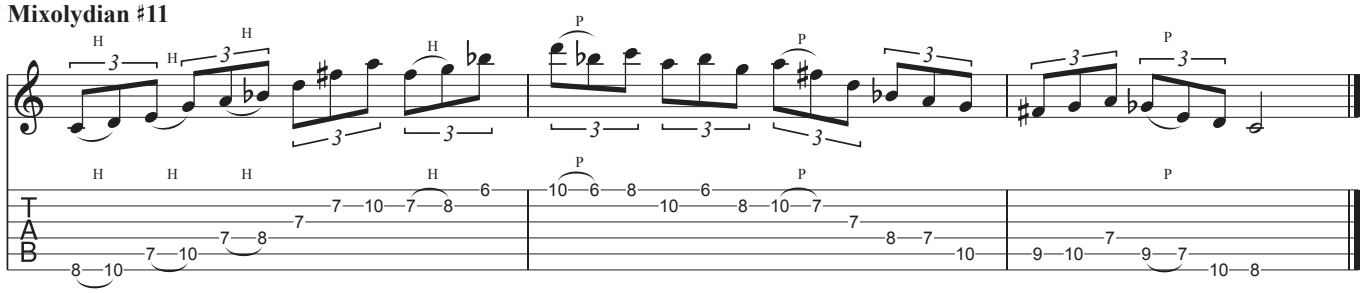
Ex. 198p

G Altered Dominant

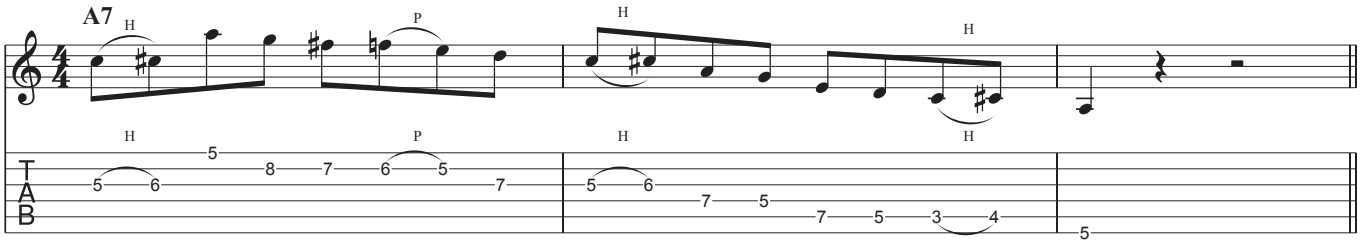


Ex. 198q

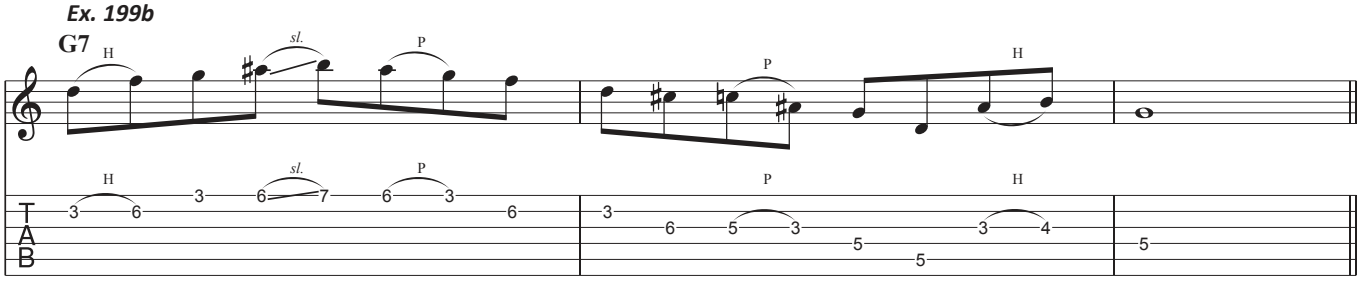
C Mixolydian ♯11



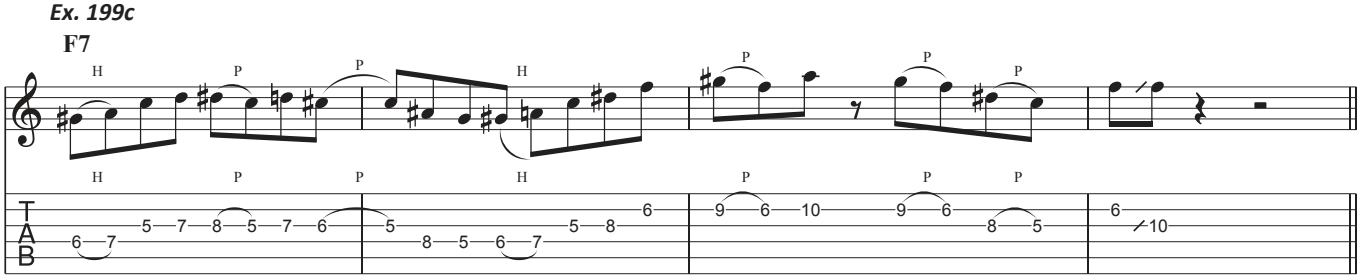
Ex. 199a



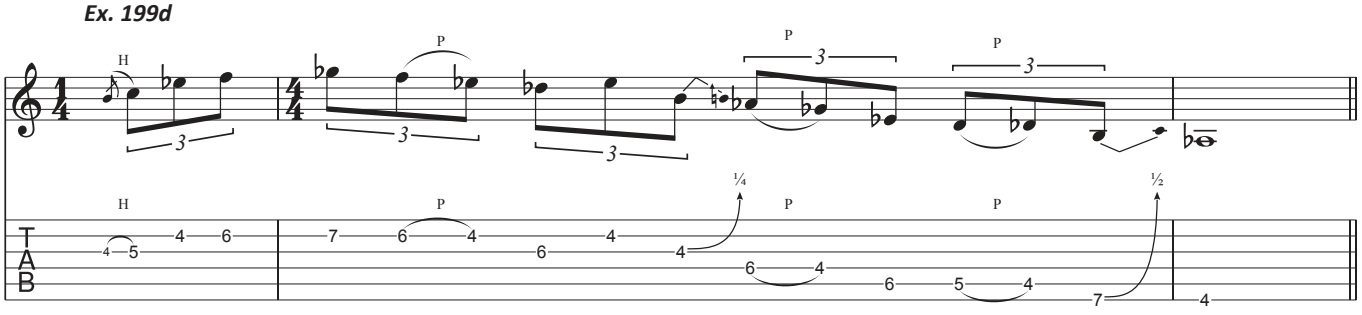
Ex. 199b



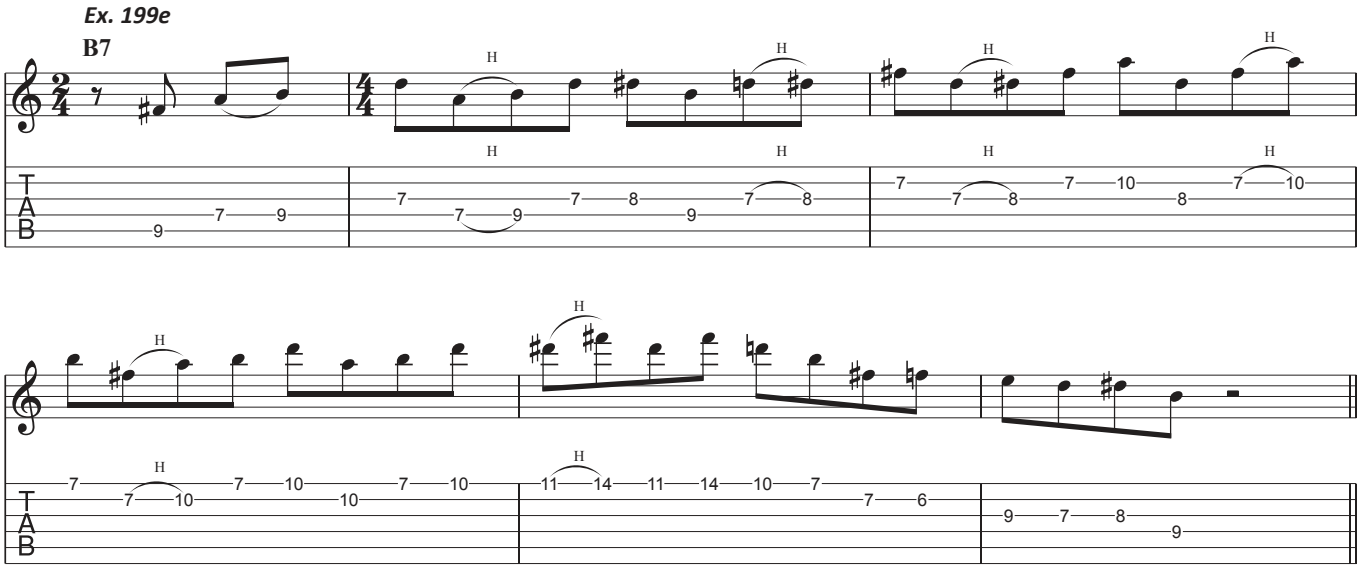
Ex. 199c



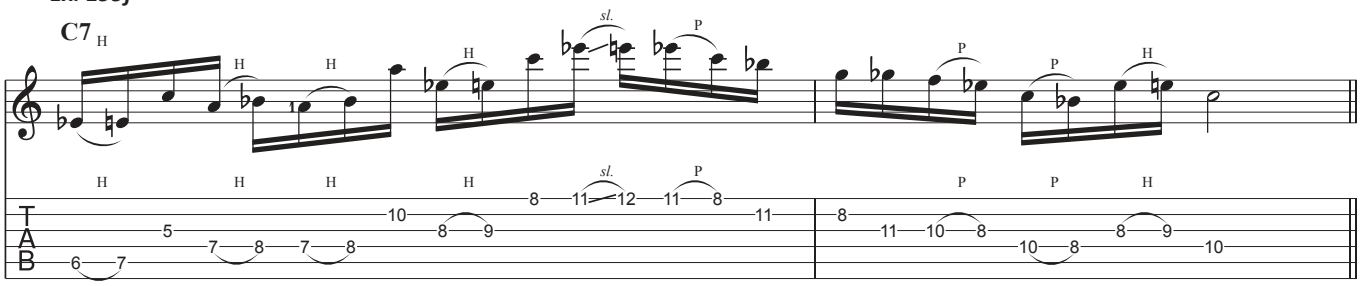
Ex. 199d



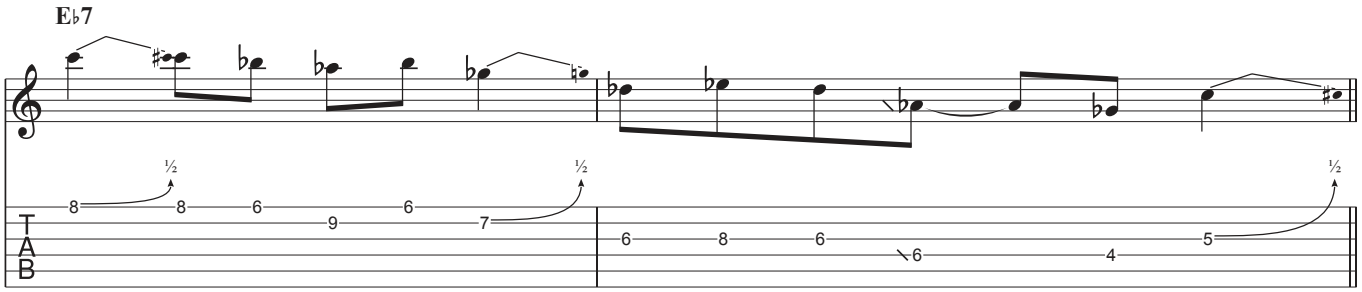
Ex. 199e



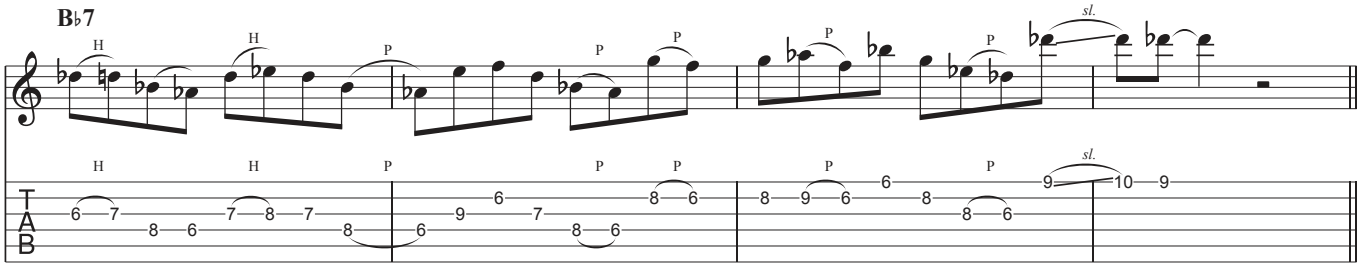
Ex. 199f



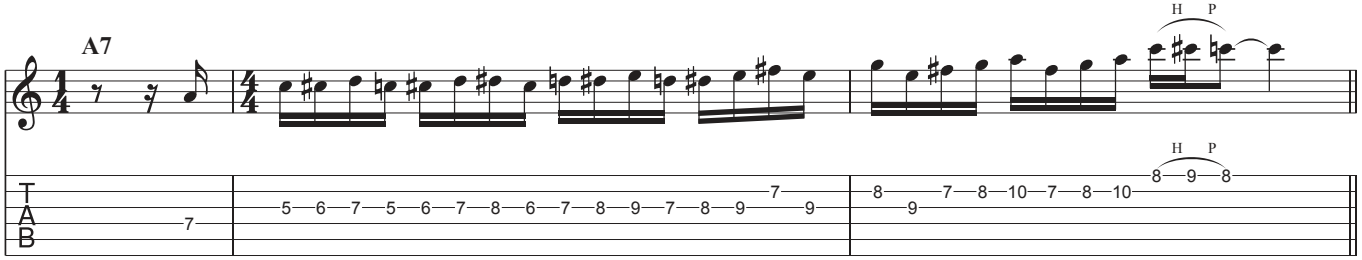
Ex. 199g



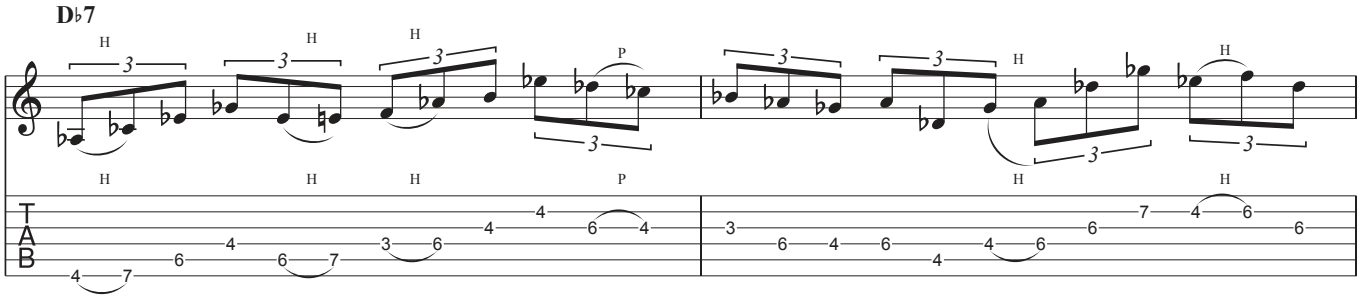
Ex. 199h

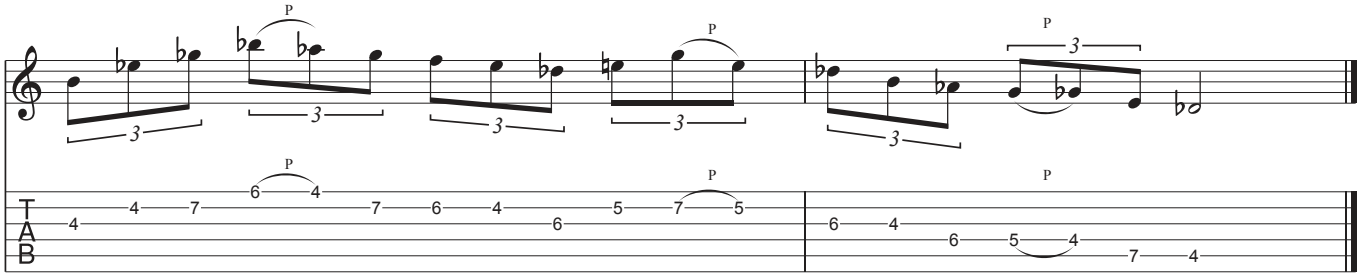


Ex. 199i



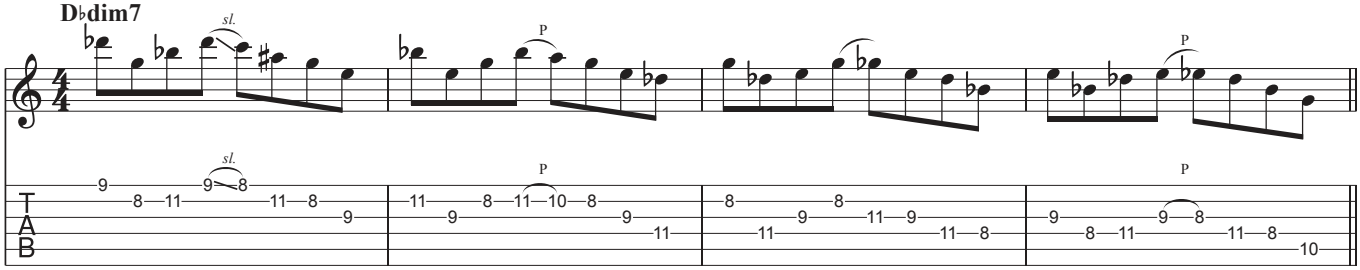
Ex. 199j




P
P
P
P
P
3
3
3
3
3
T
A
B
4 4 7 6 4 7 6 4 6 5 7 5
6 4 6 5 4 7 4

Ex. 200a

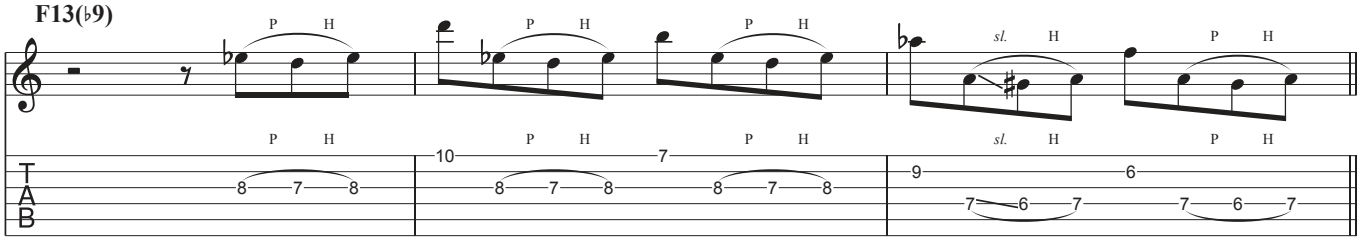
D♭dim7



Repeat down an octave

Ex. 200b

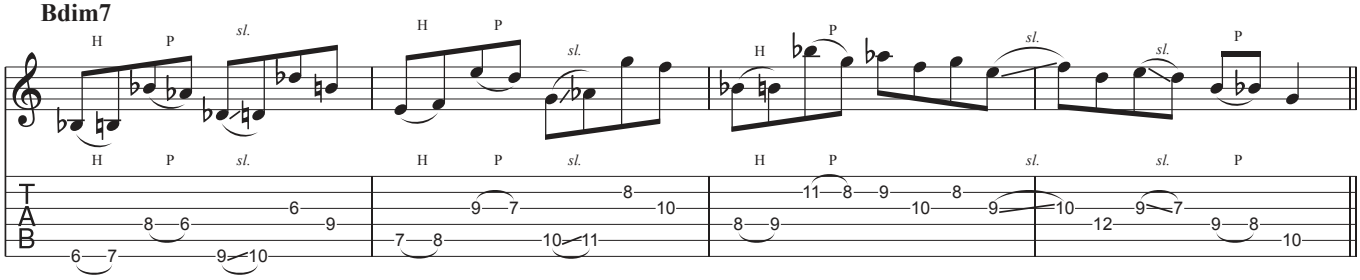
F13(♭9)



Repeat down an octave

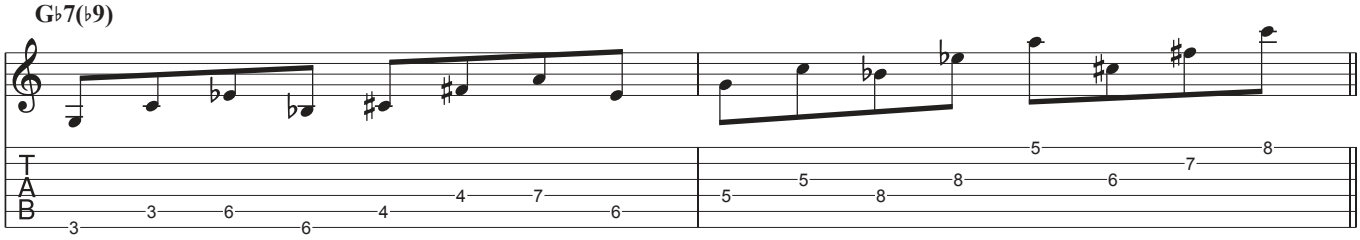
Ex. 200c

Bdim7



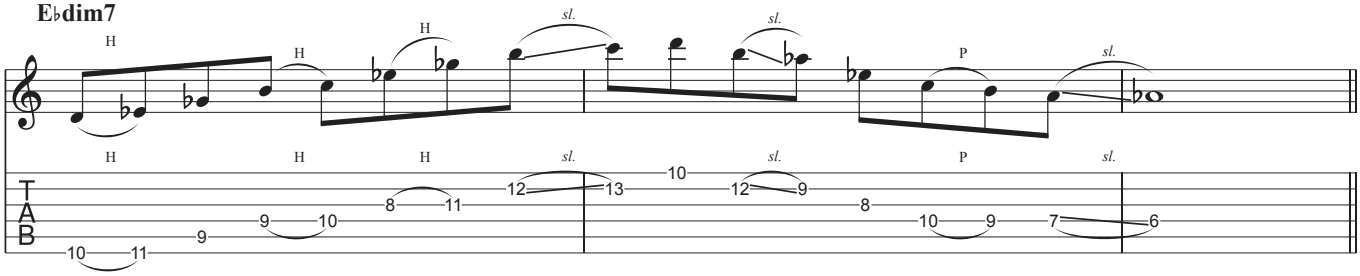
Ex. 200d

G♭7(♭9)

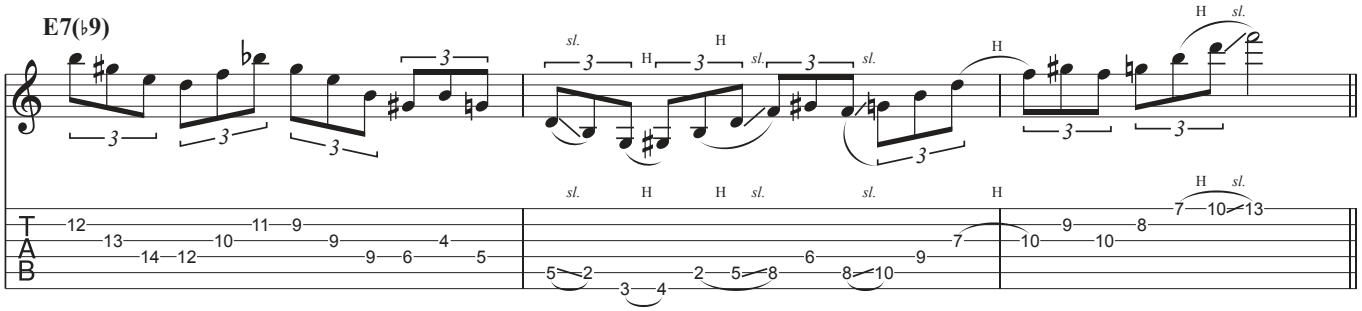


Ex. 200e

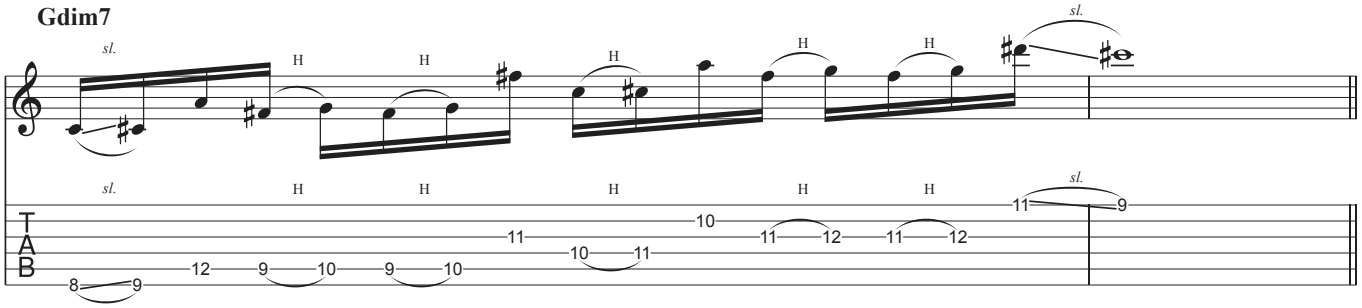
E♭dim7



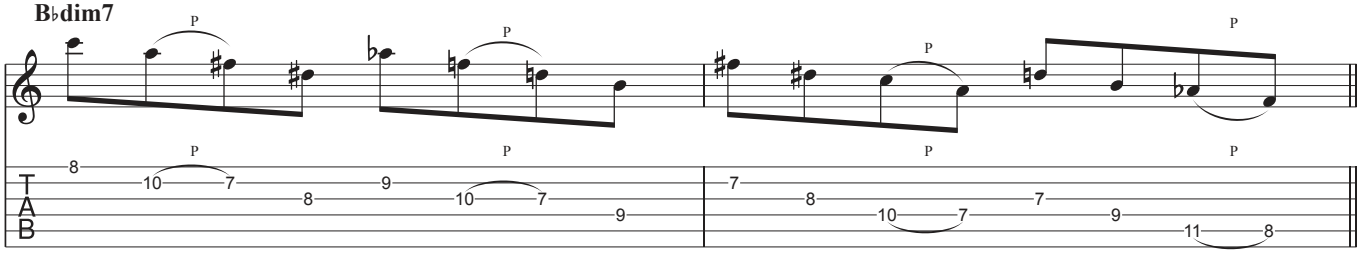
Ex. 200f



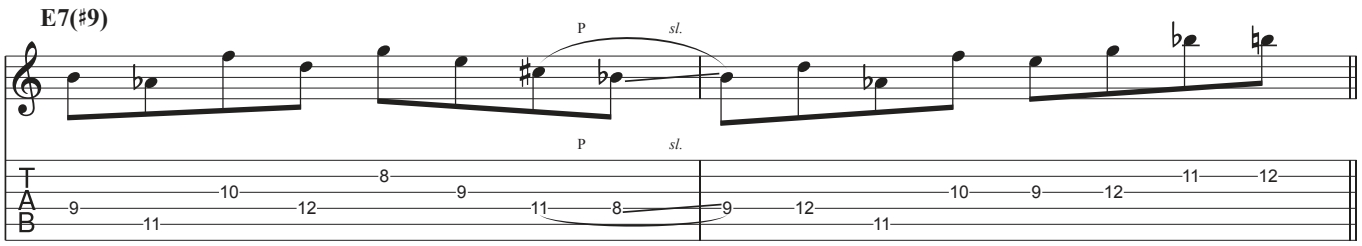
Ex. 200g



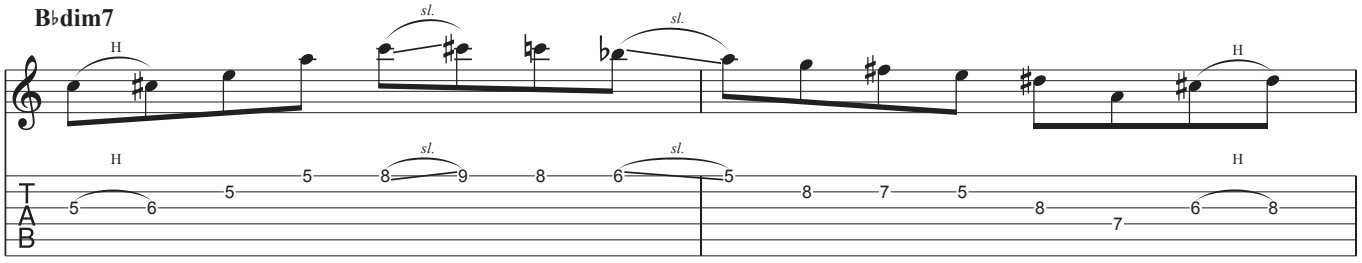
Ex. 200h

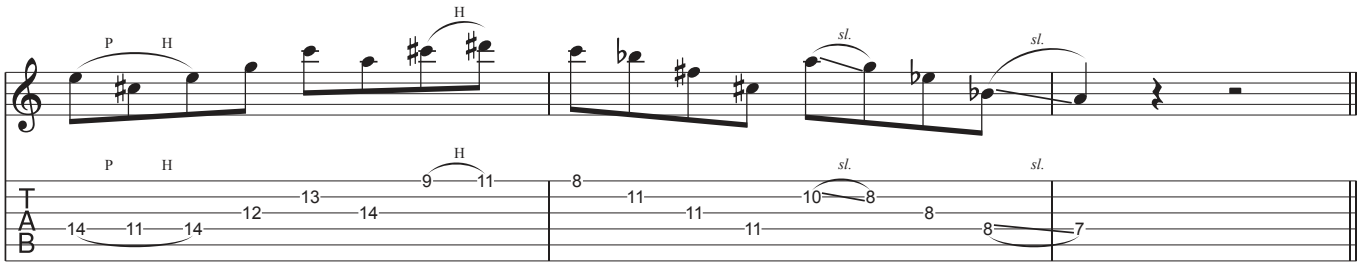


Ex. 200i



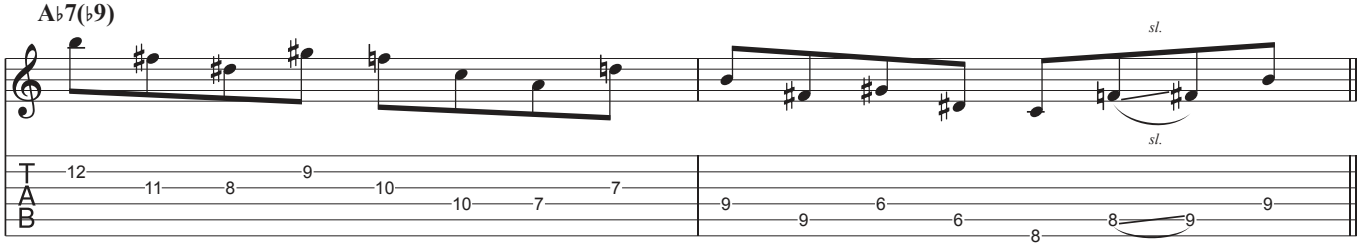
Ex. 200j





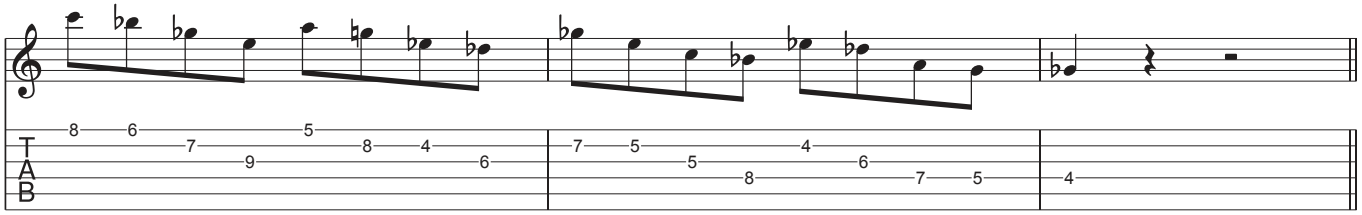
Ex. 200k

A♭7(♭9)



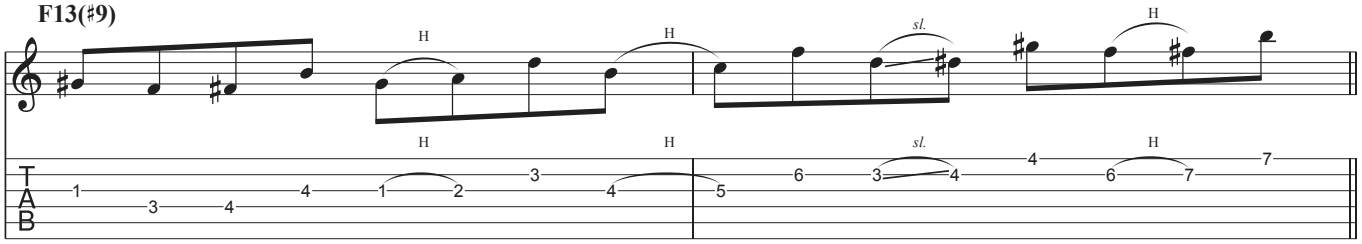
Ex. 200l

E♭dim7



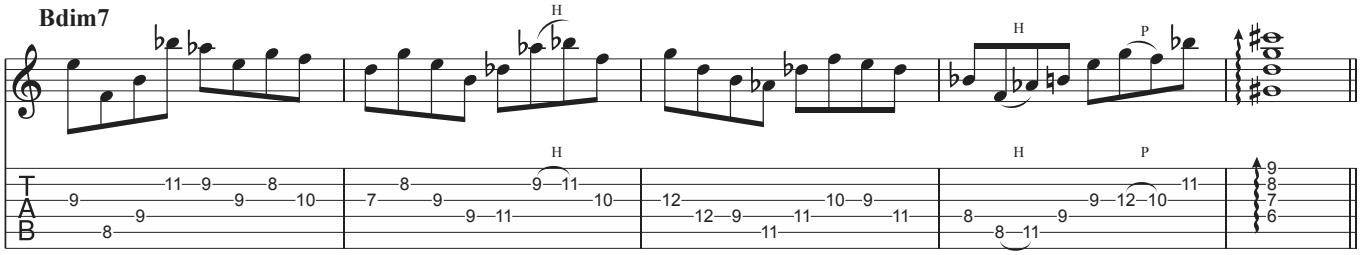
Ex. 200m

F13(♯9)



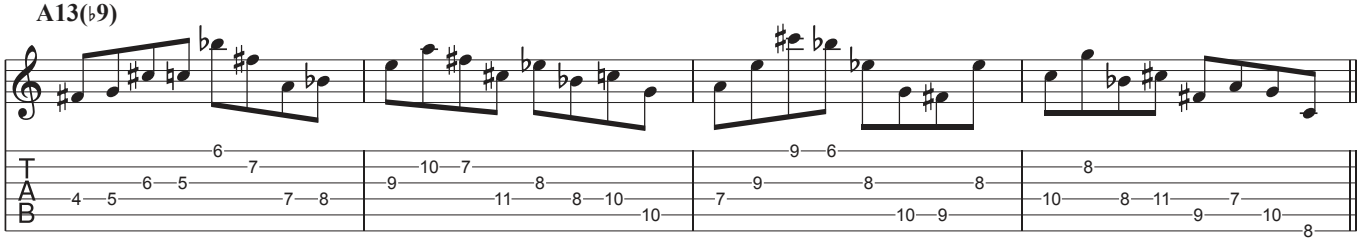
Ex. 200n

Bdim7

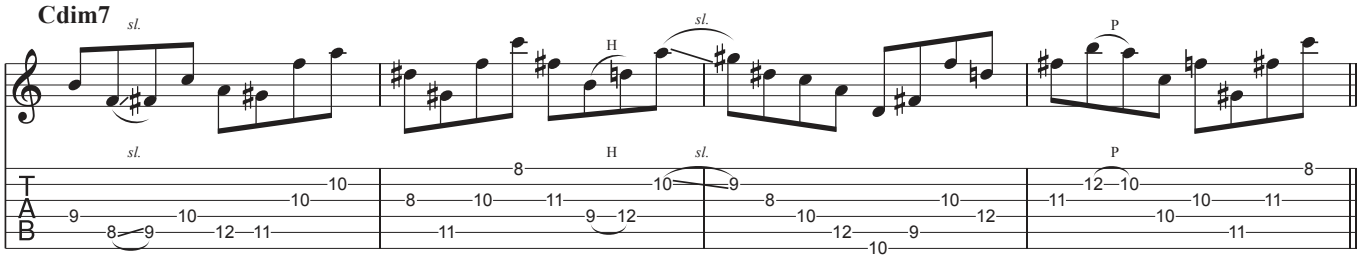


Ex. 200o

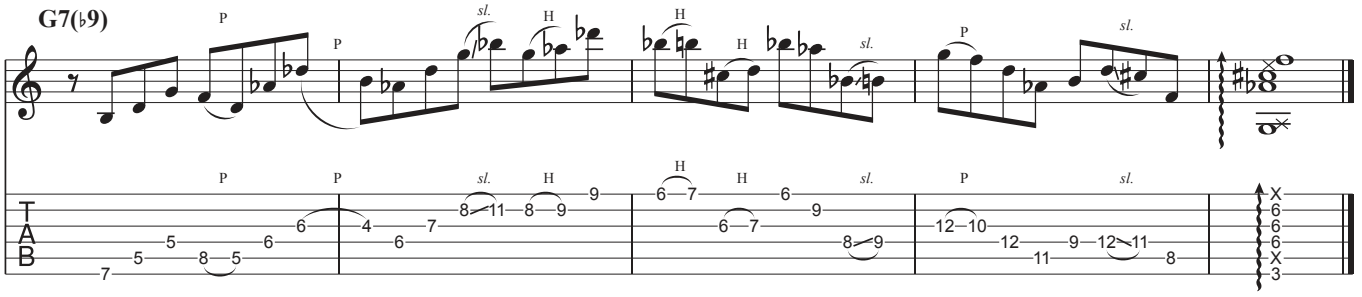
A13(♭9)



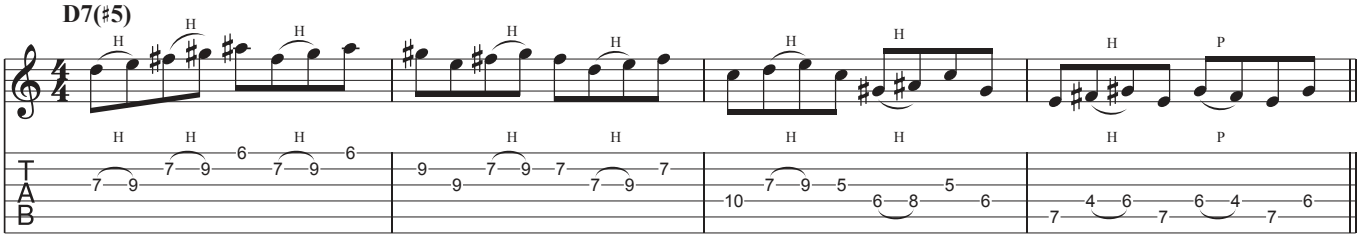
Ex. 200p



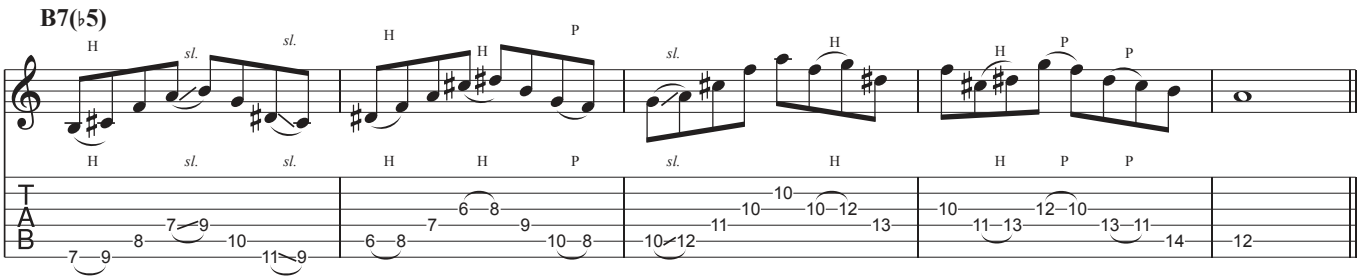
Ex. 200q



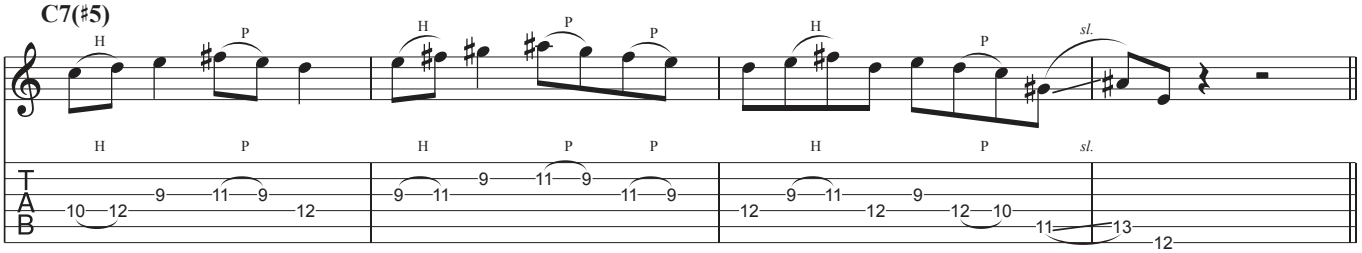
Ex. 201a



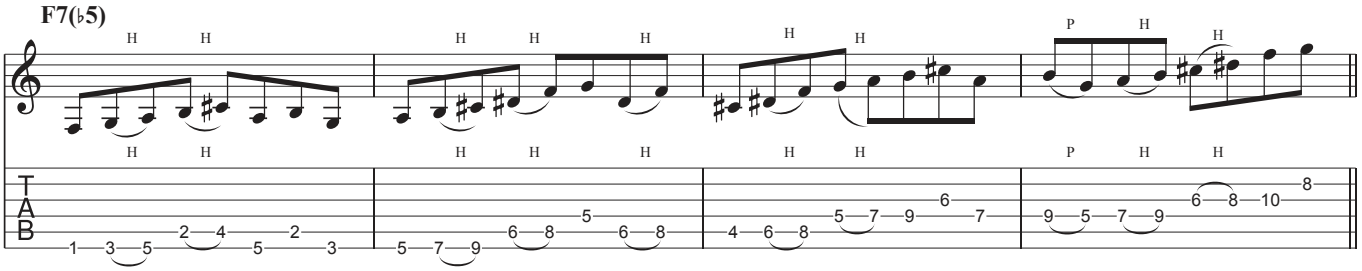
Ex. 201b



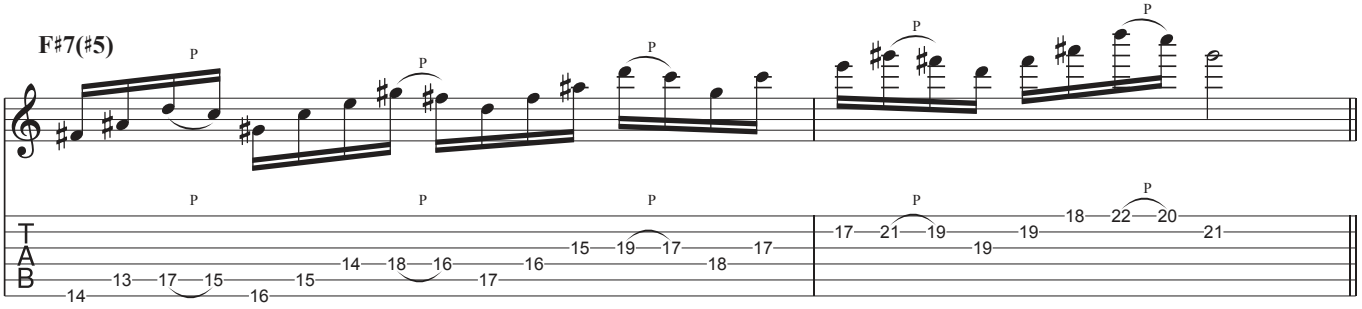
Ex. 201c



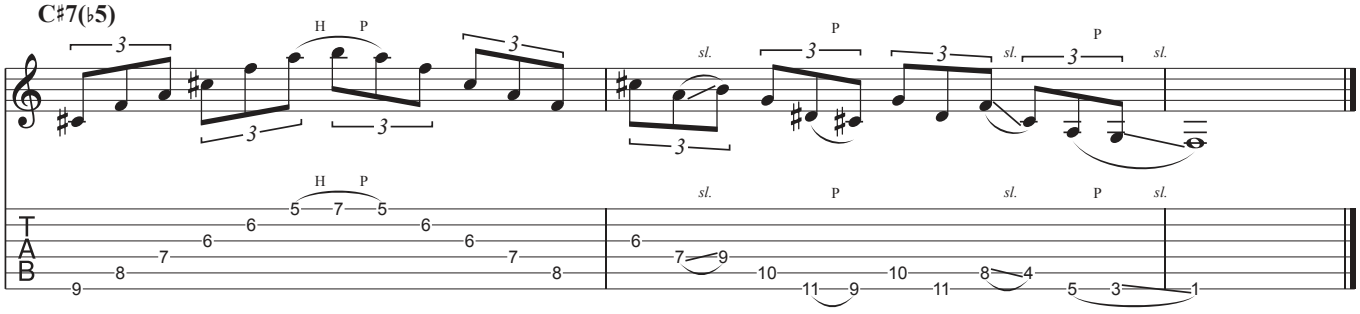
Ex. 201d



Ex. 201e

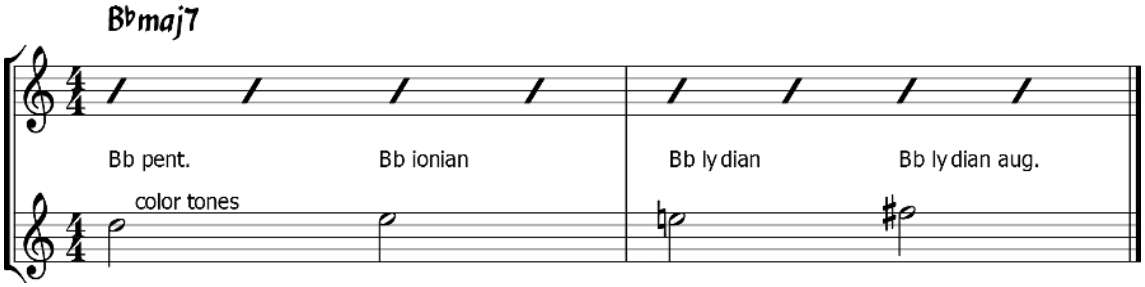
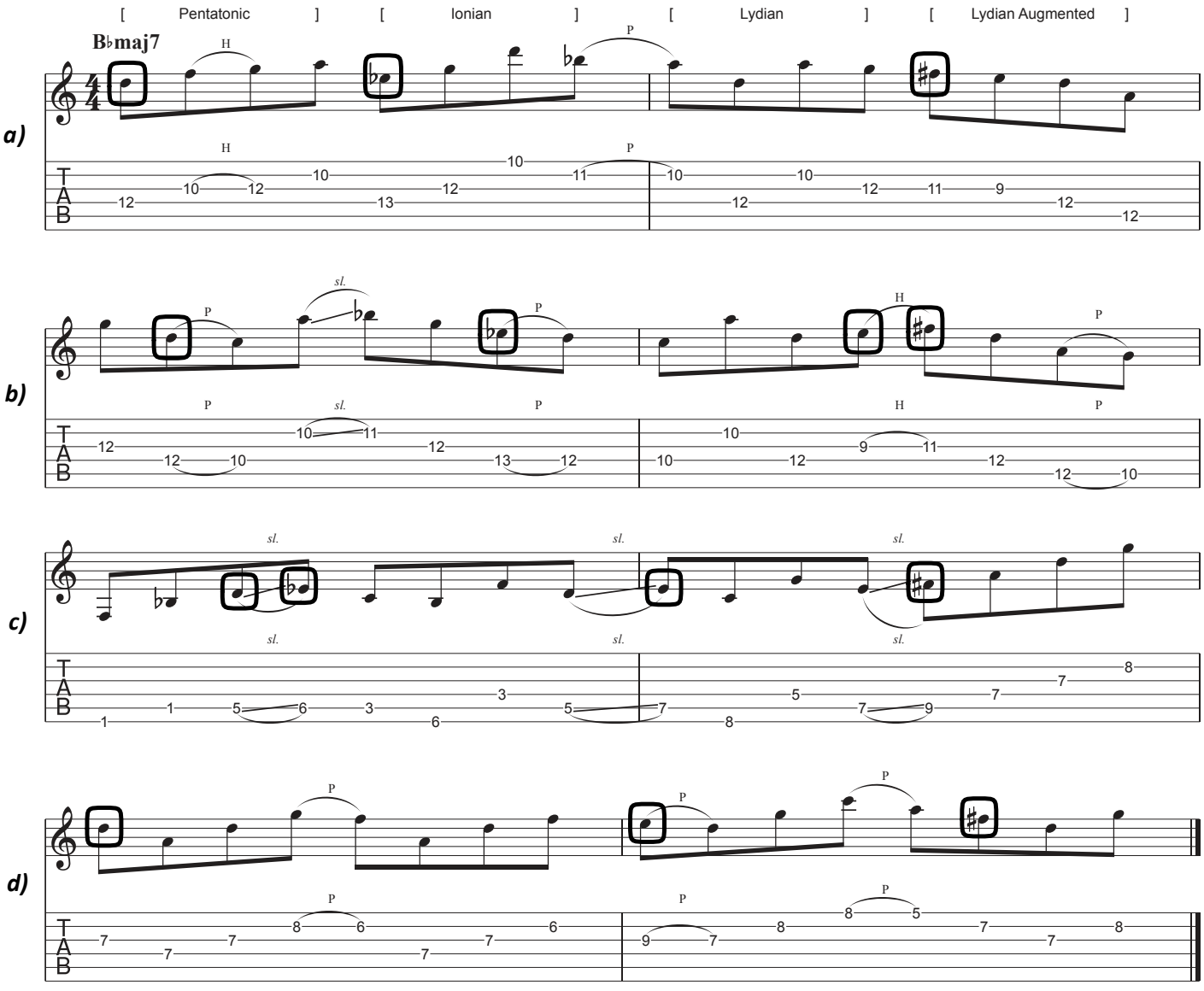


Ex. 201f



Modal Mixture

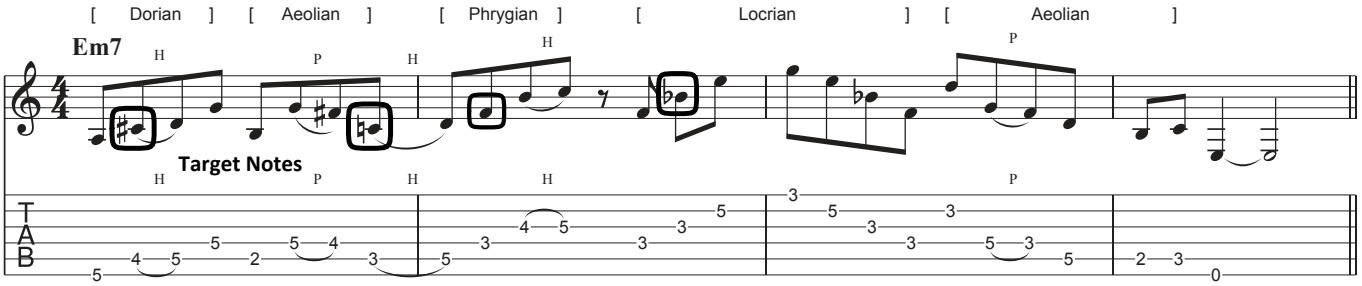
Modal Mixture is the combination of one or more modal scales over a given chord change.

Ex. 202



The color tones are the most characteristic notes of the mode. I have used them as structural points from which to build my lines.

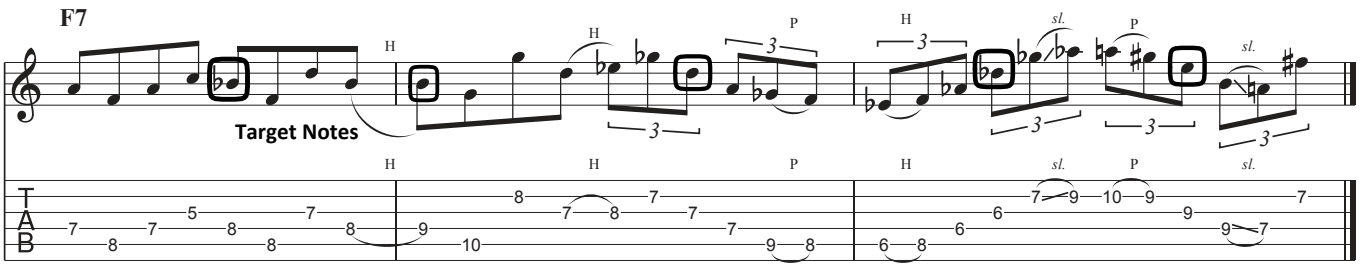
Minor Sounds

Ex. 203



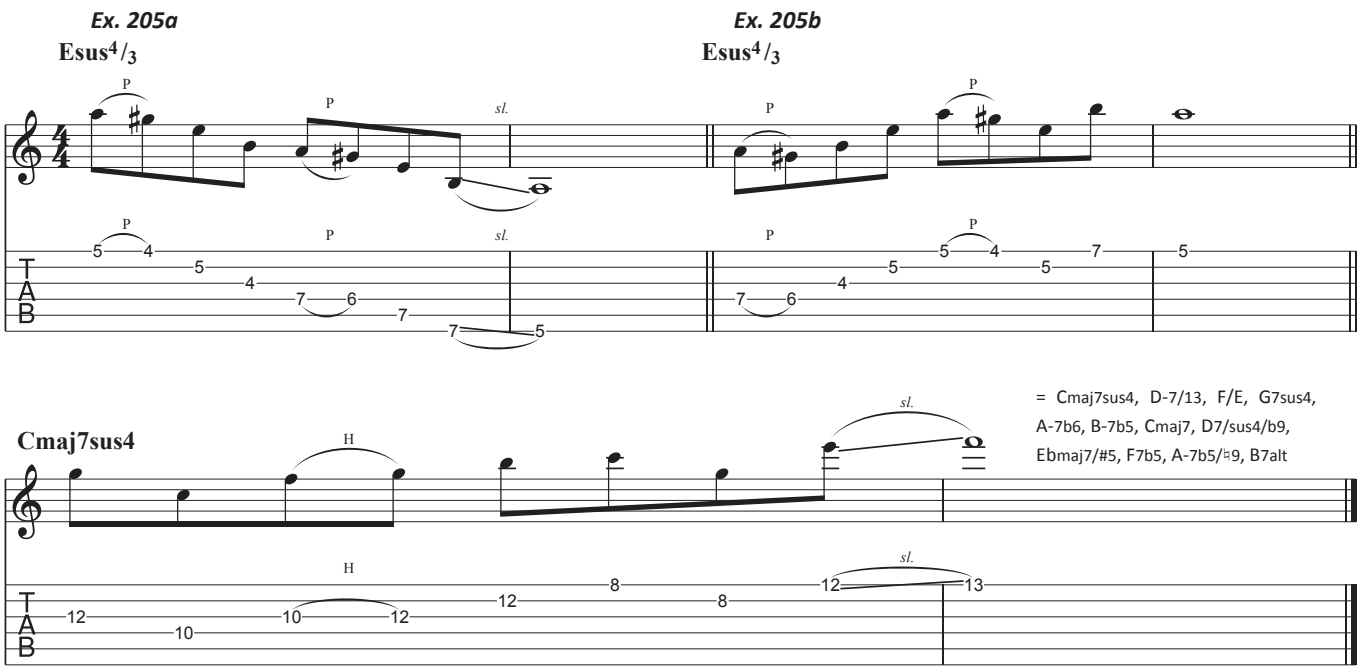
Dominant Sounds

Ex. 204



Using One Interval Set Over Many Chords

Many of the greatest players have been the ones who learned to incorporate a limited amount of melodic material into the most harmonic structures.



This particular Esus4/3 hybrid structure could be used over these chords.

E^{maj7}, E^{sus4}, $F\#^{-7/9}$, A/G#, A^{maj7}, B^{7sus4}, A/B, $C\#^{-7b6}$, $D\#^{-7b5}$, A^{-maj7}, $B^{7/sus4/b9}$, $C^{maj7\#5}$, D^{7b5}, E^{7sus4}, $F\#^{-9b5}$, $G\#^{7alt}$

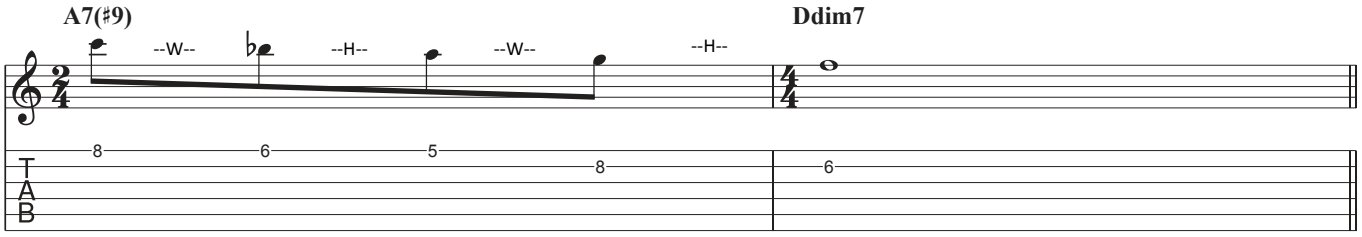
Hybrid structures are useful in these instances because many can be found in more than one scale.

This concept can be similarly applied by transposing a particular interval structure over many chord changes

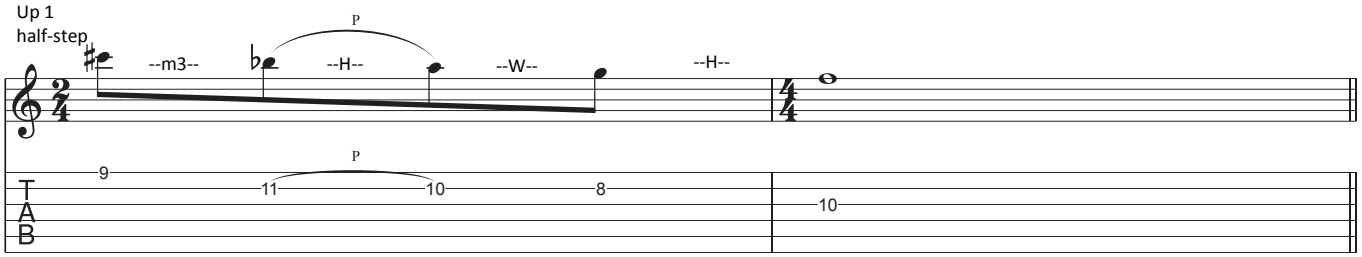
The interval structure used is comprised of a whole-step half-step alternation

--H-- = Half-Step
--W-- = Whole-Step
--m3-- = Minor Third

Ex. 206a

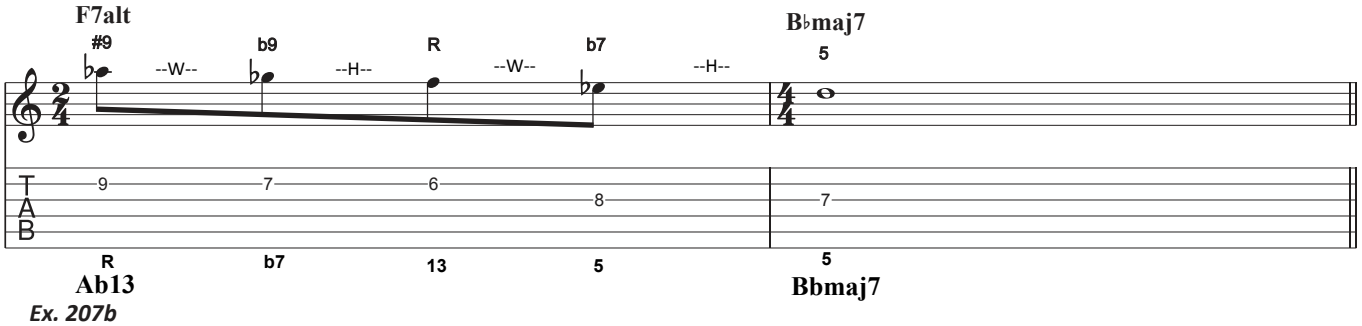


Ex. 206b For the sake of variety

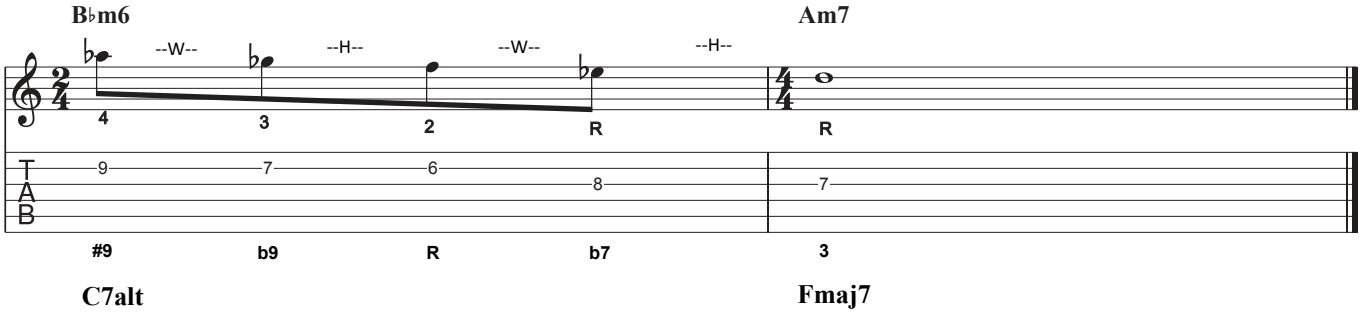


Because of its interval structure, this type of idea can function in a variety of ways

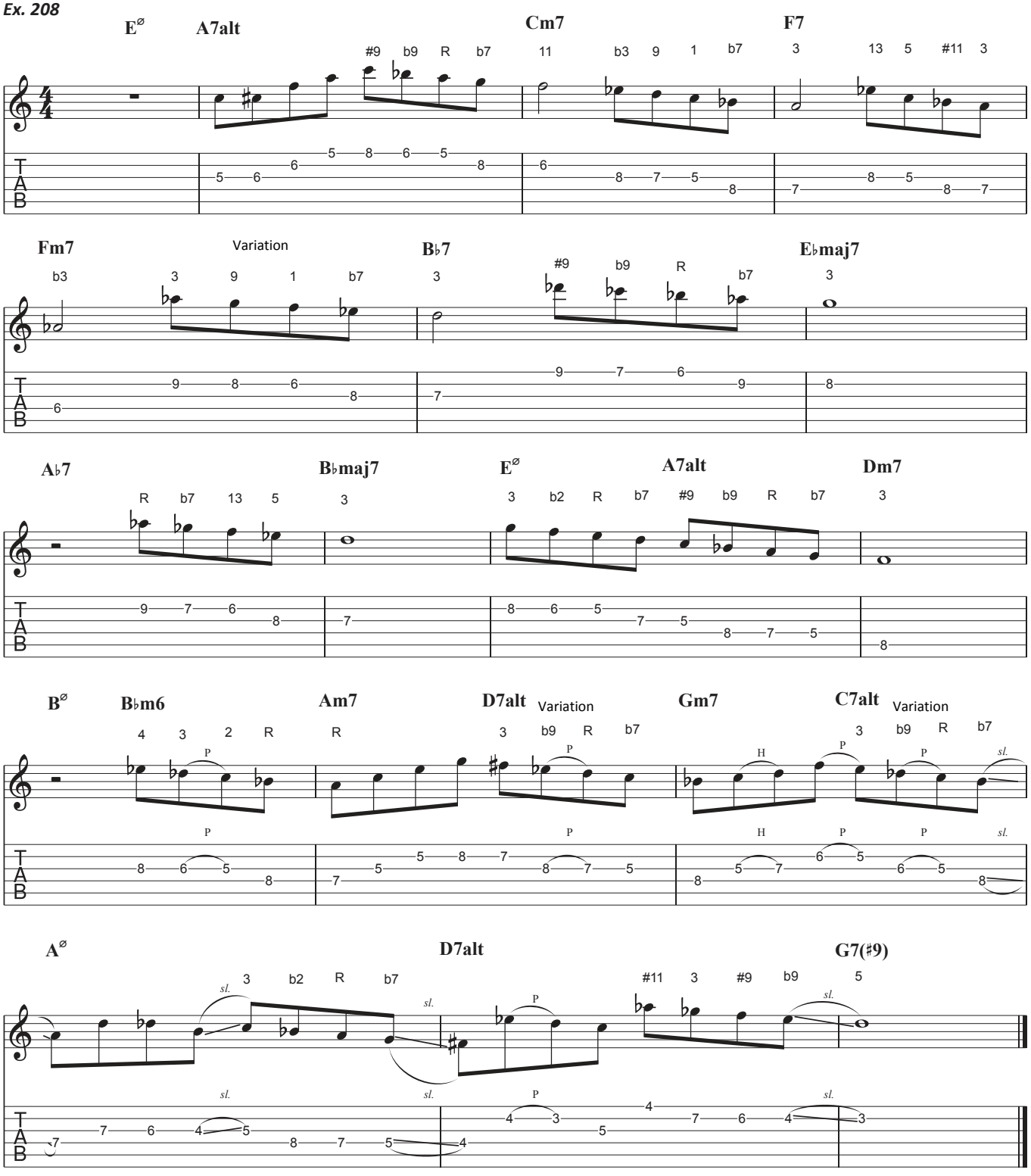
Ex. 207a



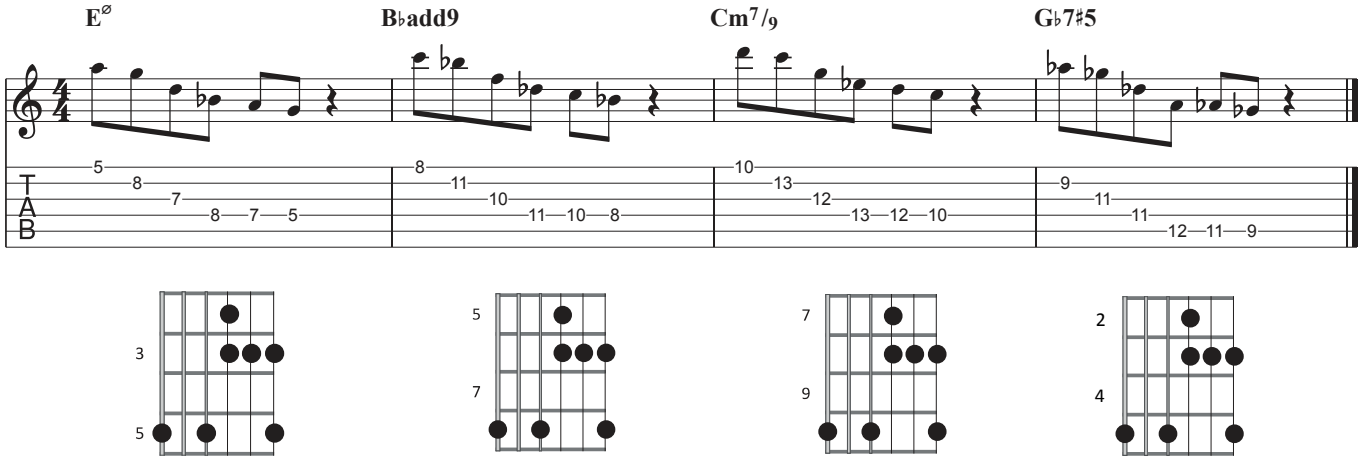
Ex. 207b



This example is over a standard progression

Ex. 208

Another great way to practice is to take a particular chord shape (triad, seventh chord, hybrid structure) and use it over as many chord changes as possible in a tune

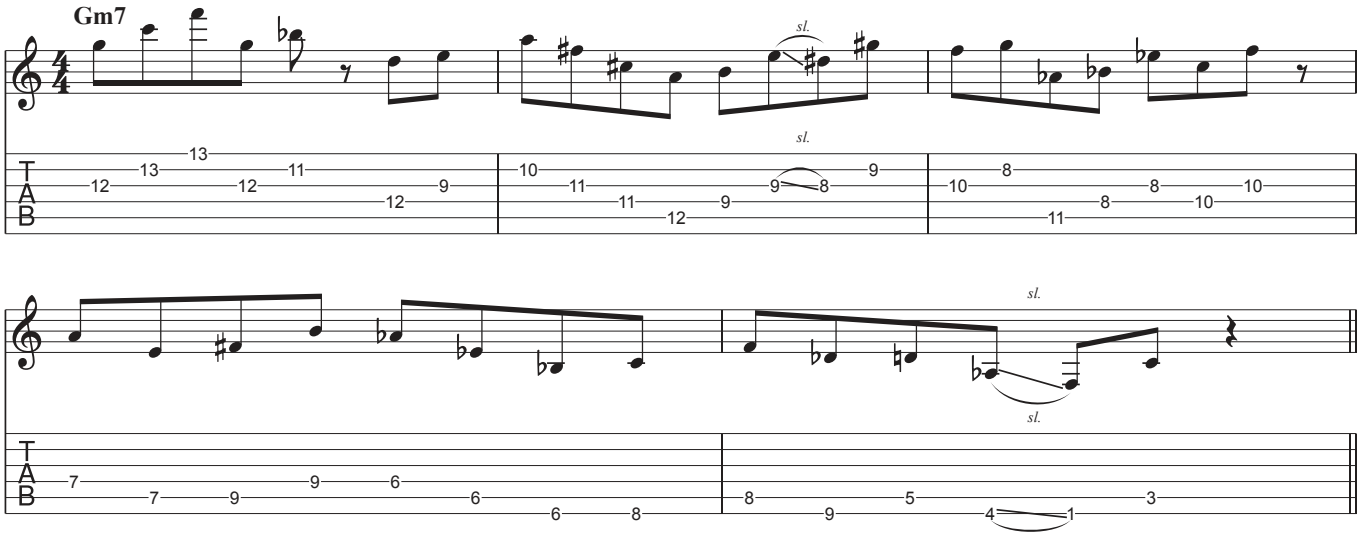
Ex. 209



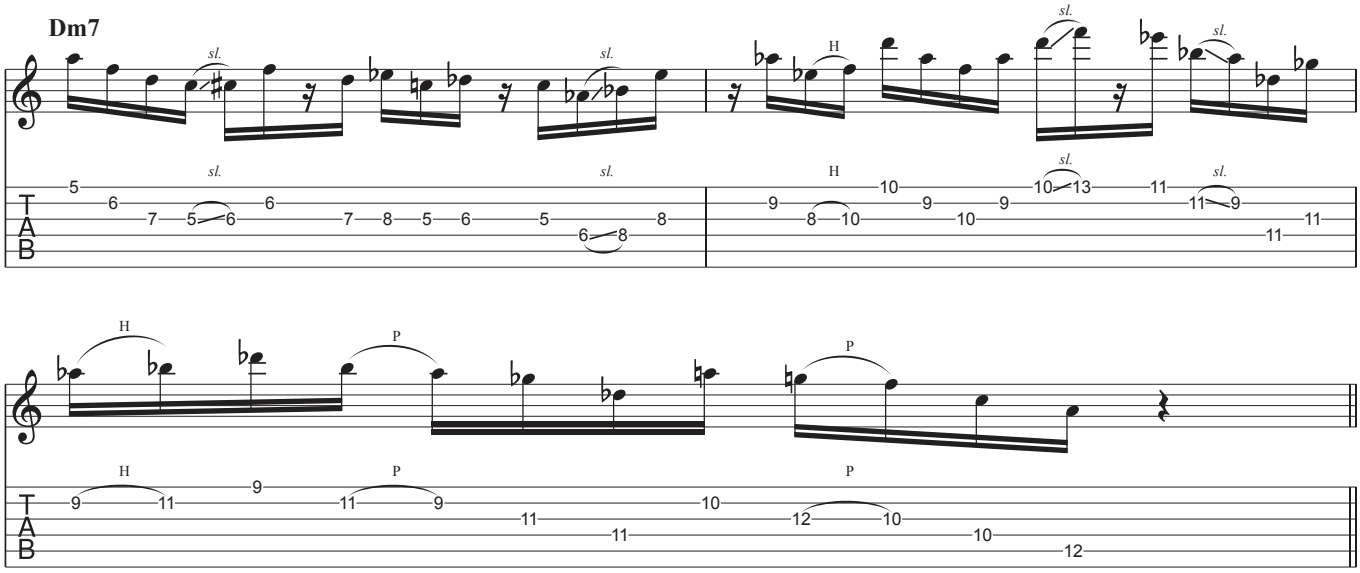
Outside Playing Over One Chord

Here are a few examples of outside playing approaches on one chord. You provide the analysis.

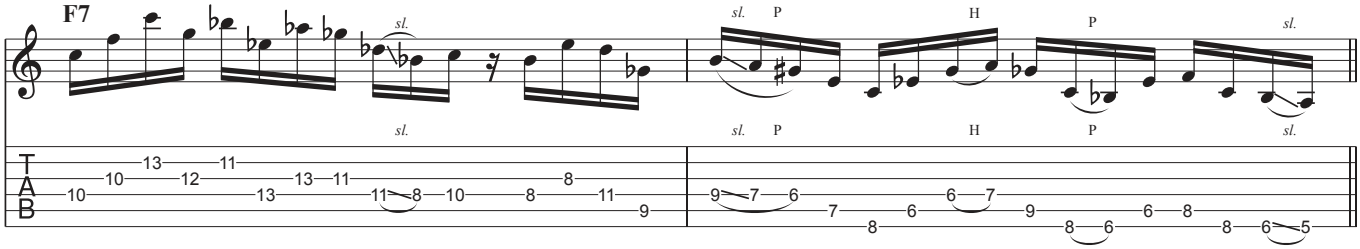
Ex. 210a



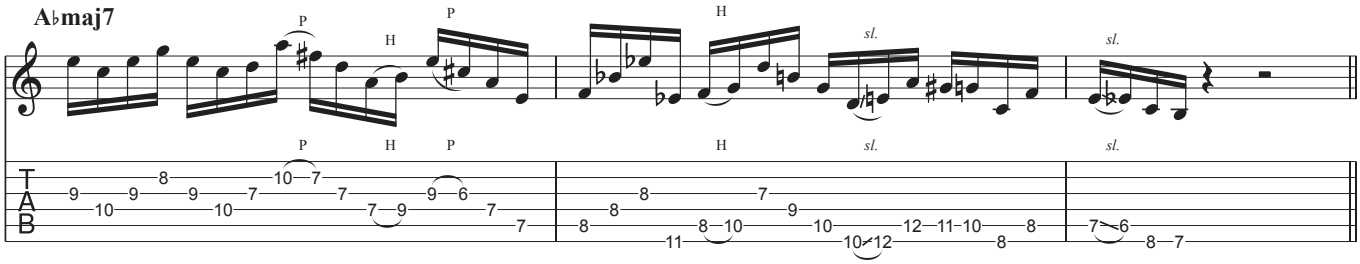
Ex. 210b



Ex. 210c

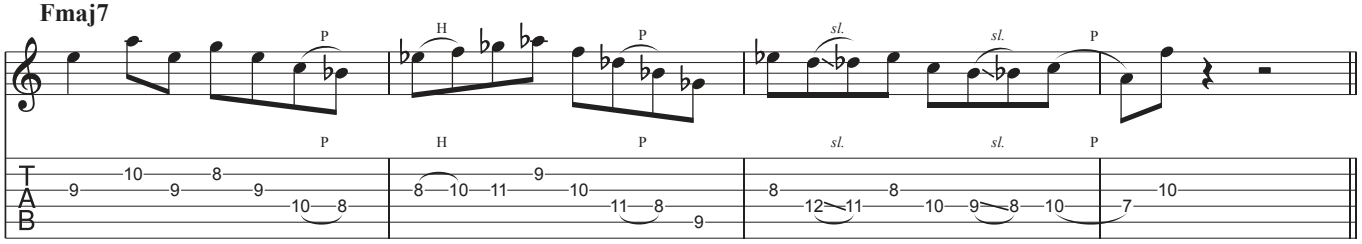


Ex. 210d



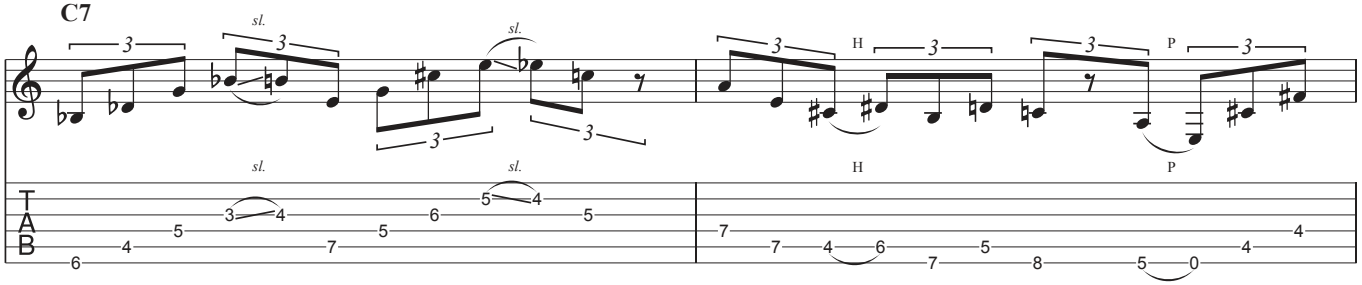
Ex. 210e

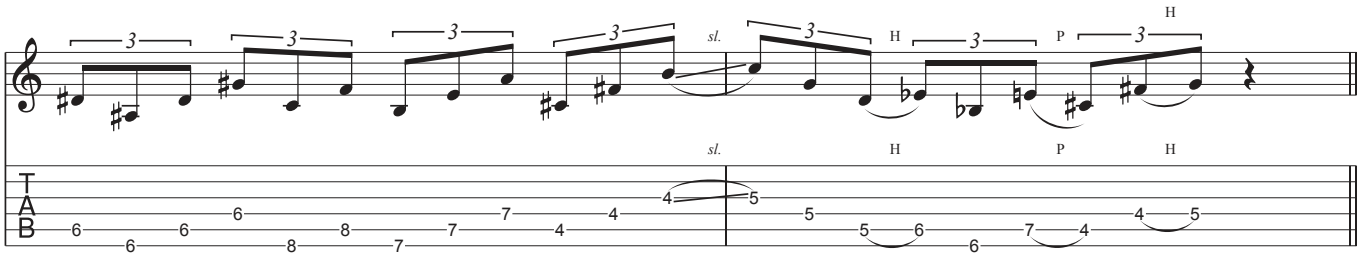
Fmaj7



Ex. 210f

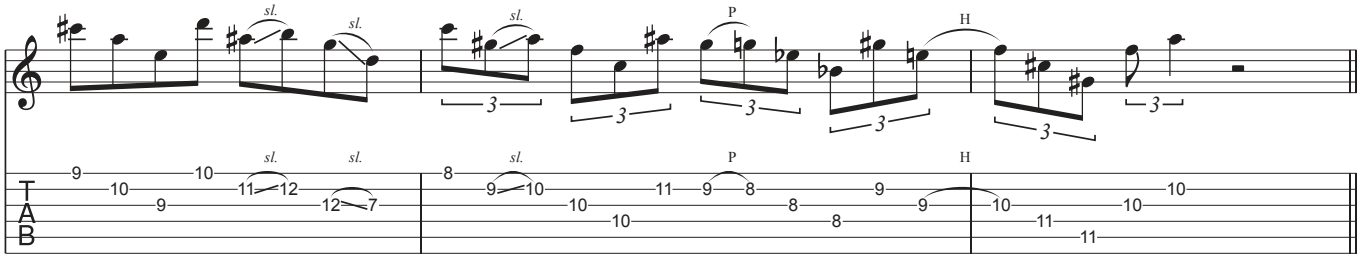
C7





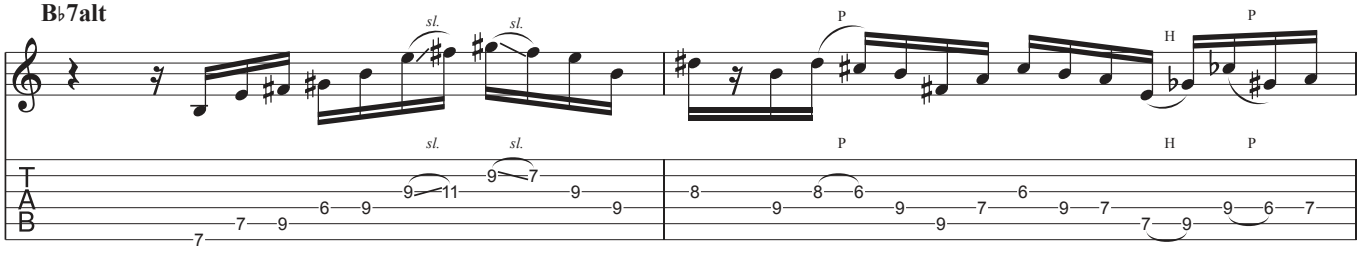
Ex. 210g

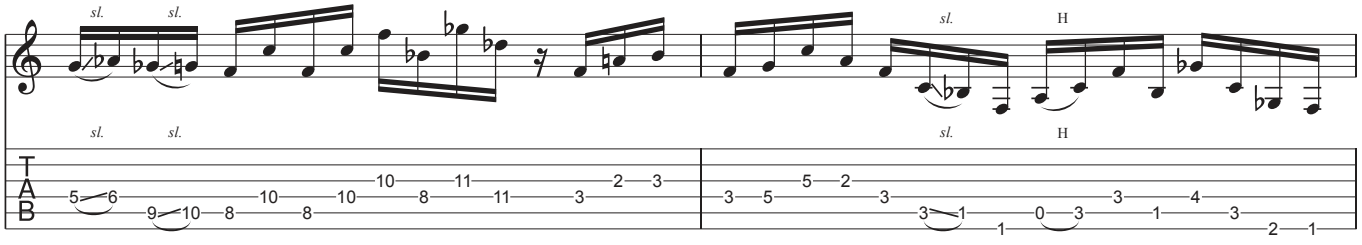
A7

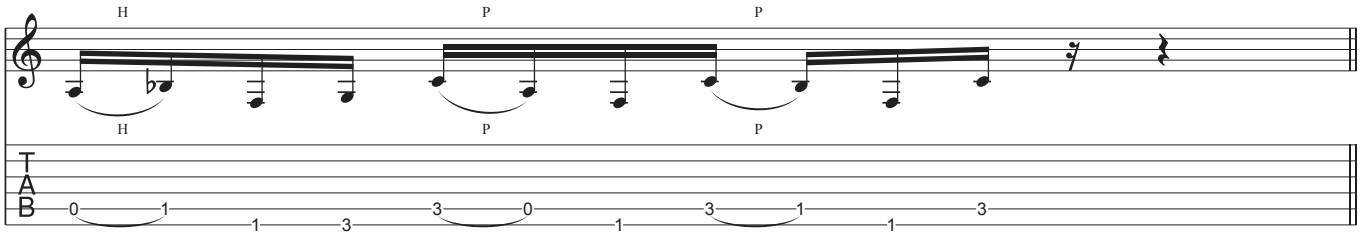


Ex. 210h

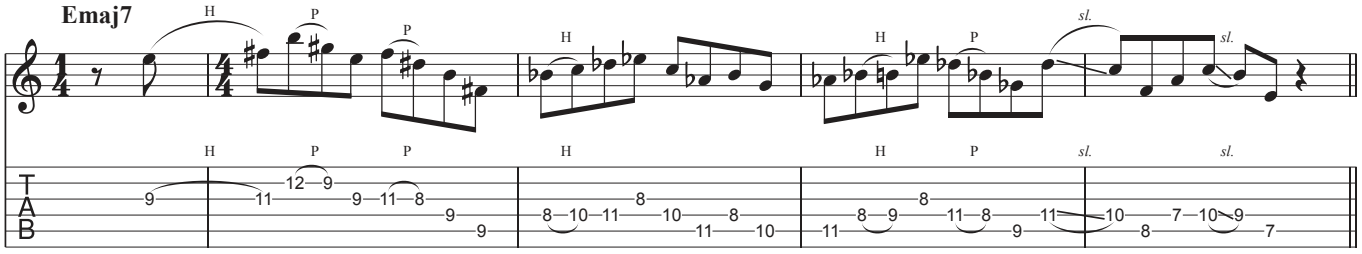
B♭7alt



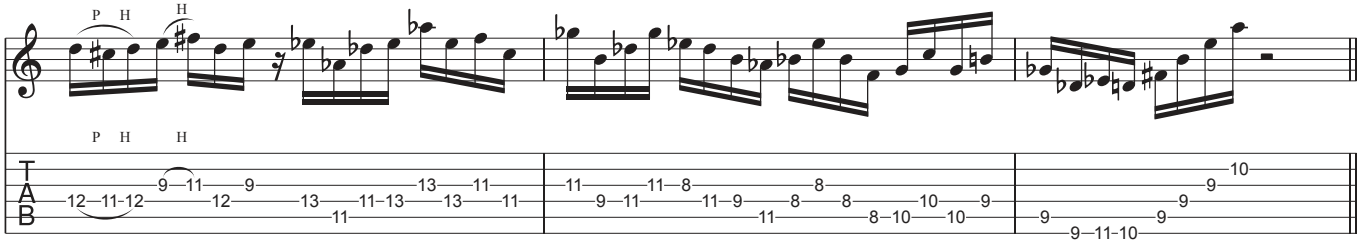




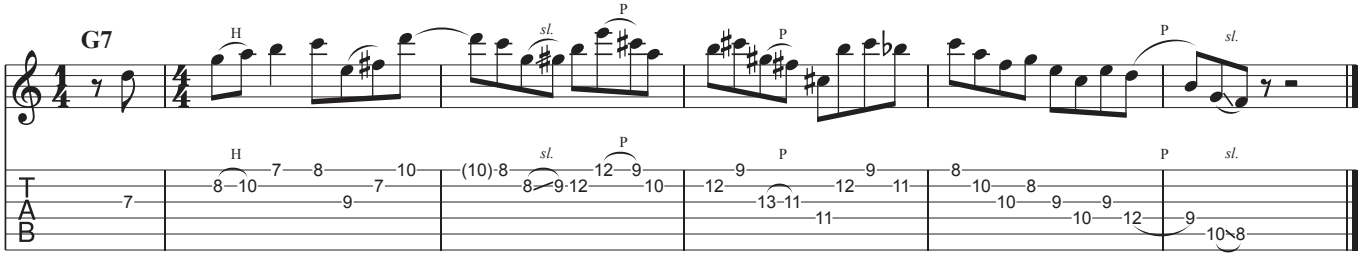
Ex. 210i



Ex. 210j



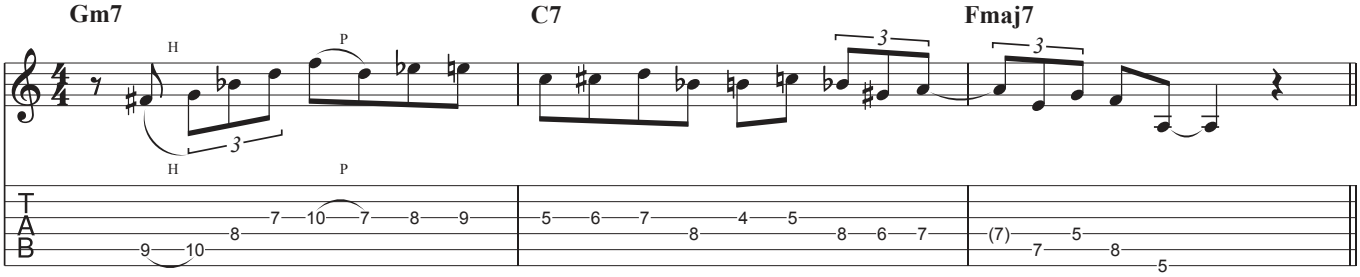
Ex. 210k



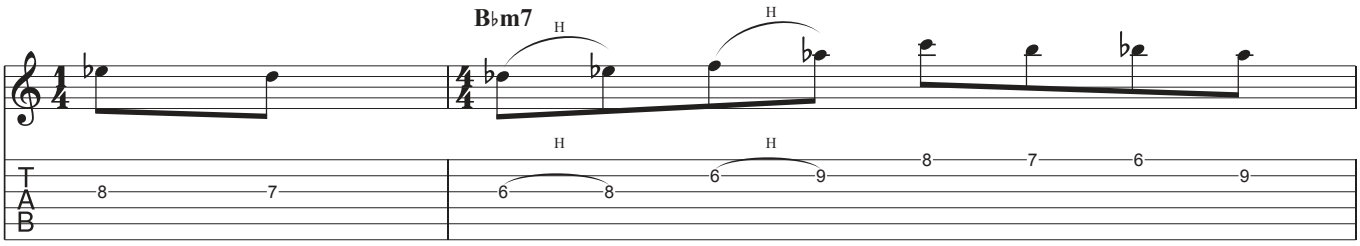
Chromaticism And Twelve-Tone Applications

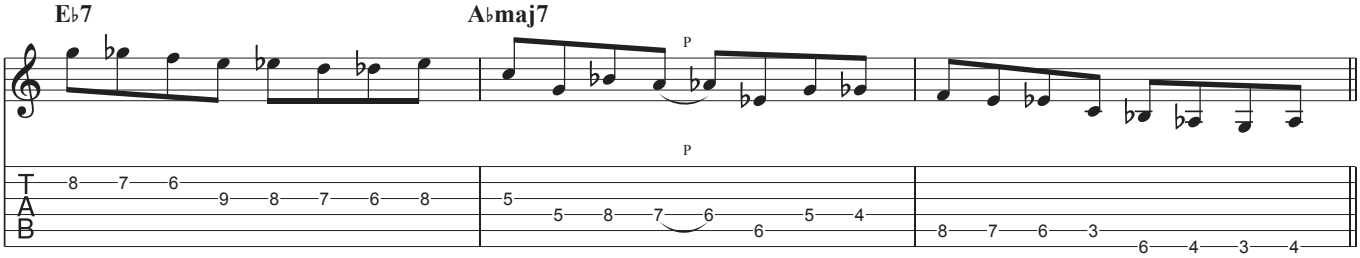
Chromaticism and twelve- tone techniques are not commonly discussed in jazz improvisation books for reasons that are unclear to me. The use of chromaticism in jazz became popular during the bebop era through the use of approach notes and chromatic enclosure of chord tones.

Ex. 211a

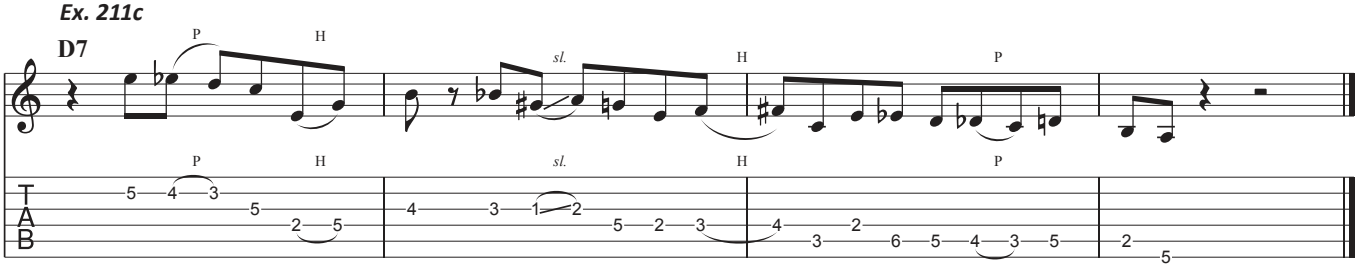


Ex. 211b





Ex. 211c

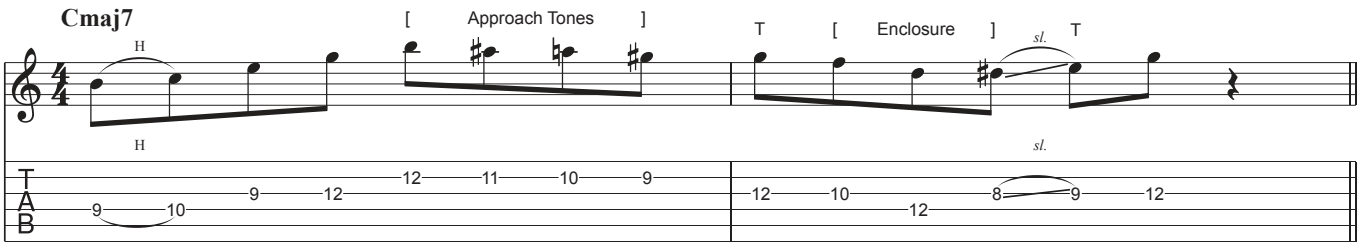


Chromaticism is best used when there is a target note, preferably a lower chord member, that falls on a strong beat.

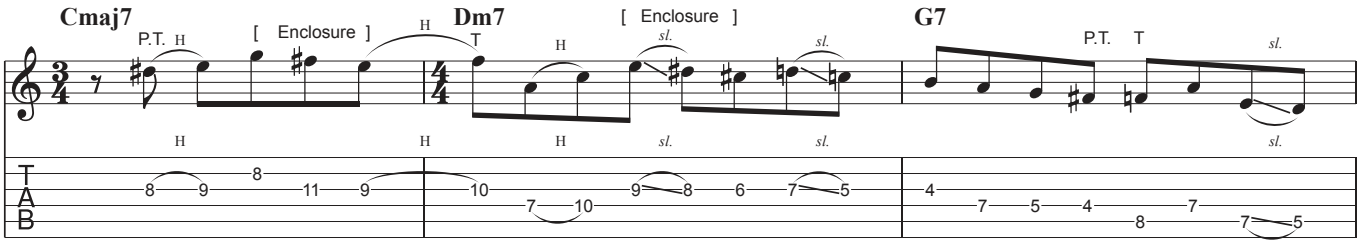
Here are a few examples of common chromatic approach usage.

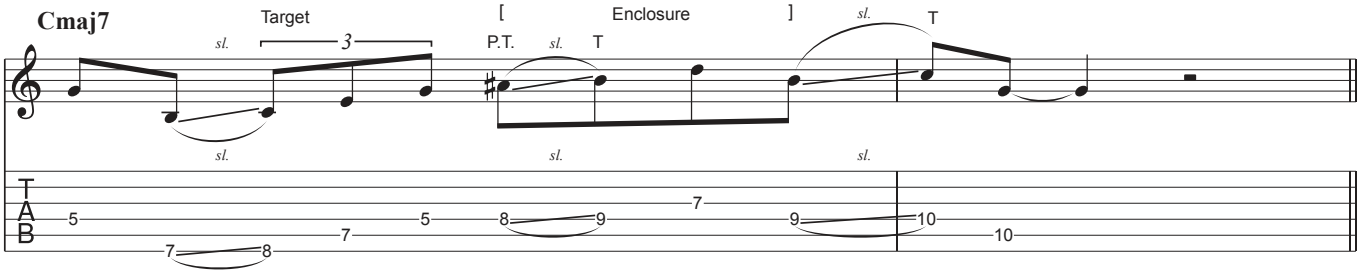
T = Target
P.T. = Passing Tone

Ex. 212a



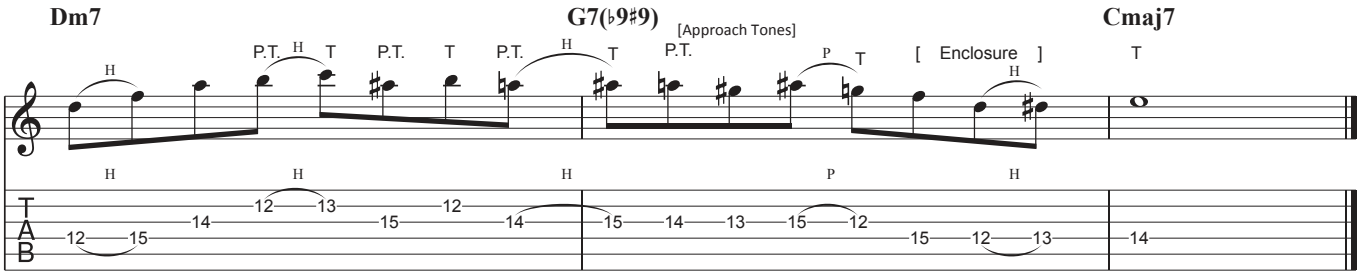
Ex. 212b





Ex. 212c

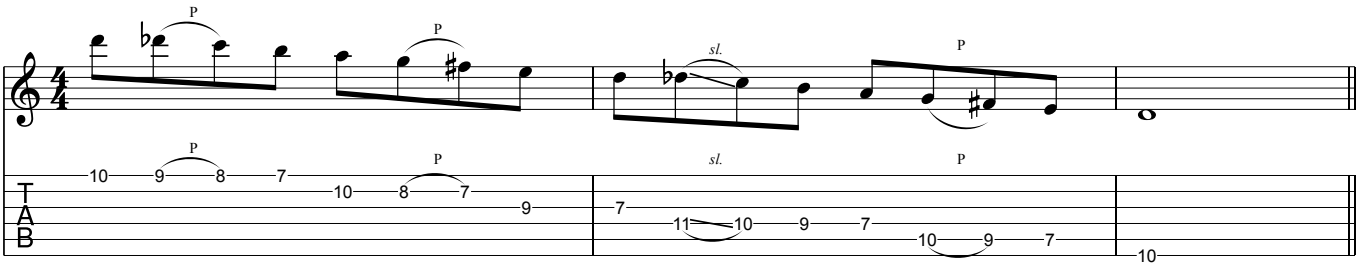
The B natural is a target note from A# while also functioning as part of a diatonic enclosure idea.



Chromatic usage in diatonic scale lines (without discussing enclosure) is usually best if confined to filling in whole steps with one chromatic passing tone. The bebop dominant scale is an example of this kind of idea.

Ex. 213

D7 (D Bebop Dom)

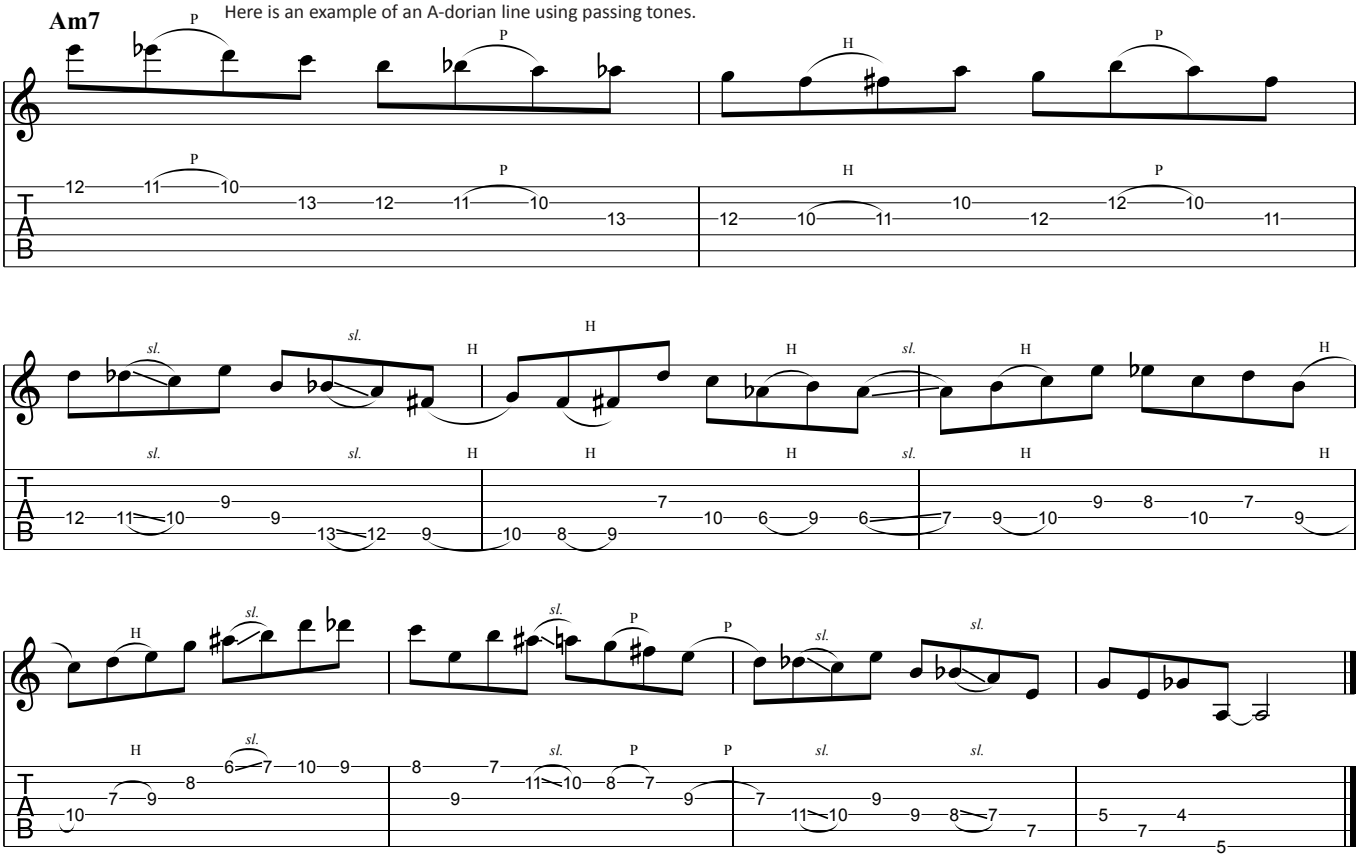


You might try the same thing with both major and minor type scales as long as the passing tone falls on a weak beat.

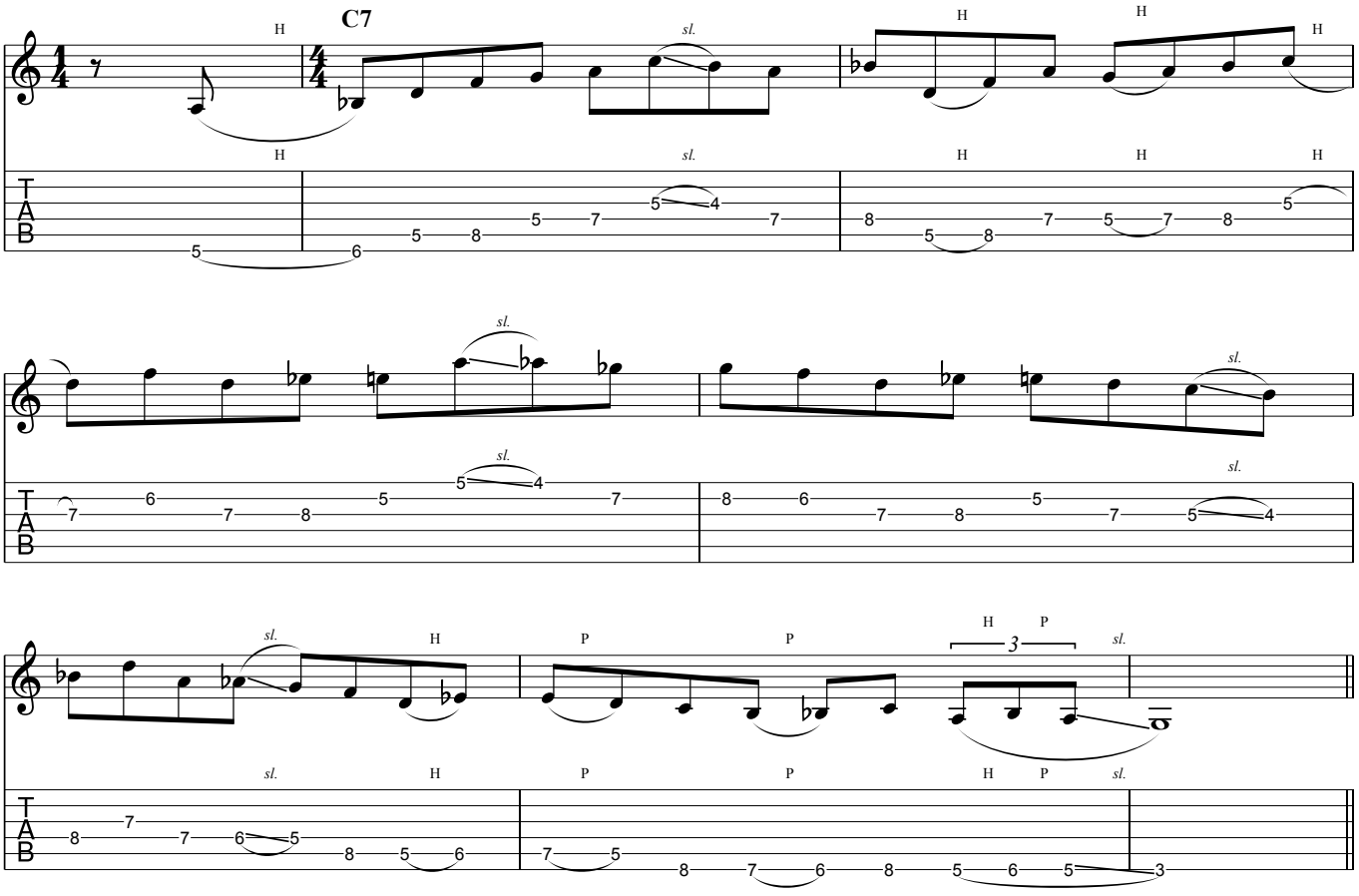
Ex. 214

Am7

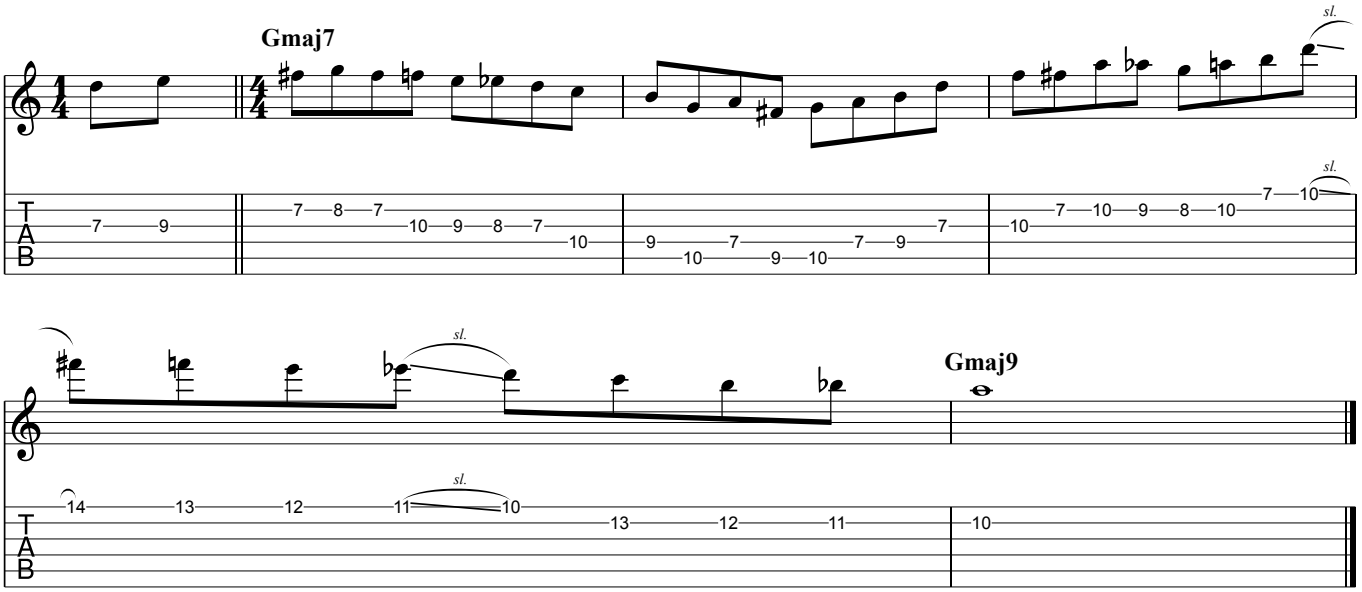
Here is an example of an A-dorian line using passing tones.



An example of usage on a Dom^{7} chord.

Ex. 215a

Ex. 215b

or a Maj^{7} chord



Twelve Tone Triadic Formulas

Here are some formulas for creating twelve tone rows comprised of triadic groupings. This is an incomplete list but should provide the basis for creating your own rows.

Group					
I		Cm	Dm	E	F#
	Formula	Min M2↑	Min M2↑	Maj M2↑	Maj TT↕
II		C^{+}	Dm	Eb^{+}	F#
	Formula	Aug M2↑	Min m2↑	Aug M2↑	Maj TT↕
III		C^{+}	Eb	F#m	B^{o}
	Formula	Aug m3↑	Maj m3↑	Min P4↑	Dim m2↑
IV		C^{o}	F^{o}	Gm	A
	Formula	Dim P4↑	Dim M2↑	Min M2↑	Maj m3↑
V		C^{+}	Db^{+}	D^{+}	Eb^{+}
	Formula	Aug m2↑	Aug m2↑	Aug m2↑	Aug M6↑
Vi		C^{sus4}	Db^{sus2}	D	E^{lyd}
	Formula	Sus4 m2↑	Sus2 m2↑	Maj M2↑	Lyd m6↑
VII		C^{sus4}	D^{lyd}	Bb^{o}	B
	Formula	Sus4 M2↑	Lyd m6↑	Dim m2↑	Maj m2↑
VIII		C^{sus4}	D^{+}	G#m	A
	Formula	Sus4 M2↑	Aug TT↕	Min m2↑	Maj m3↑

I have given the interval relationships between triads to facilitate transposition. The formula can be started on any of the triads and continued around the loop. For example:

Group III Formulas

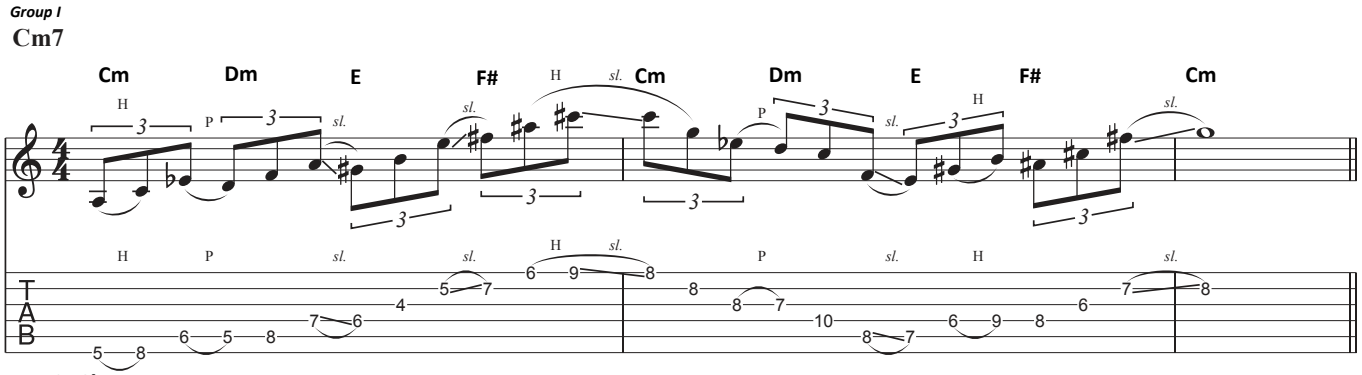
①	Aug m3↑	Maj m3↑	Min P4↑	Dim m2↑
②	Dim m2↑	Aug m3↑	Maj m3↑	Min P4↑
③	Min P4↑	Dim m2↑	Aug m3↑	Maj m3↑
④	Maj m3↑	Min P4↑	Dim m2↑	Aug m3↑

In four differnt keys it might look like this.

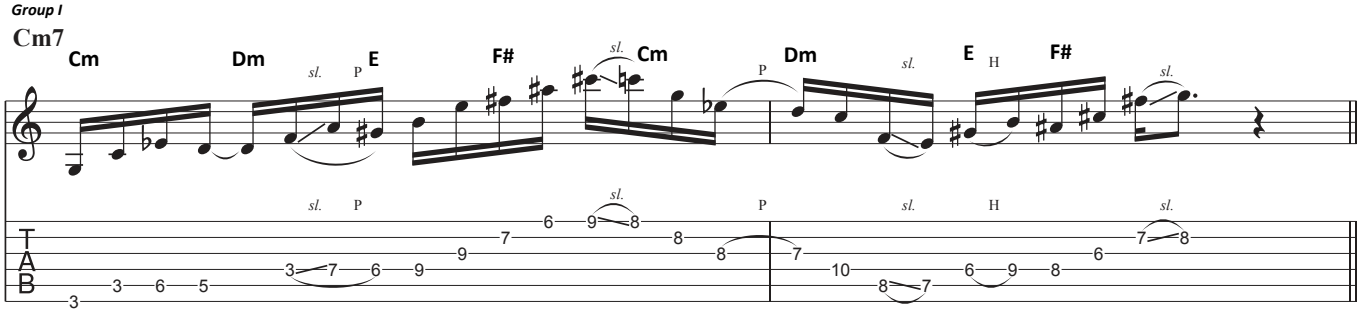
①	C^{+}	Eb	F#m	B^{o}
②	Db^{o}	D^{+}	F	Abm
③	E	Gm	C^{o}	Db^{+}
④	Am	D^{o}	Eb^{+}	F#

On the next page you ll find a few lines that demonstrate this concept.

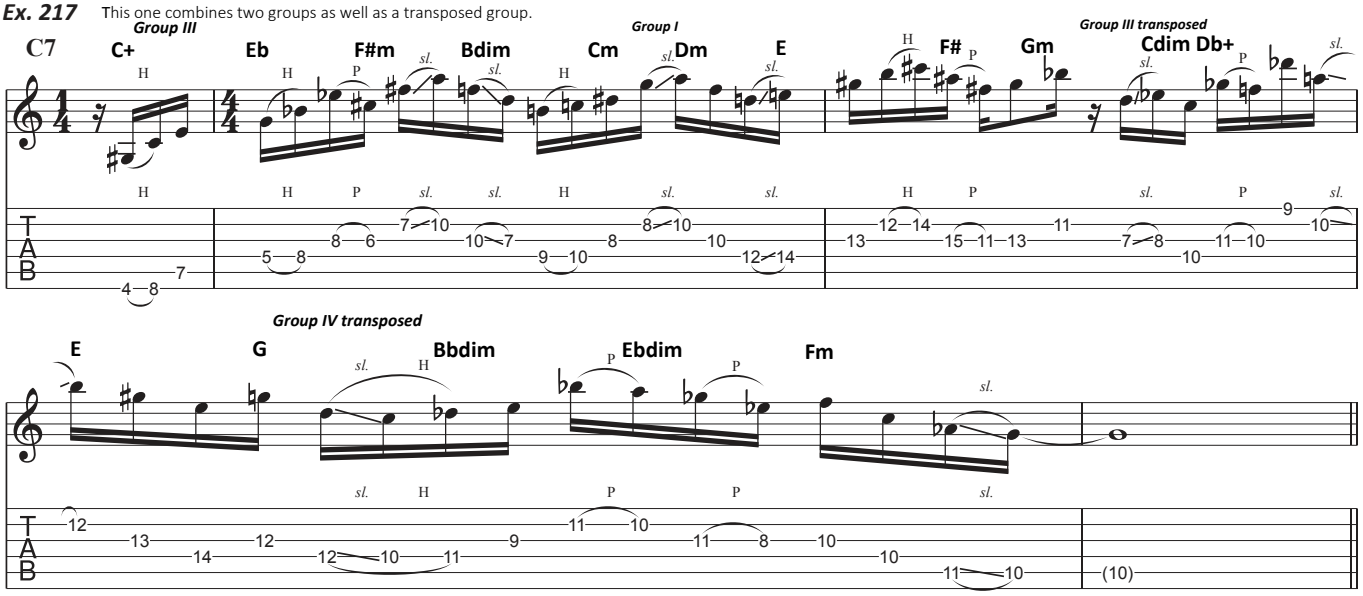
Ex. 216a



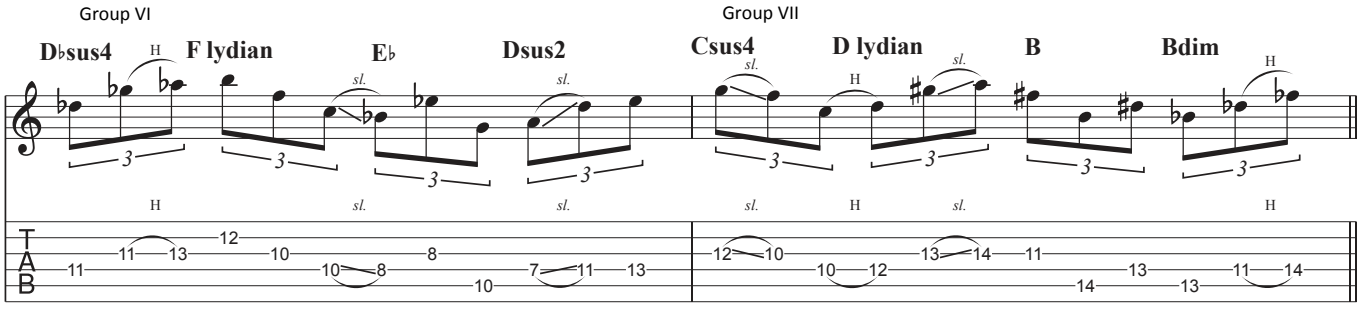
Ex. 216b



Ex. 217 This one combines two groups as well as a transposed group.



Ex. 218 Resolution notes become members of other triads and a new row is continued.



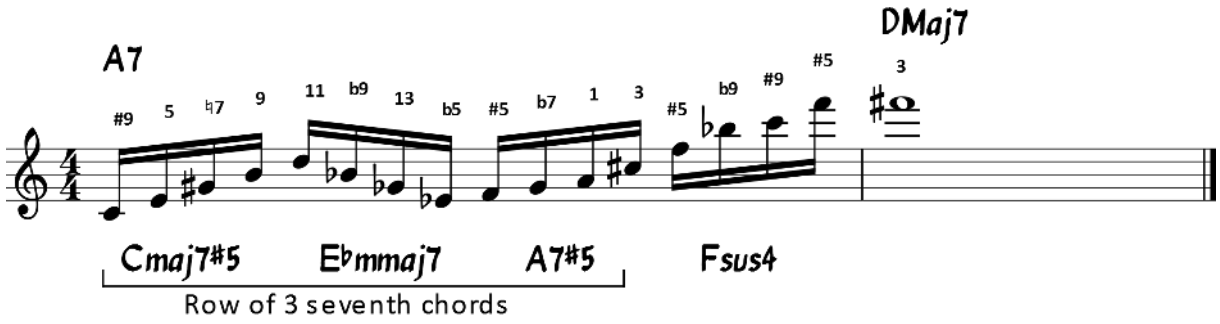
The line above might work over Db7sus4, C7alt, F7alt, G7alt etc.

Remember it is usually a good idea to connect the triads as smoothly as possible to create a more seamless effect.

Twelve-Tone Formulas Using Seventh Chords

Twelve-tone rows can also be formed by combining three seventh chords. This is an example of a twelve tone row using seventh chords:

Ex. 219

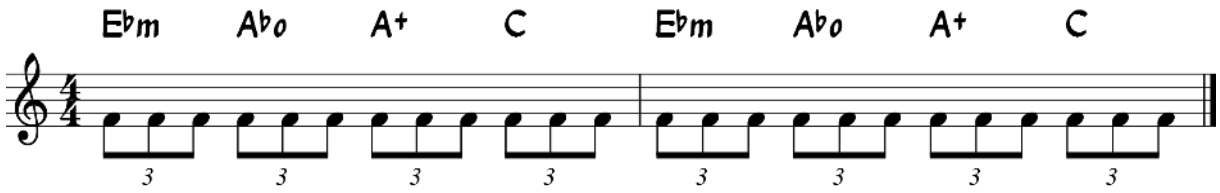


Try to come up with your own progressions and lines from this concept.

A few thoughts about triadic an seventh chord rows.

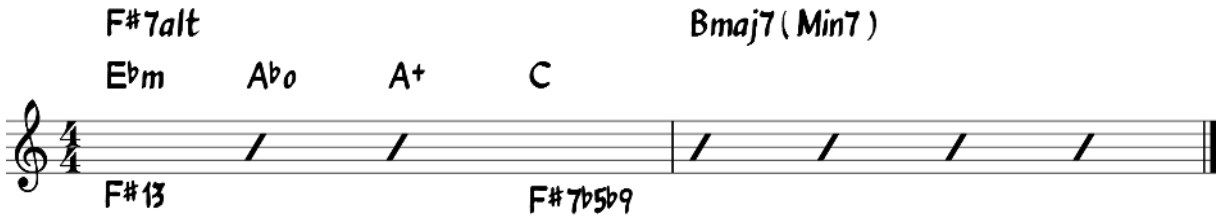
1. Because they are cycles (keep repeating) they can be continued and will keep producing twelve-tone rows.
2. In order to find chords that the rows sound best with, try to take advantage of the cyclical nature of the formulas. Begin and end on the same triads or seventh chords. In the example below, the row begins on an Eb- triad and uses Group III´s formula we get C, Eb-, Ab^{o}, A^{+}, triads whose harmonic rhythm may be broken down like this:

Ex. 220



This progression could be used over an Eb-^{7} chord because of the beginning chord Eb- or on an F#^{7alt} chord because the beginning chord Eb- produces an F#^{13} chord and the departure chord to the next measure is a C major triad which produces an F#$^{7/b5/b9}$ chord.

Ex. 221

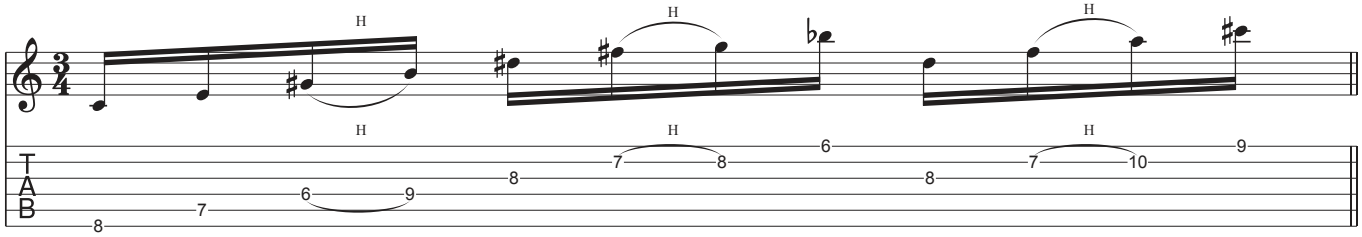


The outer chords are most important because they establish the tonality and then lead you to the next tonality.

Non-Tertian Twelve-Tone Rows

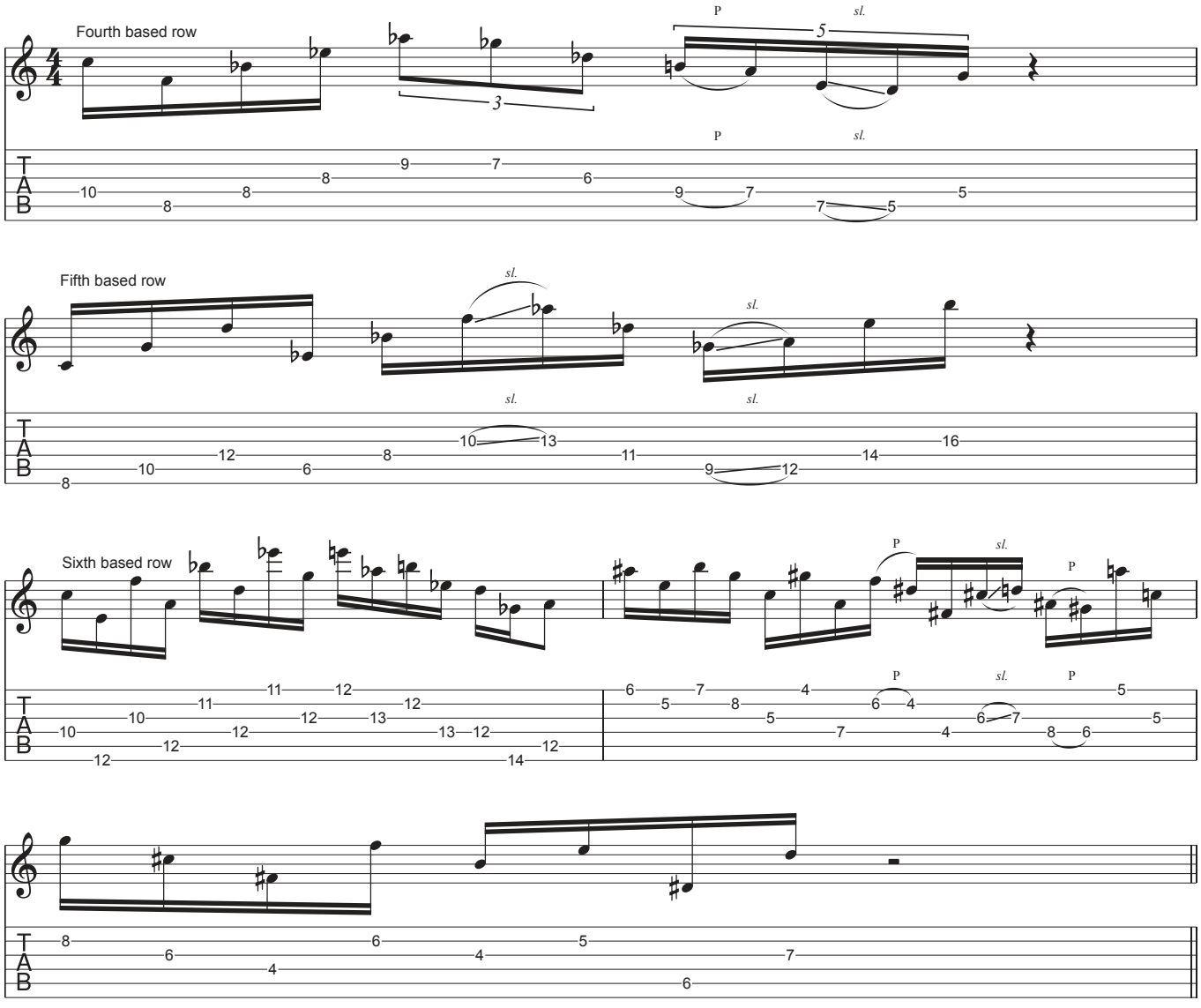
Dodecaphonic rows can be constructed through interval systems. Grouping of thirds might look like this.

Ex. 222



As we can see in the previous example, the interval relationships are not alway strict in nature (some may have extra intervals for linking purposes).

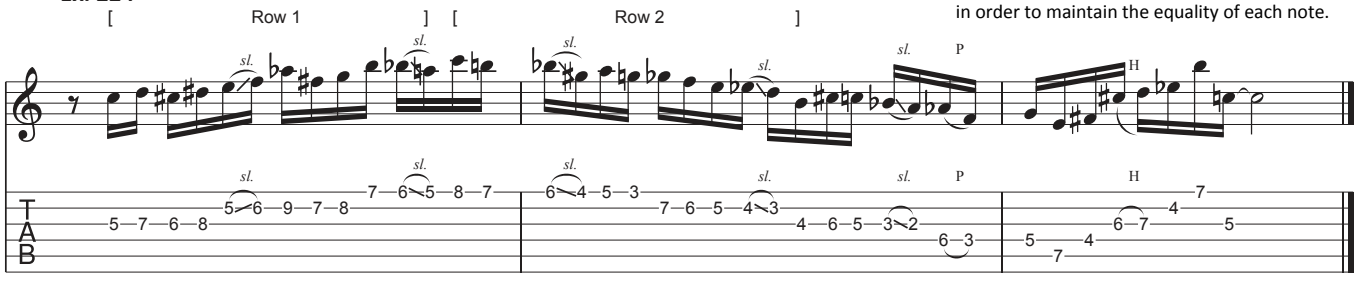
Ex. 223



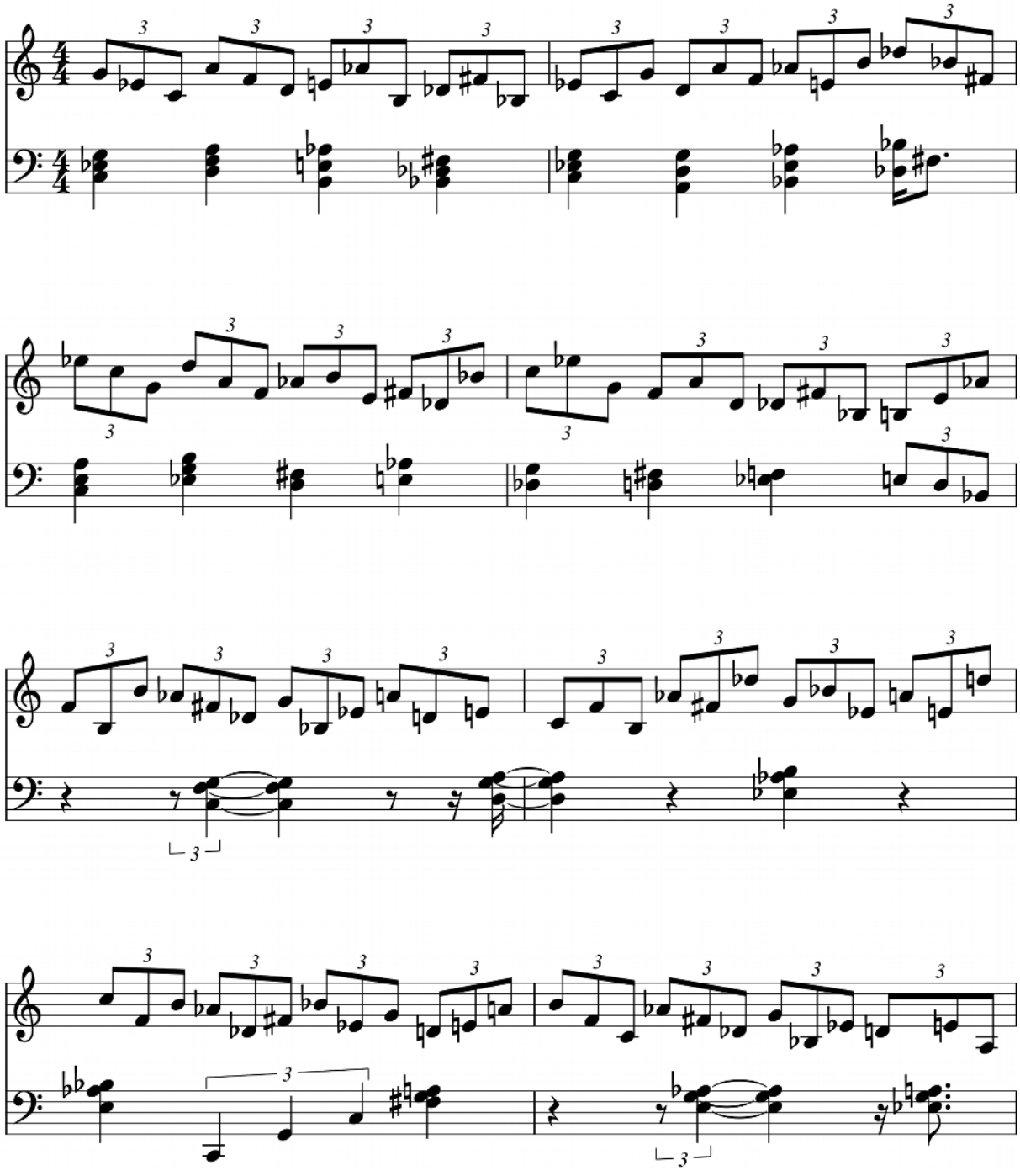
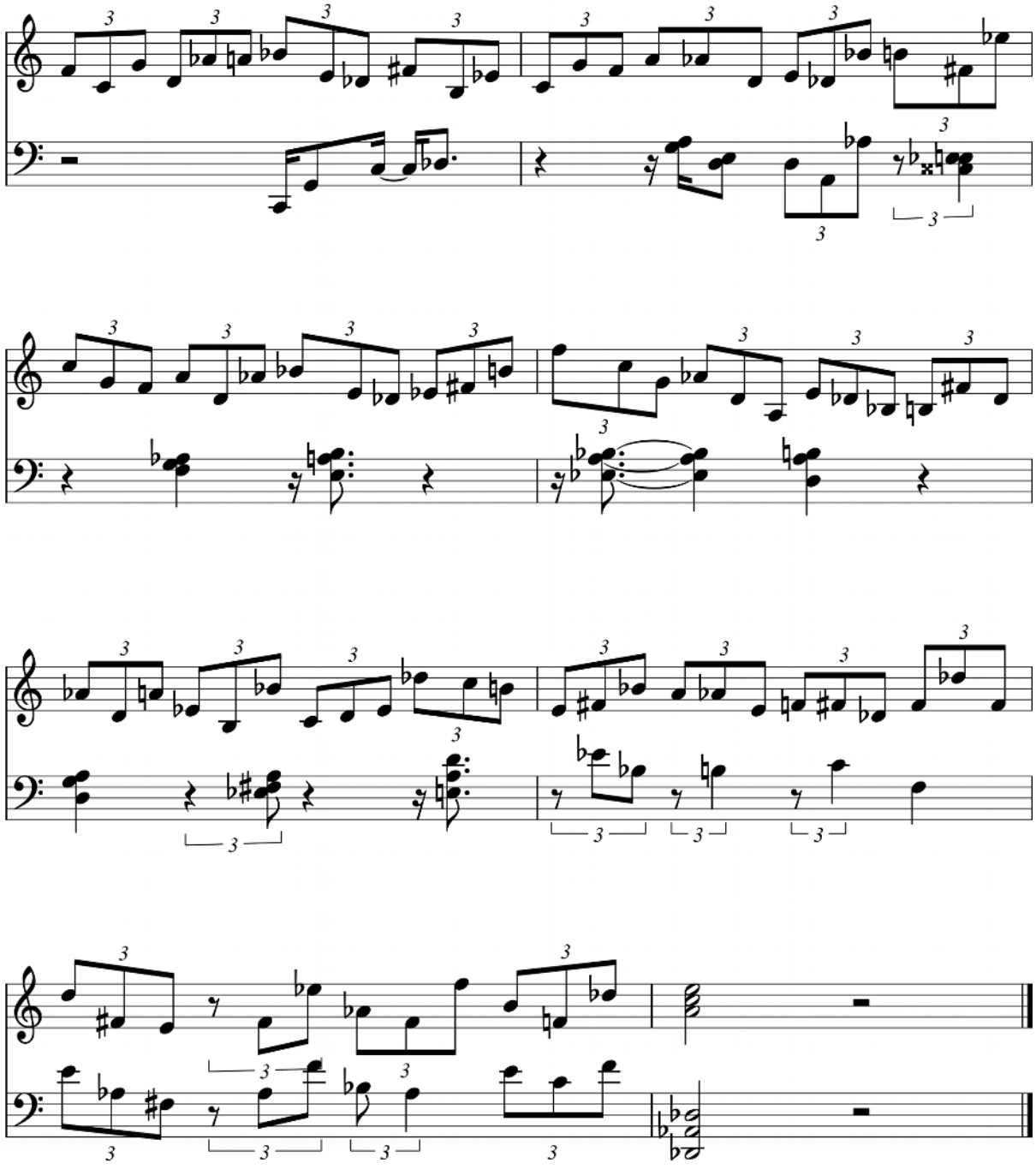
Non-Systematic Row

Non-Systematic Rows may use many interval combinations but should avoid any harmonic structures in order to maintain the equality of each note.

Ex. 224

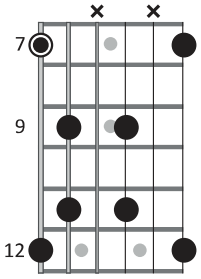


This type of line works equally well over free tunes and chords such as C7alt. Twelve-tone rows may be played in inversion (mirror of intervals), retrograde (backwards) and retrograde inversion (mirror of intervals and backwards).

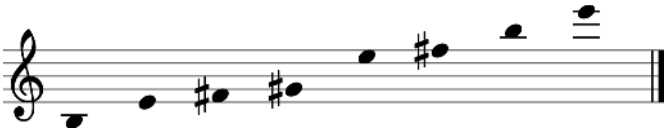
12-Tone Triad Etude No.1

Modal Shapes

A modal shape is a fingerboard pattern whose intervals produce a modal or chordal sound.

Ex. 225



This shape (Ex. 222) if played in an ascending manner would look like this:

Ex. 226

These notes form an E^{add9} hybrid structure which is derived from these scales: E-ionian, E-lydian, E-mixolydian, E-mixolydian b6 and E-mixolydian #11.

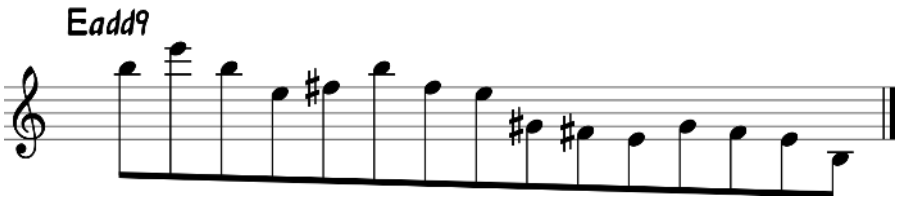
This shape may also be played over any chords derived from the parent scales B-melodic minor, E-major, B-major, A-major and A-melodic minor.

Ex. 227

Bb^{7alt}, E^{maj7}, $A^{maj7/\#11}$, $F\#m^{11}$, E/F#, D/C#, E^{7sus4}, $D^{maj7/\#11}$, $C^{maj7\#5}$, Am^{maj7}, $D^{7\#11}$, etc.

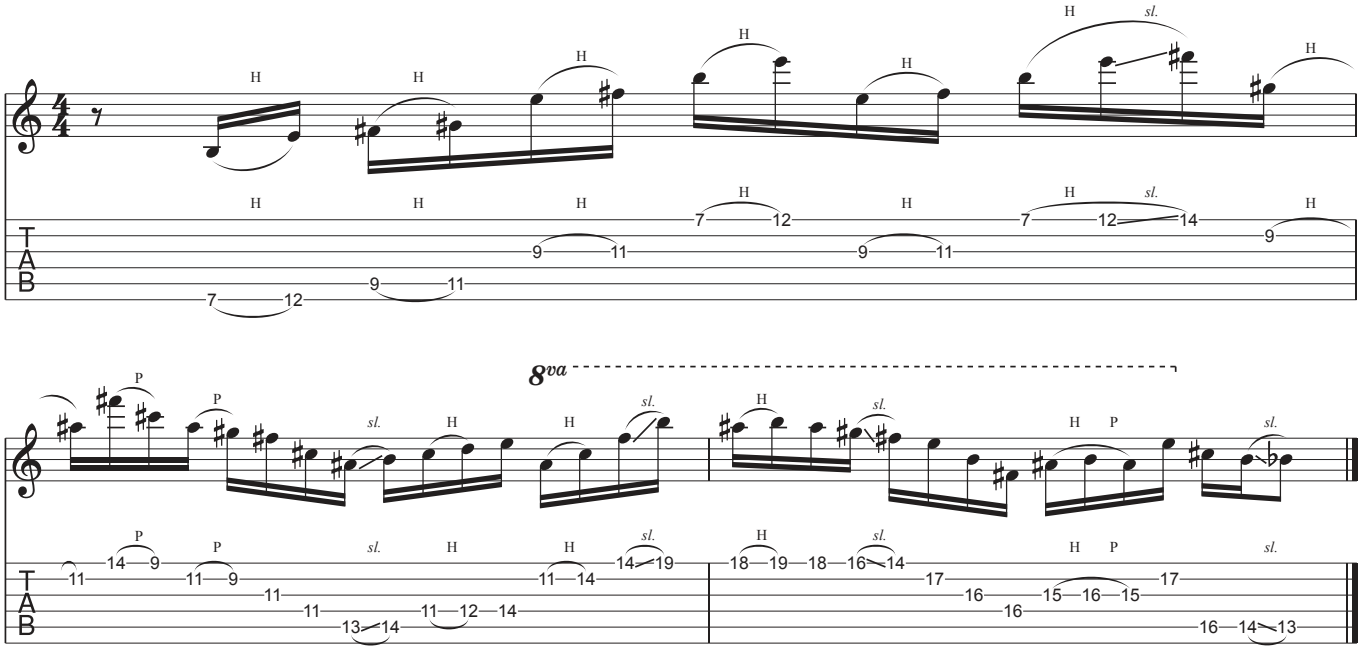
Experiment playing the shapes in different interval combinations as well as combining them together to create longer lines.

Ex. 228

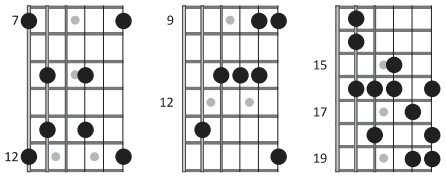


Ex. 229

Combination of two hybrid structures
Eadd9 and F#add9 over Bb7alt.

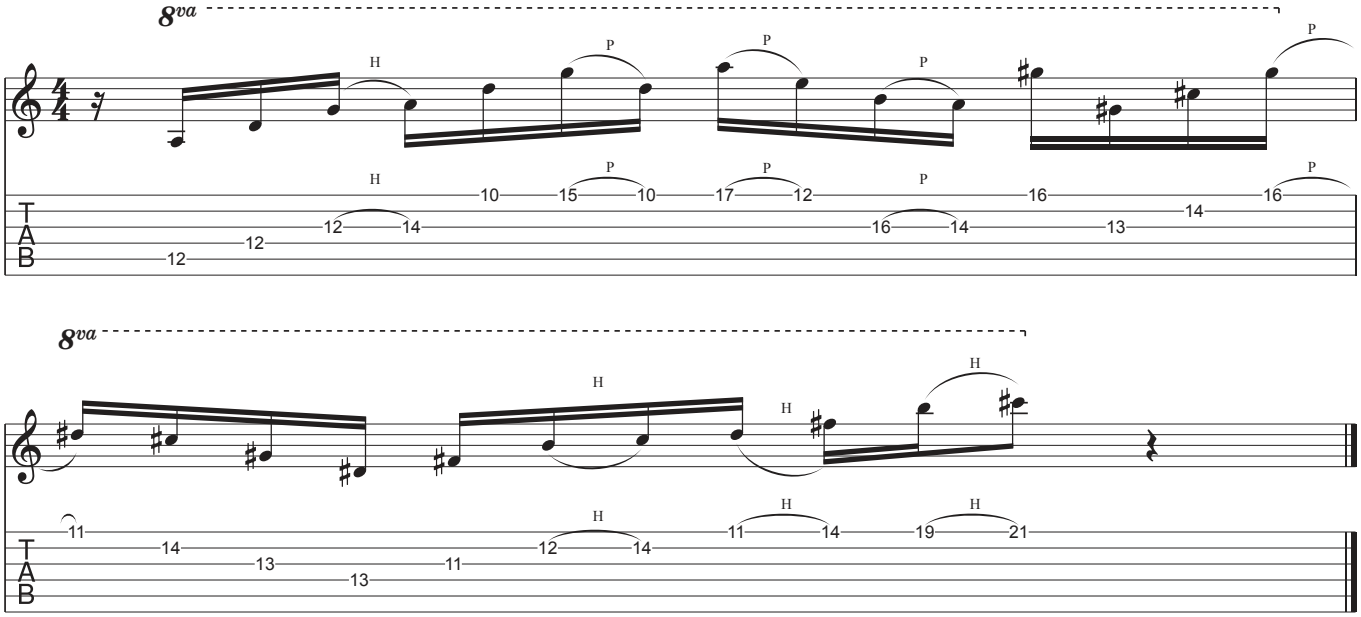


These are the three hybrid structures
involved in this line.

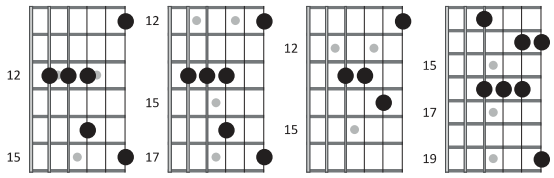


Next is an example of a free line using harmonic shapes.

Ex. 230



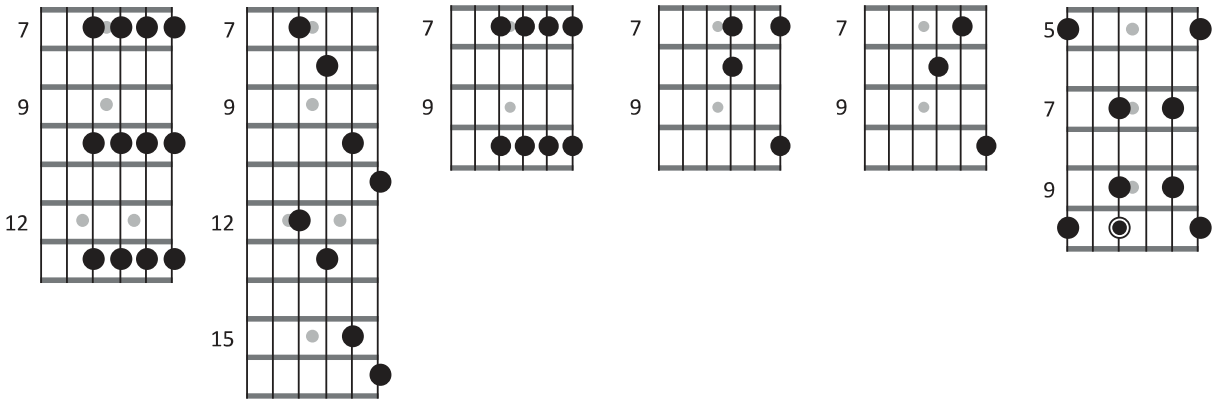
Shapes used:



Tonic and Dominant Diminished Scale Shapes

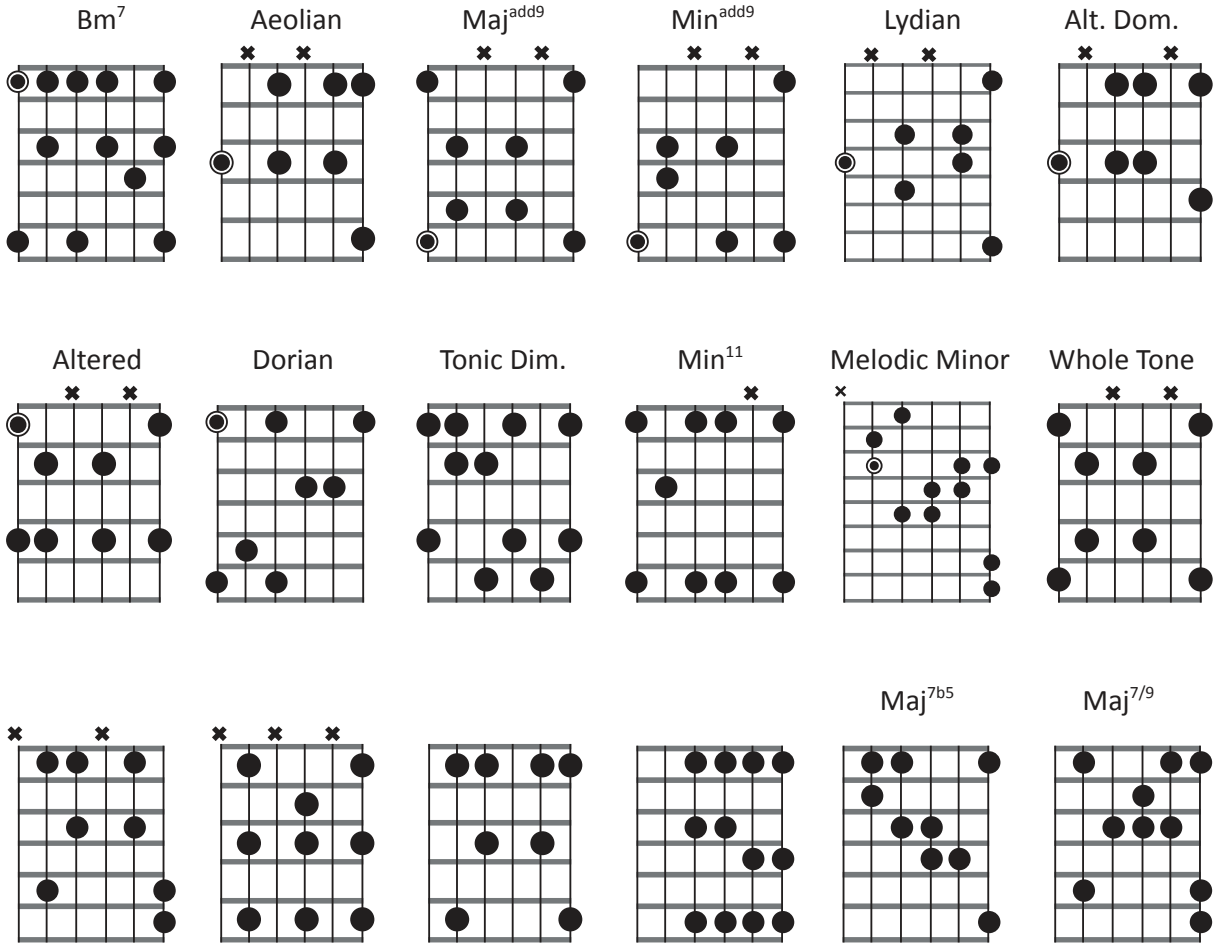
These shapes will work over the chords: F#^{o7}, A^{o7}, C^{o7}, Eb^{o7}, F^{13b9}, D^{13b9}, B^{13b9}, Ab^{13b9}

Ex. 231



Here are examples of some other shapes

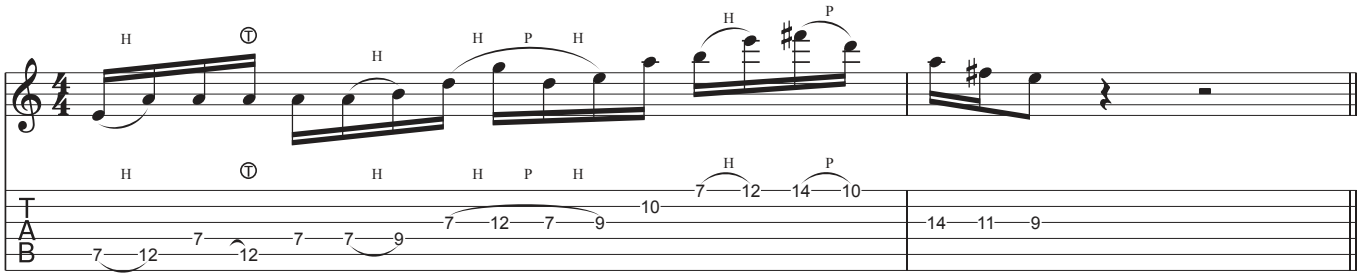
Ex. 232



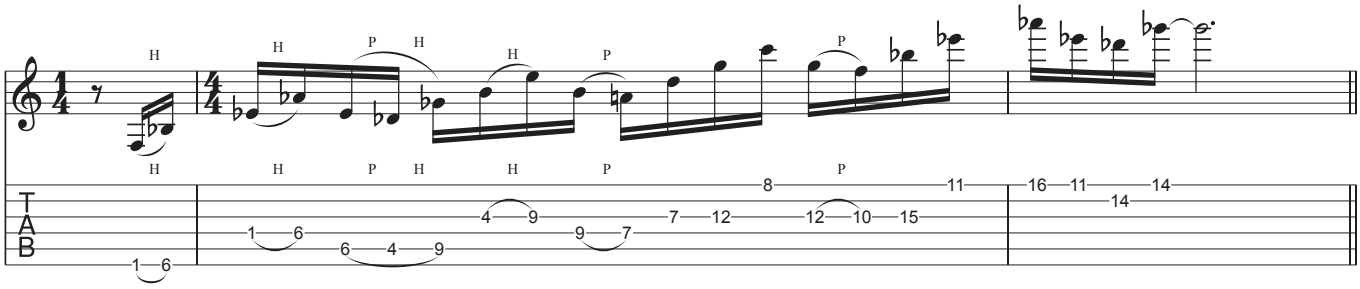
Unorthodox Left Hand Techniques

This section will deal with melodic patterns utilizing large intervals on one string. The execution of these lines will require an unusually large stretch of the left hand. Be careful not to over-practice these techniques because the wide hand stretches and may lead to injury if not approached with caution.

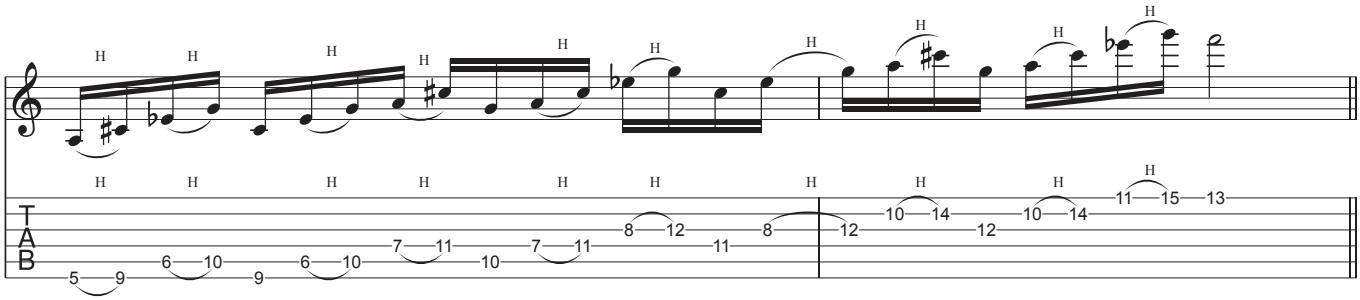
Ex. 233a



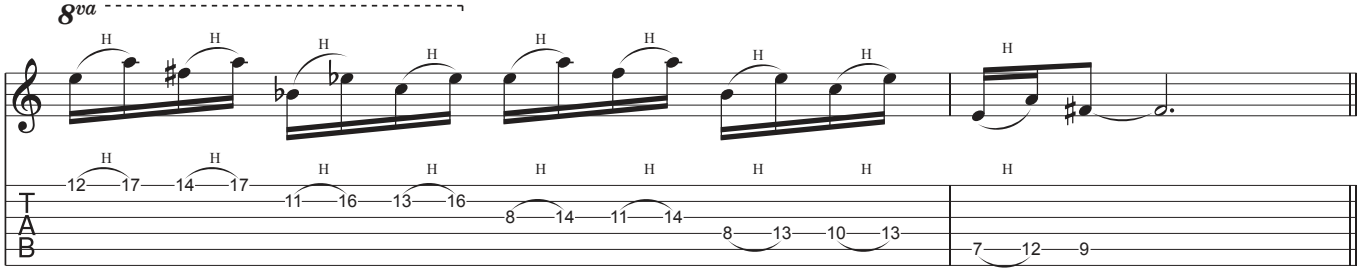
Ex. 233b



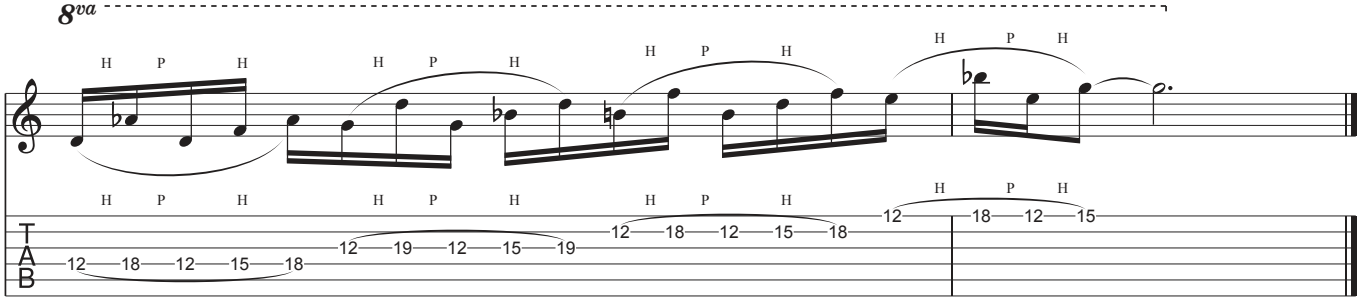
Ex. 233c



Ex. 233d



Ex. 233e

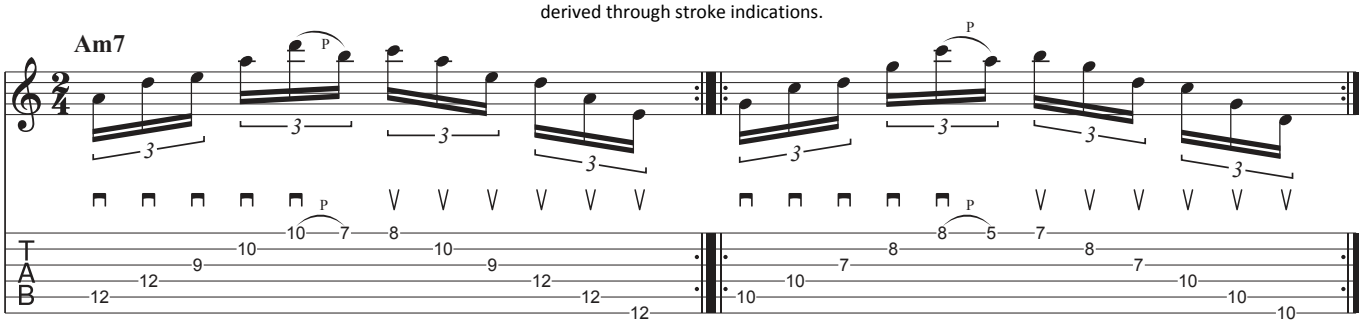


Modern Linear Examples

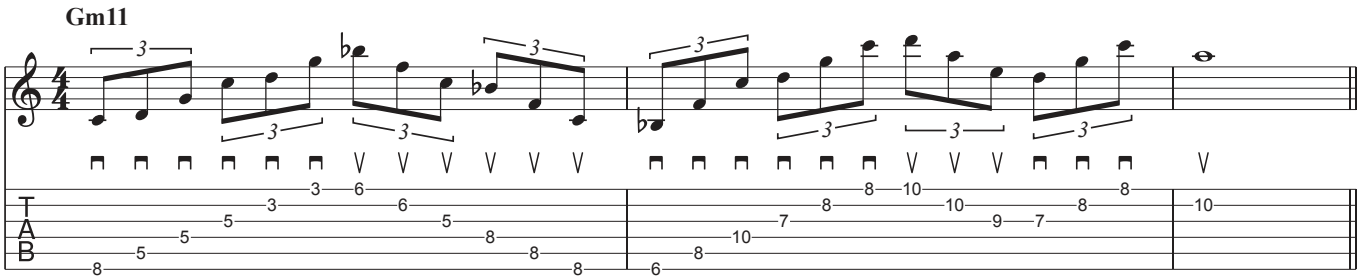
Sweep picking is an exciting technique because it enables the guitarist to execute ideas that are next to impossible with conventional alternate picking. The following ideas demonstrate some different arpeggio structures made possible through sweeping. Make sure to pay particular attention to the strokes indicated. Strict odd-even (odd number of notes on a string allow sweeping,

even number allow direction change) sweep technique is not always followed. The position changes can be derived through stroke indications.

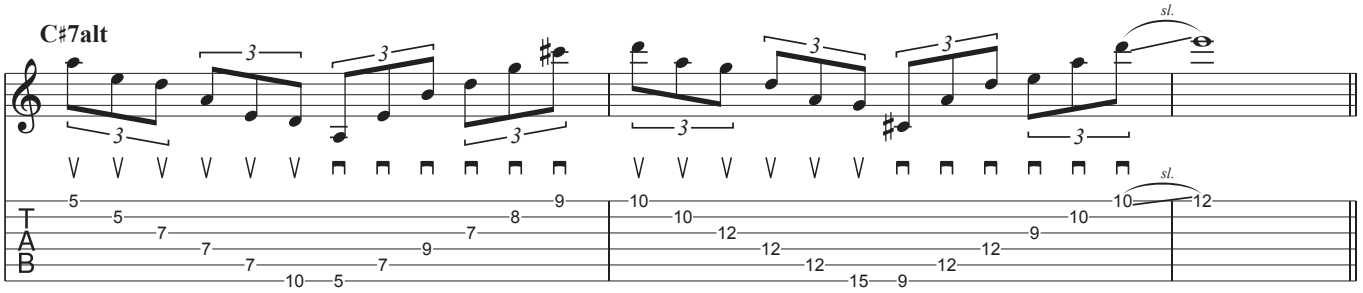
Ex. 234a



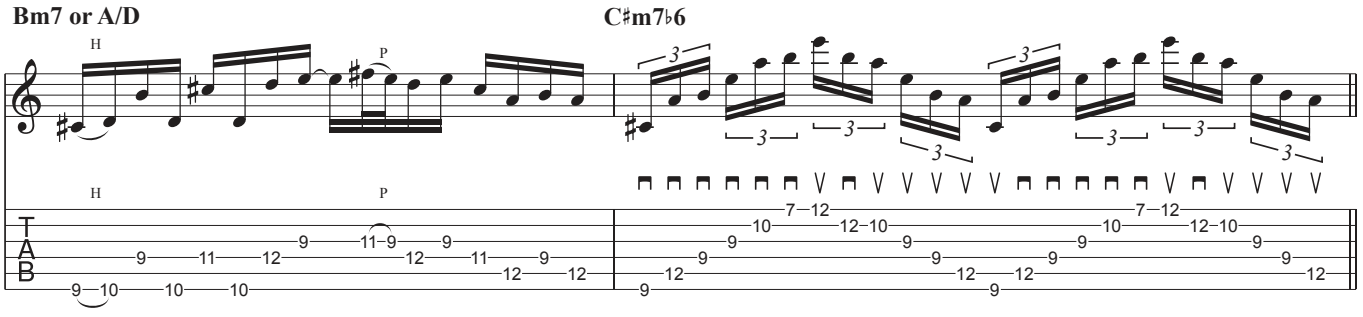
Ex. 234b



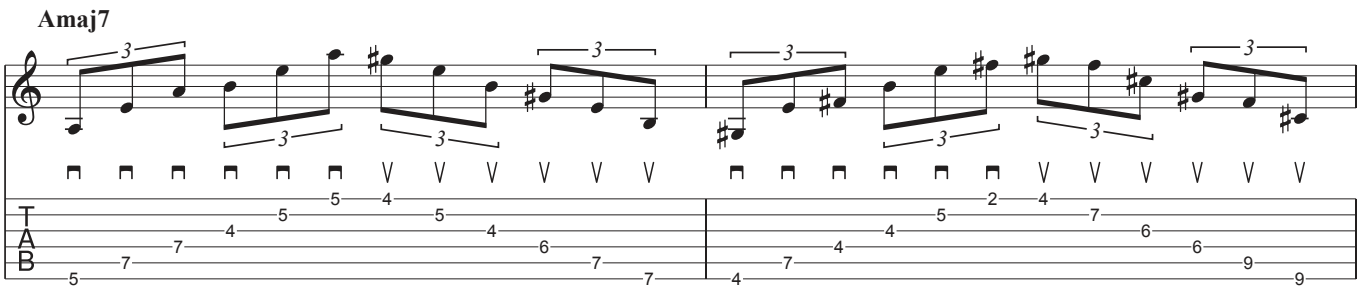
Ex. 234c

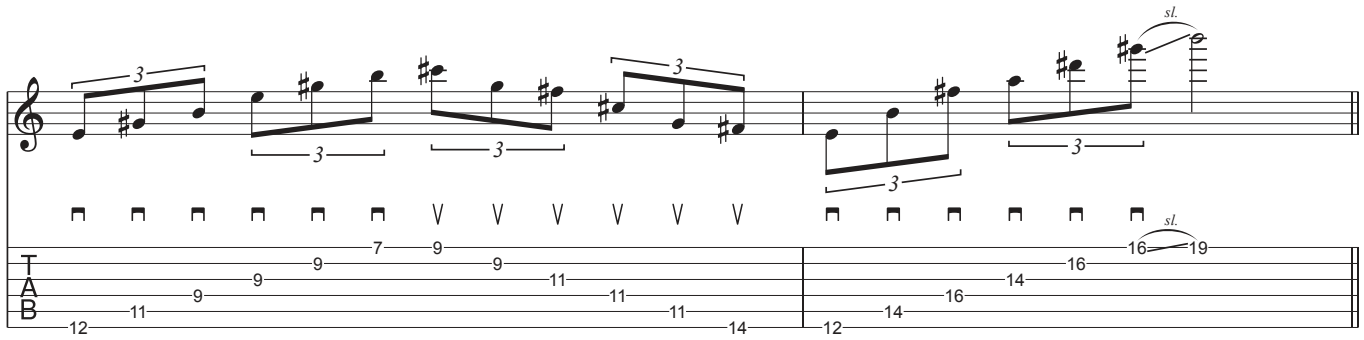


Ex. 234d



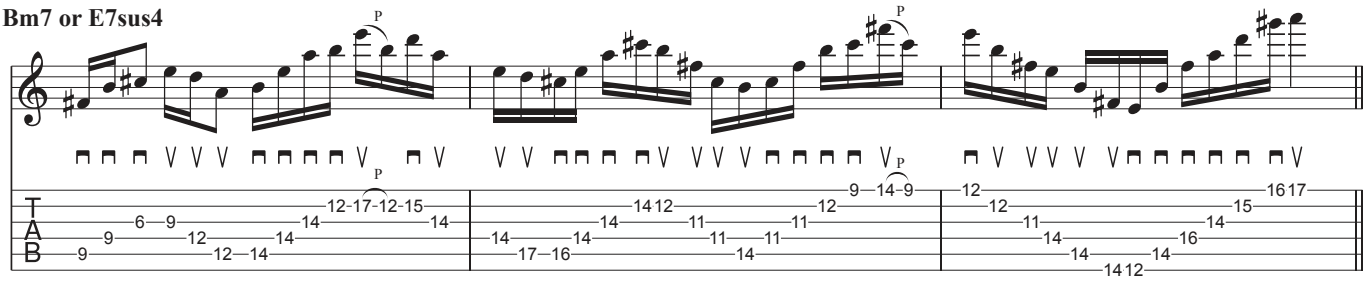
Ex. 234e





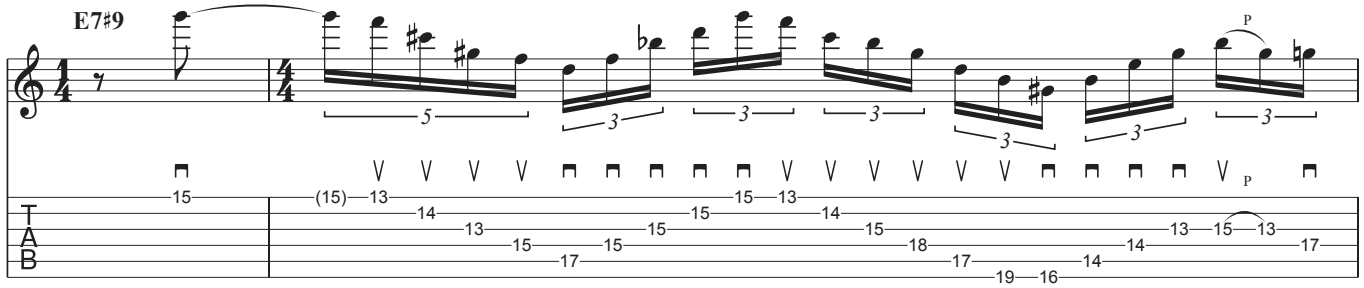
Ex. 234f

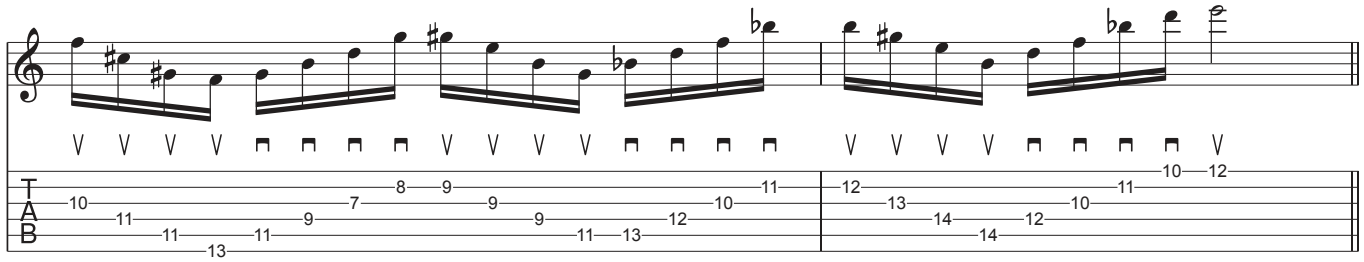
Bm7 or E7sus4



Ex. 234g

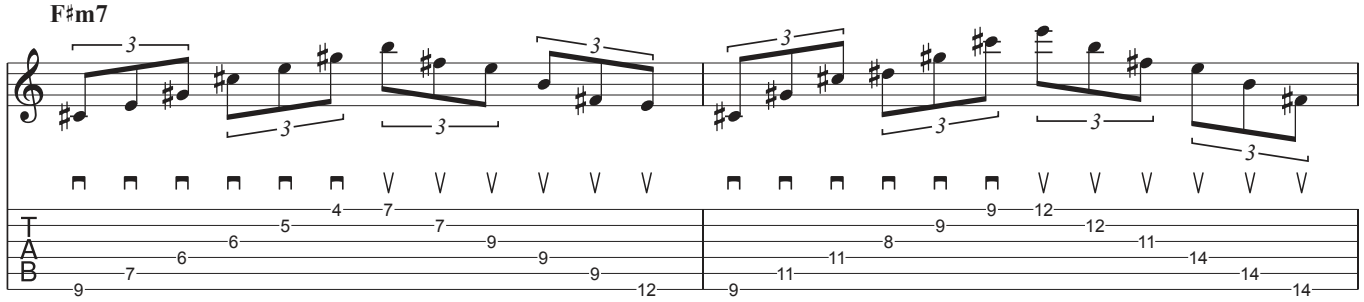
E7♯9

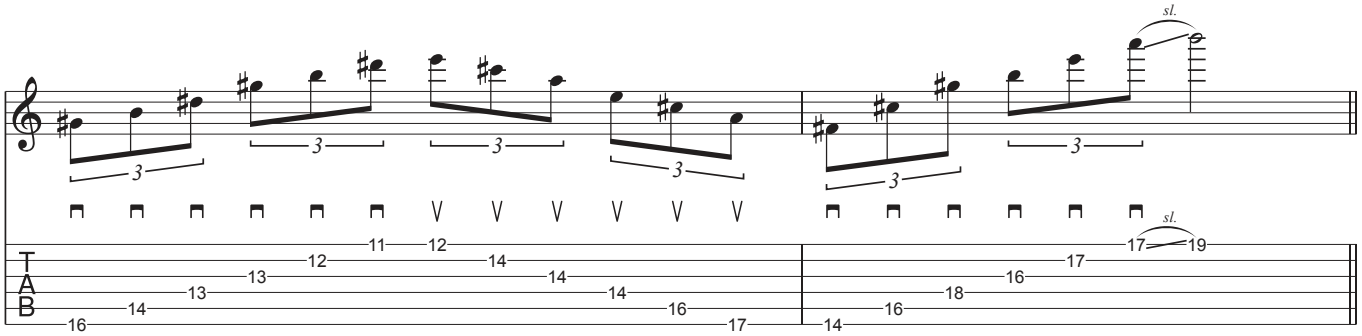




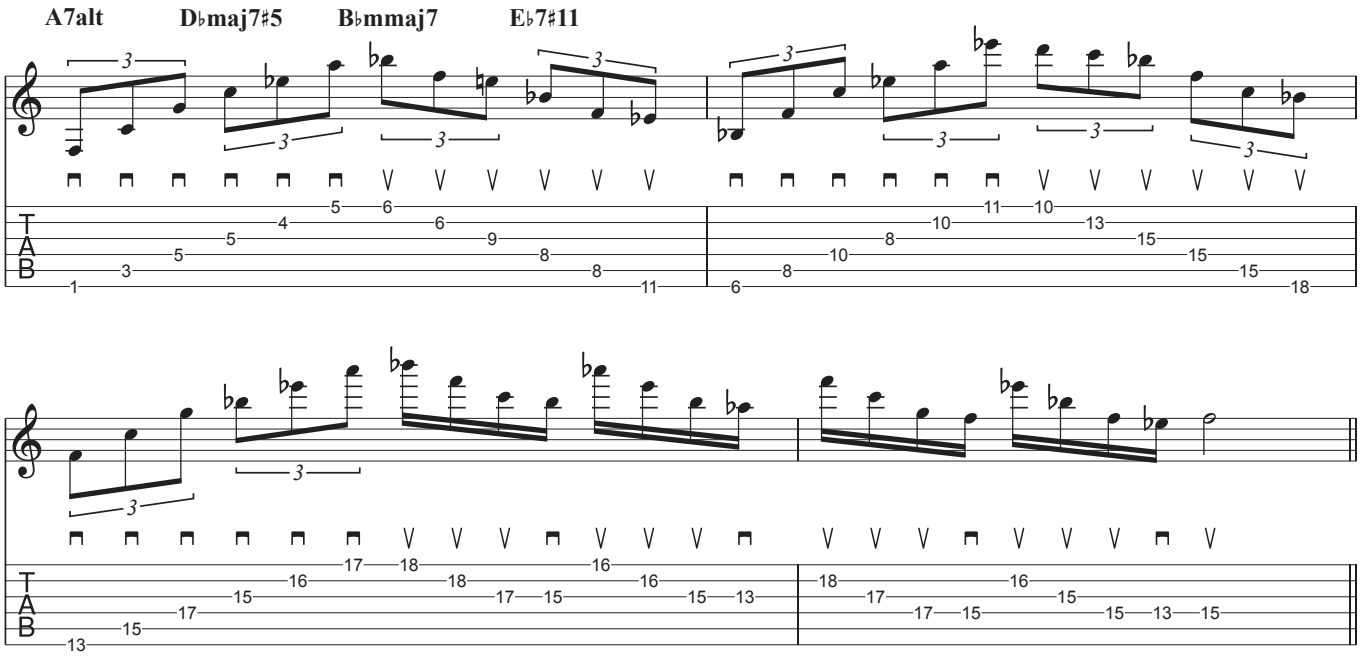
Ex. 234h

F♯m7

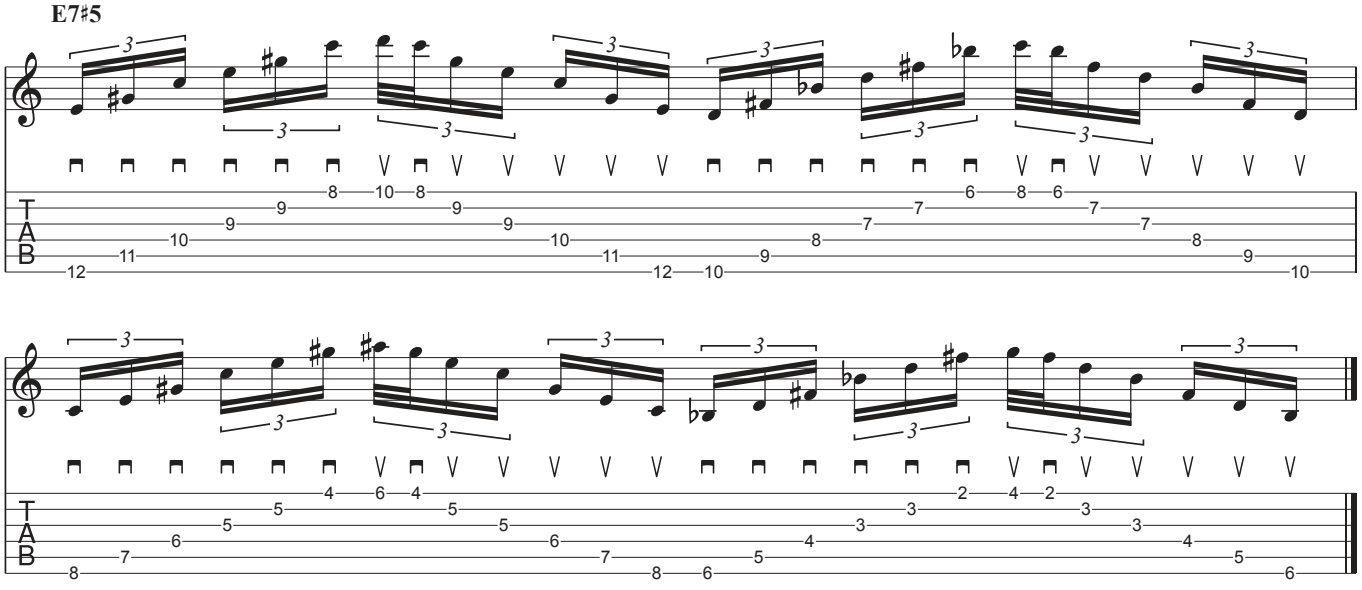




Ex. 234i



Ex. 234j



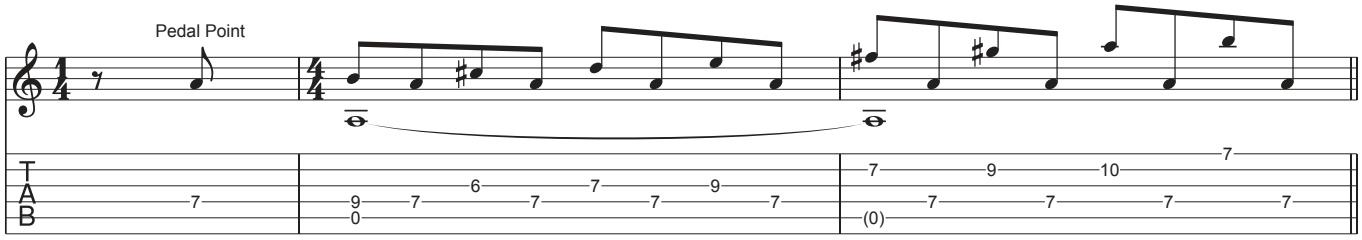
Make sure to practice the arpeggios in chapter three with the sweep articulations indicated. These will give you the technique and melodic material to develop your own sweep ideas.

Pedal Point Soloing

Pedal point can be used in two different ways in soloing.

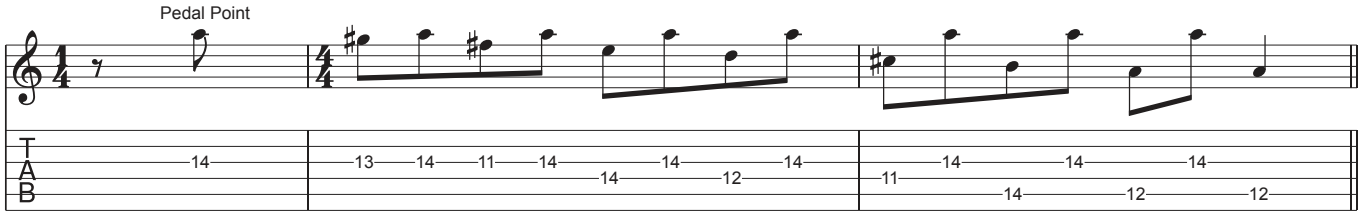
1. Lower pedal point – the pedal note occurs at the bottom of the melodic line

Ex. 235



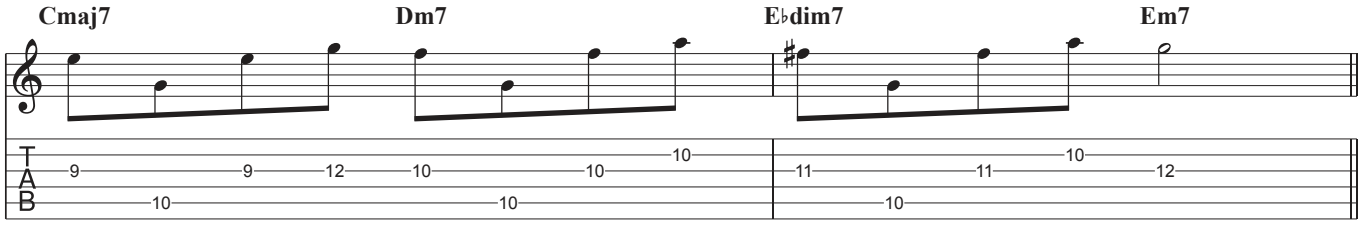
2. Upper pedal point – the pedal point note occurs at the top of the melodic line.

Ex. 236

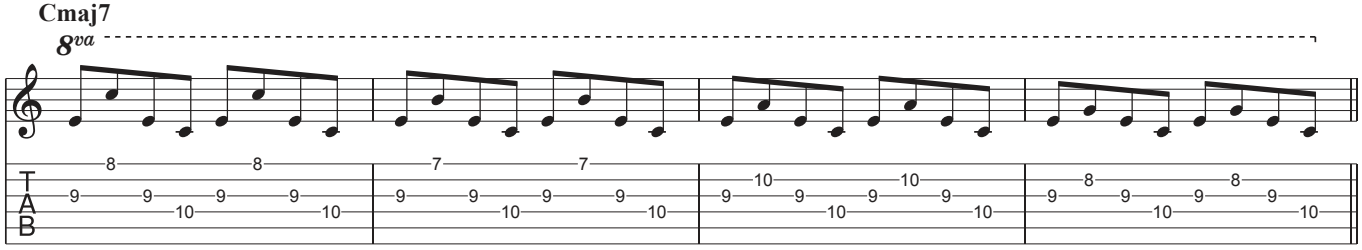


Try these few lines to get a feel for the use of this device.

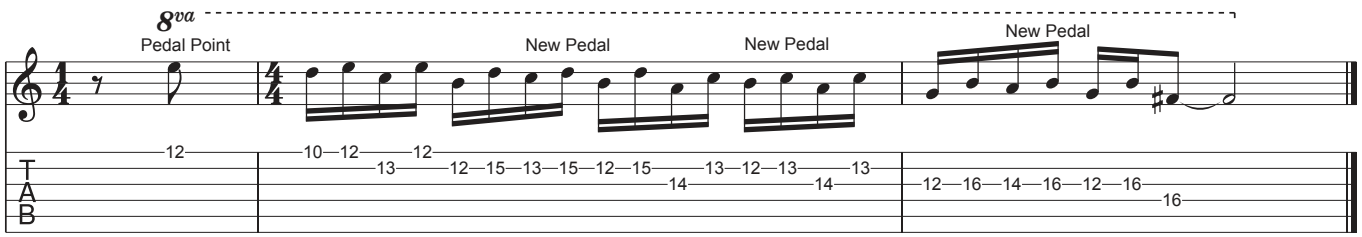
Ex. 237a



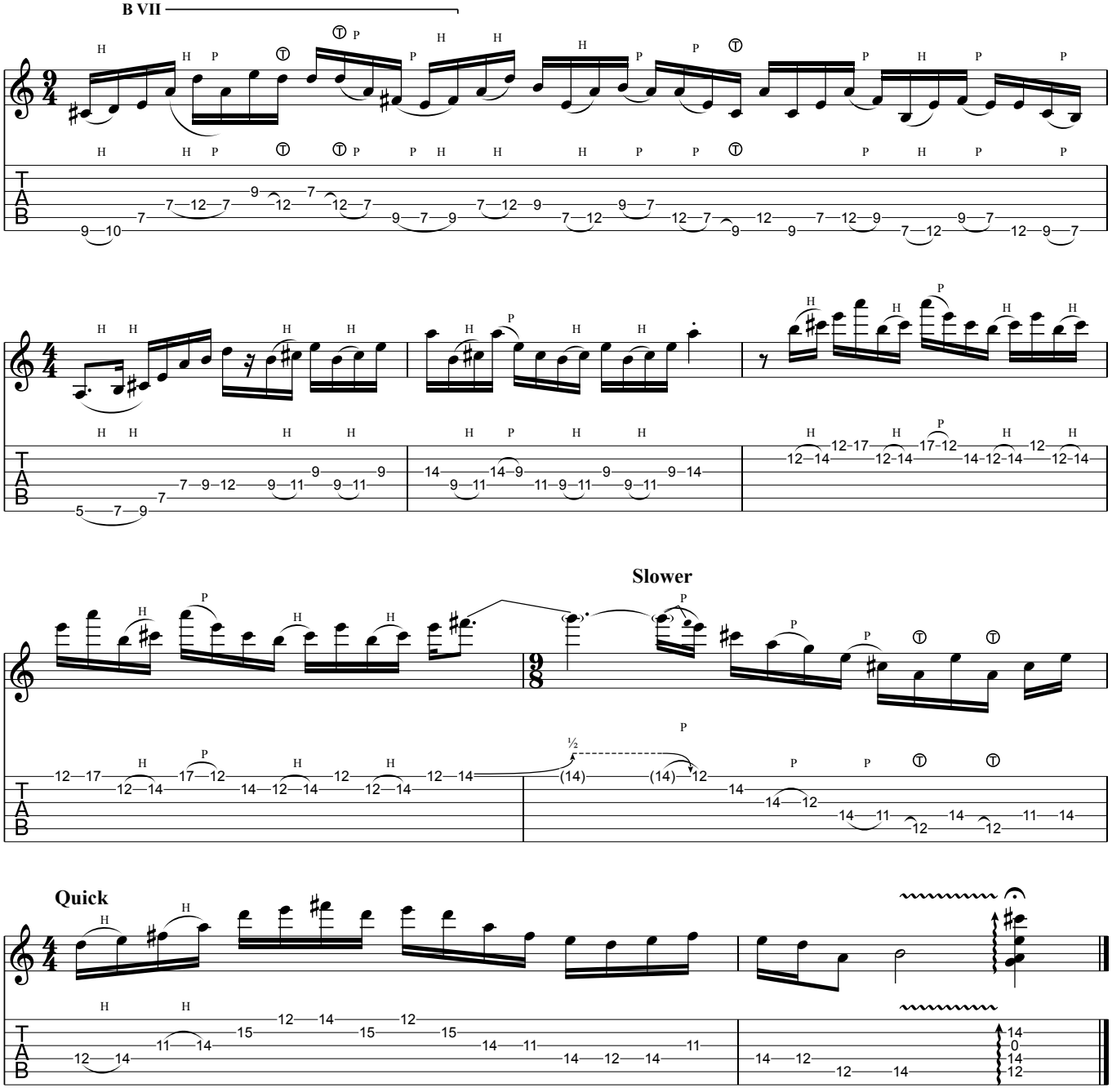
Ex. 237b

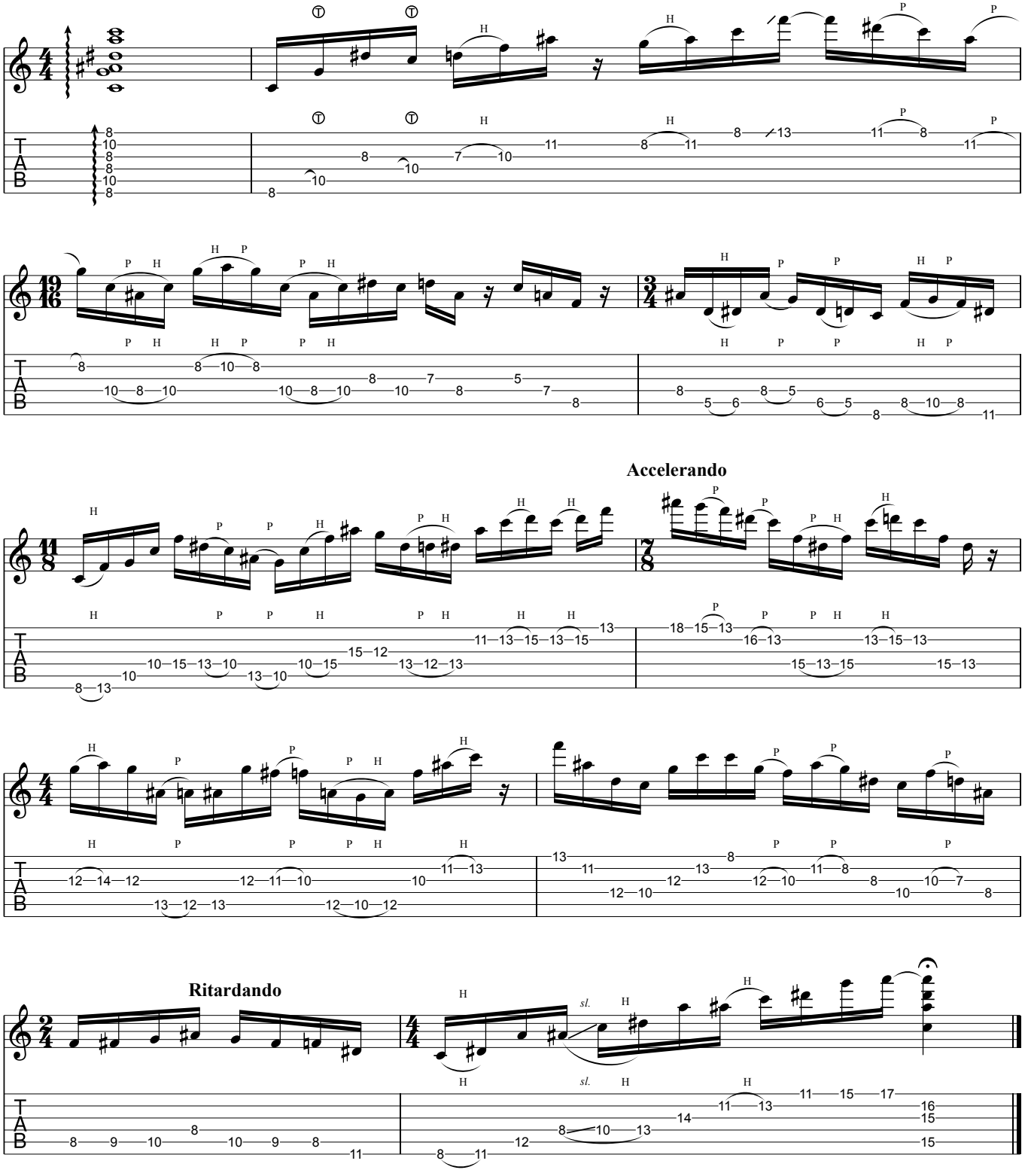


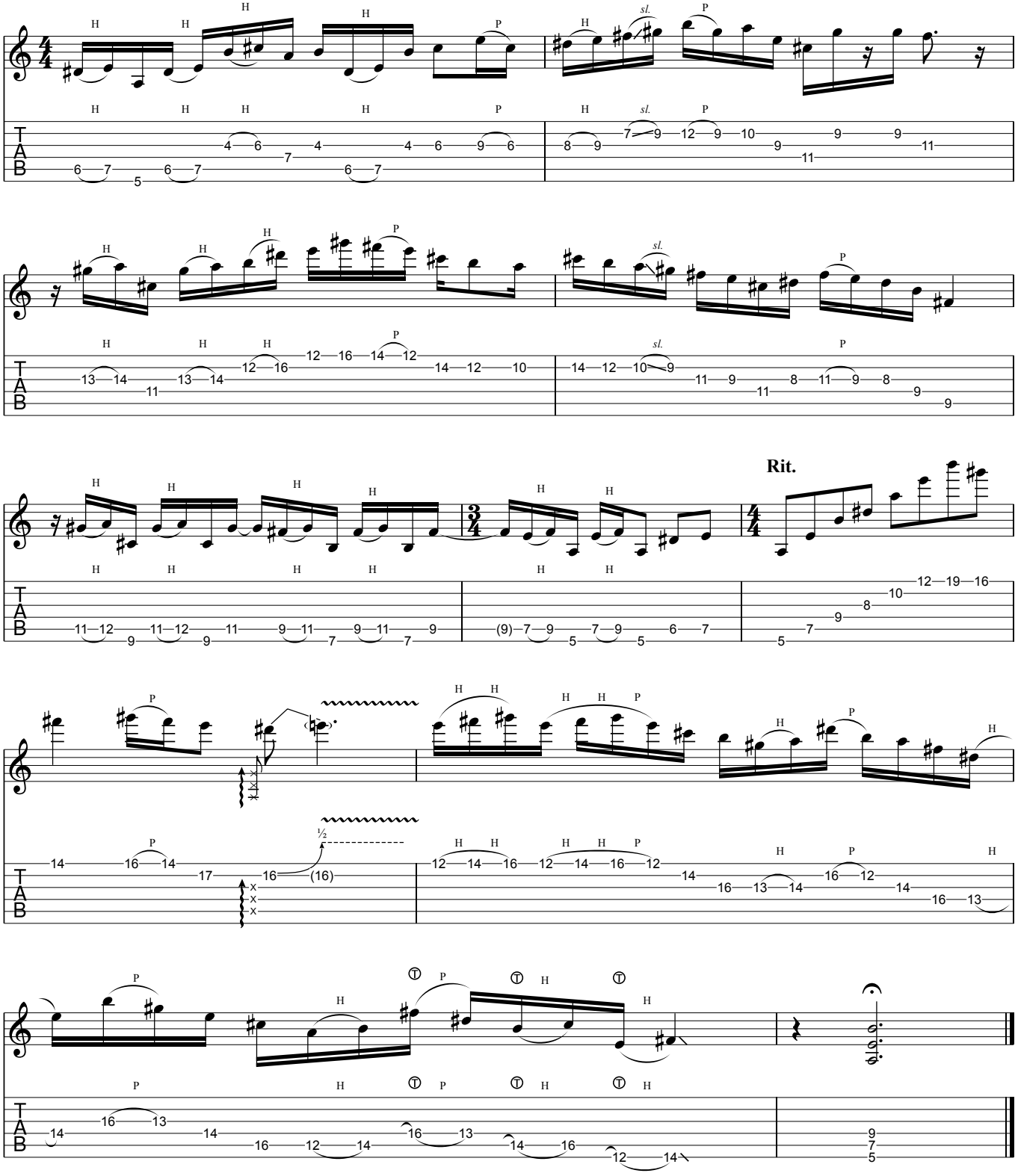
Ex. 237c



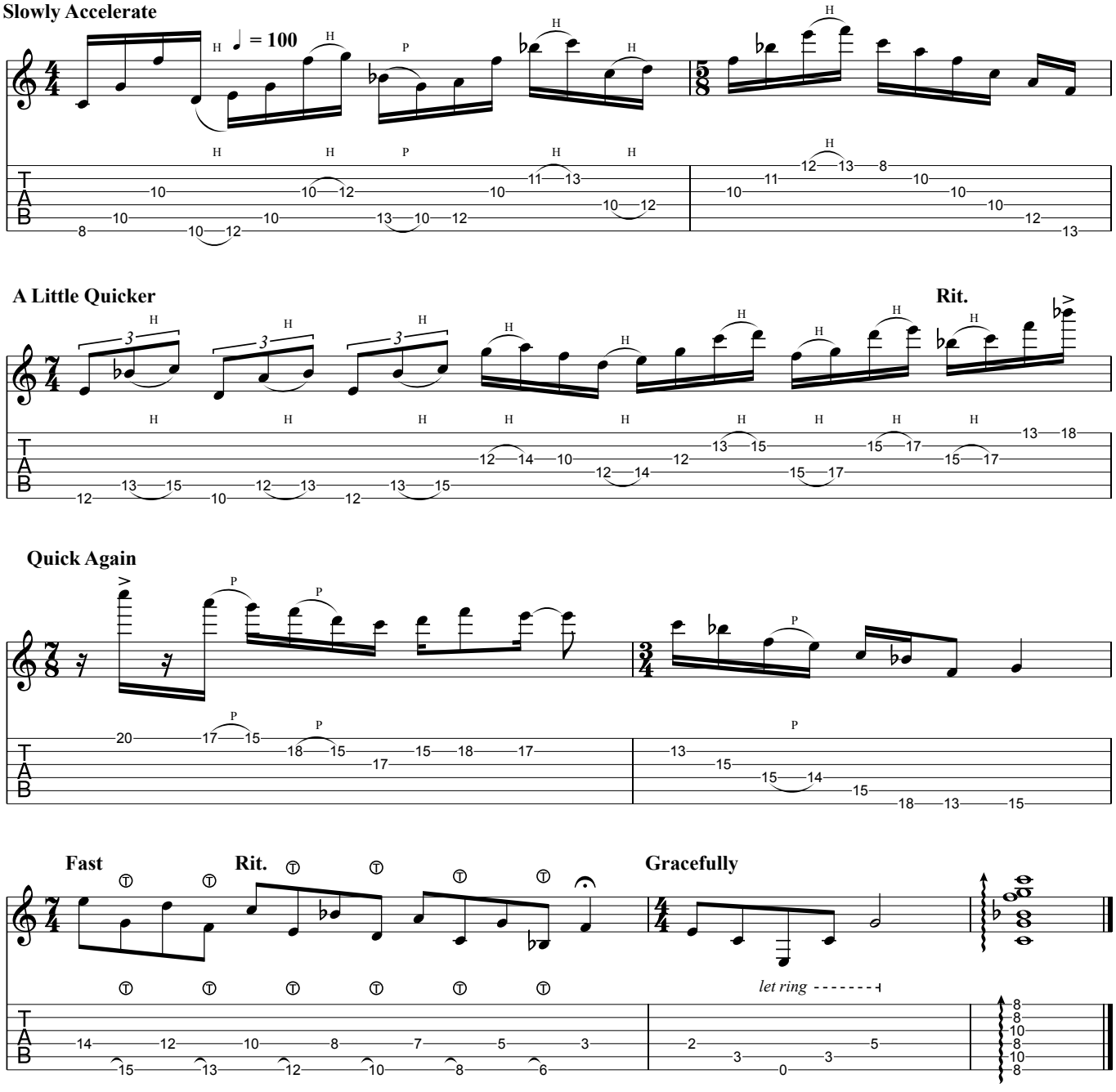
A Ionian to A Mixolydian Study

Instagram (5/25/20)

C Dorian Study

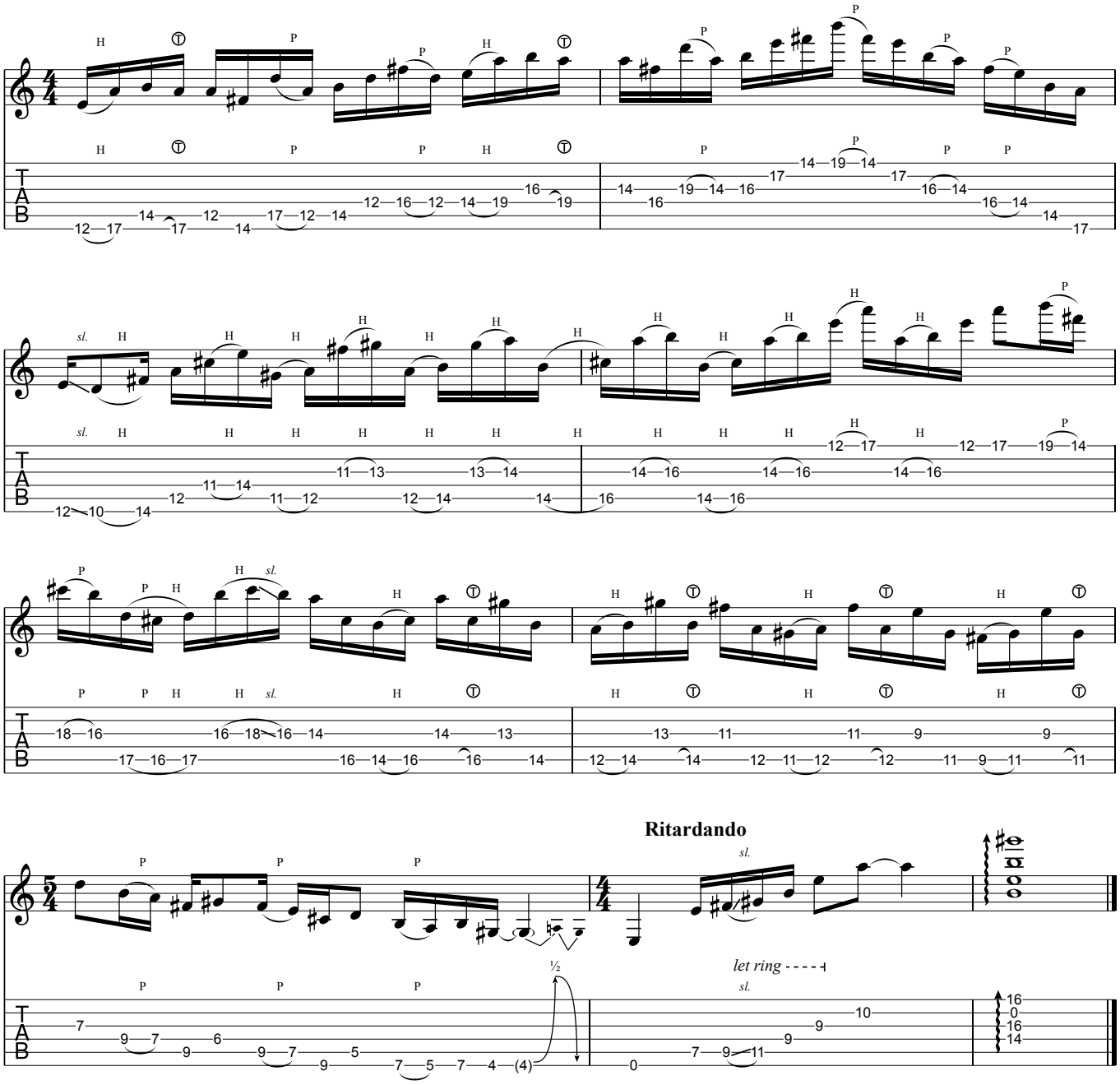
(Instagram 1/20/20)

A Lydian Study

Instagram (10/3/19)

C Mixolydian Study

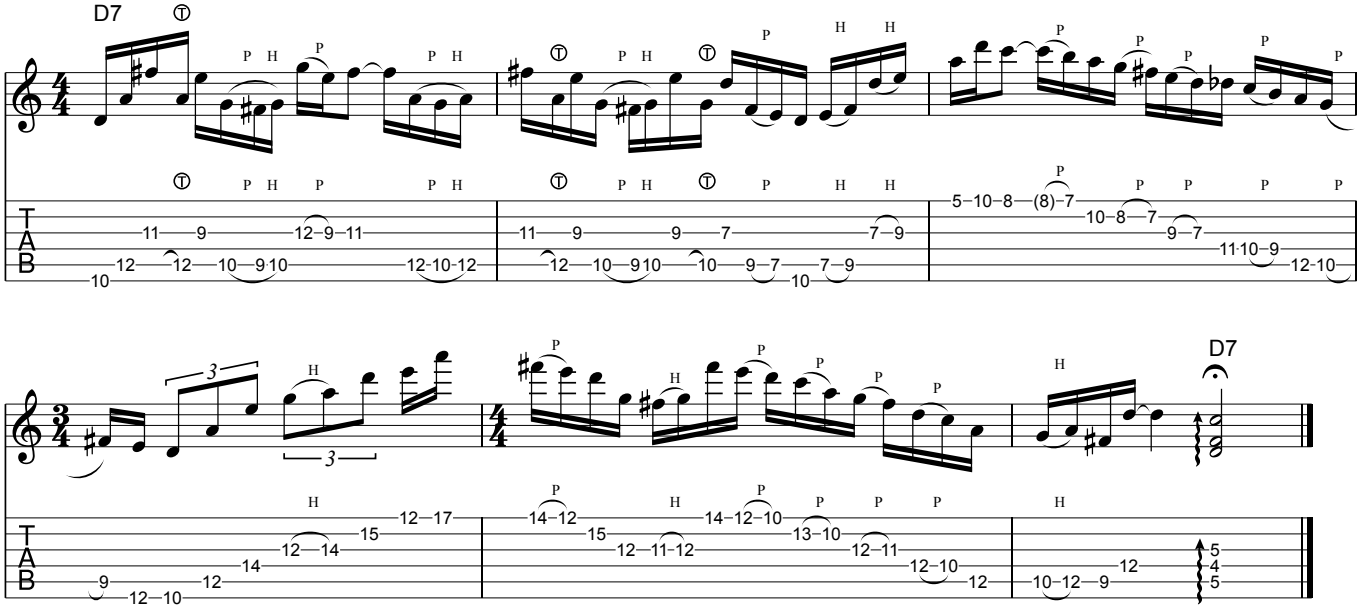
Instagram (4/30/20)



E Mixolydian Study

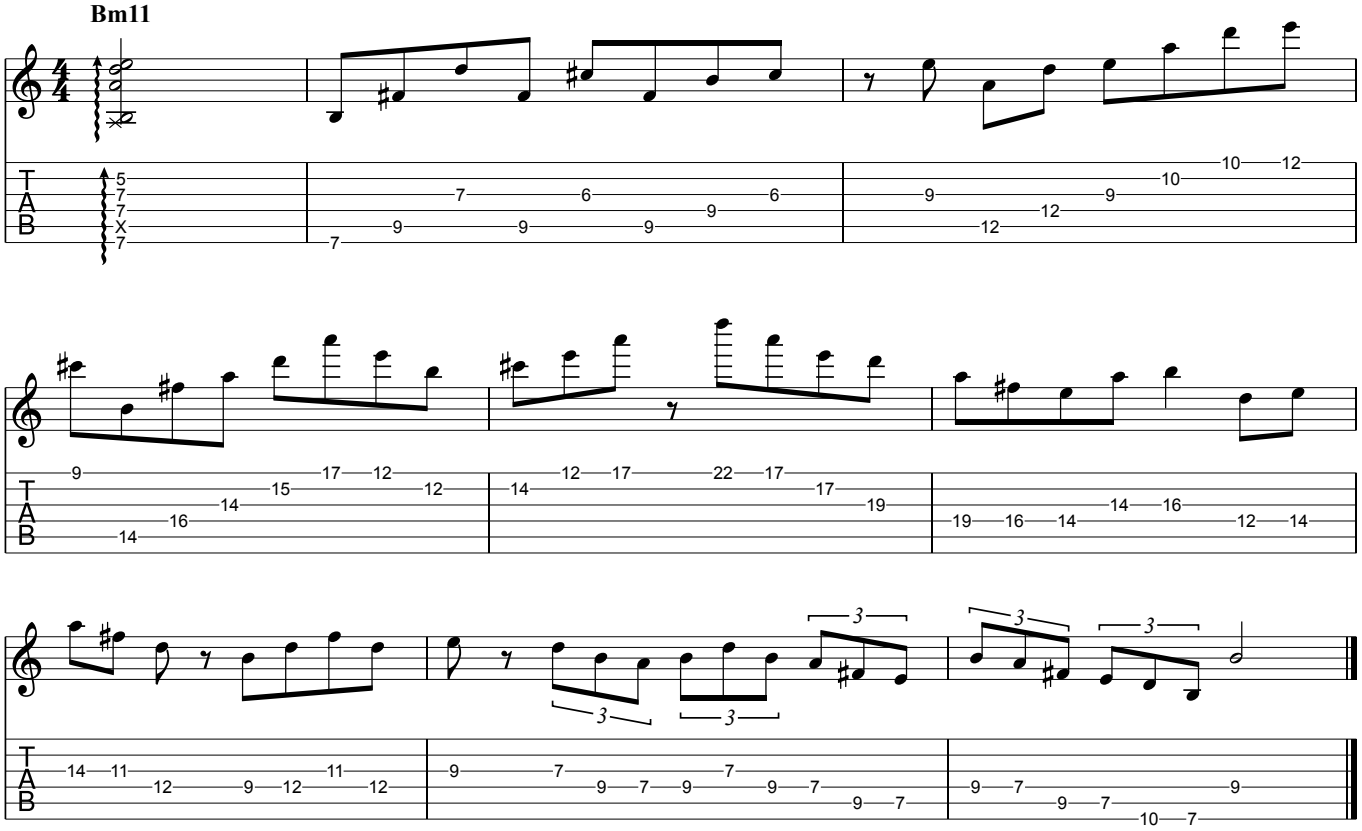
Instagram (11/23/19)

D Bebop Dominant Study

Instagram (7/23/19)



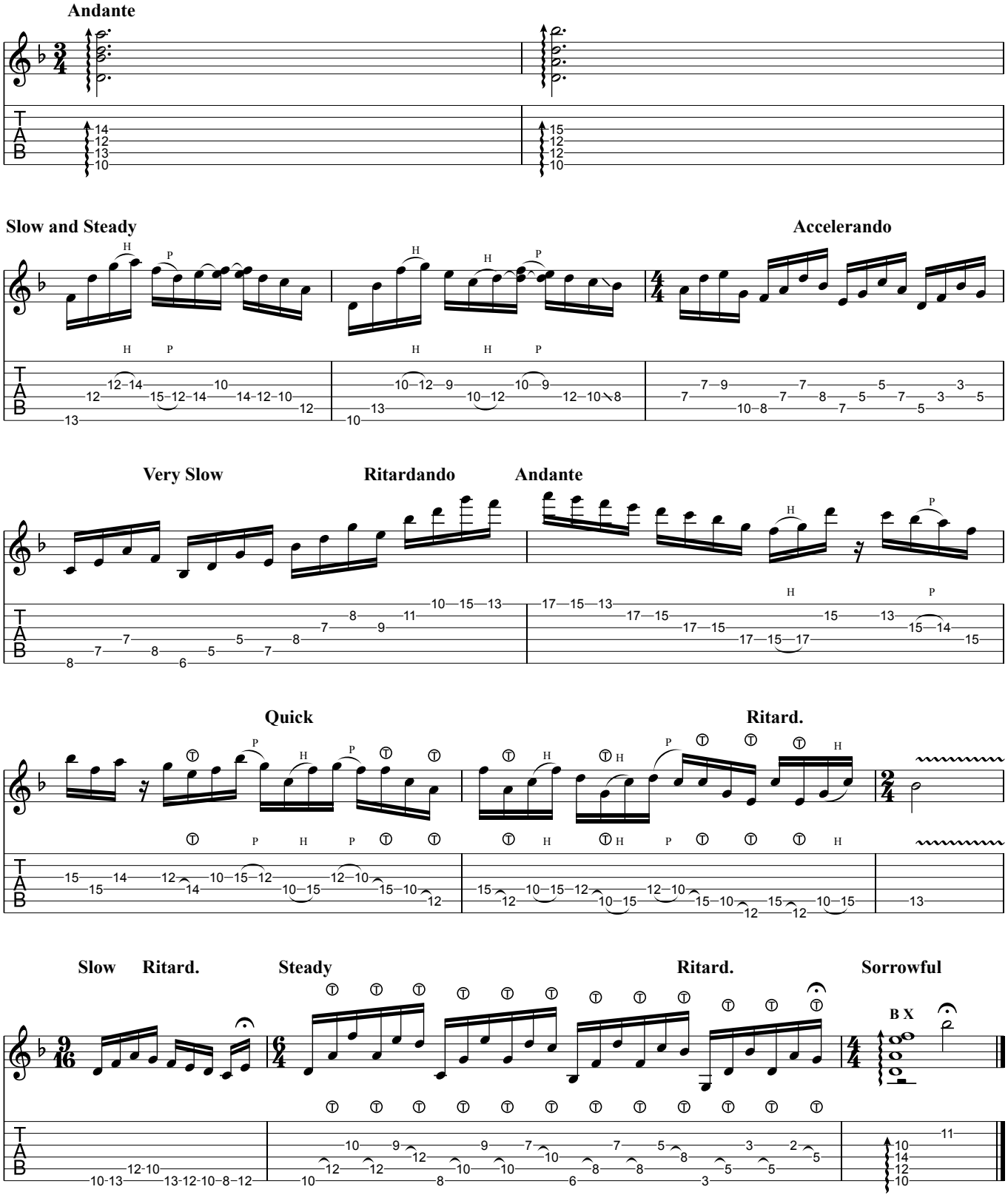
B Minor Study

Instagram (8/21/19)

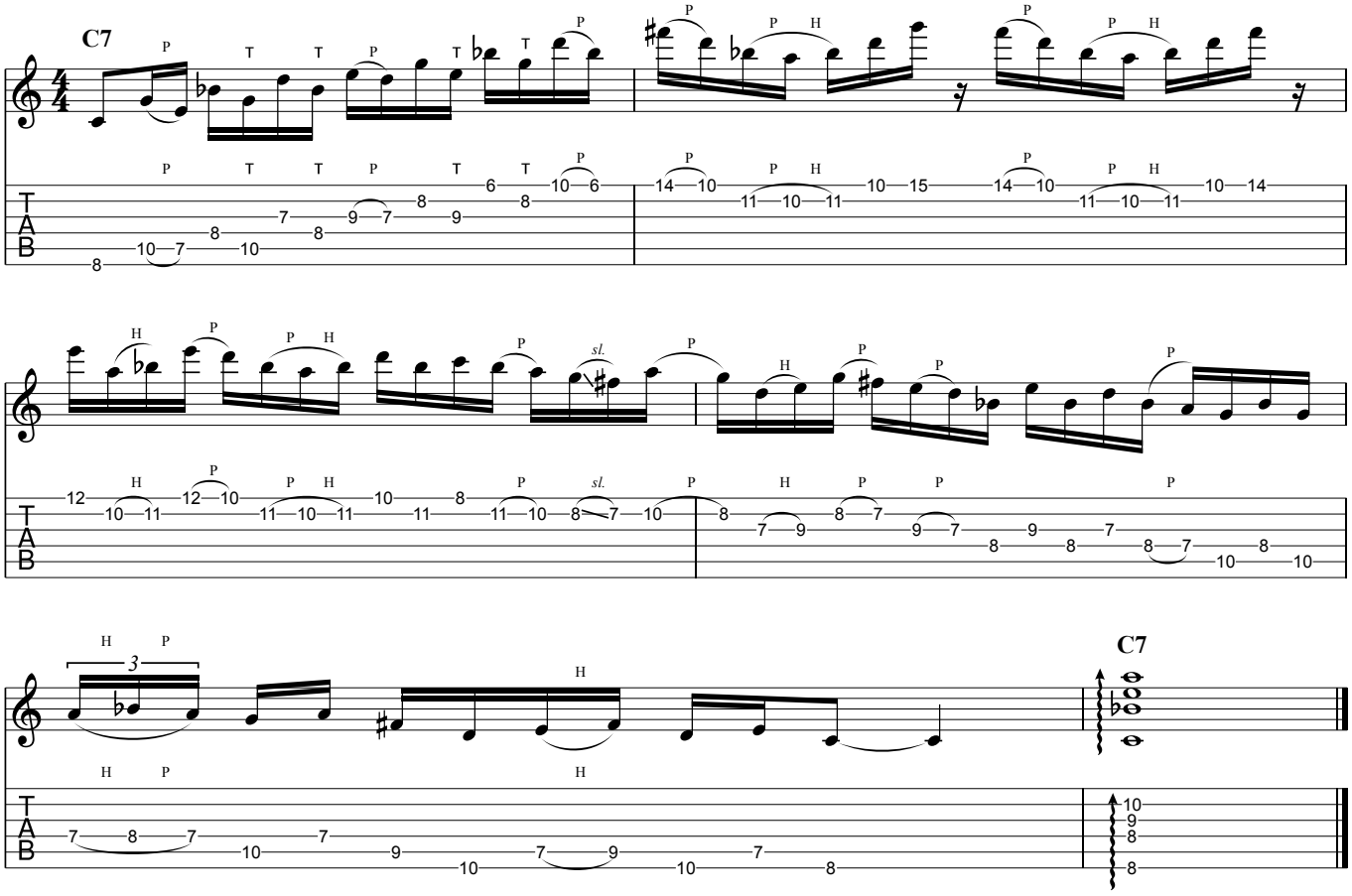


D Aeolian Study

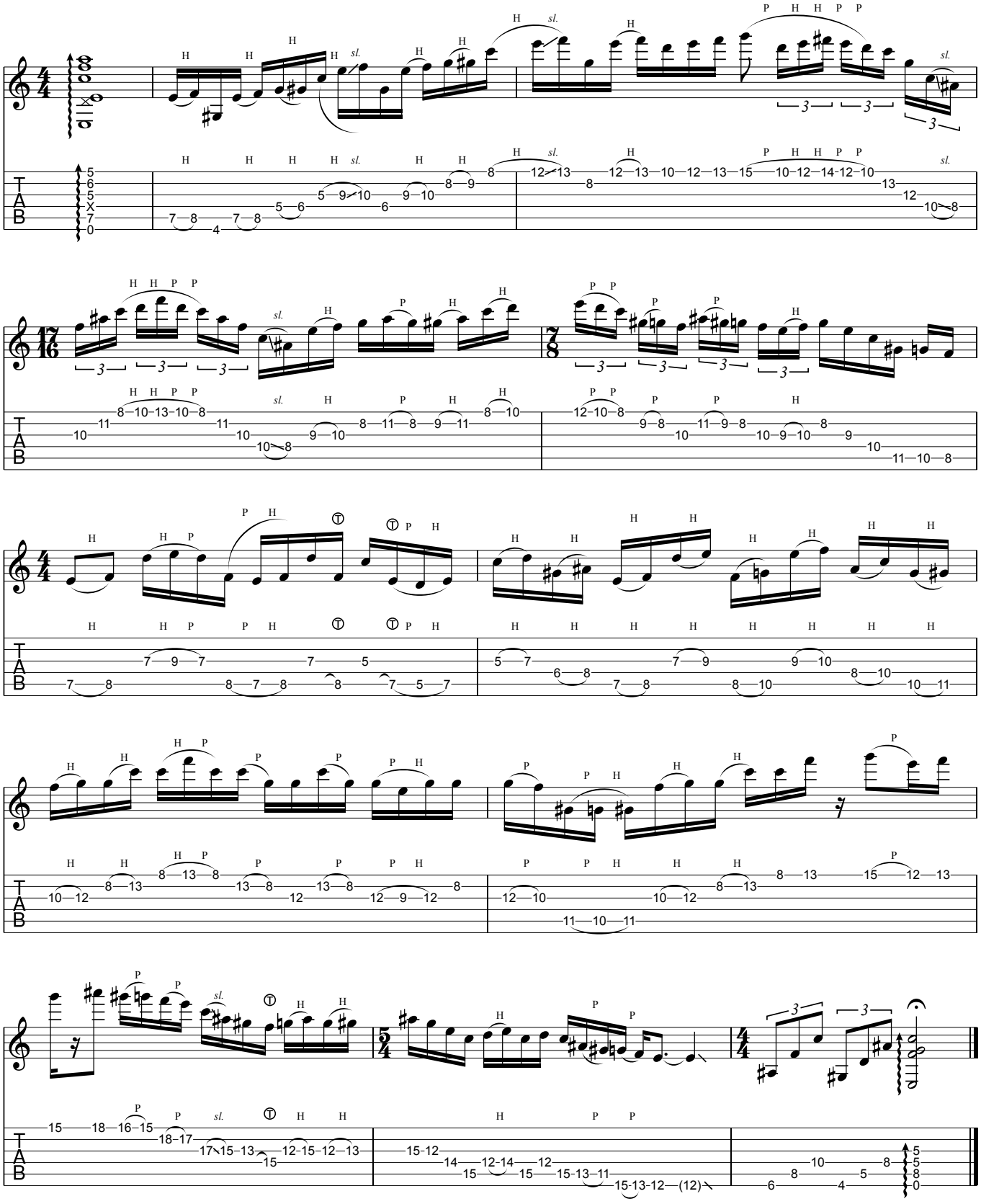
Instagram (5/30/20)

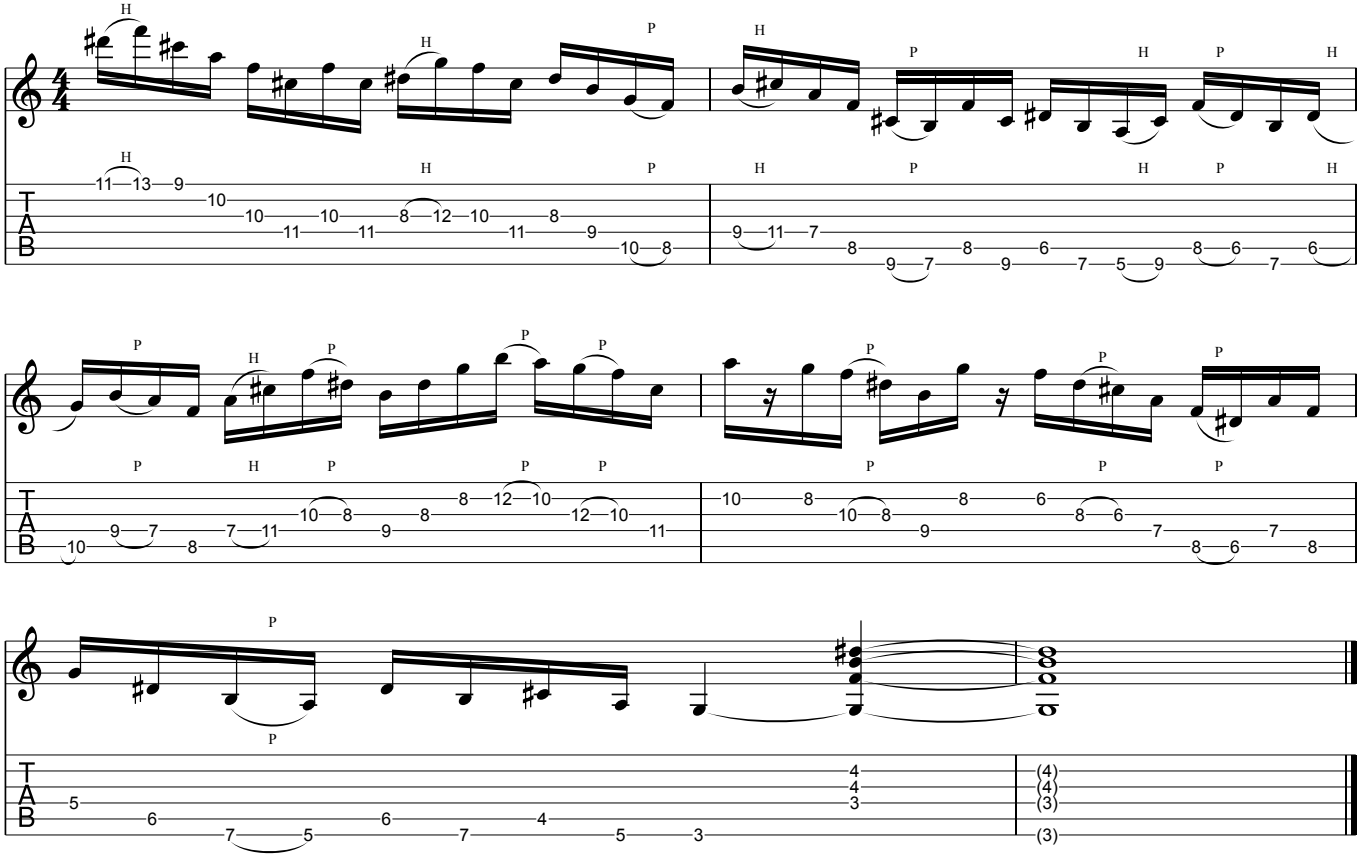


C Mixolydian #11 Study

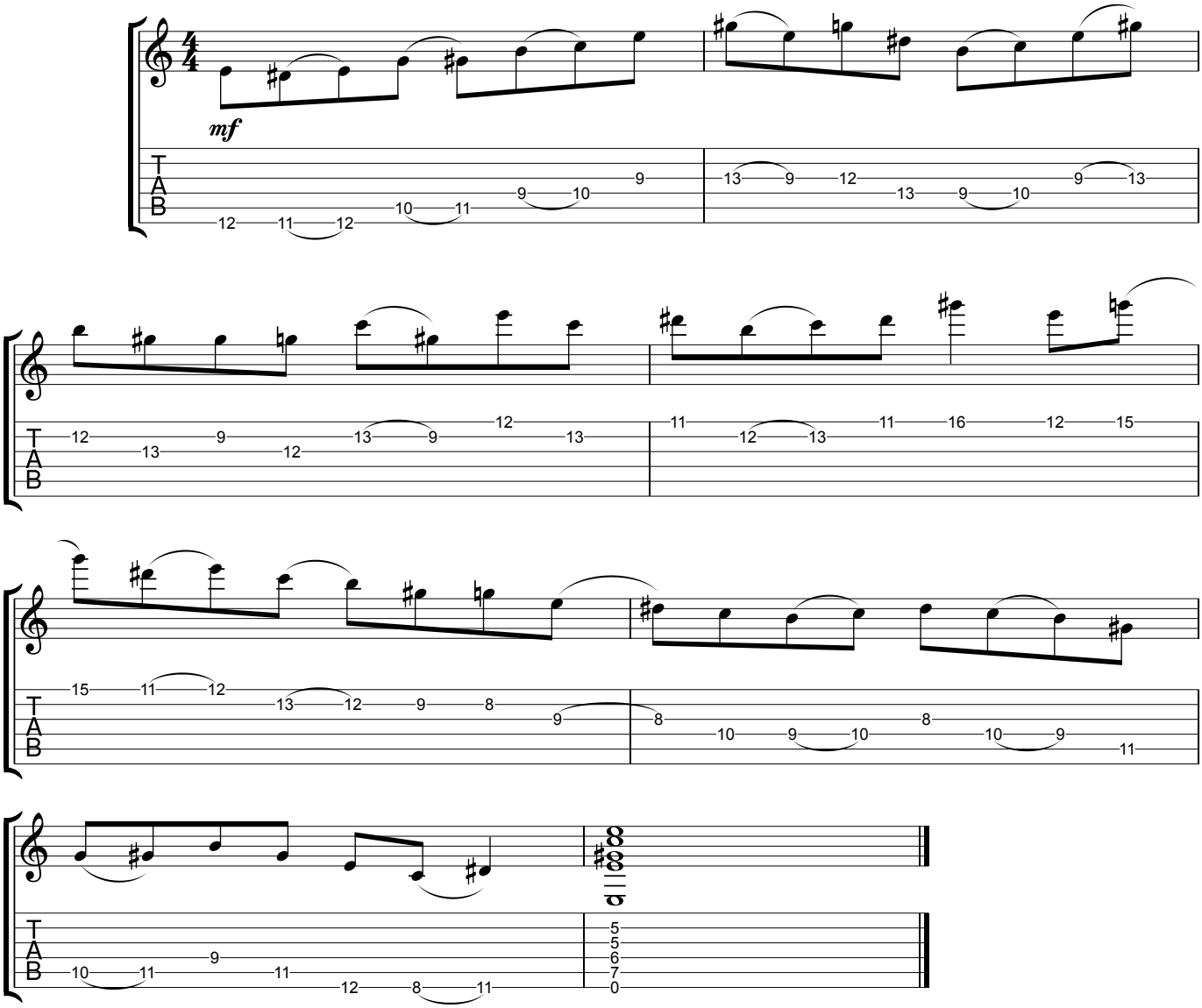


E Altered Dominant Study

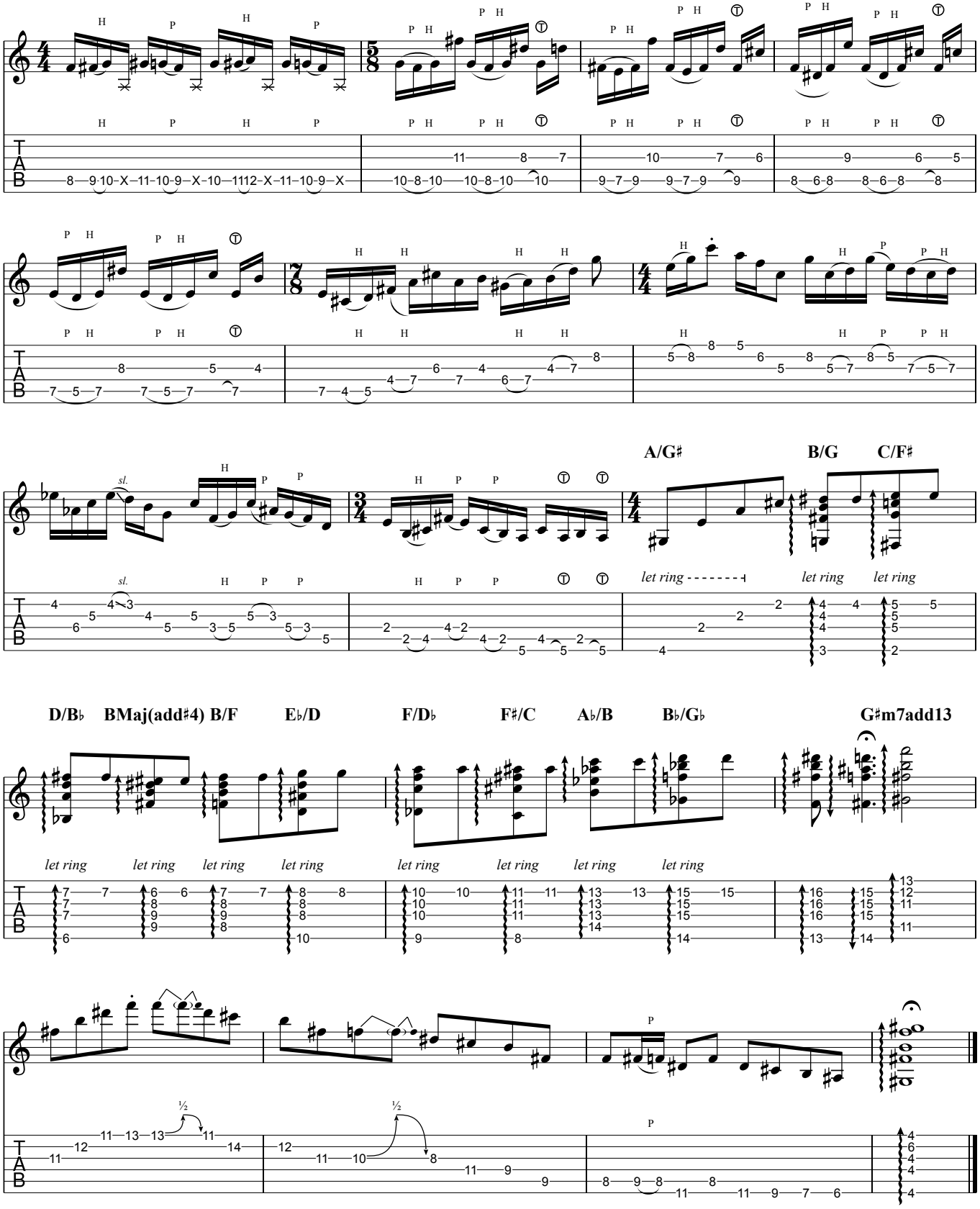
Instagram (1/11/20)

Whole Tone Study

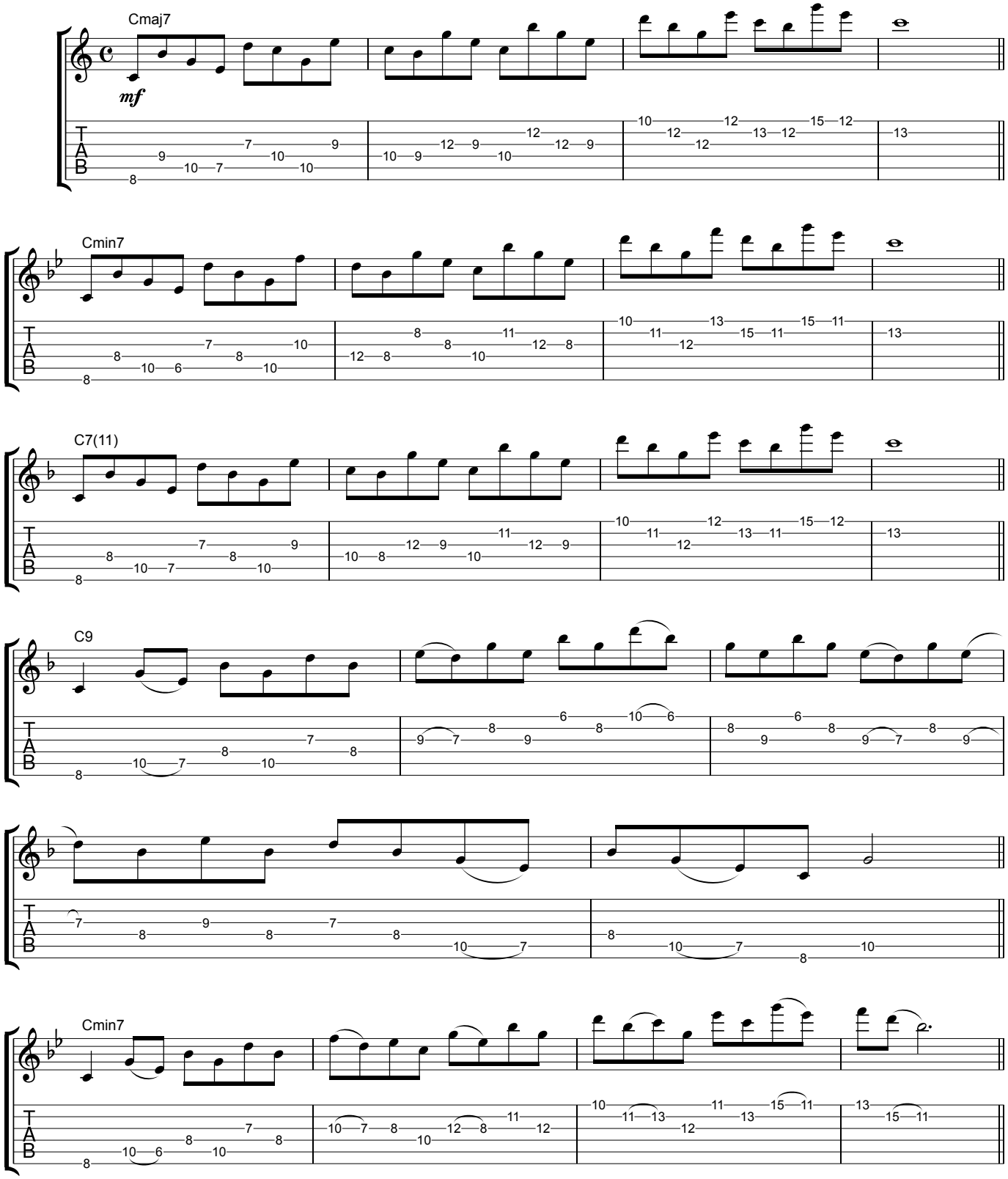
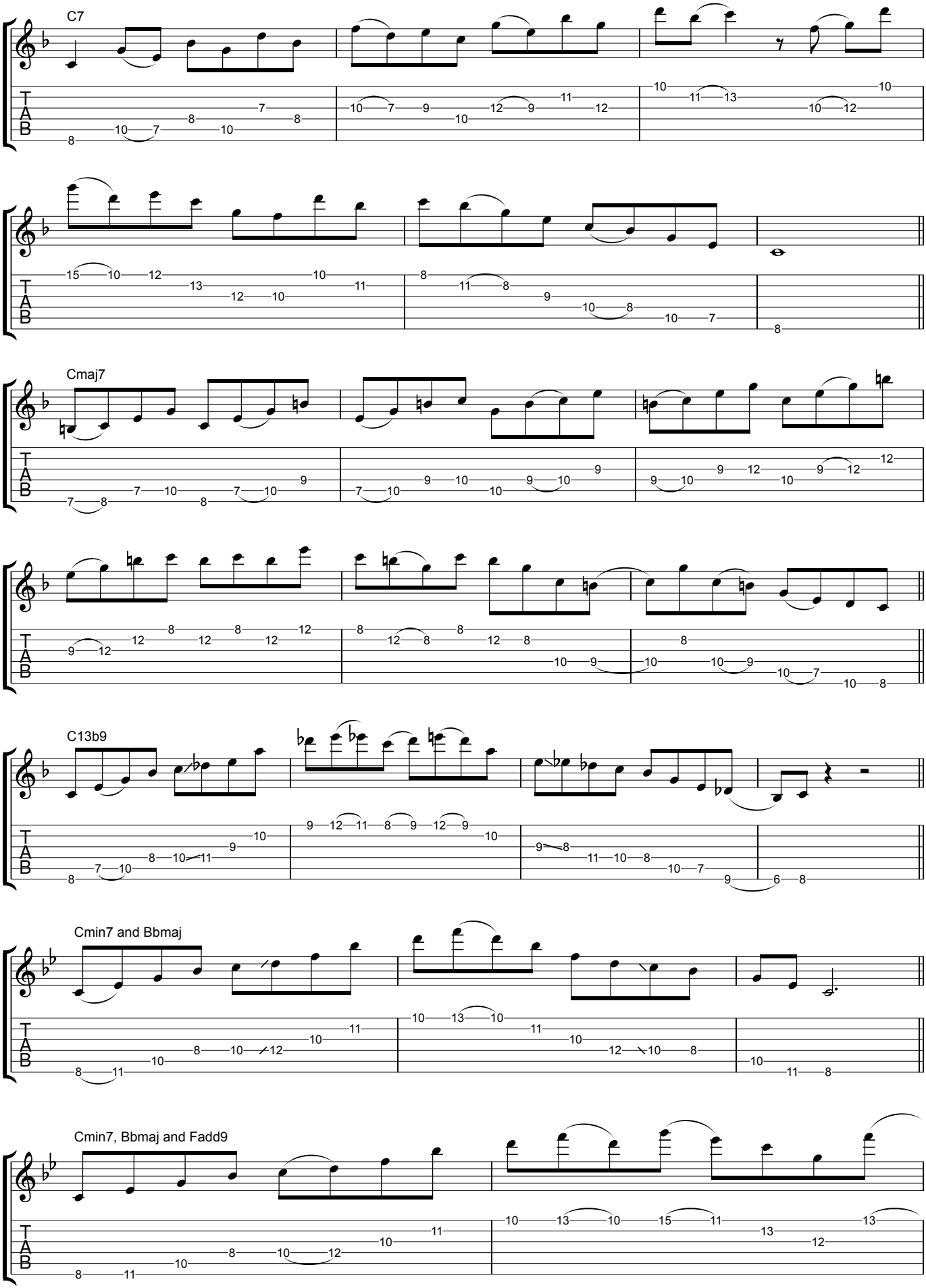
Instagram (11/1/19)

458

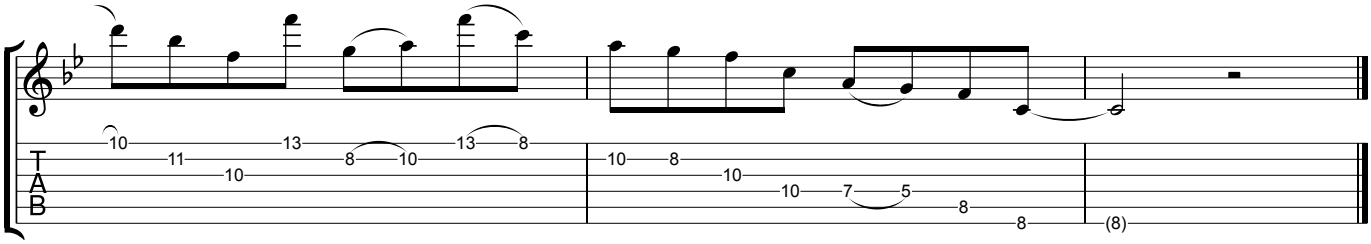
E Augmented Scale Study



Chromatic/Ploytonal Improvisation

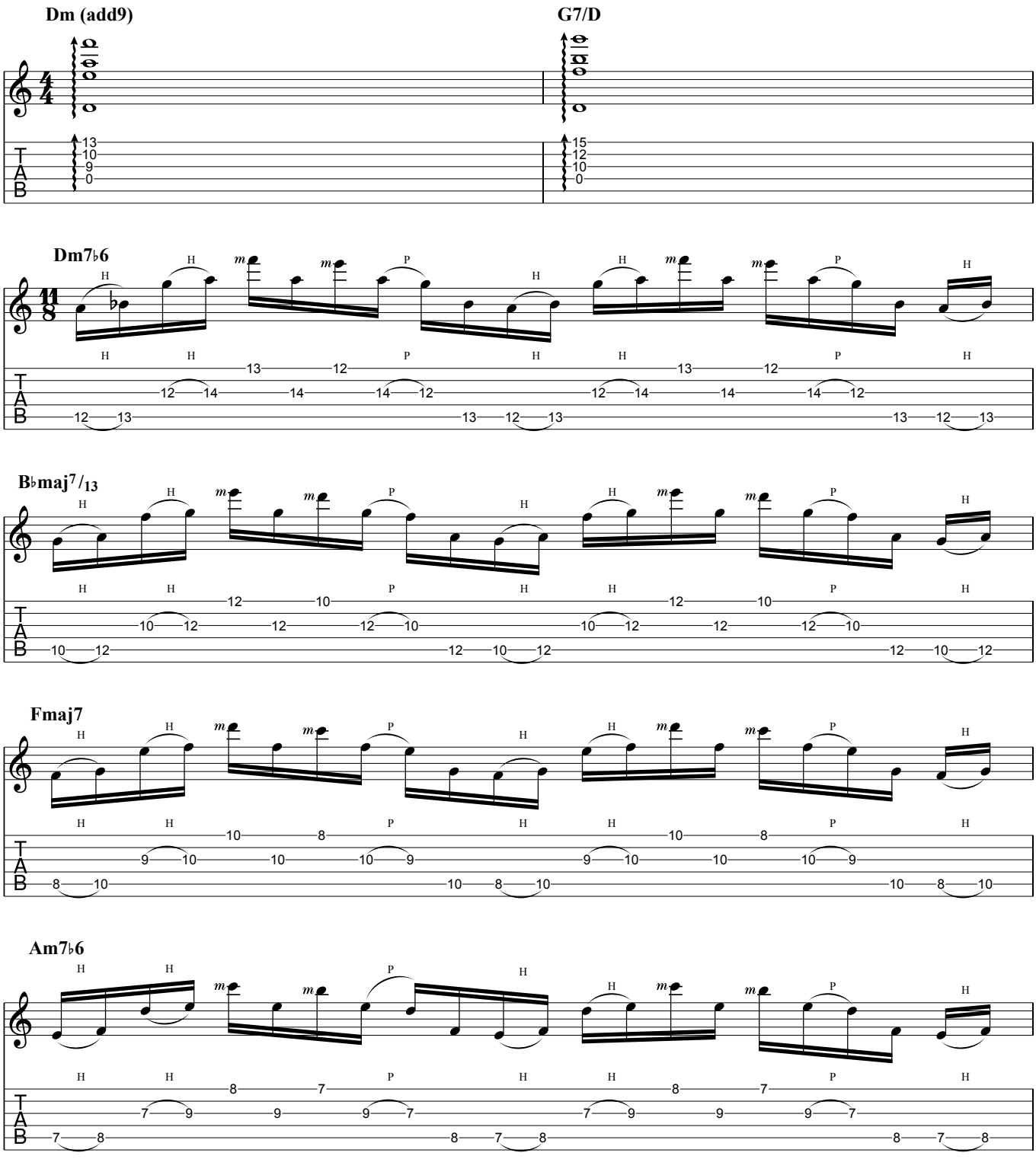
Instagram (12/22/19)

460

Ultra Mega Arpeggios

461

C7
Cmaj7
C13b9
Cmin7 and Bbmaj
Cmin7, Bbmaj and Fadd9

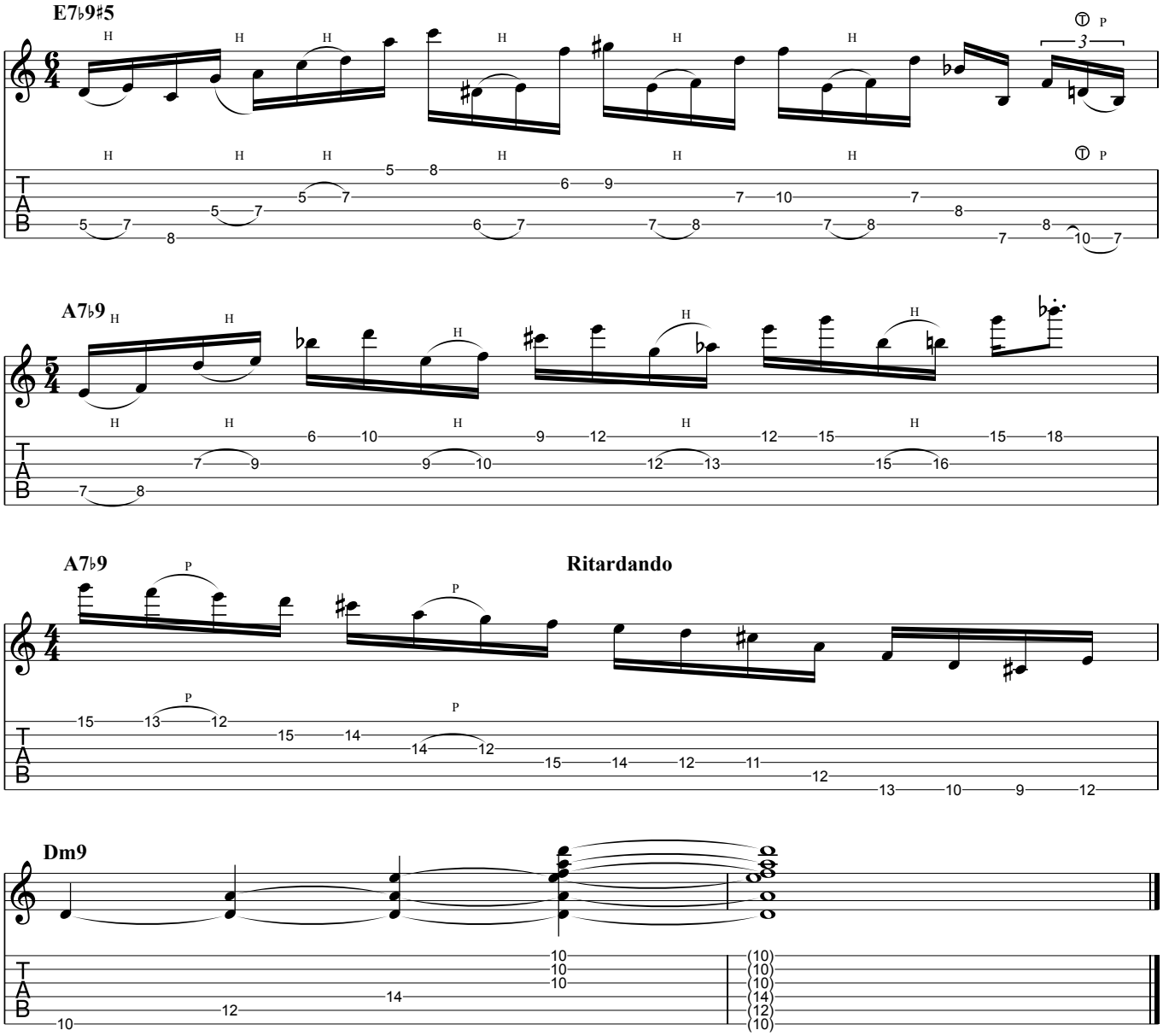


463

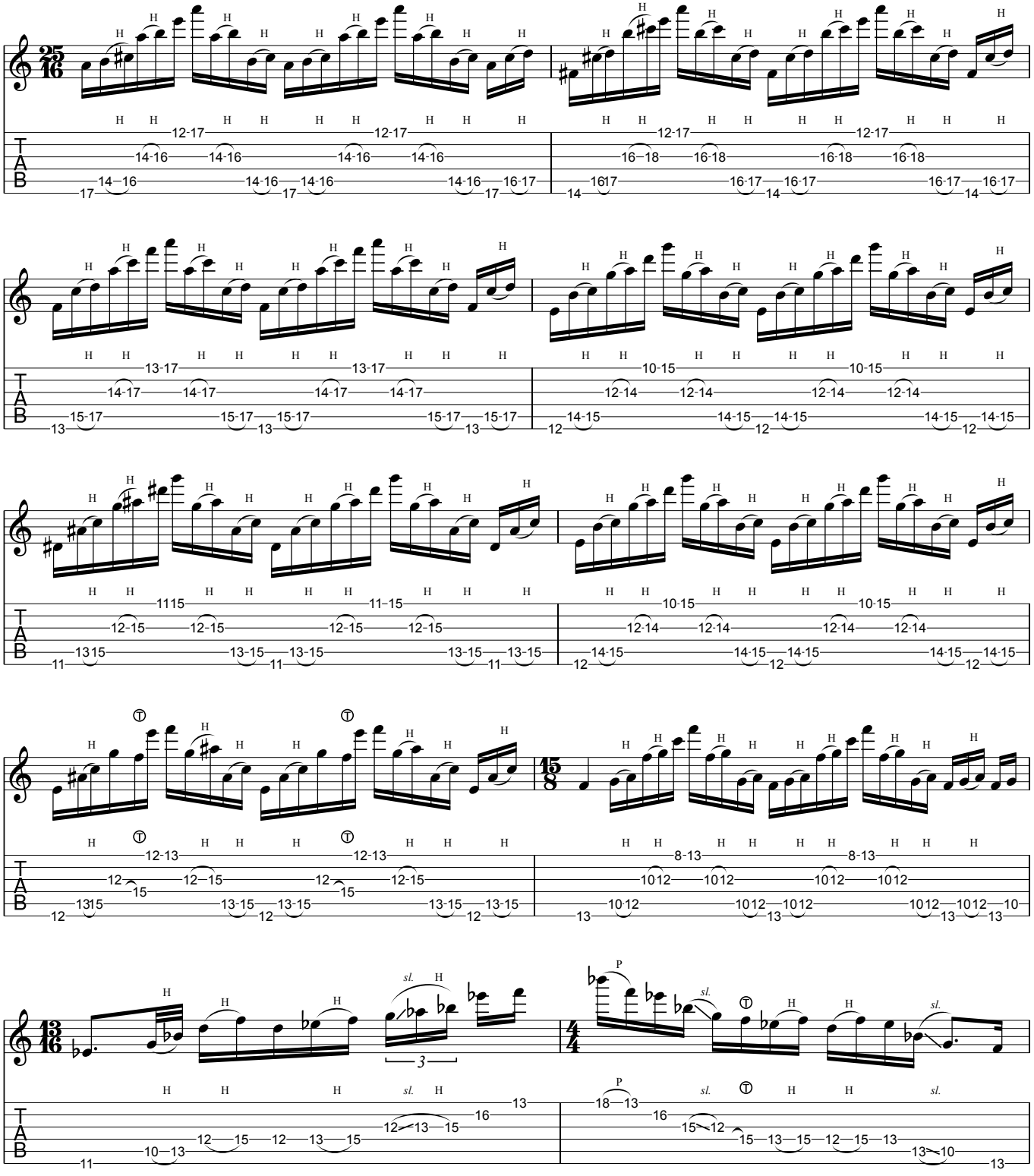
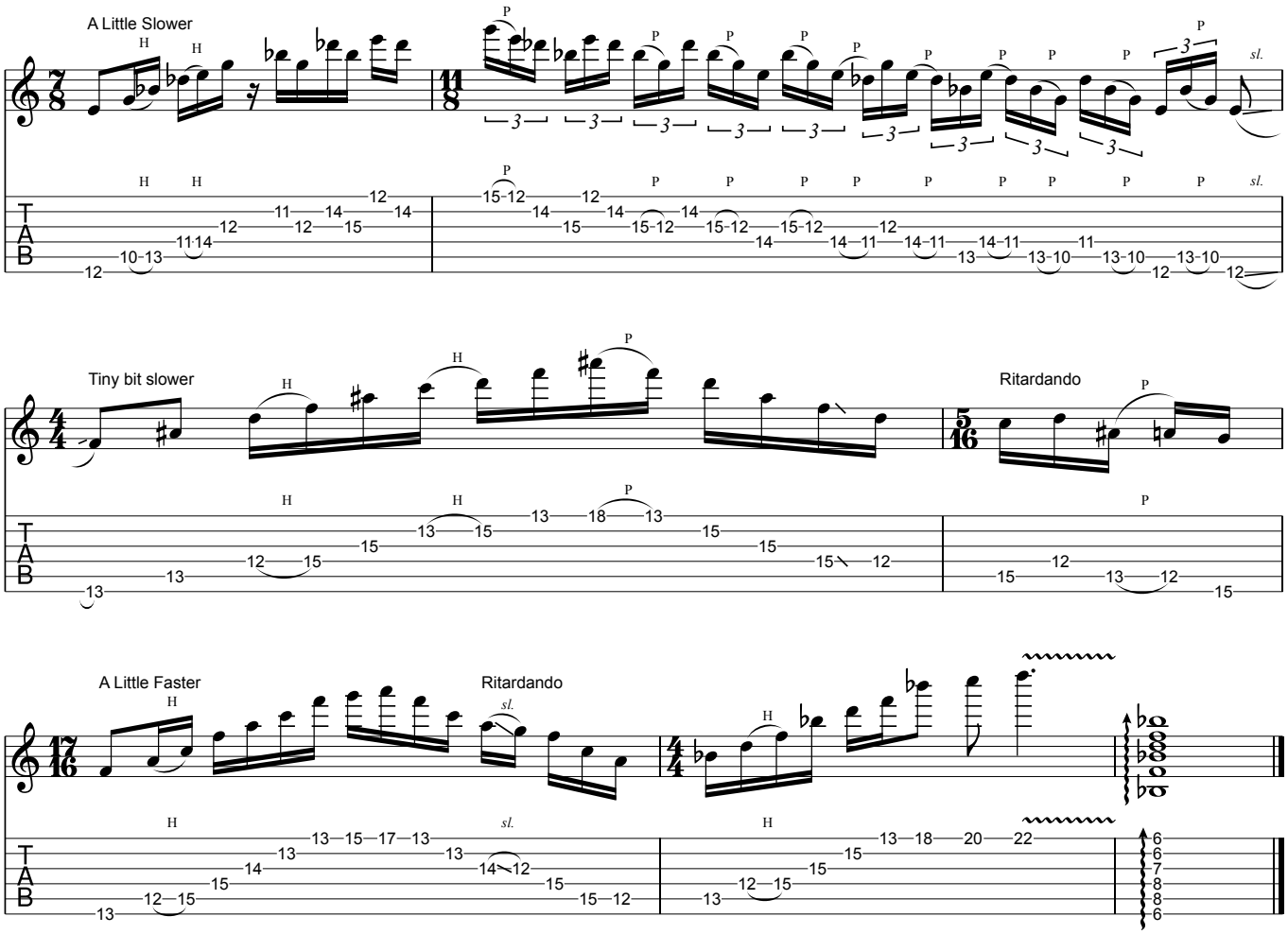
D Minor Improv



464


E7♭9♯5
A7♭9
A7♭9
Ritardando
Dm9

Sunday Morning Warmup

Instagram (2/2/20)


A Little Slower
Tiny bit slower
Ritardando
A Little Faster
Ritardando

CHAPTER 5

TECHNIQUE AND PRACTICE

Bass Lines

A good place to begin talking about bass lines is in relation to the two most common harmonic rhythms in jazz.

1. 2 chords per bar
2. 1 chord per bar

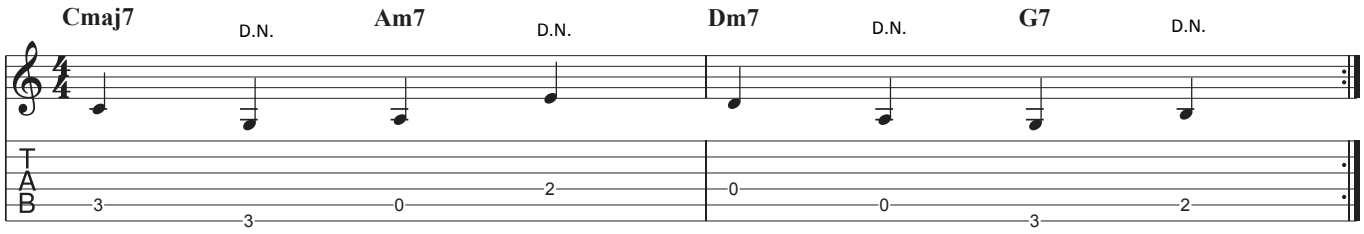
When we have two chords per bar there are three basic types of lines to choose from:

Two Chords per Bar:

D.N. = Diatonic Neighbor
C.N. = Chromatic Neighbor
I = Inversion

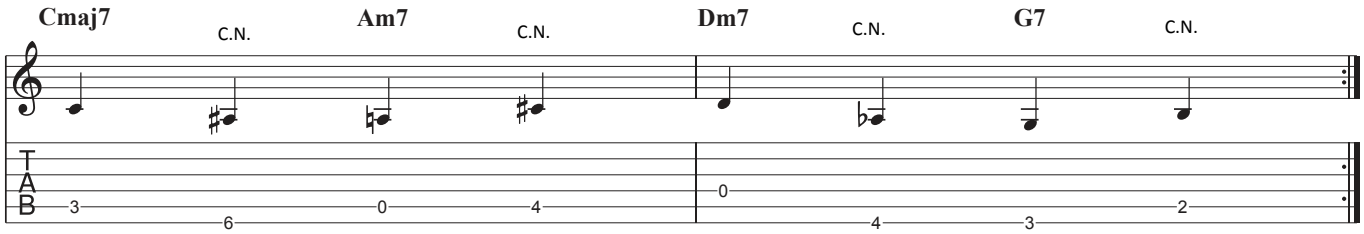
1. Roots followed by upper or lower diatonic neighbor.

Ex. 238



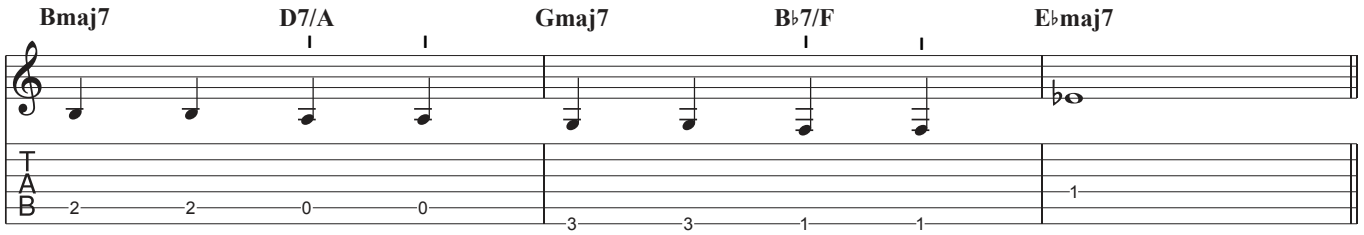
2. Roots followed by chromatic neighbor.

Ex. 239



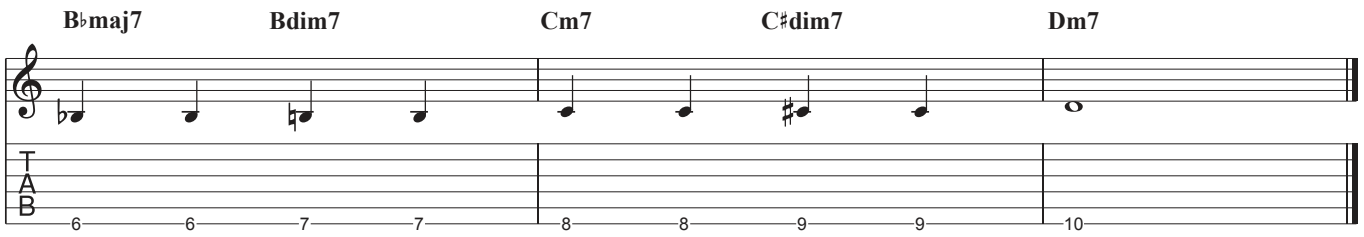
3. Ascending or descending inverted movement.

Ex. 240



4. Doubling the root on chromatic basslines.

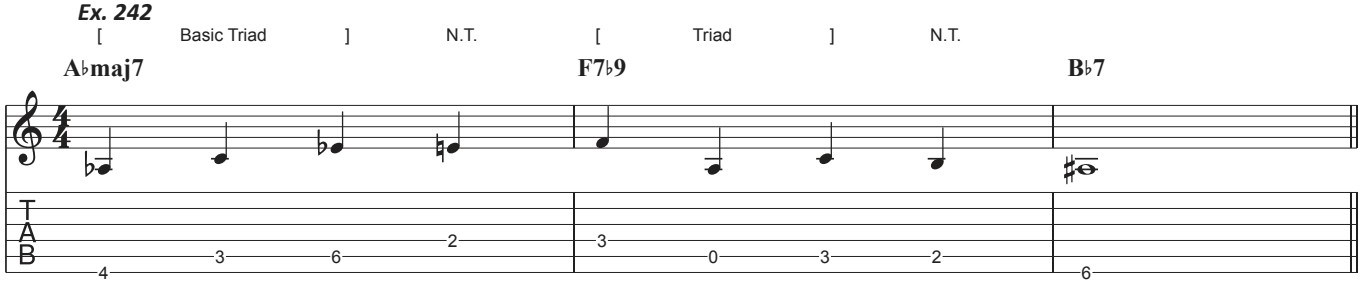
Ex. 241



One Chord per Bar:

1. Play the basic triad (in any inversion) on the first three beats and an upper or lower neighbor on the last.

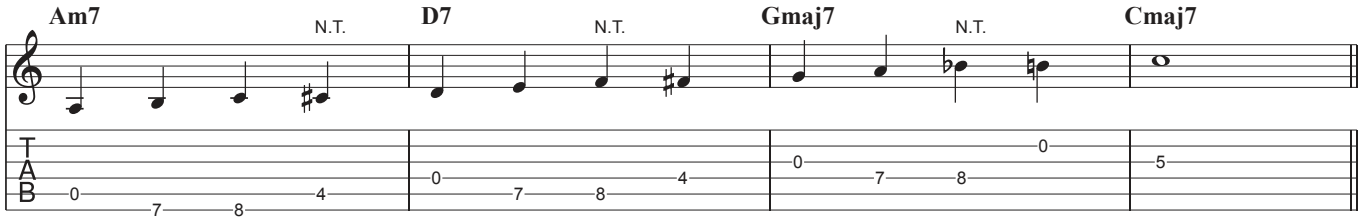
Ex. 242



2. Four chords a P5th descending apart.

Ex. 243

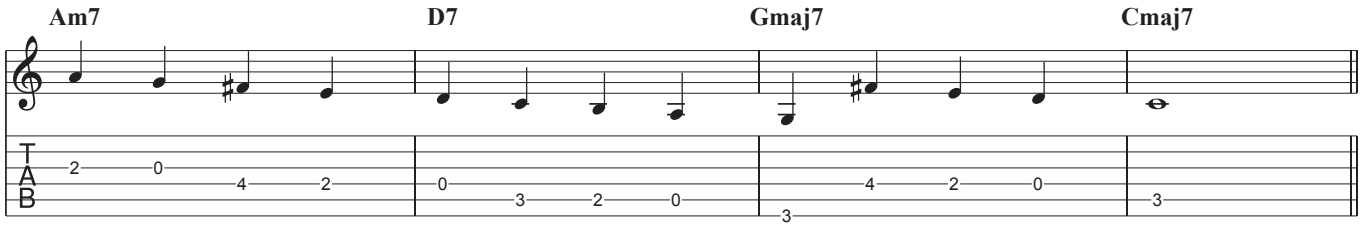
A. Scalar ascending (use appropriate modes)



On minor chords, neighbor tones fall on the $\overline{4}^{th}$ beat on Maj & Dom they fall on the $\overline{3}^{rd}$.

Ex. 244

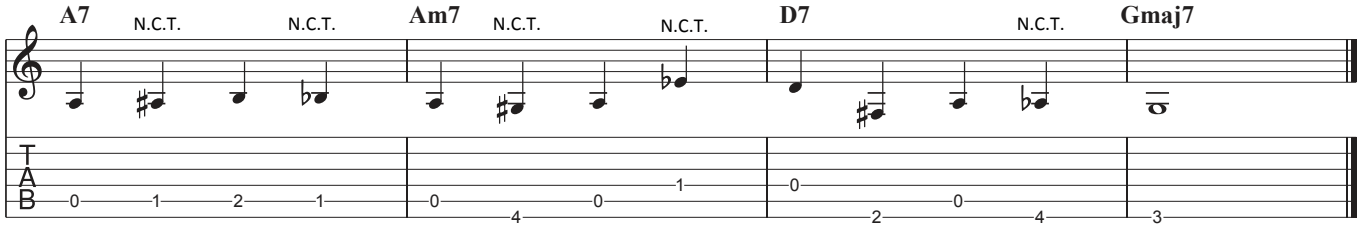
B. Scalar descending (use appropriate modes)



Note: there are no passing tones in Ex. 244 because there are five scale tones between chords.

Ex. 245

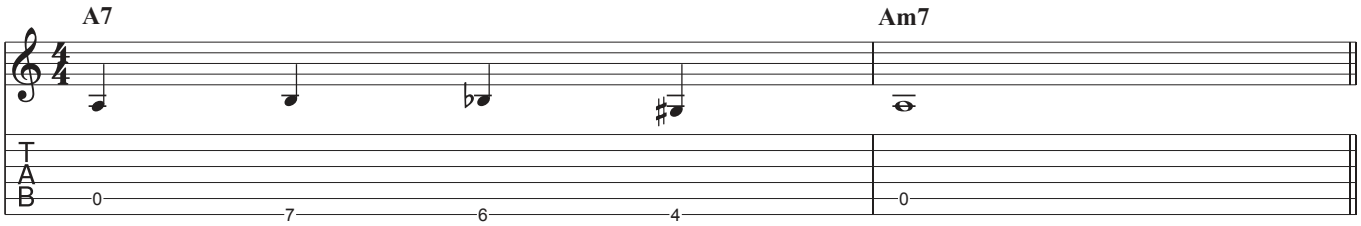
3. Chromatic



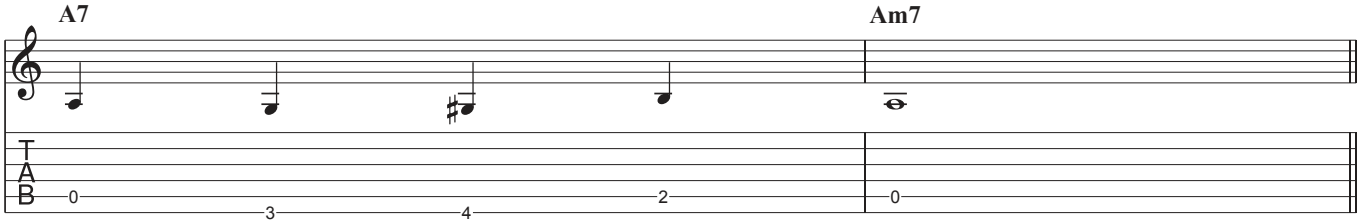
N.C.T. = Neighboring (Chromatic) Tones

Also Try:

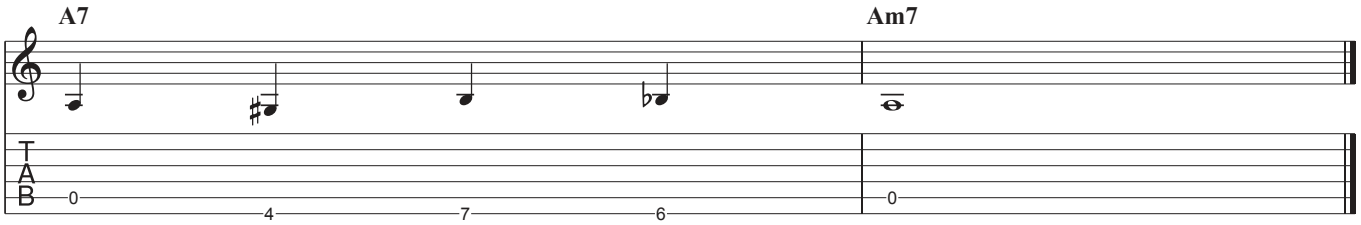
Ex. 246a



Ex. 246b



Ex. 246c



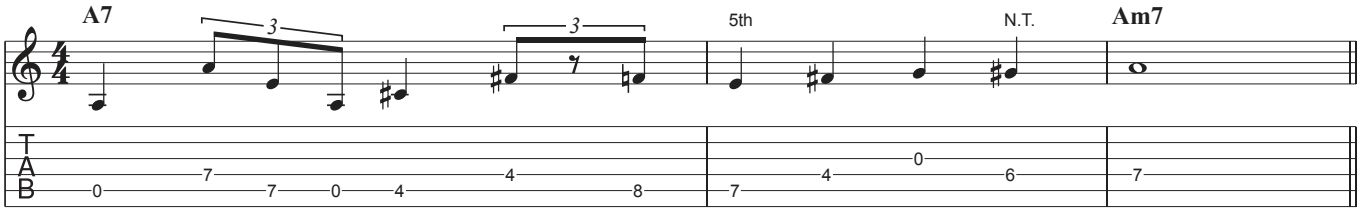
There are many combinations of chroma%c type lines. Use these as a point of departure.

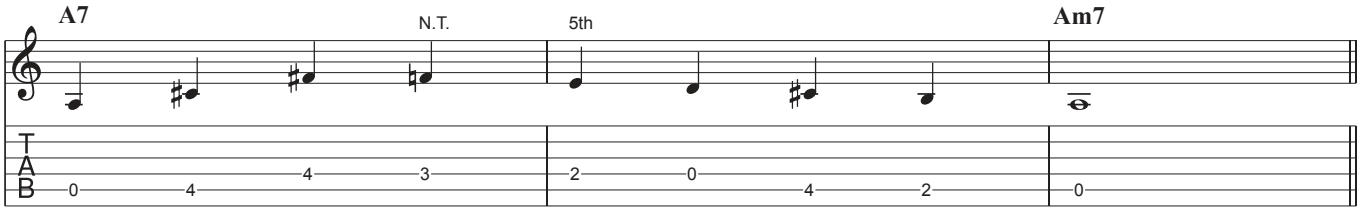
One Chord for Two Bars

In this instance we want to arrive on the fifth of A^7 at the

beginning of the second measure.

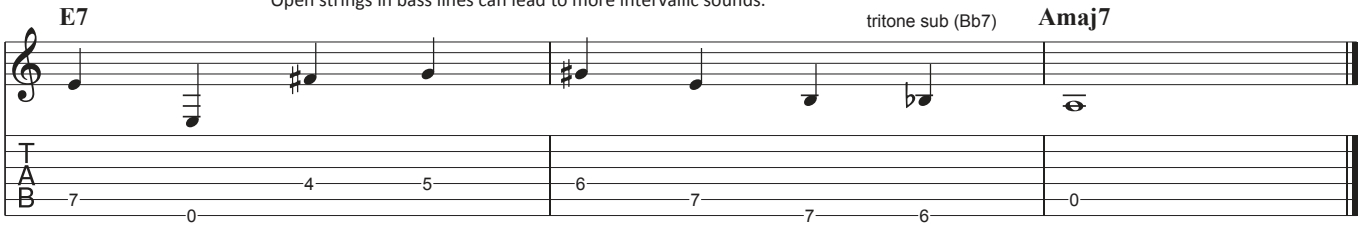
Ex. 247a



Ex. 247b

The second example used the F# (13^{th}) on the A^7 for the sake of variety.

Open strings in bass lines can lead to more intervallic sounds.

Ex. 248



Pedal Points

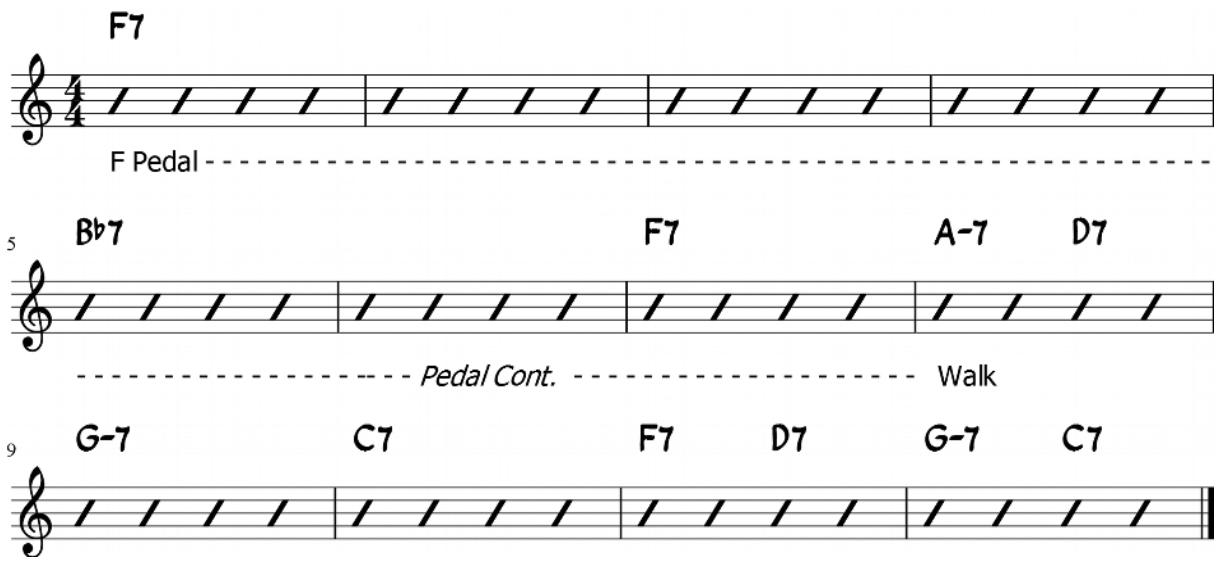
Pedal points are used in jazz to create excitement behind a solo. They many times provide a springboard to more chromatic type playing and can bring a solo or certain section of a tune to climax.

The most common type pedal points are:

Tonic Pedals

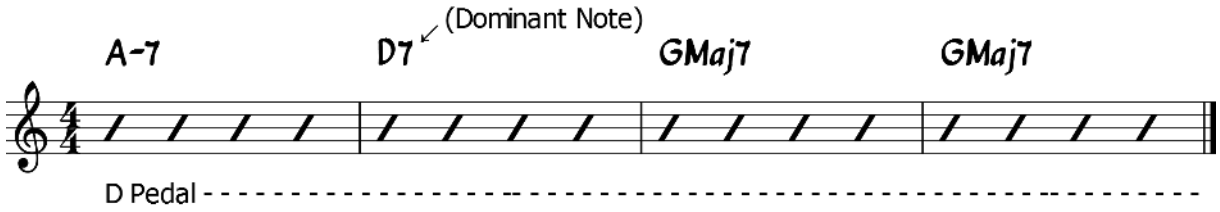
Ex. 249

F Blues



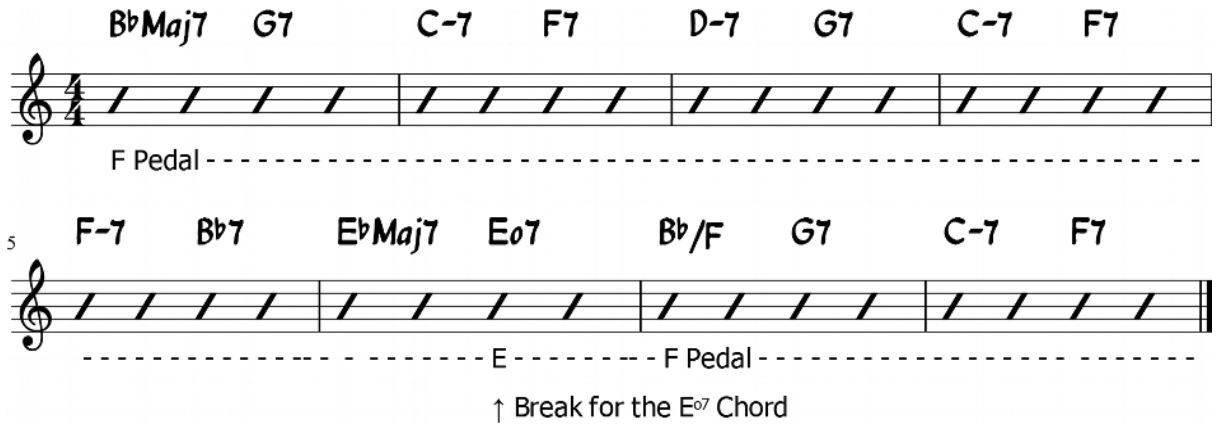
Dominant Pedals

Ex. 250



Rhythm Changes

Ex. 251

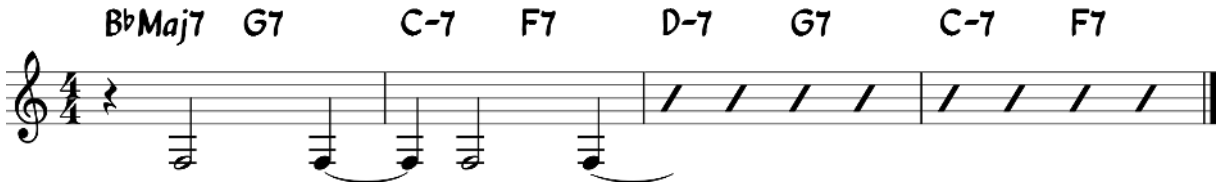


Pedals are best used

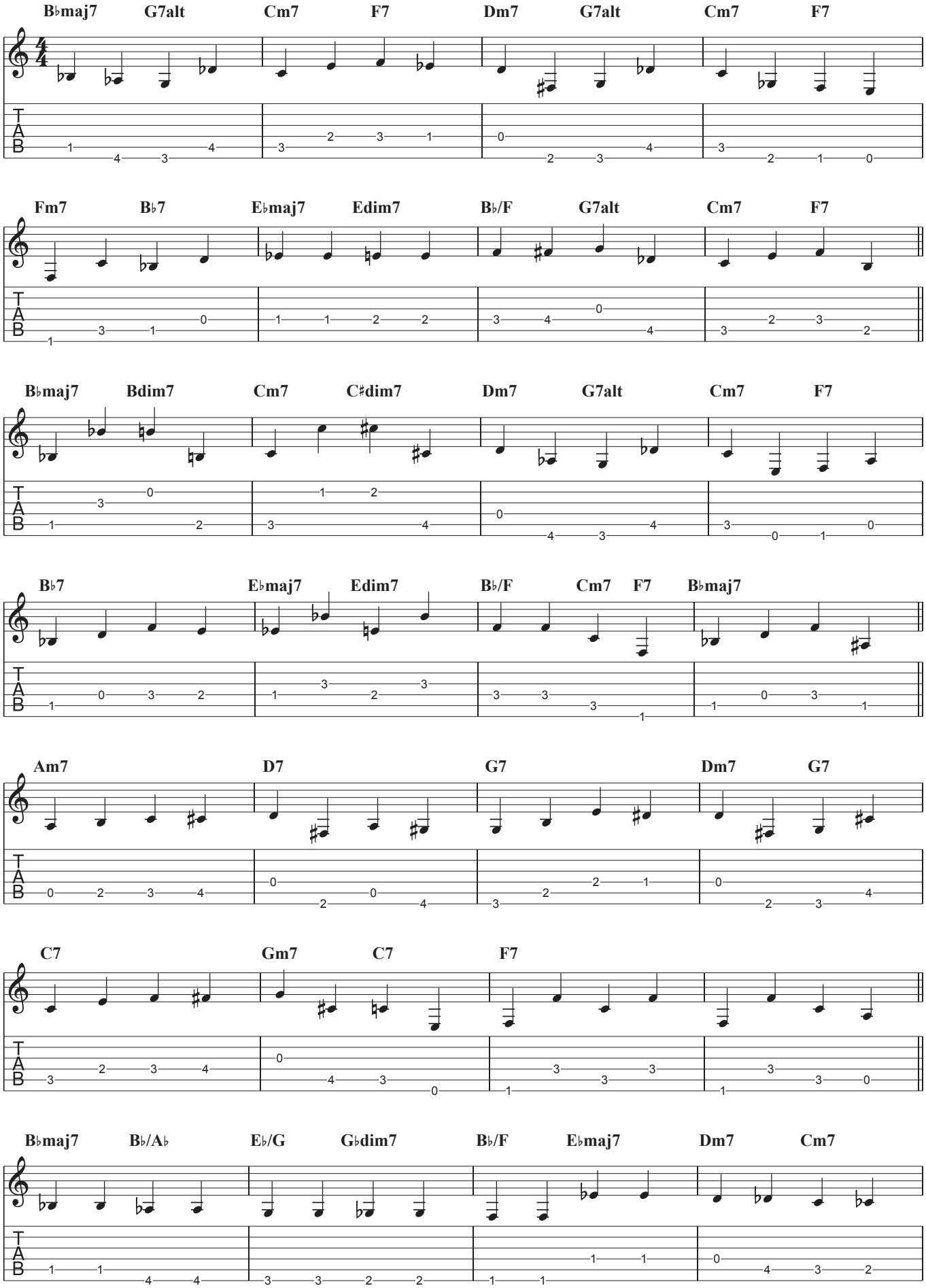
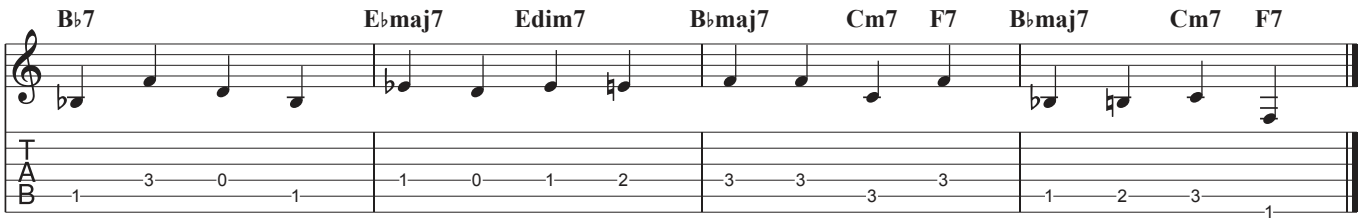
1. in even phase lengths (4,8,12,16 bars)
2. when a particular rhythmic pattern is used throughout.

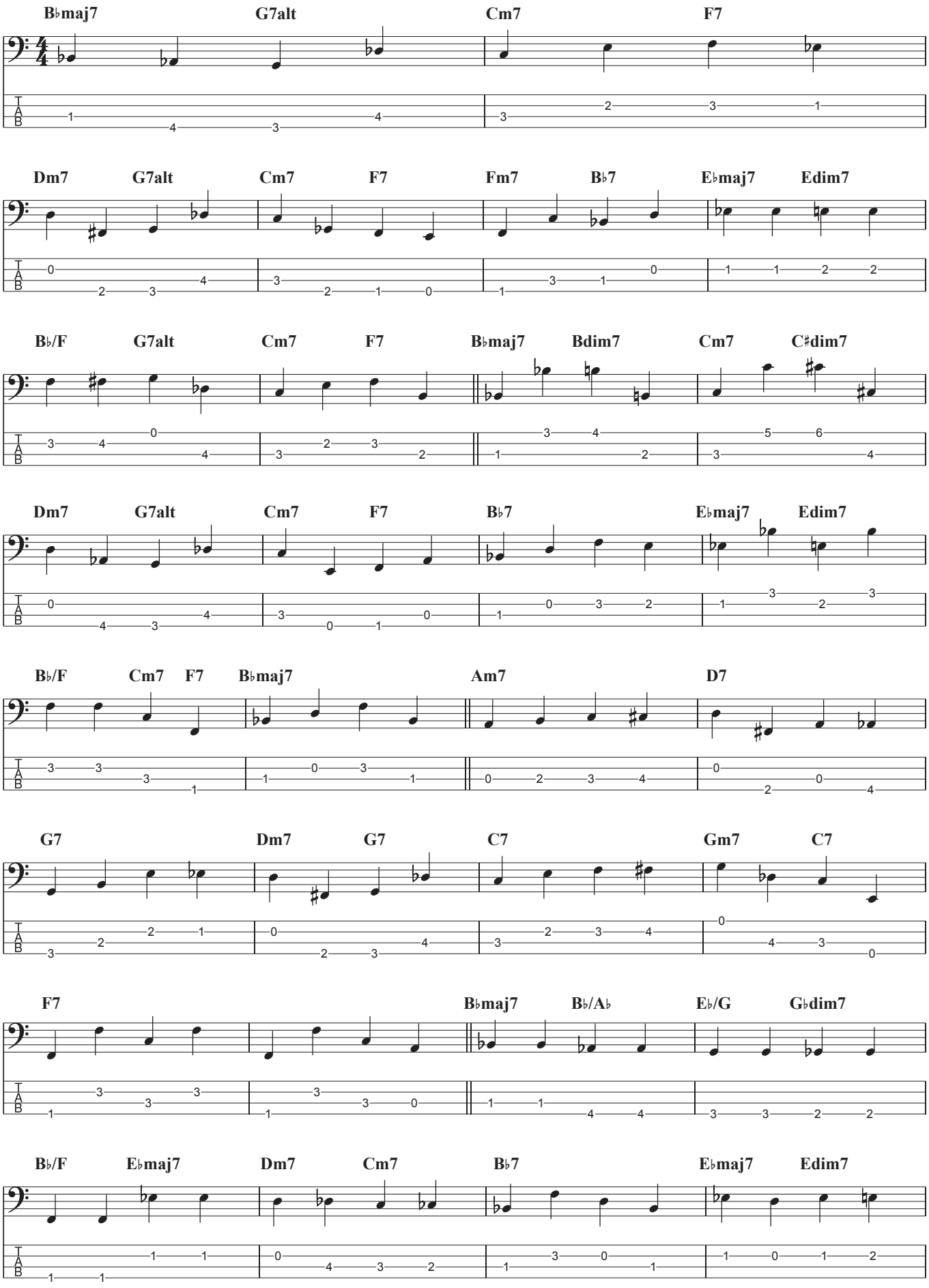
Ex. 252

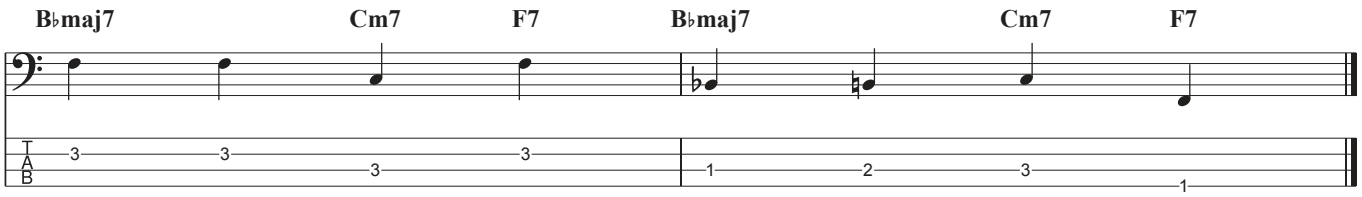
Rhythm Changes



Here is an example of a Rhythm Changes bassline.

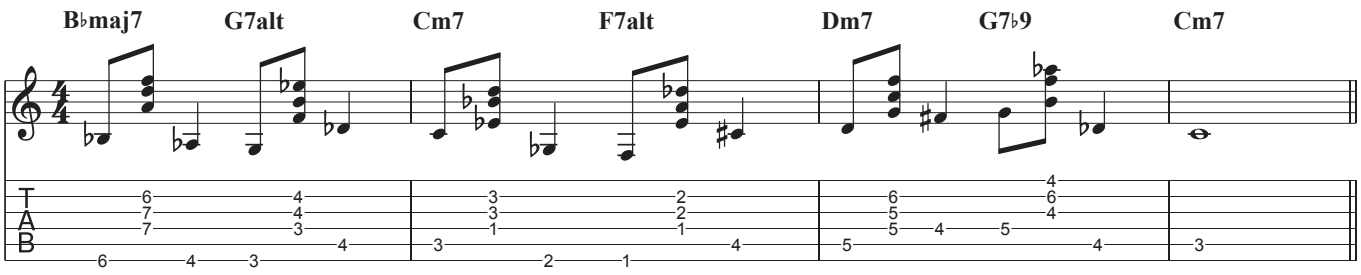
Ex. 253a


B♭7
E♭maj7
Edim7
B♭maj7
Cm7
F7
B♭maj7
Cm7
F7
T
A
B

Ex. 253b

B♭maj7 G7alt Cm7 F7
Dm7 G7alt Cm7 F7 Fm7 B♭7 E♭maj7 Edim7
B♭/F G7alt Cm7 F7 B♭maj7 Bdim7 Cm7 C♯dim7
Dm7 G7alt Cm7 F7 B♭7 E♭maj7 Edim7
B♭/F Cm7 F7 B♭maj7 Am7 D7
G7 Dm7 G7 C7 Gm7 C7
F7 B♭maj7 B♭/A♭ E♭/G G♭dim7
B♭/F E♭maj7 Dm7 Cm7 B♭7 E♭maj7 Edim7

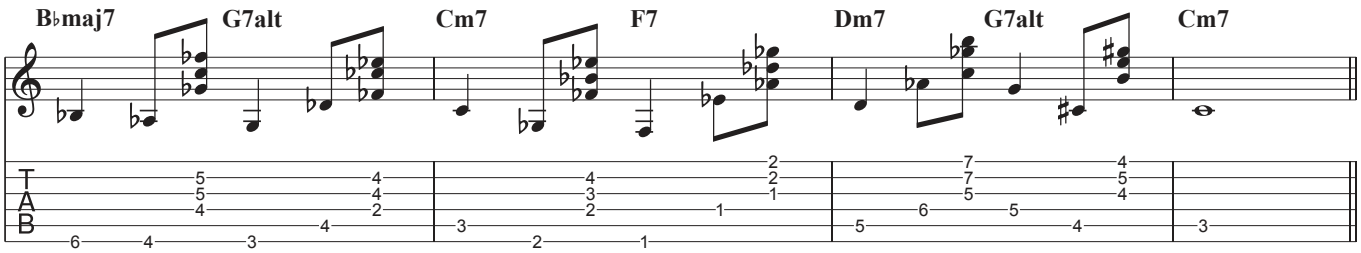

B♭maj7
Cm7
F7
B♭maj7
Cm7
F7
T
A
B
3
3
3
3
1
2
3
1

When walking a bass line, chords can be interspersed to create a more complete harmonic backdrop (especially useful in solo & duo settings).

Ex. 254a

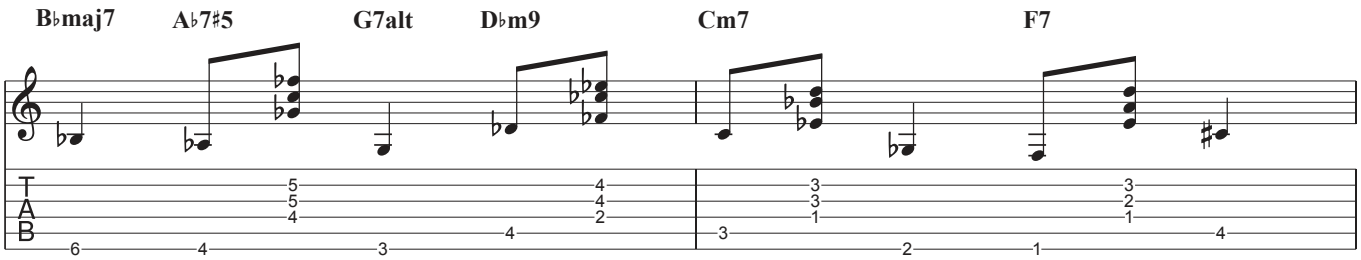


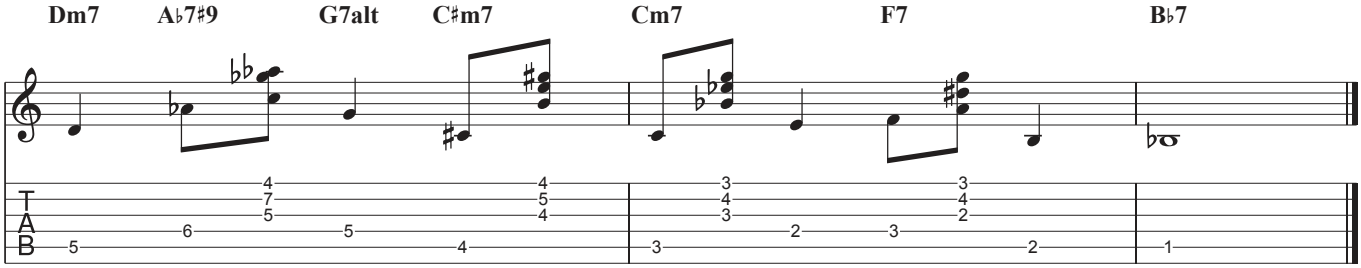
Ex. 254b



Or combination

Ex. 254c

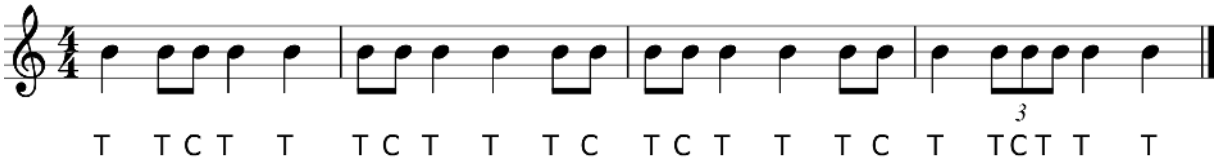




In the last example the patterns were alternated by measure.

Bassline Comping Variations

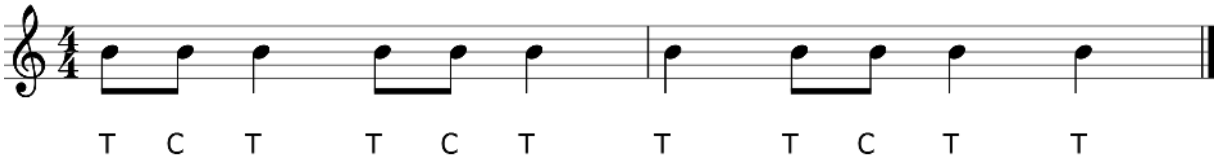
Ex. 255a

Rhythm Pattern No.1

T= Thumb

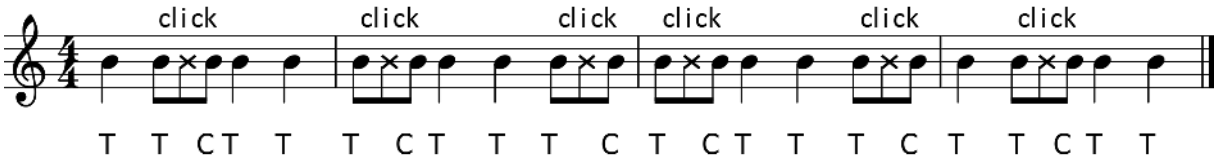
C= Chord

Practice on one chord before moving on to progressions.

Ex. 255b

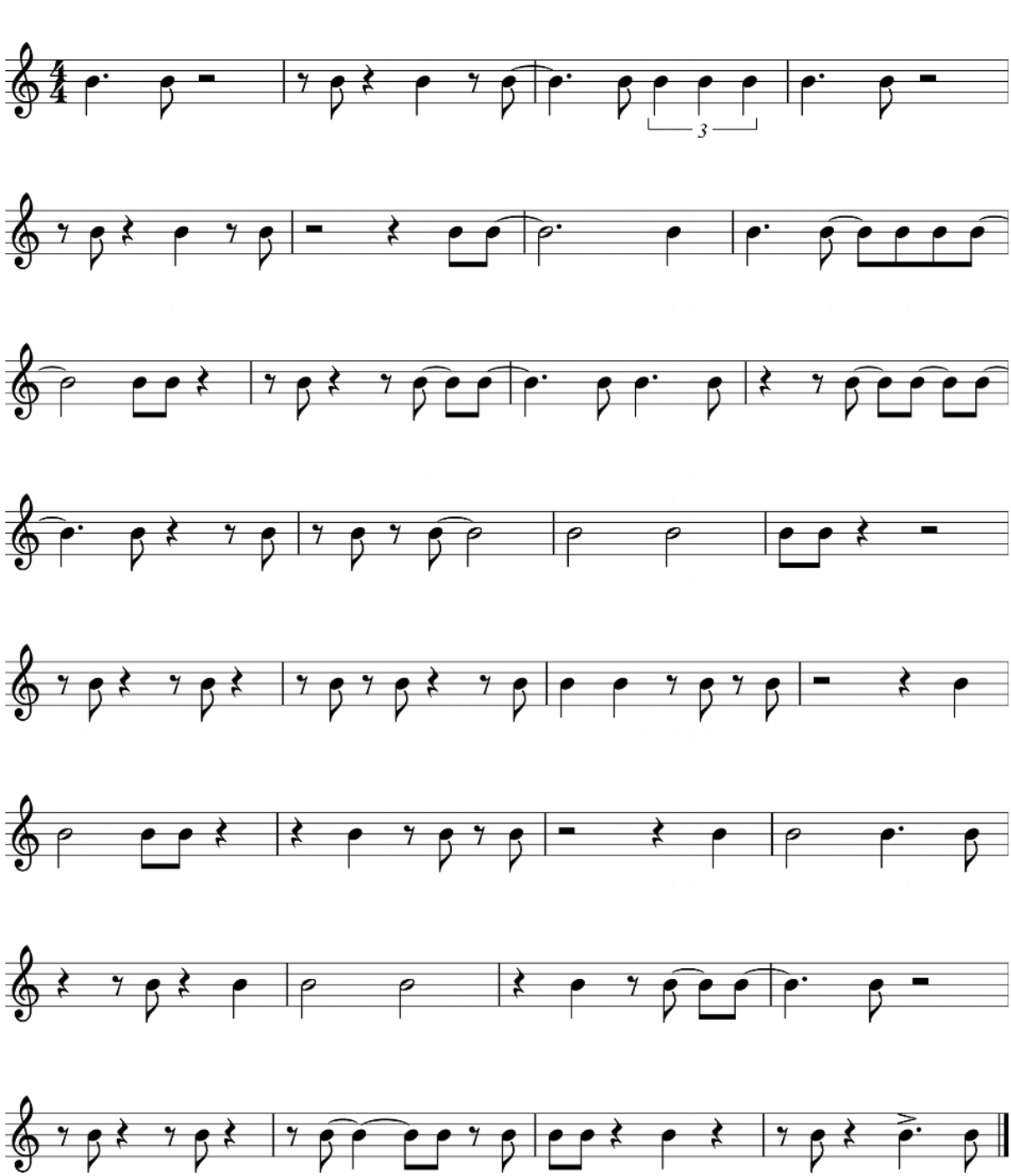
Rhythm Pattern No.2

Ex. 255c

Rhythm Pattern No.3

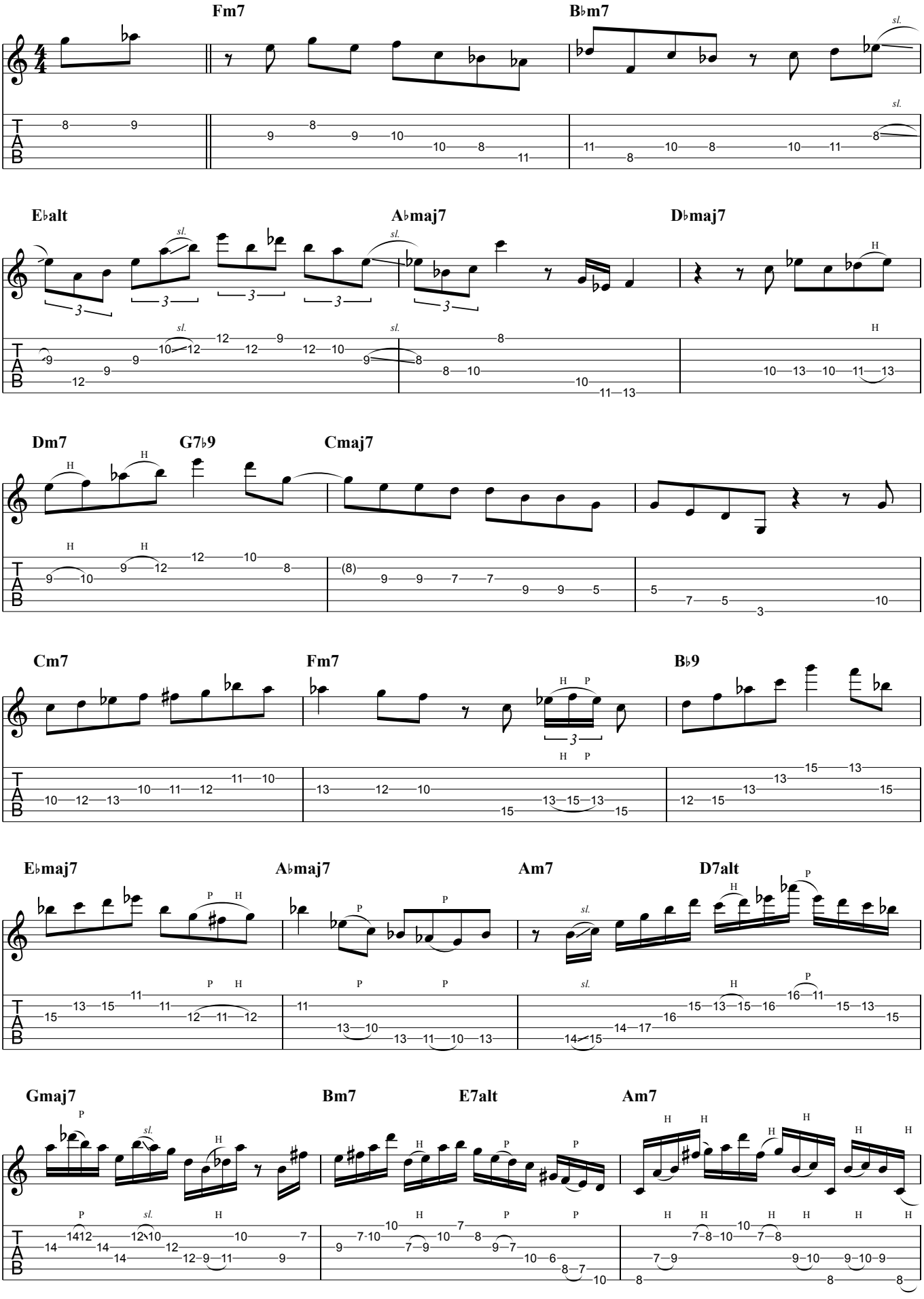
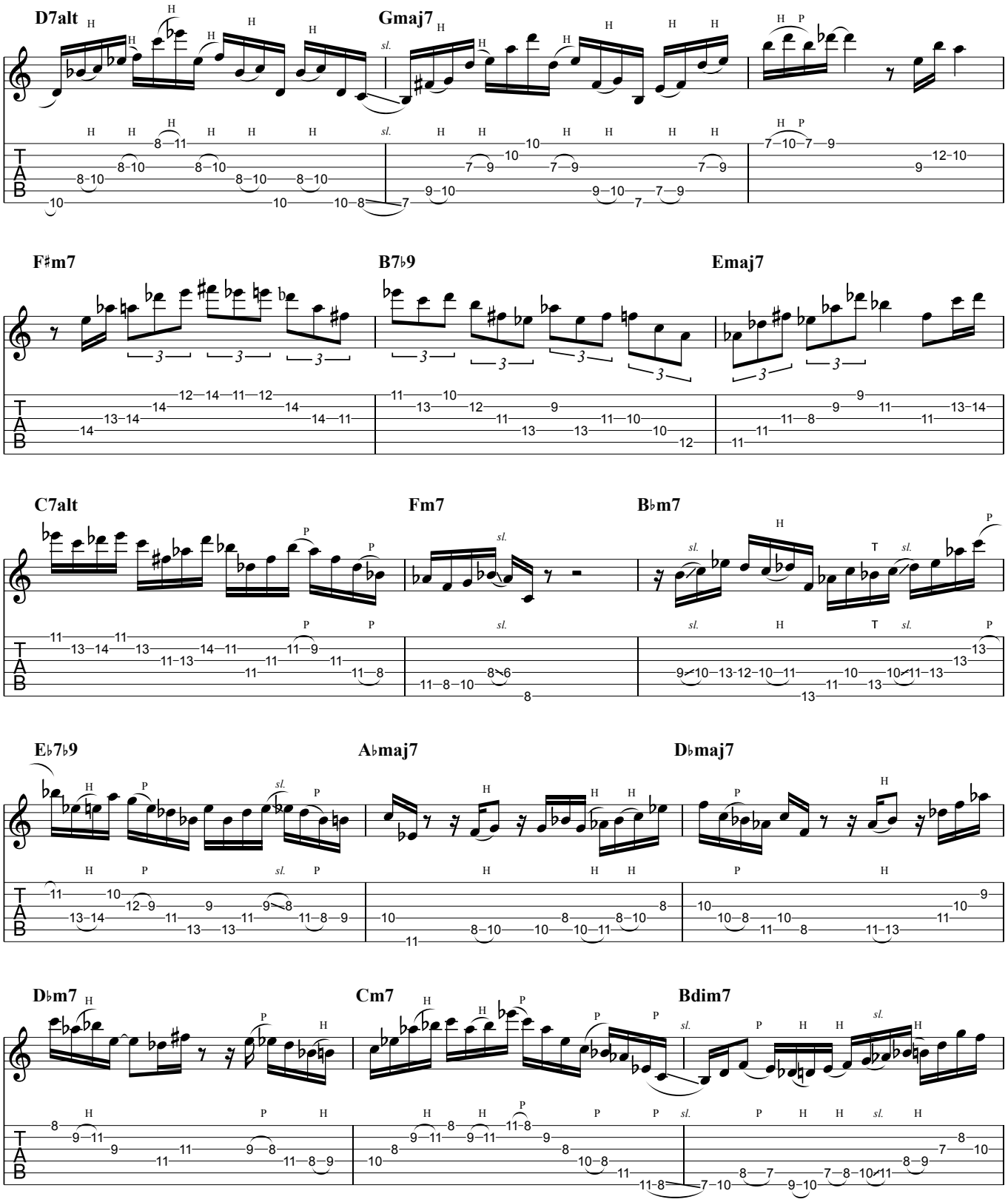


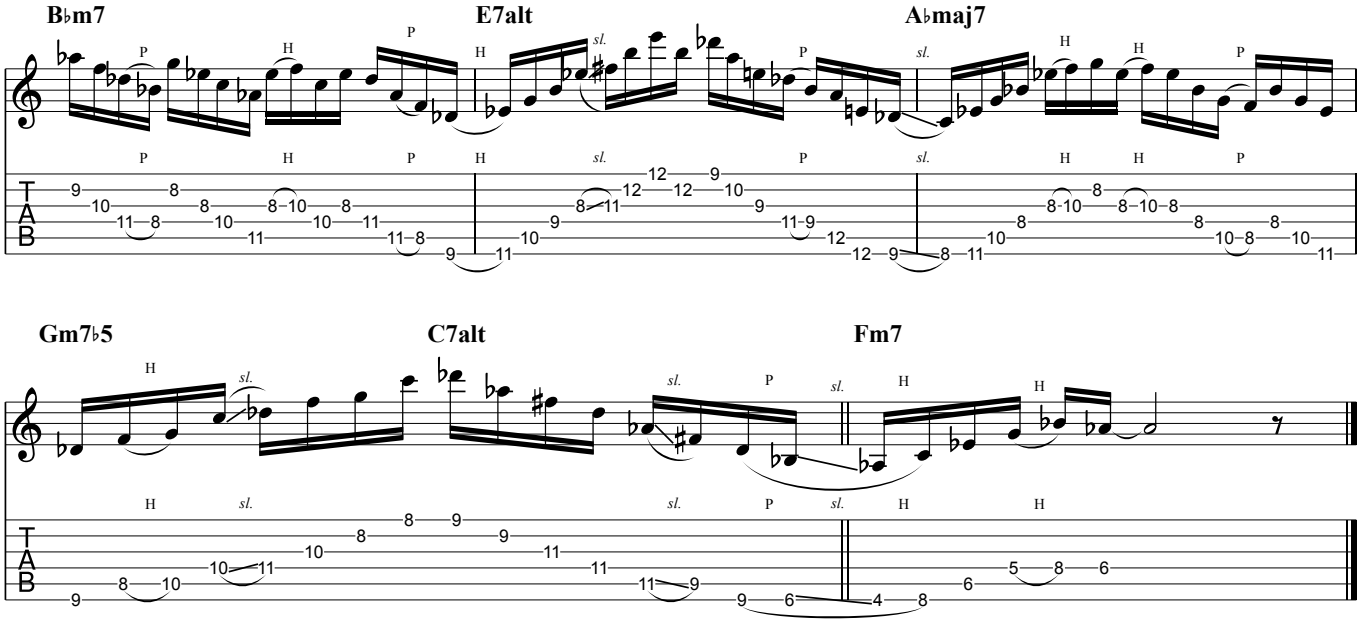
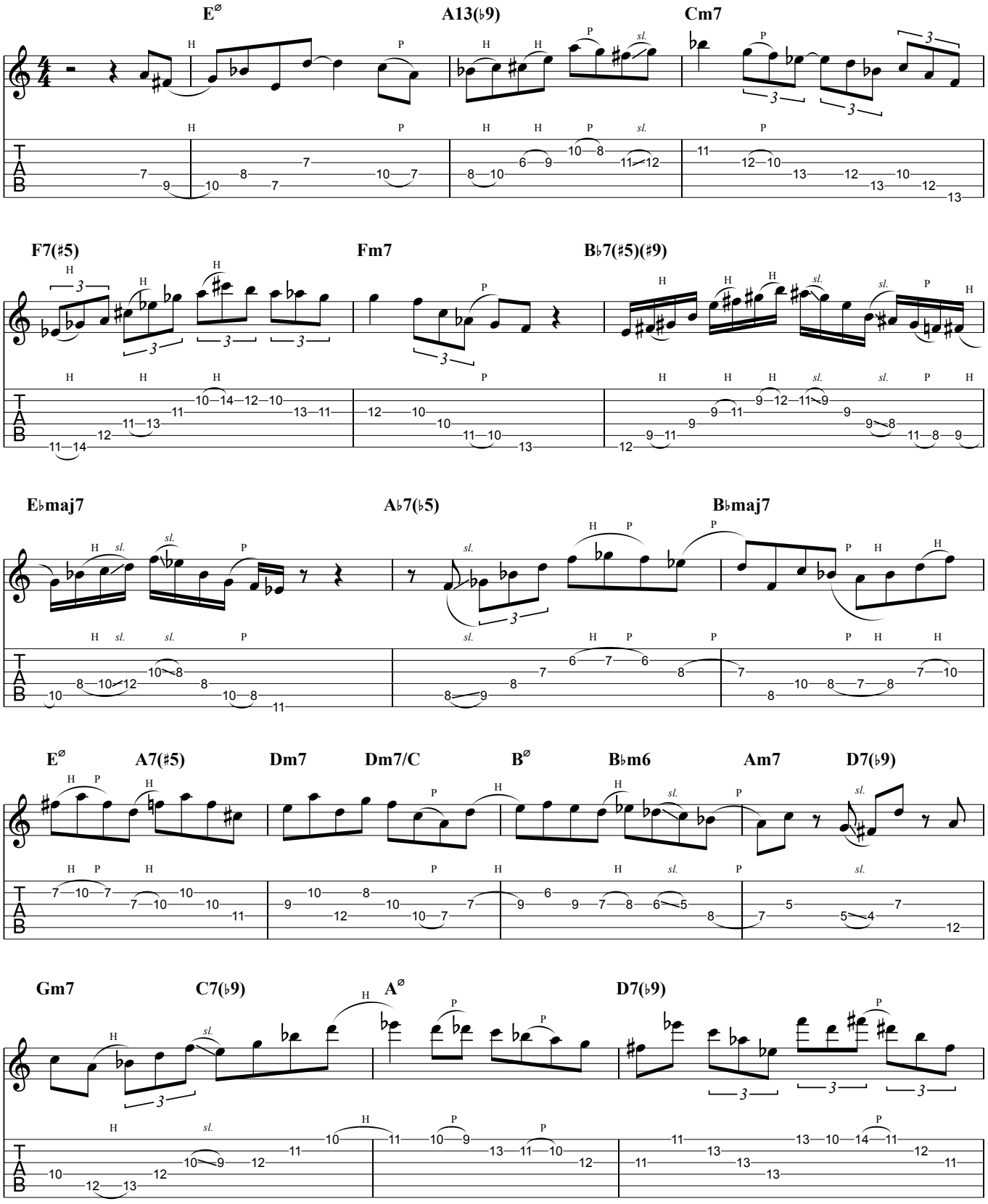
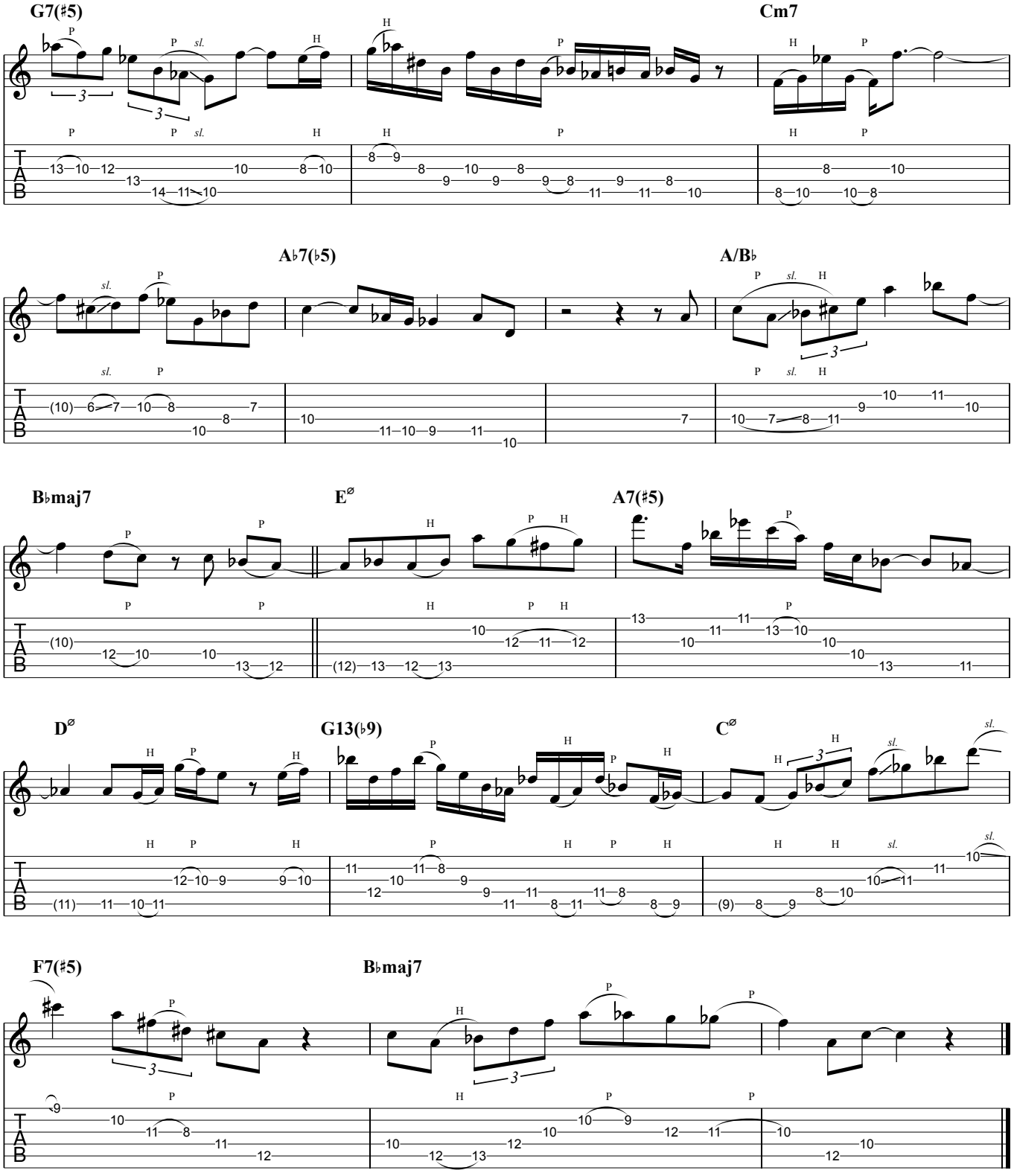
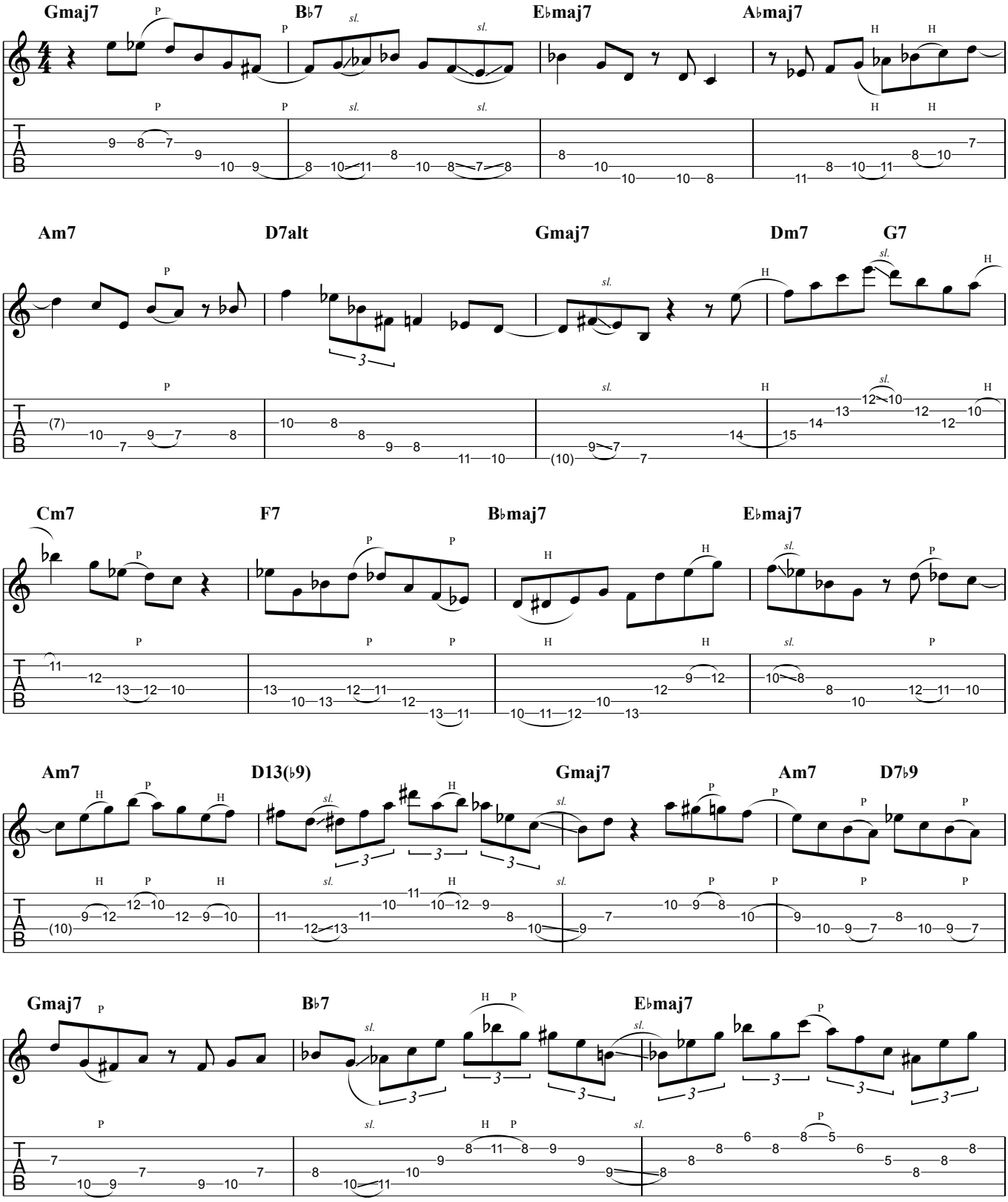
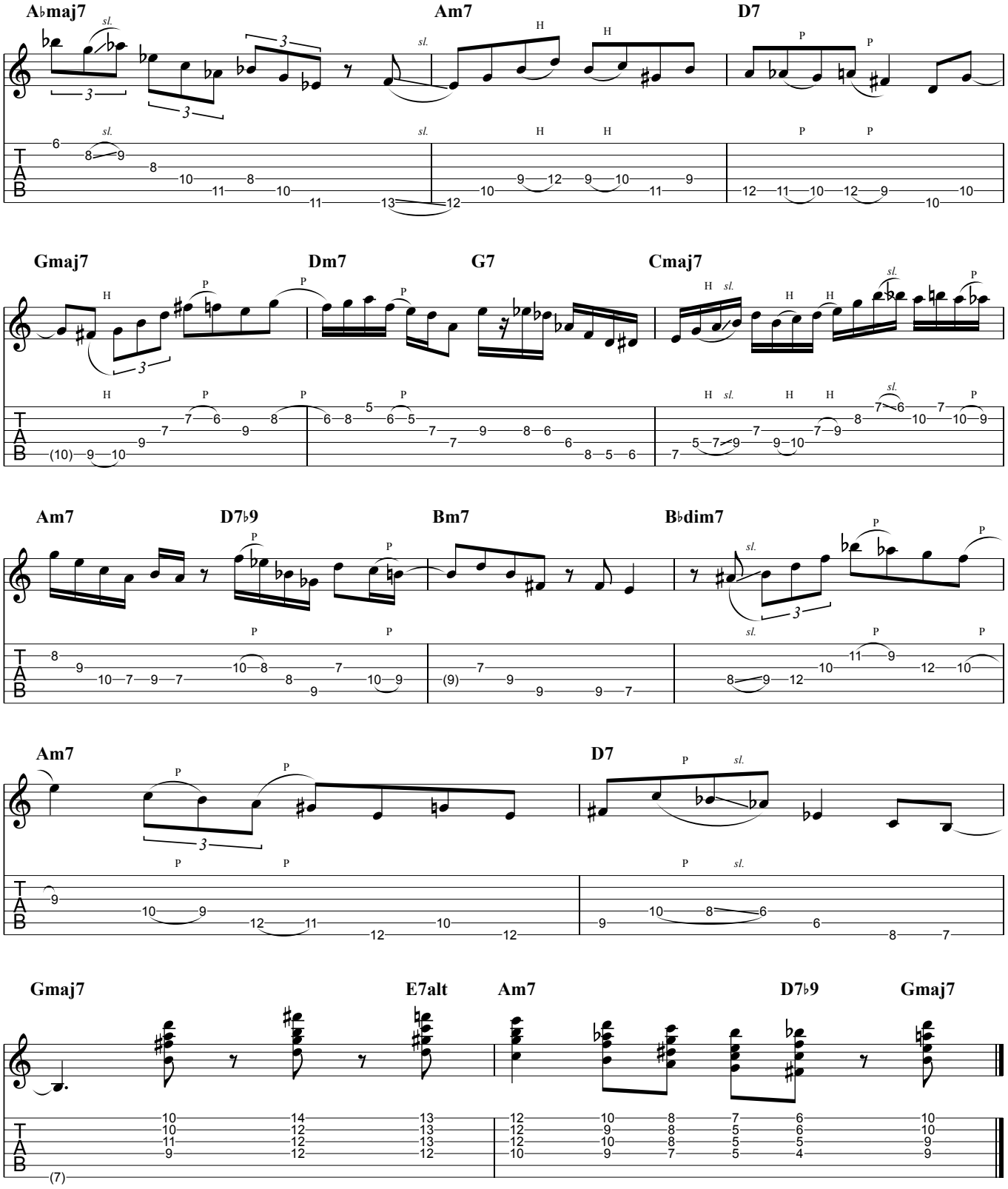
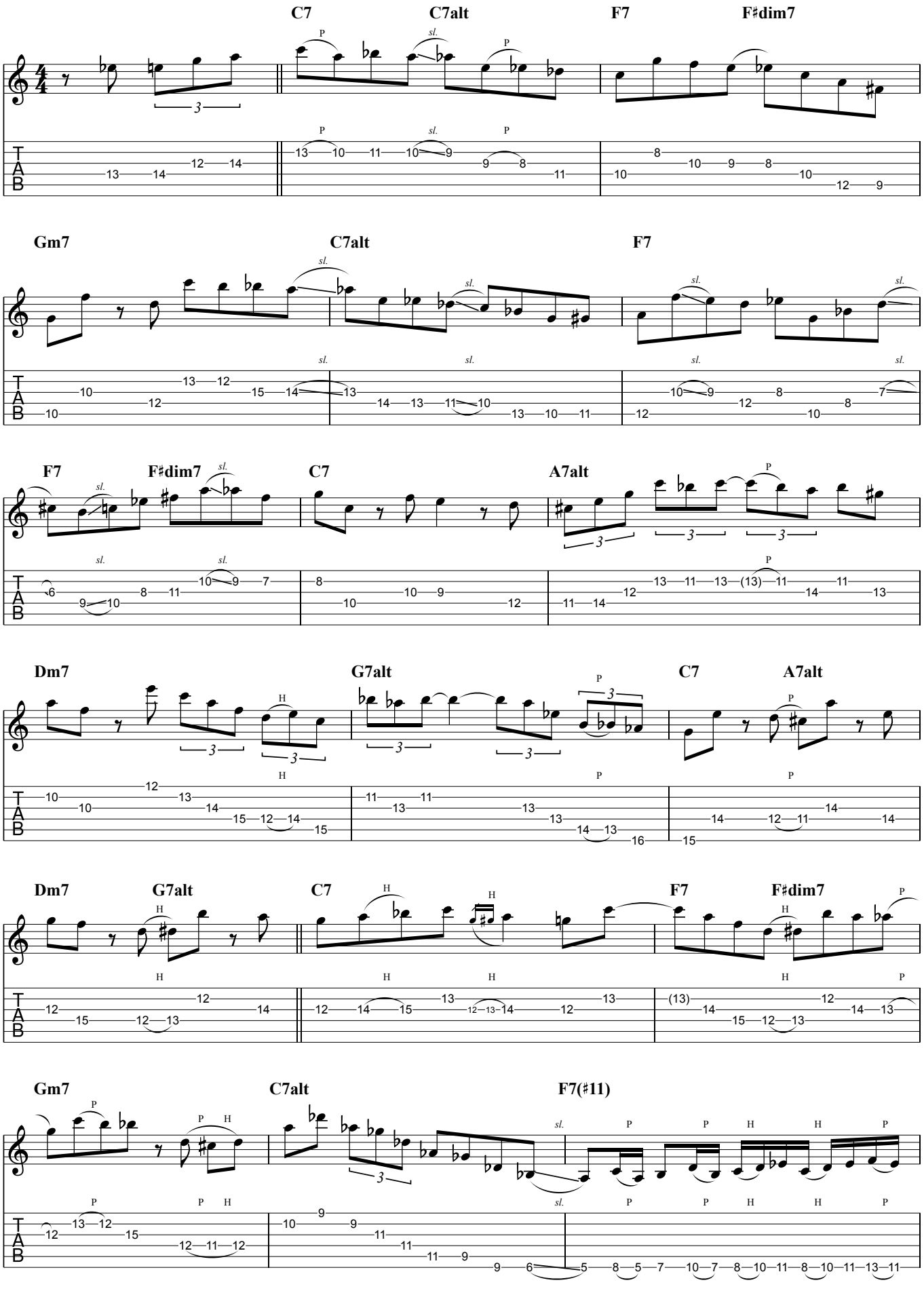
Comping Rhythms

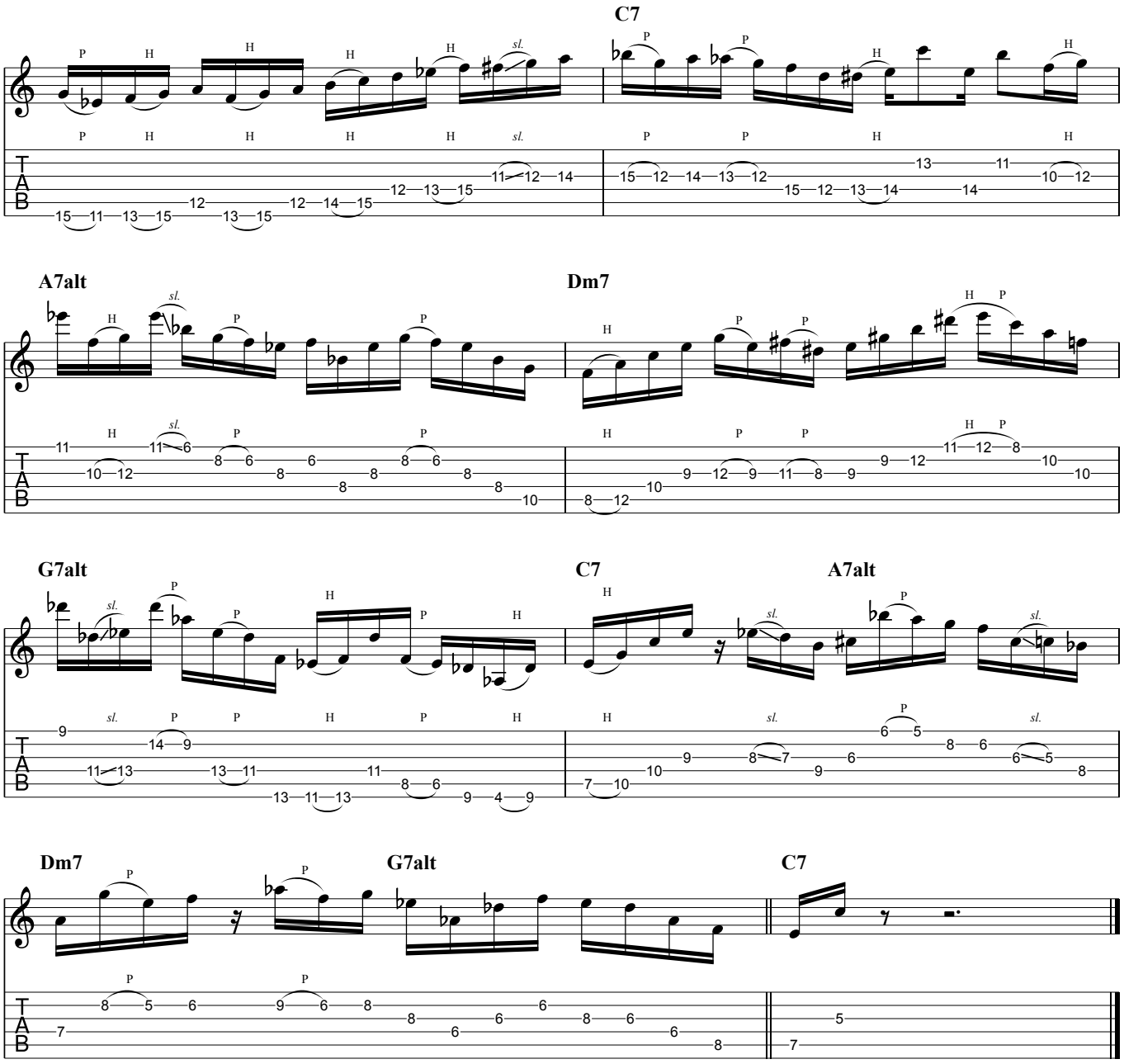
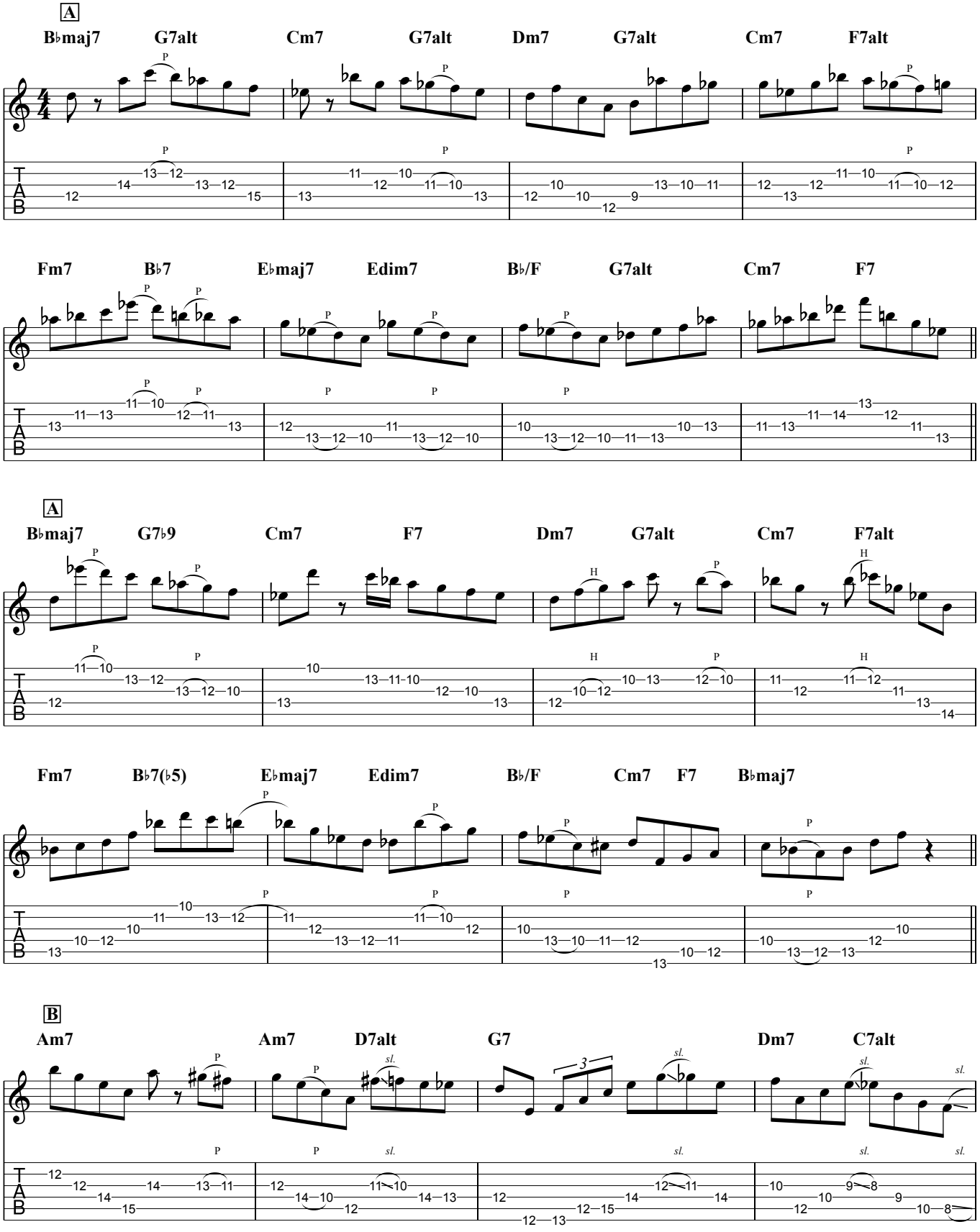
I have written some basic comping patterns over a thirty two bar form. Feel free to divide this into shorter phrases (four and eight bars) for your own use.

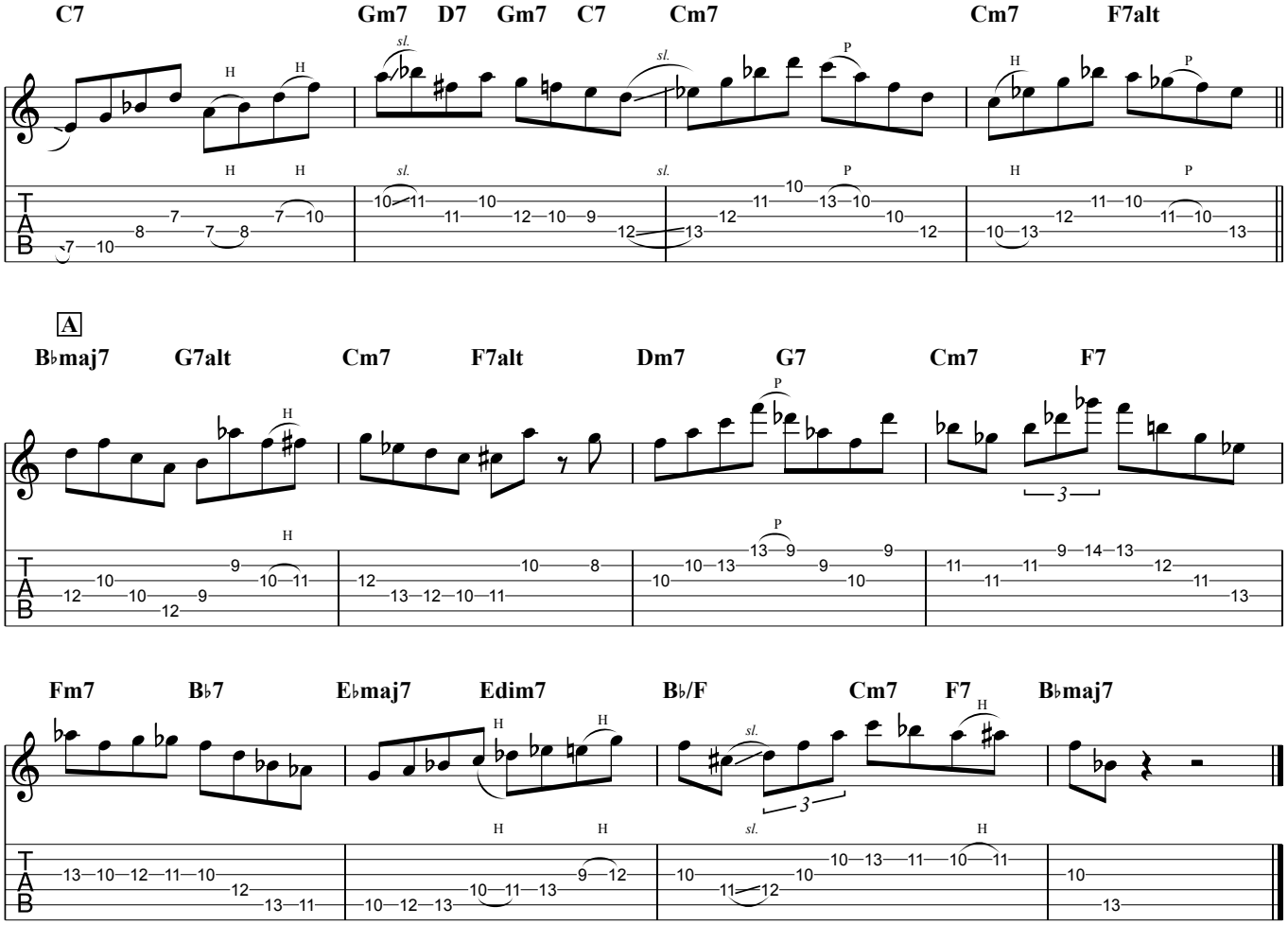
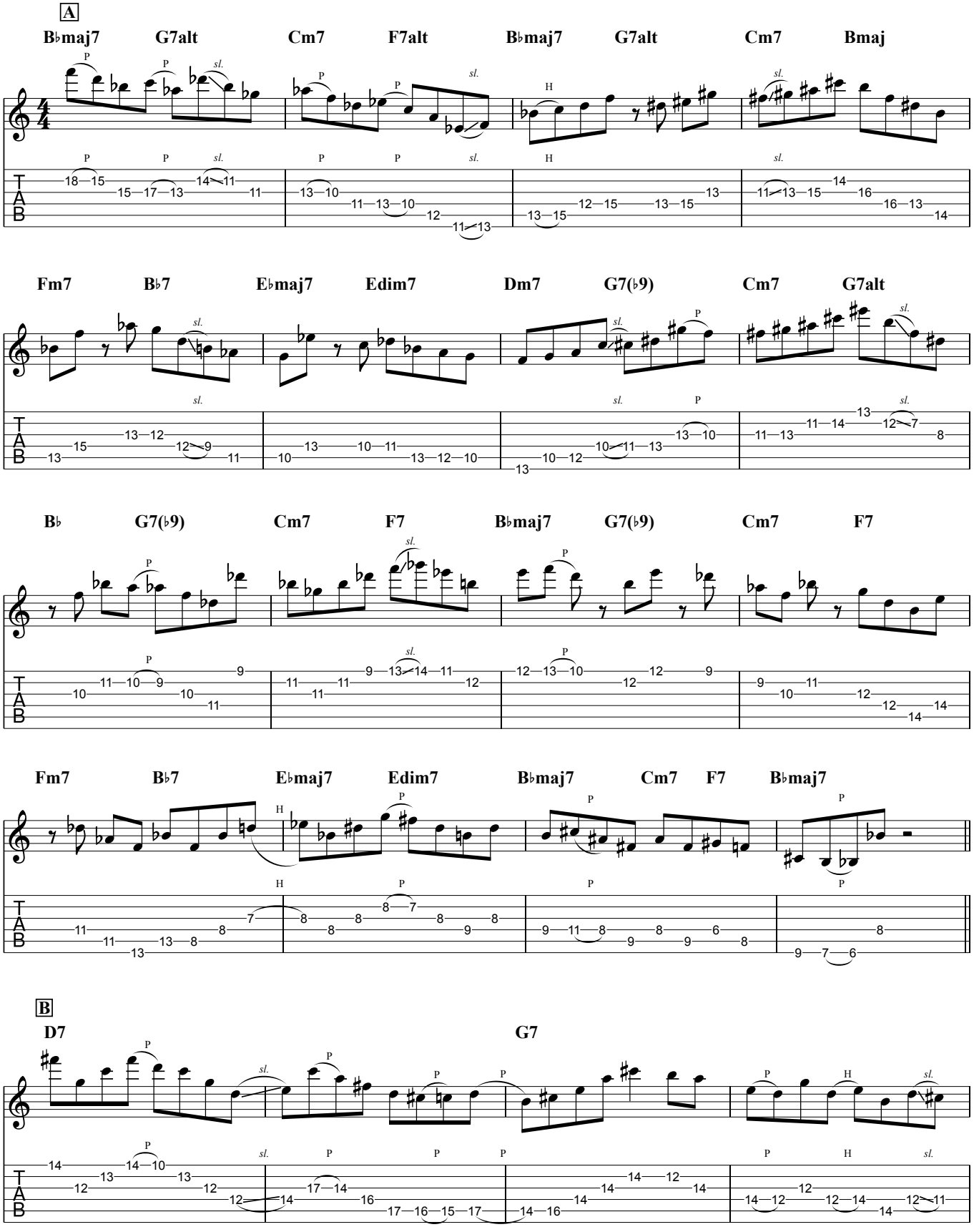
Ex. 256

Solos Over Standard Forms

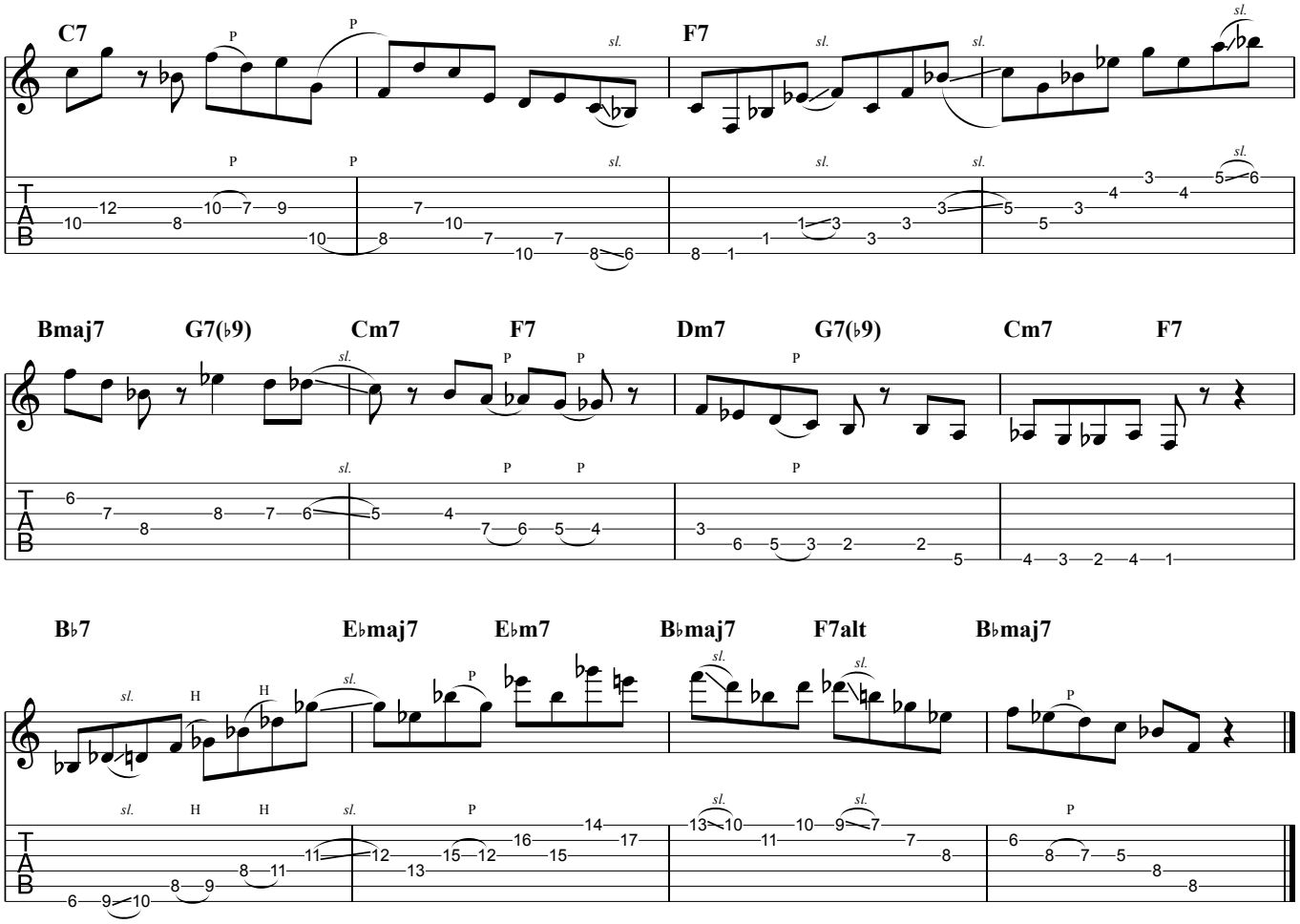
I have included a few solos over standard chord progressions in order to incorporate some of the ideas that we have learned. Practice them slowly with the metronome on two and four to develop your swing feel. I also suggest doing your own harmonic analysis to study the devices used.


Fm7
B♭m7
E♭alt
A♭maj7
D♭maj7
Dm7
G7♭9
Cmaj7
Cm7
Fm7
B♭9
E♭maj7
A♭maj7
Am7
D7alt
Gmaj7
Bm7
E7alt
Am7

D7alt
Gmaj7
F♯m7
B7♭9
Emaj7
C7alt
Fm7
B♭m7
E♭7♭9
A♭maj7
D♭maj7
D♭m7
Cm7
Bdim7


B♭m7
E7alt
A♭maj7
Gm7♭5
C7alt
Fm7
T
A
B

E∅
A13(♭9)
Cm7
F7(♯5)
Fm7
B♭7(♯5)(♯9)
E♭maj7
A♭7(♭5)
B♭maj7
E∅
A7(♯5)
Dm7
Dm7/C
B∅
B♭m6
Am7
D7(♭9)
Gm7
C7(♭9)
A∅
D7(♭9)

G7(♯5)
Cm7
A♭7(♭5)
A/B♭
B♭maj7
E⌀
A7(♯5)
D⌀
G13(♭9)
C⌀
F7(♯5)
B♭maj7

Gmaj7
B♭7
E♭maj7
A♭maj7
Am7
D7alt
Gmaj7
Dm7
G7
Cm7
F7
B♭maj7
E♭maj7
Am7
D13(♭9)
Gmaj7
Am7
D7♭9
Gmaj7
B♭7
E♭maj7

A♭maj7
Am7
D7
Gmaj7
Dm7
G7
Cmaj7
Am7
D7♭9
Bm7
B♭dim7
Am7
D7
Gmaj7
E7alt
Am7
D7♭9
Gmaj7

C7
C7alt
F7
F♯dim7
Gm7
C7alt
F7
F7
F♯dim7
C7
A7alt
Dm7
G7alt
C7
A7alt
Dm7
G7alt
C7
F7
F♯dim7
Gm7
C7alt
F7(♯11)


C7
A7alt
Dm7
G7alt
C7
A7alt
Dm7
G7alt
C7

A
B♭maj7 G7alt Cm7 G7alt Dm7 G7alt Cm7 F7alt
Fm7 B♭7 E♭maj7 Edim7 B♭/F G7alt Cm7 F7
A
B♭maj7 G7♭9 Cm7 F7 Dm7 G7alt Cm7 F7alt
Fm7 B♭7(♭5) E♭maj7 Edim7 B♭/F Cm7 F7 B♭maj7
B
Am7 Am7 D7alt G7 Dm7 C7alt


C7
Gm7 D7 Gm7 C7
Cm7
Cm7 F7alt
A
B♭maj7 G7alt
Cm7 F7alt
Dm7 G7
Cm7 F7
Fm7 B♭7
E♭maj7 Edim7
B♭/F
Cm7 F7
B♭maj7
T
A
B

A
B♭maj7 G7alt Cm7 F7alt B♭maj7 G7alt Cm7 Bmaj
Fm7 B♭7 E♭maj7 Edim7 Dm7 G7(♭9) Cm7 G7alt
B♭ G7(♭9) Cm7 F7 B♭maj7 G7(♭9) Cm7 F7
Fm7 B♭7 E♭maj7 Edim7 B♭maj7 Cm7 F7 B♭maj7
B
D7 G7


C7
F7
Bmaj7
G7(♭9)
Cm7
F7
Dm7
G7(♭9)
Cm7
F7
B♭7
E♭maj7
E♭m7
B♭maj7
F7alt
B♭maj7

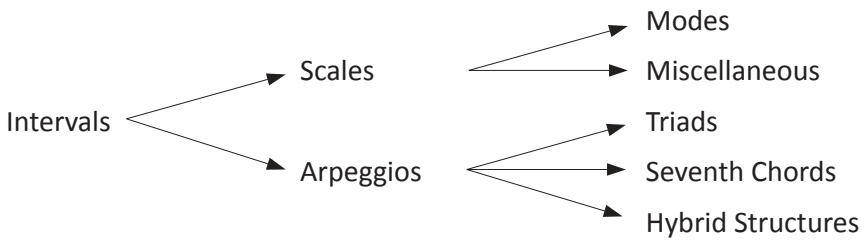
Practicing

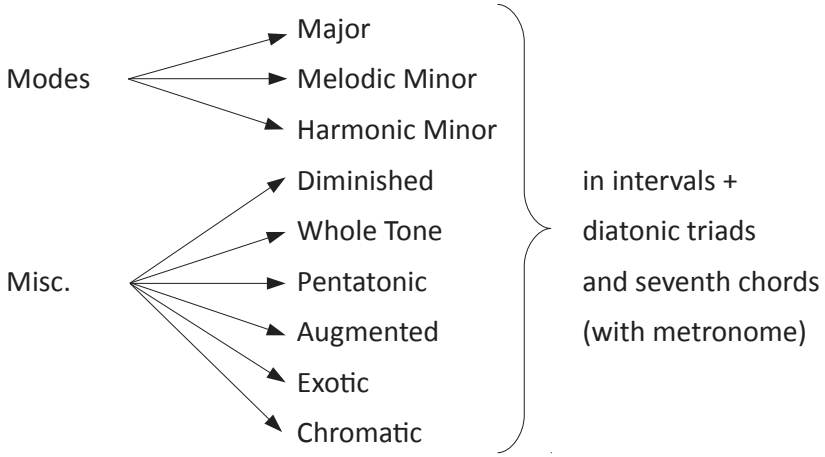
Here are some important points for practicing:

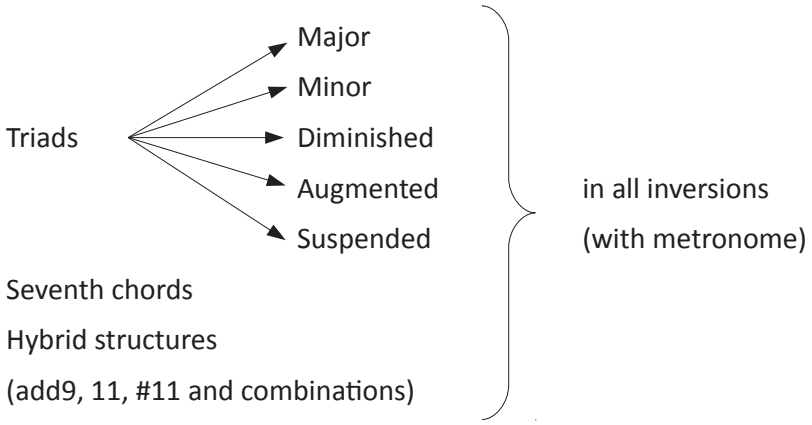
- Use your time effectively (don´t practice things that you already know).
- Divide your practice time by percentages:

1. Technique	=	15%
2. Ear Training	=	15%
3. Review	=	35%
4. New Material	=	35%
		100%

Technique







Left Hand

Finger independence drills (1,2,3,4, finger combinations)

1	2	3	4	2	3	4	1	3	4	1	2	4	1	2	3
1	2	4	3	2	3	1	4	3	4	1	2	4	1	3	2
1	3	2	4	2	1	3	4	3	2	1	4	4	2	3	1
1	3	4	2	2	1	4	3	3	2	4	1	4	2	1	3
1	4	3	2	2	4	3	1	3	1	2	4	4	3	1	2
1	4	2	3	22	4	1	3	3	1	4	2	4	3	2	1

Right Hand

I. w/pick
    1. Rhythms w/ alternate picking
    2. sweep picking
    3. cross string exercises

II. w/fingers
    1. Arpeggios studies
    2. MI alteration on scales
    3. tremolo studies IMA

Technique should be broken down into old/new material.

Review Material

- Consisting of things practiced in the past week
- Transcribed solos, new lines, voicings, tunes, compositions, etc.
- Keep a log to prevent the loss of previously learned material. Write down how, when and what was practiced using musical notation or chord diagrams when needed. Record any details such as fingerings and position changes

New Material

- New fingerings, chord substitutions, lines, transcribed solos, chord voicings, new techniques, phrasing, sight reading, etc.
- This Book!

Ear Training

- Transcribe solos
- Harmonic and melodic interval dictation (w/ a partner or tape)
- Melodic dictation
- Chord identification (EX. $Maj^{7/\#5}$)
- Mode identification (dorian, altered dominant for example)
- triad over bass note identification (Ex. Triad w/ b9 in the bass C/Db)
- Sight singing
- Play a note and sing an interval or scale above or below it
- Play three notes and identify intervals contained
- Play a chord and sing a line over it. Then write down the intervals you sang
- Try to write down harmonic progressions off recordings without your instrument (and with)
- Try to sing your favorite tunes in their original keys without your instrument and then check yourself to see if you are in the correct key
- Work on recognizing the exact pitch of a note (pitch color). You will find it easier on guitar than on a foreign instrument
- GOOD LUCK!

Rick Beato is a musician, teacher and father of three. He has a B.M. in Music Education and an M.M. in Jazz Performance from the New England Conservatory of Music. He is a record producer with numerous Platinum selling records and also co-wrote the RIAA certified Platinum selling song “Carolina” with Parmalee.
In late 2013, “Carolina” climbed to Number One for 2 weeks on the Billboard Country Music Charts.

He is currently the co-founder of Intrvyl LLC a music education company which has developed The Beato Ear Training Program a relative pitch training platform for adults. In the past years he has grown his “Everything Music” Youtube Channel to 1.5 Million Subscribers and over 175 Million views.

www.rickbeato.com

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